

CMOS DATA CONVERTERS FOR COMMUNICATIONS

**Mikael Gustavsson, J. Jacob Wikner
and Nianxiong Nick Tan**

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CMOS DATA CONVERTERS FOR COMMUNICATIONS

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*To the late Vanja Gustavsson
Mikael Gustavsson*

*To Ulrica and my family
J Jacob Wikner*

*To Regina and Diana
N. Nick Tan*

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PREFACE

“In the telecommunication era, analog design is not just a matter of isolated self-satisfying circuit tweaking, it is above all a system planning.”

Digital radio and high-speed internet access have created a great demand on high-performance data converters. The conventional view of data converters as numerical conversion blocks does not suffice for this kind of communication applications. From a communication system's perspective, the bit error rate (BER) requirement determines the signal-to-noise ratio (SNR) requirement for a given modulation and duplexing scheme. Therefore, when we design data converters for a communication system, frequency-domain measures such as the SNR are of great importance. In theory, any known distortion components in a transmit channel of a communication system can be corrected by measuring it through the use of a pilot signal or during the initialization phase. But in practice, it is difficult to predict the distortions due to the time-varying interference of adjacent channels in most communication systems. Therefore, the distortion and inter-modulation of data converters are of great importance as well. The use of discrete multi-tone (DMT) modulation for digital subscriber line (DSL) and the use of orthogonal frequency division multiplexing (OFDM) for wideband radio further complicate the data converter design in that even the single-tone frequency-domain measures do not suffice.

Most universities teach analog courses as circuit design courses with little interaction with communication or signal processing systems. Most analog designers can only design circuits specified by system engineers without much understanding of the system, and most system engineers have not much knowledge about analog imperfections. This combination usually results in non-optimal system solutions. If analog designers with the system knowledge participate in the system specification and system partitioning, we can really design the optimal system with judicious trade-offs between analog and digital, software and hardware. By realizing the industrial and educational need of analog designers with system knowledge, the Microelectronics Research Center of Ericsson started analog activities with close cooperations with Linköping University in 1995. One of the results is this book.

CMOS data converters for communications addresses analog design for communication systems. It distinguishes itself from other data converter books by emphasizing system-related aspects of data converter designs and frequency-domain performance of data converters. However, this is not a book written by system designers rather than by circuit designers. It therefore also covers circuit designs and implementations of

CMOS data converters, based on measured chips.

CMOS data converters for communications bridges the gap between the communication system requirements and the mixed signal design. It derives data converter requirements from the communication system specifications such as line codes, modulation schemes, and duplexing methods, etc. It also gives examples of relating asymmetrical DSL (ADSL) and wideband radio systems to the data converter requirements.

CMOS data converters for communications can be used as a reference book by analog circuit designers to understand the data converter requirements for communications. It can also be used by communication system designers to understand the difficulties of certain performance requirements on data converters. This book reflects the authors many year philosophy in both industrial practice and academic education. In the telecommunication era, analog design is not just a matter of isolated self-satisfying circuit tweaking, it is above all a system planning. To prepare analog students for the new challenge, this book can be an excellent resort.

This book consists of 13 chapters.

In chapter 1, we discuss the characterization of data converters with emphasis on frequency-domain measures. It also discusses the multi-tone power ratio (MTPR) measure for DMT or OFDM applications. It is shown that multi-carrier modulations usually have a higher dynamic range requirement on data converters due to the instantaneous summation of many carriers resulting in a large peak to average ratio (PAR).

In chapter 2, we first briefly discuss the fundamentals of digital communication systems and introduce the relevant terminologies such as bit error rate (BER), line codes, modulation schemes, and duplexing methods, etc. Then we derive the minimum data converter requirements as a function of the bit error rate, the line code, and the modulation. The influence of duplexing methods is discussed as well. By considering the transmit and receive power and the background noise/interference power, we also derive the optimum data converter requirements for ADSL and wideband radio systems.

Chapters 3 and 4 are devoted to the overview of data converter architectures. These two chapters cover the most data converter architectures and discuss their suitabilities for communications. Chapter 3 presents an overview of high-performance analog-to-digital converters (ADCs) while chapter 4 presents an overview of high-performance digital-to-analog converters (DACs).

Chapters 5~8 are devoted to analog circuits and circuit techniques before we discuss data converter designs and implementations.

Chapter 5 briefly compares the two existing circuit techniques, the switched-capacitor (SC) and switched-current (SI) technique. Although the SI technique offers the simplicity advantages among others, it is advised to use the SC technique for high-performance ADCs due to the higher dynamic range and lower distortion of SC circuits.

For most ADCs, sample-and-hold (S/H) amplifiers have a great impact on the frequency-domain performance such as the SNR and distortion, especially for sub-sampled

ADC systems where the signal frequency is beyond the Nyquist bandwidth determined by the sampling rate. S/H amplifiers are a special case of multiplying DACs used in pipelined ADCs and a special case of integrators used in oversampling ADCs. Chapter 6 discusses the design and characterization of these functional blocks such as multiplying DACs and integrators.

Chapter 7 discusses the circuit building blocks needed for data converters including operational amplifiers (opamps), comparators, and references, etc. It highlights how to design high-gain high-speed opamps and fast low-power comparators. It also covers common-mode feedback (CMFB) techniques and practical issues such as bandgap reference generation and reference buffering.

Chapter 8 is an extension of chapter 7, focusing on low-voltage building blocks in order to operate analog circuits at a supply voltage of 2.5 V or even lower. It includes low voltage (2.5 V) high-speed opamps, voltage doublers, and high-voltage generators, etc.

Chapters 9~11 deal with ADC designs and implementations.

Chapter 9 is devoted to pipelined ADCs, from architecture to design and implementation. It discusses different pipelining techniques, different coding techniques, digital correction techniques, and digital calibration techniques.

Chapter 10 is about how to increase the speed of ADCs by time-interleaving or parallelism. The major limitation of time-interleaved ADCs is the phase skews in the different channels which introduces distortion. A global S/H circuit at the input would eliminate this limitation but the speed and accuracy would be limited by the opamp used in the S/H circuit. Besides discussions on the conventional methods, it also presents a passive global sampling technique to increase the speed beyond the opamp limitation and reduce the phase skew distortion by 8~10 dB.

Chapter 11 covers oversampling ADCs. It discusses different trade-offs between single-stage and multi-stage architectures, between single-bit and multi-bit quantizers. A design example of a fourth-order oversampling ADC is also presented. It highlights the architecture selection as well as the SC design and implementation.

Chapter 12 discusses the modeling of Nyquist DACs for communications. Through this modeling, it is possible to relate the frequency-domain requirements to actual design parameters and therefore provide a guideline for circuit designers.

Chapter 13 presents the actual design and implementation of a current-steering DAC chipset, including high-speed (> 50 MHz) and ultra-low voltage (1.5 V) chips. Based on the discussed architecture and improved current source matching, it is demonstrated possible to deliver a spurious dynamic range (SFDR) over 75 dBc in standard digital CMOS process and achieve an SFDR over 65 dBc with a single 1.5-V supply.

This book is based on the analog research conducted at the Microelectronics Research Center of Ericsson and at the Linköping University. First we would like to thank Dr. Gunnar Björklund, director of the Microelectronics Research Center for initializing and supporting the analog activities at the research center. We would also like to thank

Prof. Lars Wanhammar for his ‘digital’ support of the analog activities in his group and for educating and helping us in different aspects of digital signal processing. We would like to thank many of our colleagues at Ericsson and at Linköping university for their contributions. Dr. Bengt Jonsson, Helge Stenström, and Dr. Svante Signell of Ericsson Radio Systems contributed significantly by working with us on ADC projects. We appreciate the help from Peter Petersson of Ericsson Radio System in measuring some of the DAC chips. The help from Pierre Dalheim-Lander, Niklas U. Andersson, and K. Ola Andersson in DAC implementations was very valuable. We thank Yonghong Gao of Royal Institute of Technology, Sweden, for the help in the design of oversampling DACs. We would like to acknowledge the invaluable discussions with Marc Delvoux of GlobeSpan, Inc., concerning general digital transmission techniques. We also would like to thank Anders Ihlström of GlobeSpan, Inc., for valuable discussions on oversampling ADC designs.

Last but not the least, we would like to thank our families for their love, patience, and support.

ABBREVIATIONS AND ACRONYMS

2B1Q	Two Binary 1 Quaternary
$\Sigma\Delta$	Sigma-Delta
A/D	Analog to Digital
ADC	Analog-to-Digital Converter
ADSL	Asymmetrical digital subscriber line
AGC	Automatic Gain Control
ASK	Amplitude-Shift Keying
AWGN	Additive White Gaussian Noise
BER	Bit Error Rate
BP	Band pass
BPSK	Binary Phase-Shift Keying
CAP	Carrierless Amplitude Phase (modulation)
CFT	Clock Feedthrough
CNR	Carrier-to-Noise Ratio
CMFB	Common-Mode Feedback
D/A	Digital to Analog
DAC	Digital-to-Analog Converter
DEM	Dynamic Element Matching
DMT	Discrete Multi-Tone
DNL	Differential Non-Linearity
DR	Dynamic Range
DSL	Digital Subscriber Line
EC	Echo Cancellation
ENOB	Effective Number Of Bits
ERB	Effective Resolution Bandwidth
FDD	Frequency-Division Duplexing
FDM	Frequency-Division Multiplexing
FIR	Finite-length Impulse Response
FM	Frequency Modulation
FSK	Frequency-Shift Keying
GMSK	Gaussian Minimum-Shift Keying

GSM	Global System for Mobile Communication
HDSL	High data-rate Digital Subscriber Line
HDTV	High-Definition Television
HP	High Pass
IF	Intermediate Frequency
IIR	Infinite-length Impulse Response
IMD	Inter-Modulation Distortion
INL	Integral Non-Linearity
ISDN	Integrated Service Digital Network
LP	Low Pass
LSB	Least Significant Bit
MDAC	Multiplying Digital-to-Analog Converter
MSB	Most Significant Bit
MTPR	Multi-Tone Power Ratio
OFDM	Orthogonal Frequency Division Multiplexing
Opamp	Operational Amplifier
OSADC	Oversampling Analog-to-Digital Converter
OSDAC	Oversampling Digital-to-Analog Converter
PAM	Phase Amplitude Modulation
PAR	Peak-to-Average Ratio
PCM	PulseCode Modulation
PGA	Programmable Gain Amplifier
PSD	Power Spectral Density
PSK	Phase-Shift Keying
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase-Shift Keying
RF	Radio Frequency
RSD	Redundant Signed Digit
S/H	Sample-and-Hold
SC	Switched Capacitor
SDD	Space-Division Duplexing
SFDR	Spurious-Free Dynamic Range
SFG	Signal Flow Graph
SI	Switched Current
SNR	Signal-to-Noise Ratio
SNDR	Signal-to-Noise-and-Distortion Ratio
SR	Slew Rate
TDD	Time-Division Duplexing

TDM	Time-Division Multiplexing
THD	Total Harmonic Distortion
VDSL	Very high data-rate Digital Subscriber Line
XDSL	all/any digital subscriber line

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1 CHARACTERIZATION OF DATA CONVERTERS

1.1 INTRODUCTION

When a data converter is used in telecommunications applications it is important to know the limitations of the converter and how they affect the performance of the entire system. Therefore measures to characterize the converters are needed. In this chapter we investigate some basic properties of data converters and introduce a number of performance measures that are commonly used. In Sec. 1.2 we review the fundamentals of data converters while quantization noise is considered in Sec. 1.3. Performance measures and error sources are treated in Sec. 1.4 - Sec. 1.6.

1.2 THE IDEAL DATA CONVERTER

The analog-to-digital converter (ADC or A/D converter) viewed as a black box, shown in Fig. 1-1 a), takes an analog input signal usually in the form of a voltage or a

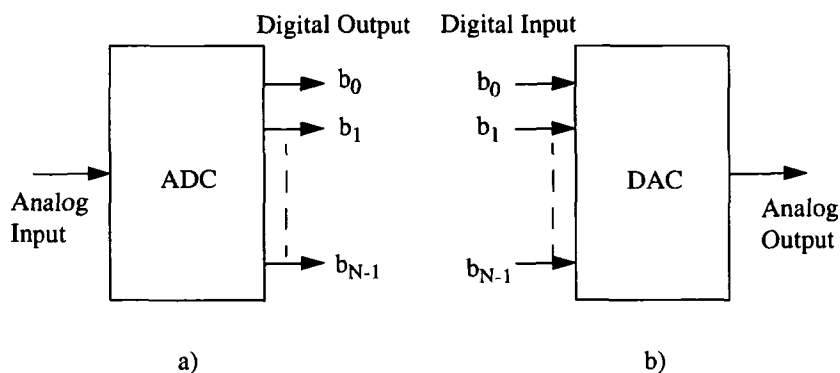


Figure 1-1 The ADC (a) and DAC (b) viewed as a black box.

current and converts it to a digital output signal. The digital-to-analog converter (DAC or D/A converter) has the opposite function and converts a digital input signal to an

analog output signal, as shown in Fig. 1-1 b). The digital signal is a binary coded representation of the analog signal using N bits. There are many different ways of coding the output signal [1]. Some of the most common coding schemes for $N = 4$ are shown in Table 1-1. The maximum number of codes when using a N bit code is 2^N .

Table 1-1 Digital Coding Schemes

Decimal	Sign-Magnitude	Two's-Complement	Offset Binary	One's-Complement	Gray
+7	0111	0111	1111	0111	1000
+6	0110	0110	1110	0110	1001
+5	0101	0101	1101	0101	1011
+4	0100	0100	1100	0100	1010
+3	0011	0011	1011	0011	1110
+2	0010	0010	1010	0010	1111
+1	0001	0001	1001	0001	1101
+0	0000	0000	1000	0000	1100
-0	1000	(0000)	(1000)	1111	1100
-1	1001	1111	0111	1110	0100
-2	1010	1110	0110	1101	0101
-3	1011	1101	0101	1100	0111
-4	1100	1100	0100	1011	0110
-5	1101	1011	0011	1010	0010
-6	1110	1010	0010	1001	0011
-7	1111	1001	0001	1000	0001
-8	-	1000	0000	-	0000

The leftmost bit of the digital word is usually called the most significant bit (MSB) and the rightmost bit is called the least significant bit (LSB). The size of the LSB compared to the total code range is sometimes referred to as the *resolution* [1] of the converter. Hence, the resolution of an N bit converter is $1/2^N$. For simplicity it is sometimes more convenient to refer to the resolution as being N bits. In some types of ADCs, e.g., flash and folding ADCs, other codes are used internally which use more than N bits to represent 2^N codes. These codes are usually converted to a more convenient representation at the output of the converter. The offset binary code is commonly used in data converters. We assume in the following that the digital code, X_d , is

$$X_d = (b_{N-1}, b_{N-2}, \dots, b_2, b_1, b_0) \quad (1-1)$$

where b_{N-1} is the MSB and b_0 is the LSB, and that all the bits are 0 or 1. The analog value of the data converter is denoted X_a . The transfer function of the data converter is a staircase function as illustrated in Fig. 1-2, where the ideal transfer function of a

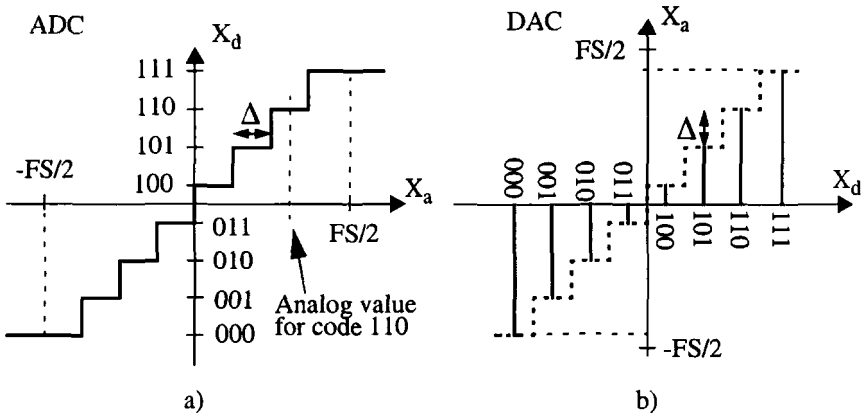


Figure 1-2 The ADC (a) and DAC (b) transfer functions for $N = 3$.

3-bit ADC is shown in Fig. 1-2a) and the transfer function of an ideal DAC in Fig. 1-2b). In both cases it has been assumed that an offset binary code is used. Notice that the ADC output saturates for large input values. Therefore the input signal should never be in the saturated range since this gives rise to large conversion errors. The step size Δ is equal to the analog value of the LSB and the conversion range without causing saturation is referred to as the Full Scale (FS) range of the converter. Hence the step size is

$$\Delta = \frac{FS}{2^N} \quad (1-2)$$

Here we have assumed that the analog signal range is $[-FS/2, +FS/2]$ but it is not uncommon to assume that it is $[0, +FS]$.

It is convenient to introduce a number of symbols based on the transfer function of the ideal converter. The digital codes are numbered by an index $k = 0, \dots, 2^N - 1$, such that $k = 0$ corresponds to the code representing the smallest analog value and $k = 2^N - 1$ corresponds to the largest analog value. For the binary offset code the index k can be calculated as

$$k = \sum_{l=0}^{N-1} b_l \cdot 2^l \quad (1-3)$$

The k -th digital code is denoted $X_{d,k}$, while the corresponding analog value is denoted $X_{a,k}$. This is illustrated in Fig. 1-3. For the ADC it is useful to introduce the analog transition levels as the analog value where the output signal changes between two digital codes. The transition levels are denoted $X_{t,k}$. The number of transition levels is $2^N + 1$. The first and last transition levels are $X_{t,0} = -FS/2$ and $X_{t,2^N} = FS/2$.

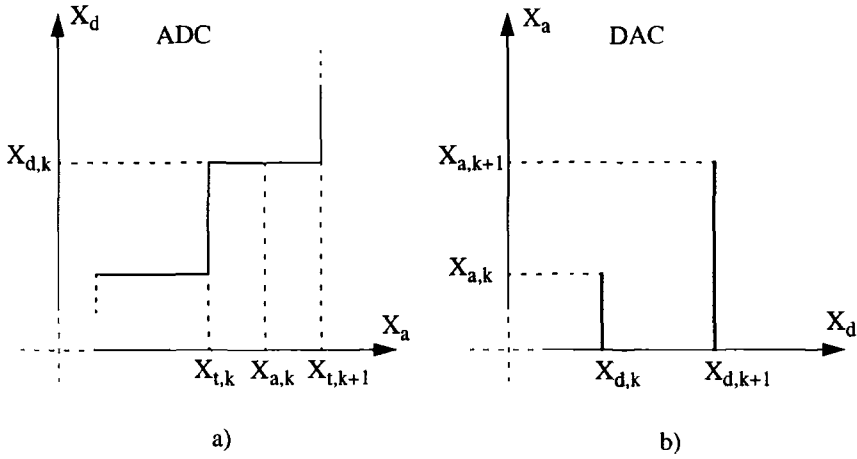


Figure 1-3 Notation for digital codes and analog levels in a) an ADC and b) a DAC.

To calculate the analog value that corresponds to a certain digital code, the reference voltage or current of the data converter must be known. The reference determines the actual analog value and is often equal to FS in volt or ampere. The analog value assuming an **offset binary code** can be calculated as

$$X_{a,k} = \Delta \cdot \sum_{l=0}^{N-1} 2^{l-1} \cdot (-1)^{(1-b_l)} \quad (1-4)$$

The analog value calculated above corresponds to the output of an ideal DAC or the midpoint between two transition levels in an ideal ADC (see Fig. 1-3).

To clarify the notation we give a short example. Assume we have a 3-bit converter and that $FS = 2$. According to (1-2) the step size is

$$\Delta = \frac{FS}{2^N} = \frac{2}{2^3} = \frac{1}{4} \quad (1-5)$$

The digital code 101 which is the 6-th code corresponds to $k = 5$ (k starts at 0). This means that $X_{a,5} = 101$, $b_2 = 1$, $b_1 = 0$ and $b_0 = 1$. According to (1-4) the analog value can now be calculated as

$$\begin{aligned} X_{a,5} &= \Delta \cdot \sum_{l=0}^{3-1} 2^{l-1} \cdot (-1)^{(1-b_l)} = \\ &= \frac{1}{4} \cdot \left(2 \cdot (-1)^0 + 1 \cdot (-1)^1 + \frac{1}{2} \cdot (-1)^0 \right) = \frac{3}{8} \end{aligned} \quad (1-6)$$

The **signed-magnitude code** uses the MSB as a sign bit. If $b_{N-1} = 0$ the analog value is positive while if $b_{N-1} = 1$ the analog value is negative. The analog value is calculated as

$$X_{a,k} = \Delta \cdot (-1)^{b_{N-1}} \cdot \sum_{l=0}^{N-2} 2^l \cdot b_l \quad (1-7)$$

There are now two codes representing the zero value, 0...00 and 10...00. This means that there are only $2^N - 1$ analog levels. The signed-magnitude code may have operand sign-dependent operations which introduces extra control logic and computation time in the hardware [2]. However, the signed-magnitude code allows the use of some special architectures for e.g. binary weighted current-steering DACs [3].

For the **one's complement code** we have

$$X_{a,k} = \Delta \cdot \left(-b_{N-1} \cdot (2^{N-1} - 1) + \sum_{l=0}^{N-2} 2^l \cdot b_l \right) \quad (1-8)$$

Also in this case we have two codes for the zero value, 0...00 and 1...11.

The **two's complement code** also utilizes a sign bit in the MSB position. The analog value is given by

$$X_{a,k} = \Delta \cdot \left(-2^{N-1} \cdot b_{N-1} + \frac{1}{2} + \sum_{l=0}^{N-2} 2^l \cdot b_l \right) \quad (1-9)$$

The two's complement code has good properties for arithmetic operations and is therefore used in several applications.

The **thermometer code** is used in flash converters. With this coding the number of ones in the code determines the corresponding analog value. This is illustrated in Table 1-2 showing the thermometer codes for $N = 4$ and the corresponding index k .

Table 1-2 Thermometer codes for $N = 4$

k	Thermometer code	Walking one
4	1111	1000
3	0111	0100
2	0011	0010
1	0001	0001
0	0000	0000

This table also shows the walking one code which is used in some data converters.

The number of codes that can be represented with the thermometer code is only $N + 1$ and it is therefore rarely used in digital signal processing (DSP).

It is a general practice to choose the code representation that is most suitable for a specific data converter. The code can easily be converted to a representation that is more suitable for the algorithms in the DSP.

1.3 QUANTIZATION

The number of codes in a data converter is approximately 2^N , if an effective coding is used. Hence only a limited number of analog amplitude levels can appear in the data converter and there is an inherent conversion error. The quantization error of a 3 bit ADC is shown in Fig. 1-4. We see that the amplitude of the error is $\pm\Delta/2$ and that it

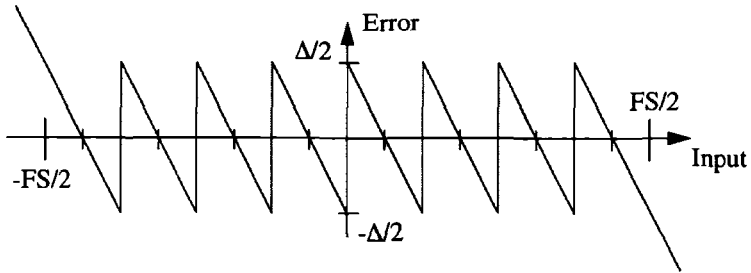


Figure 1-4 Quantization error for a 3 bit ADC.

is divided into 2^N segments. In each segment the error changes linearly as a function of the input signal. If the resolution of the data converter is high it is normally reasonable to assume that the conversion error can be treated as white noise having equal probability of lying anywhere within the range $\pm\Delta/2$. For noise signals we can calculate the power (mean square value) as [4]

$$P_n(t) = x_{rms}^2(t) = \int_{-\infty}^{\infty} e^2 \cdot p(e, t) de \quad (1-10)$$

where $x(t)$ is the noise signal, $p(e, t)$ the probability density function and e is used as an integration variable. In this case we assume that the noise has a uniform probability density function and that it is not a function of time, i.e.

$$p(e, t) = \begin{cases} \frac{1}{\Delta}, & -\frac{\Delta}{2} < e < \frac{\Delta}{2} \\ 0, & \text{all other } e \end{cases} \quad (1-11)$$

The power, i.e. the mean square value, can now be calculated as [5]

$$P_n = \int_{-\Delta/2}^{\Delta/2} e^2 \cdot \frac{1}{\Delta} de = \frac{\Delta^2}{12} \quad (1-12)$$

The same result can be achieved by using a slightly different approach which may be more intuitive for DACs [6]. Assume that we sweep through all the codes sequentially as a function of time, i.e., sweeping a ramp, with period T as shown in Fig. 1-5. Each

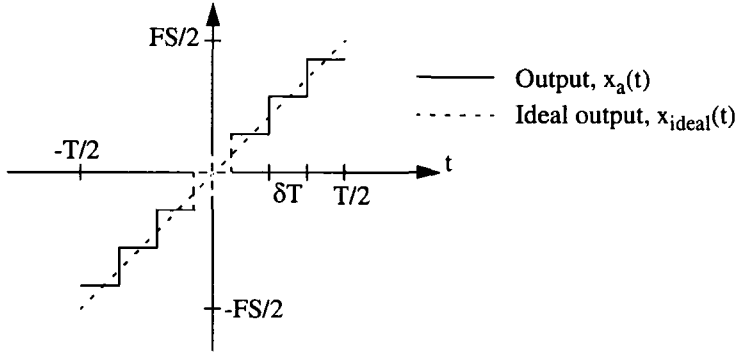


Figure 1-5 Input ramp to determine quantization noise of a DAC.

code is represented during a certain time interval given by $\delta T = T/2^N$. The actual analog output, $x_a(t)$, is a staircase function while the ideal output signal, $x_{ideal}(t)$, is a ramp function. We can now find the error power (mean square value) as

$$P_n = x_{err, rms}^2 = \frac{1}{T} \int_0^T [x_a(t) - x_{ideal}(t)]^2 dt \quad (1-13)$$

where $x_a(t) - x_{ideal}(t)$ corresponds to the quantization error. The error has the same sawtooth shape as in Fig. 1-4 but now as a function of time. Since the same pattern is repeated 2^N times it is sufficient to calculate the power within only one of the segments. The quantization error in the time interval $t \in [0, \delta T]$ is

$$x_a(t) - x_{ideal}(t) = -\frac{t \cdot \Delta}{\delta T} + \frac{\Delta}{2} \quad (1-14)$$

The error power can now be calculated as

$$P_n = \frac{1}{\delta T} \int_0^{\delta T} \left[-\frac{t \cdot \Delta}{\delta T} + \frac{\Delta}{2} \right]^2 dt = \frac{\Delta^2}{\delta T} \int_0^{\delta T} \left[\left(\frac{t}{\delta T} \right)^2 - \frac{t}{\delta T} + \frac{1}{4} \right] dt = \frac{\Delta^2}{12} \quad (1-15)$$

The result is the same as in (1-12).

The above results are valid if all the quantization levels are uniformly spaced. Assuming that the input signal is statistically equally distributed over the whole input amplitude range, this uniform quantization intuitively provides the minimum error. If the signal is not equally distributed over the amplitude range it may be better to use non-uniform quantization. This means that the step size is not equal for all codes in the data converter. Normally a higher resolution is used for small input signals. An example of a transfer function with non-uniform quantization is shown in Fig. 1-6. With this

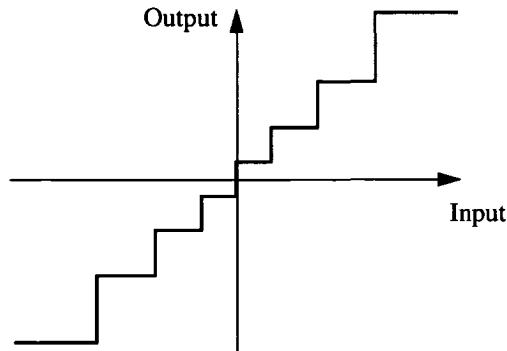


Figure 1-6 Transfer function with non-uniform quantization.

method some amplitude levels are quantized with larger errors, but statistically a minimum quantization noise is achieved since the probability for these amplitude levels is lower. Non-uniform quantization is widely used when transmitting voice signals.

The quantization noise is a fundamental limitation in data converters that represents a lower limit on how small the error power can be. Due to circuit imperfections the total error will always be larger in an actual implementation. To evaluate the performance of a data converter it is convenient to introduce a number of performance measures.

The performance measures can be divided into two groups, static and dynamic measures [3, 7, 8]. A common source of static errors is component mismatch in the implementation. The differential non-linearity (DNL) and integral non-linearity (INL) are often used as static performance measures. The dynamic performance is determined by signal-dependent errors as non-linear slewing, clock feedthrough (CFT), glitches, settling errors, etc. Both static and dynamic properties can be investigated in the frequency domain [7]. The signal-to-noise ratio (SNR), total harmonic distortion (THD), spurious-free dynamic range (SFDR) and signal-to-noise-and-distortion ratio (SNDR) are commonly used as dynamic performance measures. These measures are often determined in single-tone measurements but in many telecommunications applications a multi-tone signalling system is used and static measures or even single-tone measurements may not give all the information needed to characterize the converter.

1.4 STATIC PERFORMANCE

Due to non-ideal circuit elements in the actual implementation of a data converter the code transition points in the transfer function will be moved as illustrated in Fig. 1-7.

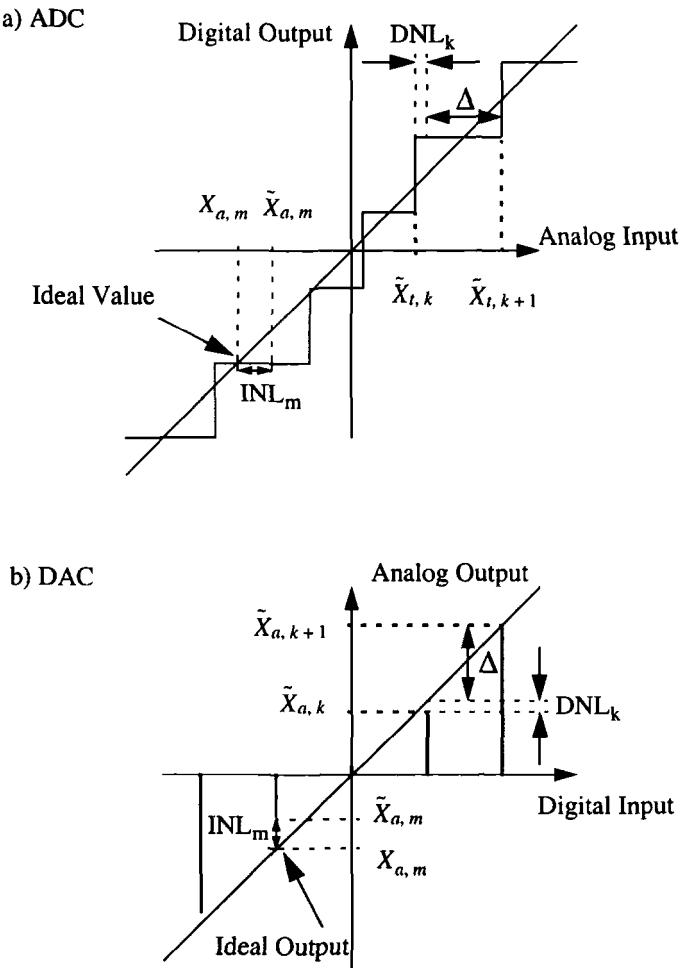


Figure 1-7 Non-ideal transfer function with INL and DNL errors (a) ADC and (b) DAC.

To distinguish between the actual and ideal values in the data converters, all actual values are indicated with a $\sim$. This means that $X_{a,k}$ corresponds to the ideal analog value for digital code $X_{d,k}$ while $\tilde{X}_{a,k}$ corresponds to the actual value.

1.4.1 Differential Nonlinearity (DNL)

The step size in the non-ideal data converter deviates from the ideal size Δ and this error is called the differential nonlinearity (DNL) error. For a DAC the DNL can be defined as the difference between two adjacent analog outputs minus the ideal step size, i.e.

$$\text{DNL}_k = \tilde{X}_{a,k+1} - \tilde{X}_{a,k} - \Delta \quad (1-16)$$

The DNL is often normalized with respect to the step size to get the relative error, i.e.

$$\text{DNL}_k = \frac{\tilde{X}_{a,k+1} - \tilde{X}_{a,k} - \Delta}{\Delta} \quad (1-17)$$

The above definitions are often most practical for DACs since the analog values can be directly measured at the output. For ADCs it may however be more practical to define the DNL based on the difference between transition points, since this corresponds to the result of a histogram test [3]. The normalized DNL of the ADC can be expressed as

$$\text{DNL}_k = \frac{\tilde{X}_{t,k+1} - \tilde{X}_{t,k} - \Delta}{\Delta} \quad (1-18)$$

1.4.2 Integral Nonlinearity (INL)

The total deviation of an analog value from the ideal value is called integral nonlinearity (INL). The normalized INL can be expressed as [9]

$$\text{INL}_k = \frac{\tilde{X}_{a,k} - X_{a,k}}{\Delta} \quad (1-19)$$

for both ADCs and DACs. The relation between INL and DNL is given by

$$\text{INL}_k = \sum_{l=1}^k \text{DNL}_l \quad (1-20)$$

The nonlinearity errors are usually measured using a low frequency input signal to exclude dynamic errors appearing at high signal frequencies. The DNL and INL are therefore usually used to characterize the static performance. In some applications offsets and linear gain errors are acceptable and it is then common to specify the INL with respect to a best fit line rather than to the ideal transfer function [1]. Hence, offsets and linear gain errors will not appear in the INL.

1.4.3 Offset Error

The offset, X_{offset} , of the converter can be found by minimizing the expression

$X_{a,k} - X_{a,k} - X_{offset}$ for all k , with the least square method. To find the minimum we first find where the derivative with respect to X_{offset} is zero, i.e.

$$\frac{\partial}{\partial X_{offset}} \sum_{k=0}^{2^N-1} (\tilde{X}_{a,k} - X_{a,k} - X_{offset})^2 = 0 \quad (1-21)$$

which gives

$$X_{offset} = \frac{1}{2^N} \cdot \sum_{k=0}^{2^N-1} (\tilde{X}_{a,k} - X_{a,k}) \quad (1-22)$$

This corresponds to a minimum and the offset is thus given by (1-22). We see that the offset corresponds to the average of all the errors in the converter. To eliminate the offset from the INL calculations, the offset should be subtracted from all the analog values, $\tilde{X}_{a,k}$.

1.4.4 Gain Error

The gain can be linear or non-linear as illustrated in Fig. 1-8. Compared to the ideal

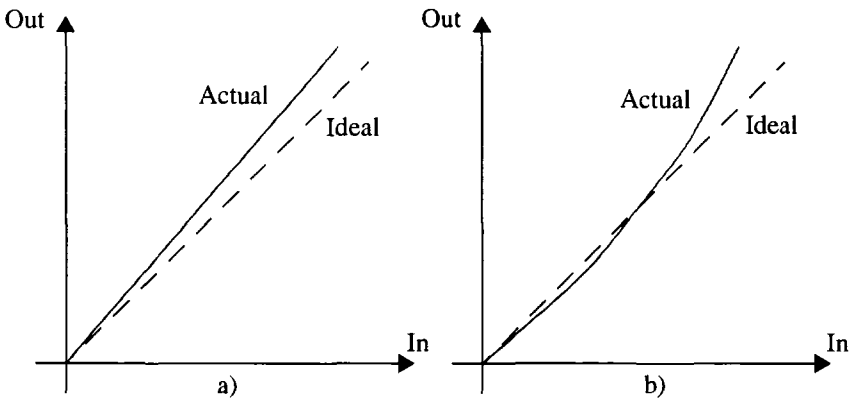


Figure 1-8 Characteristics of a) linear and b) non-linear gain error.

straight line, the actual output has a linear gain error (Fig. 1-8a) and also non-linearity (Fig. 1-8b). Linear gain error does not introduce distortion as long as the output signal does not clip. The actual output with a linear gain and offset error can be written as

$$\tilde{X}_a = A \cdot X_a + X_{offset} \quad (1-23)$$

where A is the gain error while the actual output for a non-linear gain can be expressed as

$$\tilde{X}_a = A_1 \cdot X_a + A_2 \cdot X_a^2 + A_3 \cdot X_a^3 + \dots + X_{offset} \quad (1-24)$$

The non-linear errors can sometimes be reduced by using pre-distortion [10].

The actual gain and the offset, A and X_{offset} , are found by using the least square method. We first find where the derivative with respect to X_{offset} and A are zero, i.e.

$$\frac{\partial}{\partial A} \sum_{k=0}^{2^N-1} [\tilde{X}_{a,k} - (X_{offset} + A \cdot X_{a,k})]^2 = 0 \quad (1-25)$$

and

$$\frac{\partial}{\partial X_{offset}} \sum_{k=0}^{2^N-1} [\tilde{X}_{a,k} - (X_{offset} + A \cdot X_{a,k})]^2 = 0 \quad (1-26)$$

By using (1-25) and (1-26) the gain and offset can be shown to be

$$A = \frac{\langle \tilde{X}_a \cdot X_a \rangle - \langle \tilde{X}_a \rangle \cdot \langle X_a \rangle}{\langle X_a^2 \rangle - \langle X_a \rangle^2} \quad (1-27)$$

and

$$X_{offset} = \langle \tilde{X}_a \rangle - A \cdot \langle X_a \rangle \quad (1-28)$$

where $\langle X \rangle$ indicates a mean value, i.e.

$$\langle \tilde{X}_a \rangle = \frac{1}{2^N} \sum_{k=0}^{2^N-1} \tilde{X}_{a,k}, \quad \langle X_a \rangle = \frac{1}{2^N} \sum_{k=0}^{2^N-1} X_{a,k}, \dots \quad (1-29)$$

Here we assume that the analog mid-point in the ideal transfer function is 0, which implies that the mean value $\langle X_{a,k} \rangle = 0$. The above expressions can then be simplified as

$$A = \frac{\langle \tilde{X}_{a,k} \cdot X_{a,k} \rangle}{\langle X_{a,k}^2 \rangle} \quad (1-30)$$

and

$$X_{offset} = \langle \tilde{X}_{a,k} \rangle \quad (1-31)$$

The line $X_{offset} + A \cdot X_a$ is the best-fit straight line with respect to the actual output values, $\tilde{X}_{a,k}$. The compensated values, can now be used to find the DNL and INL

errors that are not affected by offset and linear gain errors. The INL can for instance be calculated as

$$INL_k = \frac{\tilde{X}_{a,k} - (A \cdot X_{a,k} + X_{offset})}{A \cdot \Delta} \quad (1-32)$$

1.4.5 Monotonicity

If the analog amplitude level of the converter increases with increasing digital code, the converter is monotonic. An example of a non-monotonic DAC is shown in Fig. 1-9. Non-monotonicity in an ADC results in missing output codes that never appears for

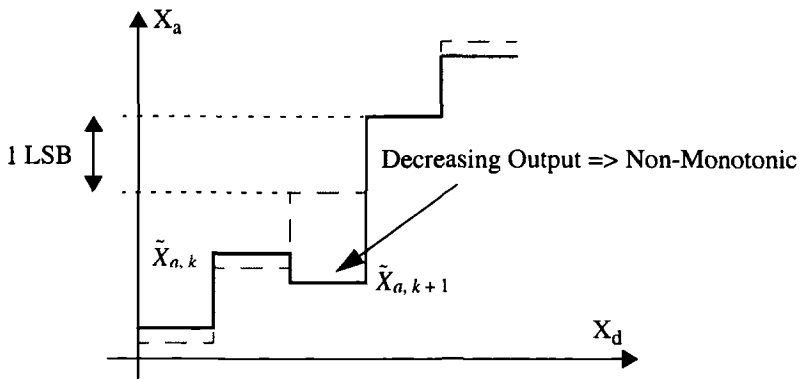


Figure 1-9 A non-monotonic DAC.

any analog input signal. Monotonicity is guaranteed if the deviation from the best-fit straight line is less than half a LSB [3], i.e.

$$|INL_k| \leq \frac{1}{2} \text{LSB for all } k \quad (1-33)$$

This implies that the DNL errors are less than one LSB [3], i.e.

$$|DNL_k| \leq 1 \text{LSB for all } k \quad (1-34)$$

It should be noted that the above relations are sufficient to guarantee monotonicity, but it is possible to have a monotonic converter that does not meet the relations in (1-33) and (1-34). There are some data converter architectures that are monotonic by design, e.g. a thermometer coded DAC.

1.5 DYNAMIC PERFORMANCE

In addition to the static errors that are caused by mismatch in the components in the

data converter, several other error sources will appear when the input signal change rapidly. These dynamic errors are often signal and frequency dependent and increases with signal amplitude and frequency. They appear in both ADCs and DACs but are usually more critical in DACs since the shape of the analog wave form determines the performance.

1.5.1 Settling Errors in DACs

In the previous sections the data converters have been regarded as discrete-time circuits that operate on analog values only at discrete time instants. For the ADC this is true, but for the DAC, however, we must also consider the shape of the analog waveform at the output. A number of dynamic effects arise when the output signal is changed between two samples. These dynamic error sources will have a large impact on the DAC performance, especially at high clock and signal frequencies. The ADC is affected by dynamic errors as well, but as long as the final value at the end of the sampling period has a low enough error the performance is not degraded.

When the input of the DAC is changed, the analog output should ideally change from the ideal start value, $X_{a,k}$, to the ideal final value, $X_{a,m}$, see Fig. 1-10. Due to circuit

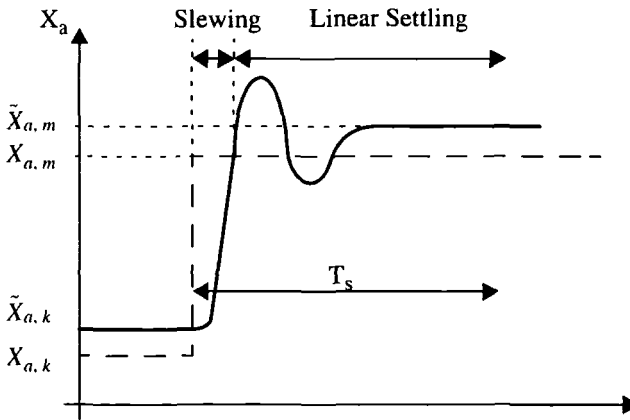


Figure 1-10 Actual output signal and ideal output signal (dashed) of a DAC.

imperfections the actual start and final values are $\tilde{X}_{a,k}$ and $\tilde{X}_{a,m}$ respectively. The output signal of an actual DAC can not change its value instantly. The time it takes for the output to settle within a certain accuracy of the final value, for instance 0.1%, is called the settling time, T_s , and determines the highest possible speed of the circuit. The settling can be divided in two phases, a non-linear slewing phase and a linear settling phase. The slewing phase should be as small as possible since it both increases the settling time and introduces distortion in the analog waveform. The slewing is normally caused by a too small bias current in the circuit driving the output and is therefore increased for large steps when more current is needed.

There may be additional dynamic error sources in the DAC that can both change the final value and the shape of the waveform, such as glitches and clock feedthrough (CFT).

1.5.2 Glitches in DACs

Glitches occur when the switching time of different bits in a binary weighted DAC is unmatched. For a short period of time a false code could appear at the output. For example if the code transition is

$$0111 \dots 111 \rightarrow 1000 \dots 000$$

and the MSB switches faster than the LSBs, the code 11... 111 may be present for a short time. This code represents the maximum value and hence a large glitch appears at the output. The glitch adds a signal dependent error to the output signal, that degrades the performance. The effect on the output signal is determined by the energy of the glitch. If the glitch is modeled as a pulse, as shown in Fig. 1-11, with a certain

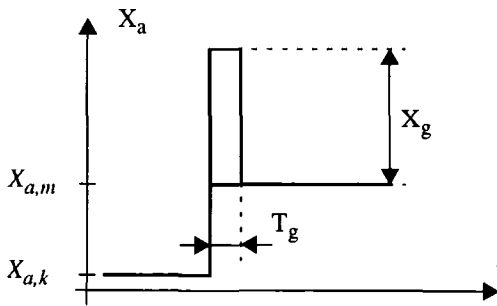


Figure 1-11 Glitch modeled as a pulse with height X_g and duration T_g .

amplitude height, X_g , and with a time duration, T_g , the normalized average power, P_g , of the glitch distributed over the shortest possible code duration, i.e., the clock period, $T_s = 1/f_s$, is

$$P_g = X_g^2 \cdot \frac{T_g}{T_s} \quad (1-35)$$

Assume that the maximum peak glitch amplitude, i.e., the amplitude of the MSB, is

$$X_{g,max} = 2^{N-1} \cdot \Delta \quad (1-36)$$

which gives the maximum glitch power over one clock cycle

$$P_{g,max} = 2^{2N-2} \cdot \Delta^2 \cdot \frac{T_g}{T_s} \quad (1-37)$$

This should be compared with the power of the quantization noise, P_Q , during the

$$P_Q = \frac{\Delta^2}{12} \quad (1-38)$$

The power of the glitch should be smaller than the quantization in order not to decrease the SNR. Hence

$$P_{g, max} = 2^{2N-2} \cdot \Delta^2 \cdot \frac{T_g}{T_s} < \frac{\Delta^2}{12} = P_Q \quad (1-39)$$

which gives a bound on the time duration of the glitch as

$$T_g < \frac{T_s}{3 \cdot 2^{2N}} \quad (1-40)$$

It should be noted that the glitches are difficult to model accurately and the above result is only a coarse approximation. Sometimes the glitch impulse is also specified by the glitch area with the unit $\text{pV} \cdot \text{s}$ [3].

1.5.3 Clock Feedthrough (CFT) in DACs

Due to capacitive coupling in switches the clock (or digital switching signals) affects the analog output signal [11]. The clock feedthrough (CFT) can introduce both harmonic distortion and distortion tones at multiples of the clock signal.

For current-steering DACs the CFT error can be modelled in a similar way as glitches, while in e.g. SC DACs the CFT will give an error in the final value. The CFT is reduced when reducing the capacitive coupling and therefore the switch transistor sizes should be small to decrease the size of the parasitic capacitances. However, with a smaller transistor, the on-resistance increases which may degrade the performance due to an increased settling time.

1.5.4 Sampling Time Uncertainty in DACs

Due to noise and other non-ideal effects in the circuit the time between two samples will change. This sampling time variation gives an error in the output that is determined by the size of the output step and the time variation. The average power of this error can be calculated as

$$P_{T_\delta} = X_\delta^2 \cdot \frac{T_\delta}{T_s} \quad (1-41)$$

where X_δ is the step size, T_δ the sampling time error and T_s the sampling period. The step size X_δ is determined by the difference between two consecutive samples, i.e.

$$X_\delta = X((n+1) \cdot T_s) - X(nT_s) \quad (1-42)$$

For a sinusoidal signal the step size is proportional to the signal frequency, and hence

the error in (1-41) will increase at higher signal frequencies since the step size gets larger. The largest possible step size is

$$X_{\delta} = \Delta \cdot 2^N \quad (1-43)$$

which gives the maximum error power

$$P_{T_{\delta}, max} = \Delta^2 \cdot 2^{2N} \cdot \frac{T_{\delta}}{T_s} \quad (1-44)$$

The error power should be smaller than the quantization noise and therefore we have

$$\Delta^2 \cdot 2^{2N} \cdot \frac{T_{\delta}}{T_s} < \frac{\Delta^2}{12} \quad (1-45)$$

which gives an upper bound on the sampling time error

$$T_{\delta} < \frac{T_s}{3 \cdot 2^{N+2}} \quad (1-46)$$

1.5.5 Dynamic Errors in ADCs

The dynamic errors in ADCs are mainly caused by the same effects as in DACs, but only their effect on the final value at the end of the sampling period is important. This means that it does not matter if the settling is non-linear or if there are glitches as long as the remaining settling error is small enough. The settling time is, however, in most cases increased when the settling is slew rate limited. The effect of dynamic errors in different ADC architectures is treated in more detail in chapter 3.

1.6 FREQUENCY DOMAIN MEASURES

For data converters used in communications applications, the INL and DNL are not sufficient to characterize the performance. It is more convenient to characterize the performance in the frequency domain using measures as the signal-to-noise ratio (SNR) and spurious-free dynamic range (SFDR). The performance is usually determined by using a single-tone sinusoidal input signal, but sometimes dual-tone [12] or multi-tone measurements are more informative. Since several communication standards use multi-tone modulation the single-tone measures may not be sufficient to characterize converters for multi-tone applications [7]. In Fig. 1-12 we show a typical FFT spectrum of a 14-bit non-ideal ADC when the input signal is a single-tone sinusoidal. The input signal appears as the fundamental in the FFT spectrum and the quantization error generates a white noise floor. The nonlinearities in the ADC cause harmonic tones to appear above the noise floor where some of the harmonics may be folded from higher frequencies due to the sampling process. Based on the FFT in Fig. 1-12 a number of measures to characterize dynamic performance of ADCs can be defined.

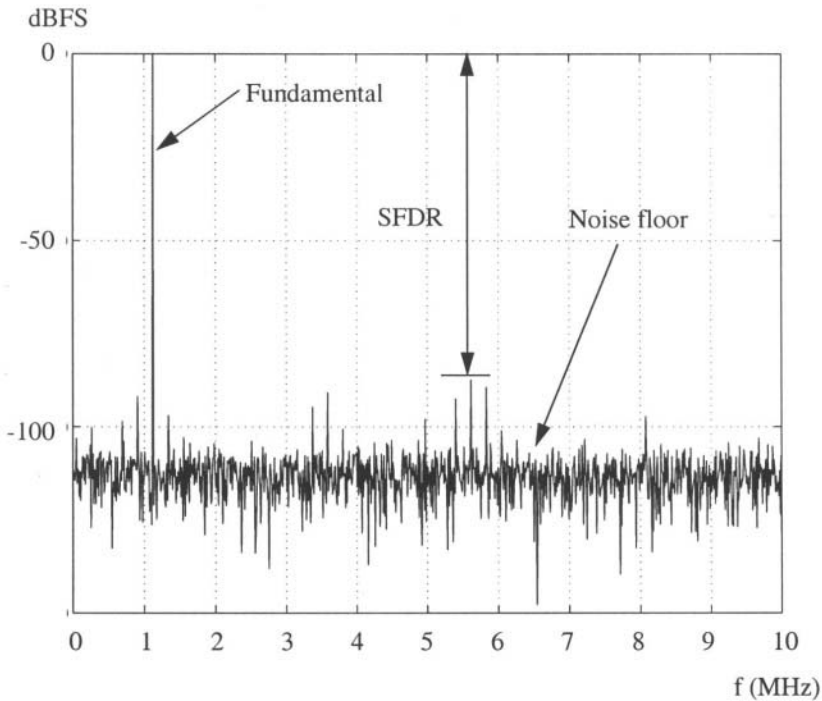


Figure 1-12 FFT spectrum of non-ideal 14 bit ADC.

1.6.1 Signal-to-Noise Ratio (SNR)

A sinusoidal signal is often used to characterize a data converter. It is therefore interesting to calculate the ideal signal-to-noise ratio of a data converter using such an input signal. The maximum amplitude without causing saturation of a sinusoidal input signal is $\Delta \cdot 2^{N-1}$ and the average power of the sine wave is given by

$$P_s = \frac{(\Delta \cdot 2^{N-1})^2}{2} \quad (1-47)$$

The signal-to-noise ratio (SNR) of an ideal ADC with a sinusoidal input signal can now be calculated as [3]

$$SNR = \frac{P_s}{P_n} = \frac{(\Delta \cdot 2^{N-1})^2 / 2}{\Delta^2 / 12} = 1.5 \cdot 2^{2N} \quad (1-48)$$

or expressed in decibels

$$SNR_{dB} = 10 \cdot \log\left(\frac{P_s}{P_n}\right) = 6.02 \cdot N + 1.76 \text{ dB} \quad (1-49)$$

Hence the SNR_{dB} is increased by 6 dB for every additional bit in the converter. The representation using decibels is more convenient. For simplicity the subindex dB is not used in the following. It should be noted that (1-49) only holds if the input is a full scale sine wave. The effect of having a non-sinusoid input signal is discussed in Sec. 1.6.11. The SNR of an actual data converter including non ideal effects can be determined by measuring the output signal. For single tone measurements the SNR is the ratio of the power of the fundamental and the total noise power within a certain frequency band, excluding the harmonic components, i.e.

$$SNR = 10 \cdot \log\left(\frac{\text{Signal Power}}{\text{Total Noise Power}}\right) \quad (1-50)$$

1.6.2 Spurious Free Dynamic Range (SFDR)

The spurious free dynamic range (SFDR) is the ratio of the power of the signal and the power of the largest spurious within a certain frequency band. SFDR is usually expressed in dBc as

$$SFDR(dBc) = 10 \cdot \log\left(\frac{\text{Signal Power}}{\text{Largest Spurious Power}}\right) = 10 \cdot \log\left(\frac{X_1^2}{X_s^2}\right) \quad (1-51)$$

where X_1 is the rms value of the fundamental and X_s the rms value of the largest spurious. In some cases it is more convenient to express the SFDR with the full scale input (dBFS) as reference rather than the input signal, i.e.

$$SFDR(dBFS) = 10 \cdot \log\left(\frac{\left(\frac{FS}{2\sqrt{2}}\right)^2}{X_s^2}\right) \quad (1-52)$$

The relation between SFDR in dBFS and dBc is given by

$$SFDR(dBc) = SFDR(dBFS) - 10 \cdot \log\left(\frac{X_1^2}{\left(\frac{FS}{2\sqrt{2}}\right)^2}\right) \quad (1-53)$$

1.6.3 Harmonic Distortion (HD_k)

The harmonic distortion with respect to the k -th harmonic (HD_k) is the power ratio between the k -th harmonic and the fundamental as

$$\begin{aligned}
 HD_k &= 10 \cdot \log \left(\frac{\text{k:th Harmonic Power}}{\text{Signal Power}} \right) = \\
 &= 10 \cdot \log \left(\frac{X_k^2}{X_1^2} \right)
 \end{aligned} \tag{1-54}$$

where X_1 is the rms value of the fundamental and X_k the rms value of the k -th harmonic component. With the above definition the harmonic distortion is a negative number. Sometimes HD_k is defined as

$$HD_k = 10 \cdot \log \left(\frac{X_1^2}{X_k^2} \right) \tag{1-55}$$

which is positive. Thus there is a sign difference between the two definitions.

1.6.4 Total Harmonic Distortion (THD)

The total harmonic distortion (THD) is the ratio of the total harmonic distortion power and the power of the fundamental in a certain frequency band, i.e.

$$\begin{aligned}
 THD &= 10 \cdot \log \left(\frac{\text{Total Harmonic Distortion Power}}{\text{Signal Power}} \right) \\
 &= 10 \cdot \log \left(\sum_{k=2}^{\infty} X_k^2 / X_1^2 \right)
 \end{aligned} \tag{1-56}$$

where X_1 is the rms value of fundamental and X_k the rms value of the k -th harmonic component. Since there is an infinite number of harmonics the THD is usually calculated using the first 10-20 harmonics or until the harmonics can not be distinguished from the noise floor. The THD is sometimes defined as

$$THD = 10 \cdot \log \left(\frac{\text{Signal Power}}{\text{Total Harmonic Distortion Power}} \right) \tag{1-57}$$

The only difference to the expression in (1-56) is the sign.

1.6.5 Signal-to-Noise and Distortion Ratio (SNDR)

The signal-to-noise and distortion ratio (SNDR) is the ratio of the power of the fundamental and the total noise and distortion power within a certain frequency band, i.e.

$$SNDR = 10 \cdot \log \left(\frac{\text{Signal Power}}{\text{Noise and Distortion Power}} \right) \tag{1-58}$$

1.6.6 Effective Number Of Bits (ENOB)

The effective number of bits (ENOB) is a measure based on the SNDR of an ADC with a full scale sinusoidal input signal, according to (1-49). The ENOB is determined by

$$ENOB = \frac{SNDR - 1.76}{6.02} \quad (1-59)$$

where SNDR correspond to the actual (non-ideal) value.

It should be noted that all the measures above are both frequency and signal amplitude dependent. At low input amplitude levels the ADC performance is usually limited by the quantization noise while the distortion will limit the performance at higher signal levels. For communications applications it is common to specify the SNDR as a function of the input amplitude as shown in Fig. 1-13.

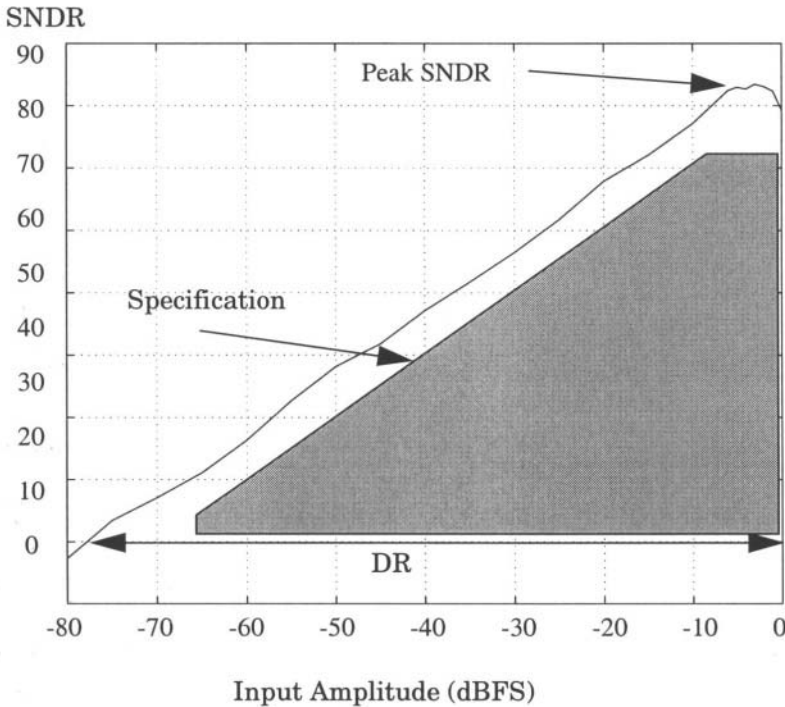


Figure 1-13 SNDR vs. input frequency.

1.6.7 Peak SNDR

The maximal SNDR in Fig. 1-13 is referred to as the peak SNDR and is usually located slightly below the full scale input (0 dBFS in the plot). For oversampling ADCs the peak SNDR may be located significantly below the full scale input. The requirements

from the application is specified as an area in the plot that must be met by the ADC (the shaded area in Fig. 1-13).

1.6.8 Dynamic Range (DR)

The range from full scale (FS) to the smallest detectable signal (usually $\text{SNDR} = 0$) is called the dynamic range (DR) of the converter, i.e.

$$DR = 10 \cdot \log\left(\frac{\text{Maximum signal power}}{\text{Smallest signal power}}\right) \quad (1-60)$$

In communications it is not uncommon to define the smallest signal as the signal which gives a certain $\text{SNR} > 0$.

1.6.9 Effective Resolution Bandwidth (ERB)

An important parameter for the data converter is the signal bandwidth that can be handled. The bandwidth is limited by the analog bandwidth of the input circuits in the ADC as well as the maximal sampling frequency of the converter. The input signal frequency must be smaller than the Nyquist frequency (half the sampling frequency) to avoid aliasing in conventional applications of ADCs. The signal bandwidth can be larger than the Nyquist frequency for sub-sampling ADCs. To specify the frequency behaviour of the converter it is common to plot the SNDR, SFDR or SNR as function of input frequency as illustrated in Fig. 1-14. The effective resolution bandwidth is the

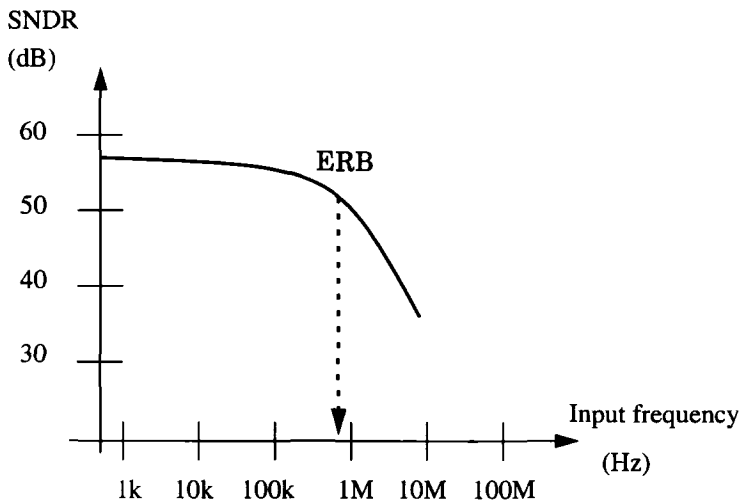


Figure 1-14 SNDR as function of input frequency.

input frequency where the SNDR has dropped 3 dB (or ENOB 1/2 bit). A well designed wideband ADC for sub-sampling applications can have an effective bandwidth well above the Nyquist frequency.

1.6.10 Inter-Modulation Distortion (IMD)

Inter-modulation distortion (IMD) appears when the input is a multi-tone signal. Assume that two tones with the frequencies f_1 and f_2 are applied to the converter with sampling rate f_s . Intermodulation distortion will appear at the frequencies

$$(k \cdot f_1 + m \cdot f_2) \bmod (f_s/2) \quad (1-61)$$

where k and m are integer numbers, and further $k \neq 0$, $m \neq 0$, and $f_1 \neq f_2$. The inter-modulation distortion is calculated as

$$IMD = 10 \cdot \log \left(\frac{\sum X_{k,m}^2}{X_0^2} \right) \quad (1-62)$$

where X_0 is the rms value of the fundamental and $X_{k,m}$ is the rms value of the tones at the frequencies given by (1-61). For some multi-tone applications the tone frequencies are multiples of a specific fundamental frequency and hence the intermodulation terms will interfere with other tones, see Sec. 1.6.12.

1.6.11 SNR with Multi-Tone Input

For multi-tone measurements we also use the peak-to-average ratio (PAR) or crest factor. The PAR gives information on how the signal is distributed over the amplitude range. A low PAR indicates a more uniform distribution, which is advantageous in most cases. The PAR can be calculated as

$$PAR = \frac{\text{peak amplitude}}{\text{rms value}} \quad (1-63)$$

For a sinusoidal signal, $PAR = \sqrt{2}$. A sinusoidal input signal is common when testing converters but in many practical communications applications there are more than one sinusoidal tone in the input signal. In such applications it is convenient to use the peak-to-average ratio (PAR) when calculating the SNR. For an N bit converter with input range FS the average power, P_s , is

$$P_s = \frac{1}{(PAR)^2} \cdot \left(\frac{FS}{2} \right)^2 = \frac{1}{(PAR)^2} \cdot (\Delta \cdot 2^{N-1})^2 \quad (1-64)$$

making the SNR of an ideal ADC

$$\begin{aligned} SNR &= 10 \cdot \log \left(\frac{\frac{1}{(PAR)^2} \cdot (\Delta \cdot 2^{N-1})^2}{\Delta^2/12} \right) = \\ &= 6.02 \cdot N + 4.77 - 20 \cdot \log(PAR) \end{aligned} \quad (1-65)$$

We see from (1-65) that a small PAR is preferred since it maximizes the SNR. The relation between the SNR of a single-tone measurement and a multi-tone measure-

ment is

$$SNR_{multi} = SNR_{single} - 20 \cdot \log\left(\frac{PAR}{\sqrt{2}}\right) \quad (1-66)$$

1.6.12 Multi-Tone Power Ratio (MTPR)

It is difficult to find the distortion for multi-tone transmission schemes (DMT, OFDM, etc.), since tones often are multiples of a fundamental frequency. This implies that distortion terms, harmonics, are added to information carriers. A method to find out the distortion is to apply a number of tones (that are multiples of a fundamental, ω_0), all with the same amplitude, A . Some tones are left out, and the distortion terms that occur at these positions determine the quality and the multi-tone power ratio (MTPR). The MTPR is defined as

$$MTPR = 10 \cdot \log\left(\frac{A^2/2}{\text{Power of tones at left-out frequency positions}}\right) \quad (1-67)$$

This is also depicted in Fig. 1-15, where 25 tones have been applied. At two frequen--

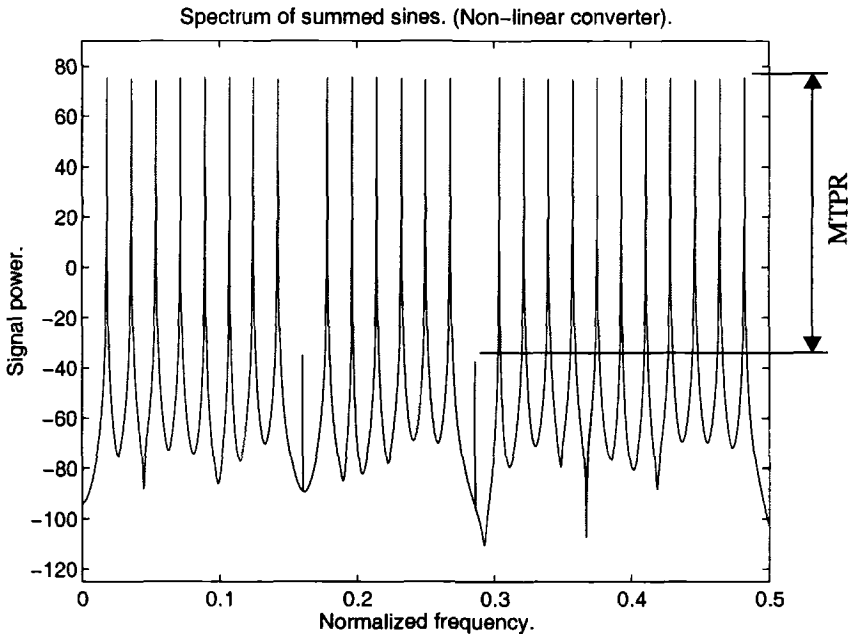


Figure 1-15 Illustration of the impact of MTPR. Two tones are left out and the non-linearity can be found by observing the power of the tones appearing at the left-out positions.

cies tones have been excluded. The non-linearity of the converter introduces harmon-

ics at these positions, and MTPR can be determined by the power of these harmonics. It should be noted that MTPR is a good measure for harmonic distortion. In certain data converters the non-harmonic may exceed the harmonic distortion. This means that $SFDR < MTPR$.

1.7 SUMMARY

In this chapter we have discussed the fundamentals and characterization of data converters. The performance of the data converter is limited by both static and dynamic errors. The static errors are usually caused by mismatch in the circuit elements of the converter and will limit the accuracy of the conversion at low speed. The static performance is often described by the differential-non-linearity (DNL) and the integral-non-linearity (INL). Quite often, offset and linear gain errors are acceptable. Therefore the these errors are sometimes removed before the DNL and INL are calculated. The static linearity measures are rarely sufficient to characterize the performance of data converters in communications. The dynamic errors, as settling errors glitches, etc. will typically increase at higher sampling and signal frequencies. It is usually easier to analyze the effect of these errors in the frequency domain. A number of frequency domain measures as SNDR, SFDR and ENOB were introduced. These measures are normally derived from the output spectrum when a sinusoidal input signal is used. In some applications it may be necessary to use several sinusoidal tones to get all the information. A common measure in such measurements is multi-tone power ratio (MTPR). We showed that the power-to-average ratio (PAR) of the input signal affects the maximum SNR of the converter. A small PAR is preferable but a multi-tone signal normally have a larger PAR than a single-tone signal.

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2 DATA CONVERTER REQUIREMENTS FOR COMMUNICATIONS

2.1 INTRODUCTION

Data converters are crucial building blocks in modern communications systems where DSPs are extensively used. A D/A converter is usually needed at the transmit side and an A/D converter is usually needed at the receive side. The most critical criteria in choosing such data converters are that the data converters shall not degrade the SNR and that the data converters shall not introduce any spurs or distortions. Depending on the actual communication system, the data converter requirements vary dramatically. There are many system-related issues that can influence the data converter requirements, among which modulation schemes, duplexing methods, and analog pre-processing are most important and less understood by data converter designers.

In this chapter, we will briefly discuss communication system-related issues that have strong impacts on data converter requirements. The data converter requirements can also be derived from the standard for a given system and this is also detailed in this chapter in treating asymmetrical digital subscriber line (ADSL) systems. Following this introduction, we will brief line coding and modulation in Sec. 2.2. Both baseband and passband codes will be discussed. We will also touch upon the excess bandwidth concept since it has significant impact on the peak-to-average ratio (PAR). In Sec. 2.3, we will discuss the relationship between the bit rate, SNR, and error probability. The maximum bit rate for a given channel is referred to channel capacity. In Sec. 2.4, we will discuss the duplexing methods. In Sec. 2.5, we will discuss multi-carrier systems and their impact on the PAR. In Sec. 2.6, we will apply the system-related constraints to deriving the minimum requirement on data converters for digital transmission while Sec. 2.7 is devoted to finding the optimum data converters for ADSL. Sec. 2.8 will address A/D converter requirements for wideband radio. In all discussions, we will concentrate on SNR without explicitly discussing distortions. The SNR should be treated as SNDR and the number of bits should be treated as the effective number of bits for data converters in this chapter. The only difference between the noise and distortion generated in a data converter is that oversampling has no impact on the distortion while it reduces the noise power spectral density.

Notice that we do not intend to treat the communication systems in detail. We only touch upon some of key parameters and concepts so that data converter designers can understand their impacts. For some wired communication systems and most radio sys-

terns, coding is used that generally reduces the error probability or reduces the SNR requirement for a given error probability. Also depending on the system trade-off, certain design margin is needed. These two factors can be easily taken into consideration by modifying the required SNR value. Since they are system-specific, we will not include them in the following discussions.

2.2 LINE CODING AND MODULATION

Line coding is the conversion of abstract symbols into real, temporal waveforms to be transmitted in the baseband while modulation is the process where the message information is added to a carrier according to radio terminology [1]. However, for wired digital transmission, e.g. for xDSL (x stands for different types of digital subscriber line), people tend to mix the terminologies of line coding and modulation. For xDSL, we sometimes refer to modulation as the conversion of bit streams into equivalent analog signals that are suitable for the transmission line [2]. The reason is that modulation is usually done digitally for xDSL while modulation is done at radio frequencies (RF) for radio communication.

In this book, we do not confuse readers with different jargons. We specify line codes as baseband or passband codes. The baseband codes are referred to the codes that have energy at DC while the passband codes are referred to the codes that do not have energy at DC, although baseband codes can be modulated to a carrier as well.

2.2.1 Baseband Codes

One of the widely used baseband codes is the phase amplitude modulation (PAM) code. A b -bit PAM has $M = 2^b$ equally spaced levels symmetrically placed about zero ($b=1, 2 \dots$). A b -bit PAM is often referred to PAM- M or M -ary PAM. In Fig. 2-1 we



Figure 2-1 An 8-PAM constellation diagram.

show a 3-bit or 8 PAM. The assignment of the b -bit information to the $M = 2^b$ possible signal amplitudes can be done in different ways. The preferred mapping or assignment is called Gray coding in which the adjacent signal amplitudes differ by only one binary digit [3]. If a PAM modulates a carrier, it is usually called amplitude-shift keying (ASK) in digital communication.

As discussed in Chapter 1, the peak-to-average ratio (PAR) has a strong impact on the data converter requirements. Assume that the distance between two adjacent levels is d , the peak amplitude of an M -ary PAM is then $d \cdot (M - 1)/2$. Under the assumption of equal probability of occurrence for all the levels, the *rms* amplitude is given by (if M is even)

$$\begin{aligned}
 & \sqrt{\left\{ \left(\frac{d}{2} \right)^2 + \left(d + \frac{d}{2} \right)^2 + \dots + \left(\frac{M}{2} \cdot d - \frac{d}{2} \right)^2 \right\} \cdot \frac{2}{M}} \\
 &= \frac{d}{2} \cdot \sqrt{\frac{M^2 - 1}{3}}
 \end{aligned} \tag{2-1}$$

Therefore, the peak-to-average ratio is given by

$$\begin{aligned}
 PAR &= \left\{ \frac{d}{2} \cdot (M - 1) \right\} / \left\{ \frac{d}{2} \cdot \sqrt{\frac{M^2 - 1}{3}} \right\} \\
 &= \sqrt{\frac{3 \cdot (M - 1)}{M + 1}} = \sqrt{\frac{3 \cdot (2^b - 1)}{(2^b + 1)}}
 \end{aligned} \tag{2-2}$$

The simulated PAR for PAM is shown in Fig. 2-2. The PAR for PAM approaches 1.73

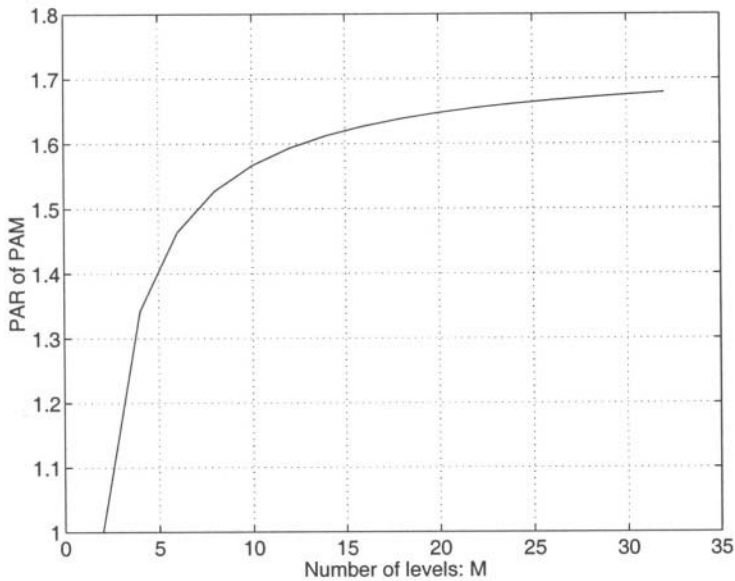


Figure 2-2 Peak-to-average ratio of PAM.

when the number of levels increases. The 4-PAM, or two bits per quaternary (2B1Q) used in integrated service digital network (ISDN) and high-bit-rate digital subscriber line (HDSL) has a PAR of about 1.34.

Binary PAM or 2-PAM is widely used in both radio communication and xDSL due to its simplicity. There are two major classes of binary line codes: level codes and transition codes. Level codes carries information in the voltage level while transition

codes carry information in the change in level appearing in the line code waveform. There are many variations based on 2-PAM. Interested readers are referred to [1,2,3]-

It is not unusual that PAM codes can have an odd number of levels. It can include a level having a value of 0. For instance, PAM-3 is used in Ethernet 100BASE-TX and PAM-5 is used in Ethernet 100BASE-T2 and 1000BASE-T [4]. In this case, the code mapping is more complex [4]

Pulse code modulation (PCM) codes are widely used in communication for transmitting voice signals. A PCM signal is obtained from the PAM signal by encoding each value into a digital word. A PCM signal can be thought of as a serial representation of a PAM signal. The actual transmission of PCM codes is binary. In order to reduce the data converter requirement, compression such as A law or μ law is used for voice communication. Interested readers are referred to [5].

2.2.2 Passband Codes

Passband codes have no energy at DC. The widely used passband code is the quadrature amplitude modulation (QAM) code. A QAM signal is constructed by the summation of an in-phase signal (I) and a quadrature signal (Q), given by

$$I + Q = \varphi(t) \cdot \cos(\omega_c \cdot t) - \varphi(t) \cdot \sin(\omega_c \cdot t) \quad (2-3)$$

where ω_c is the carrier frequency and $\varphi(t)$ is a real-time pulse like a sinc or a square-root raised cosine pulse that is determined by the digital data stream. The multiplication of the pulse by a cosine and sine moves the energy away from the DC to the carrier frequency and different pulse shapes have different properties such as bandwidth requirement, inter-symbol interference, etc.

The QAM codes are two-dimensional. With a b -bit QAM there are $M = 2^b$ symbols in the constellation. A 6-bit or 64-QAM constellation is shown in Fig. 2-3. Assume

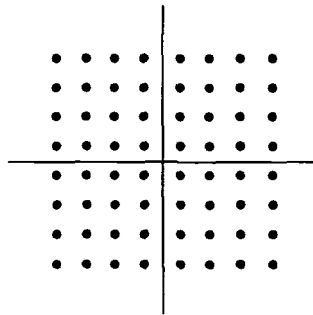


Figure 2-3 A 64-QAM constellation.

that the distance between two neighboring symbols on the x- or the y-axis is d , an $M = 2^b$ QAM constellation have the following magnitudes if b is even

$$\begin{aligned}
& \frac{d}{2} \cdot \sqrt{2}, \frac{d}{2} \cdot \sqrt{1+3^2}, \dots, \frac{d}{2} \cdot \sqrt{1 + \left(2^{\frac{b}{2}} - 1\right)^2}, \frac{d}{2} \cdot \sqrt{3^2+1}, \frac{d}{2} \cdot \sqrt{3^2+3^2} \\
& \dots \frac{d}{2} \cdot \sqrt{3^2 + \left(2^{\frac{b}{2}} - 1\right)^2}, \dots \frac{d}{2} \cdot \sqrt{\left(2^{\frac{b}{2}} - 1\right)^2 + 1}, \\
& \dots \frac{d}{2} \cdot \sqrt{\left(2^{\frac{b}{2}} - 1\right)^2 + \left(2^{\frac{b}{2}} - 1\right)^2}
\end{aligned}$$

Under the assumption of equal probability of occurrence for all the levels, the *rms* magnitude is given by

$$\begin{aligned}
& \frac{d}{2} \cdot \sqrt{\sum_{k=1}^{\frac{b}{2}} \sum_{j=1}^{\frac{b}{2}} \{(2^j - 1)^2 + (2^k - 1)^2\} \cdot \frac{4}{2^b}} \\
& = \frac{d}{2} \cdot \sqrt{\frac{2(2^{\frac{b}{2}} - 1)}{3}}
\end{aligned} \tag{2-4}$$

The maximum signal magnitude is

$$\frac{d}{2} \cdot \sqrt{\left(2^{\frac{b}{2}} - 1\right)^2 + \left(2^{\frac{b}{2}} - 1\right)^2} = \frac{d}{2} \cdot \sqrt{2} \cdot \left(2^{\frac{b}{2}} - 1\right) \tag{2-5}$$

Therefore, the peak-to-average ratio is given by

$$PAR = \frac{\sqrt{3 \cdot \left(2^{\frac{b}{2}} - 1\right)}}{\sqrt{\left(2^{\frac{b}{2}} + 1\right)}} \cdot PAR_c = \sqrt{\frac{3 \cdot (\sqrt{M} - 1)}{(\sqrt{M} + 1)}} \cdot PAR_c \tag{2-6}$$

where PAR_c is the peak-to-average ratio of the carrier. If the carrier is a sine wave, PAR_c is equal to 1.4.

In Fig. 2-4 we show the peak-to-average ratio as a function of the number of levels M . When the number of levels M increases to infinity, the PAR approaches the maximum value of 2.45.

Carrierless amplitude and phase (CAP) modulation is considered as a special case of QAM. If the carrier frequency is not significantly larger than the bandwidth, the carrier modulation in QAM is superficial because a judicious choice of two DC-free functions can realize the same function. Compared with QAM, CAP simplifies the transmitter implementation [6].

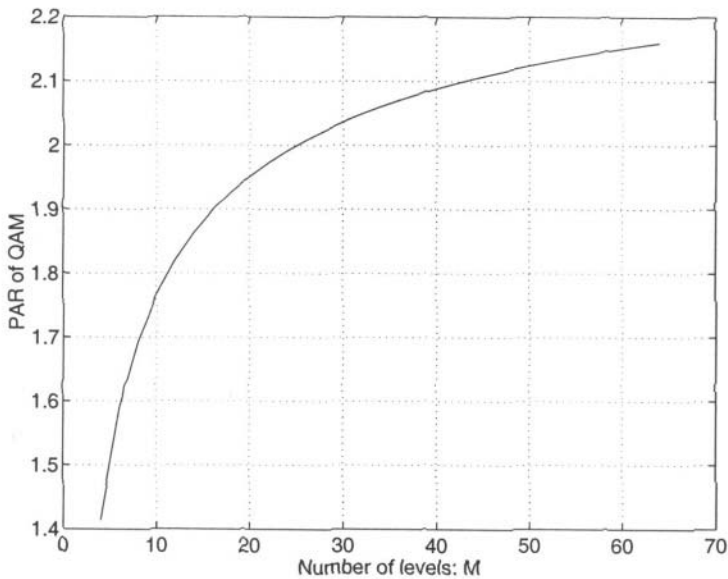


Figure 2-4 Peak-to-average ratio of QAM.

If we keep the signal amplitude constant and only change the phase according to the digital bit stream, we have MPSK, or M-ary phase-shift-keying. Binary PSK (BPSK, $M = 2$) and quadrature PSK (QPSK, $M = 4$) are widely used in radio communications. BPSK is equivalent to 2 PAM (if modulated) or 2 QAM, and QSPK is equivalent to 4 QAM.

We can also use the bit information to modulate the carrier frequencies. It is called frequency-shifting keying (FSK). Binary FSK (BFSK) is widely used in radio communications.

2.2.3 Excess Bandwidth

When we transmit data successively, interferences between the successive symbols deteriorate the performance. To minimize the interferences between successive symbols (intersymbol interferences), we can use Nyquist pulses that are orthogonal to one another. Another possibility to minimize the intersymbol interferences is to introduce a controlled amount of interference at the transmitter, which can be removed at the receiver. This technique is called partial-response signalling. Interested readers are referred to [1].

There are many Nyquist pulses, and the best known are the raised-cosine pulses, which decay with $1/t^3$ instead of $1/t$ as in sinc pulses. The fast decay in time is crucial since it reduces the timing-phase errors in the sampling clock of the receiver.

Suppose that the symbol period is T , or the symbol rate is $f_s = 1/T$. The excess band-

width is defined as the bandwidth relative to the symbol rate, i.e., $(1 + \alpha) \cdot f_s$, where α is the fraction that defines the excess bandwidth (α is usually between 0 and 1). Notice that sometimes the excess bandwidth is defined as $(1 + \alpha) \cdot 0.5 \cdot f_s$ if the spectrum is symmetrical.

With a larger excess bandwidth, the pulses ring less and it is easier to receive the signal, but the channel usage is not effective, and vice versa. The reason that we touch upon the excess bandwidth concept here is due to its influence on the peak-to-average ratio.

It is very tedious and complicated to derive the *PAR* as a function of the excess bandwidth, since it is dependent on the pulse shapes. Due to the excess bandwidth, the *PAR* usually increases. The smaller the excess bandwidth is, the larger the increase in the *PAR* is. It is not unusual for the *PAR* to increase by more than 50% due to the effect of the excess bandwidth. Also notice that with a smaller excess bandwidth, the bandwidth requirement on data converters should be reduced. This advantage can seldom be utilized since it requires a relatively complex digital decimation filter with a fractional decimation ratio.

2.3 CHANNEL CAPACITY AND ERROR PROBABILITY

2.3.1 Channel Capacity

In communication, a terminology called channel capacity is used to describe the maximum transmit throughput and to compare the efficiency of a certain coding vs. this theoretical limitation. If the channel has a bandwidth of BW , the maximum number of bits per second that the channel can support, or the channel capacity, is given [3] by

$$C = BW \cdot \log_2(1 + SNR) \quad (2-7)$$

where SNR is the received signal to noise ratio. Normalizing against the bandwidth, we have the channel capacity in bits per second per Hz, given [7] by

$$c = \log_2(1 + SNR) \quad (2-8)$$

Obviously, if we have M levels in the code, the number of bits per second per Hz is equal to $\log_2(M)$. In Fig. 2-5, we show the channel capacity vs. the SNR . It is seen that for every 1-bit increase in the channel capacity, we need to increase the SNR by 3 dB.

The above conclusion is drawn based on the assumption of optimum codes that necessarily requires infinite complexity and infinite encoding/decoding delay [8]. To achieve the same number of bits per Hertz, the SNR requirement increases above the theoretical SNR depending on many factors such as modulation schemes and coding. The actual data rate is smaller than this theoretical value and it is given by

$$\bar{c} = \log_2\left(1 + \frac{SNR}{\Gamma}\right) \quad (2-9)$$

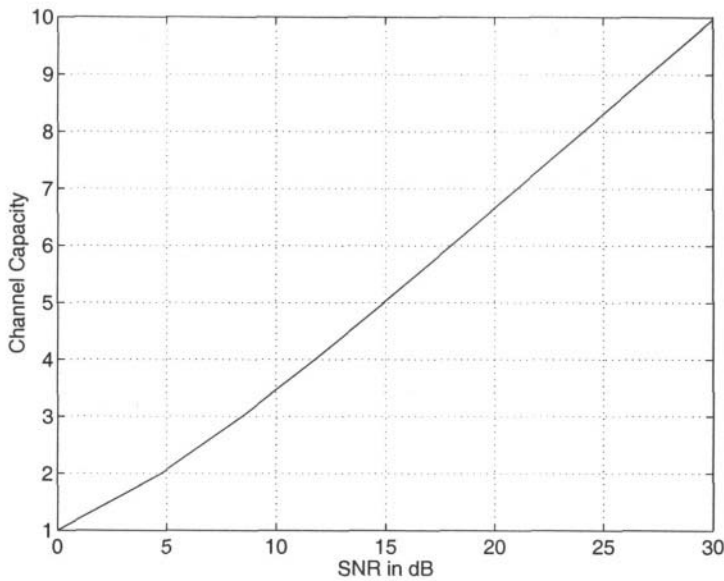


Figure 2-5 Channel capacity vs. SNR.

where Γ is referred to as the gap which is always larger than 1. The gap is a function of the error probability, modulation, and coding. With coding, we can reduce the gap. The difference in the gap Γ between the coded and un-coded transmission is sometimes referred to as coding gain. The readers are referred to [1,2,3,7].

For baseband codes that are one-dimensional, the channel capacity is defined [2] as

$$c = \frac{1}{2} \cdot \log_2(1 + SNR) \quad (2-10)$$

For baseband transmission, the channel capacity increases by 1 bit/second/Hz if we increase the SNR by 6 dB.

2.3.2 Error Probability

When we receive the signal, we need to re-construct the symbol constellation. Intuitively, the larger the minimum distance between two symbols, the easier it is to re-construct the constellation. In the presence of noise, some symbols will not be able to be re-constructed correctly. Therefore, the error probability, or the symbol error rate is determined by the minimum symbol distance and the noise. The larger the minimum distance between two symbols is, the easier it is to recover the information. To achieve the same minimum distance, the MPSK usually needs more signal energy than M -ary QAM for M larger than 4. The larger M is, the more extra energy is needed. With the same energy, QAM usually have better symbol error probability than PSK, depending on M .

In order to evaluate the symbol error rate, we have to introduce the Q function that is widely used in telecommunications. The quantity $Q(x)$ is the probability that a unit variance zero-mean Gaussian random variable exceeds the value in the argument, x , i.e.,

$$Q(x) = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{u^2}{2}} du \quad (2-11)$$

For an M -ary PAM, the error probability is given [2] by

$$P_{PAM} = 2 \cdot \left(1 - \frac{1}{M}\right) \cdot Q\left(\sqrt{SNR} \cdot \sqrt{\frac{3}{M^2 - 1}}\right) \quad (2-12)$$

Therefore, we have the SNR in dB for an M -ary PAM given by

$$SNR_{PAM} = 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{2 \cdot \left(1 - \frac{1}{M}\right)} \cdot P_{PAM} \right) \right\} + 10 \cdot \log \left(\frac{M^2 - 1}{3} \right) \quad (2-13)$$

We plot the SNR requirement as a function of M for different error probabilities in Fig. 2-6. The gap Γ can be easily found by using equations (2-9) and (2-13). It is given by

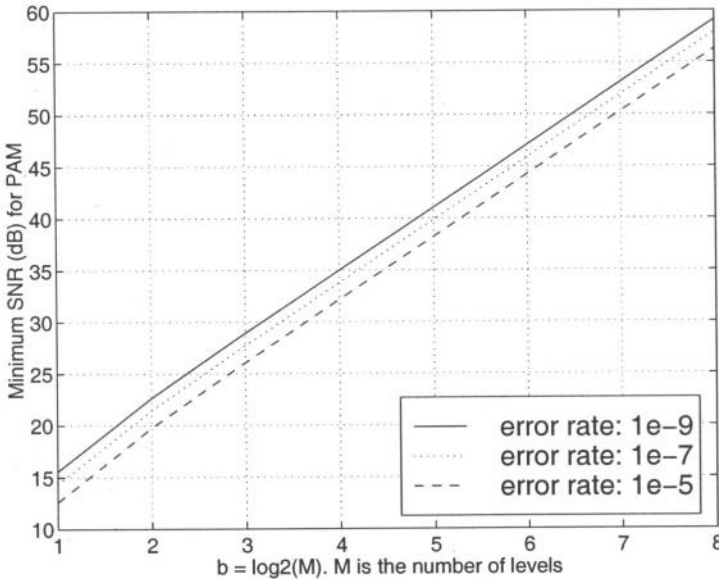


Figure 2-6 Minimum SNR for un-coded M -ary PAM for different error probabilities.

$$\Gamma = 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{2 \cdot \left(1 - \frac{1}{M}\right)} \cdot P_{PAM} \right) \right\} - 4.77 \quad (2-14)$$

In Fig. 2-7 we show the gap vs. the error probability. It is seen that the gap Γ is larger

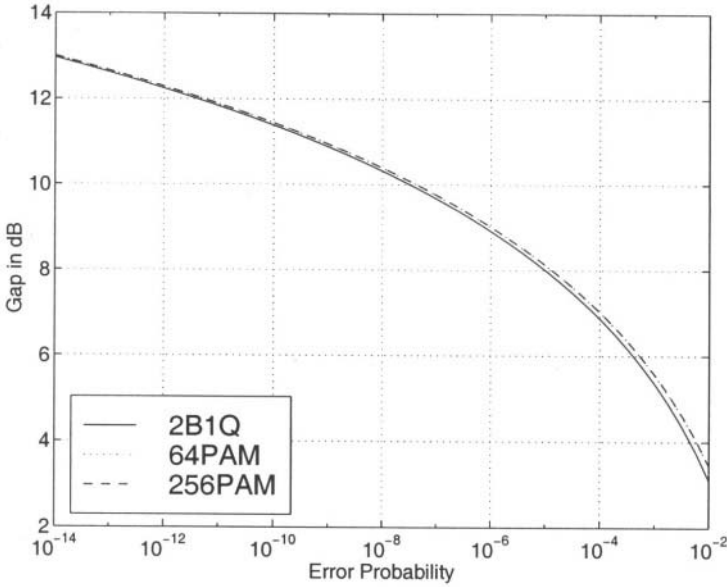


Figure 2-7 Gap vs. the error probability for PAM.

than 9 dB for an error probability less than 10^{-7} . Coding can significantly reduce the gap. For an M -ary QAM, the symbol error probability is given [2] by

$$P_{QAM} \leq 4 \cdot \left(1 - \frac{1}{\sqrt{M}}\right) \cdot Q\left(\sqrt{SNR} \cdot \sqrt{\frac{3}{M-1}}\right) \quad (2-15)$$

if $b = \log_2 M$ is even. If b is odd, we have [2]

$$P_{QAM} \leq 4 \cdot \left(1 - \frac{3}{2 \cdot M}\right) \cdot Q\left(\sqrt{SNR} \cdot \sqrt{\frac{6}{2 \cdot M - 1}}\right) \quad (2-16)$$

Therefore, we have the SNR in dB given by

$$SNR_{QAM} \geq 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{4 \cdot \left(1 - \frac{1}{\sqrt{M}} \right)} \cdot P_{QAM} \right) \right\} + 10 \cdot \log \left(\frac{M-1}{3} \right) \quad (2-17)$$

if $b = \log_2 M$ is even. If b is odd, we have

$$SNR_{QAM} > 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{4 \cdot \left(1 - \frac{3}{2M} \right)} \cdot P_{QAM} \right) \right\} + 10 \cdot \log \left(\frac{2M-1}{6} \right) \quad (2-18)$$

We plot the SNR requirement as a function of M for different error probabilities in Fig. 2-8. The gap can be easily found by using equation (2-9) and (2-17). It is given by

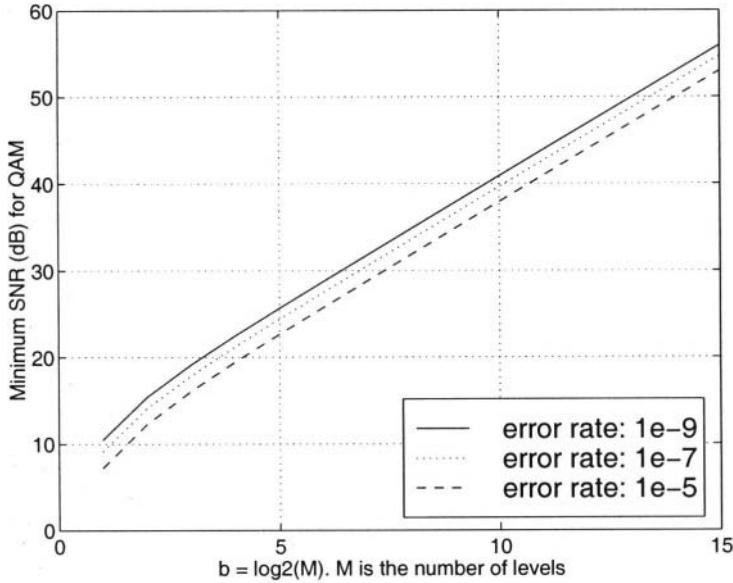


Figure 2-8 Minimum SNR for un-coded M-ary QAM for different error probabilities.

$$\Gamma = 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{4 \cdot \left(1 - 1/\sqrt{M} \right)} \cdot P_{QAM} \right) \right\} - 4.77 \quad (2-19)$$

for even $b = \log_2 M$. In Fig. 2-9, we show the gap vs. the error probability. It is seen

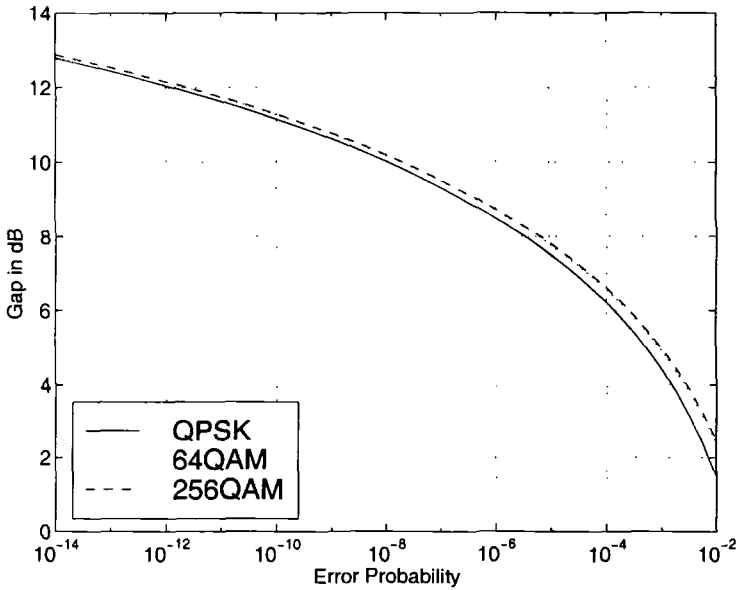


Figure 2-9 Gap vs. the error probability rate for QAM.

that the gap is larger than 9 dB for a symbol error rate less than 10^{-7} . Coding can significantly reduce the gap to 3 to 5 dB. Some extremely powerful coding schemes such as Turbo codes or Concatenated codes can reduce this gap to 1 to 3 dB.

2.4 DUPLEXING METHOD

Duplexing or multiplexing is referred to the exchange of information in both directions of a connect. Do not confuse duplexing with multiple access which is based on insulating signals used in different connections from each other in a wireless radio system.

There are several types of duplexing techniques. The space-division duplexing (SDD) uses physically different media. For instance, the T1/E1 services use different pairs for each direction of transmission. Sometimes, this kind of duplexing is referred to as dual-simplex in that there are two simplex (unidirectional) transmission channels.

The time-division duplexing (TDD) uses the same physical channel. When one-direction transmission is activated, the other direction is de-activated. One of the preliminary proposals for very high-data-rate DSL (VDSL) uses TDD. Some digital cordless telephones such as CT3 and DECT systems use TDD.

The frequency-division duplexing (FDD) or frequency-division multiplexing (FDM) separates the physical channel into two or more distinguished bands such that one direction of transmission does not coincide in frequency with the other direction of transmission. Most cellular systems such as NMT, GSM, IS-54, etc., use FDD. FDD is also supported as an ANSI and ITU standard for ADSL.

Another technique is called echo-cancellation duplexing (ECD), or full duplexing that is widely used in xDSL. Echo cancellation duplexing systems allow transmission in both directions on the same physical channel at the same time. ECD is based on the concept that the receiver always knows what the transmitter at the same end sends to the other direction. The total signal that the receiver sees is the summation of the receive signal that was sent from the other end of the line and the transmit signal (or the echo signal) that it sends itself. By subtracting the transmit signal from the total signal, the receive signal can be recovered. ECD is used for instance in ISDN, HDSL, and ADSL. ECD usually requires both an analog echo rejection (hybrid) and a digital adaptive filter. In Fig. 2-10 we show the concept of the hybrid. All impedances are

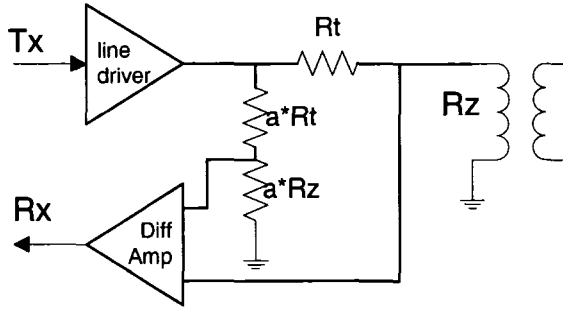


Figure 2-10 Simplified diagram for hybrid.

shown in the figure. Assume that the transmit signal at the line driver output is V_{Tx} and the receive signal at the transformer is V_{Rx} . The signal after the difference amplifier (neglecting the gain in the difference amplifier) is given by

$$\begin{aligned}
 V_{out} &= \left\{ \frac{R_z}{R_t + R_z} \cdot (V_{Tx} - V_{Rx}) + V_{Rx} \right\} - \frac{a \cdot R_z}{a \cdot R_t + a \cdot R_z} \cdot V_{Tx} = \\
 &= \frac{R_t}{R_t + R_z} \cdot V_{Rx}
 \end{aligned} \tag{2-20}$$

It is seen that the output does not contain any Tx signal or echo signal. However, the derivation above is based on the assumption that the impedance of the network perfectly matches the line impedance. Due to the vast variation in the line impedance, the echo signal is usually present at the receiver. If the hybrid does not provide a sufficient echo rejection, the data converter requirements increase significantly since the ADC needs to quantize the weak signal together with the strong echo signal. Adaptive

hybrids (capable of adjusting the impedance to match the line impedance) can increase the echo rejection if properly designed. All the received echo signal can be eliminated by digital adaptive filters, provided that the Rx path has high enough dynamic range and linearity.

2.5 MULTI-CARRIER SYSTEMS

Multi-carrier systems are systems that have more than one carrier. Orthogonal frequency division multiplexing (OFDM) is used in digital audio broadcasting (DAB) and has been suggested for use in mobile communications. For xDSL, the discrete multi-tone (DMT) modulation has been adopted as an ANSI [9] and ITU standard for ADSL.

DMT is essentially the same as OFDM. Each carrier within DMT or OFDM is usually QAM. With DMT or OFDM, the modulation and demodulation can be achieved in the discrete domain by using IFFT and FFT. One of the drawbacks of multi-carrier systems is the increased PAR which calls for more stringent requirements on the analog building blocks.

Assume that the individual carrier is a QAM signal having a PAR_i and all the carriers are un-correlated, the PAR of a DMT or OFDM consisting of m -carriers is given [7] by

$$PAR_{DMT} = \frac{1}{\sqrt{m}} \cdot \sum_{i=1}^m PAR_i \quad (2-21)$$

Assuming that all the sub-carriers have the same PAR_{sub} , the PAR of a DMT having m -sub-carrier is given by

$$PAR_{DMT} = \frac{1}{\sqrt{m}} \cdot \sum_{i=1}^m PAR_i = \sqrt{m} \cdot PAR_{sub} \quad (2-22)$$

With a maximum of 255 sub-carriers in a DMT-ADSL system, the PAR is increased by 16 or 24 dB. This is directly translated into a 24-dB increase in the dynamic range requirement for data converters. Fortunately the peak level does not occur so often.

If the number of sub-carriers is large, the DMT signal approaches a Gaussian distribution, and the probability for a certain peak level to occur can be derived. In Fig. 2-11 we show the clipping probability as a function of the PAR for a DMT signal. Based on this, we can derive the PAR given a clipping probability (i.e., the probability when the peak-to-average ratio is larger than a given PAR). In order to achieve a clipping probability less than 10^{-7} , the PAR of the ADSL DMT signal is ca 5.3. It is obvious that the single-carrier QAM has a smaller PAR than the multi-carrier DMT or OFDM.

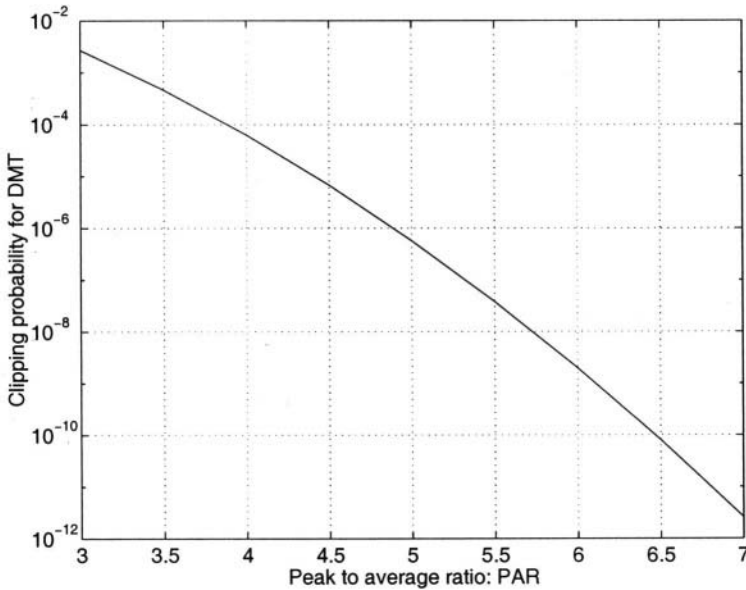


Figure 2-11 Clipping probability as a function of the *PAR* for DMT signals.

2.6 MINIMUM DATA CONVERTER REQUIREMENTS

The minimum data requirements for telecommunications are usually determined by

- the desired error probability or SNR requirement,
- the peak-to-average ratio after digital filtering (i.e, with the consideration of the influence of the excess bandwidth), and
- the duplexing method.

The term minimum refers to the fact that the data converters have to meet these criteria in order to be used in a given communication system.

2.6.1 DAC for Baseband

For baseband transmissions without digital transmit filters, the required DAC resolution is directly determined by the number of levels in the baseband signal. For instance, an $M = 2^b$ PAM needs a b -bit DAC.

If digital transmission filters are used, the DAC requirement for baseband line codes is determined by the required SNR and PAR. According to equations in Chapter 1, the SNR for the DAC (assuming a single sinusoidal input when measuring the SNR) used

in a baseband transmission system is therefore given by

$$SNR_{DAC} = SNR_{PAM} + 20 \cdot \log\left(\frac{PAR}{1.4}\right) \quad (2-23)$$

where the SNR_{PAM} requirement is determined by the error probability requirement and the number of levels in the line code signal, and the PAR is determined by the number of levels in the line code signal and the excess bandwidth.

Notice that the SNR_{PAM} requirement is given at the receive end. There are many factors that contribute to the noise. The DAC quantization noise, the channel additive white Gaussian noise (AWGN), and the ADC quantization are major contributors. If these three sources have equal contributions, the DAC SNR requirement needs to increase by at least 4.78 dB. In this chapter, all the effects are collectively considered as a design margin and are not included in the derivations for simplicity.

The SNR_{PAM} requirement is related to the error rate requirement and the number of levels in the line code by equation (2-13). The peak-to-average ratio is related to the number of levels in the line code by equation (2-2). Therefore, we have

$$\begin{aligned} SNR_{DAC} &= 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{2 \cdot \left(1 - \frac{1}{M}\right)} \cdot P_{PAM} \right) \right\} + 10 \cdot \log \left(\frac{M^2 - 1}{3} \right) \\ &\quad + 10 \cdot \log \left(\frac{3 \cdot (M - 1)}{M + 1} \right) + 20 \cdot \log(\beta) - 3 \\ &= 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{2 \cdot \left(1 - \frac{1}{M}\right)} \cdot P_{PAM} \right) \right\} \\ &\quad + 20 \cdot \log(M - 1) + 20 \cdot \log(\beta) - 3 \end{aligned} \quad (2-24)$$

where M is the number of level in the baseband code, P_{PAM} is the error probability, β is related to the excess bandwidth. Notice that the factor β is not the same as the excess bandwidth parameter α . The smaller α is, the larger β is. Deriving the factor β as a function of the excess bandwidth parameter α is tedious, depending on the transmit pulse shapes. The larger the excess bandwidth, the less stringent is the requirement on the DAC.

If the Nyquist bandwidth (half the clock frequency) is higher than the signal bandwidth, the resolution requirement on the DAC is relaxed, i.e.,

$$SNR_{DAC} = 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{2 \cdot \left(1 - \frac{1}{M}\right)} \cdot P_{PAM} \right) \right\} \quad (2-25)$$

$$+ 20 \cdot \log(M - 1) + 20 \cdot \log \beta - 10 \cdot \log OSR - 3$$

where OSR is the oversampling ratio, defined as the ratio between the Nyquist bandwidth and signal bandwidth.

The number of bits is related to SNR by

$$N_{DAC} = \frac{SNR_{DAC} - 1.76}{6.02} \quad (2-26)$$

In Fig. 2-12, we plot the DAC requirement as a function of M for PAM. β is assumed

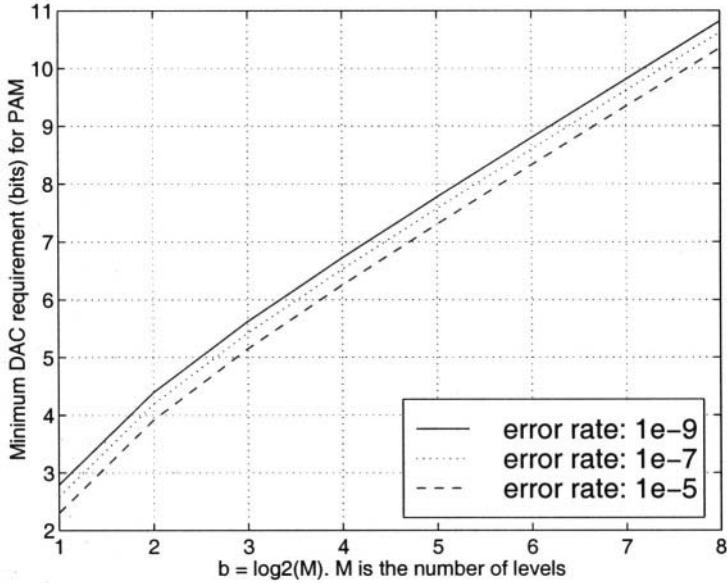


Figure 2-12 DAC requirement for PAM.

to be 2 and no oversampling is used. It is seen that for every doubling in the number of levels M for PAM, the DAC requirement increases by 1 bit. If the error probability is 10^{-7} , and the number of levels is very large, we have the approximation given by

$$N_{DAC} \approx \frac{20 \cdot \log(M) + 20 \cdot \log(\beta) - 10 \cdot \log(OSR) + 9.8}{6.02} \quad (2-27)$$

$$= b + \frac{20 \cdot \log(\beta) - 10 \cdot \log(OSR) + 9.8}{6.02}$$

2.6.2 DAC for Passband

Following the same derivation, we have the DAC requirement for an M-ary QAM given by

$$\begin{aligned}
 SNR_{DAC} = & 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{4 \cdot \left(1 - \frac{1}{\sqrt{M}} \right)} \cdot P_{QAM} \right) \right\} \\
 & + 10 \cdot \log \left\{ \frac{(M-1) \cdot (\sqrt{M}-1)}{(\sqrt{M}+1)} \right\} \\
 & + 20 \cdot \log(\beta) - 10 \cdot \log(OSR)
 \end{aligned} \tag{2-28}$$

where M is the number of levels in the QAM code, P_{QAM} is the error probability, β is determined by the excess bandwidth, and OSR is the oversampling ratio. The number of bits is related to the SNR by equation (2-26).

In Fig. 2-13, we show the DAC requirement as a function of the number of levels M .

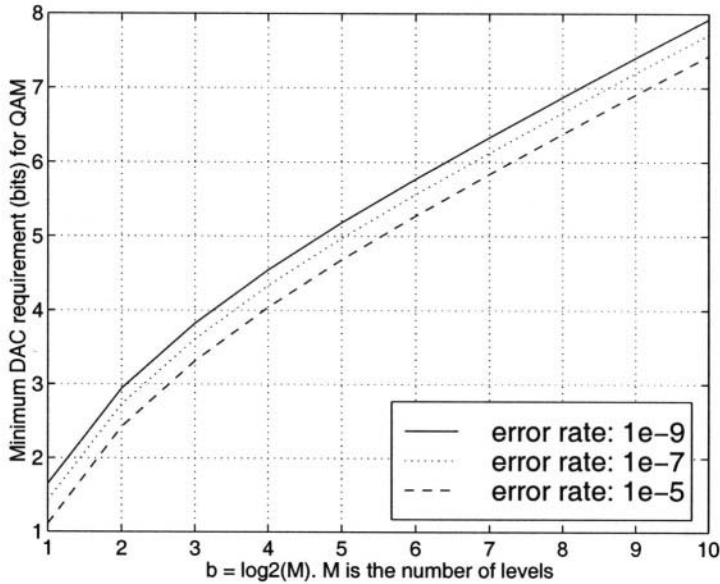


Figure 2-13 DAC requirement for QAM.

β is assumed to be 1.6 and no oversampling is used. It is seen that for every doubling in the number of levels M for QAM, the DAC requirement increases by 0.5 bit.

If the error probability is 10^{-7} , and the number of levels is very large, we have the approximation given by

$$\begin{aligned} N_{DAC} &\approx \frac{10 \cdot \log(M) + 20 \cdot \log(\beta) - 10 \cdot \log(OSR) + 12.6}{6.02} \\ &= \frac{b}{2} + \frac{20 \cdot \log(\beta) - 10 \cdot \log(OSR) + 12.6}{6.02} \end{aligned} \quad (2-29)$$

2.6.3 DAC for DMT

One of the disadvantages of DMT is the increased PAR. Suppose that there are m sub-carriers in the DMT and that every sub-carrier has the same number of levels, M . Since the PAR of the DMT increases by $\sqrt{m}$ compared to its individual sub-carrier, by using equation (2-28), we have

$$\begin{aligned} SNR_{DAC} &= 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{4 \cdot \left(1 - \frac{1}{\sqrt{M}} \right)} \cdot P_{QAM} \right) \right\} \\ &\quad + 10 \cdot \log \left\{ \frac{(M-1) \cdot (\sqrt{M}-1)}{(\sqrt{M}+1)} \right\} \\ &\quad + 10 \cdot \log(m) - 10 \cdot \log(OSR) \end{aligned} \quad (2-30)$$

where M is the number of levels in the sub-carriers, m is the total number of sub-carriers, and OSR is the oversampling ratio. The modulation and demodulation for DMT (usually using IFFT and FFT) is different from a single-carrier QAM. Therefore, the excess bandwidth parameter does not apply.

Theoretically, with 255 tones, the DAC requirement for the DMT increases by 4 bits compared to its corresponding QAM. Fortunately, when m is large, the DMT tones can be treated as AWGN. With a clipping probability of 10^{-7} , we have the approximation given by

$$\begin{aligned} SNR_{DAC} &\approx 20 \cdot \log \left\{ Q^{-1} \left(\frac{1}{4} \cdot 10^{-7} \right) \right\} + 10 \cdot \log \frac{(M-1)}{3} \\ &\quad + 20 \cdot \log(PAR) - 10 \cdot \log(OSR) - 3 \\ &\approx 10 \cdot \log(M-1) - 10 \cdot \log(OSR) + 21 \end{aligned} \quad (2-31)$$

Notice that the DAC SNR requirement is considerably less than the transmit MTPR in the ADSL standard [9]. The requirement in the ADSL standard is given by considering other aspects such as design margins and the influence of Tx path on the Rx path, etc.

The number of bits is related to the SNR by equation (2-26) and it is given by

$$N_{DAC} \approx \frac{b}{2} + 3.2 - \log(OSR) \quad (2-32)$$

In Fig. 2-14 we show the DAC requirement as a function of the number of levels in

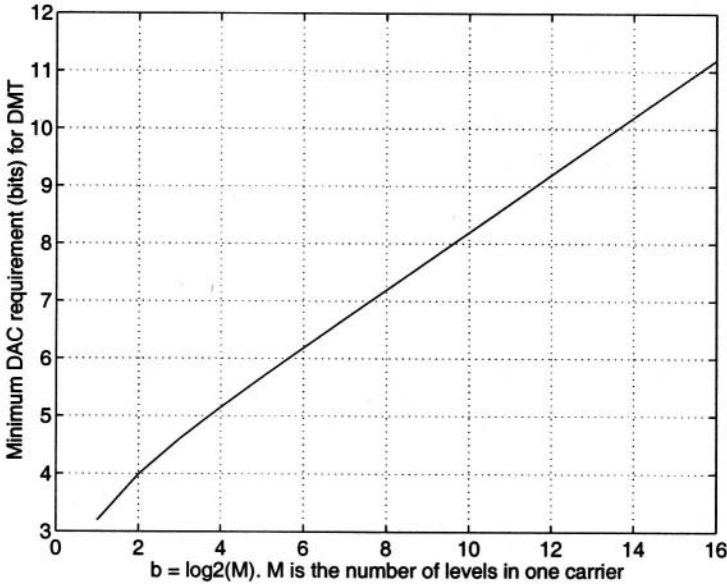


Figure 2-14 DAC requirement for DMT.

one carrier. We assume that all the carriers have been modulated with the same number of levels. In the DMT standard, the maximum number of bits that is allowed to be modulated on a single carrier is 15, i.e., 32768-QAM. This QAM has a PAR of 2.45. The DMT signal that has a PAR of 5.3 increases the DAC requirement by 1.1 bits. Due to the influence of the excess bandwidth in a real implementation, the PAR of a single carrier system, e.g. CAP can be somewhere between 3.6~4. Therefore, the use of DMT usually increase the DAC requirement by ~0.5 bits. This may not seem to be significant for DACs. However, it can increase the requirements on line drivers significantly [10].

2.6.4 ADC Requirements

The transmitted signal is attenuated by the wire or line. However, the line attenuates the noise generated in the transmitter as well. Therefore the SNR will not be degraded until the attenuated transmitter noise is comparable to the background noise on the line. For ADSL systems, the background thermal noise is -140 dBm/Hz or 32 nV/sqrt(Hz).

At the receiver, the SNR after the ADC must meet the error probability requirement for a given modulation scheme. The total noise present at the ADC output includes the

attenuated transmitter noise, the background noise, and the quantization noise generated by the ADC. Therefore, the SNR at the ADC output is given by

$$\begin{aligned}
 SNR_{total} &= 10 \cdot \log \frac{T^2 \cdot G^2 \cdot P_s}{\sigma_{ADC}^2 + T^2 \cdot G^2 \cdot \sigma_{DAC}^2 + G^2 \cdot \sigma_n^2} \\
 &= SNR_{DAC} - 10 \cdot \log \left(1 + \frac{\sigma_{ADC}^2}{T^2 \cdot G^2 \cdot \sigma_{DAC}^2} + \frac{\sigma_n^2}{T^2 \cdot \sigma_{DAC}^2} \right)
 \end{aligned} \tag{2-33}$$

where the SNR_{DAC} is the transmit (or DAC) signal-to-noise ratio, T is the line attenuation, G is the receive gain before the ADC, P_s is the transmit signal power, σ_{ADC}^2 is the ADC quantization noise power, σ_{DAC}^2 is the DAC noise power, and σ_n^2 is the background noise on the line.

It is seen that due to the presence of the background noise and the ADC quantization noise, the DAC SNR requirements must be increased. In a real implementation, the DAC and ADC usually have the same or close to the same signal swing, and therefore we usually set the receive gain to compensate for the attenuation. For most cases, we can assume that the line noise is negligible. (We will discuss later the influence of the line noise). Now we have

$$SNR_{total} = SNR_{DAC} - 10 \cdot \log \left(1 + \frac{\sigma_{ADC}^2}{\sigma_{DAC}^2} \right) \tag{2-34}$$

Using the equations above, we can find out the DAC and ADC requirements by repeating the above process. It is obvious that the requirements are dependent on the relative resolution of the ADC and DAC.

If the DAC and ADC have the same number of bits, both the DAC and ADC resolution requirements are given by the equations (2-23) to (2-31) plus 0.5 bit accounting for the contribution from both the DAC and the ADC. In a practical implementation, a 1~2 bits margin is usually needed.

2.6.5 Influence of the Duplexing Methods

In the above discussion, we assumed that the signal at the ADC is only the signal sent from the other end and the noise, i.e., the duplexing method is the FDD with an ideal analog band-splitting filter in front of the ADC. If the band-splitting filter is not perfect, the echo signal (the signal being transmitted at the same end) will enter the ADC, calling for a higher dynamic range. (The un-filtered echo signal can be filtered by a digital filter.) For EC based systems, a portion of the echo signal will be at the ADC input depending on the analog network, significantly increasing the ADC requirements. (The echo signal will be cancelled by a digital adaptive filter, or echo canceller.) However, the noise generated by the DAC cannot usually be cancelled. Therefore, the DAC requirements also increase significantly. To derive the actual requirement for EC-based systems, we usually need to find out how much echo signal relative to the receive signal will be present at the ADC. The difference between the

total signal at the ADC input and the receive signal at the input determines the increase in the requirements for the data converters.

Assume SNR_o is the data converter requirement without echo signal and SNR_e is the data converter with the presence of the echo signal. They are related by

$$SNR_e = SNR_o + 10 \cdot \log\left(\frac{P_{Rx} + P_{echo}}{P_{Rx}}\right) \quad (2-35)$$

where P_{Rx} is the receive signal power at the ADC input and P_{echo} is the echo signal at the ADC input.

If the analog band-splitting filter is not ideal for FDM systems, a portion of the Tx or echo signal will be present at the ADC input. It will increase the data converter requirements as given by equation (2-35).

For long reach where the receive signal is very weak compared with the echo signal, the data requirements is increased dramatically if the analog hybrid cannot provide enough echo rejection.

2.7 OPTIMUM DATA CONVERTERS FOR ADSL

In the above discussion, we derived the data converter requirements from the system requirements. For more advanced digital transmission systems, the modulation (including number of modulated bits), the duplexing method, and/or the signal bandwidth change according to the environment. A typical example is the ADSL. For most modern digital transmission systems, there is also a need to minimize the number of analog components. Therefore, higher performance data converters are preferred. The question is how high performance is enough to meet a specific standard for a given practical environment. This section is devoted to answering this question.

2.7.1 Optimum ADCs

Without any analog filter in front of an ADC for either FDM or EC-based system, the ADC is supposed to quantize both the echo signal and the received signal. We need to find out the requirement on the ADC. Since the transformer does not change the SNR, we can assume the transformer ratio to be 1:1 without losing any generality. If the hybrid network and some optional gain stages before the ADC do not introduce appreciable noise, the gain before the ADC does not change the SNR. We can assume a unity gain for simplicity in the derivation.

If the ADC is sampled at f_s , the power spectral density of the background noise including interference on the loop is PSD_n , the transmit power spectral density is PSD_{Tx} (bandwidth from f_{i1} to f_{i2}), the echo rejection is assumed to be $T_e(f)$, the power spectral density of the signal sent from the other side is PSD_{Rx} (bandwidth from f_{r1} to f_{r2}), and the loop attenuation of the received signal is $T_l(f)$.

The echo power at the ADC input in dBm is therefore given by

$$\begin{aligned}
P_{echo} &= 10 \cdot \log \left\{ \int_0^{f_s/2} 10^{\frac{PSD_{Tx}}{10}} \cdot 10^{\frac{-T_e(f)}{10}} \cdot df \right\} \\
&\approx 10 \cdot \log \left\{ \int_{f_{r1}}^{f_{r2}} 10^{\frac{PSD_{Tx}}{10}} \cdot 10^{\frac{-T_e(f)}{10}} \cdot df \right\} \\
&= PSD_{Tx} - T_e + 10 \cdot \log BW_{Tx}
\end{aligned} \tag{2-36}$$

where BW_{Tx} is the transmit bandwidth. In the above derivation, we have used the averaged transmit PSD in dBm/Hz and the averaged echo rejection $T_e(f)$ in dB.

The power of the received signal excluding the echo at the ADC input in dBm is given by

$$\begin{aligned}
P_{Rx} &= 10 \cdot \log \left\{ \int_0^{f_s/2} 10^{\frac{PSD_{Rx}}{10}} \cdot 10^{\frac{-T_l(f)}{10}} \cdot df \right\} \\
&= 10 \cdot \log \left\{ \int_{f_{r1}}^{f_{r2}} 10^{\frac{PSD_{Rx}}{10}} \cdot 10^{\frac{-T_l(f)}{10}} \cdot df \right\} \\
&= PSD_{Rx} - T_l + 10 \cdot \log BW_{Rx}
\end{aligned} \tag{2-37}$$

where BW_{Rx} is the received signal bandwidth. In the above derivation, we have used the averaged received PSD in dBm/Hz and the averaged line attenuation $T_l(f)$ in dB.

The total signal power in dBm at the ADC input is therefore given by

$$P_{sig} = 10 \cdot \log \left\{ 10^{\frac{P_{echo}}{10}} + 10^{\frac{P_{Rx}}{10}} \right\} \tag{2-38}$$

The total line noise power in dBm at the ADC input is given by

$$P_n = 10 \cdot \log \left\{ \int_0^{f_s/2} 10^{\frac{PSD_n}{10}} \cdot df \right\} = PSD_n + 10 \cdot \log \left(\frac{f_s}{2} \right) \tag{2-39}$$

It appears that the SNR requirement for the ADC is given by the ratio of the signal power vs. noise power at the ADC input. However, the peak-to-average ratio (PAR)

significantly increases the requirement due to the fact that the SNR measure of an ADC is based on a single sinusoidal input which has a *PAR* of 1.4. Therefore, we have

$$\begin{aligned}
 SNR_{ADC} &= P_{sig} - P_n + 20 \cdot \log\left(\frac{PAR}{1.4}\right) \\
 &= 10 \cdot \log \left\{ 10^{\frac{PSD_{Tx} - T_e + 10 \cdot \log BW_{Tx}}{10}} + 10^{\frac{PSD_{Rx} - T_l + 10 \cdot \log BW_{Rx}}{10}} \right\} \\
 &\quad - PSD_n - 10 \cdot \log\left(\frac{f_s}{2}\right) + 20 \cdot \log\left(\frac{PAR}{1.4}\right)
 \end{aligned} \tag{2-40}$$

(Strictly speaking, a margin should be included due to the DAC and ADC quantization noise.) Now, we need to find out the SNR requirement needed for the ADC as a function of the echo rejection T_e and the background noise power spectral density PSD_n .

Different lengths of the loop introduce different loop attenuation of the received signal. The smallest attenuation occurs when the loop length is zero. By assuming a zero line attenuation $T_l = 0$ dB, we can find out the maximum SNR needed for the ADC. However, the SNR or carrier to noise ratio (CNR) of each sub-carrier only needs to be large enough to demodulate a maximum 15-bit QAM signal with a given bit error rate for ADSL. By using equation (2-18), the SNR for the QAM is required to be larger than 55 dB for an error probability better than 10^{-7} . Considering a 3-dB coding gain and a 6-dB noise margin, the SNR only needs to be **55-3+6=58 dB** to guarantee an error rate less than 10^{-7} . Therefore, the received maximum signal power spectral density only needs to be $PSD_n + 58$ dBm/Hz. If the loop is extremely short, we can reduce the PSD_{Rx} as the standard suggests.

Therefore, we have the SNR requirement for the ADC for ADSL without any analog filter, given by

$$\begin{aligned}
 SNR_{ADC} &= 10 \cdot \log \left\{ 10^{\frac{PSD_{Tx} - T_e + 10 \cdot \log BW_{Tx}}{10}} + 10^{\frac{PSD_n + 58 + 10 \cdot \log BW_{Rx}}{10}} \right\} \\
 &\quad - PSD_n - 10 \cdot \log\left(\frac{f_s}{2}\right) + 11.6
 \end{aligned} \tag{2-41}$$

The SNR is related to the number of bits in an ADC given by

$$N(bits) = \frac{SNR_{ADC} - 1.76}{6.02} \tag{2-42}$$

Notice that the quantization noise contribution is not factored into the above equations for simplicity. We use the noise margin in the following discussions to account for it as well for other influences.

In the above discussion, we have assumed that the ADC is sampled at f_s and the signal bandwidth is $f_s/2$. If the signal bandwidth is much lower than $f_s/2$, the ADC needs less

dynamic range. Since the ADC noise floor is governed by the input signal, we can integrate the noise within the receive band to get the total in-band ADC noise. The peak input signal power does not vary, therefore, we have the SNR requirement on an over-sampling ADC given by

$$SNR_{ADC} = 10 \cdot \log \left\{ 10^{\frac{PSD_{Tx} - T_e + 10 \cdot \log BW_{Tx}}{10}} + 10^{\frac{PSD_n + 58 + 10 \cdot \log BW_{Rx}}{10}} \right\} - PSD_n - 10 \cdot \log(BW_{Rx}) + 11.6 \quad (2-43)$$

If the echo signal dominates, we have

$$\begin{aligned} SNR_{ADC} &\approx PSD_{Tx} - T_e + 10 \cdot \log(BW_{Tx}) \\ &\quad - PSD_n - 10 \cdot \log(BW_{Rx}) + 11.6 \\ &= PSD_{Tx} - T_e - PSD_n + 10 \cdot \log\left(\frac{BW_{Tx}}{BW_{Rx}}\right) + 11.6 \end{aligned} \quad (2-44)$$

It is seen that using a receive filter reduces the requirement on the ADC if the transmit bandwidth is larger than the receive bandwidth.

A) Optimum ADC for ADSL-CO

At the ADSL-CO, we have [9]

- $PSD_{Tx} = -40 \text{ dBm/Hz}$
- Maximum $BW_{Tx} = 1104 - 26 = 1078 \text{ KHz}$
- Maximum $BW_{Rx} = 138 - 26 = 112 \text{ kHz}$

We plot the SNR and number of bit requirements in Fig. 2-15. It is seen that with a small loop noise floor (-140 dBm/Hz as specified as the background thermal noise floor for ADSL) and poor echo rejection, the ADC requirement is formidably high. However, with good echo rejection which is usually feasible at the CO side, and large noise floor (due to the interferences from other lines and from other services), the ADC is realizable.

B) Optimum ADC for ADSL-CP

At the ADSL-CP, we have [9]

- $PSD_{Tx} = -38 \text{ dBm/Hz}$
- Maximum $BW_{Rx} = 1104 - 26 = 1078 \text{ KHz}$
- Maximum $BW_{Tx} = 138 - 26 = 112 \text{ kHz}$

We plot the SNR and number of bits requirements in Fig. 2-16. It is seen that with a small loop noise floor and poor echo rejection, the ADC requirement is formidably

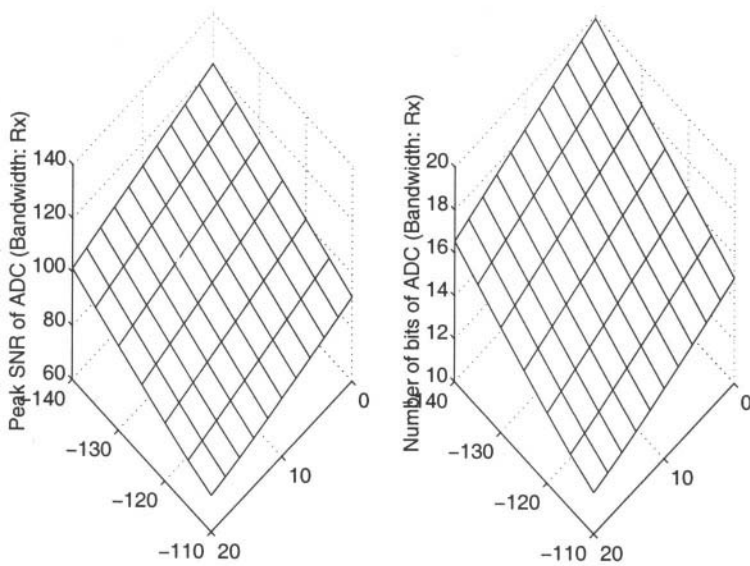


Figure 2-15 ADSL-CO ADC requirements vs. echo rejection and background noise. X-axis: background noise and interference power spectral density in dBm/Hz; Y-axis: echo rejection in dB.

high especially considering the bandwidth.

2.7.2 Optimum DACs

In the above discussions, we assumed that the echo signal will only increase the received signal power. In reality, the noise floor in the echo signal may be another limitation to achieve a high SNR. The noise at the ADC input consists of two parts, one being the contribution due to the line noise and the other being the contribution due to the DAC assuming the noise contributions from the line driver and hybrid network are negligible.

To calculate the noise contribution from the DAC, we need to calculate the voltage gain from the DAC output to the ADC input. The gain consists of the gain of the line driver, echo rejection by the hybrid network and the gain stage before the ADC. In a practical design, ADCs and DACs usually have a comparable signal range and we usually set the gain such that the ADC dynamic range is fully utilized. If the echo signal dominates at the ADC input, the total voltage gain from the DAC output to the ADC input is unity. If the receive signal is not negligible compared to the echo signal, the total gain from the DAC to the ADC is less than unity. Therefore, as long as the DAC has comparable or smaller noise floor than the ADC, the noise floor in the DAC will not degrade the receiver performancesignificantly ($< 3\text{ dB}$). Since the ADC noise floor is set by the PSD_n , the DAC noise floor should be less than PSD_n .

If we integrate the noise within the Tx band, and compare it to the total received signal

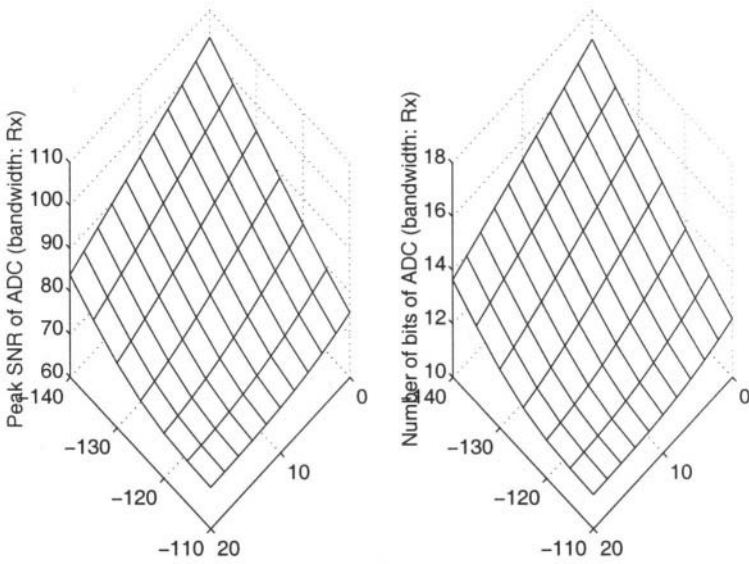


Figure 2-16 ADSL-CP ADC requirements vs. echo rejection and background noise. X-axis: background noise and interference noise spectral density in dBm/Hz; Y-axis: echo rejection in dB.

(since ADC and DAC have the same signal swing) we can derive the SNR requirement for the DAC, i.e.,

$$\begin{aligned}
 SNR_{DAC} &= P_{sig} - P_n + 20 \cdot \log\left(\frac{PAR}{1.4}\right) \\
 &= 10 \cdot \log \left\{ 10^{\frac{PSD_{Tx} - T_e + 10 \cdot \log BW_{Tx}}{10}} + 10^{\frac{PSD_{Rx} - T_l + 10 \cdot \log BW_{Rx}}{10}} \right\} \quad (2-45) \\
 &\quad - PSD_n - 10 \cdot \log(BW_{Tx}) + 20 \cdot \log\left(\frac{PAR}{1.4}\right) \\
 &\approx PSD_{Tx} - PSD_n - T_e + 11.6
 \end{aligned}$$

In the above derivation, we assume that the echo dominates at the ADC input. The SNR is related to the number of bits in a DAC is given by

$$\begin{aligned}
 N(bits) &= \frac{SNR_{DAC} - 1.76}{6.02} \\
 &\approx \frac{PSD_{Tx} - PSD_n - T_e + 9.84}{6.02} \quad (2-46)
 \end{aligned}$$

Notice that we only require that the DAC noise within the receive band is governed by

equation (2-46). Outside the received band, the noise floor is governed by how many bits are modulated on each sub-carrier according to the ADSL standard.

C) Optimum DAC for ADSL-CO

At the ADSL-CO, we have [9]

- $PSD_{Tx} = -40 \text{ dBm/Hz}$
- Maximum $BW_{Tx} = 1104-26 = 1078 \text{ KHz}$
- Maximum $BW_{Rx} = 138-26 = 112 \text{ kHz}$

The simulated DAC requirement is shown in Fig. 2-17.

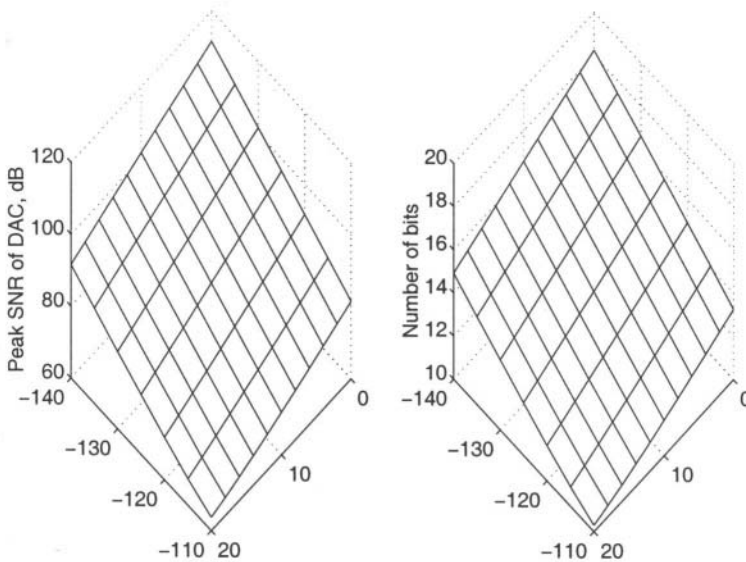


Figure 2-17 ADSL-CO DAC requirements vs. echo rejection and background noise. X-axis: background noise and interference power spectral density in dBm/Hz; Y-axis: echo rejection in dB.

D) Optimum DAC for ADSL-CP

At the ADSL-CP, we have [9]

- $PSD_{Tx} = -38 \text{ dBm/Hz}$
- Maximum $BW_{Rx} = 1104-26 = 1078 \text{ KHz}$
- Maximum $BW_{Tx} = 138-26 = 112 \text{ kHz}$

The simulated DAC requirement is shown in Fig. 2-18.

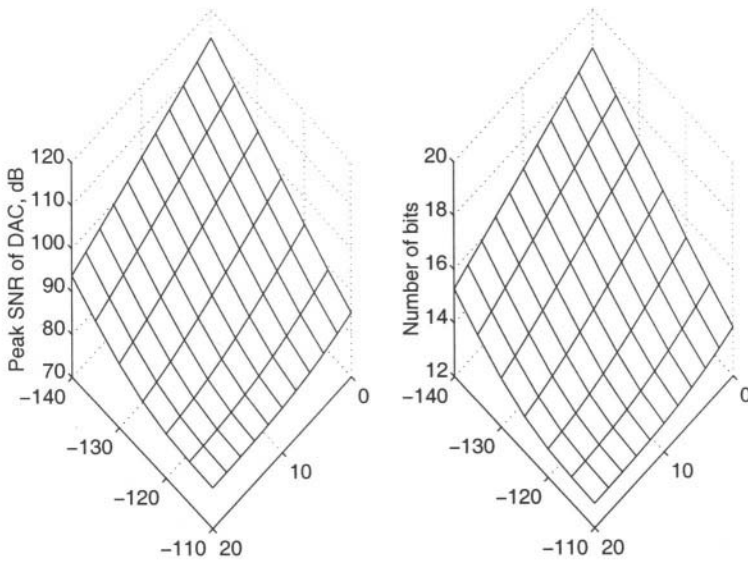


Figure 2-18 ADSL-CP DAC requirements vs. echo rejection and background noise. X-axis: background noise and interference power spectral density in dBm/Hz; Y-axis: echo rejection in dB.

2.8 ADC REQUIREMENTS FOR WIDEBAND RADIO

A digital radio (transceiver) is exactly the same as an analog radio except one difference; some of the analog functions are replaced with their digital equivalent, i.e., data converters are usually used in digital radio in the radio link. Notice that digital radio is not necessarily digital modulation and demodulation. As a matter of fact, digital radio can do an excellent job in receiving analog modulated radio as well.

We further divide digital radio into two categories, the narrow band, and the wideband. By narrowband we mean that there are sufficient analog filtering functions that eliminate or significantly reduce the un-desired signals such that only the signal of interest is present at the ADC. By wideband, we simply mean that all the signals including the adjacent channels are present at the ADC without much attenuation by analog circuitry and the filtering is done digitally in the later stages in the DSP. A wideband radio can be used to receive all the channels within an entire band such as cellular or other similar wireless services. The obvious advantage is the sharing of most hardware among all channels. In order to fully utilize the DSP power in a wideband radio, the mixer and the ADC must have a high dynamic range. And they are usually most difficult to design. A typical wideband receiver is shown in Fig. 2-19. The radio signal is received by the antenna. The radio frequency (RF) filter or tuner selects the band of interest (more than one channel is within the band). The received signal is amplified before it is fed to the mixer to generate the intermediate frequency (IF) sig-

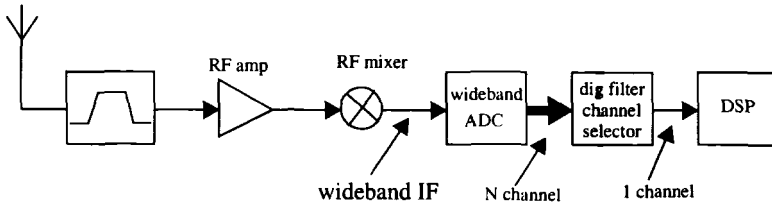


Figure 2-19 A typical wideband radio receiver.

nal. The ADC is used to convert the IF signal into a digital signal. The digital IF signals are processed by a digital decimation filter and digital channel selector before it reaches the DSP block. Depending on the noise level after the RF mixer, some analog filters might be needed for anti-aliasing purposes.

2.8.1 IF Sampling

In order to sample the wideband IF, the wideband ADC must have a high sampling rate governed by the Nyquist theory. However, since the wideband IF is bandpass filtered, the ADC data rate is only required to be twice the bandwidth, not twice the highest IF. When the ADC data rate is less than twice the highest IF, it is usually referred to as undersampling. For undersampling ADCs, the input sample-and-hold must have a bandwidth capable of sampling the highest IF signal.

One of the biggest challenges in designing a radio architecture is the placing of the IF, since the ADC tend to generate unwanted spurs that show up in the digital spectrum of the data conversion. For narrow band radio, there is only the signal of interest. If we know that a certain harmonic (for instance, the third harmonic) dominates, we can place the IF and the ADC sampling rate such that the strong harmonic is out of the signal band of interest. This can be easily achieved by frequency translation. When the IF signal is sampled at f_s , the output spectrum will appear at the following frequencies.

$$\sum_{n=-\infty}^{n=\infty} \pm f_{IF} + n \cdot f_s \quad (2-47)$$

where f_{IF} is the IF (it could be a whole band) and f_s is the ADC sampling rate. The minus sign prior to the f_{IF} indicates that this spectrum is reversed, which means that the higher frequencies are translated to the lower frequencies and vice versa. The k -th harmonic will appear at the following location

$$\sum_{n=-\infty}^{n=\infty} \pm k \cdot f_{IF} + n \cdot f_s \quad (2-48)$$

By properly choosing the IF and the data rate, the harmonics can be placed out of the band of interest. With this technique, the distortion requirement on the ADC is greatly

reduced.

For wideband radio, it is more complicated. Besides the distortion generated by each carrier, the inter-modulation may be the dominating factor, making the above-discussed technique less appealing. Therefore, the ADC converters for wideband radio must have superior linearity.

2.8.2 ADC Requirements

Since the ADC in a wideband radio receiver quantizes all the channels within the entire band, no automatic gain control (AGC) can be used to compensate for weak/strong signals. Reducing the gain for strong signals reduces the sensitivity to weak signals, and increasing the gain for weak signals overloads the ADC in presence of strong adjacent signals. In a practical implementation, some kind of AGC (actually attenuation) should be used to avoid overloading the ADC which would cause all calls to be disconnected rather than just dropping the calls with very weak signals. In principle, the ADC requirement in a wideband radio is very similar to the ADC requirement for DMT.

To find the optimum ADC, we need to know the noise floor, the signal power, and the peak-to-average ratio.

The fundamental noise floor is due to the atmospheric noise received by the antenna, its PSD in dBm/Hz is given by

$$PSD_{na} = 10 \cdot \log\left(\frac{B \cdot T}{0.001}\right) \approx -174 \text{ dBm/Hz} \quad (2-49)$$

where B is the Boltzman constant ($1.38 \times 10^{-23} \text{ J/K}$) and T is the temperature in K. The above approximation assumed T is equal to 300 K.

The RF signal passes through an RF filter or tuner with a signal bandwidth designated by BW . Therefore, the *rms* noise power within the total signal bandwidth is given by

$$P_{na}(\text{dBm}) = PSD_{na} + 10 \cdot \log(BW) \quad (2-50)$$

Assume that within the bandwidth BW , there are m channels, each having a maximum power level P_c . For most radio systems, the maximum power level is specified by the standard, e.g. it is -13 dBm for GSM. For most radio systems, constant envelop modulation is used, i.e. the carrier amplitude is not altered according to the digital bit stream. For instance, GSM uses Gaussian minimum-shift keying (GMSK) that basically is a frequency modulated (FM) signal preceded by a Gaussian low-pass filter.

Assuming that all carriers are un-correlated, the maximum signal power within the bandwidth BW is given by

$$P_{sig}(\text{dBm}) = P_c + 10 \cdot \log(m) \quad (2-51)$$

The number of channels m within the bandwidth BW can be easily found out given the channel bandwidth and the channel spacing (the distance between the centers of two

adjacent channels).

For a single carrier or single channel, the peak-to-average ratio is 1.4 since the carrier is usually a sine or cosine. Assuming that the channels are un-correlated, we have the worst case peak-to-average ratio given by

$$PAR = 1.4 \cdot \sqrt{m} \quad (2-52)$$

(If there are many channels, the total signal behaves like AWGN. The PAR is about 5.3 for a clipping factor of 10^{-7} .)

The RF amplifier and mixer will introduce noise. Assume that the total noise figure from the antenna to the ADC input is NF (dB). The total noise at the ADC input is therefore increased by NF (dB). Since the conversion gain amplifies both the signal and noise, it does not have impact on the ADC dynamic range requirement. Assuming the oversampling ratio is OSR (the sampling rate divided by twice the bandwidth BW), we have the ADC peak SNR requirement with the bandwidth BW given by

$$\begin{aligned} SNR_{ADC} &= P_{sig} - (P_{na} + NF) + 20 \cdot \log\left(\frac{PAR}{1.4}\right) - 10 \cdot \log(OSR) \\ &= P_c - (P_{na} + NF) + 20 \cdot \log(m) - 10 \cdot \log(OSR) \\ &= P_c - \left(PSD_{na} + 10 \cdot \log\frac{BW}{m}\right) - NF \\ &\quad + 10 \cdot \log(m) - 10 \cdot \log(OSR) \end{aligned} \quad (2-53)$$

Notice that BW/m is a constant, i.e, the channel spacing. The first line in the equation is the dynamic range for a single carrier. It is seen that the ADC requirement increases with the number of the channels.

To derive the minimum ADC requirement, we only need to replace the noise floor at the input due to the atmospheric noise and the RF noise figure with required minimum carrier power and its carrier-to-noise ratio for a given bit error rate.

2.9 SUMMARY

In this chapter, we have briefed communication-related terminologies and concepts and discussed the impacts of communication systems on data converter requirements. In order to meet a communication system specification, we have derived minimum data converter requirements that are mostly influenced by the error probability and the peak-to-average ratio. We have also discussed optimum data converters for ADSL that are a function of application environments such as the transmit power spectrum, the receive power spectrum, the hybrid rejection, and the background noise. Wideband radio for multi-channel applications are similar to DMT but with much fewer carriers. We have derived optimum A/D converter requirements for wideband radio as a function of the single carrier dynamic range, the number of channels, the noise figure, and the background noise. To apply these knowledge to a specific system, we usually need

to add 1~2 bit (or even more) design margin depending on the system trade-offs. Also coding can reduce the data converter requirements by 3~6 dB, which has not been treated in this chapter but can be easily added to the equations. As far as distortions are concerned, they usually follow the SNR requirements without being effected by oversampling.

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3 OVERVIEW OF HIGH-SPEED A/D CONVERTER ARCHITECTURES

3.1 INTRODUCTION

There are several well-known ADC architectures with different properties making them more or less suitable for a certain specification. To choose the proper architecture, their limitations must be investigated. The oversampling sigma-delta converter is often a suitable architecture when a relatively high resolution over a moderately high bandwidth is required. However, for instance VDSL applications, requires a large resolution (~ 12 bits) over a bandwidth larger than 10 MHz. The high bandwidth makes the sigma-delta converter less attractive for such applications since the oversampling makes the sampling frequency very high. Therefore other architectures must be considered.

In this chapter we summarize some of the most important limitations of different high-speed ADC architectures. The focus is on high performance over large bandwidths. In Sec. 3.2 the influence of the S/H circuit and sampling time uncertainty is discussed. Flash converters are the subject of Sec. 3.3 where we also briefly consider meta-stability in comparators. The semi-flash converter and its limitations are discussed in Sec. 3.4 while folding and interpolating converters are treated in Sec. 3.5. The pipelined ADC is the most popular architecture for high speed, high resolution applications. The properties of the pipelined ADC are reviewed in Sec. 3.6. The discussion in this chapter is focused on high-speed ADCs with a high signal bandwidth, but Sec. 3.7 briefly deals with oversampling sigma-delta converters and section Sec. 3.8 deals with some other types of low speed converters. For very high sampling rates it may not be possible to use a single ADC. In such applications the time-interleaved converter may be an option. The time-interleaved converter is the subject of Sec. 3.9. Finally a survey of the reported performance of high-speed ADCs is presented in Sec. 3.10.

3.2 SAMPLE-AND-HOLD (S/H) CIRCUITS

In all ADCs it is necessary to sample-and-hold (S/H) the signal. In flash converters, this is done in the digital domain after the comparators but an analog S/H circuit is used at the input of most CMOS wide band ADCs. Without a S/H at the input the performance at high signal frequencies of any ADC is usually very poor. This will become clear in the following discussions.

3.2.1 Sampling-Time Uncertainty

The dynamic performance of a S/H circuit is limited by the precision in the sampling instants [1]. Variations in the sampling instants may be caused by clock jitter, switch imperfections and signal dependent delays. The effect of the sampling time uncertainty is illustrated in Fig. 3-1 where the desired sampling time is t but due to circuit

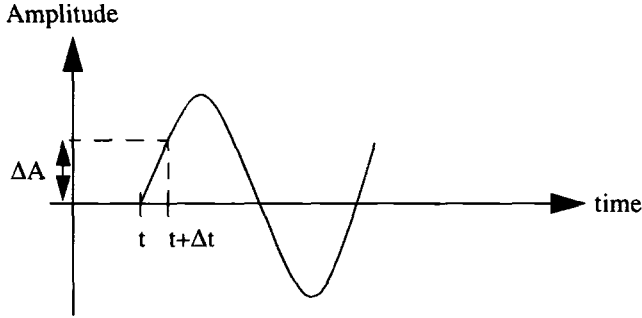


Figure 3-1 Effect of sampling time uncertainty.

imperfections the actual sampling time is $t + \Delta t$. The error ΔA caused by this delay depends on the slope of the input signal and for a sinusoidal input signal the worst case slope is at the zero crossing. The first-order approximation of the peak error is given by

$$\Delta A_{max} = \Delta t \cdot \left. \frac{\partial}{\partial t} V_{in}(t) \right|_{t=0} = \omega \cdot A \cdot \cos(\omega t) \cdot \Delta t \Big|_{t=0} = \omega \cdot A \cdot \Delta t \quad (3-1)$$

where A is the amplitude of the sinusoid. The error clearly increases at high signal frequencies but is independent of the sampling frequency. Many of the errors causing sampling time uncertainty can be reduced by using clever circuit techniques as for instance bottom plate sampling. The performance is however fundamentally limited by clock jitter in the switch signals. If the clock jitter is assumed to be random noise with variance σ_t^2 the error power can be approximated as [2]

$$v_{jn}^2 = \sigma_t^2 \cdot \frac{1}{T} \int_0^T \left(\frac{\partial}{\partial t} V_{in}(t) \right)^2 dt \quad (3-2)$$

where T is the integration time. For periodic signals the integration time can be chosen as the signal period. If the input signal is assumed to be a sinusoidal, the error power can by using (3-2) be calculated as [2]

$$v_{jn}^2 = (2\pi f_{in} A)^2 \sigma_t^2 / 2 \quad (3-3)$$

where A is the amplitude of the sinusoid and f_{in} the signal frequency.

Thus the SNDR is limited by

$$\text{SNDR} = 10 \cdot \log \frac{1}{(2\pi f_{in} \sigma_t)^2} \quad (3-4)$$

or by using the ENOB

$$\text{ENOB} = \frac{10 \cdot \log \frac{1}{(2\pi f_{in} \sigma_t)^2} - 1.76}{6.02} \quad (3-5)$$

It should be noted that the SNDR is independent of the sampling frequency. The obtainable ENOB as function of the clock jitter, σ_t , is plotted for different input frequencies in Fig. 3-2. The figure shows that the performance degrades rapidly at high

ENOB

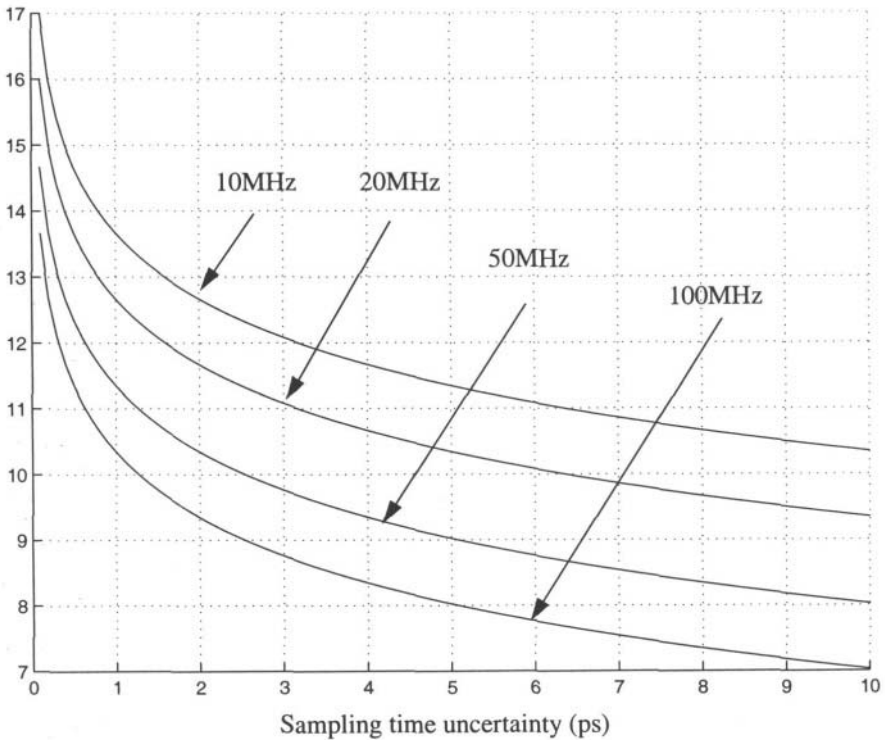


Figure 3-2 ENOB as function of sampling time uncertainty.

signal frequencies. To achieve 10 ENOB at 20 MHz the sampling time uncertainty must be in the order of 6 to 7 ps.

3.2.2 Thermal Noise

The thermal noise is a limiting factor in the S/H circuits. It usually dominates over the flicker noise (the other noise source present in MOS circuits) in wideband circuits. Due to the sampling, the thermal noise is folded back into the signal band, introducing a fundamental limitation.

Most high-speed S/H circuits use passive sampling to get a high bandwidth and a high sampling rate. In this type of sampling circuits the thermal noise power is kT/C [4]. Assuming a sinusoidal input signal the SNDR is given by

$$SNDR = 10 \cdot \log\left(\frac{V_{in}^2}{2 \cdot kT} \cdot C\right) \quad (3-6)$$

where V_{in} is the amplitude of the input signal.

Since the voltage swing is limited by the power supply voltage, the only way to increase the SNDR is to increase the capacitor. However, for every doubling of the capacitor, the SNDR only increases by 3 dB and a large capacitance implies a large power consumption. For moderate resolutions, the capacitor size calculated from the thermal noise requirement is very small and a larger capacitor may be chosen for other reasons. For high resolutions a capacitor of several pF is needed to handle the thermal noise. In addition to this the opamp noise may be in the same order as the kT/C noise, leading to further increase of the capacitor.

3.2.3 Nonlinearity

The linearity of the input S/H must be at least as good as the resolution of the ADC. The errors may be caused by nonlinear switch on resistance, clock feedthrough errors, finite amplifier gain, parasitic capacitors or any other error sources present in the circuit. It should be noted that most of the errors increases at high signal and sampling frequencies.

3.3 FLASH CONVERTERS

Flash converters can reach very high sampling rates since the only analog building block is the comparator. The principle of the flash converter is illustrated in Fig. 3-3. There are one comparator for each decision level in the converter. The reference levels for the comparators are usually generated by a resistor string. The signal at the output of the comparators is a thermometer coded representation of the input signal. A decoder is used to generate a more convenient representation at the output.

The main problem with this architecture is that the number of comparators increases exponentially with the number of bits. For N bits, $2^N - 1$ comparators are needed. Due to the large number of comparators the number of bits is usually limited to 10, since the chip area and power consumption would be too large for higher resolutions. Thus the SNDR is limited to 60 dB. One more problem caused by the large number of comparators is the large input capacitance. The circuit driving the ADC must therefore

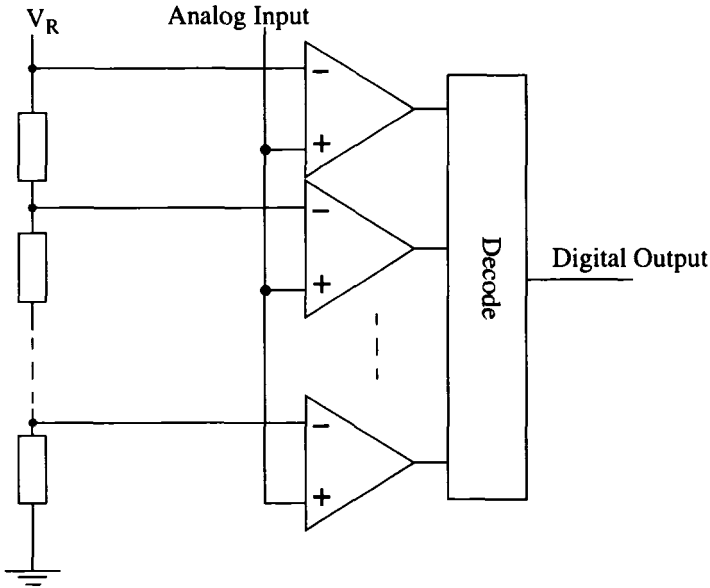


Figure 3-3 The Flash Converter.

handle a large capacitive load. The pure flash ADC is not suitable for telecommunications applications where a high resolution is required. However, a flash converter or at least a comparator is always used as a part of all ADC architectures. It is therefore necessary to know the limitations of the flash converter.

3.3.1 Static Errors

The static errors affect the accuracy in the decision levels of the converter. Any deviation from the ideal levels will cause DNL and INL errors. Errors in the reference levels are caused by resistor mismatch, which usually is in the order of 2 to 0.1 % without laser trimming [5], and offsets in the comparators. The errors must be less than [6] (no missing codes are guaranteed for $INL < \pm 1/2 \text{ LSB}$)

$$\epsilon_{ADC} = \pm \text{LSB}/2 = \pm FS/2^{N+1} \quad (3-7)$$

where FS is the full scale range of the input signal.

3.3.2 Dynamic Errors

The dynamic errors typically increase at high signal and sampling frequencies. They are mainly caused by timing errors. There are four main sources of timing errors [6] in the flash converter

1. Skew of the clock and input signal at different places on the chip.

It gives frequency dependent errors and degrades the performance at high signal fre-

quencies. It can be reduced by using an input S/H circuit.

2. Limited rise/fall times of the sampling clock.

They give signal dependent turn off times in the sampling switches. It can be reduced by, for instance, feedback [1] or bottom-plate sampling techniques [3]. The rise and fall times can to a limited extent be reduced by increasing the size of the clock drivers, but this will also increase the power consumption.

3. Signal-dependent delays.

Signal dependent delays can cause distortion in e.g. comparators (see below). These errors can be reduced by increasing the bandwidth of the circuit or in some cases by using an input S/H circuit.

4. Sampling clock jitter.

Sampling clock jitter is present in all ADCs and will act as a fundamental limitation in the circuit at high signal frequencies.

3.3.3 Signal Dependent Delay in Comparators

In [6] it is shown that the delay in the comparators is signal dependent and that this will cause third-order distortion. Using a single pole model for the comparator the SNDR due to signal dependent delay can be approximated as [6]

$$SNDR = 10 \cdot \log \left(\frac{3\pi \cdot f_b \cdot e^{\frac{V_{tr}f_b}{V_{fs}f_{in}} + 1}}{2 \cdot f_{in}} \right) \quad (3-8)$$

where f_{in} is the input frequency, f_b the -3 dB bandwidth of the comparator, V_{tr} the linear range of the comparator and V_{fs} the full scale voltage of the converter.

For a 10-bit accuracy the bandwidth of the comparator should be about 10 times the input signal bandwidth. The timing errors can be reduced by using a S/H circuit at the input of the ADC and thus moving the timing problems to S/H where they are easier to handle. A problem with this solution is that the sampling frequency is reduced since it is difficult to design a high speed S/H, especially if the number of comparators is large.

3.3.4 Meta-Stability

A severe limitation in very high-speed converters is meta-stability. It increases rapidly at high sampling frequencies. Meta-stability is caused by the finite gain in the comparators. Certain input signals generate output signals that can not be detected by the digital circuits. It is common to use comparators with a pre-amplifier followed by a regenerative latch. The metastability error probability for an N -bit converter using this type of comparators is [7]

$$P_E = \frac{2(2^N - 1)V_o}{V_R A} \cdot e^{-t_r/\tau} \quad (3-9)$$

where V_R is the analog input range, V_o the output voltage swing required for valid logic levels, A is the combined voltage gain of the pre-amplifier and the latch's gain in the transparent state, τ is the regenerative time constant for the latch and t_r is the resolution time of the latch. It is obvious from (3-9) that the performance deteriorates at high sampling frequencies since the error probability is exponentially related to the resolution time. The effect of these errors in the flash converter depends on the decoding logic and the performance can be improved by using pipelined latches [7].

3.4 SEMI-FLASH CONVERTERS

To reduce the number of comparators the semi-flash converter can be used. In this type of ADC a coarse ADC generates the most significant bits. The residue, the difference between the input signal and the D/A converted signal from the coarse ADC, is converted by a fine A/D converter (see Fig. 3-4). The conversion speed of the semi-flash

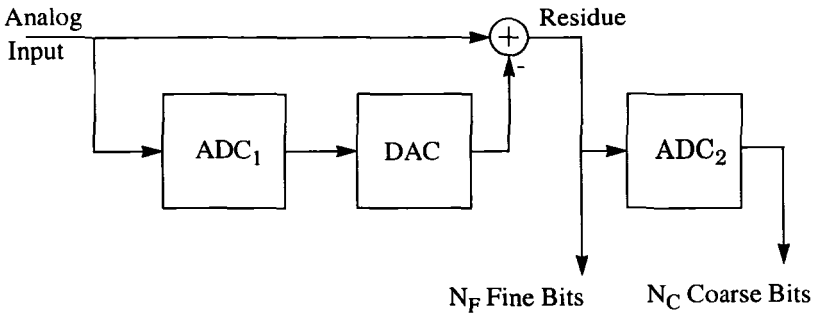


Figure 3-4 The Semi-Flash Converter.

converter is less than half compared to the flash converter but the number of comparators is only

$$2^{N_C} + 2^{N_F} - 2 \quad (3-10)$$

where N_C is the number of bits in the coarse ADC and N_F the number of bits in the fine ADC.

There are two additional building blocks in the semi-flash converter compared with the flash converter, a DAC and a subtracter. It is common to use a residue amplifier after the subtracter as shown in Fig. 3-5 to avoid small residue signals which are sensitive to noise. The gain of the amplifier is usually chosen as

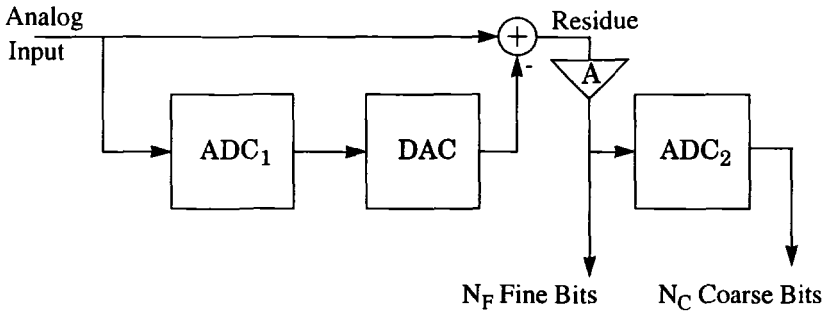


Figure 3-5 The Semi-Flash converter with residue amplifier.

$$A = 2^{N_c} \quad (3-11)$$

to make the signal swing equal for both the ADCs. This simplifies the design since the same reference ladder can be used by both ADCs.

There are in principle three ways of performing the D/A conversion and subtraction, by using voltage, current or charge. All of these circuit techniques have different properties and different accuracy-speed trade-offs [8].

3.4.1 Static Errors

To avoid missing codes and large linearity errors the total error should be less than $\pm \text{LSB}/2$. The highest accuracy is required for the ADC and DAC in the first stage and the total error must be within the total resolution of the semi-flash converter, i.e. the error must be [9]

$$|\epsilon_{DAC}| = |\epsilon_{ADC_1}| \leq \frac{\pm FS}{2^{N_c + N_f + 1}} \quad (3-12)$$

The errors introduced by the second ADC and the residue amplifier only need an accuracy corresponding to the resolution in the second stage. Hence we tolerate larger relative errors in the second stage. If the residue amplifier is assumed to have a gain of 2^{N_c} the errors are limited by [9]

$$|\epsilon_A| = |\epsilon_{ADC_2}| \leq \frac{\pm FS}{2^{N_f + 1}} \quad (3-13)$$

The influence of some errors in the first stage can be reduced by using digital correction [10]. This technique is based on the fact that the errors in the first stage cause the residue to be out of the input range of the second ADC. By reducing the gain of the residue amplifier this can be avoided. The gain factor of the residue amplifier is usually reduced by a factor 2 to handle the errors in the first stage and to make the decod-

ing of the digital output easy. The gain reduction will introduce redundancy in the converter which also means that the resolution of the converter corresponds to $N_C + N_F - 1$ bits rather than $N_C + N_F$ bits. With digital correction, the nonlinearity errors in the first ADC can be much larger without degrading the performance [9]. We now have that

$$|\epsilon_{ADC1}| \leq \frac{FS}{2^{N_C+1}} \quad (3-14)$$

The errors in the DAC must still be within the total resolution according to (3-12) even though the second stage ADC is not overloaded. This is because the actual output levels of the DAC must be known to decode the digital output correctly. The static converter linearity is ultimately limited by the linearity of the DAC and calibration for the DAC is necessary if the resolution is high. The DAC calibration can be performed in both the analog [11] and in the digital domain [12] but requires additional hardware.

3.4.2 Dynamic Errors

The dynamic errors are caused by timing errors in the building blocks. The skew between the two input signals to the subtracter is difficult to match. The effect of the skew error is frequency dependent and the dynamic performance degrades at high signal frequencies. The maximal timing error [13], assuming a full scale sinusoidal input signal, is limited by

$$\Delta t \leq \frac{1}{2^{N_C+N_F+1}\pi f_{in}} \quad (3-15)$$

where f_{in} is the input frequency. If, however, one bit of digital correction is used, larger timing errors can be tolerated and the maximal timing error is only limited by

$$\Delta t \leq \frac{1}{2^{N_C+1}\pi f_{in}} \quad (3-16)$$

i.e. the resolution in the first stage. Without digital correction the required resolution is the total resolution of the ADC. To achieve a 10 bit resolution at 20 MHz input frequency the delay skew must be in the order of 8 ps which is very difficult to achieve. If digital correction is used and the resolution in the first stage is 5 bits the maximum delay skew is about 500 ps.

To avoid the delay problems, a S/H circuit can be used at the input. However the sampling frequency may be reduced since it is difficult to design an accurate high-speed S/H circuit especially when the capacitive load at the input is large. There are also semi-flash converters having a S/H circuit both at the input and between the two stages. This type of converter can be viewed as a special case of the pipelined converter discussed Sec. 3.6.

3.5 FOLDING AND INTERPOLATING CONVERTERS

In folding and interpolating A/D converters analog preprocessing is used to reduce the power consumption and the chip area while maintaining a high sampling rate [14]. The principle behind the folding technique is to perform the conversion in two steps just as for the semi-flash converter, but the coarse and fine converters operate in parallel, as illustrated in Fig. 3-6.

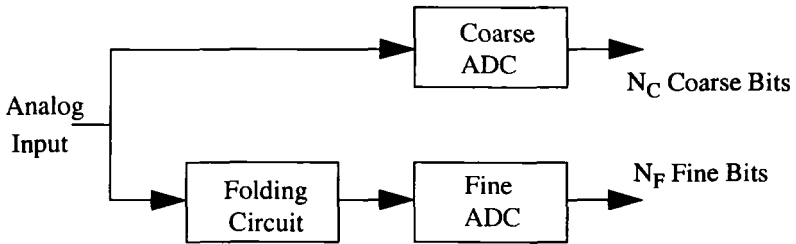


Figure 3-6 The folding ADC.

The analog folding circuit in front of the fine ADC generates the same signal as the subtractor in the semi-flash converter. Since the output from the coarse converter is not used in the fine converter the conversion rate is approximately the same as for the flash converter while the number of comparators is only $2^{N_C} + 2^{N_F} - 2$. The folding signal for $N_C = 2$ is shown in Fig. 3-7. The sawtooth waveform is difficult to gener-

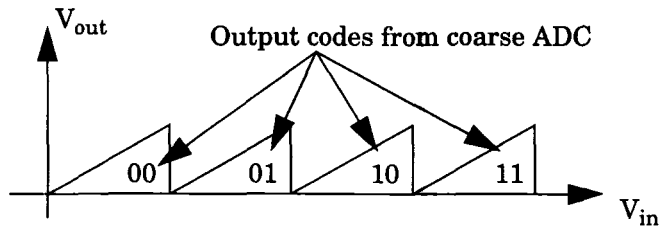


Figure 3-7 The transfer function of the folding circuit for $N = 2$.

ate and a triangular waveform is preferred in the actual implementation but it is still difficult to get sharp corners in the waveform. Therefore the linearity for large output signals of the folding circuit is poor.

This problem can be circumvented by using several folding circuits with different offsets and thus only utilizing a small linear portion of the input range. Unfortunately the number of folding circuits becomes large and the complexity of the converter may be in the same order as for the flash converter. Therefore some of the folding signals are generated by resistive interpolation. However too many interpolations will create accuracy problems [6] and most of the folding or interpolating ADCs reported so far

have a resolution less than 10 bits. The resolution can be increased by performing the folding in several stages. This technique was used in [15], [57] to achieve 12 bits resolution at 60 MS/s. The advantages of the folding-interpolating architecture are normally best utilized in a bipolar or BiCMOS process since the bipolar transistor can make the bandwidth in the folding circuit very large.

The static errors in the folding-interpolating converter are caused by comparator offsets and in-accuracies in the folding and interpolation circuits. The dynamic errors are caused by timing errors and a limiting factor is the delay mismatch between the two sub-converters. The effect of these errors can not be as easily corrected as was the case for the semi-flash converter. Another problem with the folding technique is that the required bandwidth of the folded signal is increased by the folding rate (the number of times the signal is folded) which is difficult to handle especially in CMOS. These problems can be partially solved by using a S/H circuit at the input or a distributed S/H circuit as in [16].

3.6 PIPELINED CONVERTERS

Multi-stage pipelined ADCs [9] are one of the most popular architectures for high-speed applications. They are similar to the semi-flash converters and consists of several cascaded stages each with a low resolution ADC, a S/H-amplifier and a DAC as illustrated in Fig. 3-8. The properties of the pipelined converter is similar to the semi-

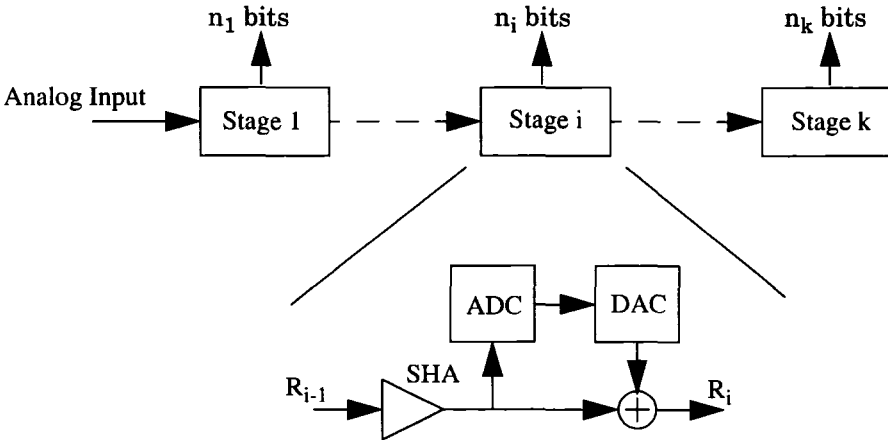


Figure 3-8 The pipelined ADC.

flash converter and will be discussed in more detail in chapter 9. The main difference between the semi-flash ADC and the pipelined ADC is that there is a S/H circuit between the stages which increases the maximum sampling rate of the converter. Due to the pipelining the output word is delayed several clock cycles but in most applications this is not a severe limitation.

3.6.1 Static Errors

The accuracy of the sub-ADCs need not be very high if digital correction is used. The only circuits needing an accuracy corresponding to the full resolution of the converter are the input S/H circuit and the DAC in the first stage. The static performance is mainly limited by the matching errors of the DAC and for a resolution of more than 10 to 12 bits it is usually necessary to use calibration.

3.6.2 Dynamic Errors

Any delay between the signal paths in the first stage will give distortion unless an input S/H is used. The delay skew is limited by the same expressions used for the semi-flash converter in Sec. 3.4. Due to the lower stage resolution, larger delays can normally be handled in the pipelined converter than in the semi-flash converter, assuming digital correction is used.

3.7 OVERSAMPLING SIGMA-DELTA ADCS

Oversampling sigma-delta ADCs [17, 18] are based on the principle that the conversion error can be highpass filtered and later removed by digital filters. The requirements on the analog parts are relaxed and high-resolutions can be achieved. The drawback of this type of converter is that for high resolutions the signal bandwidth is small due to the oversampling. A second-order sigma-delta modulator is shown in Fig. 3-9. The modulator consists of two discrete-time integrators, two DACs and one quan-

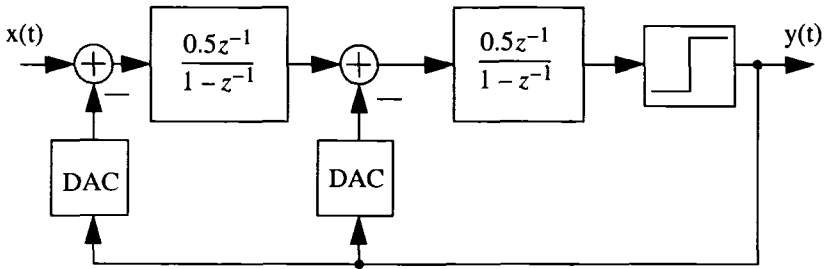


Figure 3-9 Second-order sigma-delta modulator.

tizer. If the quantization error is modelled as white noise the output signal is given by

$$Y(z) = z^{-2}X(z) + (1 - z^{-1})^2E(z) \quad (3-17)$$

where $E(z)$ is the quantization noise in the quantizer and $X(z)$ is the input signal. The quantization noise is highpass filtered while the input signal is only delayed. By removing the high frequency noise with digital filters and decimating the signal, an output signal having a small quantization noise is achieved. Every doubling of the sampling rate will provide $L + 0.5$ extra bits, where L is the order of the modulator.

To increase the resolution, the order of the modulator or the resolution of the quantizer must be increased. A problem with increasing the resolution of the quantizer is that nonlinearities in the DACs will directly limit the resolution of the ADC. A DAC with only two output levels have only offset and gain errors while a multi-bit DAC also have non-linearities.

There are also problems with increasing the order of a single-stage modulator. Special architectures to avoid instability must be used and a small input swing may be required to avoid saturation of the modulator. This may necessitate large capacitors and a large capacitance spread in the implementation which reduces the speed [19]. By using a multi-stage structure these problems are avoided, but sensitivity to matching errors between the analog and digital parts limits the resolution. A more detailed discussion on sigma-delta converters is given in chapter 11.

There are some alternatives to increasing the bandwidth. In [20] a 16-bit converter with an oversampling ratio of only 8 was designed by combining a sigma-delta modulator with a pipelined converter. The modulator uses a multi-bit quantizer and dynamic element matching must be used to reduce the distortion caused by the DAC. In [67] the modulator is pipelined in such that way the no oversampling is needed. In this case a resolution of 13-bits was achieved.

3.8 LOW SPEED CONVERTERS

There are several types of low speed converters. They are not so common in high-speed communications application since the conversion speed often is too small. They may however be used in time-interleaved converters which is the topic of the next section [22]. Here we only briefly consider some of the most common low speed converters.

The successive approximation converter [21, 22] shown in Fig. 3-10 consists of a

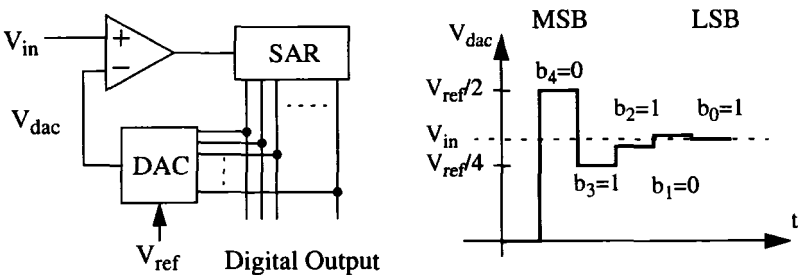


Figure 3-10 Successive approximation A/D converter.

comparator a DAC and a successive approximation register (SAR). Starting with the MSB the comparator determines whether the input signal is larger or smaller than the DAC output. The result is stored in the SAR. The process is repeated for all the bits in

the DAC. This is illustrated in Fig. 3-10 for a 5-bit converter. The number of iterations is directly determined by the number of bits in the converter. The speed is therefore reduced when the resolution is high. The static performance is limited by the DAC linearity and for high resolutions calibration is needed. To get a high resolution at high signal frequencies a S/H circuit is needed at the input of the converter.

The algorithmic or cyclic converter shown in Fig. 3-11 is based on the same algorithm

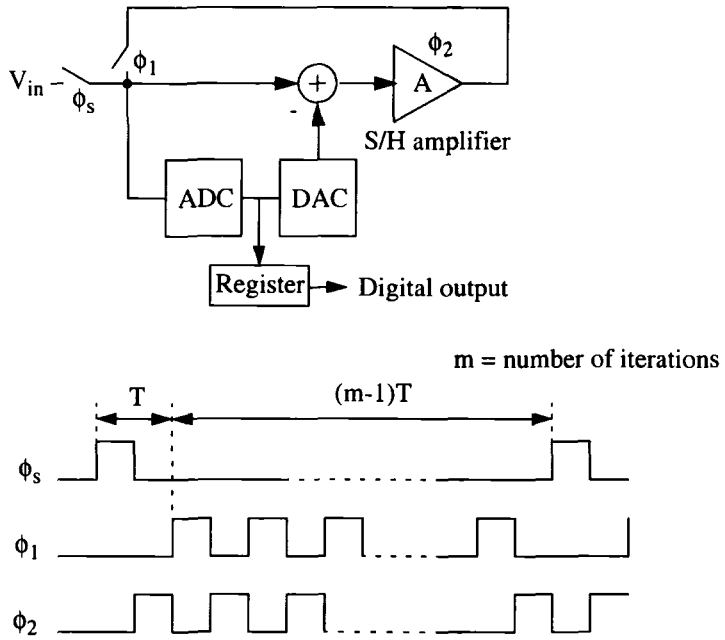


Figure 3-11 Cyclic A/D converter.

as the pipelined converter but the same stage is used to calculate all the bits. Hence the speed is reduced by the required number of iterations. The conversion starts by sampling the input signal and converting the analog input to a coarse digital value by the sub-ADC. The digital value is stored in a register. The residue is calculated by a DAC and a subtractor and stored in the S/H amplifier. In the next clock cycle the amplified residue is fed back to the input of the sub-ADC and the procedure is repeated. The properties of the cyclic converter are very similar to the pipelined converter. Digital correction can be applied by decreasing the residue gain. With digital correction the requirements on the sub ADC are relaxed.

The dual-slope integrating converter [23] shown in Fig. 3-12 is very accurate. The input signal is first integrated for a fix period of time, T_1 . Hence, the output of the integrator is proportional to V_{in} . The integrator input is then switched to a reference voltage. The time until the integrator output becomes zero, T_2 , is proportional to the input value and is measured by a digital counter. The zero crossing is detected by a comparator. The integrating converter can not be used for high signal bandwidths since the

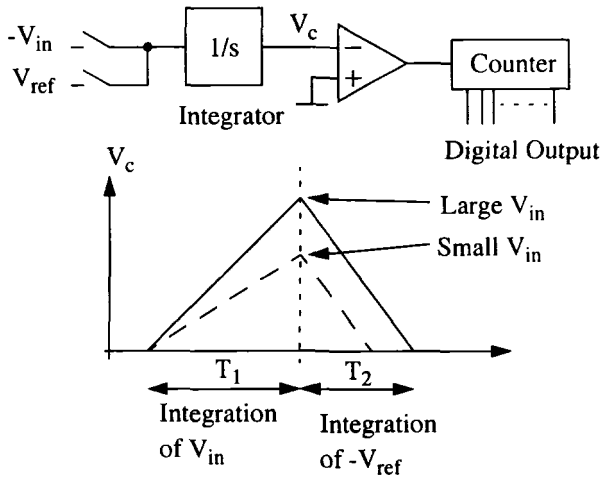


Figure 3-12 Dual-slope A/D converter.

maximum number of clock cycles is 2^N , where N is the number of bits.

3.9 TIME-INTERLEAVED (PARALLEL) CONVERTERS

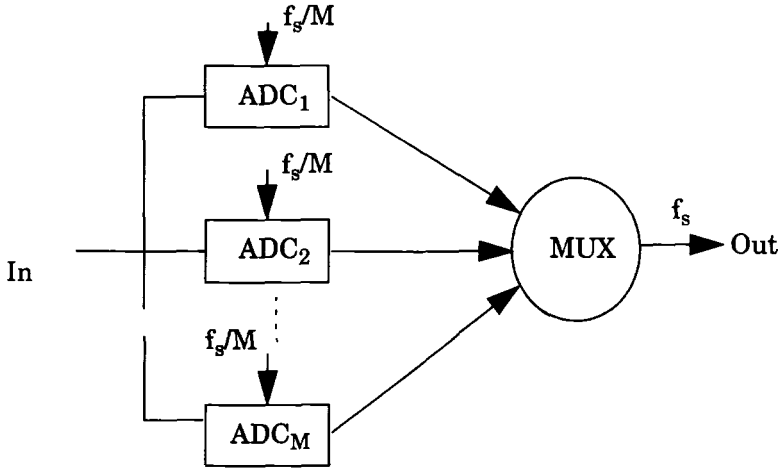
An attractive way to increase the conversion rate of ADCs is to use time-interleaving techniques where several ADCs operate on different clock phases [24]. This enables a higher total conversion speed but also introduces new problems which are mainly caused by mismatch between the channels. The concept of time-interleaving is illustrated in Fig. 3-13 where the converters can be of any type and are operated at f_s/M . The speed requirements on each converter are relaxed by a factor M but the number of converters is at same time increased by the same factor, which may result in a large chip area and power consumption. The increase in power and area is not necessarily a linear function of M compared to using only one converter, since smaller bias currents can be used due to the lower speed. It is also sometimes possible to share some of the building blocks between the channels [25]. There are three main error sources in time-interleaved systems, phase skew errors, gain errors and offset errors [24, 26, 27].

3.9.1 Offset Errors

Differences in offsets in the channels cause distortion in the output signal. The distortion is not signal dependent and will appear at (see chapter 10)

$$f_s/M \cdot m, m = 1, 2, \dots, M-1 \quad (3-18)$$

If the offset in channel k is denoted σ_k and is assumed to be normally distributed random variables with zero mean and variance σ_o^2 , the SNDR can be approximated as



Clock Phases

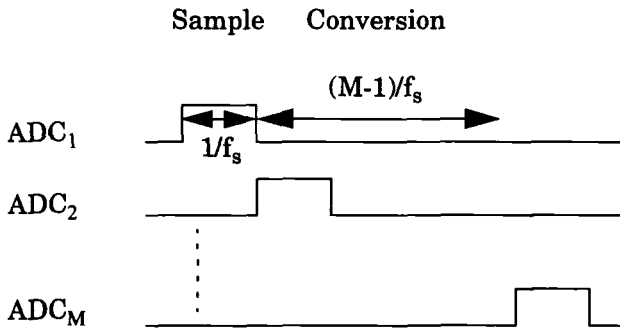


Figure 3-13 Parallel ADCs and corresponding sub-ADC timing.

(see chapter 10)

$$SNDR = 20 \cdot \log\left(\frac{1}{\sigma_o}\right) \quad (3-19)$$

3.9.2 Gain Errors

Different channel gains cause distortion in the output signal. If the gain in channel k is denoted a_k and is assumed to be a normally distributed random variable with mean a and variance σ_a^2 , the SNDR can be approximated as (see chapter 10)

$$SNDR = 20 \cdot \log\left(\frac{a}{\sigma_a}\right) - 10 \cdot \log\left(1 - \frac{1}{M}\right) \quad (3-20)$$

where M is the number of channels. The distortion tones are located at

$$f_{in} + m \cdot f_s / M, m = 1, \dots, M - 1 \quad (3-21)$$

3.9.3 Phase Skew Errors

When using parallel converters, as shown in Fig. 3-13, several clock phases, usually generated by an on-chip clock generator, are required. Any phase skew between the channel causes distortion. For sinusoidal input signals the phase skew errors will generate distortion tones at [26]

$$f_{in} + m \cdot f_s / M, m = 1, \dots, M - 1 \quad (3-22)$$

If the phase skew errors are treated as normally distributed random variables with zero mean and variance σ_t^2 the SNDR can be approximated as [26]

$$\text{SNDR} = 20 \cdot \log\left(\frac{1}{\sigma_t^2 2\pi f_{in}}\right) - 10 \cdot \log\left(\frac{(M-1)}{M}\right) \quad (3-23)$$

The last term depends on M (the number of channels) but changes only about 3 dB as M changes from 2 to infinity. Thus the number of channels have quite small effect on the total SNDR. The delay skew σ_t should be less than about 10 ps to get 10 bits resolution at 20 MHz input frequency. This is difficult to achieve in a CMOS process.

3.10 REPORTED PERFORMANCE

To compare the performance using the different ADC architectures, some high-speed ADCs are listed in Table 3-1. In the *Type* column parallel ADCs are specified as e.g. 2 Par-Pipe which means that there are 2 parallel channels and that the sub-ADCs are pipelined converters. The last two columns specify the dynamic performance at high input frequencies for the ADCs. The f_{in} column contains the input frequency and the *SNDR* column the measured SNDR. Some of the ADCs can operate at different sample frequencies and supply voltages. In such a case the values used for the dynamic measurements in the last columns are given as bold numbers. It should be noted that the converters are implemented for different supply voltages using different processes. It is therefore difficult to make a fair comparison. The intention here is only to give an overview of the achieved performance to get a rough indication on which architecture to choose for a certain application.

Table 3-1 ADC Performance

Ref.	Type	Process	Bits	f_s [MHz]	V_{dd} [V]	Power [mW]	Area [mm ²]	f_{in} [MHz]	SNDR [dB]
[28]	Flash	CMOS	6	200	5	400	2.7	low	35

Table 3-1 ADC Performance

Ref.	Type	Process	Bits	f_s [MHz]	V_{dd} [V]	Power [mW]	Area [mm ²]	f_{in} [MHz]	SNDR [dB]
[7]	Flash	CMOS	7	80	-	307	8.2	20	32
[29]	Flash	Bipolar	8	300	5.2	3300	33	100	35
[30]	Flash	CMOS	8	20	5	540	26.5	10	46
[31]	Flash	Bipolar	8	200	10.2	2000	15.2	50	35
[32]	Flash	Bipolar	8	500	5.2	3100	20.3	-	-
[33]	Flash	Bipolar	6	200	9.5	1100	9	50	34
[34]	Flash	BiCMOS	6	80	5	400	6	50	34
[35]	Semi-Flash	Bipolar	10	75	-	-	20	40	51
[36]	Semi-Flash	Bipolar	12	50	5	575	13.7	20	68
[37]	Semi-Flash	CMOS	8	50	5	600	13.8	20	44
[38]	Semi-Flash	CMOS	10	20	2	20	12	10	42
[39]	Semi-Flash	Bipolar	10	75	10	2000	20	20	54
[40]	Semi-Flash	BiCMOS	10	100	5	950	19	50	48
[41]	Semi-Flash	Bipolar	10	75	10	800	16	40	57
[42]	Semi-Flash	BiCMOS	10	50	5	500	16	20	48
[43]	Semi-Flash	Bipolar	10	50	5	750	11	20	55
[44]	Semi-Flash	Bipolar	12	128	10.2	5700	40	50	58
[45]	Semi-Flash	CMOS	8	20	5	50	4.5	10	41
[46]	Semi-Flash	CMOS	10	20	5	75	1.6	10	58.7
[47]	Semi-Flash	CMOS	10	16-25	3.3	195	0.8	14	56
[48]	Folding	CMOS	8	80-125	3.3-5	150	4	10	32
[49]	Folding	BiCMOS	8	100-200	5	575	4	20	32
[50]	Folding	Bipolar	10	300	5.2	4	39	100	43

Table 3-1 ADC Performance

Ref.	Type	Process	Bits	f_s [MHz]	V_{dd} [V]	Power [mW]	Area [mm ²]	f_{in} [MHz]	SNDR [dB]
[51]	Folding	CMOS	8	80	3.3	80	0.3	40	42
[52]	Folding	CMOS	8	70	3.3-5	110	0.7	10	32
[53]	Folding	Bipolar	8	650	4.5	850	4.2	150	48.6
[54]	Folding	Bipolar	8	100	5.2	800	12	50	45
[55]	Folding	Bipolar	8	55	5	300	6	10	46
[56]	Folding	CMOS	10	50	5	240	2	10	54
[57]	Folding	BiCMOS	12	50	5	300	8	10	<64
[58]	Folding	BiCMOS	10	40	5	65	0.8	4.1	57
[59]	Pipelined	BiCMOS	10	40	5	400	19	20	55
[60]	Pipelined	CMOS	10	20	5	50	13	10	48
[61]	Pipelined	CMOS	10	20	3.3	35	10.5	10	55
[62]	Pipelined	BiCMOS	10	20	10	1000	48	10	50
[63]	Pipelined	CMOS	10	20	5	240	8.7	10	58
[64]	Pipelined	CMOS	10	15-20	2.5	30	6.6	10	<40
[65]	Pipelined	CMOS	10	20	3	135	7	10	45
[66]	Pipelined	CMOS	10	14	1.5	36	5.7	4	54
[67]	Pipelined - SD	CMOS	13	18	5	324	47	8	71
[68]	Pipelined	CMOS	12	40	5	415	-	62	70
[69]	Pipelined	CMOS	12	25	5	280	-	20	68
[70]	Pipelined	CMOS	12	65	5	500	-	32	70
[71]	Pipelined	CMOS	12	20	5	300	-	10	68
[72]	2 Par- Pipe	CMOS	10	32-40	2.7- 5.5	85	4	10	45
[73]	14 Par- Succ	CMOS	10	40-70	5	267	8.9	-	-
[74]	2 Par- Pipe	CMOS	10	40-50	5	900	18	20	49
[75]	4 Par- Flash	GaAs, Bipolar	6	1000	-	16000	-	-	-
[24]	4 Par- Succ	CMOS	7	2.5	-	250	0.04	-	-
[3]	4 Par- Pipe	CMOS	10	95-100	5	1100	50	50	50

Table 3-1 ADC Performance

Ref.	Type	Process	Bits	f_s [MHz]	V_{dd} [V]	Power [mW]	Area [mm ²]	f_{in} [MHz]	SNDR [dB]
[25]	2 Par-Pipe	CMOS	8	52	5	250	15	20	46
[76]	4 Par-Pipe	CMOS	8	85	-	1100	25	40	40
[77]	2 Par-Semi	CMOS	12	1	5	25	14.3	1	59

From the discussion in the previous sections it is clear that it is very difficult to get good dynamic performance without a S/H circuit at the input. When a S/H or residue amplifier is needed it is usually the opamp that sets the speed limit. Since the dynamic performance over a large bandwidth is the main concern in this chapter some of the CMOS high-speed ADCs in the table above were plotted to make the comparison easier. The resolution and sampling frequency for these A/D converters is shown in Fig. 3-14. There are several publications with a resolution of 8 to 10 bits in the 10 to 100

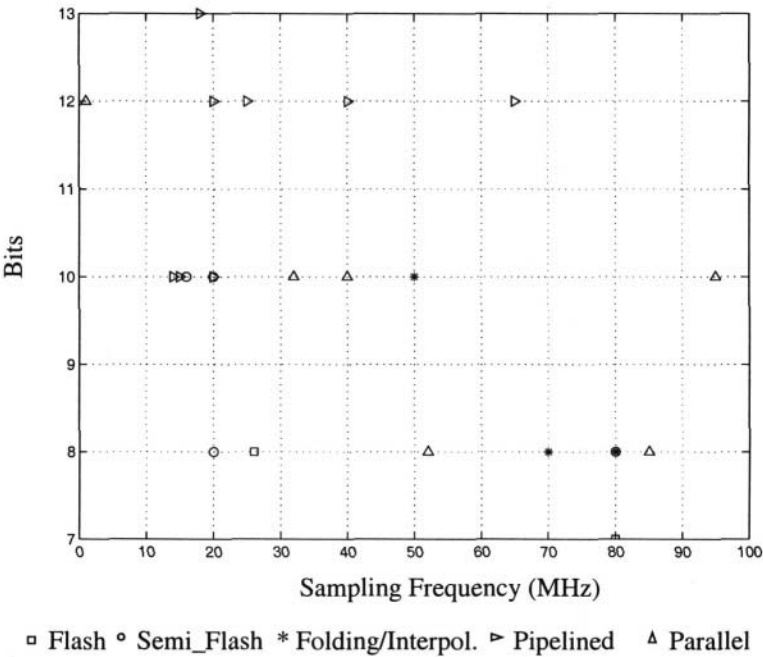


Figure 3-14 Resolution and sample frequency for CMOS A/D converters.

MHz range and a few industrial products with a resolution of 12 bits. It should be noted that the fabrication processes used for the industrial products are usually better suited for analog design at high frequencies than the processed used for many of the converters found in publications. Therefore a direct comparison of the performance is

not entirely fair. Shown in Fig. 3-15 is the ENOB for high input frequencies. It is seen

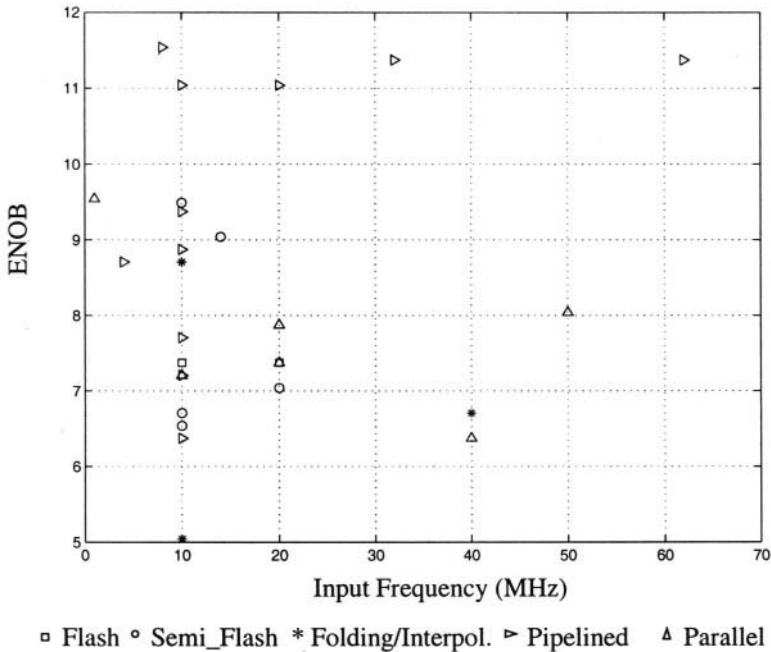


Figure 3-15 ENOB and input frequency for CMOS A/D converters.

that except for the industrial products very few converters have more than 8 ENOB at input signals larger than 10 MHz. The best performance is shown for pipelined and parallel converters with a S/H circuit at the input.

3.11 SUMMARY

In this chapter high-speed ADC architectures for telecommunications were discussed. We were mainly focused on architectures suitable for high signal frequencies and high sampling rates but both static and dynamic errors in the converters were briefly considered. In the following we summarize the most important conclusions.

1. Flash and Folding Converters

Due to large area and large power consumption flash and folding converters are best suited for resolutions < 10 bits at high sampling rates. These types of ADCs does not always have an input S/H circuit since this would limit the conversion speed. Without a S/H circuit at the input the dynamic performance at high input frequencies is poor. Some two-stage folding converters have been reported suitable for high resolutions at high sampling rates. They are however usually implemented in a bipolar or BiCMOS process.

2. Semi-Flash and Pipelined Converters

For higher resolutions pipelined or semi-flash converters are better suited than flash or folding converters. The requirements on the comparators can be relaxed by using digital correction. However, inaccuracies in the DACs necessitates calibration techniques for large resolutions. Due to the pipelining the pipelined converter can reach higher sampling rates than the semi-flash converter.

3. Time-Interleaved Converters

In the time-interleaved converter several ADCs operating at a lower speed are connected in parallel. Mismatch in the channels will cause distortion but can be reduced by calibration techniques and the dynamic performance at high input frequencies is limited by phase-skew errors between the channel. These errors are reduced by an input S/H. However, the input S/H must operate at the full speed of the ADC. Therefore it is difficult to design.

4. Oversampling Sigma-Delta Converters

Oversampling ADCs are relatively insensitive to imperfections in the analog components which makes them suitable for high resolutions. The main drawback is that the oversampling limits the signal bandwidth. The oversampling sigma-delta converter is normally used for signal bandwidths up to a few MHz.

5. Overall

Timing and delay errors are the limiting factor at high signal frequencies in all types of converters and can be reduced by using a S/H circuit at the input. The dynamic performance of the A/D converter at high signal frequencies is, thus, to a large extent determined by the input S/H circuit. A key component in an accurate S/H circuit is a high-gain, high-speed opamp. For high speed the opamp may be very power consuming, especially for high resolutions. The performance of ADCs is fundamentally limited by thermal noise and clock jitter. The effect of these noise sources can not be removed by an input S/H circuit.

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4 OVERVIEW OF D/A CONVERTER ARCHITECTURES

4.1 INTRODUCTION

In this chapter we present different techniques for converting a digital signal into an analog signal representation. The approaches differ in speed, chip area, power efficiency, achievable accuracy, etc. It is therefore necessary to understand which converter algorithms or architectures to choose for the specific application. For example, when the conversion bandwidth is relatively small, it could be advantageous to use a higher sampling ratio and some oversampling technique to reduce the noise energy within the signal band. If very high speeds are required, a flash converter architecture should be used. However, the trade-off in converter design is normally between resolution and bandwidth (or the update frequency over signal frequency ratio). The higher bandwidth the lower resolution is generally achievable.

In Sec. 4.2 we discuss the concept of Nyquist-rate conversion, and in Sec. 4.3 to Sec. 4.8 we show different implementations for Nyquist-rate DACs. There is a limited number of DAC architectures that are good candidates for high-speed and high-performance applications. One is the current-steering DAC, which uses a number of binary scaled elements which generate the output value.

In Sec. 4.9, we discuss the concept of oversampling D/A conversion. In this book we refer to an oversampling DAC (OSDAC) to a converter operating at a higher update frequency and also containing a noise shaping loop. Often, the OSDAC utilizes some kind of a Nyquist-rate converter (with a lower nominal resolution) together with digital modulators and analog filters.

To further increase the performance of DACs, some special enhancement techniques can be used, as for example dynamic randomization and dynamic element matching (DEM) techniques. This is discussed in Sec. 4.10.

In Sec. 4.11 we compare a number of DAC architectures reported in the literature and in data sheets from vendors. We present an overview of the performance in terms of linearity vs. update and signal frequencies.

4.2 NYQUIST-RATE D/A CONVERTERS

In general the Nyquist-rate converter is required for wide band applications where oversampling techniques are impossible due to the high clocking frequency. In for

example, audio applications an oversampling D/A converter (OSDAC) is preferred. Since a Nyquist-rate converter also is used in most OSDACs we first present some general and common Nyquist-rate DAC architectures. We discuss application areas and highlight some of the advantages and disadvantages with the different types of converter architectures. The architectures that are discussed, are the binary weighted, thermometer coded, and encoded DAC architectures. The different architectures can be combined into hybrid DACs where the advantages of different individual architectures are used.

We are considering DACs for telecommunication applications and therefore we focus on candidates suitable for high speed and high resolution. However, lower-speed DACs such as the algorithmic DAC are also briefly presented. These lower-speed DACs can be used in pipeline DACs where the throughput is increased by pipelining.

To illustrate the principle of the DAC, usually single-ended circuits are used. In a real implementation they are differential circuits to improve linearity and SNR. In this chapter, we consider only binary offset coded inputs. To represent a negative output of the converter, the reference elements of the most significant bit get the opposite sign, i.e., a 2's complement coded input.

We use three modes of circuit technology; *voltage-mode*, *current-mode*, and *charge-redistribution mode*. We associate *voltage-mode* with a DAC where the element values are given by voltage levels as for example in a resistor-string which divides a voltage reference level into a number of different amplitude levels. With *current-mode* the DAC elements are currents, as for example switched current sources or resistors dividing a major current into weighted subcurrents. Finally, *charge-redistribution* DACs are using the switched-capacitor technique (SC). Note that the SC technique usually is referred to as a voltage-mode technique as well.

4.2.1 Nyquist-Rate DACs

In Nyquist-rate DACs the input signal bandwidth f_B is equal to the Nyquist frequency, $f_B < f_N = f_s/2$, where f_s is the update frequency of the DAC [1, 2]. The entire possible frequency range for fully recoverable signals according to the sampling theorem.

In almost every practical DAC, the output is sample&held. The sampling period is $T_s = 1/f_s$ and, as was shown in Chapter 1, the spectrum is repeated, centered at multiples of the update frequency, f_s . There is also a sine-weighting of the signal in the frequency domain and in Fig. 4-1 we show the characteristic attenuation A (in dB) of the signal as function of f_{sig}/f_s , where f_{sig} is the output signal frequency. The attenuation of the signal at half the update frequency $f_B = f_N$ (indicated by the dashed line in the figure) is 3.9 dB. To correct for the attenuation, either anti-sine filters or an oversampling strategy is needed.

As illustrated in Fig. 4-2b) an analog continuous-time low pass (LP) filter with a cut-off frequency f_N is used to remove images. With an anti-sine filter, the attenuation from the sine-weighting is reduced. This can be done both in the analog and the digital domain.

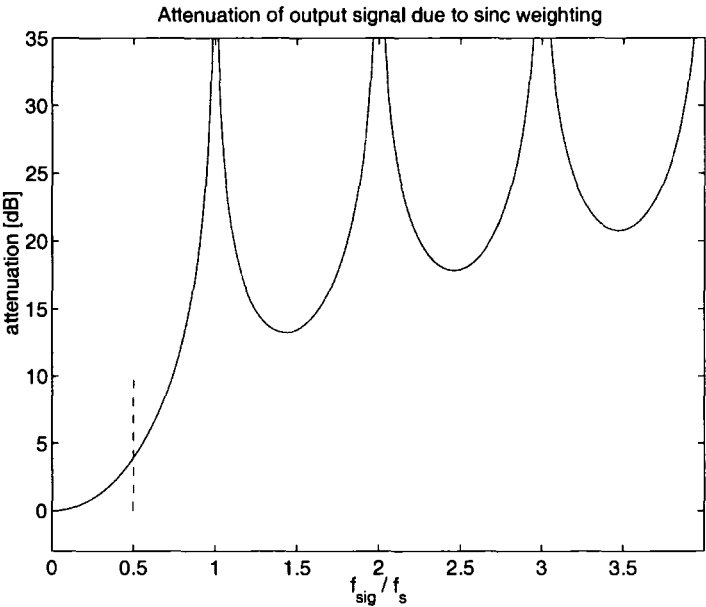


Figure 4-1 Sinc attenuation of the output signal as function of the signal to sampling frequency ratio.

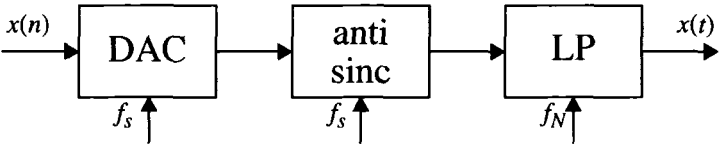


Figure 4-2 DAC with continuous-time filters, such as a low pass filter to remove images and an anti-sinc filter to reduce sinc distortion.

The digital input signal has a resolution of N bits, hence a quantization noise is introduced. This noise can for a larger number of bits be considered as white [2] and its power spectral density (PSD) is constant within the Nyquist frequency range. The influence of the noise, or the signal-to-noise ratio (SNR), was discussed in Chapter 1 and for a full-scale input sinusoid, the SNR is

$$SNR \approx 6.02 \cdot N + 1.76 \text{ dB} \tag{4-1}$$

where N is the nominal number of bits in the converter. For a lower resolution, the quantization error becomes signal correlated and hence distortion is introduced.

4.2.2 Nyquist-Rate DACs with Oversampling

To overcome the difficulties in the realization of anti-sine and low pass image rejection filters, almost all Nyquist-rate DACs are oversampled, i.e., the signal bandwidth is $f_B < f_N = f_s/2$. We could apply an input signal to the DAC creating an output signal spectrum as illustrated in Fig. 4-3 where we do not use the entire Nyquist range.

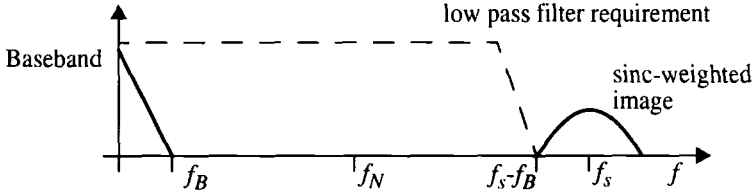


Figure 4-3 DAC input signal when not using the entire frequency range. The low pass filter requirement is illustrated with a dashed line.

If we would use a sampling frequency of $f_s = 8 \cdot f_B$ we would have a 0.22 dB attenuation of the signal at f_B . In general, the attenuation of the signal at f_B is given by

$$A(f_B/f_s) = -20 \cdot \log \left| \frac{f_s}{\pi f_B} \cdot \sin \frac{\pi f_B}{f_s} \right| \quad (4-2)$$

Therefore, in most DACs, a higher update frequency than double the bandwidth, $f_N = 2 \cdot f_B$, is used to achieve a lower sinc attenuation and to reduce the image-rejection filter order.

Oversampling can also improve the performance and the improvement by using a lower bandwidth can be found by using the definition of SNR in (4-1). Let the signal power be given by P_s and the quantization noise power spectral density be constant, i.e., $S_q(f) = q^2/\Delta f$. From (4-1) we have that

$$SNR = 10 \cdot \log \frac{P_s}{S_q(f) \cdot f_N} \text{ dB} \quad (4-3)$$

Due to the LP filter, we only consider the noise up to the frequency f_B and (4-3) can be written as

$$SNR = 10 \log \left(\frac{P_s}{S_q(f) \cdot f_B} \cdot \frac{f_B}{f_N} \right) = 10 \log \frac{P_s}{S_q(f) \cdot f_B} - 10 \log \frac{f_N}{f_B} \text{ dB} \quad (4-4)$$

We have that

$$SNR_B = SNR + 10 \cdot \log \frac{f_N}{f_B} \text{ dB} \quad (4-5)$$

where SNR_B is the signal-to-noise ratio within a smaller frequency band. From (4-5) we have the intuitive conclusion that the smaller signal bandwidth compared to the update frequency, the higher SNR. We also have from Fig. 4-3 that the transition band

of the following LP filter can be much wider and the filter order will be reduced. Since the sine-attenuation is low, there is no need for an anti-sine filter. In the frequency-domain, some of the dynamic errors, such as CFT from switches, are also moved from the signal band to higher frequencies.

Digital filters in the DSP are sometimes updated at the Nyquist rate ($2 \cdot f_B$) and therefore interpolation is used to increase the update frequency for D/A conversion. We can sample the signal at a higher frequency, but this will introduce signal images within the frequency range f_B to $f_s - f_B$. The analog image rejection filter has to remove these images, meaning that the filter order becomes high (due to the narrow transition band). A better choice is to use digital interpolation filters at the DAC input to remove these images. This is why Nyquist-rate DACs with oversampling also is referred to as interpolating DACs [3, 4, 5].

Note that the maximum achievable resolution within f_B is given by the resolution of the preceding digital circuits. The interpolation filters “compress” the frequency domain and the SNR is still given by this internal resolution. Often this resolution can be considerably larger than that of the DAC itself and hence the DAC introduces quantization noise when reducing the number of bits. Therefore, we have that the maximum achievable resolution according to (4-5) is given by the resolution of the circuits at the input of the DAC.

Higher resolution can be achieved within a small signal bandwidth if we also succeed to spectrally move the noise *introduced by the DAC* to frequencies outside the narrow signal band. This is the concept of oversampling D/A converters (OSDACs) and is further described in Sec. 4.9.

4.3 BINARY WEIGHTED DAC ARCHITECTURE

The binary weighted DAC utilizes a number of reference elements that are binary weighted. The output signal can be written as

$$X_a(nT) = A_{os} + A_0 \cdot (b_0(nT) + 2 \cdot b_1(nT) + \dots + 2^{N-1} \cdot b_{N-1}(nT)) \quad (4-6)$$

where A_0 is the reference, A_{os} is the offset, $\{b_i(nT)\}_{i=0}^{N-1}$ are the input bits, and T is the update period of the DAC. In Fig. 4-4 we show the concept of the binary weighted DAC. The total sum of all weights is $(2^N - 1) \cdot A_0$.

One of the drawbacks with this architecture is that for a large number of bits, the difference between the MSB (b_{N-1}) and LSB (b_0) weight is very large and the converter is very sensitive to mismatch errors.

The advantage is that the number of switches and digital encoding circuits become minimal in this architecture. The elements to implement the multiplication and addition operations may be current sources, resistors, or capacitors. We describe the current-steering, R-2R ladder, and charge redistribution DAC architectures below.

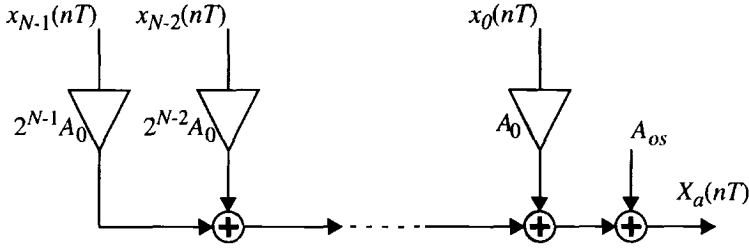


Figure 4-4 Binary weighted DAC with an offset level, A_{os} .

4.3.1 Current-Steering DAC

The switched-current (SI) technique is a natural choice in a CMOS process, since the reference and sum elements as well as switches are relatively easy to implement. The general architecture of a binary weighted current-steering DAC is shown in Fig. 4-5

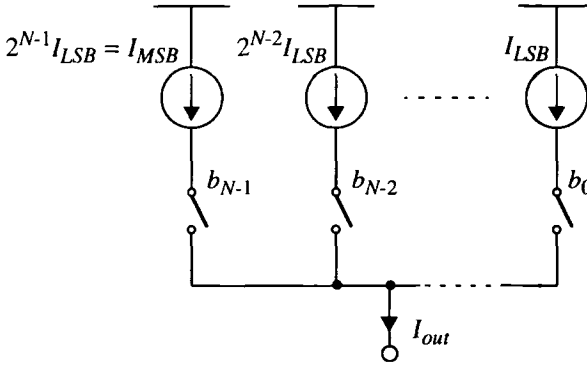


Figure 4-5 An N -bit current-steering binary weighted DAC.

and discussed in [2, 6]. The switches are controlled by the input bits, b_i , where $i = 0, 1, \dots, N-1$ and N is the number of bits. b_0 is the LSB and the corresponding current source has the DC value I_{LSB} . The source controlled by bit b_i , i.e., the i -th LSB current source, is preferably formed by connecting 2^i LSB current sources (unit current sources) in parallel, hence the MSB current source has the DC value $I_{MSB} = 2^{N-1} \cdot I_{LSB}$. The use of unit element sources increases the matching of the sources [7, 8, 9]. The output current, I_{out} , of the DAC shown in Fig. 4-5 is given by

$$I_{out}(k) = I_{LSB} \cdot b_0 + 2I_{LSB} \cdot b_1 + \dots + 2^{N-1}I_{LSB} \cdot b_{N-1} = I_{LSB} \cdot k \quad (4-7)$$

where k is the digital input given by

$$k = b_0 + 2 \cdot b_1 + \dots + 2^{N-1} \cdot b_{N-1} = \sum_{l=0}^{N-1} 2^l \cdot b_l \quad (4-8)$$

The current-steering DAC has the advantages of being quite small for resolutions below 10 bits and it is very fast. The major disadvantage is its sensitivity to device mismatch, glitches, and current source output impedance for higher number of bits. Another good property of the current-steering DAC is that it has a very high power efficiency since all power is directed to the output. The current-steering is suitable for high-speed high-resolution applications, especially when special care is taken to improve the matching of the converter. The current sources are typically implemented with cascoded NMOS or PMOS transistors.

To achieve monotonicity and reduce the influence of glitches (see Chapter 1), as well as reducing the sensitivity to matching errors, the DAC should be segmented into a coarse and fine part. The coarse part is then thermometer coded and fine part is kept binary weighted. This is referred to as a segmented converter and is further discussed in Sec. 4.6.

4.3.2 R-2R Ladder DAC

It is easy to construct a resistor-based binary-weighted DAC. But the resistance spread is very large for a large number of bits N . The better choice is the R-2R ladder architecture. This DAC architecture provides an architecture suitable in processes capable of implementing linear resistances. An N -bit R-2R ladder architecture is shown in Fig. 4-6 and discussed in [1, 2, 10, 11, 12]. The current sources are all equally large,

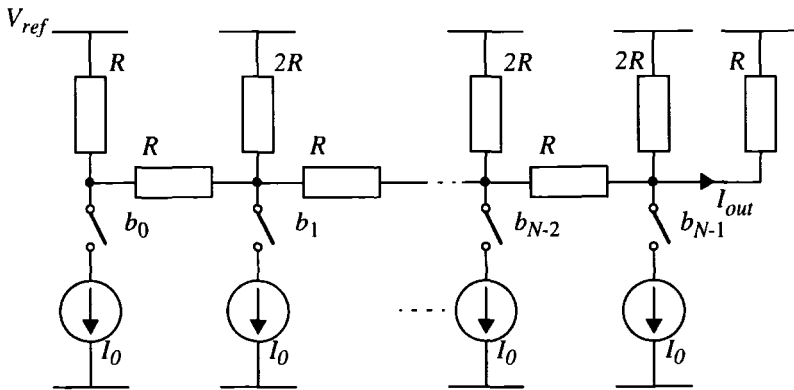


Figure 4-6 An N -bit R-2R ladder DAC.

I_0 , and the switches are controlled by the bits b_i . At every node, the impedance is always R . The resistive network divides the currents from the individual current sources to the output and the output current is given by

$$I_{out}(k) = \frac{I_o}{2^{N-1}} \cdot \sum_{l=0}^{N-1} b_l \cdot 2^l = \frac{I_o}{2^{N-1}} \cdot k \quad (4-9)$$

This DAC architecture is power inefficient, since there is a current loss through the resistive network.

Due to the fact that all current sources are equally large, the matching can be improved. In a poor process, the resistors can be non-linear and contain signal-dependent capacitive parts yielding distortion. Time-skew between switching instants generates glitches in this architecture.

In this R-2R ladder architecture there is the same amount of current through all switches, which makes the design of the switches simpler. However, the internal voltage nodes are still AC varying and therefore the current sources will have varying terminal voltages.

As a conclusion, one of the major advantages is that we only have a few number of different component sizes to implement, i.e., two resistor sizes, R and $2R$, one current source size, I_0 , and one type of current switch. This allows a more regular layout and since the current sources all have the same size. Trimming or calibration of the current sources can also be applied [2, 10]. R-2R ladder DACs are more widely used in bipolar processes where high-quality resistors are available.

4.3.3 Charge Redistribution DAC

The charge redistribution DAC is a switched capacitor (SC) circuit, where the charge stored on a number of binary weighted capacitors is used to perform the conversion. See Fig. 4-7 for an example of an N -bit converter found in [1]. The most significant

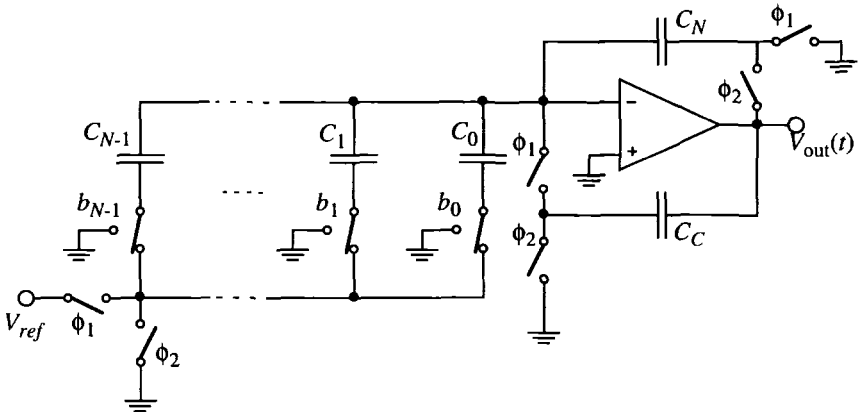


Figure 4-7 An N -bit charge-redistribution DAC.

capacitor C_{N-1} is 2^{N-1} times larger than the least significant capacitor $C_0 = C$, i.e., $C_{N-1} = 2^{N-1} \cdot C$. Typically, the weighted capacitors are created using a num-

ber of unit capacitors.

At time nT (on phase ϕ_1) the bits b_i determine which of the binary weighted capacitors that should be charged from the reference voltage, V_{ref} . During this phase, the plates of capacitor C_N are connected to ground and virtual ground at the input of the opamp, i.e., there is no charge on C_N , and $q_N(nT) = 0$. Capacitor C_C is used for offset compensation.

The total charge on the binary weighted capacitors at time point nT is given by

$$q_T(nT) = V_{ref} \sum_{l=0}^{N-1} C_l \cdot b_l = V_{ref} \sum_{l=0}^{N-1} 2^l \cdot C \cdot b_l = V_{ref} \cdot C \cdot k(nT) \quad (4-10)$$

At time point $nT + T/2$, on phase ϕ_2 , the weighted capacitors are discharged since their plates are connected to DC and virtual grounds. The charge is redistributed to ground and C_N . The charge on C_N at the end of the settling is

$$q_N(nT + \frac{1}{2}T) = C_N \cdot V_{out}(nT + T/2) \quad (4-11)$$

Charge conservation gives

$$q_N(nT + \frac{1}{2}T) = -q_T(nT) \quad (4-12)$$

Using (4-10) and (4-11) in (4-12) we have

$$V_{out}(nT + \frac{1}{2}T) = \frac{C}{C_N} \cdot k(nT) \cdot V_{ref} \quad (4-13)$$

The phases, ϕ_1 and ϕ_2 , are non-overlapping. The architecture in Fig. 4-7 is insensitive only to offset voltage and finite gain of the amplifier. The limitations of the converter are the matching of the capacitors, the switch-on resistance, and finite bandwidth of the amplifier.

4.4 THERMOMETER CODED DAC ARCHITECTURE

The thermometer coded DAC architecture utilizes a number of equally weighted elements. The binary input code is encoded into a thermometer code as illustrated in Table 4-1 for a 3-bit input code. Generally, with N binary bits, we have $M = 2^N - 1$ thermometer coded bits. The analog output at the time nT is given by

$$X_a(nT) = A_{os} + A_0 \cdot \sum_{i=1}^M c_i(nT) \quad (4-14)$$

where $c_i(nT) \in \{0, 1\}$, $1 \leq i \leq M$ are the thermometer coded bits. The signal flow graph (SFG) of the thermometer coded DAC is shown in Fig. 4-8.

Table 4-1 Decimal, binary offset, thermometer, and walking-one code representations.

Decimal	Binary offset	Thermometer	Walking-one
0	000	0000000	0000000
1	001	0000001	0000001
2	010	0000011	0000010
3	011	0000111	0000100
4	100	0001111	0001000
5	101	0011111	0010000
6	110	0111111	0100000
7	111	1111111	1000000

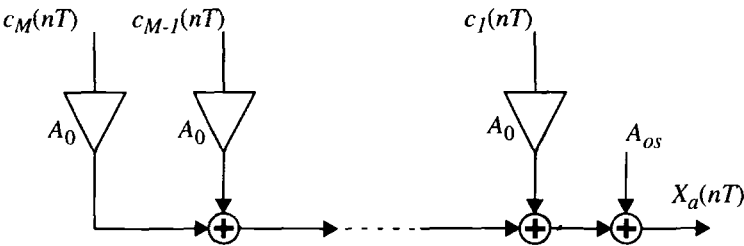


Figure 4-8 Thermometer coded DACs use a number of equally large amplification factors.

In the thermometer coded DAC the reference elements are all equally large and the matching of the individual elements becomes simpler than for the binary case. The total sum of all weights is $2^N - 1$. The transfer function of the thermometer coded converter is monotonic and the DNL and INL is improved compared to the binary version. The requirements on element matching is also relaxed. In fact, if the matching is within a 50% margin, the converter is still monotonic [13].

In Fig. 4-9 we show a current-steering implementation of a thermometer coded DAC with binary-to-thermometer encoder. All current sources are equally large, I_{unit} . For a larger number of bits, the digital circuits converting the binary code (X) into thermometer code (C) and the number of interconnecting wires become large, since the number of outputs is growing exponentially. This implies a more complex circuit layout. The encoding circuit can easily be pipelined and the propagation time through the encoder can be controlled. This is further discussed in Chapter 13.

4.5 ENCODED DAC ARCHITECTURE

An architecture very similar to the thermometer coded DAC is the direct encoded

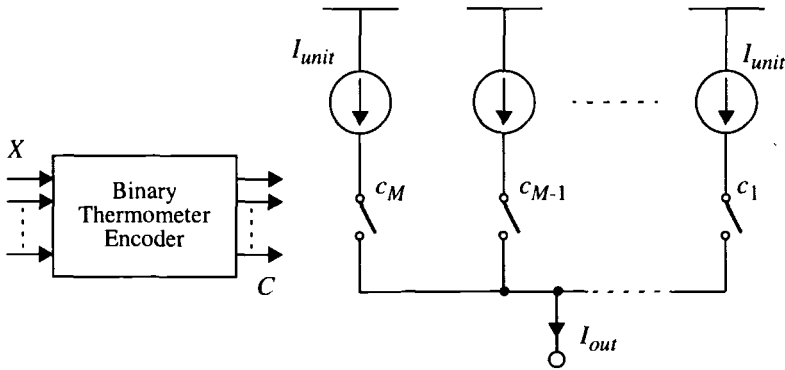


Figure 4-9 A thermometer coded current-steering DAC with encoding circuit.

DAC as the SFG illustrates in Fig. 4-10. In this architecture, linearly weighted refer-

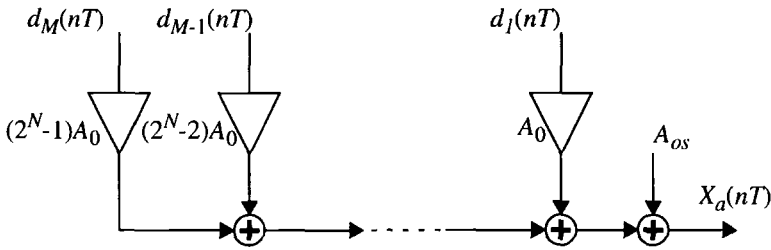


Figure 4-10 Direct encoded DAC uses a number of linearly weighted element values.

ences are used. For an N -bit converter we need $M = 2^N - 1$ references, one for each level. The output value is given by

$$X_a(nT) = A_{os} + A_0 \cdot \sum_{i=1}^M A_i \cdot d_i(nT) \quad (4-15)$$

where $d_i(nT) \in \{0, 1\}$, $1 \leq i \leq M$ are given by a “walking one” code shown in Table 4-1.

In the resistor-string DAC, the principle of this architecture becomes clear. This architecture, illustrated in Fig. 4-11 is basically a voltage-mode direct encoded DAC architecture. In this DAC, M switches are used, one for each word, i.e., $M = 2^N - 1$. The weighting elements are equally large resistors in a resistor string. If the number of bits is high, the number of resistors becomes very high. The reference voltage V_{ref} is feeding the resistors and from voltage division, we have that the voltage level V_i is given by

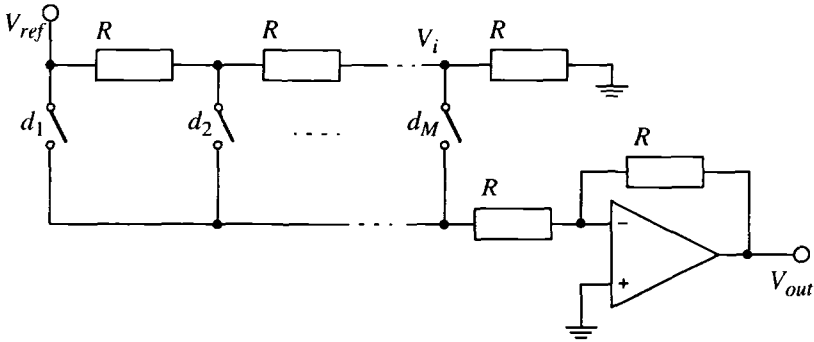


Figure 4-11 An N -bit resistor string DAC where $M=2^N-1$.

$$V_i = \frac{i \cdot R}{R_{tot}} \cdot V_{ref} = \frac{i}{2^N - 1} \cdot V_{ref} \quad (4-16)$$

The switches, implemented with for example NMOS transistors or transmission gates, connect the selected voltage V_i to the opamp and the DAC output is given by

$$V_{out}(nT) = - \sum_{i=1}^M d_i(nT) \cdot V_i = - \frac{V_{ref}}{M} \cdot \sum_{i=1}^M i \cdot d_i(nT) \quad (4-17)$$

where $M = 2^N - 1$.

This architecture has some drawbacks; for high speed, the design of the operational amplifier becomes difficult, the matching of the resistors in the resistor string is crucial for the overall accuracy, and the RC-timing through the switches and wires may also limit the bandwidth and linearity of the DAC. The encoder, converting the binary input into a “walking one” code, is a straight-forward design. But still the routing and chip area may become large and complex. The encoding circuit can be pipelined and the propagation time through the encoder is a minor problem.

Mostly, the encoded DAC shows good monotonicity behavior as well as an improved DNL and INL over the binary architecture.

The encoder can be simpler if a tree selection architecture [1] is used. This also reduces the total number of switches. The disadvantage of using such a tree architecture is the additional delay through the increased number of switches’ layers.

4.6 HYBRID DAC ARCHITECTURE

As has been addressed previously, different types of DAC architectures have their different advantages and disadvantages. An intuitive conclusion from this, is to use a combination of different architectures to improve the overall performance. This is

illustrated in Fig. 4-12 where the different sub-converters can be of completely dif-

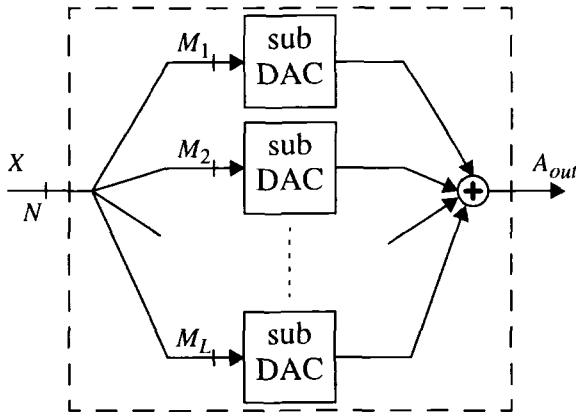


Figure 4-12 Hybrid DACs use a combination of a number of different types of DACs.

ferent types. Each sub-DAC converts M_i bits and in the simplest configuration we have that the total resolution is equal to the sum of the resolutions of the sub-converters, i.e., $N = \sum M_i$. The subconverters have different reference levels.

A popular hybrid converter architecture is the so called segmented architecture as shown in Fig. 4-13. In this N -bit architecture, the M more significant bits are encoded

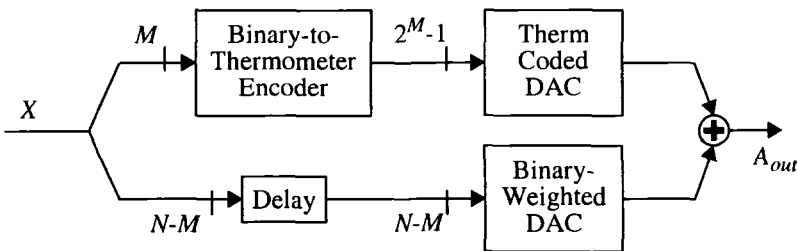


Figure 4-13 An N -bit segmented DAC where M MSBs are thermometer coded.

into a thermometer code and the $M - N$ less significant bits are binary weighted. There is a need of an encoder for the MSBs, which takes an M -bit input signal generates $2^M - 1$ one-bit output signals. When increasing M the size of the encoder grows exponentially and a larger amount of switches and interconnection are needed. However, the advantages are the improved device matching and linearity over binary weighted DACs. One of the key issues is to find the optimum number of MSBs to encode into a thermometer code. Reasonable sizes for M is in the range from 4-8,

dependent on the total number of bits N .

In most high-speed high-resolution DACs we find that segmented or multi-segmented architectures have been used (see Sec. 4.11). The multi-segmented architecture uses a thermometer code representation for clusters, e.g., the M_1 MSBs are clustered and encoded, the M_2 next MSBs are clustered and encoded, and the remaining $N - M_1 - M_2$ bits are kept binary weighted. In Chapter 13 we present the design of segmented current-steering DACs, where $N = 14$ and $M = 4$.

4.7 LOW-SPEED DAC ARCHITECTURE

Although we are mainly interested in wide band, high-resolution converters for communications, we also briefly overview some lower-speed DAC architectures, such as the algorithmic (or cyclic) DACs. In lower-speed DACs the internal clock period is smaller than the sample period. These lower-speed architectures can however be pipelined (Sec. 4.8) to increase the throughput to the cost of additional hardware.

4.7.1 Algorithmic DAC Architecture

Assuming a binary offset code, the output of an algorithmic (or cyclic) DAC is

$$A(k) = A_0 \cdot k = A_0 \cdot \sum_{l=0}^{N-1} 2^l \cdot b_l \quad (4-18)$$

where 2^l are the weights for the corresponding bit b_l and A_0 is the LSB reference value. The expression in (4-18) can be rewritten as

$$A(k) = A_0 \cdot [b_0 + 2(b_1 + 2(\dots + 2(b_{N-2} + 2b_{N-1})))] \quad (4-19)$$

where b_0 is the LSB and b_{N-1} is the MSB. (4-19) is a recursive formula and if we assume that the digital word is generated bit-wise, hence in serial with the MSB first, we can write

$$\tilde{A}(m) = 2 \cdot \tilde{A}(m-1) + x_{N-m} \quad (4-20)$$

where $m = 1, \dots, N$ and $\tilde{A}$ is an intermediate value with $\tilde{A}(0) = 0$. We need N iterations to generate an analog value, which is multiplied by the reference value and added to an offset, giving the analog output value

$$A(k) = A_0 \cdot \tilde{A}(N) + A_{os} \quad (4-21)$$

Therefore, the algorithmic DAC requires a higher internal computational speed since the output value has to be generated within one output update period. We have that $T_s \geq N \cdot T_{alg}$ where N is the number of bits, T_{alg} is the clock period of the internal accumulator, and T_s is the update period of the DAC.

In Fig. 4-14 we show a block-level implementation of an algorithmic DAC with an MSB-first operation. The structure realizes the recursive formula (4-19) and it uses a

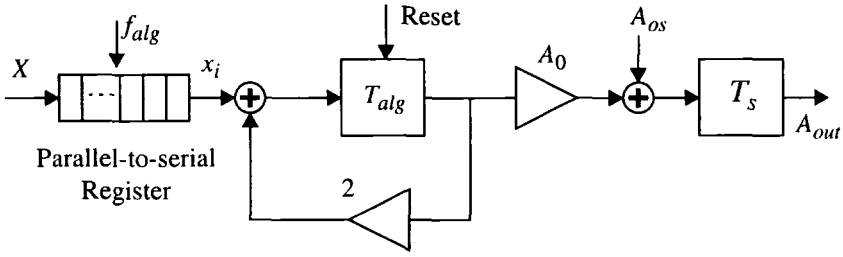


Figure 4-14 Schematic view of an algorithmic DAC.

parallel-to-serial interface where the serial values are stored in a register operating at the higher speed, $f_{alg} = 1/T_{alg}$. The digital bit values $x_i \in \{0, 1\}$ are fed into an accumulator, where the input signal is added to the output signal, which is fed back and amplified by a factor two. The output signal from the accumulator is multiplied with the LSB reference and an offset A_{os} may be added to the value. This output value is sample&held at the lower speed T_s . When each word has been processed, after $N \cdot T_{alg}$, the accumulator is reset.

The accumulator can also be fed with the LSB first and then the gain factor in the feedback becomes $1/2$ instead. This gives a lower gain in the feedback loop and is a better choice when implementing the converter.

The advantage of the algorithmic DAC is the low number of circuit components. This reduces the number of interconnections and chip area in general.

4.7.2 Switched-Current Algorithmic DAC

Algorithmic DACs have been successfully implemented in both the switched-capacitor (SC) [14] and switched-current (SI) technique [15, 16]. Since there is an accumulation phase, the algorithmic DAC is not suitable for high-speed, though, the resolution can be quite high at intermediate frequencies [15].

In Fig. 4-15 we show a current-steering algorithmic DAC as proposed in [16]. This DAC utilizes a feedback amplifier of $1/2$ instead of a factor 2 as was used in Fig. 4-14 and hence the digital input signal is given by the LSB first. The converter needs three equal subcircuits that are interconnected. Transistor M2 is biased with $I_0/2$ and therefore its size aspect ratio should be half the ratios of M1 and M3.

The circuit is controlled by four different switching signals, ϕ_1 to ϕ_4 . On phase ϕ_1 the input bit x_i is determining if the current I_{bit} should be added to the accumulator. The I_{bit} current is equal to the maximum current for all bits i . This current is added to the intermediate accumulator output current from memory cell M3. The sum is stored in memory cell M1. This sum is divided by two in the M2 transistor during phase ϕ_2 . Finally, on phase ϕ_4 the stored accumulated current in M3 is switched to the output of the DAC. The switching phase ϕ_3 is used to restore the accumulator and

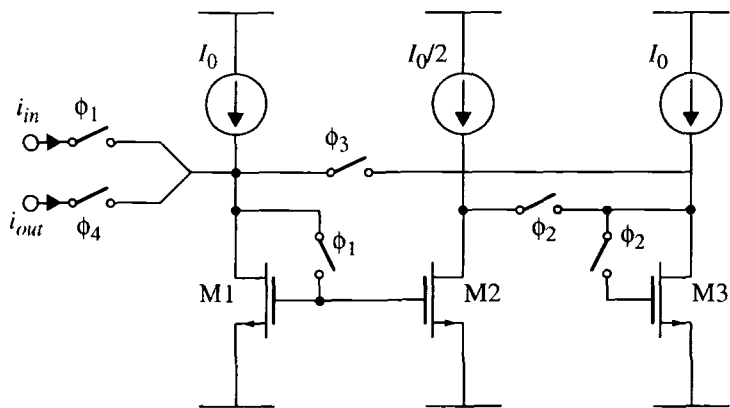


Figure 4-15 A switched-current (SI) implementation of an algorithmic DAC.

hence it should be opened when the first bit of a new word should be fed to the DAC. To reach a high resolution of the SI algorithmic converter the bias currents must be large and the transistors and switches also have to be large.

4.8 PIPELINED DAC ARCHITECTURE

Low-speed, clocked circuits can be pipelined to increase the throughput of the whole system. This is done to the cost of more hardware and power consumption.

When pipelining algorithmic DACs (previous section) we can break the loop into a pipeline as shown in Fig. 4-16a) [17]. The architecture shown in the figure uses an off-

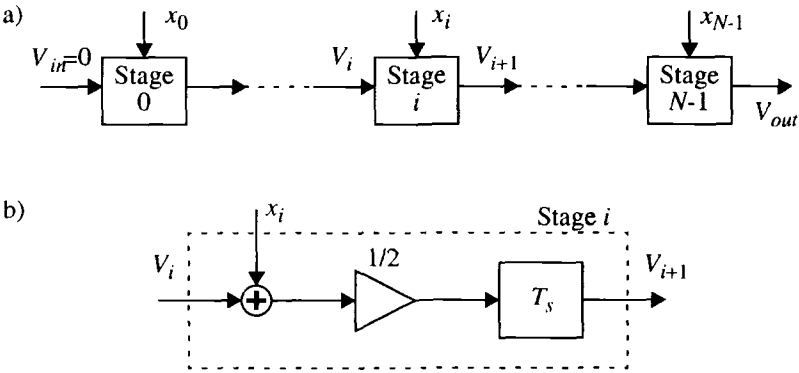


Figure 4-16 a) An N -stage pipelining DAC and b) view of the i -th stage.

set binary code, but it can easily be modified to handle 2's complement code as well. Each stage in the pipeline, see Fig. 4-16b), uses an adder, a 1/2 amplifier, and a S/H circuit.

In Fig. 4-17 we show an SC implementation of one of the stages in the pipeline. Two

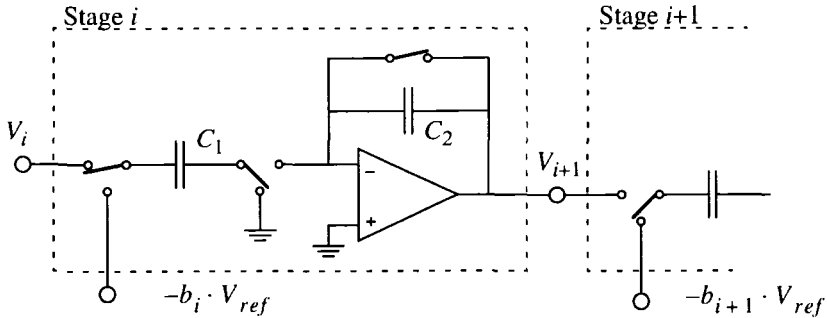


Figure 4-17 An SC implementation of the i th stage in a pipelined DAC.

capacitors and an opamp are used. The circuit is clocked with two non-overlapping clock phases, ϕ_1 and ϕ_2 . The transfer function can be derived by studying the charge redistribution. At time instant $t = nT_s$ (during phase ϕ_1) V_i is charging capacitor C_1 with

$$q_1(t) = C_1 \cdot V_i(t) \quad (4-22)$$

The opamp output is auto-zeroed and the C_2 capacitor is discharged and hence $q_2(t) = 0$. At time instant $t + T_s/2$, i.e., during phase ϕ_2 , the C_1 capacitor is charged to

$$q_1(t + \frac{1}{2}T_s) = C_1 \cdot x_i(t) \cdot (-V_{ref}) \quad (4-23)$$

and C_2 has the charge

$$q_2(t + \frac{1}{2}T_s) = C_2 \cdot V_{i+1}(t + \frac{1}{2}T_s) \quad (4-24)$$

By charge conservation we have

$$q_1(t) = q_1(t + \frac{1}{2}T_s) + q_2(t + \frac{1}{2}T_s) \quad (4-25)$$

Using (4-22), (4-23), and (4-24) in (4-25), we get

$$C_1 \cdot V_i(t) = C_2 \cdot V_{i+1}(t + \frac{1}{2}T_s) - C_1 \cdot b_i(t) \cdot V_{ref} \quad (4-26)$$

which gives

$$V_{i+1}(t + \frac{1}{2}T_s) = \frac{C_1}{C_2} \cdot [V_i(t) + V_{ref} \cdot b_i(t)] \quad (4-27)$$

The SC S/H should have the gain of 1/2, hence $C_2 = 2C_1 = 2C$. We also find from the configuration, that the next stage cannot sample V_{i+1} during the auto-zeroing phase, therefore the next stage has to have reversed clock phases. The circuits in Fig. 4-16 and Fig. 4-17 are designed for a binary offset code, but can easily be modified to cover 2's complement as well. Due to the non-idealities in the S/H amplifier (as further discussed in Sec. 6.6), this type of converter is not suitable for very high-speed and high-resolution applications.

4.9 OVERSAMPLING D/A CONVERTER (OSDAC)

We found in Sec. 4.2.2 that by increasing the update frequency with respect to the signal bandwidth, we could reduce the requirements on the continuous-time filter and improve the in-band SNR. If the input signal to the DAC is interpolated, the maximum achievable resolution is given by the preceding digital circuits. The Nyquist frequency to the signal bandwidth ratio is denoted oversampling ratio (*OSR*)

$$OSR = \frac{f_N}{f_B} = \frac{f_s}{2f_B} \quad (4-28)$$

From (4-28) we have that for a Nyquist-rate converter the oversampling ratio is $OSR = 1$. From (4-5) we have that the SNR within the signal bandwidth is expressed as

$$SNR_B = 6.02 \cdot N + 1.76 + 10 \cdot \log OSR \text{ dB} \quad (4-29)$$

We see that the effective number of bits within this frequency range is given by

$$N_B = N + \frac{10}{6.02} \cdot \log OSR \leq N_D \quad (4-30)$$

where N_D is the resolution (bits) given by the word length of the preceding digital circuits. From (4-30) we find that for each doubling of the *OSR*, the effective number of bits is increased by half a bit. To reach a high resolution we must choose a large *OSR*. For example, with a nominal DAC resolution of $N = 10$ bits and a desired resolution of $N_D = 16$ bits, an $OSR = 2^{12} = 4096$ would be required. The linearity of the converter must also meet 16 bit accuracy.

It is obvious that this approach is not feasible for high resolutions since the *OSR* becomes very large. With noise shaping, the quantization noise introduced by the DAC can be spectrally shaped. Hence, we can move noise power outside of the signal band by high pass filtering the noise. This reduces the oversampling ratio. At the output we already have an LP filter with cut-off frequency f_B which will attenuate this noise as well. This is the essence of the noise shaping oversampling D/A converter (OSDAC).

Several different OSDAC architectures can be found in the literature [1, 2, 18-23]. The OSDAC consists of an interpolator, a sigma-delta modulator, internal Nyquist-rate DAC, and continuous-time filters as illustrated in Fig. 4-18. Sometimes, discrete-time

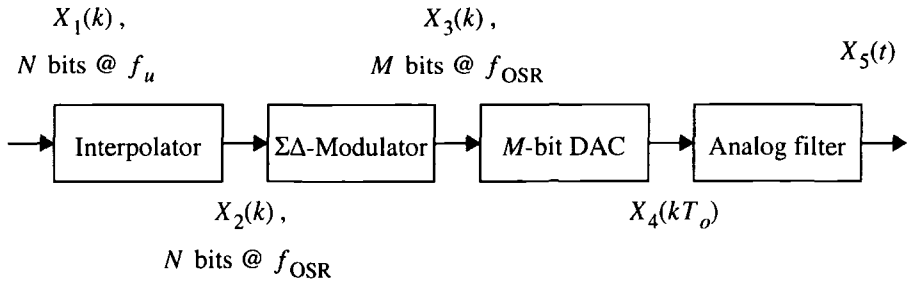


Figure 4-18 General OSDAC structure containing interpolator, sigma-delta modulator, DAC, and analog low pass filter.

filters precedes the continuous-time filter.

The $(X_1, \dots, X_5)$ signals in the OSDAC (Fig. 4-18) are illustrated by the example in Fig. 4-19 where $OSR = 4$:

The input signal (a) is using the whole bandwidth up to the frequency $f_N = f_u/2$. The interpolator increases the update frequency of the input signal to $f_{OSR} = OSR \cdot f_u$ and introduces a number $(OSR - 1)$ of images in the frequency domain as shown in b). To relax the requirements on the modulator and filters the interpolator also includes filtering, i.e., interpolation filters. In c) we show the result of using such filters.

The modulator truncates the N -bit input signal into an M -bit signal. This operation introduces a large amount of truncation error, but the modulator is also designed to high pass filter this truncation noise and hence the noise power can be moved out of the signal band.

The M -bit DAC (operating at the higher update frequency f_{OSR}) generates the analog output signal which becomes sinc weighted in the frequency domain due to the inherent S/H (e). With a lower number of bits, it may be easier to design a high-resolution, high-linearity DAC, since there is a lower number of analog reference elements that should be matched. In fact, by reducing the number of bits to only $M = 1$, linearity can be guaranteed [1, 23].

The continuous-time, low pass filter attenuates the out of band noise from the modulator and the images from the M -bit DAC (f).

Below, we briefly overview the different components of the OSDAC.

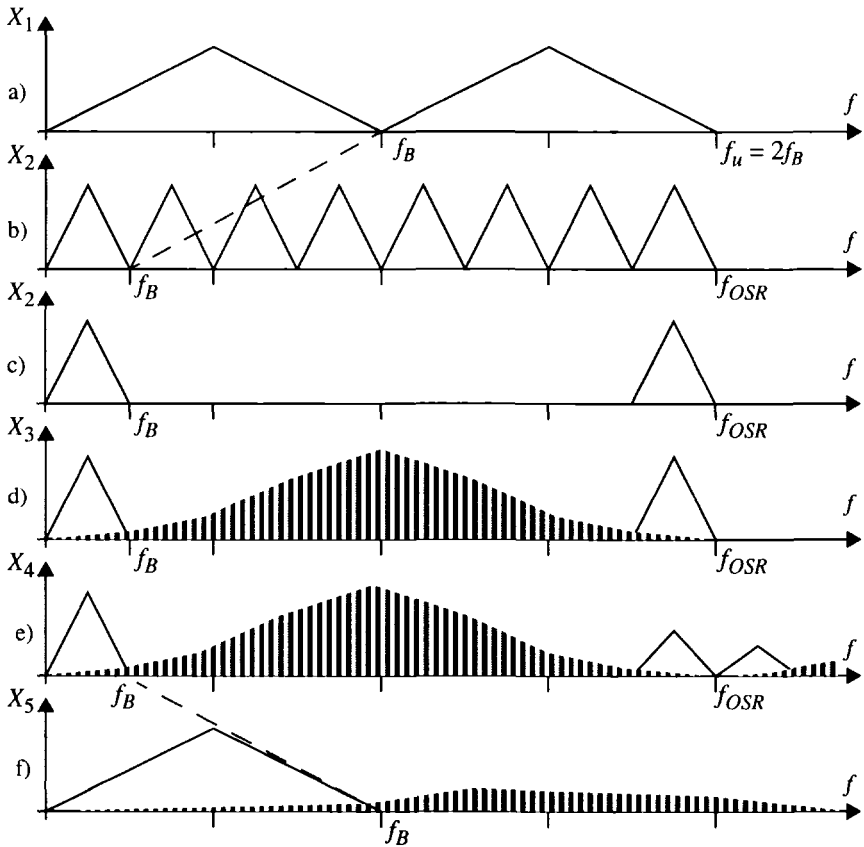


Figure 4-19 Example of the frequency power spectra for individual signals in an OSDAC with $OSR=4$: a) Original spectrum, b) interpolated spectrum, c) filtered interpolated signal, d) output of the modulator, e) output signal from DAC, and f) final output signal.

4.9.1 Interpolator and Interpolation Filters

The interpolator adds $OSR - 1$ zeros between each sample of its input signal as

$$y(n) = \begin{cases} x(\frac{n}{OSR}) & n = m \cdot OSR \\ 0 & n \neq m \cdot OSR \end{cases} \quad (4-31)$$

where x is the input signal, OSR is the oversampling ratio, and m is an integer number. $y(n)$ is updated with the oversampling frequency $f_{OSR} = OSR \cdot f_u = 1/T_o$. The z-transform of the expression in (4-31) is given by

$$Y(z) = X(z^{OSR}) \quad (4-32)$$

where $z = e^{j\omega T_o}$ on the unit circle. From (4-32) we see that the operation in (4-31) also introduces images (Fig. 4-19b). We can relax the requirements on the following modulator and filters by removing or attenuating these images with interpolation filters [24]. The spectrum is shown in Fig. 4-19c). One way to implement the interpolation filters is to use a one-stage interpolation filter, as for example the K -tap FIR filter in Fig. 4-20. The input signal $x(n)$ is updated with the frequency f_u , but the delay

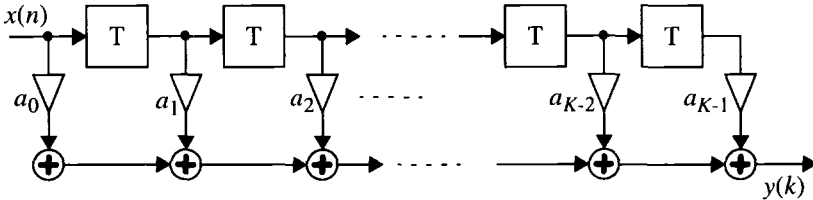


Figure 4-20 General one-stage FIR structure interpolation filter. The delay T is related to the oversampling frequency.

elements (T) are related to the oversampling frequency and hence the output signal $y(k)$ is updated with the frequency f_{OSR} .

If the transition or the signal bands of the interpolation filter have to be narrow or if the attenuation in the stop band has to be very high, the number of taps in the FIR filter is large [24, 25]. A large number of taps increase chip area and power consumption since more additions and multipliers are required. In that case, IIR filters are preferred over FIR filters. However, the design of IIR filters can be more complex and the maximum sample frequency may be lower than for the FIR filter. For IIR filters, the phase is not linear, which may not be acceptable in some applications.

Another way to implement the interpolation filters is to use multi-stage filters [18,24, 25]. In Fig. 4-21 we show the concept of a multi-rate filter of R stages and with a total

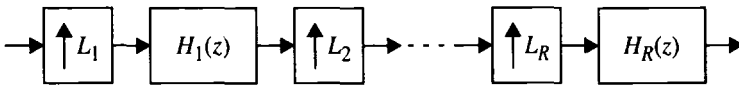


Figure 4-21 Principle description of multi-stage interpolation filtering.

interpolation or oversampling ratio of

$$OSR = L_1 \cdot L_2 \cdot \dots \cdot L_R \quad (4-33)$$

where L_i are the interpolation factors of the individual interpolators. Using (4-32) we find that the total transfer function of the filter in Fig. 4-21 becomes

$$H_{OSR}(z) = H_1(z^{L_2} \cdots L_R) \cdot H_2(z^{L_3} \cdots L_R) \cdot \dots \cdot H_R(z) \quad (4-34)$$

where $H_i(z)$ are the sub-filters of the multi-stage filter. By interpolating in stages and using frequency masking techniques the total filter $H_{OSR}(z)$ can be designed with very narrow pass and transition bands [25].

A simple filtering function is achieved by letting the modulator sample the input signal at the higher frequency f_{OSR} . This implies an interpolating function as

$$y(n) = x\left(\left\lceil \frac{n}{OSR} \right\rceil\right) \quad (4-35)$$

The corresponding weighting of the signal spectrum in the frequency domain becomes in the order of

$$\left| \text{sinc}\left(\frac{\omega T_O}{2} \cdot OSR\right) / \text{sinc}\left(-\frac{\omega T_O}{2}\right) \right| \quad (4-36)$$

However, with this operation, the signal is slightly attenuated at higher frequencies within the signal band and the attenuation of images is rather poor for a high performance OSDAC. To increase the attenuation of the out of band images, we can use a higher-order sinc filter.

4.9.2 Sigma-Delta Modulator

The sigma-delta ($\Sigma\Delta$) modulator truncates an N -bit input signal into an M -bit output signal. The truncation noise (error) *introduced by the modulator* is spectrally shaped with a high pass filtering function and the input signal is all pass (or low pass) filtered. The shaped noise is attenuated by the following continuous-time analog low pass filter.

Generally, the modulator contains some kind of a feedback loop, which can be of two kinds; *signal feedback*, where the output signal is fed back, and *error feedback*, where the truncation error is fed back.

Dependent on the order and architecture of the modulator (i.e., the order of the HP filter) the noise shaping function is different. It can be shown [1, 2, 18] that the SNR within the frequency range up to f_B is approximately given by

$$SNR_B = SNR + (20 \cdot K + 10) \cdot \log SNR + C \quad (4-37)$$

where K is the modulator order, SNR is determined by the resolution at the output of the modulator, and C is a number. (4-37) holds if there are no zeros in the noise shaping function. The improvement is approximately given by $K + 0.5$ bits for each doubling of the OSR [2]. Still the achievable SNR is given by the resolution of the DSP preceding the OSDAC. The gain in using noise shaping is obvious when comparing (4-29) with (4-37).

The signal feedback modulator feeds back the truncated output signal, $y(n)$, [1], as illustrated in Fig. 4-22a) for a first order modulator. $A(z)$ is a discrete-time accumu-

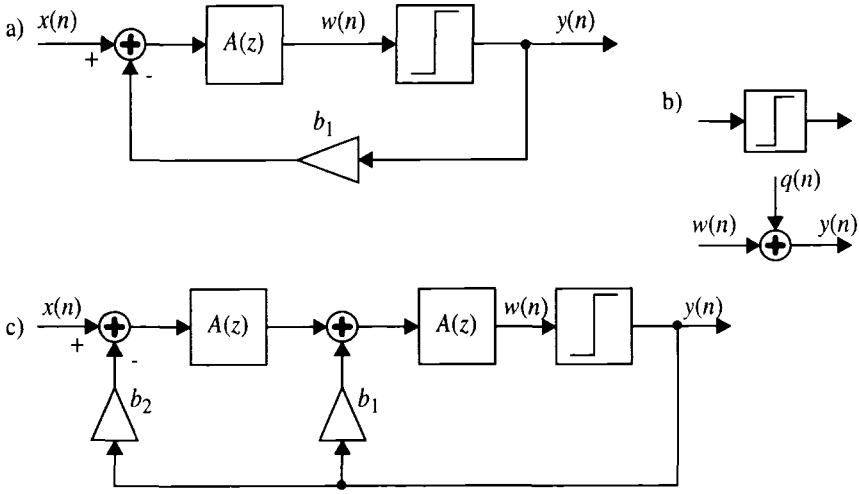


Figure 4-22 a) Principle of the 1st (a) and 2nd (c) order signal feedback sigma-delta modulator and model of the truncation error $q(n)$ (b).

tor with the transfer function given by

$$A(z) = \frac{z^{-1}}{1 - z^{-1}} \quad (4-38)$$

The output signal is subtracted from the input signal $x(n)$ and in the frequency domain, we find the transfer function

$$Y(z) = \frac{A(z)}{1 + b_1 \cdot A(z)} \cdot X(z) + \frac{1}{1 + b_1 \cdot A(z)} \cdot Q(z) \quad (4-39)$$

where $Q(z)$ is the truncation error as modeled in Fig. 4-22b). Using (4-38) in (4-39), we find the signal transfer function, $STF(z)$

$$STF(z) = \frac{z^{-1}}{1 + (b_1 - 1)z^{-1}} \quad (4-40)$$

and the noise transfer function, $NTF(z)$

$$NTF(z) = \frac{1 - z^{-1}}{1 + (b_1 - 1)z^{-1}} \quad (4-41)$$

The signal should be low or all pass filtered and the noise high pass filtered, which states the requirements on the filters $H(z)$ and $G(z)$. For a first order modulator, the STF could be given by a delay only, hence an all pass filter for $b_1 = 1$

$$STF(z) = z^{-1} \quad (4-42)$$

and the first order *NTF* is described by the high pass filter

$$NTF(z) = 1 - z^{-1} \quad (4-43)$$

For the second order modulator shown in Fig. 4-22c), we can choose the *NTF* as

$$NTF(z) = (1 - z^{-1})^2 \quad (4-44)$$

which holds for $b_1 = 2$ and $b_2 = 1$. In Fig. 4-23 the *NTFs* for the first and second

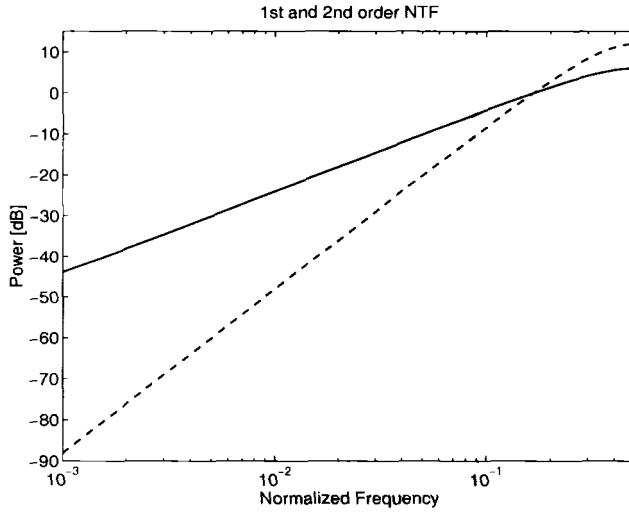


Figure 4-23 *NTF* of a first order (solid) and second order (dashed) sigma-delta modulator.

order modulator are shown. The second order modulator (dashed) shows a better noise attenuation at lower frequencies than the first order modulator (solid).

For high-performance applications we have to use higher-order modulators. As for oversampling A/D converters (OSADC) the modulator may not be stable for these cases. Therefore, we introduce zeros in the *NTF* which improves stability, but makes the noise shaping less effective. In Fig. 4-24 an example of an K -th order signal-feed-back architecture. The coefficients a_i and b_i determine the *NTF* and *STF*. The filters $A(z)$ are given by discrete-time accumulators (4-38). The accumulators do not contain multipliers and they show a good immunity against truncation noise [26]. For a fourth order modulator, the *STF* is given by

$$STF(z) = \frac{A^4}{1 + b_1 A + (a_1 + a_2 + b_2) A^2 + (a_2 b_1 + b_3) A^3 + (a_1 a_2 + a_2 b_2 + b_4) A^4} \quad (4-45)$$

and the *NTF* is given by

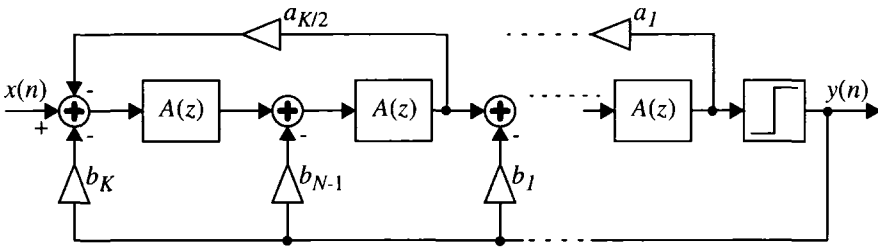


Figure 4-24 General structure of a higher order (N) modulator using signal feed-back structure.

$$NTF(z) = \frac{(1 + a_1 A^2)(1 + a_2 A^2)}{1 + b_1 A + (a_1 + a_2 + b_2)A^2 + (a_2 b_1 + b_3)A^3 + (a_1 a_2 + a_2 b_2 + b_4)A^4} \quad (4-46)$$

where $A = A(z)$ is the accumulator given by (4-38). The NTF zeros are

$$z_{1,2} = 1 \pm j\sqrt{a_1} \text{ and } z_{3,4} = 1 \pm j\sqrt{a_2} \quad (4-47)$$

The a_i 's are used to move the zeros close to the pass band edge in order to further attenuate the noise in the NTF stop band. The output spectrum of a 5th order, 14-bit input, one-bit output sigma-delta modulator is shown in Fig. 4-25. The input signal is

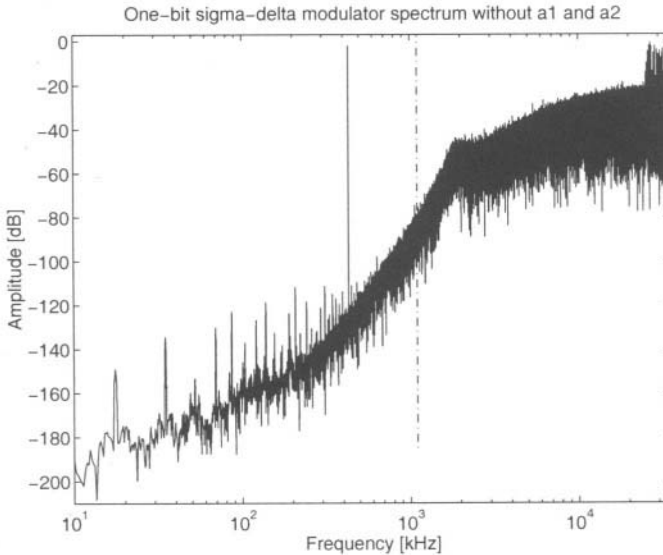


Figure 4-25 Output spectrum of the 5th order modulator without the a_i parameters.

a -6 dBFS sinusoid at the frequency 431.25 kHz, the update frequency is 70.656 MHz and the oversampling ratio $OSR = 32$. The passband edge is indicated with the dashed line. The a_i 's in the NTF are all set to zero. The SNDR within the signal band is approximately 63 dB, corresponding to 10.2 bits. With the zeros, i.e., non-zero a_i 's, included, the SNDR is improved. The zeros attenuate the noise at the signal band edge (dashed line) and a sharper noise shaping transition band is achieved. At lower frequencies the attenuation of the noise is not as strong. The simulation result when applying an -6 dBFS input sinusoid at 431.25 kHz is shown in Fig. 4-26. The SNDR

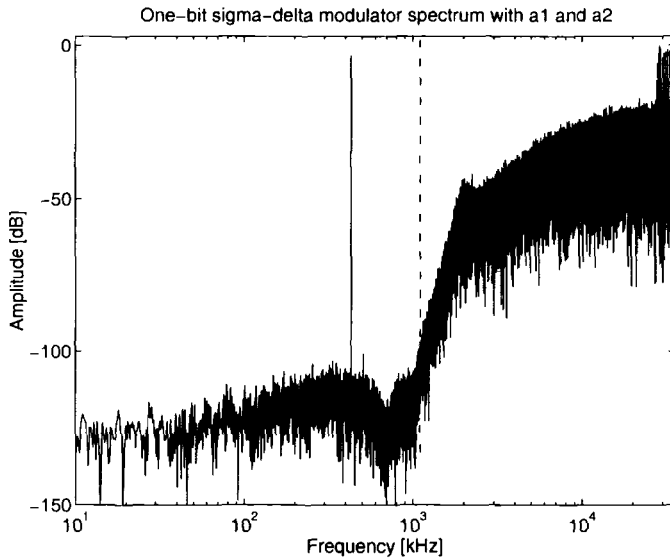


Figure 4-26 Output spectrum of the 5th order modulator with the a_i parameters.

is found to be approximately 79 dB, corresponding to 12.8 bits.

The poles are given by rather complex expressions and the complexity of the filter increases with increased modulator order. Since the output signal is truncated, the stability of the modulator has to be carefully investigated.

Dependent on the number of bits representing the output signal of the modulator, they can be divided into two groups; multi-bit and single-bit modulators. For the single-bit modulator the loop gain is high and therefore the stability of the modulator has to be carefully examined [2, 18]. For a single-bit output, the output signal is equal to the MSB of the quantizer's input.

4.9.3 M-bit DAC

With the M -bit DAC the output signal from the modulator is converted into an analog signal. The M -bit DAC has to operate at the oversampling frequency and is therefore a Nyquist-rate converter with oversampling. Naturally, the DAC has to have the same linearity accuracy as the final output. The design of a DAC with a lower number bits becomes generally less complex, since the number of circuit components are lower,

interconnection wires are shorter, etc.

To guarantee a very high linearity, a one-bit DAC can be used. The one-bit DAC has a very high linearity, since the transfer function between its minimum ("0"-input) and maximum ("1"-input) values can be expressed with a completely straight line even though there are matching errors [1]. These matching errors only give rise to gain and offset errors. However, other effects such as linearity of switches, voltage drops, and parasitics, limit the linearity.

Using a one-bit DAC implies that the continuous-time low pass filter has to have a high order due to the high noise power at high frequencies. A way to relax these requirements is to use a so called semi-digital FIR filter [23]. The semi-digital filter is very similar to a regular FIR filter and can be constructed by using a number of cascaded (in time) one-bit DACs having different reference levels as illustrated in Fig. 4-27. The reference levels, i.e., the constants a , are generated with analog components.

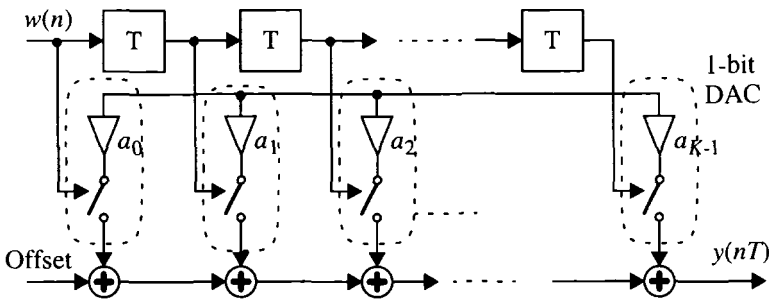


Figure 4-27 Cascaded one-bit DACs forming a K -tap FIR filter structure.

The inaccuracy in these coefficients affect the filter transfer function and does not introduce any other errors. By designing the FIR filter as a low pass filter attenuating the out-of-band modulator noise, the continuous-time filter can be implemented with a lower order.

The output signal of the FIR filter (and the M -bit DAC in general) is piece-wise linear and pulse shaped due to S/H functions, and it is given by

$$y(nT) = a_0 \cdot w(nT) + a_1 \cdot w(nT - T) + a_{K-1} \cdot w(nT - (K-1)T) \quad (4-48)$$

where K is the number of filter taps and $T = 1/f_{OSR}$ is the update cycle time. The number of coefficients is inversely proportional to the width of the transition band [25]

$$K \sim \frac{1}{\Delta\omega T} = \frac{1}{\omega_a T - \omega_c T} \quad (4-49)$$

where $\omega_c T$ is the normalized pass band frequency and $\omega_a T$ is the normalized stop band frequency. Hence, for a narrow transition band filter, which must be used for a low oversampling ratio, a long FIR filter is required.

4.9.4 Continuous-Time Filter

Due to the S/H elements in the M -bit DAC, the output contains frequency-domain images centered at multiples of the oversampling frequency. The analog continuous-time low pass filter attenuates these images and the modulator noise at higher frequencies are attenuated. In addition, especially if a one-bit DAC is used, an additional discrete-time filter e.g., switched-capacitor filter, is needed before the continuous-time filter to further attenuate the shaped noise.

4.10 SPECIAL IMPROVEMENT TECHNIQUES

Some of the architectures discussed in the previous sections prove to be very suitable for high-resolution and high-speed. However, to reach very high performance some additional techniques can or must be used. These techniques can be of several different kinds, as for example calibration, trimming of internal reference values, randomization, etc. [2, 10, 18, 27-31].

We find three different approaches to improve performance; first we can with for example laser trimming improve the matching between circuit components during the fabrication phase, secondly we can measure the DAC output with an A/D and use a feedback loop for error correction during operation, and third, we can use techniques that during operation improve the performance by smoothening distortion terms into noise without any prior knowledge about the errors.

Consider the second case; if we know the input signal and the non-linear transfer function we can in some sense predict the distortion terms in the output. In the reality, this will be very hard, since the non-linear transfer function is most likely not known. If we are able to observe and analyze the DAC output we may be able to feed back enough information to reduce an amount of the non-linearity (or errors). A compensating DAC could be used in parallel as illustrated in Fig. 4-28 to add an inverse func-

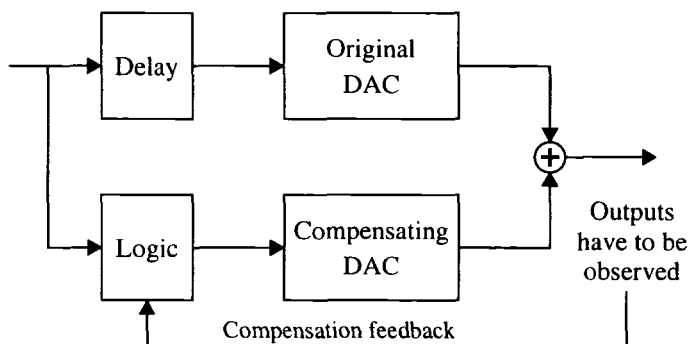


Figure 4-28 Use of an additional DAC to compensate and calibrate errors in the transfer function of the original DAC.

tion to the original DAC output. Another approach is to use an inverse non-linear function in the digital domain, hence changing the digital input data to the DAC. This is often referred to as intentional pre-distortion [29]. A third approach is instead of compensating the output signal to change the internal parameters (references) of the DAC using the fed back information. For example, the individual sizes of the weights in the DAC could be modified in order to minimize the distortion instead of compensating the output signal [32].

The drawback with these methods is obvious. An observer that is performing an analog-to-digital conversion is needed as well as an automatic compensating routine that operates at the same speed as the original converter. It should also be noted that a compensating DAC must not only compensate for static errors, since the dynamic performance determines the quality of the original DAC at higher frequencies.

Therefore, we emphasize methods to reduce non-linearity by not knowing the actual transfer function. Especially we refer to randomization techniques to smoothen distortion into noise as discussed in Sec. 4.10.1. These methods are advantageous in applications where linearity requirements are more important than the noise requirements. The use of internal reference sources to calibrate DAC weights is also similar to using a calibration routine for internal DAC references which is further described in Sec. 13.3.4.

4.10.1 Dynamic Randomization and Element Matching

In a segmented converter the specifications on performance may still not be met although an optimal number of bits has been chosen for segmentation. This is due to the influence of matching errors between reference levels in the DAC. In a conventional thermometer coded DAC, (see Fig. 4-8), the input number, $k = 1$, is represented by using one fixed reference element in the DAC, e.g., current source or capacitor, $k = 2$ is represented by using two fixed references, etc. Using this approach, the matching errors become strongly signal-dependent, since a reference is associated with specific input codes, and we will have distortion at the output.

To reduce the distortion, the binary to thermometer encoder can be designed so that, at different times, different references are chosen to represent $k = 1$, i.e., we do not fix a reference to a certain code. If we choose the references in a way that is uncorrelated with the signal, the matching errors will no longer be signal dependent and hence the error will become noise. This is achieved by using a randomizer or scrambler as illustrated in Fig. 4-29. In a real implementation the randomization can be implemented with a pseudo-random binary sequence (PRBS) generator. We can also choose to assign the references in a cyclic way, which does not give a completely uncorrelated error signal, but the improvement can be significant.

The advantage can be realized by considering the output current for a thermometer coded current-steering DAC. Assume that there are M equally large current sources where all sources, I_i , $i = 1, \dots, M$, have the same nominal size, I_{src} . For each current source a relative mismatch error, δ_i , is associated, hence

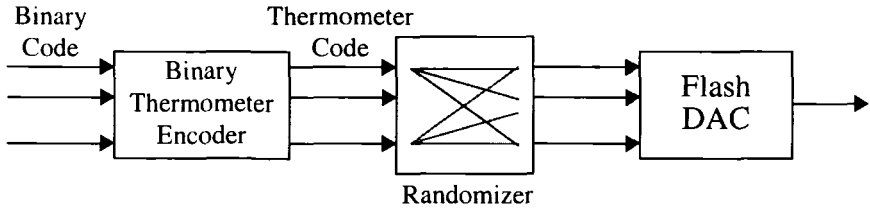


Figure 4-29 Randomization of bits in a thermometer coded DAC. The matching error becomes uncorrelated with the signal.

$$I_i = I_{src} \cdot (1 + \delta_i) \quad (4-50)$$

The errors are fixed for the specific chip and without randomization, the output current for code $k = 00011 \dots 11$ (k number of 1's) is

$$I_{out}(k) = \sum_{i=1}^k I_i = \sum_{i=1}^k I_{src} \cdot (1 + \delta_i) = I_{src} \cdot k \cdot \left(1 + \frac{1}{k} \sum_{i=1}^k \delta_i \right) \quad (4-51)$$

From (4-51) we find that with each code k we can associate a specific relative current error of magnitude

$$\delta^{(k)} = \frac{1}{k} \sum_{i=1}^k \delta_i \quad (4-52)$$

which introduces distortion.

Assume that for $I_{out}(k)$ we could use any k current sources out of the set of M sources with equal probability and independent of k . This will in mean give a relative current error which is equal for all input codes

$$\overline{\delta^{(k)}} = \frac{1}{M} \sum_{i=1}^M \delta_i \quad (4-53)$$

An equally large relative error for all k will only give rise to a gain error at the output. The total error power (for a full-scale signal) is approximately kept constant inspite of the randomization process since the error power still is dependent on the size of the matching errors, δ_k . In the frequency domain the distortion terms are reduced to the cost of a higher noise floor. To illustrate dynamic randomization, we show in Fig. 4-30. a Matlab simulation result where a full-scale sinusoid has been applied to an 8-bit thermometer coded DAC. To all current sources in the DAC a Gaussian distributed random error with standard deviation equal to 12% is applied. In Fig. 4-30a) we find clear distortion terms and in b) the distortion has been reduced and the noise floor has increased. The SNDR is in both cases approximately 41 dB.

When shifting the assignment of the sources in a cyclic way, the errors may still be

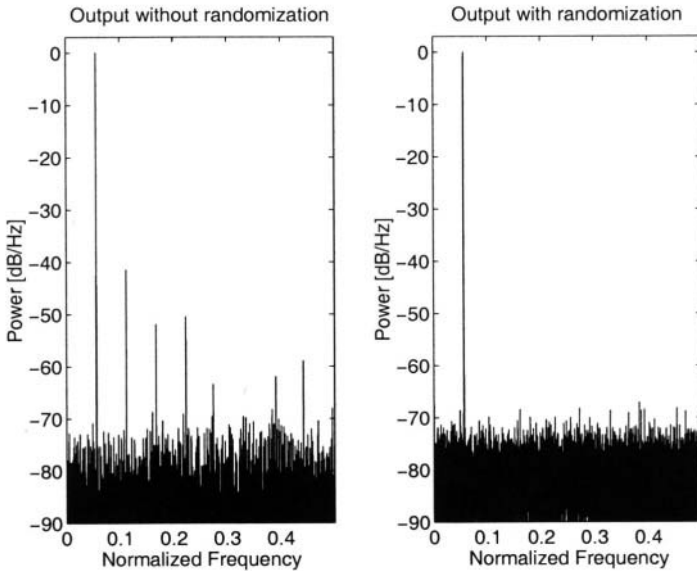


Figure 4-30 Output amplitude spectrum from a thermometer coded 8-bit DAC
a) without randomization and b) with randomization.

slightly signal-dependent, but the linearity errors are reduced. It is hard to predict and find how much the result is improved. The cyclic shift has the advantage of not needing any PRBS generator and the circuit architecture can be implemented with less complexity. As previously, the distortion is reduced to the cost of a higher noise floor.

Dynamic element matching (DEM) techniques can be of several kinds [1, 2, 30, 31, 33] and in this book we understand a DEM technique where the principle is to divide the converter into a number of subconverters with a lower resolution, as proposed in [30,31].

Basically, the method is similar to the randomization technique. In Fig. 4-31 we sketch a concept found in [30, 31]. We have an N -bit binary input and a binary switching tree of N layers. In the k -th layer we have 2^{N-k} switching blocks, $S_{k,j}$. Each of those switching block has one $(k+1)$ -bit input and two k -bit outputs. For the single switching block in the N -th layer, $S_{N,1}$ the number of input bits is increased to $N+1$ by adding a '0' as new LSB.

The objective of each switching blocks, is to direct the input signal through the tree. Dependent on implementation, a control bit for each switching block, $c_{i,j}$, determines how the input should be directed. Actually, the switching tree performs the binary to thermometer encoding. The thermometer code is fed to the $M = 2^N$ one-bit DACs terminating the tree, i.e., an N -bit thermometer coded DAC as illustrated with dashed line in Fig. 4-33.

If all control bits $c_{i,j}$ are fixed and if there are matching errors in the DACs, there will

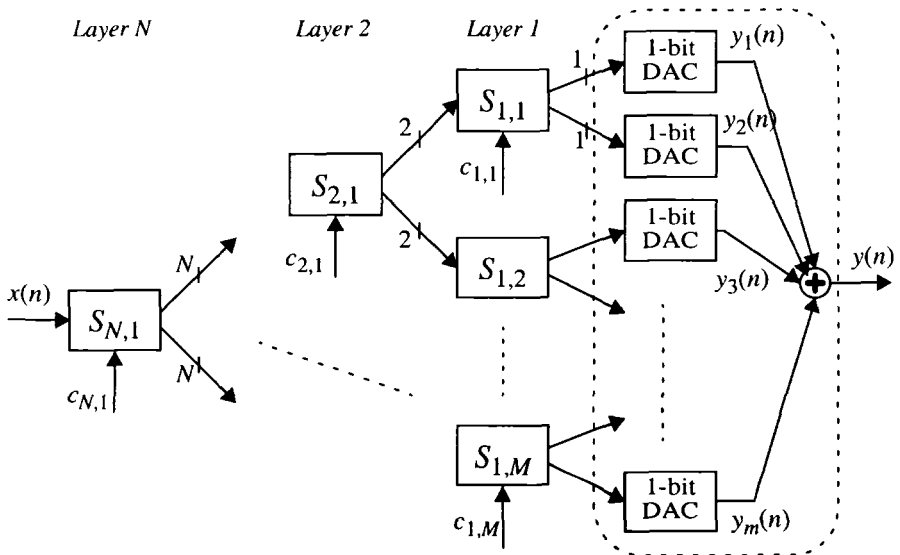


Figure 4-31 An N -bit DAC using dynamic element matching and a bank of one-bit DACs.

be distortion. If we let $c_{i,j}$ be given by a PRBS generator, the signal will pass differently through the tree and distortion is smoothed into signal-independent error as for the dynamic randomization. We can however add the possibility to spectrally shape the circuit errors to higher frequencies, outside the signal band, by letting the control bits $c_{i,j}$ be given by a $\Sigma\Delta$ modulator with a noise shaping function. This action prevents us from using the entire Nyquist range, and therefore we can, at the input of the DAC, add a lower-order $\Sigma\Delta$ modulator which also shapes the input quantization noise to higher frequencies, further increasing the dynamic range within the signal band. The out-of-band noise is attenuated by a continuous-time filter at the output of the converter.

The one-bit DACs are linear and matching errors will only give rise to linear gain and offset errors. (If we do not consider second order effects, such as signal-dependent reference element values). This property makes it possible to shape the errors spectrally and move the error power outside the signal band.

If the number of input bits is high, the tree width and routing will become very large. For a 14-bit DAC we will have 16384 one-bit DACs at the output. Therefore, to reduce the design complexity, the tree can be shortened and terminated at the L -th layer and terminated by 2^{N-L+1} L -bit DACs. Multi-bit DACs will however introduce a non-linear gain due to the matching errors, and the shaping of the error does not become as powerful. The methods show an improved performance to a modest increase of hardware [34] over the regular dynamic randomization technique.

The dynamic matching techniques increase the switching noise in the DAC. The design has to be done carefully so that the gain in dynamic range from less distortion

not is lower than the loss of dynamic range from the increase of the glitch energy. Due to the complexity (chip area, delay) of the randomizer, the technique is mostly preferred when the number of bits is small.

4.11 DAC COMPARISON

In this section we conclude the properties of the DACs discussed in this chapter. We show in tables and figures the properties of different DAC types and architectures as well as reported performance in the literature.

In Fig. 4-32 and Fig. 4-33 and we show a summary of the measured and reported performance from international transactions, journals, text books, etc., as well as information from commercial circuit providers. There are different means of performance and resolution and in most publications we find reported performance on DNL, INL, and settling time. These are insufficient for communications and instead we use measures such as SFDR and SNDR to characterize the converters. In Fig. 4-32 we use the SFDR or SNDR on the y-axis and the *sample frequency* on the x-axis. For the same converters, we use in Fig. 4-33 the SFDR and SNDR on the y-axis and the *signal frequency* on the x-axis.

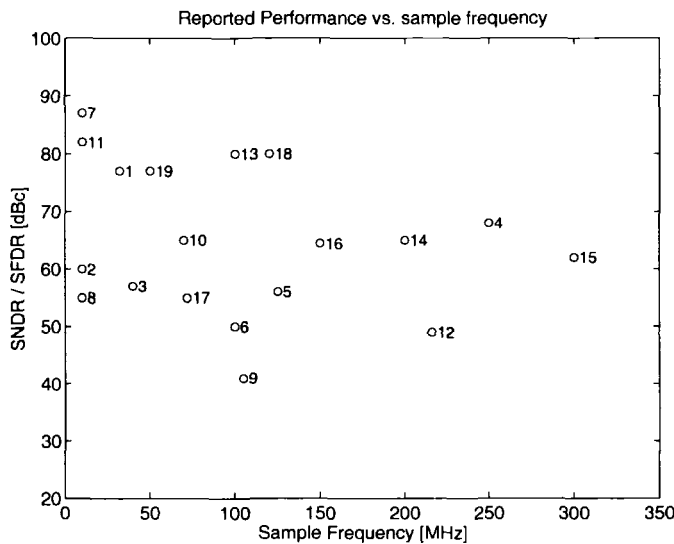


Figure 4-32 Reported measured SNDR and SFDR of different DACs as a function of the sample frequency.

In Table 4-2, we show a list of the DACs used for the summary together with a brief description of the DAC architecture, circuit technique, etc. In the literature we find mostly current-steering converters as suitable candidates for high speed and high resolution. Concluding the summaries in the figures and the table we find a rather clear trade-off between signal frequency and linearity as illustrated with the dashed line in Fig. 4-33.

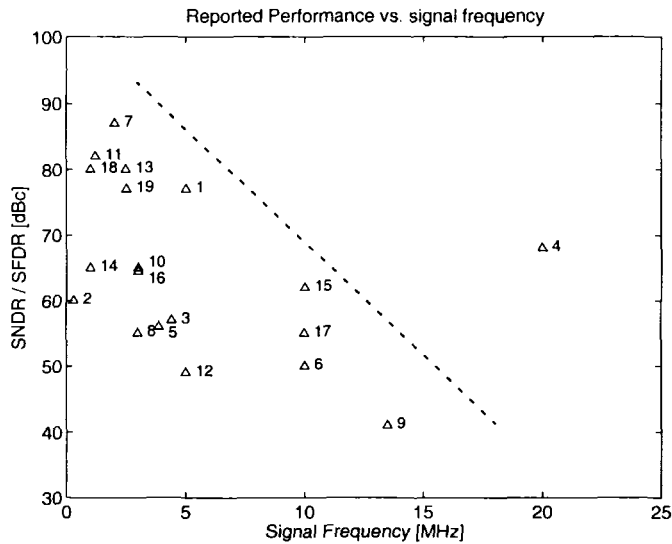


Figure 4-33 Reported measured SNDR and SFDR of different DACs as a function of the signal frequency.

Table 4-2 List of reported DAC performance.

No	Ref	Implementation	Res. [bits]	Supply [V]	Process	Fsignal [MHz]	Fsample [MS/s]	SFDR [dBc]
1	[3]	Current-steering. 4x Interpolation.	14	5	CMOS	5.01	32	77
2	[6]	Current-steering. Segmented.	10	3.3	CMOS	0.3	10	60
3	[7]	Current-steering. Binary weighted.	10	5	CMOS	4.43	40	57
4	[13]	Current-steering. Segmented.	10	3.3	CMOS	20	250	68
5	[27]	Hybrid current-steering. Thermometer coded.	10	5	CMOS	3.9	125	56
6	[35]	Current-steering. Thermometer coded.	10	5	BiCMOS	10	100	50
7	[36]	Hybrid Current-steering and R-2R ladder. Thermometer coded MSBs	14	5/-5.2	BiCMOS	2.03	10	87
8	[44]	Current-steering. Segmented.	10	1.5	CMOS	3	10	55

Table 4-2 List of reported DAC performance.

No	Ref	Implementation	Res. [bits]	Supply [V]	Process	Fsignal [MHz]	Fsample [MS/s]	SFDR [dBc]
9	[50]	Current-steering. Segmented.	8	5	CMOS	13.5	105	41
10	[51]	Current-steering. Segmented.	10	3.3	CMOS	3	70	65
11	[52]	Hybrid Current- steering and R-2R ladder. Thermometer coded MSBs	16	5	BiCMOS	1.23	10	82
12	[53]	Oversampling. Noise shaping.	8	3.3	CMOS	5	216	49 SNDR
13	[54]	Current-steering. Segmented.	14	5	CMOS	2.48	100	80
14	[55]	Current-steering. Segmented.	12	2.7	CMOS	1	200	65
15	[56]	Current-steering. Multi-segmented.	12	3.3	CMOS	10	300	62
16	[57]	Current-steering. Segmented.	14	2.7	CMOS	3	150	64.5
17	[58]	Hybrid. Segmented R- 2R ladder.	12	-5.2	Bipolar	10	72	55
18	[59]	Oversampling. Noise shaping.	14	2.5	CMOS	1	120	80 SNDR
19	[60]	Current-Steering. Segmented. 5 V supply. 14-bit.	14	5	CMOS	2.51	50	77

4.12 SUMMARY

In this chapter we have presented some of the basic DAC architectures that are suitable for high-speed and high-resolution applications. We have also outlined some possible techniques for implementation, such as the charge-redistribution, current-steering, etc. As a guidance, we have summarized a number of reported performances of DACs that are suitable for higher-speed applications in an overviewing figure and table.

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5 OVERVIEW OF CIRCUIT TECHNIQUES

5.1 INTRODUCTION

Both ADCs and DACs are sampled data systems, therefore they require the use of circuit techniques that can handle discrete-time analog signals. Depending on whether the analog signals are voltages or currents the circuits are often divided into voltage-mode or current-mode circuits. In CMOS the switched-capacitor (SC) technique is usually used for voltage-mode circuits while the switched-current (SI) technique is used for current-mode circuits.

A S/H circuit is needed in all sampled data systems. In Sec. 5.2 a general overview of errors in S/H circuits is given. Limitations of the SI technique is discussed in Sec. 5.3 while the SC technique is treated in Sec. 5.4.

5.2 SAMPLE-AND-HOLD CIRCUITS

The S/H circuit is a crucial part of most discrete-time systems. The S/H circuit samples the input signal when a hold command is given and stores the signal value for a certain amount of time. There are a number of effects that will degrade the performance of an S/H circuit. Referring to Fig. 5-1, the aperture time is the delay time from when the hold command is given until the sampling switch is actually turned off. Variations in the aperture time will cause distortion and limit the maximum sampling frequency. The sample-to-hold settling time is the time it takes for the signal to settle within a certain accuracy. The pedestal error is an offset generated by clock feedthrough from the sampling switch. This error is usually signal dependent and causes distortion. The droop rate specifies how fast the held value will decrease due to leakage currents and therefore imposes a lower limit on the sample frequency. Another source of error in the S/H circuit which is not illustrated in the figure is the feedthrough error from the input signal caused by capacitive coupling from the input to the sampling capacitor.

The acquisition time is the time from the end of the hold phase until the output signal of the S/H tracks the input signal within a certain accuracy. For high-speed operation the acquisition time must be small and the bandwidth of the S/H circuit must be large.

The errors described above limits the performance of any S/H circuit except the aperture error which is mainly a problem for S/H circuits with continuous-time input signals. A S/H circuits with a continuous-time inputs signal is sometimes referred to as a

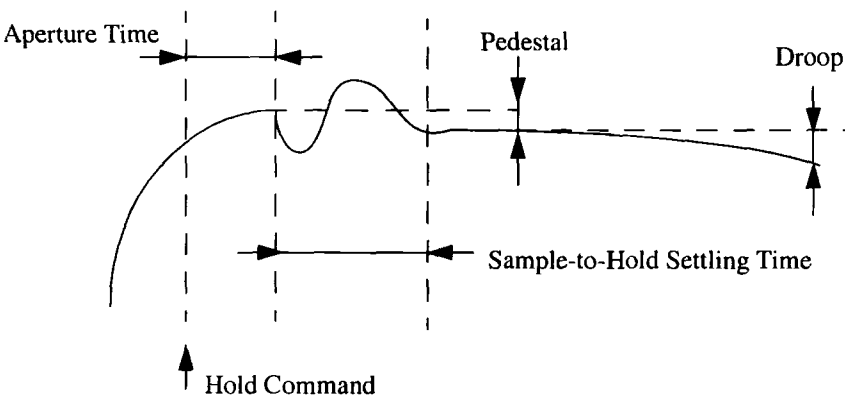


Figure 5-1 Errors in the S/H circuit.

track-and-hold (T/H) circuit and is slightly more difficult to design due to the aperture error.

5.3 SWITCHED-CURRENT TECHNIQUES

The basic SI-memory cell [1] is simple and requires no high-gain amplifiers and no linear capacitors. Therefore this technique is suitable for implementation in a digital CMOS process with a low supply voltage. The basic operation of SI circuits is illustrated in Fig. 5-2.

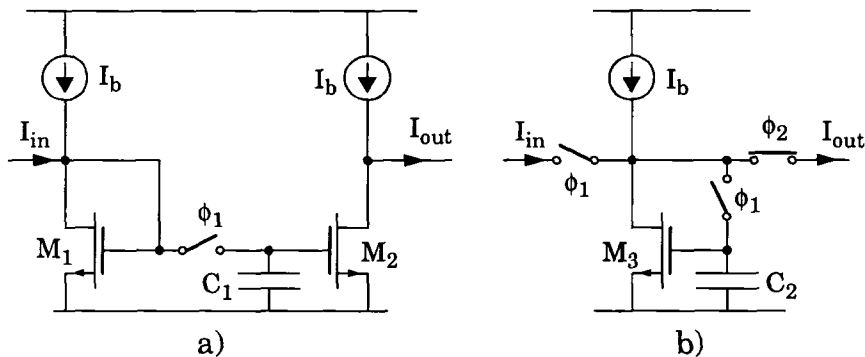


Figure 5-2 The SI-memory cell a) first generation b) second generation.

There are in principle two types of circuits referred to as the first (FG) and second generation (SG) memory cell. In the FG memory cell the input current is converted to a voltage by the diode connected transistor M_1 which is sampled-and-held on the capacitor C_1 . This capacitor can be only the gate-source capacitance of M_2 or an additional

physical capacitor. The voltage is then converted to a current by transistor M_2 and the output current is a sampled-and-held copy of the input current. The principle of the SG cell is the same as for the FG cell but the same transistor M_3 is used both for the current to voltage conversion and for generating the output current. It is therefore necessary to use two additional switches to control the direction of the current in M_3 . Hence the FG cell can provide an output current on both clock phases while the SG cell can only provide an output current on one clock phase. It is clear the both the FG and SG cell can perform a S/H or a T/H function.

In the following we assume that there are no additional capacitors to store the gate-source voltage, since this give the highest speed of the memory cell. It has also been assumed that all the bias currents are equal, i.e. $I_{b1} = I_{b2} = I_{b3}$, and that all transistors have the same size.

5.3.1 Mismatch

Since the SG cell uses only one transistor the only source of mismatch is variations in the drain-source voltage. The two transistors in the FG cell must be matched. A first order approximation of the relative error in output current caused by mismatch is given [2] by

$$\frac{\Delta I}{I} = \frac{\Delta \beta}{\beta} + \frac{2\Delta V_{GS}}{V_{GS} - V_T} - \frac{2\Delta V_T}{V_{GS} - V_T} + \frac{\Delta \lambda}{\frac{1}{V_{DS}} + \lambda} + \frac{\Delta V_{DS}}{V_{DS} + \frac{1}{\lambda}} \quad (5-1)$$

where β is the transconductance parameter, V_{GS} is the gate-source voltage, V_{DS} the drain-source voltage, λ the channel length modulation factor and V_T is the threshold voltage of the transistor. The transconductance parameter is given by

$$\beta = \mu C_{ox} \cdot W/L \quad (5-2)$$

where C_{ox} is the gate capacitance per square, μ is the mobility, W is the width and L is the length of the transistor.

To avoid large V_{GS} mismatch due to voltage drops in the power supply wires, the power supply wires should be made very wide. Variations in V_{DS} cause errors in the output signal, but these errors can be reduced by circuit techniques, as will be explained in the following.

There is an important difference between the two types of SI circuits in that the output current can be easily amplified in a FG cell by simply increasing the size of M_2 . This can not be done in the SG cell. To amplify the current in the SG cell we can either use an additional transistor to generate an amplified current in the output phase or use an additional memory cell operated on a third clock phase ϕ_3 to get a mismatch free amplification. The drawback of the second approach is that the extra clock phase decreases the speed and increases the complexity of the circuit. Another problem is that only integer gain factors can be generated by the second approach. Therefore the first approach is often used and there may be process dependent mismatch even if the

SG memory cell is used.

V_T and β mismatch are usually the dominant sources of error. It has been shown that the correlation between these mismatch errors are small and that the variance is inversely proportional to the area of the transistor [3]. In the CMOS process used in [3], the current mismatch error for a transistor pair with $WL = 1000 \mu\text{m}^2$ and $V_{GS} - V_T = 1 \text{ V}$ is 0.14% which is comparable to capacitor matching. However, for low voltage circuits it may be necessary to decrease V_{GS} and the effect of V_T mismatch will increase. For high-speed circuits smaller devices are normally used which implies that the matching can not be as good as for low-speed circuits.

5.3.2 Speed

The -3 dB bandwidth of the memory cell is determined by [1] (if the effect of the switches are ignored)

$$\omega_{-3dB} = \frac{g_m}{C_{tot}} \quad (5-3)$$

where C_{tot} is the total input capacitance at the input node. For the FG cell $C_{tot} = 2C_{ox}WL$ if both transistors are equal and for the SG cell $C_{tot} = C_{ox}WL$. The transconductance is determined [1] by

$$g_m = \mu C_{ox} \frac{W}{L} (V_{GS} - V_T) = \sqrt{2\mu C_{ox} \frac{W}{L} I_D} \quad (5-4)$$

and the pole is determined by (SG case)

$$|\omega_{-3dB}| = \frac{\sqrt{2\mu I_D}}{\sqrt{C_{ox}WL}^{3/2}} = \mu \frac{(V_{GS} - V_T)}{L^2} \quad (5-5)$$

It should be noted that the bandwidth of the FG cell is a factor 2 smaller due to the larger capacitance. In (5-5) it is seen that to increase the speed with a fixed V_{GS} the length must be reduced. To get the same speed for the FG and SG memory cell, either the length of the transistors in the FG circuit must be decreased or the bias current increased.

5.3.3 Transmission Errors

Due to the finite output impedance of the memory transistor there will be a transmission error and the low frequency transfer function is given by [2]

$$\frac{i_{out}}{i_{in}} = \frac{-1}{1 + \frac{2r_{in}}{r_{out}}} = \frac{-1}{1 + \frac{2g_{ds}}{g_m}} \quad (5-6)$$

where $g_{ds} = I_b \lambda$. The transmission error can be reduced by increasing the output

impedance or decreasing the input impedance of the memory cell. The output impedance can be increased by using a cascode transistor or the regulated cascode technique as shown in Fig. 5-3 for the SG case. The output impedance is for the cascoded mem-

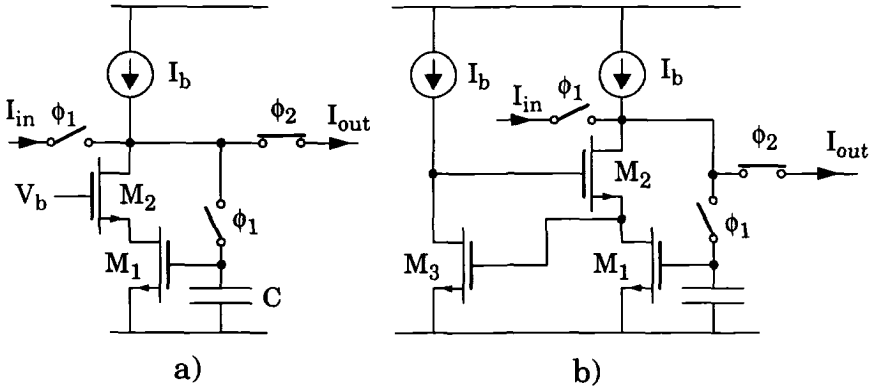


Figure 5-3 a) The cascoded memory cell b) regulated cascode memory cell.

ory cell

$$r_{out} = r_{ds1} \cdot r_{ds2} \cdot g_{m2} \quad (5-7)$$

and for the regulated cascode cell

$$r_{out} = r_{ds1} \cdot r_{ds2} \cdot g_{m2} \cdot r_{ds3} \cdot g_{m3} \quad (5-8)$$

where r_{dsi} is the output resistance of transistor M_i and g_{mi} the transconductance of transistor M_i .

The input impedance of the memory cell can be decreased by inserting an amplifier as shown in Fig. 5-4. The input impedance is now reduced to

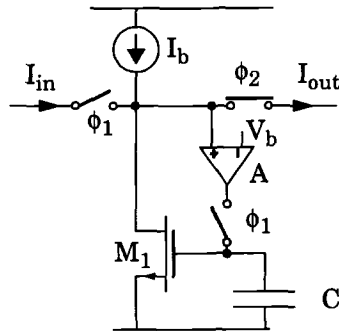


Figure 5-4 Memory cell with feedback.

$$r_{in} = \frac{1}{A \cdot g_m} \quad (5-9)$$

where A is the gain of the amplifier and g_m the transconductance of the memory transistor.

A third approach is to use the S^2I technique [4] where the input current is sampled twice as illustrated in Fig. 5-5. The input current is first sampled by a coarse memory

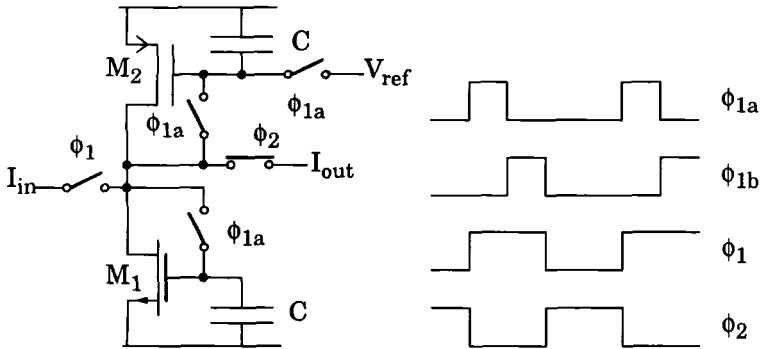


Figure 5-5 The S^2I memory cell.

(M_1) and then the remaining error is sampled by the fine memory (M_2). Since the error current sampled by M_2 is small the gate-source voltage will be almost constant. The input of this memory cell will thus act as a virtual ground. The transmission error, ϵ , of the S^2I cell is [2]

$$\epsilon = \epsilon_c \cdot \epsilon_f \quad (5-10)$$

where ϵ_c is the transmission error of the coarse memory cell and ϵ_f is the transmission error of the fine memory cell. This technique can easily be extended to a S^nI technique where n memory cells are used to further reduce the transmission error.

5.3.4 Noise

In MOS circuits there are in principle two types of noise, $1/f$ noise and thermal noise. The SG cell has an advantage over the FG cell since the inherent correlated double sampling will cancel the low frequency $1/f$ noise [1]. For high frequency circuits the thermal noise is the dominant noise source and the contribution from the $1/f$ noise is relatively small.

The output referred current noise spectral density of the memory cell is [1]

$$S_i(f) = \frac{8}{3} \cdot kT \cdot g_m \cdot (1 + k_{bias}) \quad (5-11)$$

where k is the Boltzmann's constant, T the absolute temperature, g_m the transconductance of the memory transistor and k_{bias} a factor taking the noise contribution from the current source into account. The total noise power can be calculated as the spectral density times the noise bandwidth, i.e.

$$i_n^2 = S_i(f) \cdot BW_n = S_i(f) \cdot \frac{\omega_{-3dB}}{4} = \frac{8}{3} \cdot kT \cdot (1 + k_{bias}) \cdot \frac{g_m^2}{4 \cdot C_{tot}} \quad (5-12)$$

where BW_n is the noise bandwidth and C_{tot} the total capacitance at the gate of the memory transistor. Assuming a sinusoidal input signal the signal power can be calculated as

$$i_{in,rms}^2 = \frac{m_i^2 I_b^2}{2} \quad (5-13)$$

where I_b is the bias current in the memory cell and m_i is the modulation index (the ratio of the input current amplitude and the bias current).

The dynamic range (DR) of the memory cell can now be calculated as [1]

$$DR = \frac{i_{in,rms}^2}{i_n^2} = \frac{3 \cdot m_i^2 I_b^2 \cdot C_{tot}}{4 \cdot kT \cdot g_m^2 \cdot (1 + k_{bias})} \quad (5-14)$$

The DR can be increased without affecting the speed of the circuit. This is achieved by connecting several memory cells in parallel (W is increased by a factor K and I_b is increased by a factor K). This means that for every doubling of the bias current the DR is increased by 3 dB. To make the comparison to SC circuits easier the noise current can be expressed as an input referred noise voltage with power

$$v_n^2 = \frac{i_n^2}{g_m^2} = \frac{2 \cdot kT \cdot (1 + k_{bias})}{3 \cdot C_{tot}} \quad (5-15)$$

The dynamic range can be expressed in the voltage domain as

$$DR = \frac{V_{in}^2}{8/3 \cdot kT} C_{tot} \quad (5-16)$$

where V_{in} is the input voltage swing. For current-mode circuits the input voltage swing is usually quite small (a few hundred mV) compared to voltage-mode circuits and Si-circuits are therefore relatively noisy.

5.3.5 Clock Feedthrough Errors

When the sampling switch is turned off a portion of the channel charge is transferred to the sampling capacitor. Due to capacitive coupling through the overlap capacitors the switch control signal will introduce additional charge on the sampling capacitor. The additional charge gives a pedestal error in the output signal. The error current

caused by the clock feedthrough errors can to the first order be approximated as [2]

$$\Delta i \approx \omega_{-3dB} \cdot (k_i + k_d \cdot i/I_b) = g_m/C_{tot} \cdot (k_i + k_d \cdot i/I_b) \quad (5-17)$$

where i is the input current, I_b the bias current in the memory cell, g_m the transconductance of the memory transistor k_i and k_d are coefficients determined by process parameters, overlap and channel capacitors and switching voltages. Since both g_m , k_i and k_d are signal dependent the clock feedthrough will cause both offset, gain errors and distortion. These errors will degrade the performance especially for high bandwidths. To reduce the signal dependent error the voltage swing at the sampling switch should be minimized but this conflicts with the high voltage swing required for improving the dynamic range.

5.3.6 Aperture Errors

The aperture error in the S/H is caused by signal dependent delays and signal dependent switch turn-off times. Both the source voltage and the threshold voltage of the sampling switch is signal dependent in SI circuits. Therefore aperture time errors will result and thereby distortion. Both aperture and clock feedthrough errors can be reduced by keeping the voltage swing at the sampling switch small. There are circuits techniques using feedback to improve performance ([5], [6]) but they are more difficult to design since the stability of the feedback loop must be considered. They also require the use of floating capacitors.

5.4 SWITCHED-CAPACITOR TECHNIQUES

There are different ways of realizing a S/H circuit using the SC techniques. A possible solution is the circuit shown in Fig. 5-6. The circuit is parasitic insensitive which is

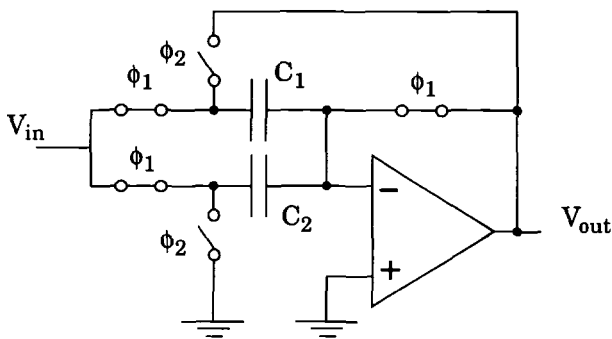


Figure 5-6 Switched-capacitor S/H circuit.

preferable for high accuracy circuits. The circuit uses two non-overlapping clock phases, ϕ_1 and ϕ_2 . On phase ϕ_1 the opamp is auto-zeroed while the input voltage is sampled by C_1 and C_2 . On phase ϕ_2 C_2 is grounded while C_1 is connected to the

output of the opamp. The output voltage on phase ϕ_2 is given by

$$V_{out} = \frac{C_1 + C_2}{C_1} \cdot V_{in} \quad (5-18)$$

5.4.1 Mismatch

The gain of a SC circuits is usually determined by a capacitor ratio. There are some exceptions, e.g. if $C_2 = 0$ in the S/H circuit shown Fig. 5-6, the gain is always unity regardless of the capacitance value. However, most SC circuits rely on accurate capacitor matching. The value of integrated circuit capacitors is given by [7]

$$C = \frac{\epsilon_{ox}}{t_{ox}} \cdot A = C_{ox} \cdot A \quad (5-19)$$

where ϵ_{ox} is the dielectric constant of the silicon dioxide, t_{ox} the thickness of the oxide, A the area of the capacitor and C_{ox} the capacitance per area. Mismatch is caused by variations in C_{ox} and A . The relative capacitance error is given by [2]

$$\frac{\Delta C}{C} = \frac{\Delta C_{ox}}{C_{ox}} + \frac{\Delta A}{A} \quad (5-20)$$

where $\Delta C_{ox}/C_{ox}$ is the relative error in C_{ox} and $\Delta A/A$ is the relative error in area. The accuracy of the capacitance value is usually not very good but for SC circuits we are more interested in the accuracy of a capacitor ratio. The relative error of a capacitor ratio is approximately [7]

$$\frac{\Delta \frac{C_2}{C_1}}{\frac{C_2}{C_1}} \approx \frac{\Delta C_2}{C_2} - \frac{\Delta C_1}{C_1} \quad (5-21)$$

Hence the accuracy of a capacitor ratio is approximately determined by the difference between the individual errors of the capacitors. It is therefore important to try to make the errors as equal as possible and to reduce the influence of systematic errors due to e.g. overetching. It is a general practice to use unit capacitors [8] to avoid systematic errors. This means that the capacitors are composed of an array of small equal unit capacitors. The capacitor matching is often expressed in SD%[FS] which is the standard deviation of the error in a capacitor ratio expressed as percent of full-scale (total capacitance in the unit-capacitor array). Experimental results in [9] indicate that the matching is inversely proportional to the square root of the unit capacitor area and directly proportional to the perimeter-to-area ratio. A quite weak dependence on the capacitor ratio was found which disagrees with the results in [10] showing a square-root dependence and the results in [11] showing virtually no dependence at all. The difference is probably explained by differences in the processes and layout techniques that were used in the measurements.

We conclude that a large area and a large perimeter-to-area ratio improves the match-

ing. It should, however, be noted that matching is usually improved only to a certain limit. If the capacitors are larger than a certain value the matching does not improve any more. Matching errors in the range of 0.1% or even better is achievable. This is better than transistors matching especially for low-supply voltages.

5.4.2 Speed

The speed of the circuit depends on the opamp architecture. Assuming a single stage opamp the unity gain frequency is determined by [8]

$$\omega_u = \frac{g_m}{C_L} \quad (5-22)$$

where g_m is the transconductance of the input transistor of the opamp and C_L the total load capacitance at the output of the opamp. For the two-stage opamp the load capacitor C_L should be replaced by the compensation capacitor C_c in the expression for the unity-gain frequency. The load capacitance is given by

$$C_L = C_{next} + \frac{C_1(C_2 + C_p)}{C_1 + C_2 + C_p} \quad (5-23)$$

where C_{next} is the input capacitance of the following SC circuit and C_p is the parasitic capacitor at the input of the opamp.

The -3dB frequency of the S/H circuit is determined by

$$|\omega_{-3dB}| = \beta \cdot \omega_u \quad (5-24)$$

where β is the feedback factor given by

$$\beta = \frac{C_1}{C_1 + C_2 + C_p} \quad (5-25)$$

Hence the -3dB frequency is given by

$$|\omega_{-3dB}| = \frac{g_m}{C_{next} + \frac{C_1(C_2 + C_p)}{C_1 + C_2 + C_p}} \cdot \frac{C_1}{C_1 + C_2 + C_p} \quad (5-26)$$

where C_p is the parasitic capacitor at the input of the opamp. The parasitic capacitor consists of the C_{gs} capacitor of the input transistor and the bottom plate-to-ground capacitors of C_1 and C_2 . If the gain is unity, i.e. $C_2 = 0$, and the parasitic capacitor C_p is small compared to C_1 , the -3dB frequency can be approximated as

$$|\omega_{-3dB}| \approx \frac{g_m}{C_{next}} \quad (5-27)$$

The expression for the speed of the SC circuit is thus in principle similar to the speed of SI circuits. The circuit in Fig. 5-6 is also similar to the SI circuits in that the speed is reduced when the gain is increased. If the two-stage opamp is used we have

$$|\omega_{-3dB}| = \frac{g_m}{C_c} \cdot \frac{C_1}{C_1 + C_2 + C_p} \quad (5-28)$$

The speed is thus independent of the load capacitance (this is only true when ignoring the high frequency poles). In theory the speed could be very high by choosing a small C_c but in practice, this is not possible since the circuit will be unstable.

The expressions above were derived on clock phase ϕ_2 since it is most likely that the speed is lowest in this clock phase especially for large gain factors. When designing SC circuits it is important to consider both clock phases with respect to speed and stability. In clock phase ϕ_1 the total load capacitance at the output is

$$C_{L,1} = C_1 + C_2 + C_p + C_{next,1} \quad (5-29)$$

where $C_{next,1}$ is the input capacitance of the following stage in clock phase ϕ_1 . The feedback factor is $\beta = 1$.

The speed of SC circuits as well as SI circuits are obviously dependent on the opamp and S/H configuration. From the simple models used here we do not see any strong indication on a significant difference in speed between the SC and SI techniques.

5.4.3 Transmission Errors

Finite opamp gain will cause additional gain errors in the total transfer function. The transfer function including the effect of finite opamp gain is given by

$$V_{out} = \frac{C_1 + C_2}{C_1 + \frac{C_1 + C_2}{A}} \cdot V_{in} \quad (5-30)$$

where A is the gain of the opamp. The effect of the finite gain is thus similar to the effect of finite output resistance in SI circuits.

5.4.4 Noise

A fundamental limitation on SC circuits is imposed by the thermal noise in the circuit. The flicker noise can in principle be made negligible small compared to the thermal noise by using circuit techniques such as correlated double sampling and chopper stabilization [12]. In the following analysis we neglect the noise from the opamp. In a more detailed analysis the opamp noise should be taken into account since it may be as large as or even larger than the switch noise. The noise voltage power spectral density of a switch is

$$S_v(f) = 4kT \cdot R_{on} \quad (5-31)$$

where R_{on} is the switch on-resistance, k is the Boltzmann's constant and T is the absolute temperature. The total noise power can be calculated as the spectral density times the noise bandwidth, i.e.

$$v_n^2 = S_v(f) \cdot BW_n = S_i(f) \cdot \frac{\omega_{-3B}}{4} = 4kT \cdot R_{on} \cdot \frac{R_{on}}{4 \cdot C_s} = \frac{kT}{C_s} \quad (5-32)$$

where BW_n is the noise bandwidth and C_s the sampling capacitor. Assuming a sinusoidal input signal the signal power can be calculated as [8]

$$v_{in, rms}^2 = \frac{v_a^2}{2} \quad (5-33)$$

where v_a is the amplitude of the sinusoid.

The dynamic range (DR) of the memory cell can now be calculated as

$$DR = \frac{v_a^2}{2kT} \cdot C \quad (5-34)$$

In conclusion, the voltage noise levels in SI and SC for the same sampling capacitor are approximately the same while the voltage swing in SC circuits usually is larger than SI circuits. Therefore SI circuits are more noisy than SC circuits [13], especially for low supply voltages.

5.4.5 Aperture and Clock Feedthrough Errors

Aperture errors and signal dependent clock feedthrough errors are caused by variations in the source and threshold voltage of the sampling switch. The signal dependent part of these errors can be reduced by using bottom-plate sampling [14]. This technique is illustrated in Fig. 5-7. There are two control signals for each sampling capac-

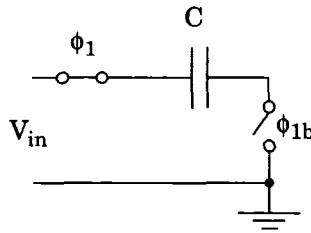


Figure 5-7 Bottom-plate sampling.

itor where ϕ_{1b} is opened slightly before ϕ_1 . When the switch controlled by ϕ_{1b} is opened the charge on C can not change. Therefore, any error caused by the switch controlled by ϕ_1 will have no impact on the stored value. Since the switch controlled by ϕ_{1b} is always connected to ground or virtual ground the only remaining errors is an offset from the clock feedthrough. This technique is not easily applied to the basic SI memory cell.

5.5 SUMMARY

We now summarize some of the main features of the two techniques discussed in this chapter [2].

SI Circuits

- Require no high-gain amplifiers and no linear capacitors which makes the technique suitable for implementation in a low-voltage digital CMOS process
- Due to their simplicity and the slightly smaller capacitive loads it is easier to design high bandwidth circuits using the SI technique.
- Due to the simple and straight forward structure of the circuits it is easier to design SI circuits and they can easily be made modular.

SC Circuits

- Since the voltage swing in SC circuits is larger than in SI, SC circuits have higher dynamic range.
- When using bottom-plate sampling, SC circuits are less sensitive to clock feedthrough errors and aperture errors. Therefore SC circuits have smaller distortion.
- The matching accuracy in SC circuits is usually better than for SI circuits.

For high performance analog circuits the SC technique is the clear choice. The SI technique could be a good candidate for medium performance analog circuits in a low voltage CMOS process.

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6 ANALOG FUNCTIONAL BLOCKS

6.1 INTRODUCTION

In chapter 5 the SC and SI circuit techniques and their properties were briefly reviewed. In this chapter we show in more detail how these circuit techniques can be used to make some important analog functional blocks such as integrators and multiplying DACs (MDAC). They are key components in sigma-delta and pipelined ADCs respectively. The S/H circuit used at the input of most ADCs can be regarded as a special case of the MDAC and is not treated separately. We also show how to derive the maximum speed and how to calculate the thermal noise in the circuits. A more detailed investigation on how these circuit limitations affects the performance of the ADC is given in chapter 9 for pipelined converters and in chapter 11 for oversampled sigma-delta converters. In Sec. 6.2 we introduce the MDAC which is an important part of the pipelined converter. SI MDACs for high speed are considered in Sec. 6.3 while the limitations on speed and the thermal noise are derived in Sec. 6.4. The discussion on SI MDACs is concluded by making a comparison on different SI architectures in Sec. 6.5. In Sec. 6.6 we consider the SC MDAC and investigate the effect of non-ideal opamps while the thermal noise in SC MDACs is discussed in Sec. 6.7. Finally the speed and thermal noise of integrators are derived in Sec. 6.8 for the SI technique and in Sec. 6.9 for the SC technique.

6.2 THE MULTIPLYING DAC

The pipelined converter contains several stages each with a low resolution ADC and an analog circuit calculating the analog output signal to the following stage. The analog circuit contains a DAC, a subtractor and a S/H amplifier and is sometimes referred to as a multiplying DAC (MDAC) [1], see Fig. 6-1. The output signal of the MDAC is

$$out_i = G_i \cdot (in_i - DAC_i) \quad (6-1)$$

where G_i is the gain of the MDAC, in_i the analog input signal and DAC_i the output signal of the DAC. If $DAC_i = 0$ the MDAC behaves as a S/H circuit while if $in_i = 0$ it behaves as a DAC. Hence the S/H circuit and the DAC can be regarded as special cases of the MDAC and are not treated separately.

The limitations of the MDAC using both the SI and the SC techniques are investigated

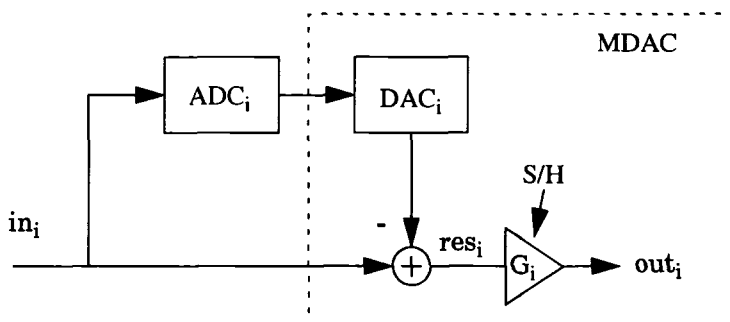


Figure 6-1 The MDAC in one stage of a pipelined ADC.

in the following sections.

6.3 MDACS IN SI PIPELINED ADCS

There are relatively few publications on SI ADCs especially for high-speed applications. Some of the publications and their performance are listed in Table 6-1. All the

Table 6-1 SI ADCs

Ref.	Process	Bits	Sampling freq. MHz	V_{dd} V	Power mW	Area mm^2	Signal freq. MHz	SNDR dB
[2]	Bi-CMOS	10	20	10	1000	48	10	50
[3]	CMOS	8	15	5	350	2.4	14	<40
[4]	CMOS	8	4.5	5	128	5.8	1	47
[5]	CMOS	10	3	3	82.5	2.7	20	40
[6]	CMOS	8	10	2.5	60	7	0.5	41

listed ADCs are pipelined converters. In [2] both current-mode and voltage-mode signals are used. The flash converters use voltage-mode signals while the signal in the main path is currents. The ADC has three-stages with one bit of digital correction in each stage. The architecture is shown in Fig. 6-2. In the figure voltage-mode signals are indicated with a V while current-mode signals are indicated with an I . The residue gain amplification is performed in two stages using two S/H amplifiers with gains of 2 and 4 respectively. The residue in the second stage is generated by only one S/H amplifier with a gain of 8. Two clock phases, ϕ_1 and ϕ_2 are used as shown in the Table 6-2. The circuit is designed in a BiCMOS process and both high-gain amplifiers and linear capacitors are used for the V/I conversion and S/H circuit.

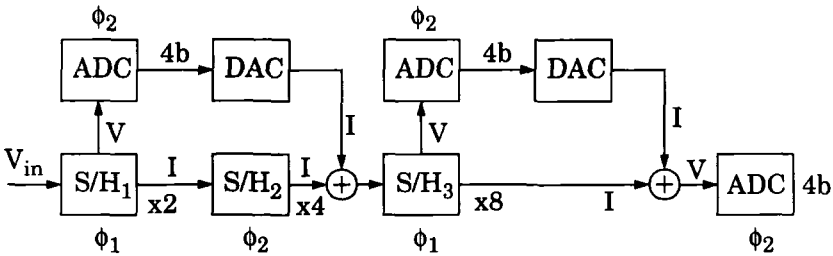


Figure 6-2 ADC architecture in [2].

Table 6-2 The operation of the stages in [2].

Stage	Block	ϕ_1	ϕ_2
1	S/H ₁	sample	hold
1	S/H ₂ , ADC	hold	sample
1	DAC	settling	-
2	S/H ₃	sample	hold
2	ADC	sample	hold
2	DAC	-	settling
3	ADC	hold	sample

In [3] the S^2I technique [7] is used for the memory cells. A high clock frequency is achieved by using unity-gain MDACs. This means that the amplification factor of the residue amplifier is one. The reference currents of the DACs must therefore be reduced by a factor two for each stage in the pipeline as illustrated in Fig. 6-3.

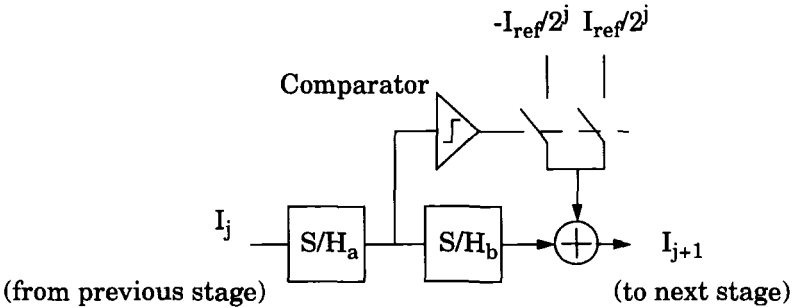


Figure 6-3 ADC architecture in [3].

There are two approaches to implementing the S/H circuits. One option is to use current mirrors to copy and amplify the currents which requires two clock phases. The second approach is to use ratio independent switching which requires more than two clock phases. In [3] the second approach was adopted and the total number of clock

phases is four. The sampling rate is thus 1/4 of the clock frequency which in this case means $f_{clk} = 60\text{MHz}$ and $f_s = 15\text{MS/s}$. The clocking of one stage in the pipeline is illustrated in Table 6-3. All the stages are clocked in the same way.

Table 6-3 The operation of the stages in [2].

Stage	Block	ϕ_1	ϕ_2	ϕ_3	ϕ_4
j	S/H _a	sample	hold to S/H _b	hold to Comp.	-
j	S/H _b	hold to I _{j+1}	sample	hold	hold
j	Comp.	-	reset	sample	hold to DAC
j	DAC	settle	-	-	-

In [4] a residue amplifiers with a gain of two is used which means that the reference currents can be equal in all the stages, see Fig. 6-4. The operation of two consecutive

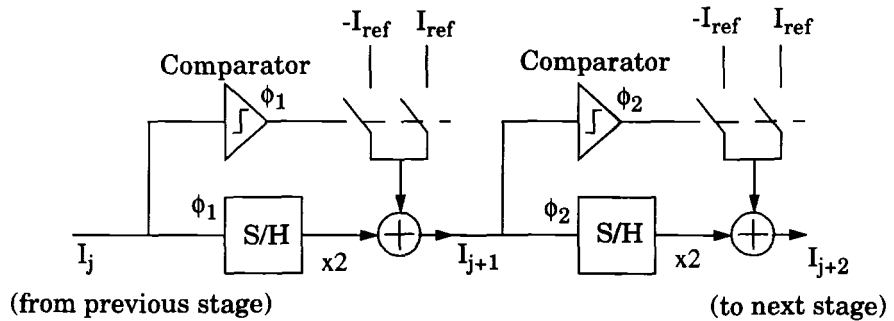


Figure 6-4 ADC architecture in [4].

stages is illustrated in Table 6-4.

Table 6-4 The operation of the stages in [4].

Stage	Block	ϕ_1	ϕ_2
j	S/H	sample	hold
j	Comp.	sample	hold
j	DAC	-	settling
j+1	S/H	hold	sample
j+1	Comp	hold	sample
j+1	DAC	settling	-

In this case the amplification by two is implemented by using current mirrors and thus only two clock phases are needed. Both the comparator and the S/H circuit are clocked on the same clock phase. The architecture in [5] is similar to the architecture in [4] but

in this case the redundant-signed-digit (RSD) algorithm [8] is used to enable digital correction. The RSD algorithm is discussed in chapter 9. The architecture in [6] is shown in Fig. 6-5. The clocking of two consecutive stages is shown in Table 6-5. In

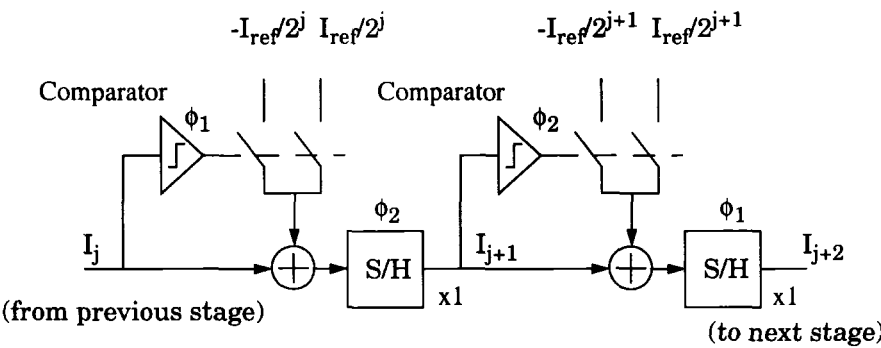


Figure 6-5 ADC architecture in [6].

Table 6-5 The operation of the stages in [6].

Stage	Block	ϕ_1	ϕ_2
j	Comp.	sample	hold
j	S/H	hold	sample
j	DAC	-	settling
j+1	Comp.	hold	sample
j+1	S/H	sample	hold
j+1	DAC	settling	-

this architecture the comparator and the S/H of one stage are clocked on different clock phases. It is thereby possible to perform the D/A conversion before the S/H, which may be advantageous as will be explained in the following sections.

6.4 PERFORMANCE LIMITATIONS IN SI MDACS

When considering the speed and the noise in the circuits we will neglect the effect of the switches. The circuits are usually designed in such a manner that the switches have a small influence, but in a more accurate analysis their effect should be taken into account.

6.4.1 Speed

The speed of the circuits is determined by the -3 db bandwidth of the memory cell and the required number of clock phases. The 3-dB cut-off frequency of a memory cell is

$$\omega_{-3dB} = \frac{g_m}{C_{tot}} \quad (6-2)$$

where C_{tot} is the total capacitance at the gate of the memory transistor and g_m the transconductance of the memory transistor. Assuming a single pole system the relative settling error is determined by

$$\varepsilon_r = e^{-\frac{t_s}{\tau}} \quad (6-3)$$

where $\tau = 1/\omega_{-3dB}$ is the time constant and t_s the available settling time. The available settling time is given by

$$t_s = \frac{T_s}{N_p} \quad (6-4)$$

where T_s is the total sampling period of the circuit and N_p is the required number of sampling phases within one sample period. We assume that all sampling phases have equal duration. The maximum sampling frequency is given by

$$f_s = \frac{1}{T_s} = \frac{1}{t_s \cdot N_p} = \frac{\omega_{-3dB}}{\ln\left(\frac{1}{\varepsilon_r}\right) \cdot N_p} \quad (6-5)$$

6.4.2 Residue Amplification

There are two approaches to amplifying a current, using current mirrors or ratio-independent SI circuits. The two approaches for a gain of two are shown in Fig. 6-6. The amplifier in Fig. 6-6a) works as a current mirror where the gate voltage of transistors M_2 and M_3 is sampled and held by a switch. The number of transistors connected in parallel at the output of the current mirror determines the gain of the amplifier. If all transistor are assumed to be equal, the bandwidth of this circuit is

$$\omega_{in,a} = \frac{g_m}{C_{gs}(1+G)} \quad (6-6)$$

where G is the residue gain, g_m the transconductance and C_{gs} the gate-source capacitor of the transistors. Two sampling phases is needed regardless of the gain factor, one sampling phase for the current stage, ϕ_1 and one for the following stage, ϕ_2 .

The ratio-independent amplifier in Fig. 6-6a) first stores the input voltage on the gate of transistor M_4 (ϕ_1) and then stores the same voltage on the gate of transistor M_5 (ϕ_2). On clock phase ϕ_3 both the current from M_4 and M_5 are sent to the output and it is sampled by the following stage. If all the stages are equal the number of sampling phases is given by

$$N_{p,b} = 2G \quad (6-7)$$

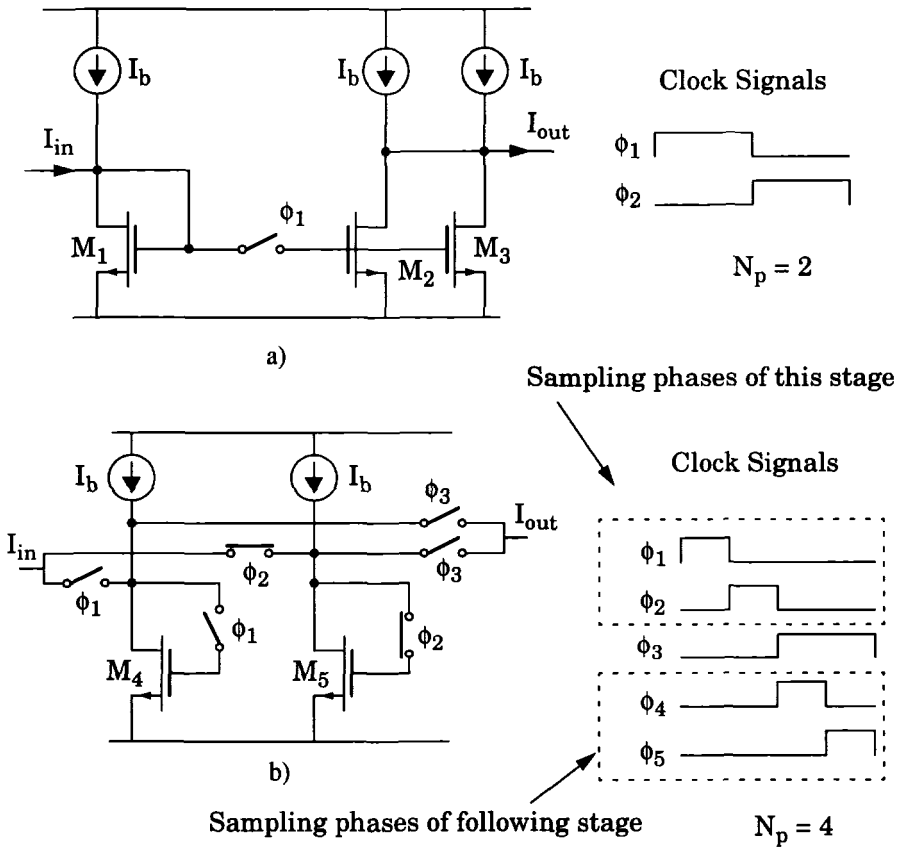


Figure 6-6 Current amplifier with gain = 2. a) current mirror b) ratio-independent.

The input bandwidth of the memory transistors is

$$\omega_{in, b} = \frac{g_m}{C_{gs}} \quad (6-8)$$

The main advantage of the ratio-independent approach is that the gain factor is not dependent on transistor matching. A disadvantages of the ratio-independent cell is that there are many clock signal which makes the circuits more complex. Another limitation is that only integer gain factors can be realized. The current mirror approach is thus more flexible. It is obvious from (6-6) and (6-8) that the speed is reduced in both cases if the residue gain factor is increased. Therefore unity gain MDACs are preferable in high speed applications.

6.4.3 Distributing Currents to Several Outputs

The output current of the MDAC must be sent to both the S/H circuit and the ADC of the following stage. Two circuits performing this function are shown in Fig. 6-7. In the first case, Fig. 6-7a), the two output currents are simply generated by using two

$$f_{s,a} = \frac{g_m}{C_{gs} \cdot \ln(1/\epsilon_r) \cdot 2 \cdot (1 + 2G)}$$

(6-10)

and

$$f_{s,b} = \frac{g_m}{C_{gs} \cdot \ln(1/\epsilon_r) \cdot 2 \cdot (1 + G)}$$

(6-11)

respectively. Hence the circuit in Fig. 6-7b) is almost twice as fast as the circuit in Fig. 6-7a).

6.4.4 Time-Interleaved Stage

From the discussion above it is clear that a small gain factor will increase the speed and if possible unity gain MDACs should be used. Two attractive features of the current mirror approach is that only two clock phases are needed and that non integer gain factors can be realized. Non-integer gain factors are needed when using the calibration technique in [9]. A third advantage of the current mirror cell is that the voltage glitches at the switching instants are usually smaller than for the ratio-independent circuits. The main disadvantages of the current mirror circuit are the higher power consumption, the larger chip area and the distortion caused by transistor mismatch. In Fig. 6-8

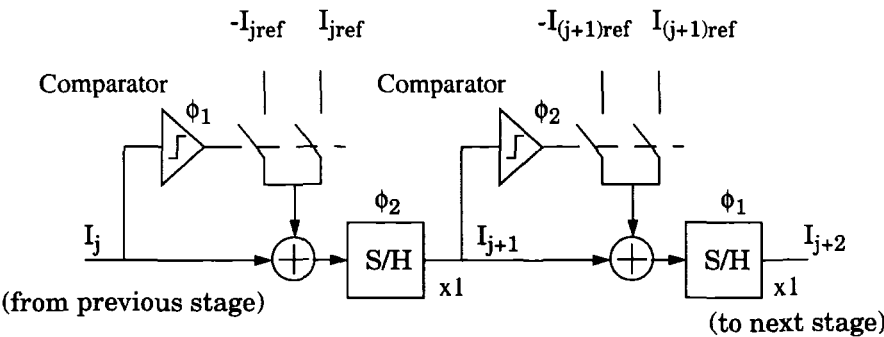


Figure 6-8 ADC architecture in [6].

we show a current mirror based stage with reduced area and power consumption [6]. The clocking of two consecutive stages is shown in Table 6-6. In this architecture the

Table 6-6 The operation of the stages in [6].

Stage	Block	ϕ_1	ϕ_2
j	Comp.	sample	hold
j	S/H	hold	sample
j	DAC	-	settling

Table 6-6 The operation of the stages in [6].

Stage	Block	ϕ_1	ϕ_2
j+1	Comp.	hold	sample
j+1	S/H	sample	hold
j+1	DAC	settling	-

comparator and S/H of one stage are clocked on different clock phases. Unity-gain MDACs are used and therefore the reference currents are scaled down by a factor of two for each stage in the pipeline i.e.

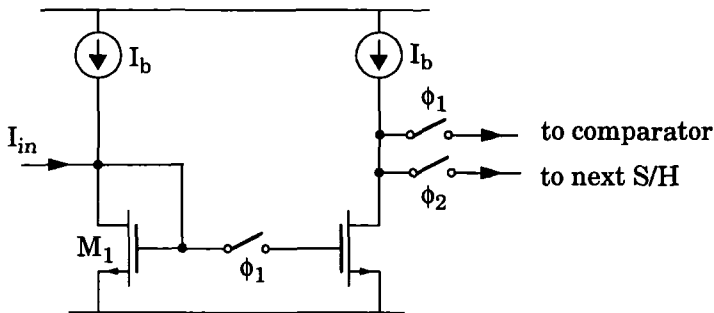
$$I_{(j+1)ref} = \frac{I_{jref}}{2} \quad (6-12)$$

where I_{jref} is the reference current in the j -th stage and $I_{(j+1)ref}$ is the reference current in the next stage. The output current of a stage is the summation of the output current of the internal D/A converter and the output current of the S/H circuit. It is given by

$$I_{j+1} = I_{in} - \sum_{k=1}^j b_k \cdot \frac{I_{ref}}{2^k} \quad (6-13)$$

where I_{in} is the input current, I_{j+1} the output current from stage j and b_k is ± 1 depending on the signal from the comparator in stage k . Due to the unity-gain MDACs the power consumption and area will be smaller in the LSB stages since the current swing gets smaller and smaller in the pipeline. The drawback with the smaller currents is that the signal gets more sensitive to noise and other error sources. However the required accuracy of the MDACs are lower for the LSB stages.

Since the comparator and the S/H circuit are time interleaved and clocked on different clock phases the memory cell can be simplified as shown in Fig. 6-9. The capacitive

**Figure 6-9** Simplified memory cell.

load is decreased and both the power consumption and area are reduced compared to the solution in Fig. 6-7a). It should be noted that when calculating the speed of the circuit we must consider that the signal now must settle through both the S/H and the comparator on the same clock phase. However, the settling time of the comparator is usually small compared to the settling time of the S/H circuit. If the speed reduction caused by the comparator is ignored the maximum speed of the MDAC is given by

$$f_{max, b} = \frac{g_m}{C_{gs} \cdot \ln(1/\epsilon_r) \cdot 2 \cdot (1 + G)} \quad (6-14)$$

which is comparable to speed of the circuit in Fig. 6-7b).

6.4.5 D/A Conversion

We have not yet considered the clocking of the DACs and the generation of the residue signal. In Fig. 6-10, two circuits to generate the residue are shown. The first circuit

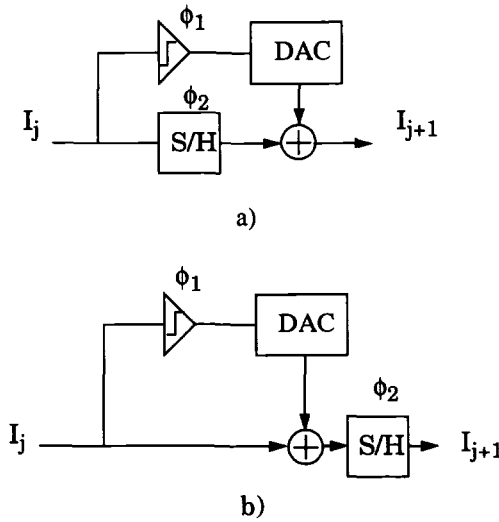


Figure 6-10 Residue generation with subtraction a) before and b) after S/H.

adds the DAC current after the S/H circuit while the second circuit adds the DAC current before the S/H circuit. The advantage of the second approach is that the bias current and transistor sizes in the S/H can be reduced by a factor 2 due to the smaller signal swing. This means that noise, area and power is a factor two smaller.

6.4.6 Noise

The thermal noise in the circuits limits the resolution of the converter. The noise current power spectral density for the thermal noise of an MOS transistor is given by [10]

$$S(f) = \frac{8}{3} \cdot kT \cdot g_m \quad (6-15)$$

where k is the Boltzmann's constant, T the absolute temperature and g_m the transconductance of the memory transistor. Assuming a single pole system and white noise the total noise power is given by [10]

$$i_n^2 = S(f) \cdot BW_N \quad (6-16)$$

where BW_N (frequency in Hz) is the noise bandwidth related to the -3 dB bandwidth as

$$BW_N = \frac{\pi f_{-3dB}}{2} = \frac{\omega_{-3dB}}{4} \quad (6-17)$$

When this noise is sampled all the high frequency noise is folded back to the signal band and the total sampled noise power is the same. By using (6-16) we have

$$i_n^2 = \frac{2}{3} \cdot kT \cdot g_m \cdot \omega_{-3dB} \quad (6-18)$$

First we calculate the noise in the circuit shown in Fig. 6-11. The gain of the circuit is

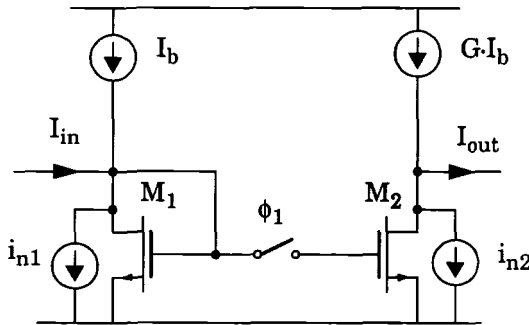


Figure 6-11 First generation memory cell with noise sources.

G . The bias current and W/L ratio of transistor M_2 is G times larger than for transistor M_1 . We neglect the noise from the bias current sources. The noise contribution from the current sources adds to the total noise in exactly the same way as the noise from M_1 and M_2 . Their effect can be included by adding a factor $1 + k_{bias}$ in the expression for the total noise [11]. k_{bias} is the relative size of the transconductance in the current source compared to the transconductance of the current mirror transistor. There are two noise sources present in the circuit which both contribute to the total noise. First the noise contribution from M_1 is sampled on the gate capacitor of M_2 . The voltage noise power is given by

$$v_{n1}^2 = \frac{i_{n1}^2}{g_{m,in}^2} = \frac{\frac{2}{3} \cdot kT \cdot g_{m,in} \cdot \omega_{in}}{g_{m,in}^2} \quad (6-19)$$

where $g_{m,in}$ is the transconductance of M_1 , and ω_{in} is the input bandwidth, i.e.

$$\omega_{in} = \frac{g_{m,in}}{C_{gs1} + C_{gs2}} = \frac{g_{m,in}}{C_{gs1} \cdot (1 + G)} \quad (6-20)$$

In the hold phase the sampled noise is transferred to the output and we have

$$i_{n1,out}^2 = v_{n1}^2 \cdot g_{m,out}^2 = \frac{g_{m,out}^2}{g_{m,in}^2} \cdot \frac{2}{3} \cdot kT \cdot g_{m,in} \cdot \omega_{in} \quad (6-21)$$

where $g_{m,out}$ is the transconductance of M_2 . In the hold phase the noise from M_2 will also contribute. The noise power is given by

$$i_{n2,out}^2 = \frac{2}{3} \cdot kT \cdot g_{m,out} \cdot \omega_{out} \quad (6-22)$$

where $g_{m,out}$ is the transconductance of M_2 and ω_{out} is the bandwidth of the following stage. Assuming uncorrelated noise sources the total noise at the output is

$$\begin{aligned} i_{n,tot}^2 &= i_{n1,out}^2 + i_{n2,out}^2 \\ &= \frac{2}{3} \cdot kT \cdot \left(\frac{g_{m,out}^2}{g_{m,in}^2} \cdot g_{m,in} \cdot \omega_{in} + g_{m,out} \cdot \omega_{out} \right) \end{aligned} \quad (6-23)$$

If we for simplicity assume that $\omega_{out} = \omega_{in}$ have that

$$g_{m,out} = G \cdot g_{m,in} \text{ and } \omega_{in} = \omega_{out} = \frac{g_{m,in}}{(1 + G) \cdot C_{gs}} \quad (6-24)$$

Hence the total noise is

$$i_{n,tot}^2 = \frac{2}{3} \cdot kT \cdot G \cdot \frac{g_{m,in}^2}{C_{gs}} \quad (6-25)$$

When using the second generation approach, see Fig. 6-12, G transistors are needed to get a gain of G . The noise for each transistor can be calculated using (6-23) and assuming $g_{m,in} = g_{m,out}$, i.e.

$$i_{n,i}^2 = \frac{2}{3} \cdot kT \cdot (g_{m,in} \cdot \omega_{in} + g_{m,in} \cdot \omega_{out}), \quad i = 1, \dots, G \quad (6-26)$$

The noise from each memory transistor adds to the total noise at the output. For simplicity we assume $\omega_{in} = \omega_{out} = g_{m,in}/C_{gs}$, which yields

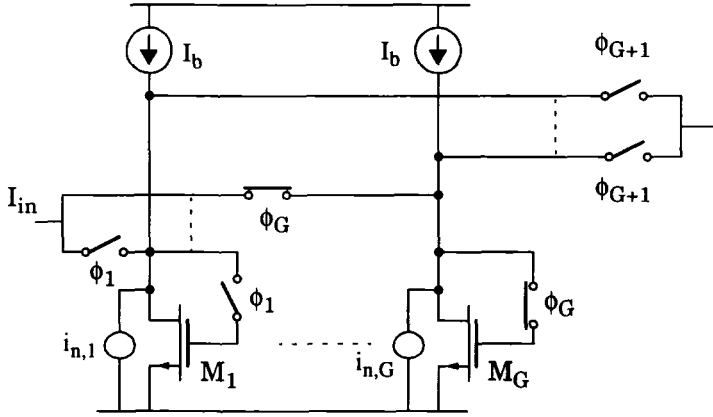


Figure 6-12 Second generation current amplifier with noise sources.

$$i_{n,tot}^2 = \sum_{i=1}^G i_{n,i}^2 = \frac{2}{3} \cdot kT \cdot 2G \cdot \frac{g_{m,in}^2}{C_{gs}} \quad (6-27)$$

Hence the noise in the ratio independent case is twice as large as in the current mirror case. Sometimes it is more convenient to calculate the input referred voltage noise since it can be directly compared to the input voltage swing. For the circuit in Fig. 6-11 we have

$$v_{n,in}^2 = \frac{i_{n,tot}^2}{g_{m,out}^2} = \frac{i_{n,tot}^2}{(g_{m,in} \cdot G)^2} = \frac{2}{3} \cdot kT \cdot \frac{1}{C_{gs} \cdot G} \quad (6-28)$$

It is interesting to note that the voltage noise only depends on the residue gain and the size of the sampling capacitor. Since the voltage swing is limited by the supply voltage the only way to improve the SNR is to increase the sampling capacitor.

6.4.7 Dynamic Range

The dynamic range at the output is determined by the signal power at the output and the total noise at the output. If the rms value of the input signal is $i_{rms,in}$ the dynamic range is given by

$$DR = \frac{i_{rms,out}^2}{i_{n,tot}^2} \quad (6-29)$$

or expressed in dB

$$DR = 20 \cdot \log\left(\frac{i_{rms,out}}{i_{n,tot}}\right) \quad (6-30)$$

It may be more convenient to refer the noise to the input as a voltage. i.e.

$$DR = \frac{v_{rms, in}^2}{v_{n, in}^2} \quad (6-31)$$

where $v_{rms, in}^2$ is the power of the input signal. The DR of the circuit in Fig. 6-11 can for instance be calculated as

$$DR = \frac{v_{rms, in}^2}{v_{n, in}^2} = \frac{v_{rms, in}^2}{\frac{2}{3} \cdot kT \cdot \frac{1}{C_{gs} \cdot G}} = \frac{v_{rms, in}^2 \cdot C_{gs} \cdot G}{\frac{2}{3} \cdot kT} \quad (6-32)$$

To improve the DR we must clearly increase the voltage swing or the sampling capacitor.

6.5 COMPARISONS OF SI MDACS

To conclude the discussion on SI MDACS in this chapter we compare the speed, DR, area and power of the different architectures. The three types considered here are shown in Fig. 6-13. The first type, shown in Fig. 6-13a), is similar to the architecture in [4] where both the comparator and the S/H circuit use the same clock phase which requires that first generation SI circuits are used. The circuit in Fig. 6-13b) corresponds to the time-interleaved stage in Sec. 6.4.4 and uses current mirror circuits with the comparator and S/H operated on different clock phases. The architecture in Fig. 6-13c) is based on ratio-independent circuits where several clock phases are required. In all cases we assume that the bandwidth of the comparator is large enough not to influence the speed. We also assume that all unit transistors are of equal size and that the transconductance is g_m and that the gate-source capacitance is C_{gs} . The different architectures are in the following referred to as Type 1, Type 2 and Type 3. The residue gain is denoted G . The speed of the circuits is determined by the number of clock phases and the bandwidth, ω_{in} . All these results have been derived in the previous sections and are summarized in Table 6-7. The speed of the Type 2 MDAC is optimis-

Table 6-7 Speed of the different MDACs.

Type	Clock Phases	ω_{in}	f_s
1	2	$\frac{g_m}{(1+2G)C_{gs}}$	$\frac{g_m}{2(1+2G)\ln(1/\epsilon_r)C_{gs}}$
2	2	$\frac{g_m}{(1+G)C_{gs}}$	$\frac{g_m}{2(1+G)\ln(1/\epsilon_r)C_{gs}}$
3	$2(1+G)$	$\frac{g_m}{C_{gs}}$	$\frac{g_m}{2(1+G)\ln(1/\epsilon_r)C_{gs}}$

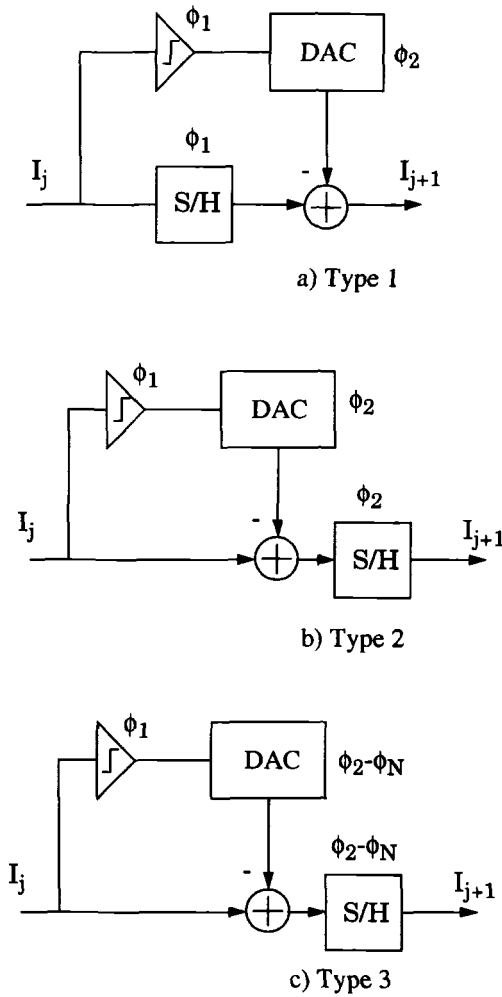


Figure 6-13 The different MDAC architectures.

tic in the sense that the speed of the comparator has been assumed to be very high not to influence the total speed of the circuit. The effect of the comparator can be included by approximating the bandwidth of the MDAC as

$$\omega_{tot} = \frac{1}{\frac{1}{\omega_{SH}} + \frac{1}{\omega_{comp}}} \quad (6-33)$$

where ω_{SH} is the bandwidth of the memory cell and ω_{comp} is the bandwidth of the comparator.

The noise in the MDACs can be derived by using (6-23). We assume $\omega_{out} = \omega_{in}$. For

the converter of type 1 we have

$$\begin{aligned}
 i_{n, tot}^2 &= \frac{2}{3} \cdot kT \cdot \left(\frac{g_{m, out}}{g_{m, in}} + 1 \right) \cdot \omega_{in} \cdot g_{m, out} = \\
 &= \frac{2}{3} \cdot kT \cdot (G + 1) \cdot \frac{g_m}{(1 + 2G)C_{gs}} \cdot g_m \cdot G
 \end{aligned} \tag{6-34}$$

For the converter of type 2 we have

$$\begin{aligned}
 i_{n, tot}^2 &= \frac{2}{3} \cdot kT \cdot \left(\frac{g_{m, out}}{g_{m, in}} + 1 \right) \cdot \omega_{in} \cdot g_{m, out} = \\
 &= \frac{2}{3} \cdot kT \cdot (G + 1) \cdot \frac{g_m}{(1 + G)C_{gs}} \cdot g_m \cdot G = \frac{2}{3} \cdot kT \cdot \frac{g_m^2}{C_{gs}} \cdot G
 \end{aligned} \tag{6-35}$$

Finally for the converter of type 3 we have

$$\begin{aligned}
 i_{n, tot}^2 &= \left(\frac{2}{3} \cdot kT \cdot \left(\frac{g_{m, out}}{g_{m, in}} + 1 \right) \cdot \omega_{in} \cdot g_{m, out} \right) \cdot G = \\
 &= \left(\frac{2}{3} \cdot kT \cdot 2 \cdot \frac{g_m}{C_{gs}} \cdot g_m \right) \cdot G = \frac{2}{3} \cdot kT \cdot \frac{g_m^2}{C_{gs}} \cdot 2G
 \end{aligned} \tag{6-36}$$

The noise performance is summarized in Table 6-8.

Table 6-8 Noise in the different MDACs.

Type	Output referred noise
1	$\frac{2}{3} \cdot kT \cdot \frac{(1 + G) \cdot G}{(1 + 2G)} \cdot \frac{g_m^2}{C_{gs}}$
2	$\frac{2}{3} \cdot kT \cdot G \cdot \frac{g_m^2}{C_{gs}}$
3	$\frac{2}{3} \cdot kT \cdot 2G \cdot \frac{g_m^2}{C_{gs}}$

It is very difficult to accurately estimate the chip area of the complete ADC since e.g. switches, wires and the digital circuits will consume a portion of the area. In the following the total area of the memory transistor (assumed to have width W and length L) is used to compare the relative area of the circuits. This is not a very accurate estimate but can give a rough indication on whether the MDAC has a large area or not. The total DC current of the S/H circuit is used as measure of the power consumption of the MDAC. Each unit transistor is assumed to use the current I_b . The area and power is summarized in Table 6-9.

Table 6-9 Noise in the different MDACs.

Type	Area	Power
1	$(1 + 2G)WL$	$(1 + 2G)I_b$
2	$(1 + G)WL$	$(1 + G)I_b$
3	GWL	GI_b

A few observations can be made about the results above.

- For all types of MDACs the speed is increased if the currents are scaled down in the pipeline. This can be achieved by reducing both the residue gain factor and the reference currents of the following stage. For a 1 bit/stage converter unity gain MDACs can be used. To avoid too small currents in the later stages of the pipeline, some of the LSB stages may use a larger residue gain factor [6].
- When comparing the speed of the circuits, Type 3 is the best choice. Type 2 is the second best choice but for high gain factors the difference is small. It should also be noted that non-overlapping clock phases must be used for the voltage switches, meaning that a portion of the sampling period can not be used for settling. Hence, increasing the number of clock phases, which is necessary in MDAC of Type 3, is likely to reduce the total available settling time.
- MDAC of Type 1 has the lowest noise but also the lowest speed, largest area and power consumption. To get the same DR for all the circuits the bias current for Type 2 and 3 can be increased. This will increase both the power and area to make them comparable to Type 1. However, an advantage of Type 2 and 3 is that the D/A conversion can be performed before the S/H circuit without decreasing the speed. This means that if unity gain MDACs are used, the bias currents can be decreased in the S/H circuits which improves the DR.
- It should also be noted that expressions above are based on very simple models and several limiting factors, such as clock feedthrough, have been neglected. The equations can be regarded as the fundamental limitation of the relative performance.

6.6 SC MDACS

In this section we discuss SC MDACs. A circuit commonly used [12] to implement the MDAC in pipelined ADCs is shown in Fig. 6-14. The circuit is single-ended but the actual implementation would normally be fully differential. The MDAC in the figure has N input bits $b_{N-1}, \dots, b_0$ where b_{N-1} is the MSB and b_0 the LSB. One bit of digital correction is used and therefore the residue gain is decreased by a factor 2. We will use this circuit in the following to illustrate how important parameters as noise and bandwidth can be calculated for SC MDACs. We assume that the DC gain of the opamp is A_0 , the parasitic capacitance at the input node of the opamp is C_p . On the

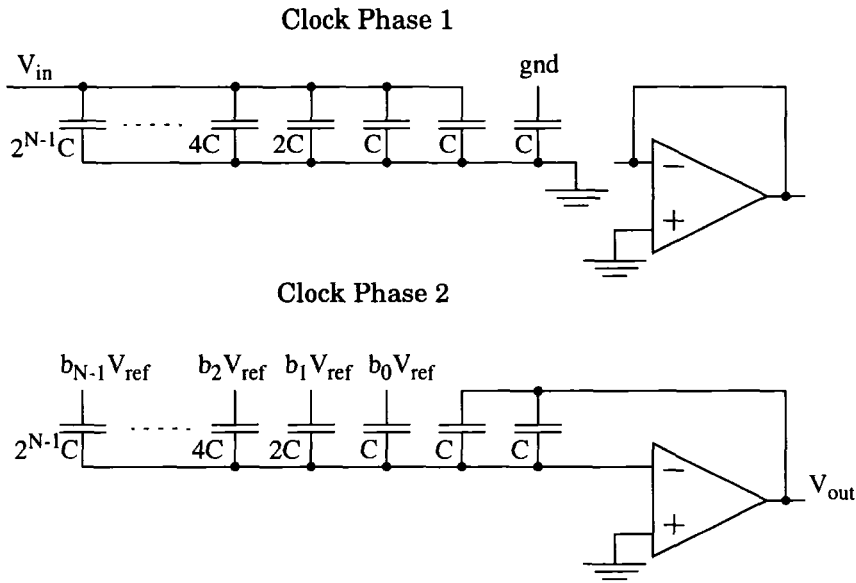


Figure 6-14 The SC MDAC of a pipelined converter.

first clock phase all capacitors but one are connected to the input voltage of the circuit, making the total charge on the top plates of the capacitors

$$q_1 = V_{in} \cdot (2^{N-1}C + \dots + C + C) \quad (6-37)$$

On the next clock phase two of the capacitors are connected to the output while the other capacitors are connected to $\pm V_{ref}$ depending on the output from the sub-ADC. The total charge on the top plates of the capacitors on this clock phase is

$$\begin{aligned} q_2 = & \left(V_{out} + \frac{V_{out}}{A} \right) \cdot 2C + V_{ref} \cdot (2^{N-1} \cdot b_{N-1} \cdot C + \dots + b_0 \cdot C) \\ & + \frac{V_{out}}{A} \cdot (2^{N-1}C + \dots + C) + \frac{V_{out}}{A} \cdot C_p \end{aligned} \quad (6-38)$$

where $b_{N-1}, \dots, b_0 = \pm 1$ are determined by the digital output bits from the sub-ADC. The total charge is conserved and the output signal can be calculated as

$$\begin{aligned} q_1 = q_2 \Rightarrow \\ V_{out} = & \frac{V_{in} \cdot 2^{N-1} - \frac{V_{ref}}{2} \cdot (2^{N-1} \cdot b_{N-1} + \dots + b_0)}{1 + \frac{C_{tot}}{2A_0C}} \end{aligned} \quad (6-39)$$

where $C_{tot} = 2^{N-1}C + \dots + C + 2C + C_p$ is the total capacitance connected to the

input node of the opamp. The reference voltages in the DACs are chosen as

$$V_{ref-} = -\frac{FS}{2} \text{ and } V_{ref+} = \frac{FS}{2} \quad (6-40)$$

where FS is the full-scale input swing of the converter.

6.6.1 Effect of Finite Opamp Gain and Parasitic Capacitors

Due to the finite opamp gain and parasitic capacitors there will be a gain error in the MDAC as can be seen in (6-39). By using the first order Taylor expansion we get the following approximation

$$\frac{1}{1+\epsilon} \approx 1-\epsilon \Rightarrow \frac{1}{1+\frac{C_{tot}}{2A_0C}} \approx 1 - \frac{C_{tot}}{2A_0C} \quad (6-41)$$

Hence the relative gain error can be approximated as

$$\epsilon_G = \frac{C_{tot}}{2A_0C} = \frac{2^N + 1 + \frac{C_p}{C}}{2A_0} \approx \frac{2^{N-1}}{A_0} \quad (6-42)$$

where the approximation is valid for large N . The relative gain error is the same for both the input signal and the DAC signal. This means that the gain error can be modelled as an error at the output of the subtracter.

6.6.2 Speed

The relative settling error assuming linear settling and a single pole opamp is determined by

$$\epsilon_r = e^{-t_s \cdot \omega_{-3dB}} \quad (6-43)$$

where ω_{-3dB} is the -3dB bandwidth and t_s is the settling time. The settling error causes a gain error at the output of the MDAC. When using two clock phases with 50% duty cycle the settling time is, limited by

$$t_s < \frac{1}{2f_s} \quad (6-44)$$

where f_s is the sampling frequency. The unity-gain bandwidth of the opamp is related to the bandwidth as

$$\omega_u = \omega_{-3dB} / \beta \quad (6-45)$$

where β is the feedback factor. The feedback factor in the evaluation phase is

$$\beta = \frac{2C}{(2^N + 1)C + C_p} = \frac{2}{(2^N + 1) + C_p/C} \quad (6-46)$$

The unity gain bandwidth of a single stage opamp is determined by

$$\omega_u = \frac{g_m}{C_{load}} \quad (6-47)$$

where g_m is the transconductance of the input device of the opamp and C_{load} is the total load capacitance at the output of the opamp. The load capacitance is determined by

$$C_{load} = \beta \cdot ((2^N - 1)C + C_p) + C_{next} \quad (6-48)$$

where C_{next} is the capacitive load of the following stage. If the following stage is identical the load capacitance is given by

$$C_{load} = \beta \cdot ((2^N - 1)C + C_p) + 2^N C \approx 2^N C \quad (6-49)$$

where the last approximation is valid for large N . When the circuit is designed, both clock phases must be considered. The feedback factor of the opamp in clock phase 1 is $\beta = 1$ while the load is only C_p . To avoid that the opamp get unstable it may be necessary to connect an additional load capacitor to the output in this clock phase.

The maximum speed of the MDAC can now be calculated as

$$\begin{aligned} f_s &= \frac{1}{2t_s} = \frac{\omega_{-3dB}}{\ln(1/\epsilon_r) \cdot 2} = \frac{\omega_u \cdot \beta}{\ln(1/\epsilon_r) \cdot 2} = \\ &= \frac{\frac{g_m}{2^N C} \cdot \frac{2}{(2^N + 1) + C_p/C}}{\ln(1/\epsilon_r) \cdot 2} \approx \frac{g_m}{\ln(1/\epsilon_r) \cdot 2^{2N} C} \end{aligned} \quad (6-50)$$

The assumption of linear settling is not valid when the gain is small, i.e. when the feedback factor is large (see chapter 7). The settling will be slew rate limited and the settling time is increased compared to the above expressions.

6.7 NOISE IN SC MDACS

There are in principle two types of noise in MOS circuits, thermal noise and $1/f$ noise. The thermal noise is white while $1/f$ noise is frequency dependent. $1/f$ noise dominates at low frequencies while the thermal noise is usually dominant for wide-band circuits. The $1/f$ can also be decreased by circuits techniques such as correlated double sampling. Therefore the $1/f$ noise is neglected in the following.

6.7.1 kT/C Noise

When a signal is sampled on a capacitor, the thermal noise generated by the switch will alias since the inherent bandwidth is wider than half of the sampling frequency. The total power is equal to kT/C , independent of the switch-on resistance [10]. The noise in the signal bandwidth, BW_s is

$$v_{n,C}^2 = \frac{k \cdot T}{C} \cdot \frac{BW_s}{f_s/2} = \frac{k \cdot T}{C} \cdot \frac{1}{OSR} \quad (6-51)$$

where k is the Boltzmann's constant, T is the absolute temperature, C the sampling capacitor, f_s the sampling frequency and OSR the oversampling ratio. If the circuit is fully differential and the sampling capacitance is C in both the positive and negative signal path the kT/C noise within the signal bandwidth is twice that of (6-51). However the effective signal swing is also doubled and therefore the DR is improved by 3 dB.

6.7.2 Thermal Noise in Opamp

The input referred noise power spectral density for a single-stage opamp is given by (see chapter 7)

$$S_{n,amp}(f) = 2 \cdot \frac{8}{3} \cdot \frac{k \cdot T}{g_m} \cdot (1 + n_t) \quad (6-52)$$

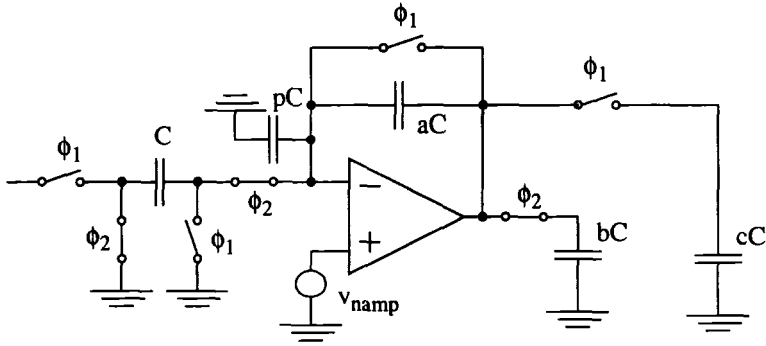
where g_m is the transconductance of the input transistor and n_t is the noise contribution factor due to the other noise sources. From equation (6-52) it is seen that we can increase the input transconductance or decrease the noise contribution factor of the other noise sources in order to decrease the input-referred noise. We will see shortly that increasing the input transconductance is hardly an option due to the increase in noise bandwidth. As we will see in chapter 7 the noise contribution factor is small for the telescopic opamp. The telescopic opamp is therefore the best choice from a noise point of view. The noise contribution from the opamp is normally the same regardless if the circuit is single-ended or fully differential. Thus the dynamic range is increased by 6 dB when using fully differential circuits if only the opamp noise is considered.

6.7.3 Input Referred Thermal Noise of SC Amplifier I

In Fig. 6-15, we show an SC amplifier configuration. The parasitic capacitance at the input is pC . The capacitor bC and cC represent the load capacitance switched to the opamp output on clock phase ϕ_2 and ϕ_1 respectively. For the moment we neglect the noise contribution from the switch on-resistance. The feedback factor on clock phase ϕ_1 is

$$\beta_1 = 1 \quad (6-53)$$

while on ϕ_2 it is

**Figure 6-15** S/H amplifier I.

$$\beta_2 = \frac{a \cdot C}{a \cdot C + C + p \cdot C} = \frac{a}{a + 1 + p} \quad (6-54)$$

The corresponding load capacitors for the opamp are

$$C_{L,1} = C \cdot p + C \cdot c \quad (6-55)$$

and

$$C_{L,2} = b \cdot C + \beta_2 \cdot (C + p \cdot C) = (b + \beta_2 \cdot (1 + p)) \cdot C \quad (6-56)$$

Assuming a single-stage opamp the unity gain bandwidth is

$$\omega_u = \frac{g_m}{C_L} \quad (6-57)$$

and the -3 dB bandwidth

$$\omega_{-3dB} = \omega_u \cdot \beta \quad (6-58)$$

The noise bandwidth is given by [10]

$$BW_n = \omega_{-3dB}/4 \quad (6-59)$$

The total noise power in the two clock phases can now be calculated by using $v_n^2 = S(f) \cdot BW_n$ [10]

$$v_{namp,1}^2 = \frac{4}{3} \cdot \frac{k \cdot T}{C_{L,1}} \cdot \beta_1 \cdot (1 + n_t) = \frac{4}{3} \cdot \frac{k \cdot T}{C \cdot (p + c)} \cdot (1 + n_t) \quad (6-60)$$

and

$$\begin{aligned}
 v_{namp,2}^2 &= \frac{4}{3} \cdot \frac{k \cdot T}{C_{L,2}} \cdot \beta_2 \cdot (1 + n_t) \\
 &= \frac{4}{3} \cdot \frac{k \cdot T}{(b \cdot (a + 1 + p) + a \cdot (1 + p)) \cdot C} \cdot a \cdot (1 + n_t)
 \end{aligned} \tag{6-61}$$

It is seen from equation (6-60) and (6-61) that the input transconductance has no effect on the input referred noise. This is because when we increase the input transconductance to decrease the input-refer noise spectral density, we increase the noise bandwidth, making the noise power constant. In order to evaluate the impact of the thermal noise generated in the opamp, we must find the input-referred noise of the SC amplifier. The z-domain transfer function from the opamp input can be derived by observing that the total charge on capacitors pC , C and aC is conserved from clock phase ϕ_1 to ϕ_2 . The output signal is given by

$$V_{out}(z) = \frac{1 + a + p}{a} \cdot V_{namp,2}(z) - \frac{p}{a} \cdot V_{namp,1}(z) \cdot z^{-1/2} \tag{6-62}$$

where the first term corresponds to the output referred opamp noise generated during clock phase ϕ_2 and the second term to the output referred opamp noise generated during clock phase ϕ_1 . Since all the samples of the thermal noise are uncorrelated the two noise terms in (6-62) will add. However if the noise samples are correlated as for $1/f$ noise, low frequency noise would be slightly attenuated due to the minus sign on the second term of (6-62). Since the gain of the SC circuit is approximately $1/a$ the total input referred noise caused by the opamp can be calculated as

$$\begin{aligned}
 v_{n,in}^2 &= (1 + a + p)^2 \cdot v_{n,amp2}^2 + p^2 \cdot v_{n,amp1}^2 + \frac{kT}{C} \\
 &= \frac{4}{3} \cdot kT \cdot \frac{(1 + n_t)}{C} \left(\frac{(1 + a + p)^2 \cdot a}{b \cdot (a + 1 + p) + a \cdot (1 + p)} + \frac{p^2}{c + p} \right) + \frac{kT}{C}
 \end{aligned} \tag{6-63}$$

In the above expression we have also included the kT/C -noise stored on the sampling capacitor. We see that if the noise contribution factor, n_t , is small and the capacitors at the output are large the kT/C -noise dominates.

6.7.4 Input Referred Thermal Noise of SC Amplifier II

Another popular SC amplifier configuration is shown in Fig. 6-16. The difference is that the feedback capacitor is also used to sample the input. The noise can in this case be derived in similar way as was done for S/H amplifier I in the previous section. Both the feedback factors and the z-domain transfer function are the same. The only difference is that in this case the gain of the S/H amplifier is $1 + 1/a$. Hence the input-referred noise is

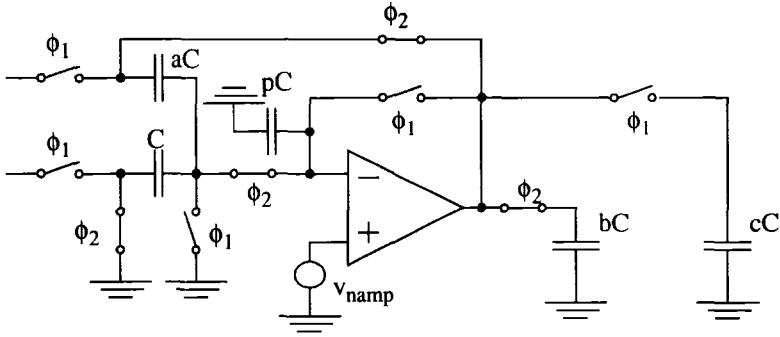


Figure 6-16 S/H amplifier II.

$$\begin{aligned}
 v_{n, in}^2 &= \frac{(1+a+p)^2}{(1+a)^2} \cdot v_{n, amp2}^2 + \frac{p^2}{(1+a)^2} \cdot v_{n, amp1}^2 + \frac{kT}{(a+1)C} \\
 &= \frac{4kT}{3} \cdot \frac{(1+n_t)}{C} \left(\frac{\left(1 + \frac{p}{(1+a)}\right)^2 \cdot a}{b(a+1+p) + a(1+p)} + \frac{p^2}{(1+a)^2(c+p)} \right) \\
 &\quad + \frac{kT}{(a+1)C}
 \end{aligned} \tag{6-64}$$

We see that if the noise contribution factor, n_t , is small and the capacitors at the output are large the kT/C -noise dominates.

6.7.5 Thermal Noise in MDAC

In this section we calculate the thermal noise in the MDAC in Fig. 6-14. We consider the thermal noise from the switches and the opamp. To simplify the analysis the parasitic capacitor at the input of the opamp is ignored and the opamp gain is assumed to be infinite. On clock phase ϕ_1 only the noise from the switches contributes. The noise voltage stored on the capacitors is [10]

$$v_n^2 = \frac{kT}{(2^N + 1) \cdot C} = \frac{kT}{C_{tot}} \tag{6-65}$$

where C_{tot} is the sum of all capacitors in the MDAC. The total noise charge can be calculated as

$$q^2 = \frac{kT}{C_{tot}} \cdot C_{tot}^2 = kT \cdot (2^n + 1) \cdot C \tag{6-66}$$

On clock phase ϕ_2 this charge will be converted to a voltage. Hence the total noise generated during ϕ_1 referred to the output of the MDAC is

$$v_{n, \phi_1}^2 = \frac{q^2}{(2C)^2} = \frac{kT \cdot (2^N + 1)}{4C} \approx \frac{kT}{C} \cdot 2^{N-2} \quad (6-67)$$

where the last approximation is valid for large N . The thermal noise sources on clock phase ϕ_2 are shown in Fig. 6-17. Noise sources v_{n1} and v_{n2} represent the thermal

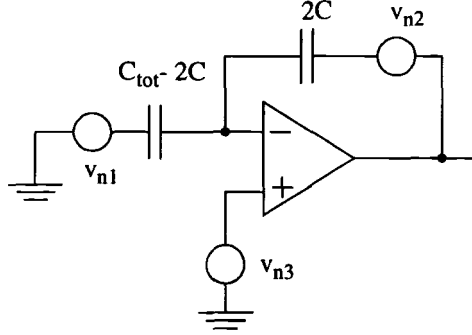


Figure 6-17 Noise sources in clock phase 2.

noise from the switch on-resistance while v_{n3} represents the thermal noise from the opamp. On clock phase ϕ_2 all three noise sources will contribute. The noise contribution from a noise source referred to the output of the circuit can be calculated as

$$v_{out, k}^2 = S_k(f) \cdot |H_k|^2 \cdot BW_{N, k}, \quad k = 1, 2, 3 \quad (6-68)$$

where $S_k(f)$ is the spectral density of the noise source k , $|H_k|$ the gain from noise source k to the output of the MDAC and $BW_{N, k}$ the noise bandwidth. The noise bandwidth is for all the noise sources

$$BW_{N, k} = \frac{\beta \cdot \omega_u}{4} \quad (6-69)$$

where β is the feedback factor and ω_u the unity gain bandwidth of the opamp. The factors $|H_k|$ can be shown to be

$$|H_1| = \frac{C_{tot} - 2C}{2C}, \quad |H_2| = 1, \quad |H_3| = \frac{C_{tot}}{2C} \quad (6-70)$$

and the spectral densities are

$$S_1(f) = 4kT \cdot R_1, \quad S_2(f) = 4kT \cdot R_2, \quad S_3(f) = 2 \cdot \frac{8kT}{3g_m} \cdot (1 + n_t) \quad (6-71)$$

where g_m is the transconductance of the input device of the opamp, n_t the noise contribution factor, R_1 the switch resistance of the switch at the input and R_2 the switch

resistance of the switch in the feedback loop of the opamp. It is common that $1/g_m$ is larger than the on-resistance of the switches. If this is the case the contribution from the opamp will dominate and we have

$$v_{n,\phi_2}^2 = \frac{16kT}{3g_m} \cdot \left(\frac{C_{tot}}{2C}\right)^2 \cdot \frac{\beta \cdot \omega_u}{4} \cdot (1 + n_t) \quad (6-72)$$

The unity-gainbandwidth, ω_u , is determined by the transconductance, g_m , and the total capacitive load on the opamp output, C_{load} . Hence we have

$$\begin{aligned} v_{n,\phi_2}^2 &= \frac{16kT}{3g_m} \cdot \left(\frac{C_{tot}}{2C}\right)^2 \cdot \frac{\beta \cdot \frac{g_m}{C_{load}}}{4} \cdot (1 + n_t) = \\ &= \frac{16}{3}kT \cdot \left(\frac{2^N + 1}{2}\right)^2 \cdot \frac{2}{4C_{load}} \cdot (1 + n_t) = \frac{2kT}{3C_{load}} \cdot (2^N + 1) \cdot (1 + n_t) \end{aligned} \quad (6-73)$$

The total noise at the output is the sum of the noise contributions in both the clock phases, i.e.

$$v_{tot}^2 = v_{n,\phi_1}^2 + v_{n,\phi_2}^2 \approx \left(\frac{kT}{C} \cdot 2^{N-2} + \frac{2kT}{3C_{load}} \cdot (2^N + 1) \cdot (1 + n_t)\right) \quad (6-74)$$

If the load capacitance is in the same order as the sampling capacitor, i.e. $C_{load} \approx 2^N C$ the noise is, at least for large N , dominated by v_{n,ϕ_1}^2 , i.e. the kT/C -noise in the sampling capacitor. If this is the case the input referred noise of the MDAC can be calculated by dividing the output referred noise by the MDAC gain squared, i.e.

$$v_{tot,in}^2 \approx \frac{kT}{C} \cdot \frac{2^{N-2}}{(2^N - 1)^2} = \frac{kT}{2^N C} \approx \frac{kT}{C_{tot}} \quad (6-75)$$

It is interesting to note that the total noise only depends on the capacitors in the circuit, not g_m . It should also be noted that the above expressions represent the total noise power. If the output of MDAC is sampled by another SC circuit all the noise power is folded into the Nyquist band. However if the circuit is used as a DAC and the analog wave form is not sampled, the noise power generated during clock phase ϕ_2 is evenly distributed up to the noise bandwidth. Since the noise bandwidth usually is much larger than the sampling frequency, only a small part of the opamp noise power falls in the signal band.

6.8 SI INTEGRATOR

The integrator is an important part of oversampled sigma-delta converters. In this section we briefly investigate fundamental limitations on speed and noise of SI integrators. A more detailed analysis on SI oversampled ADCs can be found in [13]. A

lossless integrator using the SI technique is shown in Fig. 6-18. The z-domain transfer

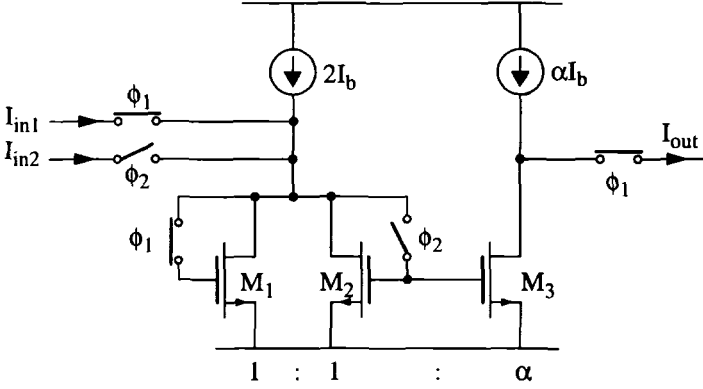


Figure 6-18 Noise sources in clock phase 2.

function of the integrator is given by [13]

$$I_{out}(z) = \frac{\alpha}{1 - z^{-1}} \cdot (z^{-1} \cdot I_{in1} - z^{-1/2} \cdot I_{in2}) \quad (6-76)$$

The gain of the integrator, α , is controlled by the dimension ratio of transistor M_2 and M_3 while the sign is determined by how the input current is switched.

6.8.1 Speed

The bandwidth of the circuit in Fig. 6-18 is different in the two clock phases. On clock phase ϕ_1 the bandwidth is determined by transistor M_1 , i.e.

$$\omega_{-3dB} = \frac{g_m}{C_{gs}} \quad (6-77)$$

where g_m is the transconductance of transistor M_1 and C_{gs} the gate-source capacitance of M_1 . On clock phase ϕ_2 the bandwidth is lower since transistor M_3 adds extra capacitive load. Assuming M_3 to be α times larger than M_1 and M_2 we have

$$\omega_{-3dB} = \frac{g_m}{C_{gs}(1 + \alpha)} \quad (6-78)$$

Thus the speed will be limited by (6-78). A small gain factor gives a higher bandwidth.

6.8.2 Noise

There are three noise sources due to transistors M_1 , M_2 and M_3 in the integrator. These noise sources will appear between the source and drain of the transistors as shown in Fig. 6-19. To simplify the analysis the two noise sources for M_1 and M_2 are added to

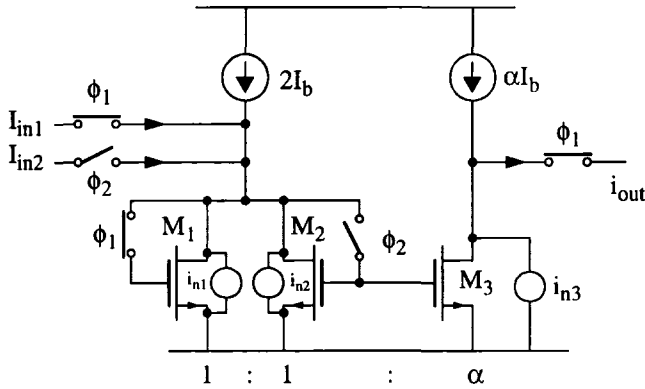


Figure 6-19 Noise sources in SI integrator.

one noise source i_n , where $i_n = i_{n1} + i_{n2}$. The effect of the noise source can now be derived as follows [11]. We assume that the input current is zero. The signal current through M_1 is denoted i_1 and the signal current through M_2 is denoted i_2 . The sampling period is T . The noise sources are here treated as sampled noise sources but they are actually continuously changing their output current. The noise currents are not sampled until at the end of each clock phase when the switches are opened. Strictly this would make the delay of the noise signal one half clock period less [11]. However, it has no influence on the results of the noise analysis. We start the analysis at time $t = (n-1)T$ and assume that the switches controlled by ϕ_1 are closed. Transistor M_1 is now diode-connected and its current is

$$i_1(n-1) = i_n(n-1) - i_2(n-1) \quad (6-79)$$

where $i_x(n-1)$ denotes a current at time $t = (n-1)T$.

During the next clock phase $t = (n-1/2)T$ and the above current is held in transistor M_1 , i.e.

$$i_1(n-1/2) = i_1(n-1) \quad (6-80)$$

while the current in transistor M_2 , now diode-connected, is

$$\begin{aligned} i_2(n-1/2) &= i_n(n-1/2) - i_1(n-1/2) = \\ &= i_n(n-1/2) - i_n(n-1) + i_2(n-1) \end{aligned} \quad (6-81)$$

In the following clock phase the current in M_2 is held and we have

$$i_2(n) = i_2(n-1/2) \quad (6-82)$$

This current is mirrored to M_3 and sent to the following stage and we have the total noise current at the output as

$$i_{out}(n) = \alpha \cdot i_2(n) + i_3(n) \quad (6-83)$$

By using the z-transform on the above equations we get the z-domain output current as

$$i_{out}(z) = \frac{\alpha \cdot i_n(z) \cdot (z^{-1/2} - z^{-1})}{1 - z^{-1}} + i_3(z) \quad (6-84)$$

The contribution from i_n has a zero at $z = 1$. This means that low frequency noise as $1/f$ noise from the transistors is attenuated in the circuit. If the noise is AWGN all samples are independent and the noise contributions from the two clock phases are added as powers. We can now write the output noise as

$$i_{out}(z) = \frac{\alpha \cdot i_{n,tot}(z)}{1 - z^{-1}} + i_3(z) \quad (6-85)$$

where $i_{n,tot}(z)$ is the total noise contribution from $\mathbf{M}_1$ and $\mathbf{M}_2$. If the noise is AWGN the total noise power of $i_{n,tot}(z)$ is

$$i_{n,tot}^2 = i_{n,\phi_1}^2 + i_{n,\phi_2}^2 \quad (6-86)$$

i_{n,ϕ_1}^2 is the noise power in clock phase ϕ_1 and i_{n,ϕ_2}^2 is the noise power in clock phase ϕ_2 . These noise power can be calculated as the product of the spectral density and the noise bandwidth, i.e.

$$i_{n1}^2 = \frac{8}{3} \cdot kT \cdot g_m \cdot BW_{noise} = \frac{8}{3} \cdot kT \cdot g_m \cdot \frac{\omega_{-3dB}}{4} = \frac{2}{3} \cdot kT \cdot \frac{g_m^2}{C_{gs}} \quad (6-87)$$

and

$$i_{n2}^2 = \frac{2}{3} \cdot kT \cdot \frac{g_m^2}{(1 + \alpha)C_{gs}} \quad (6-88)$$

Hence the noise power is

$$i_{n,tot}^2 = \frac{2}{3} \cdot kT \cdot \frac{g_m^2}{C_{gs}} \cdot \left(1 + \frac{1}{\alpha}\right) \quad (6-89)$$

The noise power generated by $\mathbf{M}_3$ depends on the bandwidth of the load connected to the output. It is given by

$$i_{n3}^2 = \frac{2}{3} \cdot kT \cdot g_m \cdot \alpha \cdot BW_n \quad (6-90)$$

where BW_n is the noise bandwidth of the load at the output of the integrator. It is convenient to refer the noise to the input. This is achieved by dividing the expression in (6-85) with the z-domain transfer function of the integrator (see (6-76)). The input current noise would be

$$i_{in}(z) = i_{n,tot}(z) + \frac{i_3(z)}{\alpha} \cdot (1 - z^{-1}) \quad (6-91)$$

The noise from M_3 is high-pass filtered. If the integrator is used in an oversampled data converter the noise from M_3 can be neglected. It is interesting to calculate the corresponding input referred noise voltage power since it can be directly compared to the maximum voltage swing to get an upper limit in the SNR. We get the noise voltage by dividing the current noise by the g_m^2 . If the contribution from M_3 is neglected we get

$$v_{n,in}^2 = \frac{i_{n,tot}^2}{g_m^2} = \frac{2}{3} \cdot kT \cdot \frac{1}{C_{gs}} \cdot \left(1 + \frac{1}{\alpha}\right) \quad (6-92)$$

Again we see that the voltage noise power is only dependent on the sampling capacitor and not the transconductance of the memory transistors.

6.9 SC INTEGRATOR

In Fig. 6-20. we show a lossless SC integrator configuration that is widely used in

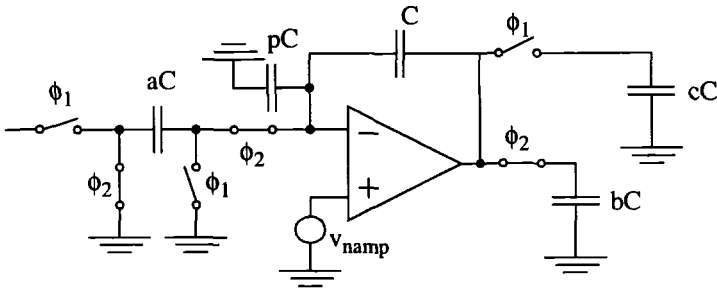


Figure 6-20 SC Integrator.

oversampling ADCs. The transfer function of this integrator is (assuming infinite opamp gain)

$$H(z) = \frac{a \cdot z^{-1}}{1 - z^{-1}} \quad (6-93)$$

It should be noted that the sampling capacitor of the following stage, here represented by bC and cC can be clocked on either ϕ_1 or ϕ_2 . By clocking on ϕ_1 the speed of the circuit is slightly larger since the load in the integration phase is smaller.

6.9.1 Speed

The speed of the circuit is limited by the bandwidth in both clock phases. Normally

the integration phase is most critical. The -3dB bandwidth is determined by the unity gain frequency of the opamp and the feedback factor. The feedback factor on ϕ_1 is given by

$$\beta_1 = \frac{C}{pC + C} = \frac{1}{p + 1} \quad (6-94)$$

while on ϕ_2 it is given by

$$\beta_2 = \frac{C}{pC + aC + C} = \frac{1}{a + p + 1} \quad (6-95)$$

For a single-stage opamp the unity-gain frequency is

$$\omega_u = \frac{g_m}{C_{load}} \quad (6-96)$$

The load capacitance on ϕ_1 is given by

$$C_{load,1} = cC + \beta_1 \cdot p \cdot C \quad (6-97)$$

while on ϕ_2 it is

$$C_{load,2} = bC + \beta_2 \cdot (a + p) \cdot C \quad (6-98)$$

Thus the -3dB bandwidth on ϕ_1 is

$$\omega_{-3dB,1} = \omega_u \cdot \beta_1 = \frac{g_m}{(c \cdot (1 + a) + p) \cdot C} \quad (6-99)$$

and on ϕ_2

$$\omega_{-3dB,2} = \omega_u \cdot \beta_2 = \frac{g_m}{(b \cdot (a + p + 1) + (a + p)) \cdot C} \quad (6-100)$$

We can now use (6-3) to find the relative settling errors in the two clock phases as

$$\epsilon_{r,1} = e^{-\omega_{-3dB,1} \cdot t_s} = e^{-\frac{g_m}{(c \cdot (1 + a) + p) \cdot C} \cdot t_s} \quad (6-101)$$

and

$$\epsilon_{r,2} = e^{-\omega_{-3dB,2} \cdot t_s} = e^{-\frac{g_m}{(b \cdot (a + p + 1) + (a + p)) \cdot C} \cdot t_s} \quad (6-102)$$

where t_s is the available settling time and ϵ_r is the relative settling error. The settling in the integration phase is usually most critical. Ideally all the charge on the input capacitor aC should be transferred to the integration capacitor C . Due to the finite bandwidth only a part of the charge will be transferred. The remaining portion of the

charge on aC is determined by the relative settling error. Hence the transfer function of the integrator is

$$H(z) = \frac{(1 - \epsilon_{r,2}) \cdot a \cdot z^{-1}}{1 - z^{-1}} \quad (6-103)$$

Thus we can conclude that the finite bandwidth introduces a gain error in the integrator.

6.9.2 Finite Opamp Gain

We assume that the DC gain of the opamp is A_0 , the parasitic capacitance at the input node of the opamp is pC . In the first clock phase ($t = (n-1)T$) capacitor aC is connected to the input voltage of the circuit, making the total charge on the top plates of the capacitors

$$q_1 = V_{in}(n-1) \cdot aC + V_{out}(n-1) \cdot C + \frac{V_{out}(n-1)}{A_0} \cdot (pC + C) \quad (6-104)$$

where $V_x(n-1)$ denotes a voltage at time $t = (n-1)T$. In the following clock phase ($t = (n-1/2)T$) the total charge is

$$q_2 = V_{out}\left(n - \frac{1}{2}\right) \cdot C + \frac{V_{out}\left(n - \frac{1}{2}\right)}{A_0} \cdot (pC + C + aC) \quad (6-105)$$

The total charge is conserved and the output signal at $t = (n-1/2)T$ can be calculated as

$$q_1 = q_2 \Rightarrow V_{out}\left(n - \frac{1}{2}\right) = \frac{V_{in}(n-1) \cdot aC + \left(\frac{pC + C}{A_0} + 1\right) \cdot V_{out}(n-1)}{C + \frac{pC + C + aC}{A_0}} \quad (6-106)$$

In the following clock phase the output is held by capacitor C , i.e.

$$V_{out}(n) = V_{out}\left(n - \frac{1}{2}\right) \quad (6-107)$$

By using (6-106) and (6-107) the z-domain transfer function can be calculated to be

$$H(z) = \frac{z^{-1} \cdot a}{\left(1 + \frac{p+1+a}{A_0} - \left(\frac{p+1}{A_0} + 1\right) \cdot z^{-1}\right)} \quad (6-108)$$

The expression can be rewritten as [14]

$$H(z) = \frac{r_2 \cdot z^{-1} \cdot a}{1 - \frac{r_2}{r_1} \cdot z^{-1}} \quad (6-109)$$

where

$$r_1 = \frac{\beta_1 \cdot A_0}{1 + \beta_1 \cdot A_0} = \frac{1}{\left(\frac{p+1}{A_0} + 1\right)} \quad (6-110)$$

$$r_2 = \frac{\beta_2 \cdot A_0}{1 + \beta_2 \cdot A_0} = \frac{1}{\left(\frac{p+1+a}{A_0} + 1\right)}$$

Hence the finite opamp gain will cause a gain error and introduce leakage in the integrator.

6.9.3 Effect of Finite Bandwidth and Finite DC Gain

In the previous two sections the transfer function of the integrator was derived considering finite opamp gain and finite bandwidth separately. If both these are taken into account simultaneously the derivation gets slightly more complicated. Using a single pole model of the integrator the transfer function is given by [14]

$$H(z) = \frac{r_2 \cdot (1 - \epsilon_{r,2}) \cdot z^{-1}}{1 - \frac{r_2}{r_1} \cdot \left(1 - \epsilon_{r,2} \cdot \left(1 - \frac{r_1}{r_2}\right)\right) \cdot z^{-1}} \quad (6-111)$$

where $\epsilon_{r,2}$ is the relative settling error on clock phase ϕ_2 . According to Sec. 6.6.2 $\epsilon_{r,2}$ is given by

$$\epsilon_{r,2} = e^{-\omega_{-3dB,2} \cdot t_s} = e^{-\frac{g_m}{(b \cdot (a+p+1) + (a+p)) \cdot C} \cdot t_s} \quad (6-112)$$

We conclude that when the finite opamp gain is included the finite bandwidth will introduce both gain errors and integrator leakage.

6.9.4 Noise

In Sec. 6.7 the noise of S/H amplifiers were derived. The noise in the integrator can be calculated in a similar way. The z-domain output noise from the opamp noise source (v_{namp} , see Fig. 6-20) can be derived based on charge conservation. We assume in the following the opamp gain is infinite and we neglect the influence of finite bandwidth in the opamp on the transfer function. On clock phase ϕ_1 ($t = (n-1)T$) we have the total charge

$$q_1 = V_{out}(n-1) \cdot C - v_{namp}(n-1) \cdot (pC + C) \quad (6-113)$$

where $V_x(n-1)$ denotes a voltage at time $t = (n-1)T$. In the following clock phase ($t = (n-1/2)T$) the total charge is

$$q_2 = V_{out}\left(n - \frac{1}{2}\right) \cdot C - v_{namp}\left(n - \frac{1}{2}\right) \cdot (pC + C + aC) \quad (6-114)$$

The total charge in the two phases is conserved and the output signal at $t = (n-1/2)T$ can be calculated as

$$\begin{aligned} q_1 = q_2 \Rightarrow \\ V_{out}\left(n - \frac{1}{2}\right) &= V_{out}(n-1) - v_{namp}(n-1) \cdot (p+1) + \\ &+ v_{namp}\left(n - \frac{1}{2}\right) \cdot (p+1+a) \end{aligned} \quad (6-115)$$

In the following clock phase the output is held by capacitor C , but we also have a noise contribution from the noise source. The total charge on C and pC is conserved, i.e.

$$V_{out}\left(n - \frac{1}{2}\right) - v_{namp}\left(n - \frac{1}{2}\right) \cdot (p+1) = V_{out}(n) - v_{namp}(n) \cdot (p+1) \quad (6-116)$$

We can now get an expression for the output signal at $t = nT$ as

$$\begin{aligned} V_{out}(n) &= V_{out}\left(n - \frac{1}{2}\right) + v_{namp}(n) \cdot (p+1) - v_{namp}\left(n - \frac{1}{2}\right) \cdot (p+1) \\ &= V_{out}(n-1) + (v_{namp}(n) - v_{namp}(n-1)) \cdot (p+1) + \\ &+ a \cdot v_{namp}\left(n - \frac{1}{2}\right) \end{aligned} \quad (6-117)$$

The z-domain transfer function is

$$V_{out}(z) = z^{-1} \cdot v_{namp,1}(z) \cdot (1+p) + v_{namp,2}(z) \cdot \frac{z^{-1/2}}{1-z^{-1}} \cdot a \quad (6-118)$$

where $v_{namp,1}(z)$ is the noise contribution on ϕ_1 and $v_{namp,2}(z)$ is the contribution on ϕ_2 . The noise can be referred to the input by dividing the output noise by the transfer function of the integrator, (6-93), i.e.

$$v_{nin}(z) = v_{namp,1}(z) \cdot \frac{(p+1)}{a} \cdot \frac{(1-z^{-1})}{z^{-1}} + v_{namp,2}(z) \cdot z^{1/2} \quad (6-119)$$

If the signal frequency is much lower than the sampling frequency, which is the case in an oversampled sigma-delta modulator, the second term in (6-119) will dominate since the first term is high-pass filtered. Assuming a single-stage opamp the input

referred noise is (see Sec. 6.7)

$$\begin{aligned}
 v_{nin}^2 &= \frac{4}{3} \cdot \frac{kT}{C_{load}} \cdot \beta \cdot (1 + n_t) \\
 &= \frac{4}{3} \cdot \frac{kT}{bC + \frac{(a+p)}{a+p+1} \cdot C} \cdot \frac{1}{a+p+1} \cdot (1 + n_t) \\
 &= \frac{4}{3} \cdot \frac{kT \cdot (1 + n_t)}{(b(a+p+1) + a+p)C}
 \end{aligned} \tag{6-120}$$

where n_t is a factor taking the noise contributions from the current sources in the opamp into account. In the above analysis only the opamp noise was considered. The kT/C noise from the passive sampling should also be considered. The total noise would then be

$$v_{nin,tot}^2 = \frac{4}{3} \cdot \frac{kT \cdot (1 + n_t)}{(b(a+p+1) + a+p)C} + \frac{kT}{C} \tag{6-121}$$

Again it is interesting to note that the noise only depends on the capacitor sizes and not the transconductance of the input device. If the noise contribution factor, n_t , is small and the load capacitance at the output is large, the kT/C -noise from the switch resistance will dominate.

6.10 SUMMARY

In this chapter we have discussed two important analog functional blocks in data converters, the MDAC and the integrator. Both SC and SI circuits were considered. We showed how to calculate the noise and maximum speed of the circuits based on first order models. The maximum speed of the circuits is determined by the bandwidth and the required number of clock phases. The maximum speed normally decreases as the gain of the circuit increases. The thermal noise of the circuits can be referred to the input as a noise voltage. We showed that the noise voltage power for all the circuits is only determined by the capacitors in the circuit.

The results in this chapter will be used in chapter 9 and chapter 11 to investigate the effect of circuit imperfections on pipelined and oversampled sigma-delta converters respectively.

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7 BASIC ANALOG CIRCUIT DESIGN

7.1 INTRODUCTION

For SC data converters, opamps and voltage comparators are the fundamental building blocks. For SI data converters, SI memory circuits and current comparators are the fundamental building blocks. We have already touched upon SI memory circuits in the previous chapters. In this chapter we will concentrate on opamps and comparators. For most data converter products, references are usually required to be generated on the same chip. We will also discuss voltage and current references in this chapter.

Following this introduction, we will discuss opamps in Sec. 7.2, with emphasize on high-speed design. Only fully differential architectures will be covered since high performance SC data converters usually use fully differential opamps. In Sec. 7.3, we will discuss voltage comparators. Current comparators are briefed in Sec. 7.4. References will be discussed in Sec. 7.5.

7.2 OPERATIONAL AMPLIFIERS

We can categorize opamps into single-stage and multi-stage architectures. Multi-stage opamp architectures are not well-suited for high-speed high-accuracy SC applications due to the compensation needed to improve the phase margin. Non-accurate compensation may introduce doublets with frequencies lower than the closed-loop -3-dB bandwidth. Low-frequency doublets are hazard for achieving the high-accuracy and high-speed settling. Therefore, we focus us on single-stage opamp architectures. However with a reduced supply voltage, we have to resort to multi-stage architecture due to the limited voltage head room. Two-stage opamps will be covered in the following chapter for low-voltage operation.

There are three popular architectures for single-stage opamps. They are the current-mirror based, folded-cascode, and telescopic architectures. In comparing the architectures, we only focus on speed, noise, and power. We do not consider the DC gain since it can be effectively improved by using the gain boosting technique which will be discussed in this chapter as well.

7.2.1 Slew Rate vs. Unity-Gain Bandwidth

If a single-stage opamp with only one dominant pole is used in a close loop, the output

setting responding to a step input is given [2] by

$$V_{out}(t) = V_{step} \cdot (1 - e^{-t \cdot \beta \cdot \omega_u}) \quad (7-1)$$

where V_{step} is the input step size, β is the feedback factor of the closed loop, and ω_u is the unity-gain bandwidth of the opamp.

To avoid slewing, the slew rate must be larger than the maximum slope at the output, i.e.,

$$SR \geq \left. \frac{d}{dt} V_{out}(t) \right|_{max} = V_{step} \cdot \beta \cdot \omega_u \quad (7-2)$$

Without slewing in the opamp, the non-complete settling only introduces a linear error. However, for most practical opamps such as the ones to be discussed below, the slew rate and the unity-gain bandwidth are closely related. We cannot simply increase the slew rate and the unity-gain bandwidth independently. Higher slew rate usually makes the unity-gain bandwidth higher. For SC data converter applications, the step size, i.e., the difference between two successive values can be very high. Therefore, it is almost practically impossible to avoid slewing in the opamp unless the gate voltage of the input transistor is excessively large or the feedback factor is very small. (This will become evident in the following discussions.) Therefore, we have to guarantee the settling accuracy. Any non-complete settling will introduce distortion, not just a linear error in some practical SC data converters.

7.2.2 Current-Mirror Based Opamp

In Fig. 7-1 we show the schematic of the current-mirror based single-stage opamp [1].

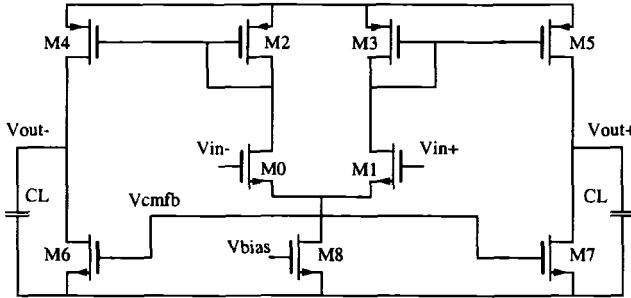


Figure 7-1 Current-mirror based single-stage opamp.

The unity gain bandwidth is given by

$$\omega_u = \frac{n \cdot g_{m0}}{C_L} \quad (7-3)$$

where g_{m0} is the transconductance of the input device M0 (M1), C_L is the capaci-

tive load and n is the dimension ratio of the current mirrors M2 and M4 (M3 and M5).

The first non-dominant pole is due to the current mirror, and is given by

$$\omega_n \approx \frac{g_{m2}}{C_{gs2} + C_{gs4} + C_{gd4} + C_{gd0} + C_{b2}} \quad (7-4)$$

where g_{m2} is the transconductance of transistor M2 (M3), C_{gs2} is the gate-source parasitic capacitance of transistor M2 (M3), C_{gs4} is the gate-source parasitic capacitance of transistor M4 (M5), C_{gd0} is the gate-drain parasitic capacitance of transistor M0 (M1), and C_{b2} is the total substrate capacitance at the drain of transistor M2 (M3). (Notice that the Miller effect associated with transistor M4 (M5) is not considered here in that practical opamps use a cascode output stage to increase the DC gain.)

The slew rate is given by

$$SR = \frac{2 \cdot I_{bias6}}{C_L} = \frac{n \cdot I_{bias8}}{C_L} \quad (7-5)$$

where I_{bias6} is the quiescent bias current in transistor M6 (M7), and I_{bias8} is the bias current in transistor M8. The factor of 2 is due to the fully differential structure. The maximum slewing current is I_{bias6} for one side while the maximum slewing current is $I_{bias5} = I_{bias6}$ for the other side.

The transconductance of an MOS transistor in the saturation region is given [2] by

$$g_m = \sqrt{2 \cdot \mu \cdot C_{ox} \cdot \frac{W}{L} \cdot I_D} = \frac{2 \cdot I_D}{V_{eff}} \quad (7-6)$$

where μ is the charge mobility, C_{ox} is the gate capacitance per unit area, W/L is the transistor dimension, I_D is the drain current, and V_{eff} is the effective gate voltage (the difference between the gate-source voltage and the threshold voltage). To avoid slewing governed by equation (7-2), we have

$$\frac{V_{eff}}{V_{step}} \geq \beta \quad (7-7)$$

This condition cannot be met in data converters such as oversampling ADCs and pipeline ADCs with low gain stages. For instance, V_{step} (differential) can be as high as 4~6 V, with a feedback factor of 0.5, the effective gate voltage of the input devices must be in the neighborhood of 2~3V. This is too high in most practical designs where the effective gate voltage is designed to be ca 200~500 mV within the whole operation range and for all the process corners. Therefore, slewing will occur in data converter applications. We have to guarantee the settling accuracy in order not to introduce distortion. Both the slew rate and the unity-gain bandwidth must be high to increase the settling accuracy. Increasing n (without increasing the input bias current) seems to increase both the slew rate and the unity gain bandwidth, but it decreases the non-dominant pole frequency, reducing the phase margin. We usually have to increase both the

input and output currents to improve the settling.

To have a large phase margin for monotonic settling, we require that the non-dominant pole must be much larger than the unity-gain bandwidth. Since the gate-source capacitance is much larger than the gate-drain capacitance in a saturated MOS transistor, the non-dominant pole is considerably lower than the non-dominant pole in the folded-cascode opamp to be discussed. The lower non-dominant pole is one of the reasons for the rejection of the current mirror based opamp for high-speed applications.

For high performance data converters, the thermal noise generated by the opamp could be another limiting factor besides the kT/C noise. The input referred noise power spectral density of an MOS transistor in the saturation region is given [2] by

$$S(f)|_{MOS} = \frac{8}{3} \cdot k \cdot T \cdot \frac{1}{g_m} \quad (7-8)$$

where k is Boltzmann constant, T is the Kelvin temperature, and g_m is the transconductance.

In the current-mirror based opamp, all the transistors except the tail bias current transistor M8 contribute to the total noise. The noise introduced by transistor M8 is only a common-mode noise seen by the input pair, having no impact on the differential signals. Therefore, the input referred noise spectral density of the current-mirror based opamp is given by

$$\begin{aligned} S(f) &= \frac{16}{3} \cdot k \cdot T \cdot \frac{1}{g_{m0}} \cdot \left\{ 1 + \frac{g_{m2}}{g_{m0}} + \frac{g_{m4} + g_{m6}}{g_{m0}} \cdot \frac{1}{n^2} \right\} \\ &\approx \frac{16}{3} \cdot k \cdot T \cdot \frac{1}{g_{m0}} \cdot \left\{ 1 + \frac{g_{m2}}{g_{m0}} \cdot \left(1 + \frac{1}{n} \right) + \frac{g_{m6}}{g_{m0}} \cdot \frac{1}{n^2} \right\} \end{aligned} \quad (7-9)$$

where g_{m0} , g_{m2} , g_{m4} , and g_{m6} are the transconductance of transistor M0, M2, M4, and M6, respectively, and n is the current mirror ratio. It is seen that by increasing the current mirror ratio, we can reduce the noise contribution from the output devices M4 and M6. It also increases the unity gain bandwidth according to equation (7-3), but it decreases the non-dominant pole frequency according to equation (7-4). The phase margin, and therefore the settling time degrades significantly with a large current mirror ratio n .

7.2.3 Folded-Cascode Opamp

In Fig. 7-2 we show the folded-cascode opamp [1]. The unity-gain bandwidth is given by

$$\omega_u = \frac{g_{m0}}{C_L} \quad (7-10)$$

where I_{bias8} is the quiescent bias current in transistor M8. To avoid slewing governed by equation (7-2), we have

$$\frac{V_{eff}}{V_{step}} \geq \beta \quad (7-13)$$

This condition can hardly be met in some data converters such as oversampling ADCs and pipelined ADCs with low gain stages. Therefore, we have to guarantee the settling accuracy in order not to introduce distortion due to the inevitable slewing in these applications. For high-speed applications, we need both a high unity-gain bandwidth and a high slew rate. This means that we need to bias both the input and output devices at a high current and therefore the architecture is not power-efficient just as the current-mirror based opamp.

Since the impedance at the sources of cascode transistors is low, the noise contributions from the cascode transistors can be neglected. Therefore, the input referred noise spectral density of the folded-cascode opamp is given by

$$S(f) = \frac{16}{3} \cdot k \cdot T \cdot \frac{1}{g_{m0}} \cdot \left\{ 1 + \frac{g_{m2} + g_{m6}}{g_{m0}} \right\} \quad (7-14)$$

where g_{m0} is the transconductance of the input device M0 (M1), g_{m4} is the transconductance of transistor M4 (M5), g_{m6} is the transconductance of transistor M6 (M7). In order to reduce the noise, we have to minimize the noise contribution from the output devices M4(M6).

7.2.4 Telescopic Opamp

In order to get rid of PMOS transistors in the signal path, the telescopic opamp [3] shown in Fig. 7-3 can be used. The unity gain bandwidth is given by

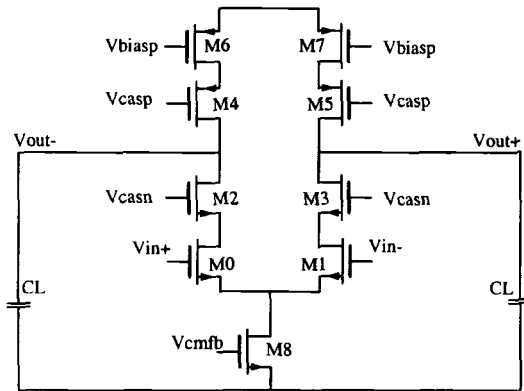


Figure 7-3 Telescopic single-stage opamp.

$$\omega_u = \frac{g_{m0}}{C_L} \quad (7-15)$$

where g_{m0} is the transconductance of the input device M0 (M1), C_L is the capacitive load.

The first non-dominant pole is created by the cascode transistor M2 (M3), and is given by

$$\omega_n \approx \frac{g_{m2}}{C_{gs2} + C_{gd0} + C_{b2}} \quad (7-16)$$

where g_{m2} is the transconductance of transistor M2 (M3), C_{gs2} is the gate-source parasitic capacitance of transistor M2 (M3), C_{gd0} is the gate-drain parasitic capacitance of transistor M0 (M1), and C_{b2} is the total substrate capacitance at the source of transistor M2 (M3).

Due to the use of NMOS transistors in the signal path and less parasitic capacitance at the source of the cascode transistors, the non-dominant pole is higher than that in the current-mirror based opamp and in the folded-cascode opamp.

The slew rate is given by

$$SR = \frac{I_{bias8}}{C_L} \quad (7-17)$$

where I_{bias8} is the quiescent bias current in transistor M8. To avoid slewing governed by equation (7-2), we have

$$\frac{V_{eff}}{V_{step}} \geq \beta \quad (7-18)$$

This condition hardly be met in some data converters. For instance, V_{step} (differential) can be as high as 4V, with a feedback factor of 0.5, the effective gate voltage of the input devices must be in the neighborhood of 2V. This is too high in most practical designs where the effective gate voltage is designed to be ca 200~500 mV within the whole operation range and for all the process corners. Therefore, the slewing will occur in some applications. We have to guarantee the settling accuracy in order not to introduce distortion. By increasing the bias current, we can increase both the slew rate and the unity-gain bandwidth.

The input referred noise spectral density is given by

$$S(f) = \frac{16}{3} \cdot k \cdot T \cdot \frac{1}{g_{m0}} \cdot \left\{ 1 + \frac{g_{m6}}{g_{m0}} \right\} \quad (7-19)$$

where g_{m0} is the transconductance of the input device M0(M1), and g_{m6} is the

transconductance of transistor M6 (M7).

Besides the advantages of having a higher non-dominant pole, the telescopic opamp is more power efficient, since no extra currents are needed other than the bias current for the input devices to provide a high slew rate and a high unity-gain bandwidth. The large bias current improves both the slew rate and the unity-gain bandwidth as well reduces the thermal noise.

There is a misconception that there is a drawback due to the three NMOS transistors in series. This makes the output signal swing small. However, for most applications, we only care about the SNR. With the reduced signal swing, we can increase the capacitors to maintain the SNR.

Assume that the average of the maximum source-drain voltage for every transistor is ca 400 mV across process corner and temperature range. We also assume that the supply voltage is 5 V. For the current-mirror based (usually have cascode transistors in the output stage) or the regulated cascode opamps, the fully differential output signal swing is $2 \times (5 - 4 \times 0.4) = 6.8 \text{ V}_{p-p}$. For the telescopic opamp, there is one more cascode transistors. Therefore the output signal swing becomes $2 \times (5 - 5 \times 0.4) = 6 \text{ V}_{p-p}$. By using the telescopic opamp, the signal swing is reduced by 6.8/6. In order to keep the same SNR in a kT/C limited environment, we have to increase the capacitor to $(6.8/6)^2$, i.e., by 28%. In order to keep the speed, we have to increase the input transconductance g_m and slew rate by 28% and therefore we need to increase the bias current by 28% to 64%. (This large range is due to the fact that we can increase the bias current and the transistor size at the same time or only increase the bias current.) From a pure power dissipation's view point, the telescopic opamp still consumes 18% to 36% less power than other types of opamps. (It is usually a common practice for high speed design that both input and output have comparable bias currents for the current-mirror based and the regulated cascode opamps.) Besides the power advantage, the telescopic opamp is usually faster due to the higher slew rate and the higher non-dominant pole and it is less noisy due to the fewer noise-contributing transistors. Therefore, for SC applications, telescopic opamps should be the top choice.

To utilize the advantages of the telescopic opamp, we need to use different common-mode voltages for the input and output. This can be easily achieved by using a level shifter, either SC or continuous-time.

7.25 Gain Boosting Technique

As discussed in previous chapters, we need a very high DC gain in the opamp for high accuracy applications. For example, for a 12-bit settling in an SC amplifier with a gain factor of 2, we need a DC gain over 85 dB. For high-speed applications, short channel transistors are used in opamps which result in a DC gain less than 60 dB. Gain boosting techniques must be employed.

The most popular gain boosting technique is to use a cascode output stage and the gain of the cascode transistors is further increased by employing a feedback [4]. The gain boosting technique applied to the folded-cascode opamp is shown in Fig. 7-4. It can also be readily applied to the current-mirror based opamps and telescopic opamps

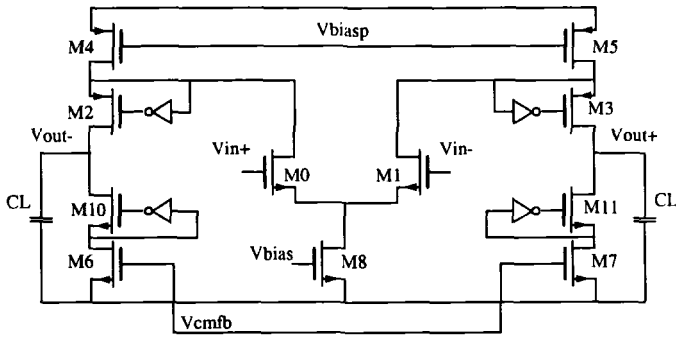


Figure 7-4 Folded-cascode opamp with gain boosting amplifiers.

without much modification. The DC gain is given by

$$A = \frac{g_{m0}}{\left(\frac{g_{ds0} \parallel g_{ds4}}{A_{M2} \cdot A_p} \right) \parallel \left(\frac{g_{ds6}}{A_{M10} \cdot A_n} \right)} \quad (7-20)$$

where g_{m0} is the transconductance of the input transistor M0 (M1), g_{ds0} is the output conductance of the input transistor M0 (M1), g_{ds4} is the output conductance of the current source transistor M4 (M5), g_{ds6} is the output conductance of the current source transistor M6 (M7), A_{M2} is the gain of the cascode transistor M2 (M3), A_{M10} is the gain of the cascode transistor M10 (M11), A_p is the gain of the gain boosting amplifier for the P-type cascode transistor M2 (M3), and A_n is the gain of the gain boosting amplifier for the N-type transistor M10 (M11).

With a gain for every transistor around 30-40 dB, it is fairly easy to achieve a DC gain of over 90 dB with the architecture shown in Fig. 7-4.

However, due to the gain boosting, a doublet is introduced. In order to have a single-pole settling, the doublet's frequency should be larger than the -3-dB bandwidth of the closed loop. For the stability concern, this doublet's frequency must be less than the non-dominant pole of the main opamp [1] [4]. Therefore, we have

$$\beta \cdot \omega_u < \omega_{gu} < \omega_n \quad (7-21)$$

where β is the feedback factor, ω_u is the unity-gain bandwidth of the opamp, ω_{gu} is the unity-gain bandwidth of the gain-boosting amplifier, and ω_n is the non-dominant pole frequency of the opamp. This equation can be easily guaranteed since the load capacitance of the gain-boosting amplifier is considerably smaller than the load capacitance of the main amplifier. Also due to the fact that much smaller capacitance loads the gain-boosting amplifiers, considerably smaller currents and smaller transistors can be used for the gain-boosting amplifiers.

7.2.6 Common-Mode Feedback

Fully differential circuits process signals fully-differentially without any control over the common-mode potential. It is the common-mode feedback circuit that keeps the common-mode potential stable. The generalized fully differential opamp with a CMFB [5] is shown in Fig. 7-5

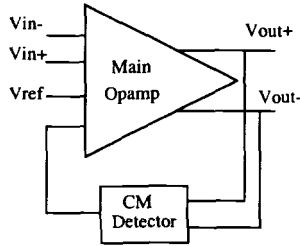


Figure 7-5 Fully differential opamp with CMFB.

The main opamp processes the fully differential input signals. The common-mode (CM) detector detects the common-mode voltage in the outputs. This detected voltage is compared to a desired common-mode reference voltage. Through the feedback action, the detected common-mode voltage is kept close to the common-mode reference voltage. There are several critical factors in designing CMFB circuits. The CM detector should not load the main opamp. The CM detector should have a larger input signal range. The interaction between the CMFB loop and the fully-differential signal processing should be minimized. The CMFB loop should be fast with a moderate to high gain. Therefore it is preferable to feed the CMFB signal to the output devices directly as indicated in Fig. 7-1 and Fig. 7-2

There are two main categories of CMFB circuits. One is the continuous-time and the other is the SC. In continuous-time CMFB circuits [2][5], current mirrors are needed. The extra poles introduced in the current mirrors (and other parts) in the continuous-time CMFB decrease the bandwidth and increase the settling time. Even when NMOS current-mirrors are used, the settling time still degrades if very large bias currents are not used to push the pole frequencies high. Besides the slow speed in continuous-time CMFB circuit, the input signal range of the CM detector is limited and the interference of the CMFB with the fully-differential signal processing is also high. It is therefore a much better choice to use an SC CMFB for high-speed high-accuracy SC circuits. Such an SC CMFB [6] is shown in Fig. 7-6.

Capacitors C_{cm} generate the average of the output voltages (V_{out+} and V_{out-}) of the opamp. The generated signal V_{cmfb} is used to control the current sources in the opamps (e.g., see Fig. 7-1 - Fig. 7-4). The DC voltage across C_{cm} is determined by capacitor C_s which is switched between a bias voltage V_b and the desired common-mode level (V_{cm}). This circuit acts as a simple SC low-pass filter having a DC output signal.

The use of SC CMFB only results in a little bit more capacitive load for the opamp.

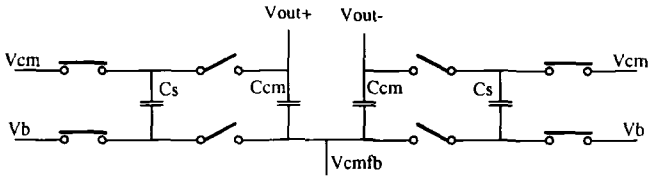


Figure 7-6 SC CMFB.

The bandwidth is determined by the transconductance of the current sources in the opamp which are controlled by V_{cmfb} and the capacitive load. It is therefore a good choice to use NMOS current source controlled by V_{cmfb} as shown in Fig. 7-1-Fig. 7-4 to increase the bandwidth of the CMFB.

Another advantage of using the SC CMFB is the freedom to choose a different common-mode voltage for the input and for the output. This is very important in order to fully utilize the advantages of the telescopic opamp that usually has a smaller input common-mode voltage than the output common-mode voltage.

7.3 VOLTAGE COMPARATORS

High-performance comparators need to amplify a small input voltage (or the difference between the input voltage and a reference voltage) to a level large enough to be detected by digital logic circuits within a very short time. Therefore, we need both high gain and high bandwidth in the comparators. Increase in the gain and in the bandwidth is usually contradictory. High gain yields low bandwidth, and vice versa.

The input to a comparator can also be very large and it overdrives the comparator. If the recovery time is too long, the accuracy of the next comparison will be influenced. The large positive gate voltage of an MOS transistor can also introduce the charge trapping mechanism where electrons are trapped via tunneling (electrons tunnel to oxide traps close to the conduction band). The release of them takes an excess time. Therefore, the voltages within a comparator needs to be clipped for high-speed comparison.

7.3.1 Amplifier-Type Comparator

The natural choice to design a comparator is to use an amplifier. A small voltage at the input is then amplified to a value large enough to be detected by the following digital logic circuits.

Suppose the amplifier consists of n stages each stage being a single-pole system as shown in Fig. 7-7.

The total gain is therefore given by

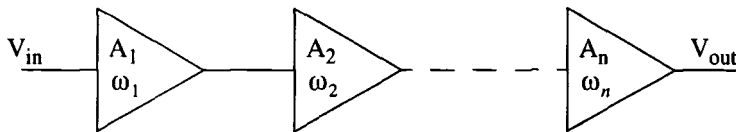


Figure 7-7 An amplifier type comparator.

$$A = A_1 \cdot A_2 \cdot \dots \cdot A_n = \prod_{i=1}^n A_i \quad (7-22)$$

where A_i is the gain of the i -th stage.

In a single-pole system, the settling time constant is related to the -3-dB bandwidth by

$$\tau_i = \frac{1}{\omega_i} \quad (7-23)$$

where τ_i is the settling-time constant of the i -th stage and ω_i is the -3-dB bandwidth of the i -th stage. And the open-loop -3-dB bandwidth is related to the unity gain bandwidth by

$$\omega_i = \frac{\omega_{ui}}{A_i} \quad (7-24)$$

where ω_{ui} is the unity-gain bandwidth of the i -th stage.

Therefore, the settling time constant of the cascaded amplifier can be approximated by

$$\tau = \sum_{i=1}^n \tau_i = \sum_{i=1}^n \frac{1}{\omega_i} = \sum_{i=1}^n \frac{A_i}{\omega_{ui}} \quad (7-25)$$

From equations (7-22) and (7-25) it is clear that it is better to cascade more low-gain stages than to use fewer high-gain stages in order to reduce the settling time.

Suppose that each stage has the same DC gain A_0 and the same unity-gain bandwidth ω_{u0} , equations (7-22) and (7-25) become

$$A = \prod_{i=1}^n A_i = A_0^n \quad (7-26)$$

and

$$\tau = \sum_{i=1}^n \frac{A_i}{\omega_{ui}} = \frac{n \cdot A_0}{\omega_{u0}} \quad (7-27)$$

If a single stage is used to achieve the same gain, the settling time constant would become A_0''/ω_{u0} , which is much larger than the settling time given by (7-27). This is why under no circumstance should an opamp be used as a high-performance comparator.

Suppose we need to design a comparator for a 12-bit accuracy, we need to amplify an LSB-voltage to a voltage that can be detected by digital logic circuits. If the LSB-voltage is assumed to be 0.1 mV and the voltage detectable by digital logic circuits is assumed to be 2 V, the small signal gain of the comparator must be larger than 86 dB and the comparator must accomplish the amplification within a given time interval. Recent study has shown that the comparator gain must be much larger than the above given value if metastability is considered in a pipeline A/D converter [7]. If we consider bit error rate, the gain has to be much larger if no positive feedback latch is used. Depending on the use of the comparator in the whole A/D system, the settling of the comparator might be coupled with the settling of the preceding gain amplifier (as in certain type of pipeline A/D converters). Therefore, we cannot assume that we can allocate half of the clock period for the comparator to settle as if it was driven by a voltage source. The settling time must be much shorter than half of the clock period. Even with a multiple stage design, it is difficult to achieve the performance needed for a 12-bit 50-MS/s A/D converter. (Notice that if the settling of comparison is not coupled with the settling of the preceding gain amplifier, the settling of auto-zeroing will be coupled in an SC pipeline A/D converter.)

7.3.2 Latch-Type Comparator

To have a fast comparator, we can use latch-type comparators. It consists of two phases. During the first phase (track mode), the output is reset and the input voltage is tracked, and during the second phase (latch mode), the output is toggled by using a positive feedback (we will show the circuit realization later on). Thanks to the positive feedback, a very fast comparison can be achieved. The time constant of the latch in the latch mode can be analyzed by using two back-to-back gain amplifiers as shown in Fig. 7-8.

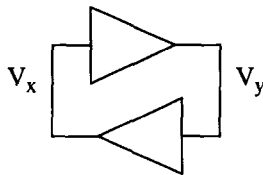


Figure 7-8 An latch type comparator in its latch mode.

Assume that the two amplifiers (single pole) are the same having an input transcon-

ductance g_m , output impedance R_{out} , and load capacitance C_L .

A linearized model gives that

$$g_m \cdot V_x + \frac{V_y}{R_{out}} = -C_L \cdot \frac{dV_y}{dt} \quad (7-28)$$

and

$$g_m \cdot V_y + \frac{V_x}{R_{out}} = -C_L \cdot \frac{dV_x}{dt} \quad (7-29)$$

Rearranging equations (7-28) and (7-29), we have

$$A \cdot V_x + V_y = -\tau \cdot \frac{dV_y}{dt} \quad (7-30)$$

and

$$A \cdot V_y + V_x = -\tau \cdot \frac{dV_x}{dt} \quad (7-31)$$

where A is the DC gain given by $A = g_m \cdot R_{out}$, and τ is the settling time constant of the amplifier given by $\tau = C_L \cdot R_{out} = A/\omega_u = 1/\omega_{-3dB}$, (where ω_u is the unity gain bandwidth and ω_{-3dB} is the -3-dB bandwidth).

Subtracting equation (7-30) from (7-31), we have

$$\Delta V = \frac{\tau}{A-1} \cdot \frac{d}{dt} \Delta V \approx \frac{\tau}{A} \cdot \frac{d}{dt} \Delta V = \omega_u \cdot \frac{d}{dt} \Delta V \quad (7-32)$$

where $\Delta V = V_x - V_y$.

Solving equation (7-32), we have

$$\Delta V = \Delta V_0 \cdot e^{\omega_u \cdot t} = \Delta V_0 \cdot e^{\frac{t}{\tau_l}} \quad (7-33)$$

where ΔV_0 is the initial voltage and τ_l is the settling time constant of the latch given by

$$\tau_l = \frac{1}{\omega_u} = \frac{\tau}{A} = \frac{C_L}{g_m} \quad (7-34)$$

Now it is seen that the settling time constant of the positive feedback is the settling time constant of the individual amplifier divided by the gain of the amplifier. By minimizing the load capacitance and increasing the input transconductance, a very fast toggling can be achieved.

In order to generate a voltage difference ΔV_{logic} , necessary to be processed by the following digital circuits, we find the time given by

$$T_{\text{latch}} = \tau_l \cdot \ln\left(\frac{\Delta V_{\text{logic}}}{\Delta V_0}\right) \quad (7-35)$$

If the time is larger than the allowed time for the latch phase, metastability occurs. Even the noise at the input can cause metastability [7].

7.3.3 MetalStability and Error Probability

A comparator has a limited time to settle. If at the end of the available time, the output of the comparator has a voltage that the following logic circuit cannot detect, metastability occurs. When the metastability occurs, the digital output is neither 1 nor 0. This introduces errors in the communication system. The error probability can be approximated by the LSB voltage and the minimum input voltage that the comparator can generate an output that the following digital circuit can generate a deterministic value 1 or 0.

By using equation (7-33), the minimum voltage needed to generate a voltage ΔV_{logic} to toggle the following digital circuit is given by

$$V_{\min} = \Delta V_{\text{logic}} \cdot e^{-\frac{t_0}{\tau_l}} \quad (7-36)$$

where t_0 is the maximum time settling time for the latch-type comparator and τ_l is the settling time constant given by equation (7-34).

If there is a pre-amplifier preceding the latch-type comparator, the error probability in a comparator due to metastability is given by

$$\begin{aligned} P(e) &= \frac{V_{\min}}{V_{\text{LSB}}} \cdot \frac{1}{A} = \frac{\Delta V_{\text{logic}}}{V_{\text{LSB}}} \cdot \frac{1}{A} \cdot e^{-\frac{t_0}{\tau_l}} \\ &\approx 2^n \cdot \frac{1}{A} \cdot e^{-\frac{t_0}{\tau_l}} \end{aligned} \quad (7-37)$$

where $V_{\min}$ is the minimum voltage at the comparator input, V_{LSB} is the LSB voltage, A is the pre-amplifier gain, n is the number of bits of the ADC, t_0 is the maximum time settling time for the latch-type comparator and τ_l is the settling time constant given by equation (7-34). If a two-phase clock is used, the settling time for the latch can be approximated by half of the clock period. We have the following approximation

$$P(e) = 2^n \cdot \frac{1}{A} \cdot e^{-\pi \cdot \frac{BW}{f_s}} \quad (7-38)$$

where BW is the -3dB bandwidth of the latch-type comparator during the latch phase and f_s is the sampling frequency of the ADC. The same equation holds true for an amplifier-type comparator. However its -3dB bandwidth is much smaller than the -3dB bandwidth of a latch-type comparator as can be seen by comparing equations (7-25) and (7-34). The -3dB bandwidth of a latch-type comparator is equal to the unity-gain bandwidth of the back-to-back gain amplifiers, while the -3dB bandwidth of an amplifier-type comparator is equal to its unity gain bandwidth divided by its DC gain.

In Fig. 7-9, we show the simulated error probability as a function of the ratio of the -

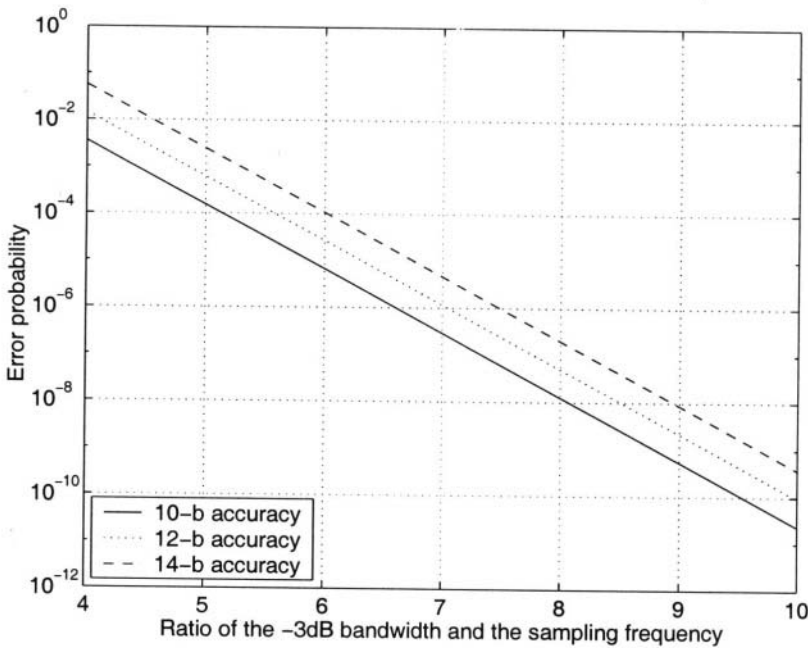


Figure 7-9 Simulated error probability vs. the ratio of the -3-dB bandwidth and the sampling frequency. The gain is assumed to be 1.

3-dB bandwidth and the sampling frequency. The pre-amplifier gain is assumed to be unity. It is seen that by increasing the bandwidth of the latch-type comparator, we can reduce the error probability drastically due to the exponential relationship.

In Fig. 7-10, we show the simulated error probability vs. the comparator gain. The -3-dB bandwidth is assumed to be 4 times larger than the sampling frequency. It is seen that it is not effective to reduce the error probability by increasing the gain due to the linear relationship. Therefore for high-speed ADCs, latch-type comparators are much preferred.

Notice that the error due to the metastability in a comparator does not necessarily introduce an error in the ADC, depending on coding and error correction. Also notice that an error in the ADC does not necessarily imply an error in the whole communication system, i.e., the bit error rate in the ADC does not necessarily impose a lower limit

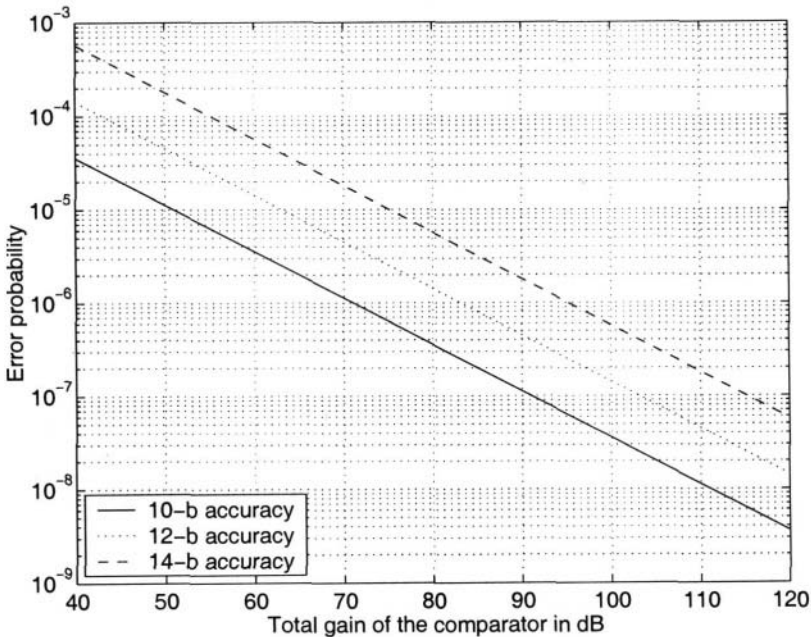


Figure 7-10 Simulated error probability vs. the comparator gain. The -3dB bandwidth is assumed to be 4 times larger than the sampling frequency.

on the bit error rate of the communications system, depending on the ADC architecture and the specific nature of the communication system. As a general rule, however, we should make the error probability in an comparator several order of magnitude lower than the required system bit error rate. For latch-type comparator, this can be easily achieved without much penalty in power dissipation due to the exponential relationship between the error probability and the -3-dB bandwidth.

7.3.4 Offset Cancellation Techniques

Depending on the A/D converter architecture, the offset within a comparator may limit the A/D converter performance. To reduce the offset, offset cancellation or auto-zeroing techniques need to be used. There are two major offset cancellation techniques [8]. One is based on input offset storage, and the other is based on output offset storage as shown in Fig. 7-11.

Both techniques need two non-overlapping clock phases as indicated in the figure. Each topology consists of a gain stage (pre-amplifier), an offset storage capacitor, and a second-stage comparator that could be a latch. With the input offset storage technique, the cancellation is performed by closing a unity-gain loop around the pre-amplifier and storing the offset on the input coupling capacitor. With the output offset storage technique, the offset is cancelled by short-circuiting the pre-amplifier inputs and storing the amplified offset on the output coupling capacitor.

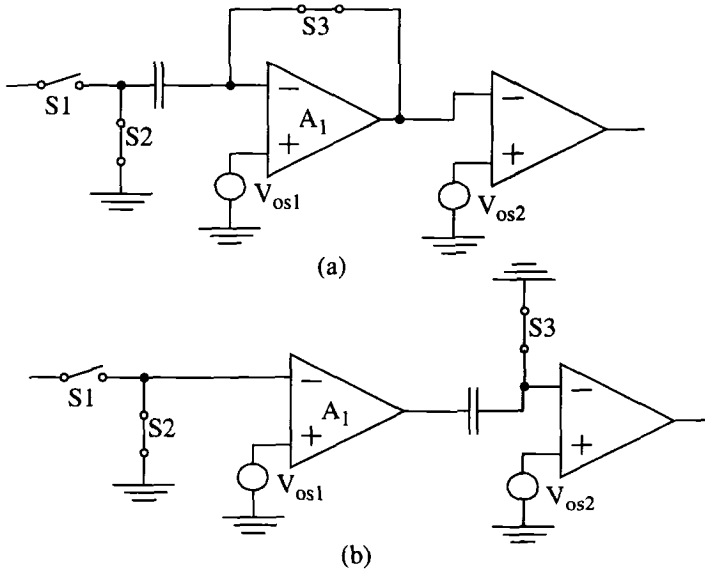


Figure 7-11 Offset cancellation techniques: (a) input offset storage and (b) output offset storage.

With the input offset storage technique, the input referred offset is given by

$$V_{os} = \frac{V_{os1}}{1 + A_1} + \frac{\Delta Q}{C} + \frac{V_{os2}}{A_1} \quad (7-39)$$

where V_{os1} and A_1 are the input offset and the gain of the pre-amplifier, respectively, ΔQ is the charge injection from switch S3 to the capacitor, and V_{os2} is the offset in the second-stage comparator.

With the output offset storage technique, the input referred offset is given by

$$V_{os} = \frac{\Delta Q}{A_1 \cdot C} + \frac{V_{os2}}{A_1} \quad (7-40)$$

where V_{os1} and A_1 are the input offset and the gain of the pre-amplifier, respectively, ΔQ is the charge injection from switch S3 to the capacitor, and V_{os2} is the offset in the second-stage comparator.

Comparing equations (7-39) and (7-40), we see that for similar pre-amplifiers, the input referred offset using the output offset storage is smaller than that using the input offset storage. However, there are some fundamental differences between these two techniques which may make it less attractive to use the output offset storage technique.

With the input offset storage technique, the input to the whole comparator is AC coupled and therefore the input common-mode range is not of a main concern. Since the pre-amplifier is configured in a closed loop fashion, a high gain in the pre-amplifier can be employed to reduce the offset, though it will slow down the comparison as dis-

cussed in the previous section.

With the output offset storage, the input to the whole comparator is DC coupled and therefore the input common-mode range is limited. Since the pre-amplifier is configured as a open loop operation, the gain has to be limited (to usually less than 20 dB) to ensure operation in the active region across the process corner and temperature range.

Both techniques can be mixed and used repeatedly for multi-stage comparators.

7.3.5 Circuit Example

All the techniques discussed above can be combined to design high-speed, high resolution comparators. An example is shown in Fig. 7-12. (If the offset in the comparator is of less concern as in A/D converters with digital correction, the latch or the latch with a pre-amplifier to prevent kickback can be used.)

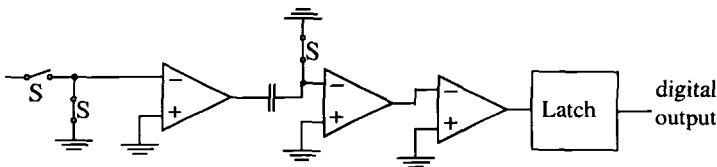


Figure 7-12 A high-speed high-accuracy comparator architecture.

The comparator consists of three pre-amplifiers followed by a positive feedback latch. The input of the first pre-amplifier is DC coupled while the coupling between the first and second pre-amplifier is AC coupled.

In order to have a high-accuracy comparison, the offset needs to be compensated. We use the output offset storage technique to cancel the offset. If the gain of the pre-amplifier is not too large so that the gain of the pre-amplifier remains the same on both phases and the pre-amplifier is not saturated on the amplification phase, this arrangement eliminates the offset error [9] [10]. For a comparator, the offset is only of concern when the input signal is very small, and when the input signal is very small, the pre-amplifier usually does not saturate and it makes this technique an attractive option.

The clock feedthrough on the AC coupling capacitors between the first and the second pre-amplifier is also reduced by the gain of the first pre-amplifier since the input signal has already been amplified by the first pre-amplifier. In order to reduce the metastability of the positive-feedback latch, two pre-amplifiers precede the latch.

Notice that if the pre-amplifiers are overdriven, it may take an excess time to recover during the reset phase. Clipping is sometimes needed. However, proper designed pre-amplifiers can automatically clip the output voltages.

When the noise is of concern, extra capacitors are needed to be placed at the input of the comparator. Since the comparator is usually preceded by a sample-and-hold ampli-

fier, the noise bandwidth due to the sample-and-hold amplifier is considerably smaller than that given by the capacitance and switch-on resistance. Therefore, the total noise power is less than kT/C and we can use capacitors in the gain amplifier.

Due to the use of the output offset storage, the pre-amplifier must have a low gain. In order to have a wide bandwidth, a fully differential pair with a diode load is one of the choices. Such an example is shown in Fig. 7-13.

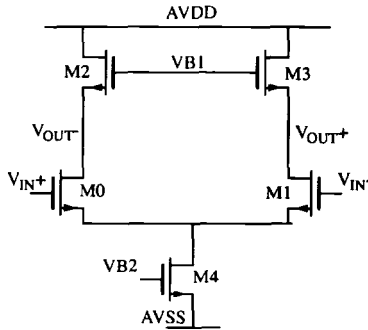


Figure 7-13 The pre-amplifier circuit.

The pre-amplifier consists of the fully differential pair M0 and M1. The load is an NMOS pair contrary to the general practice of using PMOS transistors. The use of the NMOS pair as the load eliminates the need of a level-shifting circuit in order to drive another pre-amplifier. The output voltage is automatically clipped below the positive supply voltage AVDD. The output voltage is also automatically clipped above the negative supply voltage AVSS due to the fact that one of the fully differential pair conducts all the current provided by the current source transistor M4 and therefore the output voltage cannot be smaller than the drain potential of M4. The current source transistor M4 is controlled by an SC common-mode feedback circuit (not shown here) in order for the pre-amplifier to be able to work within a wide common-mode range. If the common-mode feedback is not used, transistor M4 can be biased at a proper current.

The gain of the pre-amplifier is given by

$$A_{pre} \approx \frac{g_{m0}}{g_{m2}} \approx \sqrt{\frac{(W/L)_{M0}}{(W/L)_{M2}}} \quad (7-41)$$

where g_{m0} and g_{m2} are the transconductance of transistor M0 (M1) and M2 (M3), $(W/L)_{M0}$ and $(W/L)_{M2}$ are the dimension ratios of transistor M0 (M1) and M2 (M3), respectively. The practical value of the gain is usually around 4 to 10.

The pole frequency at the source of transistor M2 (M3) determines the -3-dB frequency of the pre-amplifier. It is given by

$$\omega_{-3dB|pre} = \frac{g_{m2}}{C_L} \quad (7-42)$$

where g_{m2} is the transconductance of transistor M2 (M3) and C_L is the total load capacitance. To have a high bandwidth with some gain given by equation (7-41), a large bias current is needed.

One realization of the positive feedback latches [11] is shown in Fig. 7-14. It only con-

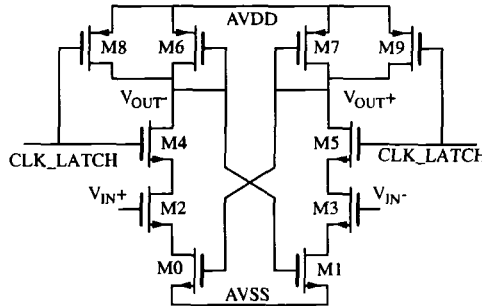


Figure 7-14 The positive-feedback latch circuit.

sumes power during the transition.

The latch basically consists of a back-to-back inverter pair consisting of transistors M0 and M6, and transistors M1 and M7, respectively. When the latch signal `clk_latch` is high, transistors M4 and M5 open, the potential difference at the outputs due to the input voltage is amplified by the positive feedback. When the latch signal `clk_latch` goes low, transistors M4 and M5 cut off. To reduce hysteresis, transistors M8 and M9 are used to set the two outputs high.

It is also possible to connect the input transistors M2 and M3 in parallel with transistors M0 and M1, respectively [12], the latch then will consume power in the latch mode.

As discussed in section 7.3.2 and given by equation (7-34), the speed of the latch is determined by the transconductance and the capacitive load. By properly dimensioning the transistor size, the latch is capable of flipping in less than 0.65 ns for a 0.1 mV input in a typical CMOS process. Though the designed positive-feedback latch is fast enough for most high speed A/D converters, without pre-amplifiers the latch will influence the proceeding track-and-hold circuit due to the charge transfer into or out of the latch input when the latch goes from the track mode to the latch mode (a mechanism called kickback). Also the offset voltage in a well-laid out differential pair is still in the tens and hundreds of mV range, far larger than the minimum voltage needed to toggle the latch, the offset compensation in the pre-amplifiers is necessary for A/D converters without digital correction. With digital correction, however, relatively large offset errors in comparators are tolerable, making the use of only the latch such as shown in Fig. 7-14 acceptable.

7.4 CURRENT COMPARATORS

Current comparators or quantizers are used in current-mode circuits. The input to cur-

rent quantizers is usually a current rather than a voltage. For current inputs, we can directly connect the inputs together to realize summation or subtraction of currents. Different current quantizer structures exist. In Fig. 7-15, we show the widely used current quantizers [13].

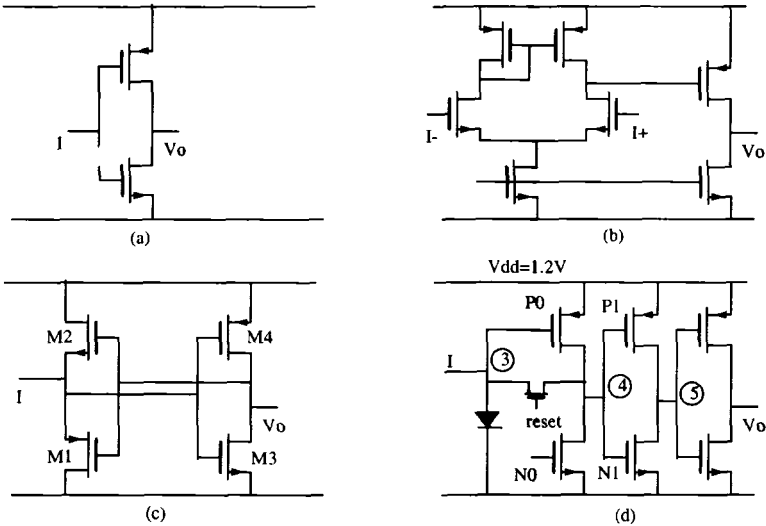


Figure 7-15 Current comparators. (a) inverter as a current quantizer; (b) fully differential voltage comparator as a current quantizer; (c) low-impedance current quantizer; and (d) ultra low-voltage current quantizer.

For the current quantizer of Fig. 7-15 (a), the input current charges or discharges the input parasitic capacitance and causes the potential at the inverter input to change. For positive input current I , i.e., the current is forced into the inverter, the voltage at the inverter input ramps up until it hits the upper voltage rail V_{dd} causing the inverter output to go low. For negative input current I , the current is pulled to ground and the inverter output goes high. For the current quantizer of Fig. 7-15 (b), fully differential SI circuits can directly drive it. The currents from the preceding SI circuits charge or discharge the gate parasitic capacitance of the input transistors of the comparator depending on the direction of the currents. When one branch of the differential outputs from the preceding SI circuits charges the gate parasitic capacitance of one input transistor, the other discharges the gate parasitic capacitance of the other input transistor. The voltages at the differential nodes ramp in opposite direction towards V_{dd} or zero, respectively. The differential voltage at the comparator inputs thus changes according to the direction of the current flows and the output is produced according to the differential voltage. The current quantizers of Fig. 7-15 (a) and (b) are not true current quantizer in the sense that the input impedance is high. The potential change at the current quantizer inputs is large. For high speed applications, low-impedance current quantizers are needed.

The low-impedance current quantizer is shown in Fig. 7-15 (c). The drawback is the

large quiescent power consumption. When the input current I flows into the current quantizer of Fig. 7-15 (c), all the current must be sunk by the p-transistor M1, making its gate-source voltage less than the value of its threshold voltage. When the input current I flows from the current quantizer, all the current must be provided by the n-transistor M2, making its gate-source voltage larger than its threshold voltage. Therefore, the gates of M1 and M2 experience a large potential change when the input current changes directions. Due to the low input impedance, the potential change at the input is small. Transistors M3 and M4 act as a voltage amplifier. Small voltage variation at the input due to the input current is amplified. The speed of the current quantizer is determined by the input impedance, the input parasitic capacitance, and the response time of the voltage amplifier consisting of M3 and M4. After optimization, a settling time of 10 ns is achieved with a current step input of $0.5 \mu\text{A}$. To use the current quantizer of Fig. 7-15 (c), a fully differential to single ended converter is needed before driving the current quantizer.

To achieve a high speed and high resolution current comparison with a ultra low supply voltage (e.g., 1.5 V), we can use a resettable current quantizer shown in Fig. 7-15 (d). When Reset is high, transistor P0 is diode connected and transistor N0 provides the bias current. We design the DC voltage at node 4 close to the corresponding DC voltage of the driving circuit. Transistors N1 and P1 form an inverter and are dimensioned to have high speed. A minimum sized inverter is used as the load. Due to the very low supply voltage, only the branch consisting of transistors N0 and P0 conducts DC current. This DC current can be designed very small ($< 10 \mu\text{A}$). Therefore, the current quantizer does not consume much static power. When Reset is high and there is a current I_{in} flowing into the current quantizer, the potential of nodes 3 and 4 changes. When Reset goes low, the gate of P0 is isolated from its drain and the gate voltage changes the potential at node 4. At the same time, the gate voltage keeps changing due to the input current. This makes the potential change at node 4 even faster. Therefore, a fast and accurate comparison can be accomplished. To reduce the settling time when Reset goes high, we use a diode at the input to limit the potential. With an input current of 50 nA, a comparison time of less than 0.25 μs is achieved. When the input current increases to 1 μA , the comparison time is less than 15 ns. And the power consumption is less than 2.4 μW for the nominal process parameters at the room temperature. To use the current quantizer of Fig. 7-15 (d), a fully differential to single ended converter is needed before driving the current quantizer.

As discussed above, a latch-type voltage comparator usually follows the current comparator to reduce the error probability.

7.5 VOLTAGE AND CURRENT REFERENCES

7.5.1 Voltage Reference Generation

For high-performance data converters, high-quality reference voltages or currents are needed. Bandgap reference circuits can generate a supply-voltage independent, process independent, and temperature independent voltage [14]. The principle behind a bandgap reference is to utilize the negative temperature coefficient of the base-emitter

voltage of a bipolar transistor and the positive temperature coefficient of the thermal voltage (kT/q). Therefore, for high-performance data converters, we must have access to bipolar transistors.

In CMOS process, we can resort to parasitic PNP transistors. A bandgap circuit in CMOS process is shown in Fig. 7-16.

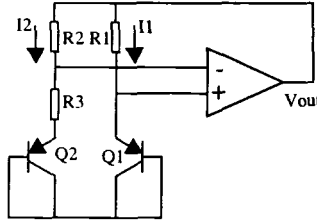


Figure 7-16 Bandgap reference in CMOS process.

Due to the high DC gain in the amplifier, the voltage across the resistors R_1 and R_2 have equal voltage, forcing the currents ratioed by the ratio of R_1 and R_2 . Therefore the difference between the base-emitter voltages is given by

$$\Delta V_{BE} = V_T \cdot \ln\left(\frac{I_1}{I_2} \cdot \frac{A_2}{A_1}\right) = V_T \cdot \ln\left(\frac{R_2}{R_1} \cdot \frac{A_2}{A_1}\right) \quad (7-43)$$

where V_T is the thermal voltage (kT/q , ~ 26 mV at 300 K), A_1 and A_2 are the emitter area of transistors Q_1 and Q_2 , respectively. This voltage difference is also the voltage across resistor R_3 . Since R_2 and R_3 have the same current, the voltage across R_2 is given by

$$V_{R2} = \frac{R_2}{R_3} \cdot V_{BE} = \frac{R_2}{R_3} \cdot V_T \cdot \ln\left(\frac{R_2}{R_1} \cdot \frac{A_2}{A_1}\right) \quad (7-44)$$

Since resistors R_1 and R_2 have the same voltage, we have

$$V_{out} = V_{BE1} + \frac{R_2}{R_3} \cdot V_T \cdot \ln\left(\frac{R_2}{R_1} \cdot \frac{A_2}{A_1}\right) \quad (7-45)$$

It is seen that the output voltage is determined by the base-emitter voltage that has a negative temperature coefficient and the thermal voltage that have a positive temperature coefficient. By choosing the constant that multiplies with the thermal voltage, we can have a temperature coefficient of the output voltage close to zero at a given temperature. With one accurate temperature independent voltage source, we can generate other voltage references by using a gain amplifier. Since the offset in the amplifier introduces an error in the output voltage, large input devices should be used to minimize the offset.

Since the parasitic PNP transistors in CMOS have a gain of 2-4, the Darlington configuration can be used to provide a higher gain. It is also critical to bias the PNP tran-

sistors in the high current gain region.

Bandgap circuits usually have two stable operation points. A start-up circuit is needed. Less known for high performance A/D converters is the noise filtering. All the references should be low-pass filtered by R-C filters with R being poly resistor and C being MOS capacitor.

7.5.2 Current Reference Generation

Accurate currents can not be generated solely on chip. We have to resort to precision external resistors. To generate an accurate current reference, we first need to generate an accurate voltage on chip using the bandgap circuit. By adding an external precision resistor with a low temperature coefficient, we can generate an accurate current reference. An example is shown in Fig. 7-17

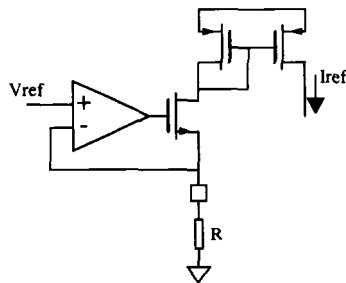


Figure 7-17 Current reference generation.

Assume that the PMOS current mirror transistors are matched and that the gain in the amplifier is infinite. The reference current is then given by

$$I_{ref} = \frac{V_{ref}}{R} \quad (7-46)$$

7.5.3 Reference Buffer

The voltage references in an SC data converter need to charge and discharge a capacitor. Assuming that the capacitor is charged to V_{ref} and then completely discharged, the capacitive load has an equivalent impedance given by

$$R_{eq} = \frac{T}{C} \quad (7-47)$$

where T is the clock period and C is the capacitance. Notice that if the capacitor is charged to V_{ref} and then charged to $-V_{ref}$ in more advanced configurations discussed in this book, there is a factor of 0.5.

If the clock period is small and the capacitive load is large, the equivalent impedance is low. Driving a low impedance needs a low-impedance voltage sources. Therefore, buffering the reference voltage is a must. In Fig. 7-18, we show a block diagram of the

reference buffer.

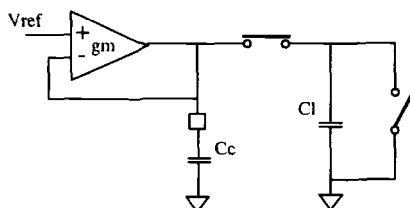


Figure 7-18 Voltage reference buffer.

The output impedance of the buffer is determined by the transconductance of the input devices. If the charging of the load is data independent (all high-performance data converters should have this kind of configuration.), the finite output impedance of the buffer does not introduce distortion but an offset voltage. And this offset voltage can be easily compensate by providing a DC current to the output. It is also possible to reduce the output impedance by using a two-stage amplifier with compensation.

Besides the equivalent low impedance of the SC load, discharging and discharging of the load call for instantaneous action, i.e, large instantaneous currents are needed. To address this issue, the most effective way is to provide a large compensation capacitor C_c as a charge reserve. This large compensation capacitor also limits the noise bandwidth. It is common to use an external capacitor to achieve the best performance. When external capacitors are used, damping of the resonance due to the bond wire and packaging inductance is necessary.

7.6 SUMMARY

In this chapter, we have discussed the basic building blocks for data converters including opamps, voltage and current comparators, reference generators, and reference buffers. For high-performance SC data converters, the telescopic opamp with the SC CMFB is the recommended top choice since it is most power-efficient, least noisy, and fastest. To boost the gain, the gain-boosting technique is recommended. For voltage comparators, the best choice is the positive-feedback latch preceded by low-gain pre-amplifier(s). When a low-offset is needed, auto-zeroing is recommended. The key to high-performance current comparator is the low-input impedance such that current sources having high output impedance can directly drive it. A complete data converter needs on-chip band-gap reference using the parasitic PNP transistors. With a high-accuracy voltage generated on-chip, we can derive an accurate current by resorting to an external precision resistor. To buffer the voltage reference for high performance data converters, we recommend a large external capacitor.

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8 LOW-VOLTAGE ANALOG TECHNIQUES

8.1 INTRODUCTION

The general trend in CMOS fabrication is to make the devices smaller and smaller to increase the density and speed of digital circuits. It is also common to decrease the thickness of the gate oxide to increase the driving capability of the transistor which implies that the supply voltage must be decreased to avoid that the electric field in the devices gets too high. The reduced power supply voltage is normally not an advantage for analog design and a low supply voltage may require some special circuit techniques. Such circuit techniques are discussed in this chapter. In Sec. 8.2 we briefly consider the impact of scaling of CMOS processes. opamp suitable for low supply voltages are considered in Sec. 8.3 and techniques to improve the performance of switches are the subject of Sec. 8.4. Finally the switched opamp technique is reviewed in Sec. 8.5.

8.2 IMPACT OF SCALING

The speed of an MOS process is indicated by the cut-off frequency of the MOS transistor. This is the frequency where the current gain is unity. This frequency can be considered to be an upper speed limit for the process and can be approximated as [1]

$$f_c = \frac{\mu \cdot (V_{GS} - V_T)}{2\pi L^2} \quad (8-1)$$

where μ is the channel mobility, V_{GS} the gate-source voltage, V_T the threshold voltage and L is the channel length of the transistor. It is seen from (8-1) that the speed is increased by making the MOS transistor length smaller. It should be noted that for very small lengths the speed is proportional to $1/L$ rather than $1/L^2$.

In digital circuits where the minimum length is used for the transistors (this is not always the case in analog circuits) the speed is increased and the density is increased if the transistor dimensions are scaled down. Therefore the trend is to make the MOS transistors smaller and smaller.

As the dimensions of the CMOS process is scaled down to increase speed and circuit density, the supply voltage must also decrease since the gate oxide is made thinner.

This makes the design of high-performance analog circuits in such a process more difficult. There are also a number of short channel effects that will degrade the intrinsic gain of the transistor, again making the analog design more difficult for small dimensions [2, 3].

For digital circuits the dynamic power consumption can be approximated as [4]

$$P = \alpha f_{clk} C_L V_{dd}^2 \quad (8-2)$$

where α is the activity factor, f_{clk} the clock frequency, C_L the total load capacitance and V_{dd} the power supply voltage. In digital circuits reducing the power supply voltage is a way of trading speed for power which in principle can be compensated for by increasing the parallelism. For analog circuits the dynamic range (DR) must be considered. A fundamental limit on the DR in discrete-time circuits is imposed by the thermal noise in the circuits. The thermal noise voltage power for discrete-time circuits is given by [2]

$$v_n^2 = \frac{kT}{C} \quad (8-3)$$

where k is the Boltzmann's constant, T the absolute temperature and C the sampling capacitor. The dynamic range with a sinusoidal input signal with peak-to-peak amplitude V_{dd} and rms power $V_{dd}^2/8$ is given by

$$DR = \frac{V_{dd}^2 C}{8kT} \quad (8-4)$$

If the power supply voltage is reduced the capacitor must be increased to keep the DR and it is necessary to increase the bias currents to maintain the speed of the circuits. The number of transistors that can be stacked on top of each other will be reduced due to the reduced supply voltage. Therefore it may be necessary to use several cascaded gain stages to get a high DC gain which further increase the power consumption. The power consumption in analog circuits may therefore increase when the power supply is reduced. The reduction in feature size is thus not advantageous for analog circuits. But using low supply voltages is necessary if analog and digital circuits are to be fabricated on the same chip or if they share the same supply to reduce system cost. Therefore, low voltage analog techniques are of interest, although we prefer to use a 5-V supply for high performance analog circuits.

8.3 LOW VOLTAGE OPAMPS

High accuracy SC circuits usually require a high-gain opamp and a simple differential stage is rarely sufficient. For low supply voltages the design of the op amp becomes troublesome. The gain can easily be increased without increasing the power consumption by using cascode transistors as in the telescopic opamp [5], but the voltage swing, especially for low supply voltages, is small. To achieve a large gain at low supply voltages we must resort to two-stage opamps architectures. For very low supply voltages

even a three stage configuration may be necessary. The highest output swing is achieved if the cascodes are used in the first stage as shown in Fig. 8-1. The voltages

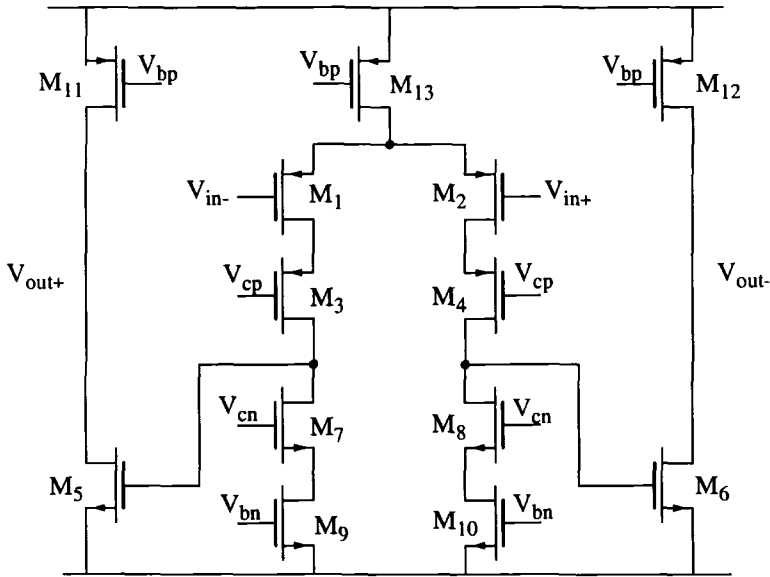


Figure 8-1 Two-stage opamp with cascodes in the first stage.

V_{bp} , V_{cp} , V_{cn} and V_{bn} are bias voltages generated by a bias circuit not shown in the figure. The voltage swing at the output of the first stage is relatively small which makes it possible to use supply voltages as low as ~ 2 V. By using a folded cascode in the first stage as low as 1.5 V supply voltage can be used. However the power consumption and the noise is large for the folded cascode case. The drawback with the two stage solution is that there are two poles in the transfer function significantly below the unity-gain frequency, which makes frequency compensation mandatory. In the following we consider some compensation techniques which can be applied to the opamp architecture shown in Fig. 8-1. When analyzing the frequency characteristics and the pole positions of the opamp, several approximations are usually required to get useful expressions. These simplified expressions does not accurately model the behavior when the second order-parasitic elements are taken into account. An analytical analysis is very difficult and therefore the effect of second order parasitic elements are investigated by simulations.

8.3.1 Compensation Techniques

A suitable small signal model of the opamp in Fig. 8-1 is shown in Fig. 8-2. If the bulk of all N-transistors are connected to V_{ss} and the bulk of all P-transistors to V_{dd} the component values in the small signal model are given by

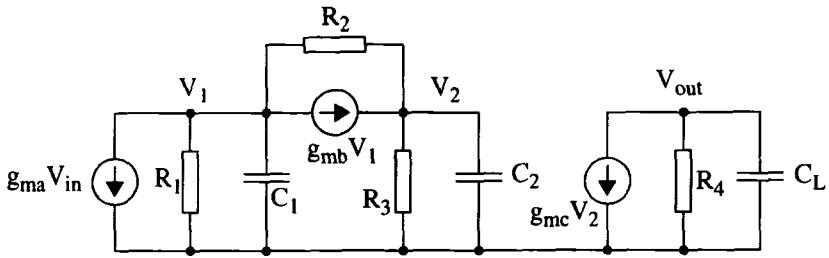


Figure 8-2 Small signal model of the opamp.

$$\begin{aligned}
 g_{ma} &= g_{m1}, \quad g_{mb} = g_{m3} + g_{mb3}, \quad g_{mc} = g_{m5} \\
 R_1 &= r_{ds1}, \quad R_2 = r_{ds3}, \quad R_3 \approx r_{ds7} r_{ds9} g_{m7} \\
 R_4 &= r_{ds5} \parallel r_{ds11} \\
 C_1 &= C_{gs3} + C_{bs3} + C_{bd1}, \\
 C_2 &= C_{gs5} + C_{gd7} + C_{gd3} + C_{bd7} + C_{bd3}
 \end{aligned} \tag{8-5}$$

where g_{mi} is the transconductance of transistor M_i , r_{dsi} is the output resistance of transistor M_i , C_{gsi} is the gate-source capacitor of transistor M_i and C_{bdi} is the bulk-drain capacitance of transistor M_i .

In the small signal model the C_{gd} capacitors of transistors M_1 and M_5 have been neglected as well as the parasitic capacitors at the drain of M_9 . The effect of transistor M_{13} has also been neglected. The total capacitive load at the output of the opamp is modeled by C_L . The transfer function of the small signal model have three poles, two in the first stage and one in the second stage.

If $g_m \gg g_{ds}$ and R_3 (output resistance of PMOS current sources in the first stage) is assumed to be very large the poles are approximately given by [2]

$$\begin{aligned}
 p_1 &\approx \frac{-1}{C_2 \cdot g_{mb} \cdot R_1 \cdot R_2} \\
 p_2 &\approx \frac{-1}{R_4 C_L} \\
 p_3 &\approx \frac{-g_{mb}}{C_1}
 \end{aligned} \tag{8-6}$$

If the effect of R_3 is neglected the DC gain of the opamp is

$$A_0 = g_{ma} g_{mb} g_{mc} R_1 R_2 R_4 \tag{8-7}$$

There are no zeros in the transfer function. If the C_{gd} capacitors of M_1 and M_5 are taken into account, two zeros will appear at

$$z_1 = \frac{g_{ma}}{C_{gd1}}$$

$$z_2 = \frac{g_{mc}}{C_{gd3}} \quad (8-8)$$

The zeros are located at high frequencies and can usually be neglected. It is obvious from (8-6) that there are two poles at quite low frequencies which results in a poor phase margin. Hence frequency compensation is necessary if the opamp is to be used in a feedback configuration with a large feedback factor. Compensation can be performed by using the Miller compensation technique in the second stage [5]. This is done by connecting a compensation capacitor between the input and output of the second stage and results in the small signal model shown in Fig. 8-3. The dominant pole

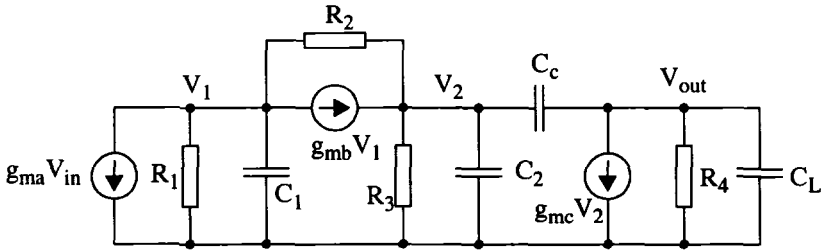


Figure 8-3 Small signal model with Miller compensation.

is now determined by the output resistance of the first stage, $r_{out1} = R_1 R_2 g_{mb}$, and the equivalent Miller capacitor, $C_M \approx C_c R_4 g_{mc}$. Due to the Miller capacitor p_2 has been moved to a high frequency and there is now a right half-plane-zero (RHP) [5]. Assuming R_3 to be very large and $g_{ds} \ll g_m$ the poles and zeros with Miller compensation are approximately given by [2]

$$p_1 \approx \frac{-1}{C_c R_4 R_2 R_1 g_{mc} g_{mb}}$$

$$p_2 \approx \frac{-C_c g_{mc}}{C_2 C_c + C_2 C_L + C_c C_L} \approx \frac{-g_{mc}}{C_L}$$

$$p_3 \approx \frac{-g_{mb}}{C_1}$$

$$z_1 = \frac{g_{mc}}{C_c} \quad (8-9)$$

The RHP zero can be located at a low frequency if g_{mc} is small, giving a small phase margin. By adding a resistor in series with the compensation capacitor the zero can be moved to the left half plane to cancel p_2 [2]. If the pole-zero doublet cancel each other perfectly the frequency behavior is similar to that of the single-stage cascoded opamp.

When the pole-zero doublet is not perfectly cancelled a slow settling component appears for the closed loop system. Due to the doublet the relative settling error as function of time in a unity gain configuration is given by [6]

$$\varepsilon_d = \frac{\Delta f_{pz}}{\omega_u} \cdot \exp\left(-\frac{t}{\tau_{pz}}\right) \quad (8-10)$$

where Δf_{pz} is the difference between the pole and zero frequency, ω_u the unity gain frequency and $\tau_{pz} = 1/2\pi f_{pz}$ corresponds to the time constant of the doublet. If the doublet is located below the unity-gain frequency the settling behavior may be sensitive to mismatch in the doublet.

A drawback with the Miller compensation technique is that for high frequencies the gate and drain of the second stage gain transistor will track one another due to the compensation capacitor C_c . This can give a poor power supply rejection ratio (PSRR) of single ended opamps [5]. Therefore compensation techniques that improves the PSRR have been developed. These techniques may also improve the settling of the opamp compared to Miller compensation. One way of improving the PSRR is to connect the compensation capacitor to the output of the second stage and the source of the cascode transistor in the first stage [6] which results in the small signal model shown in Fig. 8-4. Unless g_{mb} is very large two of the poles are in many cases complex which is not

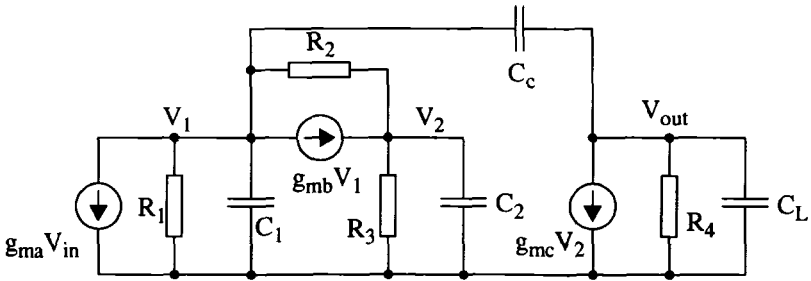


Figure 8-4 Small signal model with the compensation capacitor connected to the cascode of the first stage.

necessarily a disadvantage. This is illustrated in Fig. 8-5. and Fig. 8-6., where the simulated performance of a linear system with one left-half-plane (LHP) zero and three poles is shown. The open loop DC gain in this example was chosen to 1000 and the dominant pole was placed at 1 kHz. Hence the ideal (ignoring parasitic poles and zeros) unity-gain frequency is 1 MHz and the settling time for a relative settling error of 0.1% is

$$t_s = \frac{-\ln(0.001)}{2\pi \cdot 10^6} = 1.1 \mu s \quad (8-11)$$

The real part of all parasitic poles and zeros in the example were chosen to be three times the unity-gain frequency, i.e.

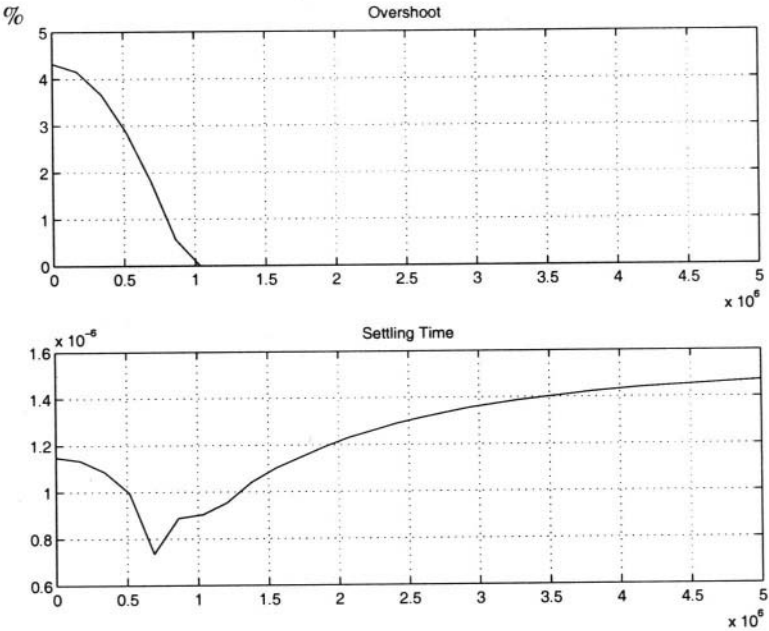


Figure 8-5 Simulated settling time and overshoot as function of the imaginary part of the complex pole pair.

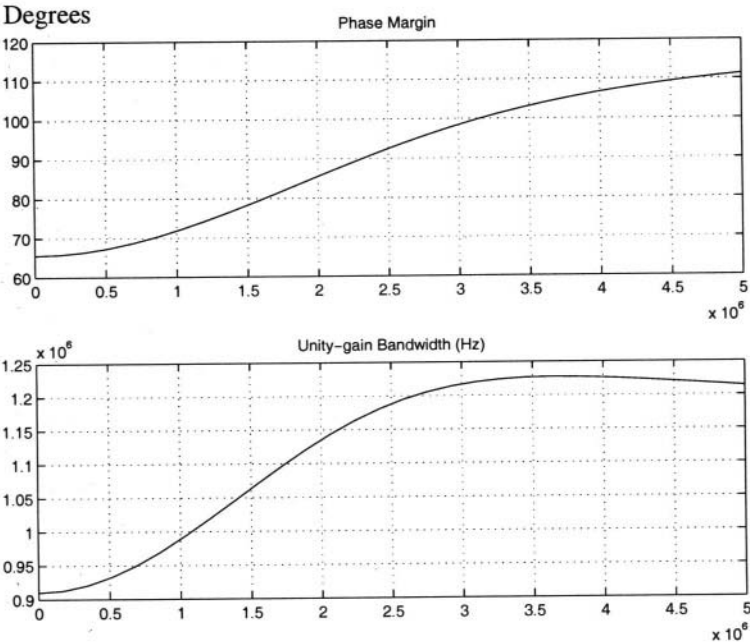


Figure 8-6 Simulated phase margin and unity-gain frequency as function of the imaginary part of the complex pole pair.

$$\begin{aligned}
 p_1 &= -2\pi 1000 \\
 p_{2,3} &= -2\pi \cdot 3 \cdot 10^6 \pm jx \\
 z_1 &= -2\pi \cdot 3 \cdot 10^6
 \end{aligned}
 \tag{8-12}$$

The imaginary part of the poles, x , is swept from 0 to $2\pi \cdot 5 \cdot 10^6$. The system was simulated in a unity-gain configuration using MATLAB and the overshoot and settling time are shown in Fig. 8-5. There is no overshoot in the step response when the imaginary part x is larger than 1 MHz and in the range 0.7 to 1.5 MHz the settling time is slightly smaller than in the single-pole case. The corresponding phase margin and unity-frequency are shown in Fig. 8-6. It should be noted that for large x , both the phase margin and unity-gain frequency is large, but the settling time is not improved. Thus the phase margin and unity-gain frequency are not always good measures of how the amplifier behaves in a closed loop configuration.

The effect of making the poles complex is not always easy to predict especially when there is a large number of parasitic poles and zeros. The slew-rate requirements may increase and the step response may not increase monotonously even though there is no overshoot. However the example shows that from a settling-time point of view complex poles in the open-loop transfer function are not a drawback. The compensation technique proposed in [7] is also a good candidate when a high PSRR is desirable. The principle behind this technique is to feed the current through the compensation capacitor back to the first stage. This can be done by connecting the compensation capacitor between the output of the second stage and the gate of the cascode transistor in the current source in the first stage, i.e. the gates of transistors M_7 and M_8 in Fig. 8-1. If the transconductances of these transistors are large we obtain the small signal model in Fig. 8-7. When using this technique the RHP zero that appeared in the Miller compen-

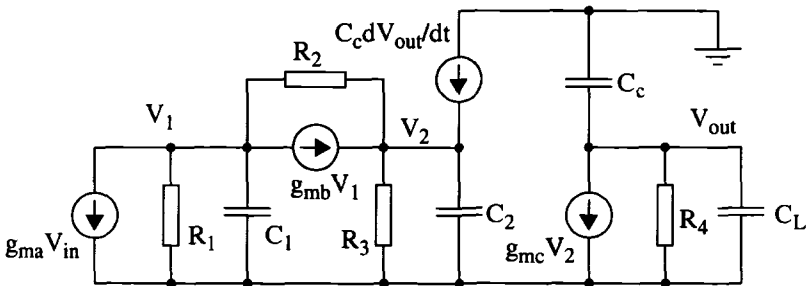


Figure 8-7 Small signal model when using the compensation technique in [7].

sated amplifier is removed and there are three poles approximately determined by [7]

$$\begin{aligned}
 p_1 &\approx \frac{-1}{C_c R_4 R_2 R_1 g_{mb} g_{mc}} \\
 p_2 &\approx \frac{-g_{mb}}{C_1} \\
 p_3 &\approx \frac{-g_{mc} C_c}{C_2 \cdot (C_L + C_c)}
 \end{aligned} \tag{8-13}$$

By keeping g_{mb} , g_{mc} large and C_1 , C_2 small the parasitic poles can be moved to high frequencies. If the transconductances of the cascode transistors M_7 and M_8 are in the same order as g_{mb} and g_{mc} , one LHP zero and one more pole will appear in the transfer function. Two of the poles are in many practical cases complex.

In some cases, especially when three or more gain stages are used in the opamp, it may be necessary to use several compensation capacitors to get high performance, such as the nested Miller compensation technique [8] or the compensation technique in [9]. The opamp in Fig. 8-1 is usually considered as a two-stage opamp but there are some advantages with using several compensation capacitors also for this architecture [8]. The compensation technique considered here is illustrated in Fig. 8-8, where a com-

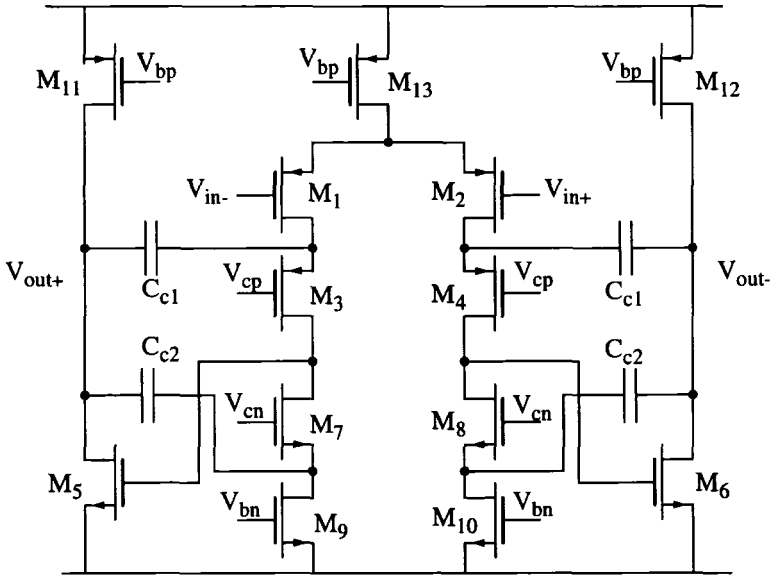


Figure 8-8 Nested compensation of the opamp in Fig. 8-1.

penation capacitor is connected to both the cascode transistors of the first stage [8]. The corresponding small signal model is shown in Fig. 8-9, where the effect of finite transconductance in M_7 and M_8 has been neglected. The poles and zeros are approximately given by

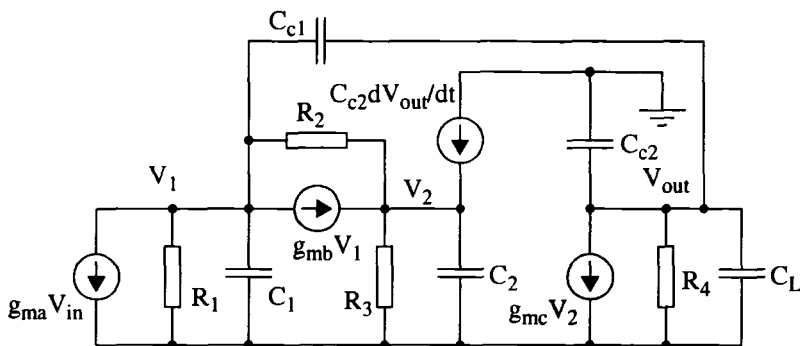


Figure 8-9 Small signal model with the nested compensation technique.

$$\begin{aligned}
 p_1 &= \frac{-1}{(C_{c1} + C_{c2})R_4R_2R_1g_{mb}g_{mc}} \\
 p_{2,3} &= -\frac{C_{c2}g_{mc}}{2C_2(C_{c2} + C_L)} \pm \\
 &\pm \sqrt{\frac{C_{c2}^2C_{c1}^2g_{mc}^2}{(2C_2(C_{c2} + C_L))^2} - \frac{C_{c1}(C_{c2} + C_{c1})g_{mb}g_{mc}}{C_2(C_{c2} + C_L)}} \\
 z_{1,2} &= \pm \sqrt{\frac{g_{mb}g_{mc}}{C_2C_{c1}}}
 \end{aligned} \tag{8-14}$$

We assumed that $g_m \gg g_{ds}$ and that the compensation capacitors C_{c1} and C_{c2} are large compared to the parasitic capacitors C_1 and C_2 .

For a large phase margin the transconductances g_{mb} and g_{mc} should be large while C_2 should be small, since this moves both the parasitic poles as well as the parasitic zeros to higher frequencies. It should be noted that the expressions in (8-14) are simplified expressions that are useful to get some basic knowledge on how different parameter influence the poles and zeros. For more accurate results all the parasitics, such as the finite transconductance of transistors M_7 and M_8 , in the circuit should be taken into account. When these parasitics are included the transfer function will have at least one more pole and one more zero. Thus it is very difficult to analyze the circuit using analytical methods and numerical simulations are more convenient. In the following section we show by an example that the above compensation method can be useful.

8.3.2 Simulations

To verify the effectiveness of the compensation techniques in the previous section we first add a few more parasitic elements to the small signal model as shown in Fig. 8-10.

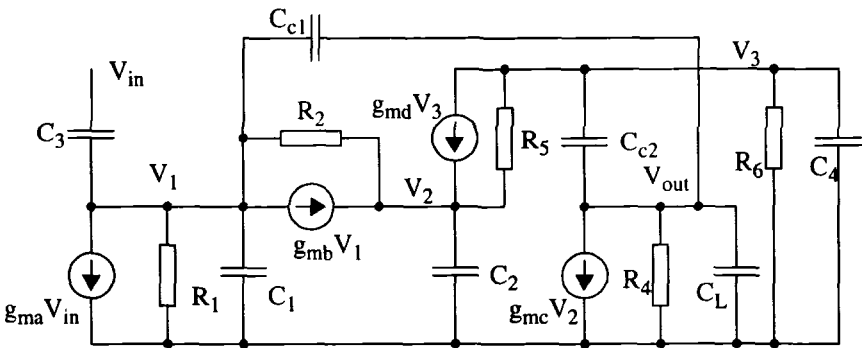


Figure 8-10 Small signal model including parasitic elements.

The new parasitic elements are determined by (referred to the transistor numbering in Fig. 8-8)

$$\begin{aligned} g_{md} &= g_{m7}, R_5 = r_{ds7}, R_6 = r_{ds9} \\ C_3 &= C_{gd1} \\ C_4 &= C_{db9} + C_{gs7} + C_{sb7} \\ C_5 &= C_{gd5} \end{aligned} \tag{8-15}$$

The small signal parameters were taken from a design in a 0.6 μm CMOS process. They are shown in Table 8-1.

Table 8-1 Small Signal Parameters

Param.	Value	Param.	Value
g_{ma}	2.6 mA/V	R_5	40 k Ω
g_{mb}	3 mA/V	R_6	40 k Ω
g_{mc}	1.7 mA/V	C_1	376 fF
g_{md}	3.8 mA/V	C_2	239 fF
R_1	11.5 k Ω	C_3	82 fF
R_2	10.5 k Ω	C_4	547 fF
R_3	∞	C_5	4 fF
R_4	47 k Ω	C_L	4 pF

Ignoring all high frequency poles the unity-gain bandwidth is given by

$$\omega_u = \frac{g_{ma}}{C_{c1} + C_{c2}} \quad (8-16)$$

In a unity-gain configuration and assuming 0.025% settling accuracy (approximately 12 bits accuracy) the settling time is

$$t_s = \frac{-\ln(0.0025)(C_{c1} + C_{c2})}{10^{-3}} \quad (8-17)$$

In Fig. 8-11 is shown the simulated settling time (using MATLAB) as function of the

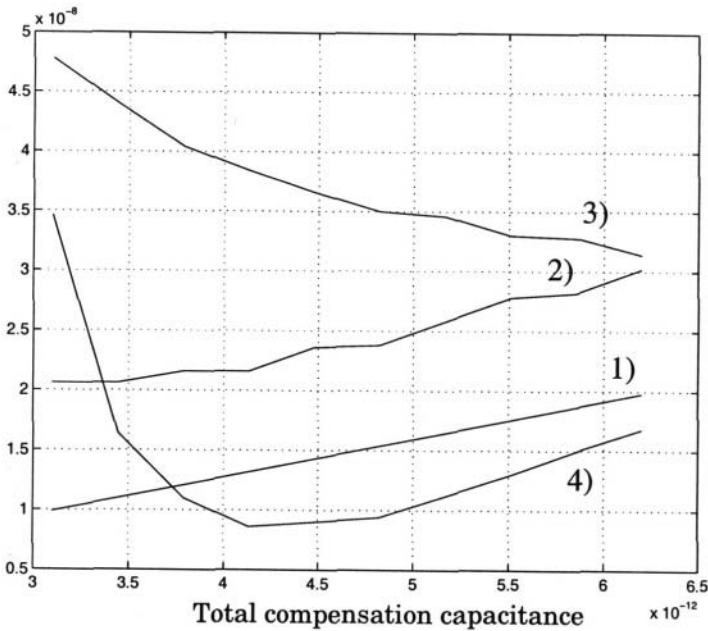


Figure 8-11 Settling times with different compensation techniques.

total compensation capacitance for four different cases:

1. Only the dominant pole is used and the settling time is given by (8-17).
2. Parasitics taken into account and $C_{c1} = 0$ while C_{c2} is changed.
3. Parasitics taken into account and $C_{c2} = 0$ while C_{c1} is changed.
4. Parasitics taken into account and $C_{c1} = 3 \text{ pF}$ while C_{c2} is changed.

The nested compensation technique can in this case give a smaller settling time compared to the other alternatives. By choosing $C_{c2} = 2 \text{ pF}$ the settling time is approximately 10 ns. If C_{c2} is decreased a faster settling time is achieved but the step response will then exhibit some overshoot and therefore C_{c2} should not be too small. The step responses for case 1) and case 4) are shown in Fig. 8-12a. The settling time

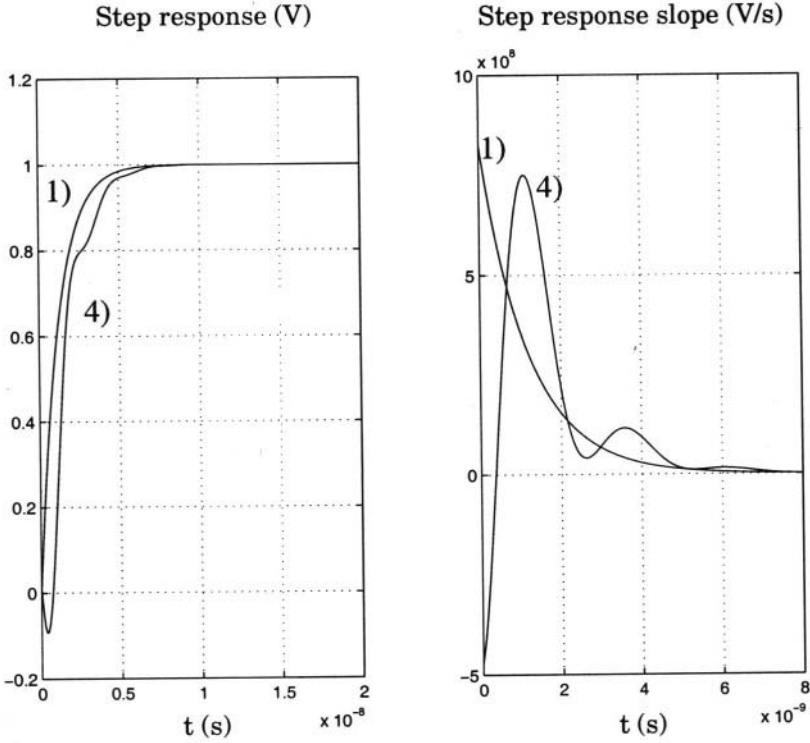


Figure 8-12 a) Step responses and b) step response slope.

is in both cases approximately 10 ns. Typically the slope of the step response of case 4) is larger than for case 1), increasing the slew rate requirements. However in this particular case the maximum slopes are approximately the same as can be seen in Fig. 8-12b, showing the slope of the step responses as a function of time. For case 2) and 3) there was no improvement in the settling time when the Miller capacitor C_5 was increased. The DC gain of the opamp is 97dB, the unity gain frequency is 84 MHz and the phase margin 72° . The poles and zeros of the transfer function are shown in Fig. 8-13a. Compared to the simplified model in Fig. 8-9 there are two more zeros and one more pole in the transfer function. The high frequency RHP zero is caused by C_3 and is approximately given by

$$z_4 \approx \frac{g_{ma}}{C_3} \tag{8-18}$$

The finite value on g_{md} (the transconductance of the NMOS cascodes, M_7 and M_8) introduces a pole-zero pair in the left half plane that almost cancel each other. The complex poles are moved compared to the model in Fig. 8-9. Hence the simplified model in Fig. 8-9 does not accurately model the behavior of the circuit when g_{md} is small.

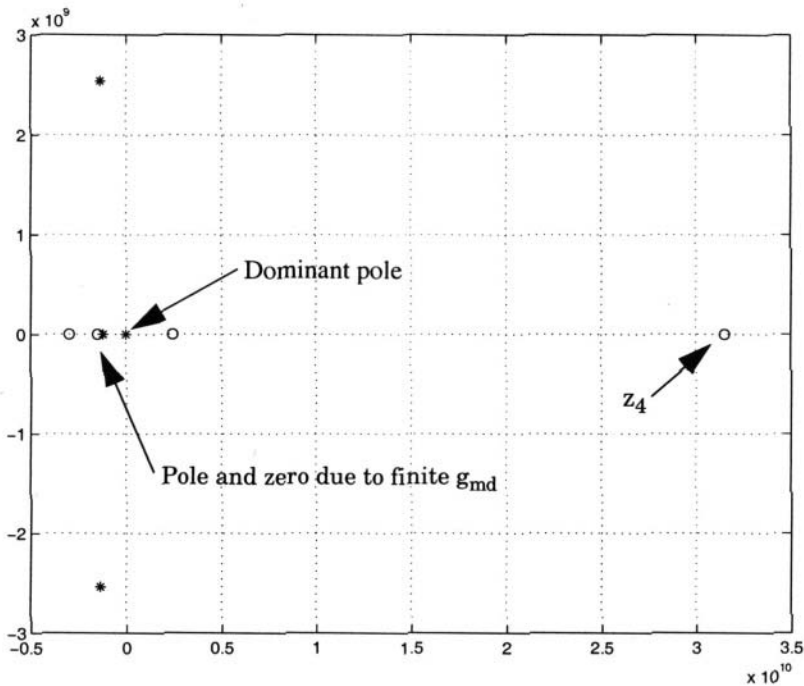


Figure 8-13 Poles and zeros of open loop transfer function.

8.3.3 Common-Mode Feedback

When the opamp is used in a fully differential circuit it is important to have a circuit to regulate the common-mode voltages in the circuit. The most important aspects concerning common-mode feedback (CMFB) is the speed and linearity, while moderate gain is usually acceptable. There are in principle two CMFB techniques. One is the continuous-time CMFB and the other is SC CMFB. Continuous-time CMFB usually suffers from strong nonlinearities. It is usually preferable to use SC CMFB for SC circuits, even though it usually introduces extra capacitive load for the main amplifier. For a two-stage fully-differential opamp, the common-mode voltages in both stages must be controlled. It is possible to sense the common-mode voltage in the second-stage, i.e. at the output, and feed the control signal to the first stage. This kind of CMFB is slow since it has to go through both stages of the main amplifier. A better choice is to use of two CMFBs applied to the individual stages since the individual CMFBs have a much wider bandwidth. For the opamp in figure Fig. 8-8, the effectiveness of the compensation technique is sensitive to the capacitance at the input of the second stage. It is therefore not a good choice to use SC CMFB for the first stage. Continuous-time CMFB is a better choice. The nonlinearity does not present any problem in the two-stage opamp since the second stage has a high gain, making the signal swing at the first stage output small. For the second stage, an SC CMFB is applied. The opamp with compensation capacitors is shown in Fig. 8-14. The CMFB circuit of the first stage consists of transistors M_{c1} and M_{c2} . The transistors are operated in the

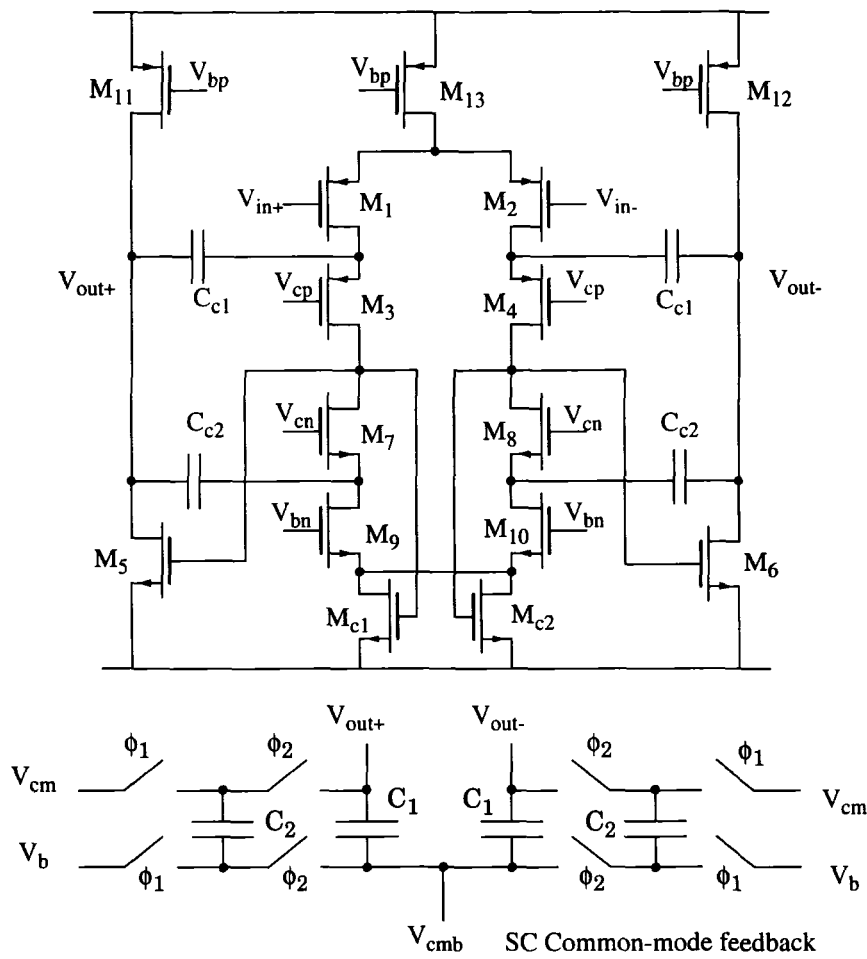


Figure 8-14 The opamp with common-mode feedback circuits.

linear region. The SC CMFB circuit in the second stage consists of four capacitors and eight switches [10] and generates the bias voltage V_{cmb} for the load transistors, M_{11} and M_{12} , in the second stage. V_{cm} is the desired output common-mode voltage and V_b is the desired control voltage in this case the same voltage as is used to bias the current source in the first stage, M_{13} .

8.4 CLOCK VOLTAGE DOUBLERS

For very low supply voltages the switches in a discrete-time circuit becomes a problem. The switch resistance is simply too large due to the low V_{gs} of the switch transistors. Therefore it is necessary to generate switch control signal with a larger swing than V_{dd} . In Fig. 8-15, we show two voltage doubler circuits that can be used for this

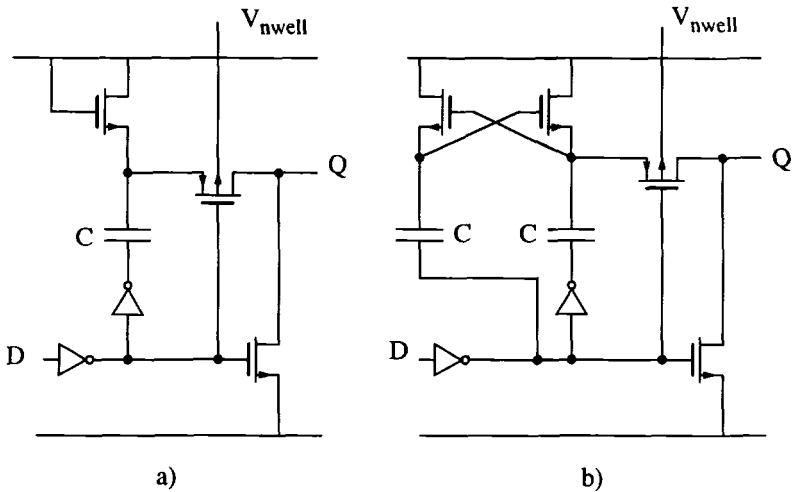


Figure 8-15 Clock doubler circuits. a) Traditional and b) cross-coupled.

purpose. Shown in Fig. 8-15a) is the conventional clock voltage doubler circuit. The drawback of this circuit is the voltage drop across the NMOS transistor. Shown in Fig. 8-15b) is a clock doubler circuit using a cross-coupled pair. Due to the cross-coupling, the sources of the cross-coupled transistor pair are charged to the supply voltage without any voltage drop. By applying a square wave input signal with voltage swing V_{dd} at the input D , the two capacitors are self-charged to the supply voltage through the cross-coupled NMOS transistors [11,12,13], and a square wave is generated at the output Q . The high voltage of the square wave output signal is given by

$$V_Q = 2V_{dd} \cdot \frac{C}{C_{load} + C + C_p} \quad (8-19)$$

where C is the capacitor connected to the cross-coupled pair, C_{load} the load capacitance at the output, i.e. the gate capacitance of the switch, and C_p is the parasitic capacitance at the output.

In Fig. 8-16 we show the SPECTRE simulation results, of the two clock doubler circuits with a supply voltage of 1.2 V. In the simulation, both load and parasitic capacitances are considered. It is seen that the cross-coupled circuit pumps the voltage to around 2.1 V while the conventional circuit only increases the voltage to about 1.6 V.

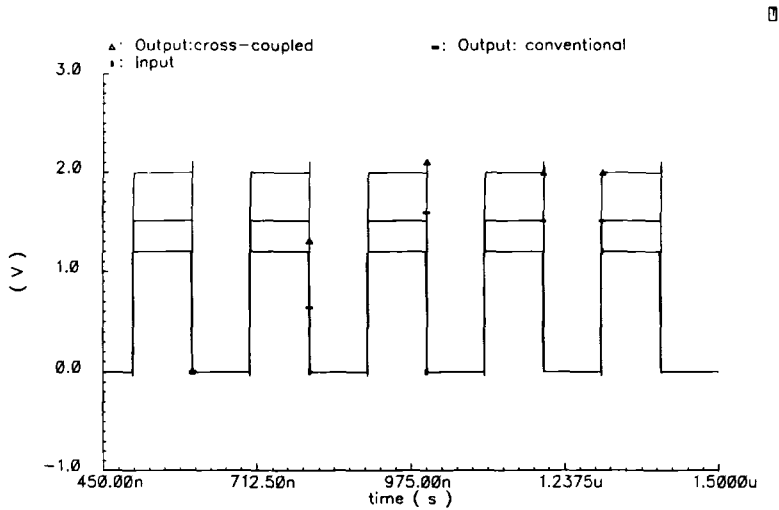


Figure 8-16 Simulation results of the voltage doubler circuits.

8.4.1 High Voltage Generator

The bias voltage for the n wells of the PMOS transistors in the clock double circuits in the previous section must be higher than the supply voltage. This voltage can be supplied by the high voltage generator [11,12] in Fig. 8-17. It needs two square input sig-

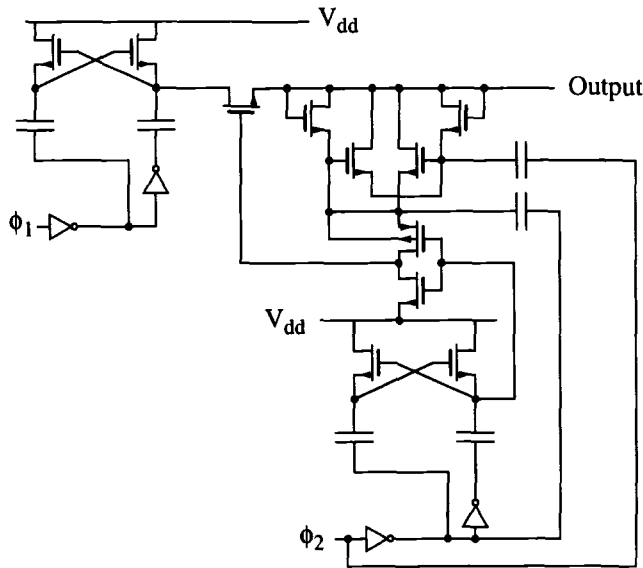


Figure 8-17 High Voltage generator.

nals. The key feature of the high voltage generator is to use feedback to pump the voltage across the capacitors to V_{dd} in the same way as in the cross-coupled clock doubler circuit of Fig. 8-15. Controlled by non-overlapping clock phases (ϕ_1 and ϕ_2), this high voltage generator is capable of generating a voltage of over 2 V if the clock phases have a 1.2 V swing. This voltage can be used to bias the nwell for the PMOS transistors in the clock double circuits of Fig. 8-15 to avoid latch up. Since the current to the nwell is very small, except during start up, the power consumption of the circuit is small. With a 1.2 V supply voltage the output is pumped to about 2.2 V. The simulated output voltage during start-up is shown in Fig. 8-18.

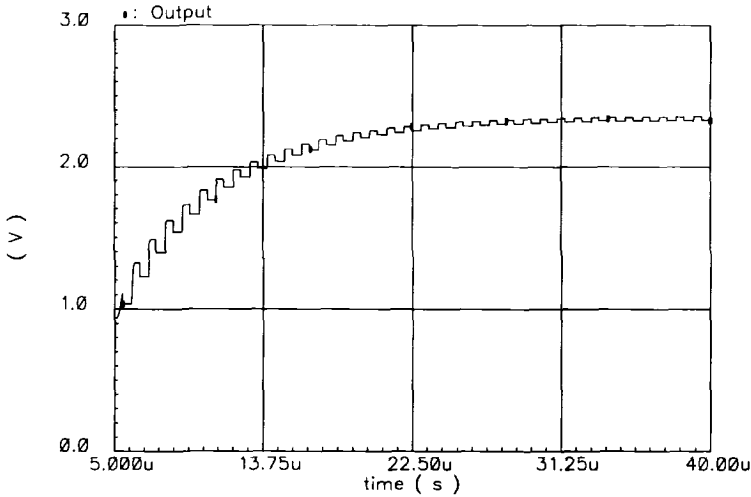


Figure 8-18 Simulation result of high voltage generator.

8.4.2 Bootstrap Switching

The voltage doublers in Fig. 8-15 gives a constant output swing. For switches that need to handle a large voltage swing it would be better if the V_{gs} voltage of the switch tracks the signal. This can be achieved by using a bootstrap switching circuit. Fig. 8-19 shows a bootstrap switching circuit [14]. The switch is represented by transistor M_{11} in this figure. The circuit is controlled by one clock signal ϕ and its inverse. In the off phase when ϕ is low the switch clock voltage, V_g , is low since the gate of M_{11} is discharged through M_7 and M_{10} . Capacitor C_3 is in this clock phase charged to V_{dd} . When ϕ goes high M_9 opens and the lower node of C_3 will track the source voltage of the switch transistor M_{11} . Since ideally no charge can leave capacitor C_3 the gate voltage of the switch transistor is its source voltage plus V_{dd} . Hence the gate voltage tracks the source voltage. This is illustrated in Fig. 8-20. Due to parasitic capacitors in the circuit the voltage across C_3 will be smaller than V_{dd} . The gate voltage can be calculated as [14]

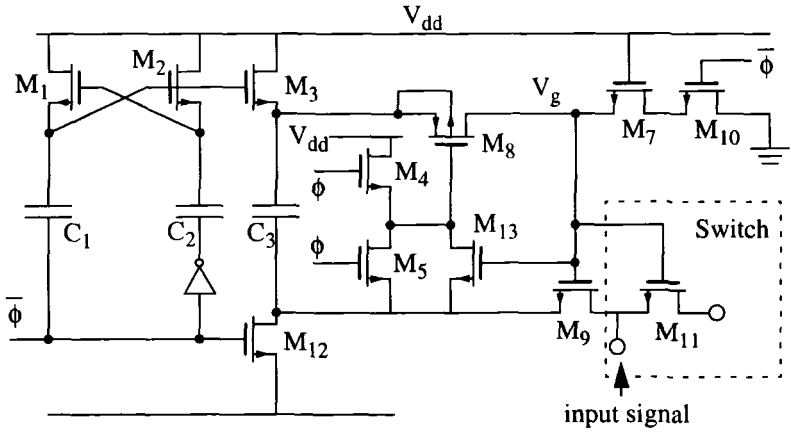


Figure 8-19 Bootstrap switching circuit.

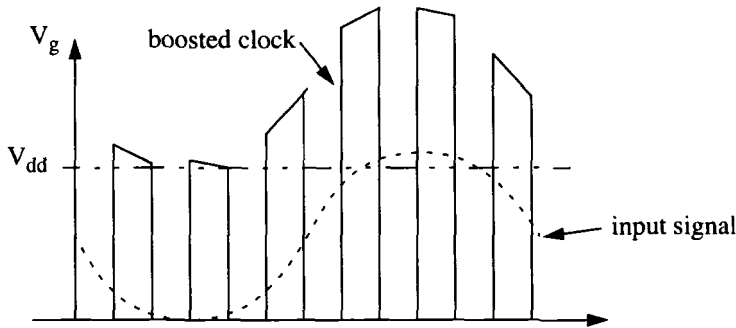


Figure 8-20 Bootstrap circuit output.

$$V_g = V_s + \frac{C_3}{C_3 + C_p} \cdot V_{dd} \quad (8-20)$$

where V_s is the source voltage of the switch and C_p is the parasitic load at the gate of the switch transistor when ϕ is high.

8.4.3 Level Conversion Circuit

It is sometimes necessary to convert low voltage digital signals to a higher voltage to interface with other digital circuits using a higher supply voltage. In Fig. 8-21 we show a level-conversion circuit [15]. The input digital signal has a voltage swing of 1.2 V and the output digital signal has a voltage swing equal to V_{ddH} . The PMOS cross-coupled pair (P_0 and P_1) are connected to the high supply voltage, driven differentially (via NMOS pair N_0 and N_1) by the low-voltage swing signal. A minimum-size inverter is used as load. The pull-down NMOS devices, N_0 and N_1 , are DC ratioed against the cross-coupled pull-up PMOS devices, so that a low swing input guarantees a correct output transition.

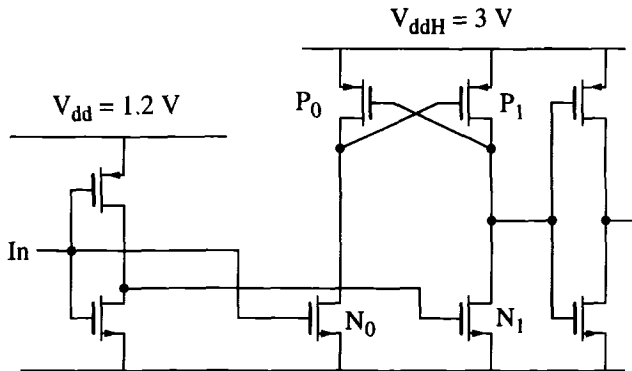


Figure 8-21 Digital level-conversion circuit.

In Fig. 8-22 we show the simulation result of the level-conversion circuit. It is seen

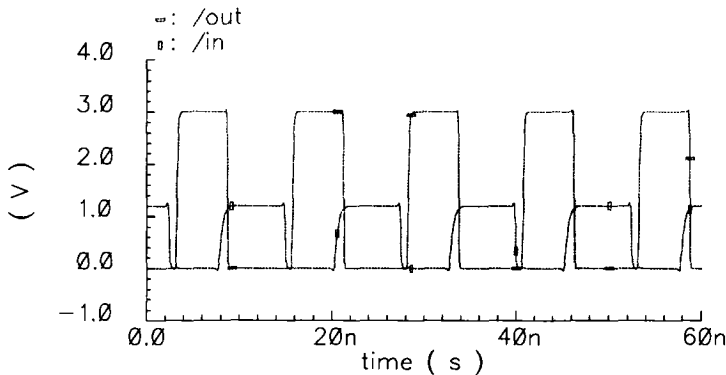


Figure 8-22 Simulation result of digital level conversion circuit.

that we can generate a 3 V square wave output with a 1.2 V square wave input. This level conversion circuit consumes power only during transitions and it consumes no DC power. This level-conversion circuit only works with nwell or twin-well CMOS processes since it requires isolated wells for the PMOS devices that are connected to the high voltage supply and the ones that are connected to the low voltage supply.

8.5 SWITCHED OPAMP TECHNIQUE

The high voltage generators from the previous section can be avoided by using the switched opamp technique [16]. The most difficult switches to realize in a SC circuit are the switches that need to handle a large voltage swing. In the switched opamp technique these switches are removed and we can disconnect the capacitor from the opamp

output by turning off the bias current in the opamp. In [16] the switched opamp technique is used to implement a SC biquad lowpass filter. A standard solution for a SC biquad is shown in Fig. 8-23. In Fig. 8-24 the corresponding switched opamp biquad

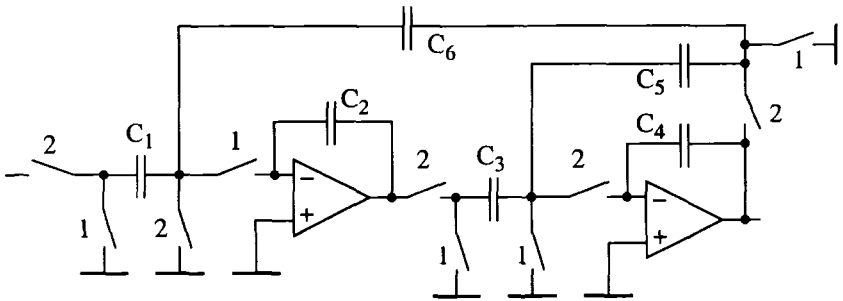


Figure 8-23 Standard SC lowpass biquad.

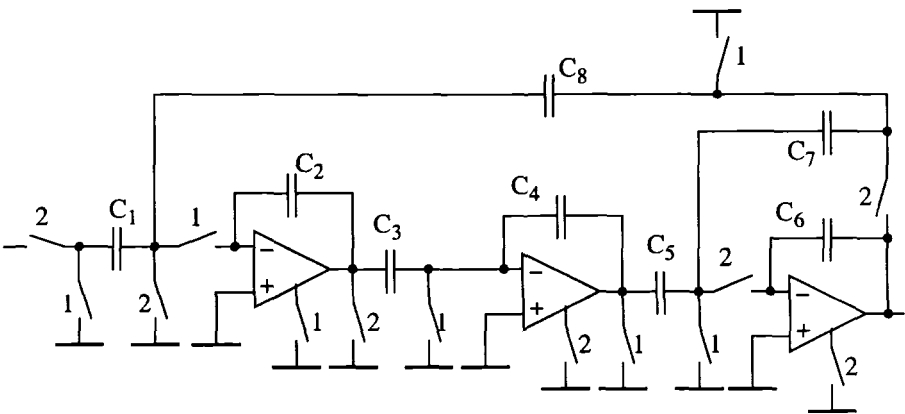


Figure 8-24 Switched opamp biquad.

is shown. With the switched opamp technique the required number of opamps is increased since the output of the integrator can not be used while the opamp is turned off.

The switched opamp in [16] is shown in Fig. 8-25. It is a two stage opamp with Miller compensation. When the clock signal goes low the currents in both the output devices M_5 and M_6 are turned off by when the switch transistors M_9 and M_{10} are opened. The switched opamp technique may not be suitable for very high speed since even though the switches with low voltage swing has been eliminated the switches at the opamp input still have a large resistance due to the small V_{gs} . Since the opamp bias currents can not be turned on instantly it takes a while until it operates properly which further limits the maximum speed of the circuit.

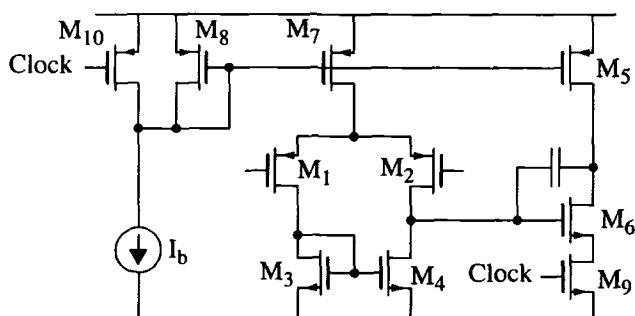


Figure 8-25 Switched opamp.

8.6 SUMMARY

Driven by the need for high-speed and high density in digital circuits the feature size in modern CMOS processes are scaled down. To avoid too large electric fields in the devices which causes reliability problems, the supply voltage must be decreased. This is usually not advantageous for the design of analog circuits. In this chapter we have considered some circuits suitable for low supply voltages. A two stage opamp with a telescopic first stage and a simple common-source output stage was shown to be a good choice for supply voltages down to about 2V. By replacing the first stage with a folded cascode gain stage the opamp can be made to work for supply voltages as low as 1.5V. Different frequency compensation techniques were considered and a nested compensation using two compensation capacitors were shown to have some advantages over single capacitor compensation techniques. For a fully differential implementation the common-mode voltage must be controlled by a common-mode feedback (CMFB) circuit. To avoid a slow common-mode feedback circuit the opamp uses a continuous-time CMFB in the first stage and an SC CMFB circuit in the second stage.

In discrete-time circuits the switches becomes a problem if the supply voltage is low. The effective clock swing can be increased by clock doubler circuits, and thereby reduce the resistance. Some different implementations of such circuits were considered.

An alternative to increasing the swing of the clock signals is to use the switched opamp technique. With this technique switches that need to handle a large voltage swing are removed. Instead the capacitors are disconnected from the opamp output by turning off the bias current in the opamp. A drawback with using the switched opamp technique is that the number of opamps will normally increase since it can only be used on one clock phase. It may also be difficult to use this technique for high speed applications. Therefore, voltage doubler circuits are usually preferred.

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9 PIPELINED A/D CONVERTERS

9.1 INTRODUCTION

The pipelined ADC is suitable for applications where a relatively high bandwidth and a high resolution are required. In Sec. 9.2 we review the fundamentals of pipelined ADCs and show how the total resolution of the ADC is determined and we also show how the digital output of the converter is calculated. Digital correction is used to relax the requirements on the sub-ADCs. The principle behind digital correction is explained in Sec. 9.3. For very high resolutions it may be necessary to use calibration to achieve the desired performance. Digital self calibration algorithms are reviewed in Sec. 9.4. Errors in pipelined converter are treated in general terms in Sec. 9.5, while a more detailed investigation of errors sources in an SC MDAC is considered in Sec. 9.6.

9.2 THE PIPELINED CONVERTER

The pipelined ADC [1] consists of several pipelined stages, each containing a sub ADC, a DAC, a subtractor and a residue gain amplifier as shown in Fig. 9-1. The last

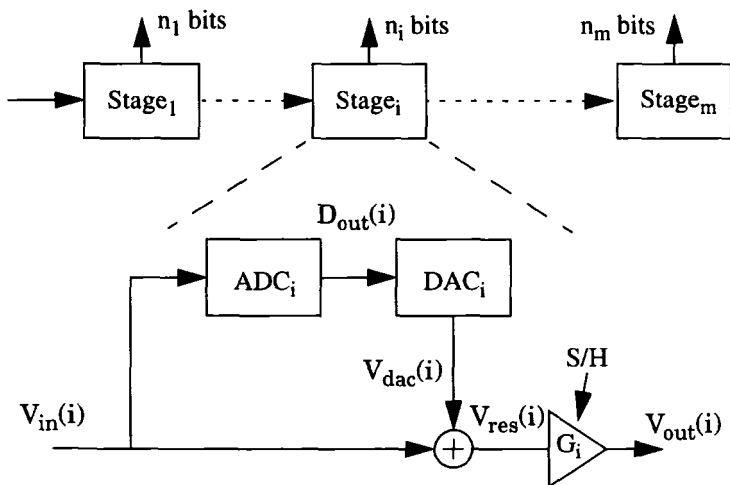


Figure 9-1 The pipelined converter.

stage needs only an sub ADC. Sometimes an additional S/H circuit is used at the input to avoid delay skew errors in the two input signals to the subtractor. The symbols listed in Table 9-1 are used in this chapter.

Table 9-1 Symbols

ADC_i	The sub ADC in stage i
DAC_i	The sub DAC in stage i
$V_{res}(i)$	Residue in stage i (before the amplifier)
$V_{in}(i)$	Analog input to stage i
$V_{out}(i)$	Analog output of stage i . Hence $V_{out}(i) = V_{in}(i + 1)$
$D_{out}(i)$	The digital output of stage i
D_{out}	Total digital output of entire ADC
$V_{dac}(i)$	The output of the DAC in stage i
FS	Full scale input range of ADC_i (assumed to be equal for all stages)
n_i	Number of output bits in stage i
n_{tot}	Total number of bits in the pipelined converter
N_i	Number of output codes in ADC_i
N_{tot}	Total number of output codes of the pipelined ADC
G_i	Residue gain in stage i
m	Number of stages

The number of output codes of a stage is usually a power of two, i.e. $N_i = 2^{n_i}$, where n_i is the number of bits in stage i but any number of codes can be used. In for example redundant-signed-digit (RSD) converters the number of output codes is 3.

9.2.1 Sub ADC

The sub ADC in stage i is assumed to have N_i output codes denoted $D_{out}(i)$. $D_{out}(i)$ is an integer number ranging from 0 to $N_i - 1$. In the following we assume that an offset binary code is used. Thus the digital output of stage i can be calculated as

$$D_{out}(i) = \sum_{l=0}^{n_i-1} b_{i,l} \cdot 2^l \quad (9-1)$$

where $b_{i,l}$ is the l -th bit in the binary output of stage i . We assume that the input sig-

nal range from $-FS/2$ to $+FS/2$ and that the decision levels in ADC_i are equally spaced over the entire input range. This is illustrated in Fig. 9-2 where the input range

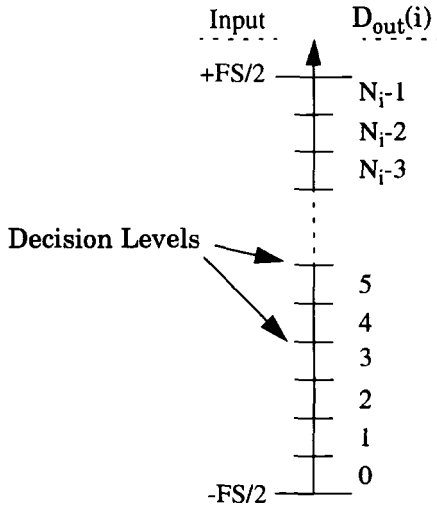


Figure 9-2 Input range and output codes of stage ADC.

has been divided into N_i segments each corresponding to one output code. The analog value corresponding to one LSB is determined by

$$LSB_i = \frac{FS}{N_i} \quad (9-2)$$

9.2.2 Sub DAC

The analog output of the sub DAC for a certain code is determined by the expression

$$V_{dac}(i) = \left(D_{out}(i) - \frac{N_i - 1}{2} \right) \cdot \frac{FS}{N_i} \quad (9-3)$$

9.2.3 Residue and S/H Amplifier

The residue is the difference between the analog input of the pipelined stage and the DAC output, i.e.

$$V_{res}(i) = V_{in}(i) - V_{dac}(i) = V_{in}(i) - \left(D_{out}(i) - \frac{N_i - 1}{2} \right) \cdot \frac{FS}{N_i} \quad (9-4)$$

The transfer function from the input of the stage to the output of the subtractor is sawtooth shaped with the amplitude $FS/2N_i$. This is illustrated in Fig. 9-3. The swing of the residue is thus N_i times smaller than the swing of the input signal to the stage. Here

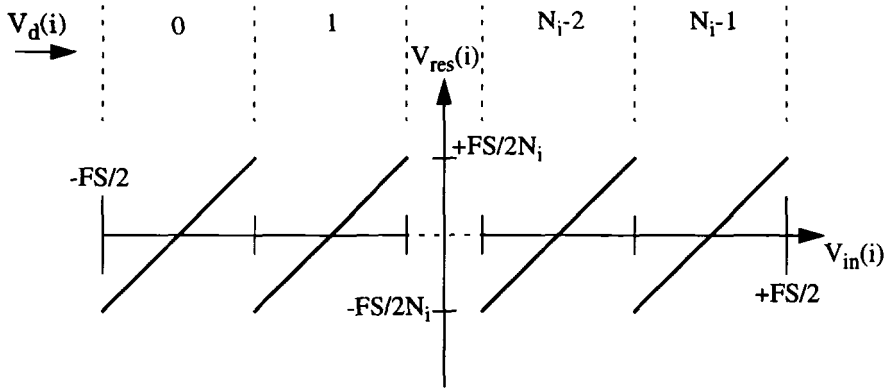


Figure 9-3 Transfer function from input to output of subtractor of one stage in the pipeline.

we have assumed that the full scale range of all the stages are the same. It is therefore necessary to amplify the residue in order to utilize the entire swing of the following stage. Choosing the gain as N_i makes the swing equal to the FS of the following stage. The analog output signal of one stage in the pipeline can be expressed as

$$V_{out}(i) = V_{res}(i) \cdot G_i = \left(V_{in}(i) - \left(D_{out}(i) - \frac{N_i - 1}{2} \right) \cdot \frac{FS}{N_i} \right) \cdot G_i \quad (9-5)$$

or if the signal swing of all the stages are made equal the output signal can be written as

$$V_{out}(i) = \left(V_{in}(i) - \left(D_{out}(i) - \frac{N_i - 1}{2} \right) \cdot \frac{FS}{N_i} \right) \cdot N_i \quad (9-6)$$

9.2.4 Digital Output

The digital outputs of the stages, $D_{out}(i)$, must be combined to generate the total output code of the pipelined converter. The output signal of the first stage is the input signal to the second stage, i.e. $V_{out}(i) = V_{in}(i + 1)$. Hence each segment of the sawtooth shaped output signal of the first stage will be quantized by the second stage. This is illustrated by an example in Fig. 9-4, for a two stage converter where both stages have a resolution of 2 bits. The input range is clearly divided into 16 sections each corresponding to a unique code combination $\{D_{out}(1), D_{out}(2)\}$. The number of codes in a pipelined ADC where $G_i = N_i$ can be calculated as

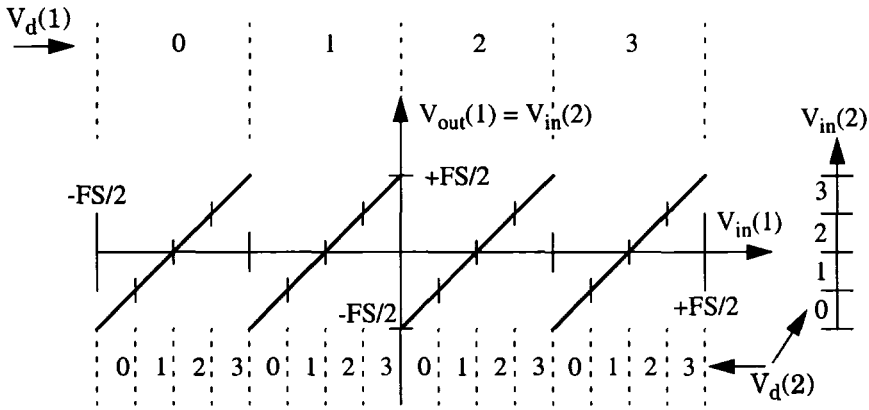


Figure 9-4 Analog output of first stage and corresponding digital output codes of a two stage ADC.

$$N_{tot} = \prod_{i=1}^m N_i \quad (9-7)$$

or if all the resolutions are chosen as $N_i = 2^{n_i}$ the total number of bits is given by

$$n_{tot} = \sum_{i=1}^m n_i \quad (9-8)$$

By using that

$$V_{out}(i) = V_{res}(i) \cdot G_i = (V_{in}(i) - V_{dac}(i)) \cdot G_i \quad (9-9)$$

the input signal to the first stage of the pipeline can be written as

$$V_{in}(1) = V_{dac}(1) + \frac{V_{out}(1)}{G_1} \quad (9-10)$$

Since $V_{in}(2) = V_{out}(1)$ we can use (9-9) one more time to get

$$V_{in}(1) = V_{dac}(1) + \frac{V_{dac}(2)}{G_1} + \frac{V_{out}(2)}{G_1 \cdot G_2} \quad (9-11)$$

By repeatedly using (9-9) for all the stages we finally arrive at the expression

$$\begin{aligned}
 V_{in}(1) = & V_{dac}(1) + \frac{V_{dac}(2)}{G_1} + \dots + \frac{V_{dac}(i)}{G_1 \dots G_{i-1}} + \dots \\
 & \dots + \frac{V_{dac}(m)}{G_1 \dots G_{m-1}} + V_{res}(m)
 \end{aligned} \tag{9-12}$$

where the residue of the last stage, $V_{res}(m)$, corresponds to the quantization error of the converter. It should be noted that the residue of the last stage need not be generated in an implementation of a pipelined converter. By using (9-3) and (9-12) we get

$$V_{in}(1) = \sum_{i=1}^m \left[\left(D_{out}(i) - \frac{N_i - 1}{2} \right) \cdot \frac{FS}{N_i} \cdot \frac{1}{i-1} \cdot \prod_{k=1}^{i-1} G_k \right] + V_{res}(m) \tag{9-13}$$

We are now interested in finding an expression to calculate the total output code of the converter. The total output should start at 0 and increase by 1 as the digital code in the last stage change by 1. We can rewrite (9-13) as

$$V_{in}(1) = \sum_{i=1}^m \left(D_{out}(i) \cdot \frac{N_m}{N_i} \cdot \prod_{k=i}^{m-1} G_k \right) \cdot \frac{\frac{FS}{N_m}}{\prod_{k=1}^{m-1} G_k} + V_{res}(m) + \alpha \tag{9-14}$$

where α is a constant given by

$$\alpha = \sum_{i=1}^m \left[-\frac{N_i - 1}{2} \cdot \frac{FS}{N_i} \cdot \frac{1}{i-1} \cdot \prod_{k=1}^{i-1} G_k \right] \tag{9-15}$$

The first term in (9-14) is clearly 0 if all the digital outputs, $D_{out}(i)$, are 0, which is what we desired for the total output code. We also see that the last term in the sum, i.e. when $i = m$, is equal to $D_{out}(m)$. Hence, the sum changes by 1 as the code in the last stage changes by one. Therefore, we can use the expression

$$D_{out} = \sum_{i=1}^m \left(D_{out}(i) \cdot \frac{N_m}{N_i} \cdot \prod_{k=i}^{m-1} G_k \right) \tag{9-16}$$

to calculate the total output code of the converter.

When choosing the residue gains as $G_k = N_i$ the expression simplify to

$$D_{out} = \sum_{i=1}^m \left(D_{out}(i) \cdot \prod_{k=i+1}^m N_k \right) \quad (9-17)$$

We will illustrate how to use (9-16) and (9-17) by an example. Assume that we have a three stage pipelined converter with the stage resolutions 4, 3 and 2 bits. This means that $N_1 = G_1 = 16$, $N_2 = G_2 = 8$ and $N_3 = G_3 = 4$ respectively. According to (9-17) the output code of the pipelined converter can now be calculated as

$$\begin{aligned} D_{out} &= N_2 \cdot N_3 \cdot D_{out}(1) + N_3 \cdot D_{out}(2) + D_{out}(3) \\ &= 32 \cdot D_{out}(1) + 4 \cdot D_{out}(2) + D_{out}(3) \end{aligned} \quad (9-18)$$

Since multiplication by 2 is easily performed in the digital domain by shifting, the total output code in this example can be calculated by shifting $D_{out}(1)$ 5 steps, shifting $D_{out}(2)$ 2 steps, not shifting $D_{out}(3)$ and adding all three together. This is illustrated in Fig. 9-5.

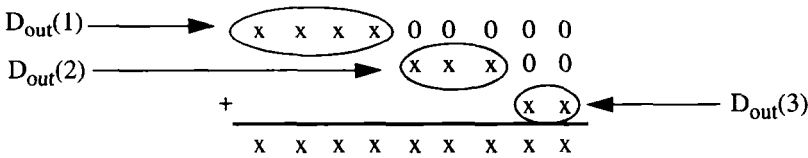


Figure 9-5 Calculation of the output code of a pipelined ADC.

We see by this simple example that no digital processing except shifting is necessary to get the total output code. It is therefore very easy to generate the output code for this type of converter.

9.3 DIGITAL CORRECTION

In a pipelined converter where $G_i = N_i$, the decision levels in the sub ADCs, and thereby also the comparators, must be very accurate if the total resolution of the converter is high. If a decision level in a sub-ADC is moved, the transfer function of the stage will be affected as shown by an example in Fig. 9-6. We see that the output signal swing now is larger than FS. If the input signal to a sub-ADC is larger than +FS/2 or smaller than -FS/2 the output will saturate as shown in Fig. 9-7. Hence there is a large conversion error and the effective resolution of the converter is reduced. To relax the requirements on the comparators, digital correction can be used [2].

In a pipelined ADC with digital correction the residue gain is reduced to introduce redundancy. This is illustrated in Fig. 9-8 for an ADC with two stages where the first

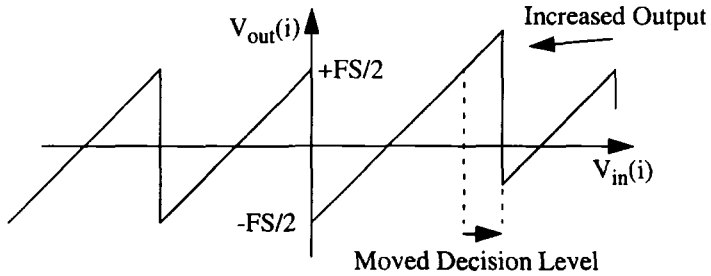


Figure 9-6 Moved ADC decision level increases signal swing.

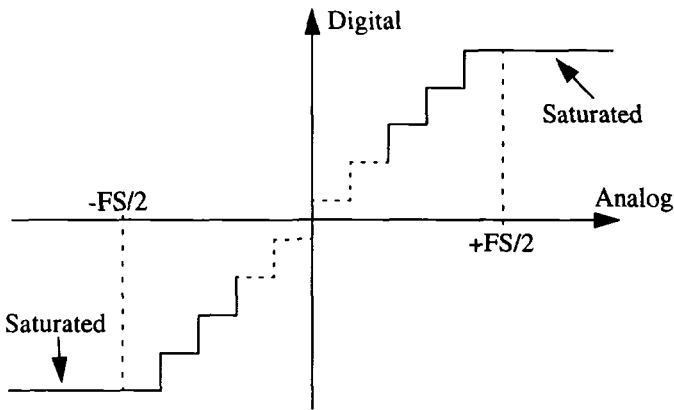


Figure 9-7 The output of an ADC saturates for large inputs.

stage has 4 codes and the second stage has 8 codes. The residue gain has been reduced from 4 to 2. The output signal swing of the first stage is now only half the input range of the following stage. This means that the codes **0,1,6** and **7** will never be used. The gain is usually reduced by a factor 2 but can in principle be chosen arbitrarily. However, it is usually desirable to have the same step size for all the codes. The gain is then restricted to values that give the correct step size at the decision levels in the first stage. This is illustrated in Fig. 9-9 showing the output when one of the decision lines has been moved. In this figure it is seen that moving the decision level will not cause saturation in the following stage since there are now the redundant codes **0,1,6**, and **7** in stage 2. The digital correction can correct errors in the comparators as long as the residue is within the FS range of the following stage. The smaller the residue gain, the larger errors can be accepted. The maximum decision line deviation is given by

$$\Delta V = \pm \frac{FS}{2} \left(\frac{1}{G_i} - \frac{1}{N_i} \right) \quad (9-19)$$

If the gain factor is reduced by a factor 2 we get

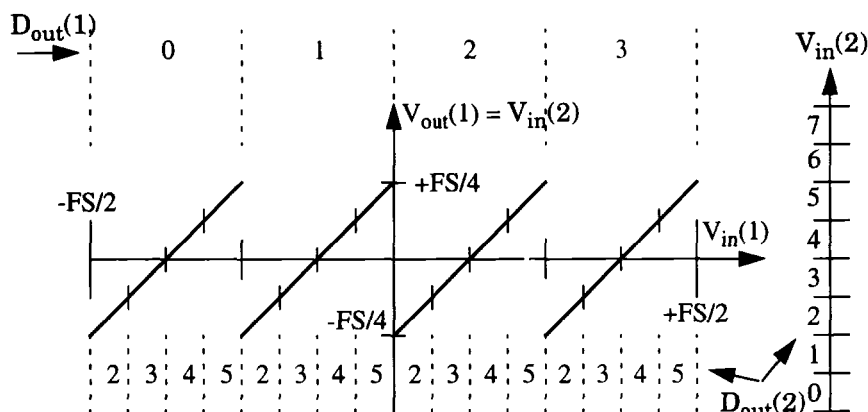


Figure 9-8 Analog output of first stage and corresponding digital output codes for a two stage ADC with reduced residue gain.

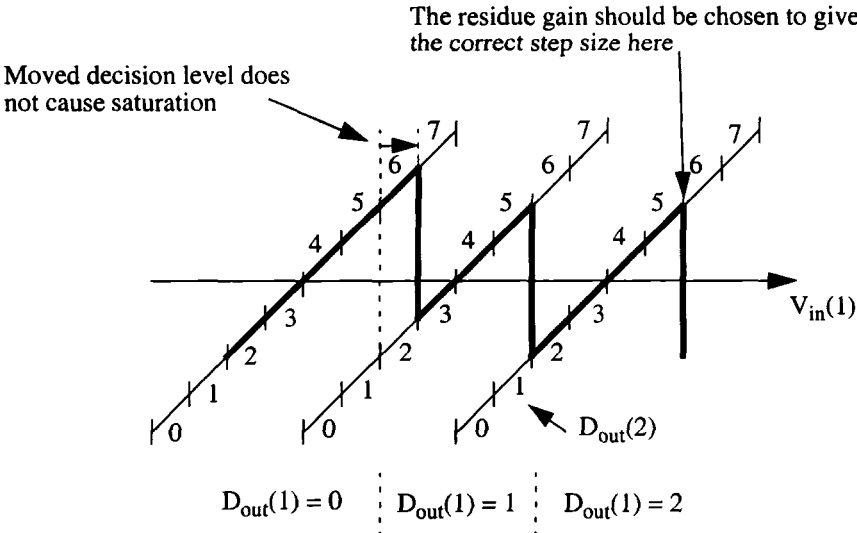


Figure 9-9 A moved decision line does not cause saturation in the following stage when the residue gain is reduced.

$$\Delta V = \pm \frac{FS}{2N_i} \tag{9-20}$$

Hence, the error in ADC_i can be $\pm LSB_i/2$, without causing a large conversion error. A drawback of the digital correction is, as was illustrated by the example above, that several code combinations give the same total output code. The total resolution of the converter is thus decreased when digital correction is introduced unless more stage are

added.

9.3.1 Output Codes and Decoding

Due to the reduced residue gain factor in converters with digital correction the total number of codes is reduced compared to a converter without digital correction. All the codes in the first stage are used but only $N_2 \cdot G_1 / N_1$ codes are used in the second stage and only $N_i \cdot G_{i-1} / N_{i-1}$ are used in the i -th stage. The total number of codes in the converter can thus be calculated as

$$N_{tot} = N_1 \cdot \prod_{i=2}^m N_i \cdot \frac{G_{i-1}}{N_{i-1}} = \prod_{i=1}^{m-1} G_i \cdot N_m \quad (9-21)$$

With digital correction the digital output code can still be calculated using (9-16). For example, assume that we have a three stage converter with the stage resolutions 3, 4 and 5 bits and that the gain in the two first stages is reduced by a factor two to introduce digital correction. This means that we now have $N_1 = 8$, $G_1 = 4$, $N_2 = 16$, $G_2 = 8$, $N_3 = 32$ and $G_3 = 16$. By using (9-16) the output code can now be calculated as

$$\begin{aligned} D_{out} &= \frac{N_3}{N_1} \cdot G_1 \cdot G_2 \cdot D_{out}(1) + \frac{N_3}{N_2} \cdot G_2 \cdot D_{out}(2) + D_{out}(3) \\ &= 128 \cdot D_{out}(1) + 16 \cdot D_{out}(2) + D_{out}(3) \end{aligned} \quad (9-22)$$

The total output code in this example can be calculated by shifting $D_{out}(1)$ 7 steps, shifting $D_{out}(2)$ 4 steps, not shifting $D_{out}(3)$ and adding all three together. This is illustrated in Fig. 9-10. In this case the digital codes from the stages overlap when they

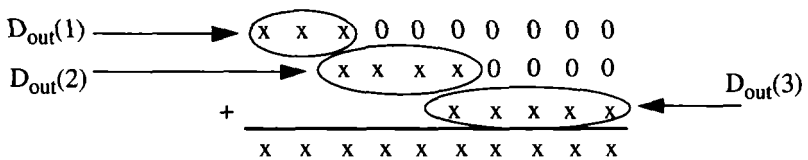


Figure 9-10 Calculation of the output code of a pipelined ADC.

are added. This means that the generation of the digital output code is slightly more complicated since the digital logic must handle carry propagation in the addition. This is not a severe drawback and most high resolution pipelined ADCs use some form of digital correction. There may however be a small problem with directly using (9-16) to calculate the output code, as will be discussed in the following section.

9.3.2 Modified Converter

In the two stage converter used as an example in Fig. 9-8, four codes in the second stage were used for digital correction. This means that for the smallest possible input signal, $-FS/2$, the output codes from the two stages will be 0 and 2 respectively. By using (9-16) the total output code of the converter is

$$D_{out} = \sum_{i=1}^m D_{out}(i) \cdot \frac{N_m}{N_i} \cdot \prod_{k=i}^{m-1} G_k = \frac{8}{4} 2 \cdot D_{out}(1) + D_{out}(2) \quad (9-23)$$

$$= 4 \cdot 0 + 2 = 2$$

The largest possible input, $+FS/2$, gives the output codes 3 in the first stage and 5 in the second stage. By again using (9-16) the total output code is 17. Hence there are 16 output codes which corresponds to a 4 bit converter, but since the smallest code is 2 there is a digital offset. Clearly we must remove the digital offset, since codes 16 and 17 can not be represented using 4 bits. One way of doing this would be to change the coding of the second stage as shown in Fig. 9-11.

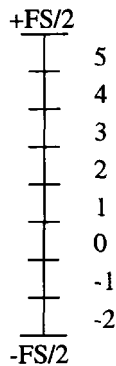


Figure 9-11 Changed coding of the second stage ADC.

If there are no errors in the decision levels of the first stage the coding is very simple. According to (9-23) the code in stage 1 should be multiplied by 4 (corresponds to two binary shifts) and added to the code in the stage 2. Since the codes in stage 2 without errors now range from 0 to 3 there will be no carry propagation in the addition. However if a decision level in the first stage is moved up significantly the codes 4 and 5 are used for some input signals and a carry bit must be propagated. On the other hand if a decision level is moved down, subtraction must be performed since we have the negative codes -1 and -2. Thus there are three cases, do nothing, addition of carry and subtraction of carry. This makes the testing of the correction logic more difficult [3].

One way to simplify the testing is to introduce offsets in the stages as shown in Fig. 9-12 ([3]). Due to the offsets the transfer function of the stage is shifted to the right as

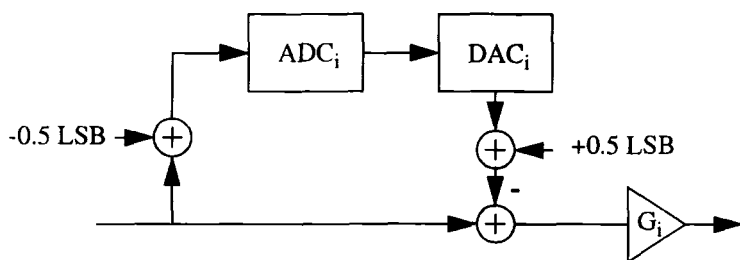


Figure 9-12 One stage with offset.

shown in Fig. 9-13. If we keep the old coding in the second stage with codes ranging

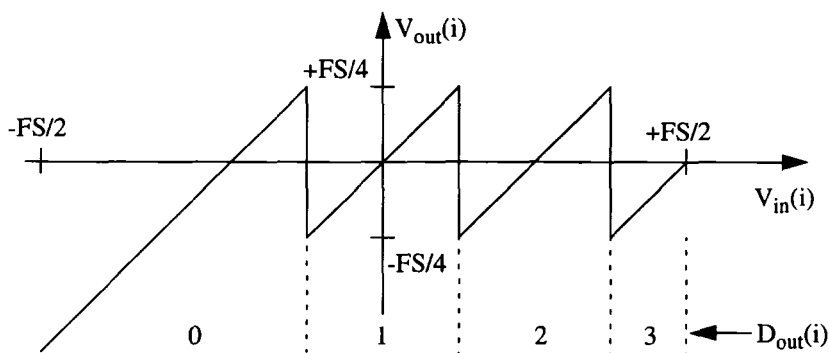


Figure 9-13 Output of a stage with offsets ($N_i = 4$).

from 0 to 7, the total output code will now range from 0 to 15 which can be represented with four output bits. It is also easier to test the correction logic since subtraction is no longer needed and carry addition will occur even if there are no errors in the decision levels. There is one more advantage of introducing the offsets in that the decision level in the zero crossing has now been moved, improving the linearity for small input signals [3]. The transfer function of the stage with offsets is no longer symmetric. A symmetric transfer function can be achieved by removing the rightmost decision line in Fig. 9-13. The resulting transfer function is shown in Fig. 9-14. There are now only 3 codes in the stage but the output range has been increased, making the total number of codes the same as before. A converter containing stages with the output shown in Fig. 9-14 is usually referred to as a redundant signed digit (RSD) converter [4]. The same technique with introducing offsets and removing one comparator can be used for any number of bits in the stages. The digital output can still be calculated using (9-16) but we must remember to use the original number of codes for N_i before one comparator was removed. There is one small difference depending on the last stage of the pipeline. The last stage contains only a sub ADC since the residue does not need to be generated. It is therefore not necessary to add the offset to the last sub ADC to get a correct

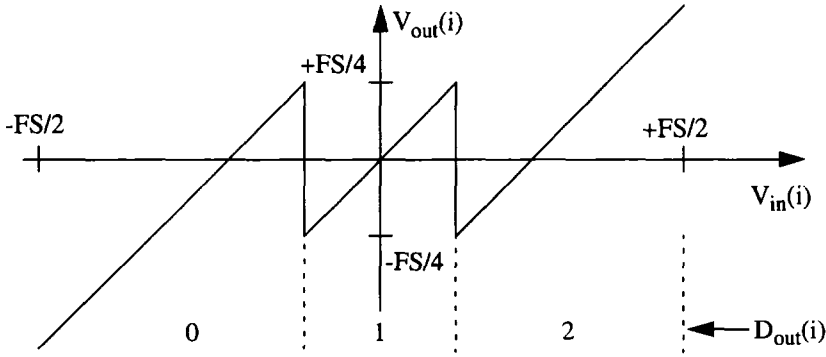


Figure 9-14 Symmetric output with one comparator removed.

coding. The maximum output code in this case is

$$D_{out, max1} = \sum_{i=1}^{m-1} (N_i - 2) \cdot \frac{N_m}{N_i} \cdot \prod_{k=i}^{m-1} G_k + N_m - 1 \quad (9-24)$$

Since the minimum code is 0, the total number of codes is $D_{out, max1} + 1$. However if the offset is introduced also in the last stage (this would save one comparator), only $N_m - 1$ codes are available in the last stage making the maximum code in this case

$$D_{out, max2} = \sum_{i=1}^{m-1} (N_i - 2) \cdot \frac{N_m}{N_i} \cdot \prod_{k=i}^{m-1} G_k + N_m - 2 \quad (9-25)$$

There is thus one code less in this case. If the total resolution of the converter is large this will not affect the resolution of the converter.

As an example we calculate the total number of codes in an RSD converter where all the stages are equal as

$$\begin{aligned} N_{tot} &= D_{out, max2} + 1 = \sum_{i=1}^{m-1} (N_i - 2) \cdot \frac{N_m}{N_i} \cdot \prod_{k=i}^{m-1} G_k + N_m - 1 \\ &= \sum_{i=1}^{m-1} 2 \cdot 1 \cdot \prod_{k=i}^{m-1} 2 + 2 - 1 = 2^{m+1} - 1 \end{aligned} \quad (9-26)$$

Hence, to get n_{tot} bits in a RSD converter $n_{tot} - 1$ stages are needed.

9.4 DIGITAL CALIBRATION

By using digital correction, large errors in the sub ADCs can be corrected. But the digital correction only works if there are no errors in the residues of the stages. The DAC of the first stage must therefore have a linearity which corresponds to the full resolution of the pipelined ADC. The linearity of the DAC is limited to about 10 to 12 bits by component matching and for high resolutions some calibration algorithm is needed. There are both analog [5] and digital [6] calibration techniques to reduce the effect of DAC errors. The pure digital techniques are usually preferred since almost no additional analog hardware is needed and the speed of the converter is therefore not sacrificed because of the calibration. The idea behind the digital calibration is to estimate the DAC errors in one stage by using the other stages of the converter.

In e.g. [7] the stages in the converter are connected in a circular structure to enable calibration of all the stages, starting with the LSB stage. The actual residue gains and DAC outputs of each stage are in this manner approximated by the other stages of the converter. By repeating this procedure the calibration algorithm will finally converge to a solution where the desired parameters have been approximated with sufficient accuracy to compensate for the errors. It is however not always necessary to calibrate all the stages in the pipeline. It may be sufficient to calibrate only a few of the MSB stages, since the MSB stages are most sensitive to DAC errors. In the following we will briefly describe this calibration technique ([6], [8], [9]).

Assume first that there are errors in the first stage DAC of a pipelined ADC while all the following stages are error-free. When using (9-16) to calculate the output code of the converter it is assumed that the DAC output will always change its value by a certain amount when the input code is changes by 1 and that the residue gain is exactly the nominal value. Any errors in the DAC or residue gain amplifier will cause the correction logic to calculate the wrong output code and the transfer function of the pipelined ADC will have discontinuities as illustrated in Fig. 9-15. The errors causing the non-linearities, $\delta(k)$, can now be corrected in the digital domain if we know their sizes. Their sizes can be estimated by using the following stages in the converter. When the digital code to the DAC in the first stage is changed from $k-1$ to k , the digital output of stages 2 to m will change by an amount here denoted $\Delta(k)$. The errors can now be calculated as

$$\delta(k) = \Delta(k) - \Delta_{ideal} \quad (9-27)$$

where Δ_{ideal} is the ideal change in the digital code when the DAC code in stage 1 is changed from $k-1$ to k . The ADC can be linearized by adding correction terms in the digital domain. These correction terms can be calculated as

$$CORR(k) = \sum_{l=1}^k \delta(k) \quad (9-28)$$

and they are stored in a memory from which they can later be retrieved during normal operation. We illustrate this with an example. Assume we have an ADC where the first

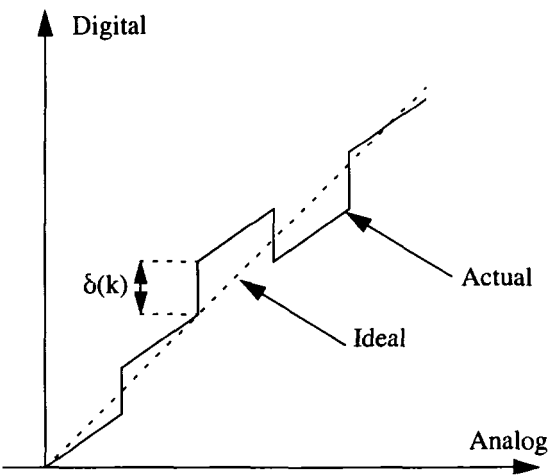


Figure 9-15 Transfer function of pipelined ADC with errors in the first stage.

stage needs calibration, see Fig. 9-16. The first stage has 2 output bits but offsets have

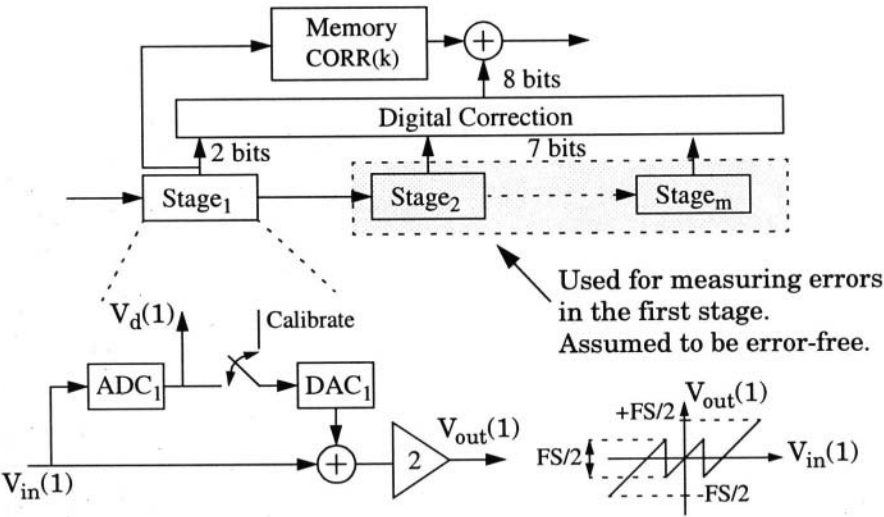


Figure 9-16 Calibration of first stage in a pipelined converter.

been introduced as was described in the previous section such that one comparator can be removed (Fig. 9-12). The number of codes in the first stage ADC is thus only 3 and the analog output signal is described by Fig. 9-14 also shown in Fig. 9-16. The residue gain factor is 2 to enable digital correction. The remaining stages of the pipeline has a total resolution of 7 bits. In Fig. 9-16 we see that increasing the digital code by 1 will decrease the analog value by $FS/2$, This corresponds to a change of 2^6 codes in the 7 bit converter following stage 1, i.e.

$$\Delta_{ideal} = 2^6 \quad (9-29)$$

The calibration now proceed as follows.

1. Turn the input of the DAC to calibrate (see Fig. 9-16) and force the input code to 0. Apply an input offset such that the output ends up in the upper part of the range (exactly how this is done depends on the implementation). Measure the digital output code from stage 2 to m .
2. Change the input code to the DAC to 1. Measure the corresponding digital output and calculate the difference from the previous measurement. This difference is $\Delta(1)$.
3. Repeat the above procedure when changing the code from 1 to 2. This gives $\Delta(2)$. If more codes are used in the first stage the measurements must be repeated for all the codes to obtain all the $\Delta(k)$.
4. Assume $\delta(0) = 0$ and $CORR(0) = 0$. Calculate $CORR(1)$ and $CORR(2)$ by using (9-27) and (9-28). Store the correction terms in a memory.

During normal operation the correction terms are added as illustrated in Fig. 9-17. In

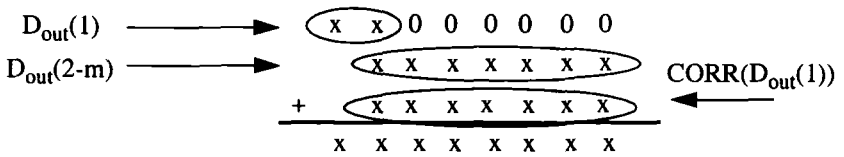


Figure 9-17 Calculation of the output code of a pipelined ADC with calibration.

this way DAC errors and linear gain errors in the first stage can be corrected. The purpose of the above example is to show the principle behind the calibration. The exact procedure to measure the errors depends on the circuit architecture and which type of errors that needs to be calibrated (see e.g. [6], [10]). If more than one stage needs to be calibrated the LSB stages must be calibrated before the MSB stages. The number of stages that needs to be calibrated depends on the total resolution of the ADC and the size of the DAC errors. It may be necessary to add a few more bits at the end of the pipeline to avoid digital truncation errors when the correction terms are measured. The final result can easily be truncated to the desired number of bits.

9.4.1 Other Calibration Techniques

Digital calibration can also be applied if the stages have only one bit per stage [11]. To avoid that the residue gets too large when errors are introduced a non-integer residue gain factor must be used. In [11] the gain is 1.96.

A drawback with the calibration algorithms considered so far is that the ADC can not

be used during calibration. In [12] an interpolation technique is used to calculate some of the output samples. The interpolated samples need not be calculated by the ADC and it is therefore possible to perform the calibration while the ADC is running. In [13] DAC errors are measured by using an additional sigma-delta converter. Due to the circuit architecture the calibration can be performed in the background. Another background calibration algorithm is presented in [14].

There are also some analog circuit techniques that do not really calibrate the ADC but they can still improve the linearity, e.g. capacitor error-averaging [15] and the commutated feedback-capacitor switching (CFCS) technique [16].

9.5 ERRORS IN PIPELINED ADCS

In the following we will discuss how different error sources affect the performance of the pipelined converter [1]. The stage resolution $n_i = \log_2(N_i)$ may be any number of bits but for simplicity we assume that all stage resolutions are equal, i.e.

$$n = n_i, 1 \leq i \leq m \quad (9-30)$$

We assume that x bits in every stage is used for digital correction and will not contribute to the final resolution of the ADC. If we assume that all the stages except the last use x bits for correction the resolution of the total ADC is [1]

$$n_{tot} = (m-1)(n-x) + n \quad (9-31)$$

To get the same full-scale conversion range for all the stages the residue amplifier gain should be chosen as

$$G = 2^{n-x} \quad (9-32)$$

It is convenient to introduce r_i as the resolution remaining to be determined from stage i to stage m

$$r_i = n_{tot} - (i-1)(n-x) \quad (9-33)$$

If the error in stage i is denoted e_i , the input referred error e_{in} can be expressed as

$$e_{in} = e_1 + \sum_{i=1}^{m-1} \frac{e_{i+1}}{G^i} \quad (9-34)$$

To keep the errors smaller than LSB/2 (guarantees no missing codes) we have

$$e_i \leq \frac{FS}{2^{n_{tot}+1}} G^{i-1} \quad (9-35)$$

where FS is the full-scale conversion range of the ADC. This means that larger rela-

tive errors are acceptable for the LSB stages of the converter. Assuming equal errors e in all the stages gives

$$e_{in} = e \left(1 + \sum_{i=1}^{m-1} \frac{1}{A^i} \right) = F \cdot e \quad (9-36)$$

The factor F is minimized for large residue gain factors, G , which indicates that large resolutions in the stages are preferable. However sometimes the errors increase as the stage resolution increases, e.g. it may be more difficult to design a high-speed S/H amplifier with a large gain factor.

The errors in one stage of the pipelined converter can in principle appear at four different places as shown in Fig. 9-18. The error e_1 appearing at the input of the subtract-

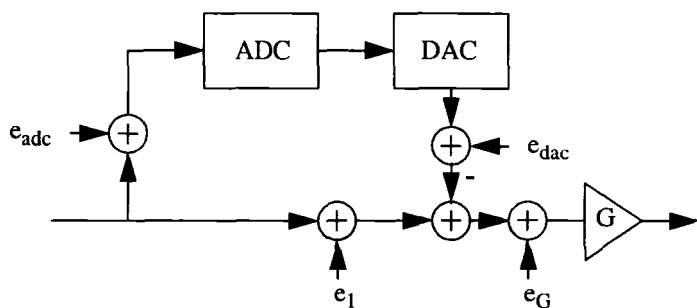


Figure 9-18 Error sources in one stage.

tor will have the same effect as a DAC error and we need not consider this error source separately.

9.5.1 ADC Errors

For most practical situations $x = 1$ is sufficient to correct errors in the sub-ADCs and errors as large as $\pm LSB/2$ at an n -bit level are then acceptable. Therefore, any static errors in the sub-ADCs, e_{adc} , in this type of pipelined converters are of minor concern.

9.5.2 S/H Amplifier Errors

The relative gain errors, e_G , in the S/H amplifiers (SHA) must be smaller than [1]

$$e_{Gi} \leq \frac{1}{2^{r_{i+1}}} \quad (9-37)$$

where r_{i+1} is the resolution remaining in the stages after stage i . This shows again that the MSB stages must have a more accurate gain compared to the LSB stages. The first stage is the most critical and we have that

$$e_{G1} \leq \frac{1}{2^{n_{tot} - (n - x)}} \quad (9-38)$$

If an input S/H circuit is used it must as accurate as the total resolution of the converter, i.e.

$$e_0 < \frac{1}{2^{n_{tot}}} \quad (9-39)$$

However, gain errors in the input S/H circuit is normally acceptable and it is only the distortion that needs to meet the relation in (9-39).

9.5.3 DAC Errors

There are three types of errors in the DACs that should be considered separately, linearity, offset and gain errors.

1. Offset Errors

Offset errors in the DACs can be replaced by offsets in the sub-ADCs and an offset at the input of the ADC. Due to the digital correction this does not limit the resolution of the ADC.

2. Gain Errors

Gain errors in the DACs can be replaced by gain errors in the sub ADCs and gain errors in the S/H amplifiers. We only consider the first stage gain errors here since the first stage is most critical. Gain errors in **ADC**₁ are corrected by the digital correction and is not a problem. Gain error at the input SHA will only change the gain of the ADC and will not introduce distortion. The gain error at the output of the first stage is equivalent to a gain error in the S/H amplifier of the first stage and must meet the relation in (9-38).

3. Linearity Errors

Linearity errors in the DACs must be small enough to meet the relation [1]

$$e_{DACi} \leq \frac{FS}{2^{r_i + 1}} \quad (9-40)$$

where FS is the full-scale range of the converter. The highest requirements are on the first stage DAC, and we have

$$e_{DAC1} \leq \frac{FS}{2^{n_{tot} + 1}} \quad (9-41)$$

It is seen that the linearity must be at least as good as the total ADC linearity. However exactly how component mismatch influence the performance of the converter depends on the implementation. As will be shown in Sec. 9.6.5 a high stage resolution may be better. For high resolutions it may be necessary to use calibration as discussed earlier

in this chapter.

From the above discussion it is clear that it is preferable to have only gain and offset errors in the DACs to having linearity errors. A 1-bit DAC has no non-linearity errors, but a low resolution DAC is in conflict with the high stage resolution needed to reduce the effect of SHA gain errors. It is therefore difficult to find the optimum resolution of the stages since it depends on both the specification and the circuit technique used for the implementation. In the following we investigate how circuit limitations in SC circuits influence the performance of the ADC.

9.6 IMPACT OF NON-IDEAL SC CIRCUITS

In this section we discuss the effect of performance limitations in SC pipelined converters. The circuit performing the function of the DAC and the S/H amplifier is sometimes referred to as a multiplying DAC (MDAC). We will in the following use the SC MDAC from chapter 6 as an example. The SC MDAC is shown in Fig. 9-19 and it has

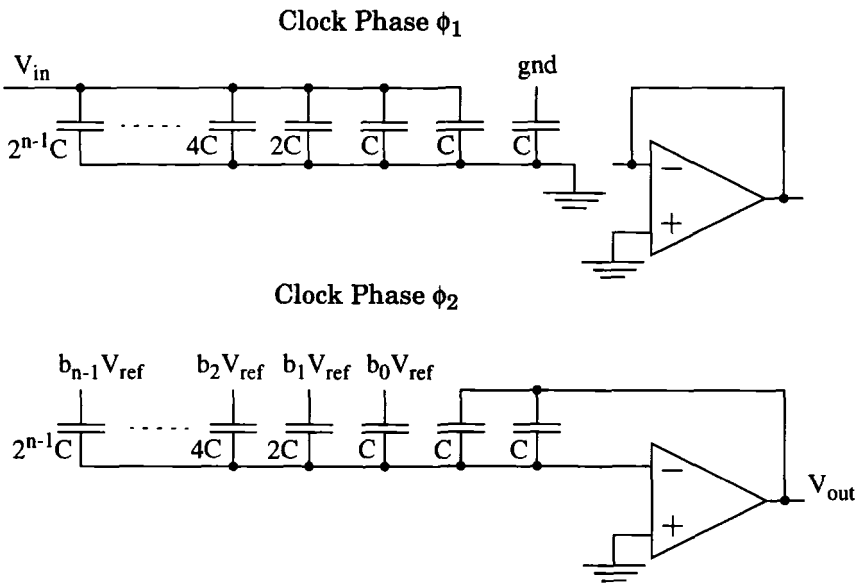


Figure 9-19 The SC MDAC of a pipelined converter.

a gain corresponding to one bit of digital correction.

9.6.1 Finite Opamp Gain

From chapter 6 we have that the transfer function with finite opamp gain is

$$V_{out} = \frac{V_{in} \cdot 2^{N-1} - \frac{V_{ref}}{2} \cdot (2^{N-1} \cdot b_N + \dots + b_1)}{1 + \frac{C_{tot}}{2A_0C}} \quad (9-42)$$

where A_0 is the DC-gain and C_{tot} the total capacitance at the input of the opamp on ϕ_2 . Hence a finite DC-gain will introduce a gain error. According to chapter 6 the relative gain-error can be approximated as

$$\varepsilon_G \approx \frac{C_{tot}}{2A_0C} = \frac{2^n + 1 + \frac{C_p}{C}}{2A_0} \quad (9-43)$$

The error must be smaller than $LSB/2$ referred to the required resolution of the remaining stages in the pipeline. The highest DC-gain is thus needed for the first stage. Due to the digital correction the remaining resolution after the first stage is $n_{tot} - n_1 + 1$ bits, where n_{tot} is the total number of bits in the converter and n_1 the number of bits in the first stage. Hence the required opamp gain in the first stage is determined by

$$\varepsilon_G < \frac{1}{2^{n_{tot}-n_1+1} \cdot 2} \Rightarrow \quad (9-44)$$

$$A_0 > 2^{n_{tot}-n_1+1} \left(2^{n_1} + 1 + \frac{C_p}{C} \right) = 2^{n_{tot}+1} \left(1 + \frac{1 + \frac{C_p}{C}}{2^{n_1}} \right)$$

For a fixed C_p/C ratio the required opamp gain is reduced if the resolution in the first stage is increased. However the unit-capacitor, C , is normally smaller for higher stage resolutions, making the C_p/C ratio larger. Therefore the opamp gain requirement is weak function of the stage resolution.

9.6.2 Noise

In chapter 6 we showed that the performance is fundamentally limited by the kT/C noise in the sampling capacitor. There is also a noise contribution from the opamp but here we ignore this noise source. This is usually a fair assumption when the stage resolution is large. The requirements on the first stage is most critical. The input referred noise of the first stage MDAC can be approximated as

$$v_{n,1}^2 \approx \frac{kT}{2^{n_1}C_1} \quad (9-45)$$

where C_1 is the unit capacitor size in stage 1, n_1 the number of bits in the first stage and. To avoid that the thermal noise has a large impact on the resolution the ratio it should be smaller than the quantization noise. The quantization noise power is (see chapter 1)

$$v_q^2 = \frac{\Delta^2}{12} = \left(\frac{FS}{2^{n_{tot}}} \right)^2 \cdot \frac{1}{12} = \frac{FS^2}{2^{2n_{tot}+2}} \cdot \frac{1}{3} \quad (9-46)$$

By making the thermal noise smaller than the quantization noise we have

$$v_{n,1}^2 < \frac{FS^2}{2^{2n_{tot}+2}} \cdot \frac{1}{3} \Rightarrow C_1 \geq \frac{3 \cdot 2^{2 \cdot n_{tot} + 2} \cdot kT}{2^{n_1} \cdot FS^2} \quad (9-47)$$

The total sampling capacitor in the first stage is thus independent of the stage resolution. We have for the i -th stage

$$v_{n,i}^2 < \frac{FS^2}{2^{2r_i+2}} \cdot \frac{1}{3 \cdot 2} \Rightarrow C_i \geq \frac{3 \cdot 2^{2 \cdot r_i + 3} \cdot kT}{FS^2 \cdot 2^{n_i}} \quad (9-48)$$

where r_i is the resolution of the stages from i to the last stage. Hence the total sampling capacitor can be smaller for large stage resolutions (except for $i = 1$).

When deriving the above limits we considered only one stage at a time. All the thermal noise sources will contribute to the total noise in the converter. The rms value of the total thermal noise referred to the input can be expressed as [18]

$$\begin{aligned} v_{n,tot}^2 &= \sqrt{\frac{v_{n,1}^2}{1} + \dots + \frac{v_{n,i}^2}{(G_1 \dots G_{i-1})^2} + \dots + \frac{v_{n,m}^2}{(G_1 \dots G_{m-1})^2}} = \\ &= \sqrt{\frac{v_{n,1}^2}{1} + \dots + \frac{v_{n,i}^2}{2^{2(n-1)(i-1)}} + \dots + \frac{v_{n,m}^2}{2^{2(n-1)(m-1)}}} \end{aligned} \quad (9-49)$$

where $v_{n,i}^2$ is the noise power contribution from the MDAC in stage i , G_i is the residue gain factor in stage i and m is the number of stages. If the gain factors, G_i are large the thermal noise in the later stages is attenuated when referred to the input. Hence the capacitor sizes according to (9-48) will be very small and must be larger for other reasons such as clock feedthrough. Therefore, if the stage resolution is large the noise in the first stage will dominate. For small stage resolutions several stages contribute to the total noise and the capacitor sizes must be larger than the size given by (9-48). It should also be noted that when the thermal noise power is equal to the quantization noise power the SNR decreases by 3 dB. The capacitors should therefore in the actual implementation be larger to give some margin for other errors sources.

From the above discussion we conclude that a large stage resolution implies that we can use small capacitors in all but the first stage. This is advantageous from a power consumption point of view.

9.6.3 Limited Bandwidth

The relative settling error assuming a single pole system is given by

$$\varepsilon_r = e^{-t_s \cdot \omega_{-3dB}} \quad (9-50)$$

where t_s is the available settling time and ω_{-3dB} is the -3dB bandwidth of the circuit.

The first stage is most critical since it needs the highest accuracy. The closed loop -3 dB bandwidth of the MDAC in the first stage, assuming a single-stage opamp is

$$\omega_{-3dB} = \omega_u \cdot \beta = \frac{g_m}{C_{load}} \cdot \beta = \frac{g_m}{C_{load}} \cdot \frac{2}{(2^{n_1} + 1) + C_p/C_1} \quad (9-51)$$

where C_{load} is the total load capacitance at the opamp output, n_1 is the number of bits in the first stage, C_1 is the unit-capacitor in the first stage MDAC and C_p the parasitic capacitance at the input of the opamp. The size of the load capacitor C_{load} has a large impact on the speed of the circuit. It consists of parasitic capacitors at the opamp output, the load due to the capacitors in the MDAC and the sampling capacitor of the following stage. The load capacitance is given by

$$C_{load} = 2^{n_2} \cdot C_2 + \frac{2(C_1(2^{n_1} - 1) + C_p)}{(2^{n_1} + 1) + C_p/C_1} + C_{p, out1} \quad (9-52)$$

where C_2 is the unit-capacitor in stage 2, n_2 the number of bits in stage 2, and $C_{p, out1}$ the parasitic capacitor at the output of the opamp in the first stage. For simplicity we ignore the parasitic capacitors. The bandwidth can now be approximated as

$$\omega_{-3dB} \approx \frac{2g_m}{2^{n_2} \cdot C_2 \cdot (2^{n_1} + 1) + 2C_1(2^{n_1} - 1)} \quad (9-53)$$

The smallest possible size on C_2 is determined by the thermal noise requirement (9-49) and we have

$$C_2 > \frac{2^{n_1} \cdot C_1}{2^{n_2} \cdot G_1} = \frac{2^{n_1} \cdot C_1}{2^{n_1-1} \cdot 2^{n_2}} = \frac{C_1}{2^{n_2-1}} \quad (9-54)$$

Sometimes when the power consumption is not critical all the stages in the pipeline can be identical to save design time. The sampling capacitor is now much larger than required by the thermal noise requirement. The sampling capacitor in stage 2 should never be larger than in stage 1 and we therefore have an upper limit given by

$$C_2 < 2^{n_1-n_2} \cdot C_1 \quad (9-55)$$

1. The sampling capacitors are as small as required by the thermal noise requirement (see (9-54))

By using (9-50), (9-53), (9-54) and requiring an error smaller than LSB/2 referred to the remaining resolution in the pipeline, we have

$$\begin{aligned}
\omega_{-3dB} &> \ln(2^{n_{tot}-n_1+1})2f_s \Rightarrow \\
\ln(2^{n_{tot}-n_1+1})2f_s &< \frac{g_m}{C_1 \cdot 2^{n_1+1}} = \frac{g_m \cdot FS^2}{3 \cdot 2^{2n_{tot}+3} \cdot kT} \Rightarrow \\
f_s &< \frac{g_m \cdot FS^2}{3 \cdot 2^{2n_{tot}+4} \cdot kT \cdot \ln(2^{n_{tot}-n_1+1})}
\end{aligned} \tag{9-56}$$

where f_s is the sampling frequency.

2. All stages are equal making the sampling capacitor in stage 2 have its largest value.

By using (9-50), (9-53), (9-55) and requiring an error smaller than LSB/2 referred to the remaining resolution in the pipeline, we have

$$\begin{aligned}
\omega_{-3dB} &> \ln(2^{n_{tot}-n_1+1})2f_s \Rightarrow \\
\ln(2^{n_{tot}-n_1+1})2f_s &< \frac{2g_m}{(2^{2n_1} + 3 \cdot 2^{n_1} - 2) \cdot C_1} = \\
&= \frac{2g_m \cdot FS^2}{(2^{n_1} + 3 - 2^{1-n_1}) \cdot 3 \cdot 2^{2n_{tot}+2} \cdot kT} \Rightarrow \\
f_s &< \frac{g_m \cdot FS}{(2^{n_1} + 3 - 2^{1-n_1}) \cdot 2^{2n_{tot}+2} \cdot 3 \cdot kT \cdot \ln(2^{n_{tot}-n_1+1})}
\end{aligned} \tag{9-57}$$

We see that if the load capacitance on the MDAC is small the maximum sampling frequency increases as the stage resolution increases, but the increase is slow due to the log function. If the load capacitance is big the maximum sampling frequency decreases as the stage resolution increases. For many practical cases the load capacitance is somewhere between the two extreme values. Therefore there is usually no strong impact on the speed when the stage resolution is changed. It should be noted that the above derivations assumes a first order linear settling model. In many cases this is not an accurate assumption.

9.6.4 Limited Slew Rate

The results in the previous section are valid if the settling is entirely linear. For most practical cases however the settling is slew rate (SR) limited. It can be shown that the output voltage of the MDAC with slew rate limited settling can be calculated as [17]

$$v(v_s) = \begin{cases} v_s(1 - e^{-t_s \omega_{-3dB}}), & |v_s| \leq v_{lin} \\ v_s - \text{sgn}(v_s)v_{lin}e^{\left(\frac{|v_s|}{v_{lin}} - t_s \omega_{-3dB} - 1\right)}, & v_{lin} < |v_s| \leq v_{lin} + t_s \cdot SR \\ \text{sgn}(v_s) \cdot t_s \cdot SR, & v_{lin} + t_s \cdot SR < |v_s| \end{cases} \tag{9-58}$$

where v_s is the step size of the output voltage as $t \rightarrow \infty$, SR is the slew rate, ω_{-3dB} is the -3dB bandwidth, t_s is the available settling time and v_{lin} is the maximum step size that gives linear settling. The maximum step size for linear settling can be calculated as

$$v_{lin} = SR \cdot \omega_{-3dB} = SR \cdot \omega_u \cdot \beta \quad (9-59)$$

If the settling is SR limited the -3dB bandwidth must be higher compared to if the settling is linear. The smaller the SR the larger the required bandwidth. If the bandwidth is too high it becomes very difficult to achieve a large phase margin for the opamp. It is easier to get linear settling if the feedback factor, β , of the MDAC is small (see chapter 7). Therefore the settling is more linear for higher stage resolutions due to the smaller feedback factor as seen in (9-58).

9.6.5 Matching

The capacitor matching in the MDAC ultimately limits the linearity of the converter. A conclusion from Sec. 9.5 was that the DAC linearity in the first stage must be as good as the total resolution of the entire ADC. It may therefore seem unlikely that anything would be gained by changing the stage resolution. However the effect of the component matching on the performance depends on the implementation. This is best illustrated by an example.

We assume that the capacitor matching errors are normally distributed and statistically independent (in reality there is usually some correlation between the errors). We compare the performance of pipelined converters with different resolutions in the first stage and we assume that all stages but the first are error free. In all cases the total resolution of the converter is 15 bits. Fig. 9-19 show the result of MatLab simulations where the resolution in the first stage was changed from 3 to 5 bits and the relative matching was changed from 0.5 - 2%. In the figure the fail percentage, i.e. the percentage of the samples that do not meet a certain ENOB requirement, as function of the ENOB is plotted. The results in the figure indicate that the achievable resolution increases with 0.5 ENOBs as the stage resolution is increased by one bit while the resolution is increased by 1 ENOB if the capacitor matching is improved by a factor 2.

The achievable capacitor matching is of course dependent on the layout and the process, but for a fixed relative capacitor matching we conclude that a converter with a large resolution in the first stage is better. In the example we intentionally made the matching errors large to avoid that the quantization errors influence the result. A relative capacitor matching of 0.1% or better can normally be achieved in an analog CMOS processes.

9.6.6 How to Choose Stage Resolution

The optimum stage resolution is a trade-off between many different factors and it is not easy to find the ADC with the smallest area and power for a certain specification. An attempt to optimize the performance is presented in [18]. In this paper a 5 MS/s 15 bit converter is used as a design example. Choosing the resolution in the two first stages as 5,4 bits or 4,4 bits while the following stages have a smaller resolutions opti-

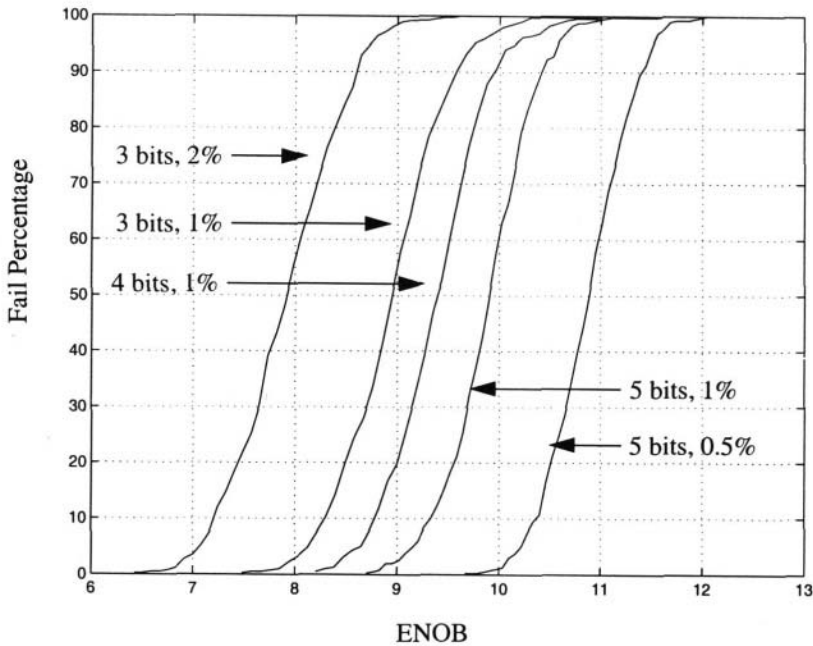


Figure 9-20 Effect of capacitor mismatch in SC MDAC of a pipelined converter.

mizes the power. The area is optimized for a converter where all stages have a resolution of 3 bits. In general a good rule of thumb is to use a low stage resolution of <2 -3 bits for converters up to about 10 bits. For higher resolutions the first stages should have a higher resolution of 4-5 bits.

9.7 SUMMARY

In this chapter we have reviewed the principle of pipelined converters, digital correction and calibration. With digital correction, the influence of errors in the comparators is significantly reduced. Hence the dominating error sources are DAC errors and errors in the residue amplifier. For very high resolutions it may be necessary to use calibration. The principle of self-calibration algorithms in pipelined converters was explained in this chapter. It is normally difficult to calibrate dynamic errors such as settling errors.

The effect of the errors in the LSB stages of a pipelined ADC is reduced by the residue gain factors, making the errors in the last stages almost negligible. The most critical stage is thus the first stage. For errors in the first stage MDAC we concluded that

- Errors appearing at the output of the subtractor in the first stage must be within the resolution of the remainder of the pipeline
- Errors at the input of the subtractor must be within the total resolution of the con-

verter.

Whether an error should be modeled as appearing at the input or the output of the subtractor depends on the circuit solution.

We have also presented expressions showing how the stage resolution affects the influence of circuit limitations such as finite opamp gain and limited bandwidth in an SC MDAC. These expressions can be used as guide-lines when designing the circuits. The optimum stage resolution is a trade-off between many different factors and it is not easy to find the ADC with the smallest area and power for a certain specification. The general rule for high-speed ADC however seems to be that a larger stage resolutions should be used in the first stages while the resolution should be reduced in the LSB stages of the pipelined ADC.

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10 TIME-INTERLEAVED A/D CONVERTERS

10.1 INTRODUCTION

By using time-interleaving techniques very high sampling rates can be achieved. Unfortunately any mismatch between the time-interleaved channels will give rise to distortion. At high signal frequencies the most difficult mismatch error to handle is the delay skews between the clock signals to the S/H circuits of the different channels. In Sec. 10.2 and Sec. 10.3 we discuss the basic principle and review the effect of the error sources in time-interleaved ADCs. Solutions to reduce these errors are considered in Sec. 10.4. In Sec. 10.5 and Sec. 10.6 we discuss a passive sampling technique in more detail. This technique can reduce the effect of delay skew errors between the channel. Tedious derivations are covered in Sec. 10.7.

10.2 PRINCIPLE OF TIME-INTERLEAVED ADCS

An attractive way to increase the conversion rate of ADCs is to use time-interleaving techniques where several ADCs, using different clock phases, are operated in parallel [1]. This enables higher total conversion speed but mismatch between the channels introduces distortion.

The concept of time-interleaving is illustrated in Fig. 10-1. The converters can be of any type and are operated at f_s/M where f_s is the total sampling frequency of the time-interleaved ADC and M is the number of channels. The speed requirements on each converter are relaxed by a factor M but the number of converters is at the same time increased by the same factor. This may result in a large chip area and a high power consumption. The increase in power and area is not necessarily a linear function of M compared to using only one converter, since smaller bias currents can be used in the opamps (if SC circuits is used) due to the lower speed. It is also sometimes possible to share opamps between the channels [2]. The performance of the time-interleaved converter is of course limited by the accuracy of the channel ADCs but there are additional errors caused by mismatch between the channels. There are three main sources of error in time interleaved systems, phase skew errors, gain errors and offset errors [1], [3], [4].

In the analysis below the SNDR is used as a measure on how the errors effect the performance of the parallel converter. When characterizing ADCs it is common to use

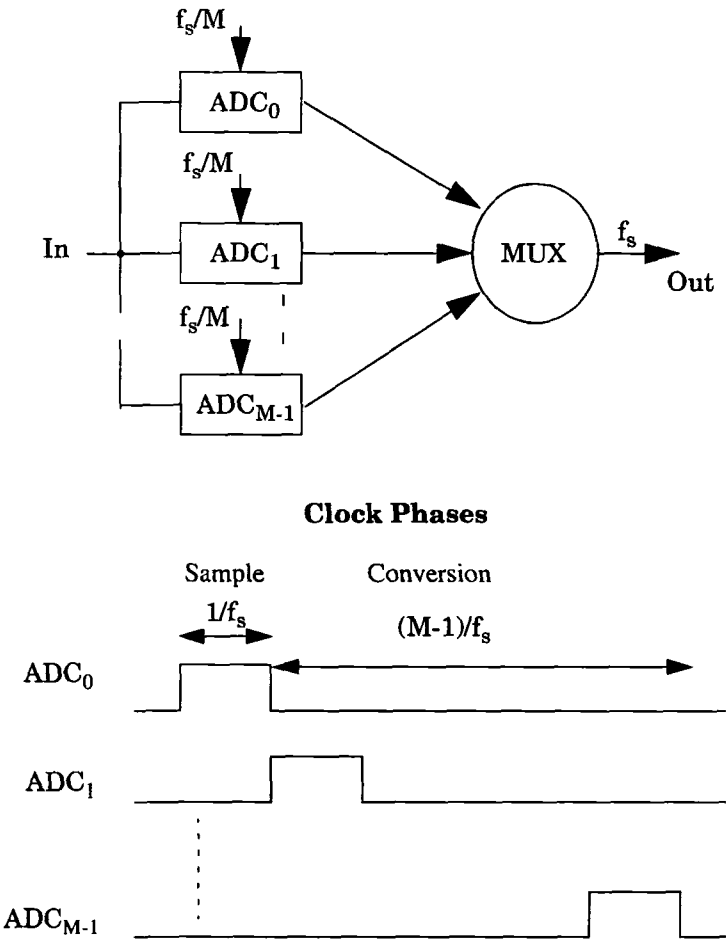


Figure 10-1 Parallel ADCs and corresponding sub-ADC timing.

histogram methods. It should be noted that errors in time-interleaved converters will not be detected in such a test. Therefore FFT spectrum testing methods are better suited [5].

The samples in channel $m = 0, \dots, M - 1$ is ideally sampled at time instants (see Fig. 10-2)

$$t_m, t_m + M, t_m + 2M, \dots \tag{10-1}$$

where $t_m = mT$ and T is the sampling period of the time-interleaved ADC. Due to imperfections in the sampling circuits the actual sampling instant will deviate from the ideal sampling instant. We assume that the delay skew error in channel m is constant,

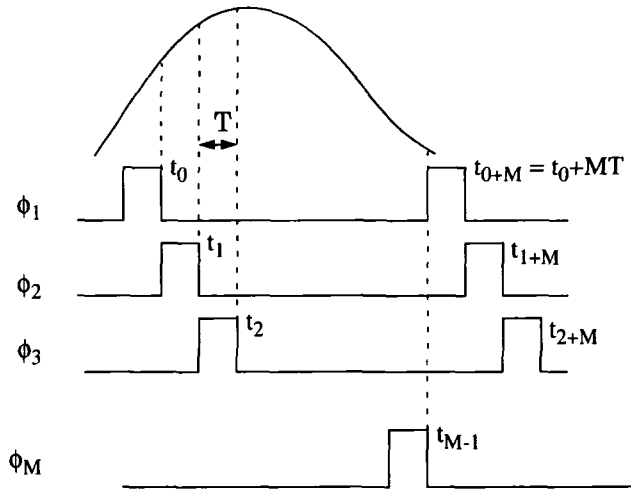


Figure 10-2 Sampling of the input signal by a time-interleaved ADC.

i.e. it does not change with time. The actual sampling instant in channel m can be expressed as

$$t_m = mT - r_m T \quad (10-2)$$

where r_m is the relative error in the sampling instant with respect to the sampling period of the time-interleaved ADC.

Assuming the analog input signal to have the spectrum $G_a(\omega)$, the digital spectrum of the time-interleaved system can be expressed as [3]

$$G(\omega) = \frac{1}{MT} \sum_{m=0}^{M-1} \left(\sum_{k=-\infty}^{\infty} G_a\left(\omega - k \frac{2\pi}{MT}\right) e^{-j\left(\omega - k \frac{2\pi}{MT}\right)t_m} \right) \cdot e^{-jm\omega T} \quad (10-3)$$

which can be rewritten as

$$G(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \left(\frac{1}{M} \sum_{m=0}^{M-1} e^{-j\left(\omega - k \frac{2\pi}{MT}\right)r_m T} e^{-jmk \frac{2\pi}{M}} \right) G_a\left(\omega - k \frac{2\pi}{MT}\right) \quad (10-4)$$

If there are no errors in the time-interleaved ADC we have [3]

$$G(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} G_a(\omega - 2\pi k) \quad (10-5)$$

which corresponds to a signal sampled at sampling rate T .

10.3 ERRORS IN TIME-INTERLEAVED ADCS

Based on the equations in the previous section we can derive the effects of mismatch in the channels.

10.3.1 Offset Errors

If we assume that there are no delay skew or gain errors in the channels the digital spectrum is expressed by

$$G(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \left(\frac{1}{M} \sum_{m=0}^{M-1} G_a\left(\omega - k\frac{2\pi}{M}\right) \cdot e^{-jmk\frac{2\pi}{M}} \right) \quad (10-6)$$

With a sine wave input signal, $A \cdot \sin(\omega_0 \cdot t)$, and channel offsets the Fourier transform of the analog signal, sampled by channel m is given by

$$G_a(\omega) = \pi j A (\delta(\omega + \omega_0) - \delta(\omega - \omega_0)) + 2\pi o_m \delta(\omega) \quad (10-7)$$

where o_m is the offset in channel m and A is the amplitude of the sine wave. By using (10-6) and (10-7) the digital spectrum can be expressed as

$$\begin{aligned} G(\omega) &= G_s(\omega) + \frac{1}{T} \sum_{k=-\infty}^{\infty} A(k) \cdot 2\pi \delta\left(\omega - k\left(\frac{2\pi}{MT}\right)\right) \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} \pi j A (\delta(\omega + \omega_0 - 2\pi k) - \delta(\omega - \omega_0 - 2\pi k)) + \\ &\quad + \frac{1}{T} \sum_{k=-\infty}^{\infty} A(k) \cdot 2\pi \delta\left(\omega - k\left(\frac{2\pi}{MT}\right)\right) \end{aligned} \quad (10-8)$$

where $G_s(\omega)$ is the digital spectrum of the sinusoidal sampled at the sampling rate $f_s = 1/T$ and

$$A(k) = \sum_{m=0}^{M-1} \left[\frac{1}{M} \cdot o_m \right] \cdot e^{-j2\pi \frac{km}{M}} \quad (10-9)$$

The first term of (10-8) corresponds to the input signal while the second term correspond to the distortion caused by channel offsets. The distortion is not signal dependent and appears at

$$f_s/M \cdot m, m = 0, 1, \dots, M-1 \quad (10-10)$$

This is illustrated by an example shown in Fig. 10-3. The input signal is a 10 MHz sine

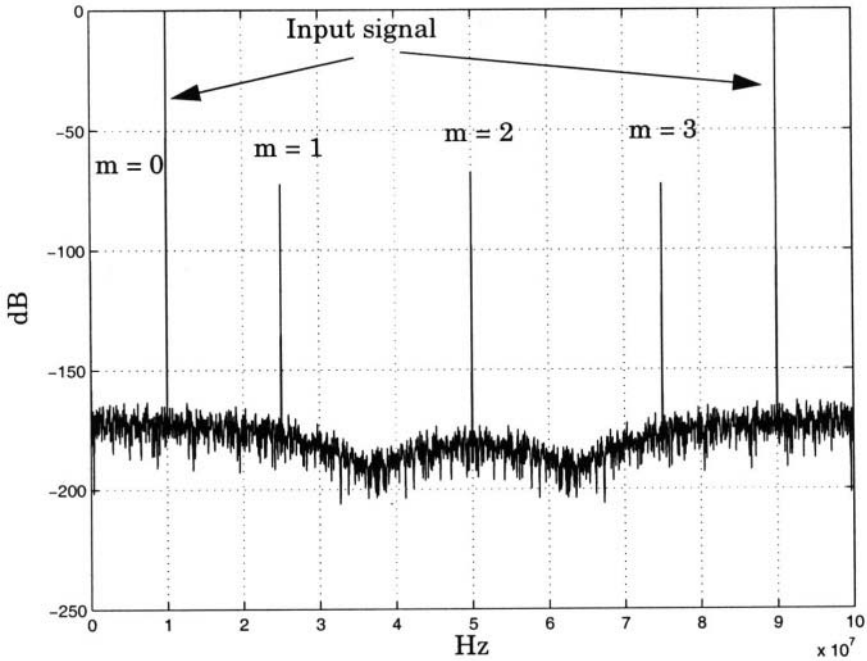


Figure 10-3 Output spectrum with offset errors.

wave sampled at 100 MS/s and the time-interleaved ADC has 4 channels. The simulation was done in MATLAB and the channel offsets were one set of samples of random variables with zero mean and a standard deviation of 0.001. The distortion tones appear at 0, 25, 50, 75 MHz while the input signal shows as two tones at 10 MHz and 90 MHz.

The signal power of the input signal is

$$P_s = A^2/2 \quad (10-11)$$

We see in (10-8) that the distortion consists of a sum of impulses. Each impulse corresponds to a complex exponential signal in the time domain, i.e. $e^{j\omega}$. The power of the exponential signal is 1. The factors $A(k)$ determines the amplitude of the signal and we can therefore calculate the total distortion power as

$$P_d = \sum_{k=0}^{M-1} |A(k)|^2 \quad (10-12)$$

The factors $A(k)$ in (10-9) can be interpreted as the Discrete Fourier Transform (DFT) of the sequence o_m/M , $m = 0, \dots, M-1$. From Parseval's relation we have

$$\sum_{m=0}^{M-1} (o_m/M)^2 = \frac{1}{M} \sum_{m=0}^{M-1} |A(m)|^2 \Rightarrow \sum_{m=0}^{M-1} |A(m)|^2 = \frac{1}{M} \sum_{m=0}^{M-1} o_m^2 \quad (10-13)$$

Assuming that the channel offsets are Gaussian random variables with zero mean and a variance of σ_o^2 the expected value of the distortion power is

$$P_d = E \left[\sum_{k=0}^{M-1} |A(k)|^2 \right] = E \left[\frac{1}{M} \sum_{m=0}^{M-1} o_m^2 \right] = \sigma_o^2 \quad (10-14)$$

Hence the distortion power is independent of the signal amplitude. The SNDR for a sinusoidal input signals is limited by

$$SNDR < 10 \cdot \log \left(\frac{P_s}{P_d} \right) = 10 \cdot \log \left(\frac{A^2}{2\sigma_o^2} \right) \quad (10-15)$$

The ENOB as function of σ_o is shown in Fig. 10-4. In the plot we assumed that

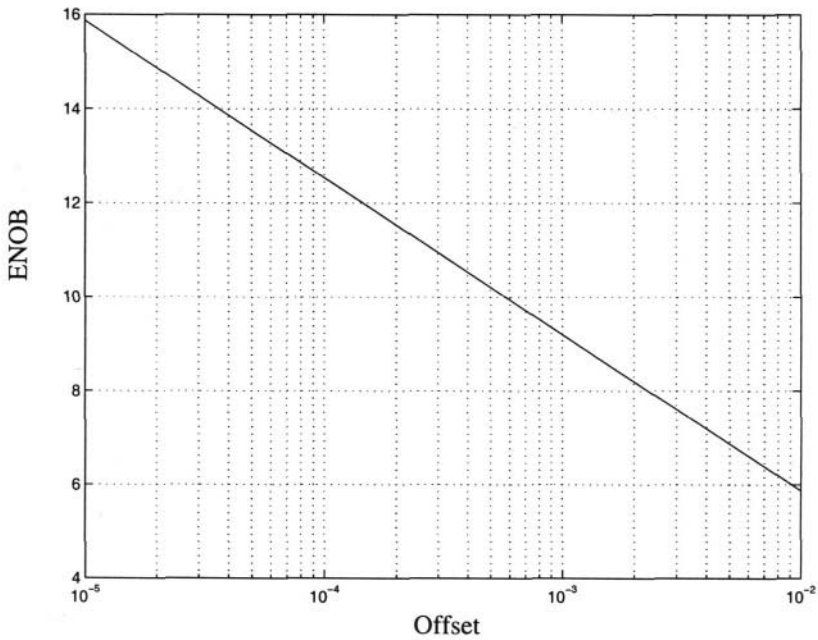


Figure 10-4 ENOB as function of offset errors.

$A = 1.$

10.3.2 Gain Errors

If we assume that there are no delay skew or offset errors between the channels the digital spectrum is expressed by

$$G(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \left(\frac{1}{M} \sum_{m=0}^{M-1} a_m G_a \left(\omega - k \frac{2\pi}{M} \right) \cdot e^{-jmk \frac{2\pi}{M}} \right) \quad (10-16)$$

where a_m is the gain in channel m . The Fourier transform of a sinusoidal signal with an amplitude of 1 is given by

$$G_a(\omega) = \pi j (\delta(\omega + \omega_0) - \delta(\omega - \omega_0)) \quad (10-17)$$

The digital spectrum can now be rewritten as

$$G(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} A(k) \cdot j\pi \left(\delta \left(\omega + \omega_0 - \frac{2\pi k}{MT} \right) - \delta \left(\omega - \omega_0 - \frac{2\pi k}{MT} \right) \right) \quad (10-18)$$

where $A(k)$ is determined by

$$A(k) = \sum_{m=0}^{M-1} \left[\frac{1}{M} \cdot a_m \right] \cdot e^{-j2\pi \frac{km}{M}} \quad (10-19)$$

If there are no gain errors $A(k)$ is

$$A(k) = \begin{cases} a & \text{if } k = 0, M, 2M, \dots \\ 0 & \text{for all other } k \end{cases} \quad (10-20)$$

where a is the nominal gain in the channels. This means that there is no distortion in the output signal since all tones appear at frequencies corresponding to the input signal. However, with gain errors $A(k) \neq 0$ for all k and according to (10-18) there are tones at

$$f_{in} + k \cdot f_s / M \text{ and } (f_s - (f_{in} + k \cdot f_s / M)), k = 1, \dots, M-1 \quad (10-21)$$

This is illustrated by an example in Fig. 10-5, showing an FFT of a time-interleaved ADC. The input signal is a 10 MHz sinusoid sampled at 100 MS/s and the time-interleaved ADC has 4 channels. Random gain errors with a mean value of 1 and a standard deviation of 0.001 were introduced. The fundamental obviously corresponds to $k = 0$ while $k = 1, \dots, M-1$ corresponds to the distortion. All distortion tones are folded into the Nyquist band 0 to $f_s/2$. The signal amplitude is determined by $A(0)$ while the distortion amplitudes are determined by $A(m)$, $m = 1, \dots, M-1$.

The factors $A(k)$ can be interpreted as the Discrete Fourier Transform (DFT) of the sequence a_m/M , $m = 0, \dots, M-1$. From Parseval's relation we have the total out-

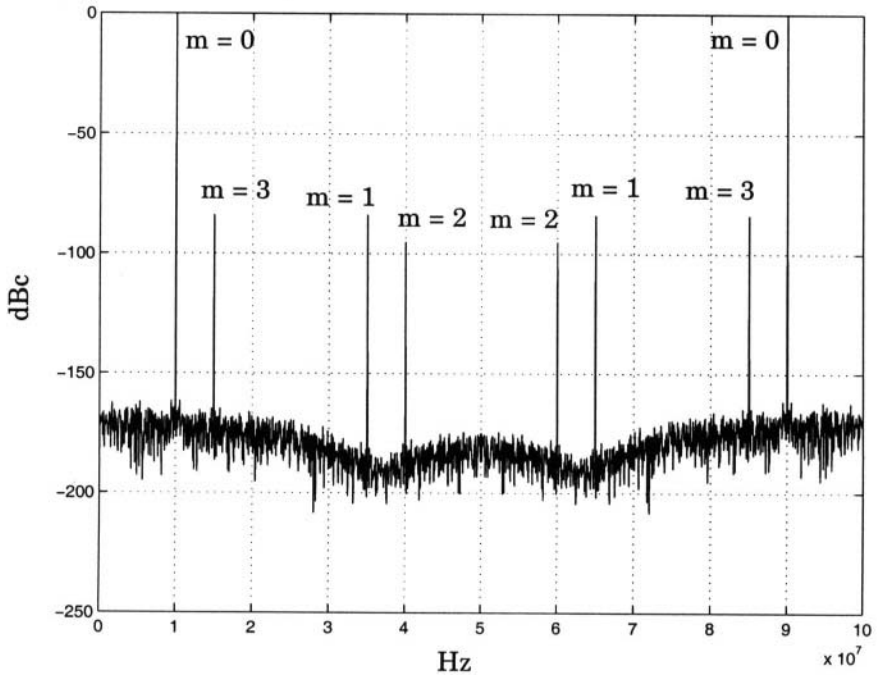


Figure 10-5 Output spectrum with gain errors.

put signal power is given by

$$P_{tot} = \sum_{m=0}^{M-1} \frac{|A(m)|^2}{2} = \frac{1}{2M} \sum_{m=0}^{M-1} a_m^2 \quad (10-22)$$

The fundamental corresponds to $k = 0$ and the signal power is given by

$$P_s = \frac{|A(0)|^2}{2} = \frac{1}{2} \sum_{m=0}^{M-1} \left[\frac{1}{M} \cdot a_m \right] \quad (10-23)$$

Assuming the gain errors to be random variables with a mean value of a and a standard deviation of σ_a the expected value of the total power is

$$P_{tot} = E \left[\frac{1}{2M} \sum_{m=0}^{M-1} a_m^2 \right] = \frac{1}{2M} \sum_{m=0}^{M-1} E[a_m^2] = \frac{E[a_0^2]}{2} = \frac{\sigma_a^2 + a^2}{2} \quad (10-24)$$

and the expectation of the signal power is

$$\begin{aligned}
P_s &= \frac{E[A(0) \cdot A^*(0)]}{2} = \frac{1}{2M^2} \sum_{m=0}^{M-1} \sum_{n=0}^{M-1} E[a_m \cdot a_n] = \\
&= \frac{1}{2M} E[a_m^2] + \frac{1}{2M^2} (M^2 - M) (E[a_m])^2 \\
&= \frac{1}{2M} (\sigma_a^2 + a^2) + \frac{1}{2M^2} (M^2 - M) a^2 = \frac{1}{2M} (\sigma_a^2 + M \cdot a^2)
\end{aligned} \tag{10-25}$$

The distortion power is the total power minus the signal power and the SNDR is limited by

$$\begin{aligned}
SNDR &= 10 \cdot \log \left(\frac{P_s}{P_{tot} - P_s} \right) = 10 \cdot \log \left(\frac{\frac{1}{2M} (\sigma_a^2 + M \cdot a^2)}{\frac{(\sigma_a^2 + a^2)}{2} - \frac{1}{2M} (\sigma_a^2 + M \cdot a^2)} \right) \\
&= 10 \cdot \log \left(\frac{\frac{1}{M} (\sigma_a^2 + M \cdot a^2)}{\left(1 - \frac{1}{M}\right) \cdot \sigma_a^2} \right)
\end{aligned} \tag{10-26}$$

Since the standard deviation of the gain error usually is much smaller than the nominal gain, (10-26) can be simplified to

$$SNDR \approx 20 \cdot \log(a/\sigma_a) - 10 \cdot \log \left(1 - \frac{1}{M} \right) \tag{10-27}$$

The last term in (10-27) depends on the number of channels, but will only change about 3 dB as M goes from 2 to ∞ . The ENOB as function of gain error is shown in Fig. 10-6. The figure shows the result for $M = 2, 4, 8$ and $a = 1$ in all cases. The approximation in (10-27) is based on that linear gain errors are allowed which is common in many applications. If gain errors are not allowed the SNDR can be approximated as the ratio of the input signal power and the difference in power between the input and output signal. The expected value of the SNDR is then given by

$$SNDR \approx 10 \cdot \log \left(\frac{a^2}{a^2 - (\sigma_a^2 + a^2)} \right) = 20 \cdot \log \left(\frac{a}{\sigma_a} \right) \tag{10-28}$$

which corresponds to the result in [4].

10.3.3 Phase Skew Errors

If we assume that there are no gain or offset errors between the channels the digital spectrum is expressed by (10-4) repeated here for convenience

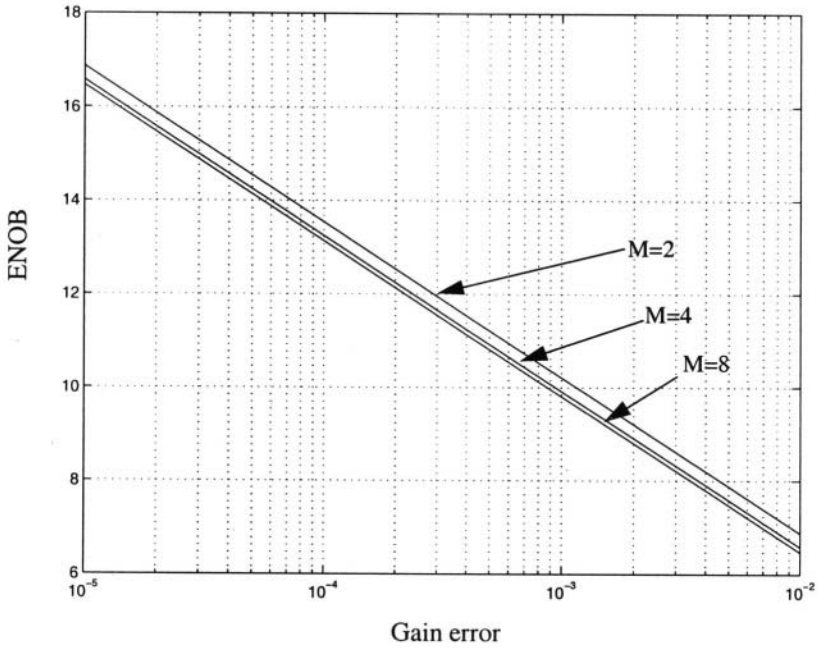


Figure 10-6 ENOB as function of gain errors.

$$G(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \left(\frac{1}{M} \sum_{m=0}^{M-1} e^{-j\left(\omega - k\frac{2\pi}{MT}\right)r_m T} e^{-jmk\frac{2\pi}{M}} \right) G_a\left(\omega - k\frac{2\pi}{M}\right) \quad (10-29)$$

The Fourier transform of a sinusoidal signal with an amplitude of 1 is given by

$$G_{a,m}(\omega) = \pi j(\delta(\omega + \omega_0) - \delta(\omega - \omega_0)) \quad (10-30)$$

The digital spectrum can now be rewritten as [3]

$$G(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} A(k) \cdot j\pi \left(\delta\left(\omega + \omega_0 - k\frac{2\pi}{MT}\right) - \delta\left(\omega - \omega_0 - k\frac{2\pi}{MT}\right) \right) \quad (10-31)$$

where $A(k)$ is determined by

$$A(k) = \sum_{m=0}^{M-1} \left[\frac{1}{M} \cdot e^{-j\omega_0 r_m T} \right] \cdot e^{-j2\pi \frac{km}{M}} \quad (10-32)$$

This is a similar expression as for gain errors and the distortion tones for phase skew errors will appear at signal frequencies

$$f_{in} + m \cdot f_s / M \text{ and } (f_s - (f_{in} + m \cdot f_s / M)), m = 1, \dots, M - 1 \quad (10-33)$$

If the phase skew errors are treated as Gaussian random variables with zero mean and a variance of σ_t^2 the same principle as for gain errors can be adopted to derive the SNDR. The SNDR can be approximated as [3]

$$SNDR = 20 \cdot \log\left(\frac{1}{\sigma_t 2\pi f_{in}}\right) - 10 \cdot \log\left(1 - \frac{1}{M}\right) \quad (10-34)$$

Just as for gain errors the last term in (10-34) will disappear if the SNDR is calculated as the ratio of the input signal power and the total error power. Fig. 10-7 shows how

ENOB

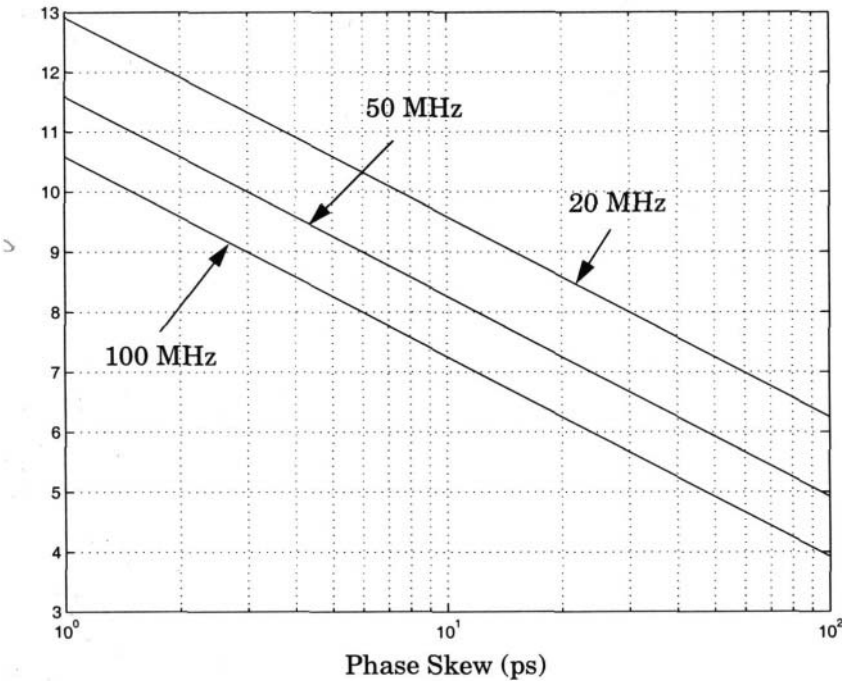


Figure 10-7 ENOB versus phase skew in a four channel ADC.

the ENOB is affected by phase skew for $M = 4$ at some different input signal frequencies. At 20 MHz and a resolution of 10 bits σ_t should be less than 8 ps. This is difficult to achieve in a CMOS process.

10.4 ERROR REDUCTION TECHNIQUES

There are several techniques to reduce the effect of mismatch in the channels. Offset

and gain errors can be removed by calibration. Phase skew errors are usually more difficult to handle since it is difficult to measure the actual skews.

10.4.1 Two-Rank Sample-and-Hold

The phase skew can almost be eliminated by using a S/H circuit at the input as shown in Fig. 10-8. The output of this S/H circuit is a discrete-time analog signal and it is not

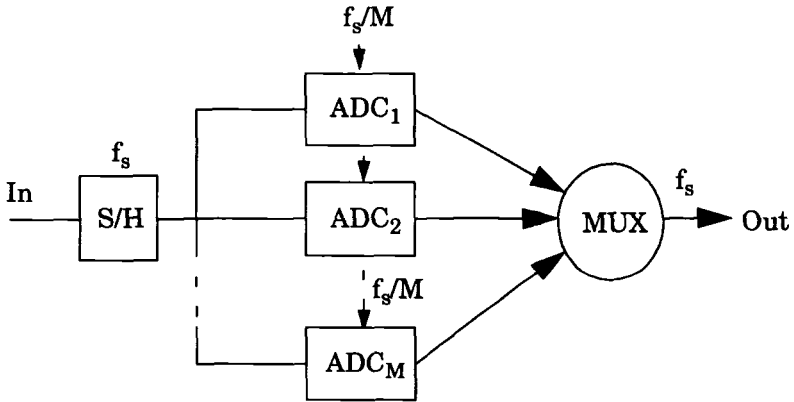


Figure 10-8 Two-Rank S/H circuit.

continuously changing. The phase skew errors have no effect on the result as long as the input S/H settles to its final value. The obvious disadvantage of this method is that the S/H circuit must be operated at the full speed and it is difficult to design high-speed, high-accuracy S/H circuits in CMOS. This technique was used in [6] and [7] to get 10 bits at 100 MHz in CMOS and 6 bits at 1 GHz in GaAs. The two-rank sampling will only improve delay skew errors and have no effect on gain and offset errors.

10.4.2 Averaging

If the input signal is oversampled, averaging can be used to reduce the channel mismatch [8]. The concept of averaging for $M = 2$ is illustrated in Fig. 10-9. The transfer function of the averaging block is

$$H(z) = \frac{1}{2}(1 + z^{-1}) \quad (10-35)$$

and the Fourier transform

$$|H(j\omega)|^2 = \frac{1}{2}(1 + \cos(\omega T_s)) \quad (10-36)$$

There is a zero at $f_s/2$ and the channel mismatch errors, located in the frequency band $[f_s/4, f_s/2]$ will be low-pass filtered. The distortion term at $f_s/2$ caused by offset errors is completely removed while gain and phase skew errors are reduced by a factor

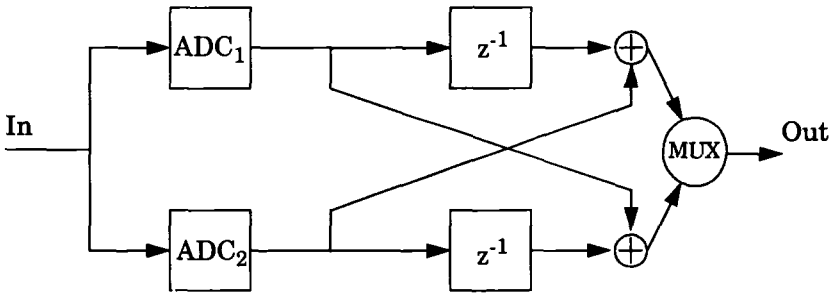


Figure 10-9 The principle of averaging.

$4/(\omega_{in} T_s)$. The concept of using digital processing after the time interleaved ADC can be generalized. With an oversampling ratio of M and M time-interleaved channels no distortion terms appear in the signal band. The distortion can thus be removed by digital filters. The drawback of this method is that even though we have increased the sampling rate by parallelism the input bandwidth is not increased due to the oversampling.

10.4.3 Randomization

By randomly choosing the channel to convert the next sample, the distortion caused by gain and offset mismatch will be reduced [9] since the correlation between the channel errors and the input signal is reduced. However the noise floor will increase since the distortion is randomized to noise. This method can not reduce the effect of phase skew errors and a serious drawback is that the individual converters must use a clock signal which is at least as fast as the total sampling speed of the time-interleaved ADC. Another problem is that when having many time-interleaved channels it may be difficult to do the layout.

10.4.4 Calibration

There are several calibration techniques to improve the linearity of pipelined ADCs, e.g. [10], [11]. A drawback with these techniques is that the conversion must be interrupted when the ADC is calibrated. It is sometimes preferable to use a calibration algorithm that works in the background [12], [13]. When using these calibration techniques the linearity of the converter can be very good but in time-interleaved converters the matching between the channels must also be considered. Both analog and digital calibration techniques for time-interleaved ADCs have been reported which can reduce the effect of gain and offset errors [14]. The phase skew errors are more difficult to calibrate, but in [15], a method of estimating the phase skew errors is proposed. However this method can not easily be implemented as a background calibration technique.

10.4.5 Filter Banks

By introducing signal processing before and after the individual ADCs the performance can be improved. In [16] a quadrature mirror filter (QMF) was used to reduce the effect of gain and offset mismatch as shown in Fig. 10-10. The discrete-time filters

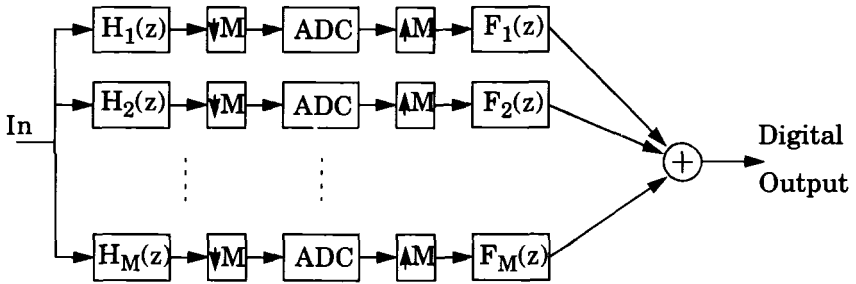


Figure 10-10 Time-interleaved ADCs with a QMF bank.

$H_i(z)$ are SC filters that are operated at the full speed of the time-interleaved ADC. This is a serious drawback since it makes the SC filters design too difficult. However by using polyphase structures it is possible to run the filters at the lower channel speed which makes a practical implementation feasible. Channel gain and offset errors appearing after the filters $H_i(z)$ will be suppressed and have a small effect on the performance, but unfortunately the error suppression in the transition band of the filters is very small. Another limitation is that delay skew errors will not be improved by this type of filter bank. The discrete-time filters at the input can be replaced by continuous-time filters [17]. These hybrid filter banks have the advantage that also delay skew errors will be suppressed (except in the transition bands). When using passive filters very high speeds can be achieved. However for monolithic implementations it is very difficult to make high performance passive components. Similar methods for sigma-delta ADCs where filter banks and modulation have been used to improve performance with several parallel channels have been proposed in [18] and [19].

10.5 PASSIVE SAMPLING TECHNIQUE

One limitation of time-interleaved ADCs is the sampling phase skew distortion. One way to overcome this problem is to place a track-and-hold amplifier at the input. This is not desirable since it requires a high-gain operational amplifier driving a large capacitive load at a very high frequency. In this section, we discuss a passive sampling technique for SC circuits to reduce the influence of the sampling phase skew. Since it does not require operational amplifiers, it is suitable for high-speed applications and yet simulations indicate that it can reduce the sampling-phase-skew-related distortion by 10~20 dB in a high-speed time-interleaved SC ADCs.

10.5.1 Sampling in SC Time-Interleaved ADCs

The time-interleaved ADC shown in Fig. 10-11 consists of M identical sub ADCs

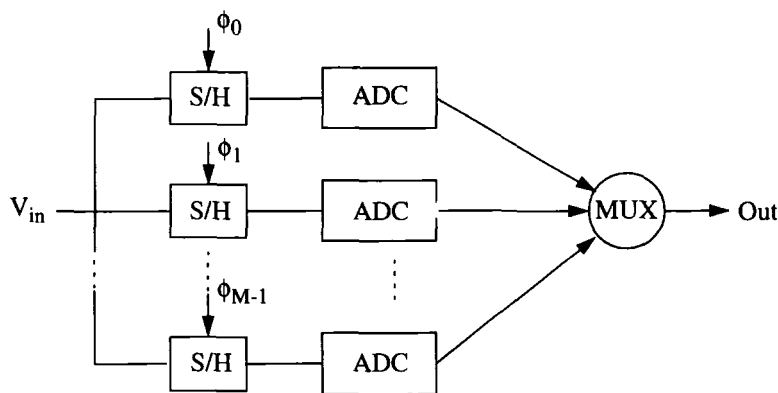


Figure 10-11 Time-interleaved ADC, the block diagram.

preceded by a passive sampling circuit. The passive sampling circuit and its corresponding clock phases are shown in Fig. 10-12. The first sub-ADC samples the input

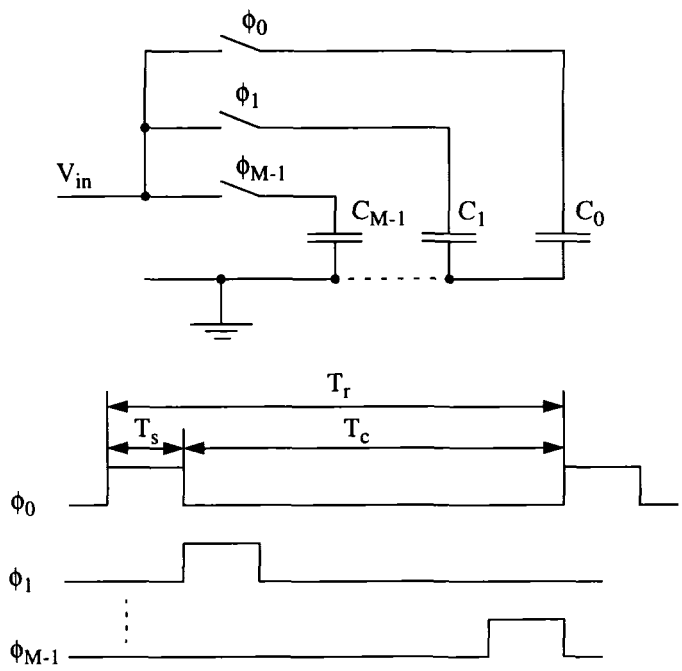


Figure 10-12 Input passive sampling circuit, and clock phases.

voltage V_{in} on clock phase ϕ_1 , the second sub-ADC samples the input voltage V_{in} on clock phase ϕ_2 , and the M -th sub-ADC samples the input voltage V_{in} on clock phase ϕ_M .

Assume that the sampling duration for every phase is T_s , and the repetition period for every phase is T_r . The time for every sub-ADCs to convert the sampled analog value is T_c , given by

$$T_c = T_r - T_s = (M - 1) \cdot T_s \quad (10-37)$$

The only high-speed part is the passive sampling circuit, consisting of capacitors and switches, which needs to track and sample the analog input during the time interval T_s . This passive sampling circuit can be designed to be very fast and can handle a very large input bandwidth. The problem with phase skew errors arises when the M different clock phases shown in Fig. 10-12 is generated. Very small differences in delay between the clock phases generates large distortion for high signal frequencies.

10.5.2 Improved Passive Sampling Technique

The use of two-rank sampling solves the problem with delay skews distortion. But it naturally requires an opamp. This opamp must run at the full speed of the ADC and it is therefore difficult to design and power consuming. If we can have a global sampling technique without an opamp we would improve performance without much penalty. This leads to the improved passive sampling technique where a single clock phase is used to define the sampling instant. This is illustrated in Fig. 10-13.

The improved sampling circuit is controlled by a global clock phase ϕ and it defines the sampling instant. When the clock phase ϕ is high and ϕ_m is high, the input is sampled by the i -th sub-ADC. When the clock phase ϕ goes low, the analog value is sampled by the sampling capacitor since one plate of the sampling capacitor is floating. The clock phases ϕ_i always goes low after the clock phase ϕ goes low. Even if there are large phase skew between successive clock phases ϕ_i , they do not have any influence on the sampling instant and therefore the effect of the phase skew is eliminated. However, due to the parasitic capacitance, the charge stored on the sampling capacitor still changes with the analog input even when the clock phase ϕ is low.

10.5.3 Effect of Parasitic Capacitors

When an extra switch is introduced in the sampling, the parasitic capacitance has influence on the sampling instant even for a single channel. We show the parasitic capacitors associated with one channel in Fig. 10-14. All the switches are assumed to be NMOS transistors for simplicity. C_{p1} represents the parasitic capacitance at the right hand side of the sampling capacitor, and C_{p2} represents parasitic capacitance between switch transistors M_2 and M_3 . Fig. 10-15 shows the circuit after the sampling switch (M_3) is opened while the other switches (M_1 and M_2) controlled by ϕ_m are still closed. The switch-on resistance of the switches is neglected for simplicity. The time instant when the sampling switch (M_3) is opened is denoted t and the instant when switch M_2 is opened is denoted $t + \tau$. The charge stored on C_{p2} will cause a signal dependent error on the output signal. At time t the total charge on the right hand side

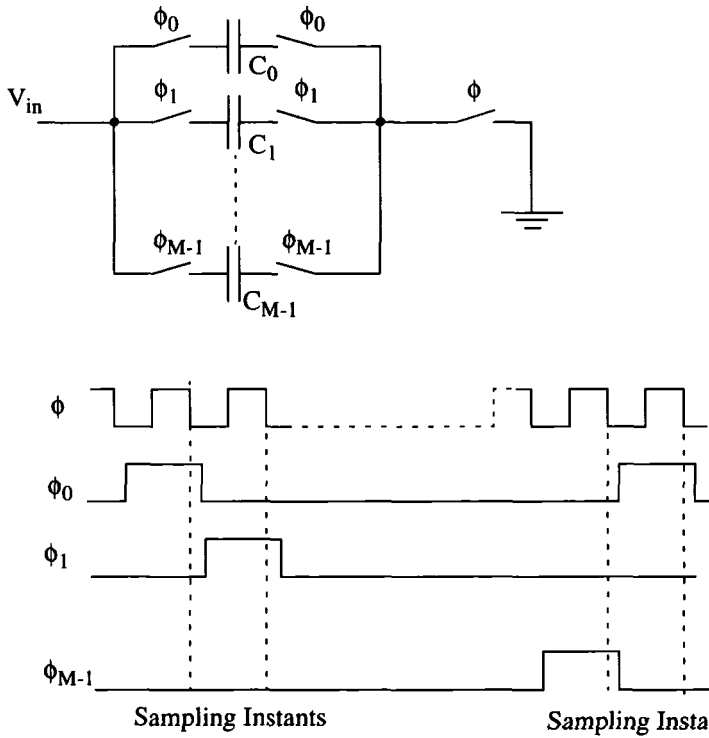


Figure 10-13 Proposed passive sampling technique for time-interleaved ADCs.

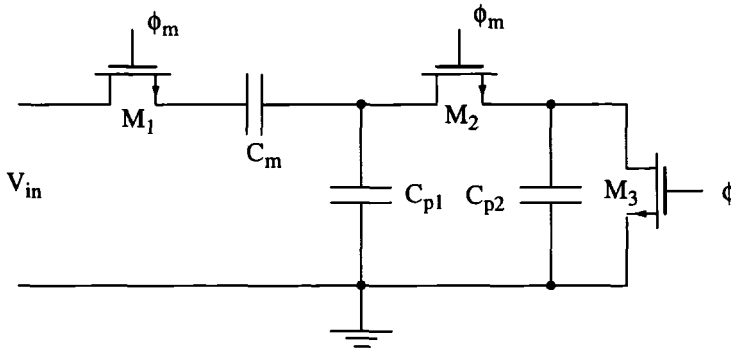


Figure 10-14 Parasitic capacitors in one channel.

node of C_m is

$$\begin{aligned}
 q(t) &= q_{C_m}(t) + q_{C_{p1}}(t) + q_{C_{p2}}(t) = \\
 &= -C_m V_{in}(t) + 0 + 0 = -C_m V_{in}(t)
 \end{aligned}
 \tag{10-38}$$

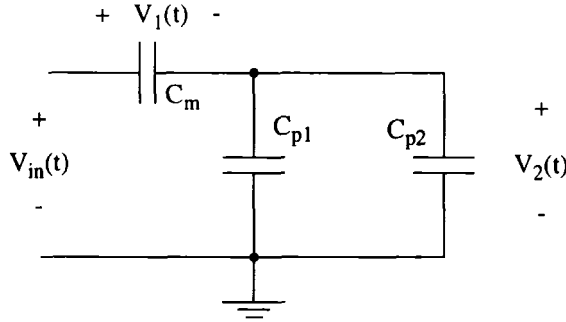


Figure 10-15 The circuit when the sampling switch M_3 is open.

At $t + \tau$, when switch M_2 is opened, the total charge on the right hand side of the sampling capacitor C_m is given by

$$\begin{aligned} q(t + \tau) &= q_{C_m}(t + \tau) + q_{C_{p1}}(t + \tau) + q_{C_{p2}}(t + \tau) \\ &= (V_2(t + \tau) - V_{in}(t + \tau)) \cdot C_m + V_2(t + \tau) \cdot (C_{p1} + C_{p2}) \end{aligned} \quad (10-39)$$

Due to charge conservation, the charge in equations (10-38) and (10-39) should be equal. Therefore, we have the voltage across the parasitic capacitors at $t + \tau$ given by

$$V_2(t + \tau) = (V_{in}(t + \tau) - V_{in}(t)) \cdot \frac{C_m}{C_{p1} + C_{p2} + C_m} \quad (10-40)$$

The charge stored on C_{p2} is given by

$$\begin{aligned} q_{C_{p2}}(t + \tau) &= C_{p2} \cdot V_2(t + \tau) = \\ &= (V_{in}(t + \tau) - V_{in}(t)) \cdot \frac{C_i \cdot C_{p2}}{C_{p1} + C_{p2} + C_m} \end{aligned} \quad (10-41)$$

After the switch M_2 opens, the charge stored on C_{p2} will be lost while all the charge stored on the sampling capacitor C_m and the parasitic C_{p1} will be transferred during the hold phase when an opamp is used. Assuming an ideal opamp, all the charge stored on the sampling capacitor C_m and the parasitic C_{p1} will be completely transferred. The only error source is due to the lost charge stored on C_{p2} at $t + \tau$. Therefore the analog output voltage after the sampling is given by

$$V_{out} = -\frac{q(t + \tau) - q_{C_{p2}}(t + \tau)}{C_m} = V_{in}(t) \cdot (1 - \alpha) + \alpha \cdot V_{in}(t + \tau) \quad (10-42)$$

where

$$\alpha = \frac{C_{p2}}{C_{p1} + C_{p2} + C_m} \quad (10-43)$$

It is seen that using this sampling technique for one channel, an error is introduced due to the parasitic capacitance. When we apply the passive sampling technique to a time-interleaved ADC, mismatch in the parameters α and τ between the channels will introduce distortion. Referring to Fig. 10-13 and Fig. 10-14, we assume that there are M time-interleaved channels and that the switch $\mathbf{M}_3$ controlled by clock phase ϕ opens at time instants

$$T_s \cdot n, n = 0, \dots, \infty \quad (10-44)$$

and that the switch controlled by clock phase ϕ_m in channel m ($m = 0, \dots, M-1$) opens at

$$m \cdot T_s + n \cdot M \cdot T_s + \tau + t_{skew, m}, n = 0, \dots, \infty \quad (10-45)$$

where T_s is the sampling period, τ is the average delay between the turn off of switch $\mathbf{M}_3$ and switches $\mathbf{M}_2$ in the m -th channel and $t_{skew, m}$ is the relative clock skew of clock phase ϕ_m . If the parasitic capacitors and sampling capacitors for all the channels are assumed to be equal, i.e. the factor α is equal for all the channels, and the time skews are assumed to be independent random variables with normal distribution and variance σ_t^2 the SNDR can be approximated as (see Sec. 10.7)

$$SNDR = 20 \cdot \log\left(\frac{1}{\sigma_t \cdot 2\pi f_{in}}\right) - 10 \cdot \log\left(1 - \frac{1}{M}\right) - 20 \cdot \log(\alpha) \quad (10-46)$$

for small α and small values on $f_{in}\tau$, where f_{in} is the input signal frequency. Eq.(10-46) shows that with parasitic capacitors the effect of phase skew errors is not completely removed but it is reduced by the factor $1/\alpha$ compared to time-interleaved ADCs using the ordinary sampling techniques. To investigate the effect of mismatch in the capacitors it is assumed that there are no phase skew errors and that α_m is the capacitor ratio factor for channel m . If the factors α_m are independent random variables with normal distribution, mean value α and variance σ_α^2 the SNDR due to capacitor mismatch can be approximated as (see Sec. 10.7)

$$SNDR = 20 \cdot \log\left(\frac{1}{\sigma_\alpha}\right) - 10 \cdot \log\left(1 - \frac{1}{M}\right) - 10 \cdot \log(|e^{j2\pi\tau f_{in}} - 1|^2) \quad (10-47)$$

for small σ_α and small $f_{in}\tau$. In order to maintain a large SNDR, we need a small τ .

A conventional sampling technique for two time-interleaved channels is shown in Fig. 10-16. Two non-overlapping clock phases are used. We also use bottom plate sampling, i.e. the rightmost switch is opened slightly before the leftmost switch, since this will minimize the signal dependent clock feedthrough. In the clock phase when the capacitors are disconnected from the input they are usually connected to an opamp before they are converted by a sub ADC. Here the capacitors are simply connected to AC-ground for simplicity. The AC-ground voltage is chosen to 1 V while. All the

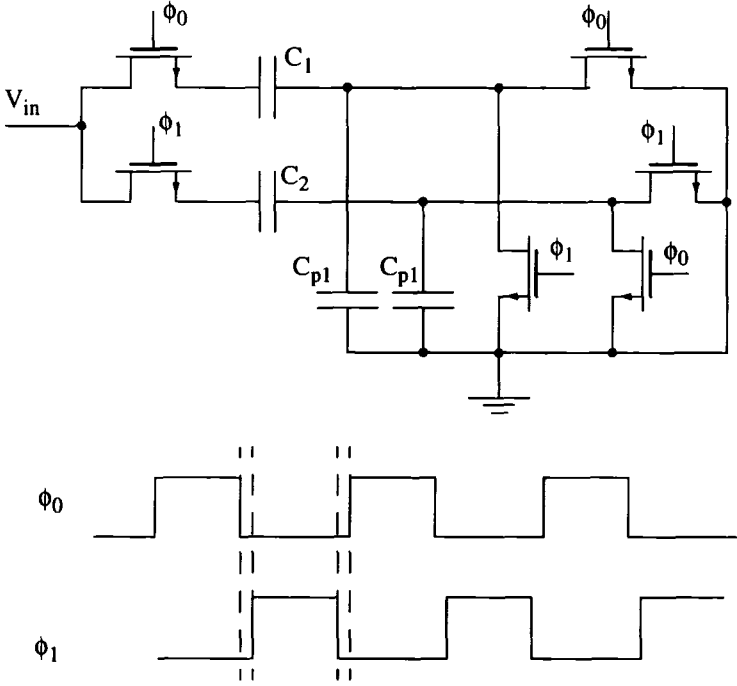


Figure 10-16 The circuit when the sampling switch M_3 is open.

bulks of the transistors are connected to ground. The time gap between the non-overlapping clock phases is 2 ns and the clock voltage swing is 0 to 5 V. The clock phase for the right most switches in Fig. 10-16 is delayed 1 ns to reduce clock feedthrough errors. The rise and fall times for all clock phases are 200 ps. The sampling capacitors are 1 pF and all the transistor sizes are $50 \mu\text{m} / 0.6 \mu\text{m}$. The parasitic capacitors C_{p1} are approximately equal to C_{gs} of the switches, here approximately 50 fF. The passive sampling circuit was simulated using a sampling rate of 50 MS/s, i.e. each channel samples at 25 MS/s, and a 18 MHz input signal. A 50 ps delay skew was introduced in one of the channels. From [3] we have that for a two channel S/H with delay skew errors the SNDR is limited by

$$\begin{aligned}
 SNDR &= 10 \cdot \log \left(\frac{\cos^2(\pi \cdot t_{skew} \cdot f_{in})}{\sin^2(\pi \cdot t_{skew} \cdot f_{in})} \right) \\
 &= 10 \cdot \log \left(\frac{\cos^2(\pi \cdot 50p \cdot 18M)}{\sin^2(\pi \cdot 50p \cdot 18M)} \right) = 51\text{dB}
 \end{aligned} \tag{10-48}$$

A 500-point FFT plot from the simulation both with and without delay skew is shown in Fig. 10-17 and Fig. 10-18. The distortion caused by the delay skew appears at 7 MHz and is approximately 51 dB. The other distortion tones in the FFT is harmonic distortion from clock feedthrough errors and nonlinearities in the transistors.

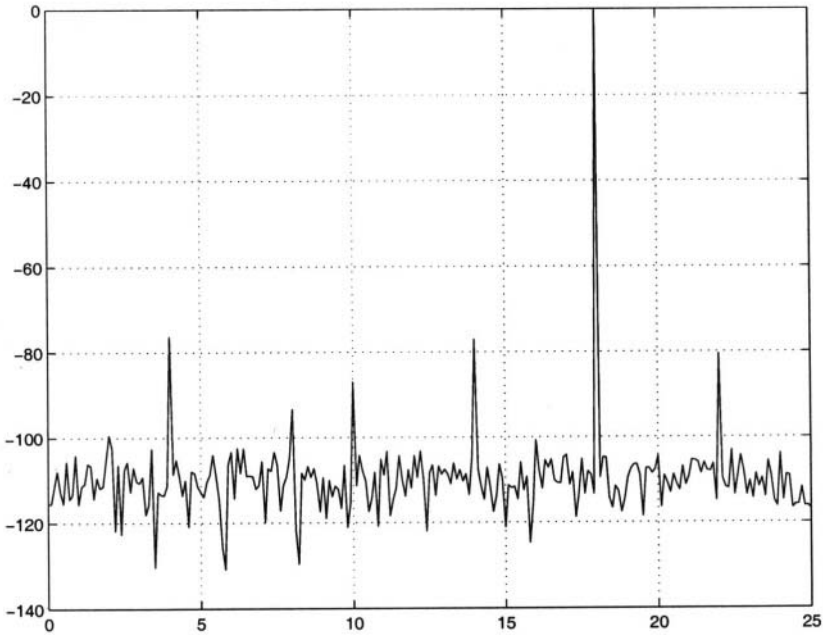


Figure 10-17 Simulated spectrum without phase skew. The conventional sampling technique is used.

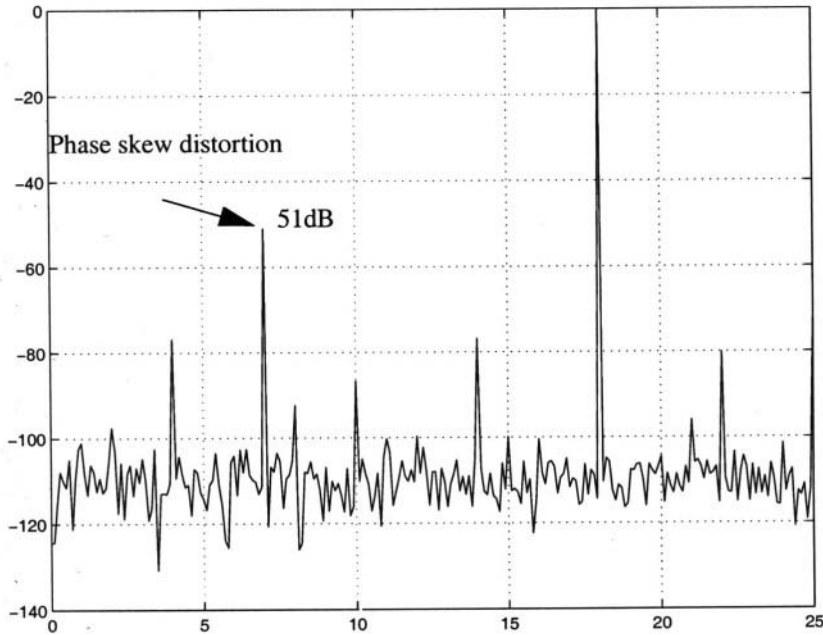


Figure 10-18 Simulated response with 50ps phase skew. The conventional sampling technique is used.

The circuit used for simulating the improved switching technique and the corresponding clock phases are shown in Fig. 10-19. The clock phases for the conventional sam-

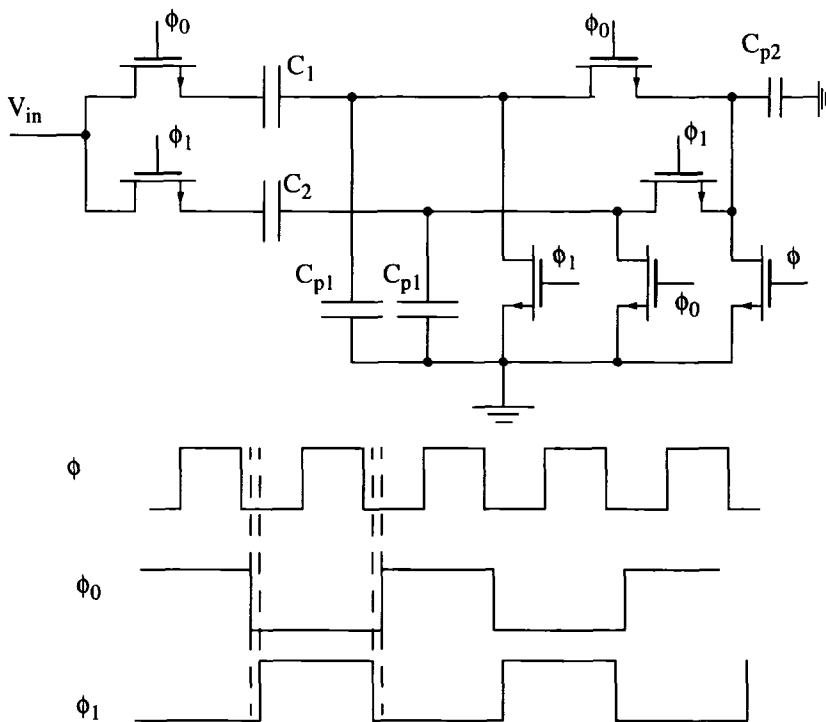


Figure 10-19 Improved sampling technique applied to 2-channel ADC.

pling technique are used. The only difference is the clock phase used for the additional sampling transistor. The additional sampling switch is turned off 1 ns before the bottom plate switch in the channels. The parasitic capacitors are approximately $C_{p1} = 50$ fF and $C_{p2} = 150$ fF where C_{p2} is the gate-source capacitance of a switch transistor and an additional 100 fF capacitor to account for parasitic elements in the layout. According to (10-46) we can expect the distortion to be reduced by a factor α . It is given by

$$\alpha = \frac{C_{p2}}{C_{p1} + C_{p2} + C_m} = \frac{150}{50 + 150 + 1000} = 0.125 \quad (10-49)$$

which corresponds to an 18 dB improvement. Thus we expect the phase skew distortion to be approximately 69 dB.

A 500-point FFT both with and without a 50 ps phase skew is shown in Fig. 10-20 and Fig. 10-21. The phase skew distortion is approximately 68.5 dB which is close to the predicted value. Fig. 10-20 shows that the harmonic distortion is larger than for the conventional sampling technique. This is mainly caused by the additional 100 fF

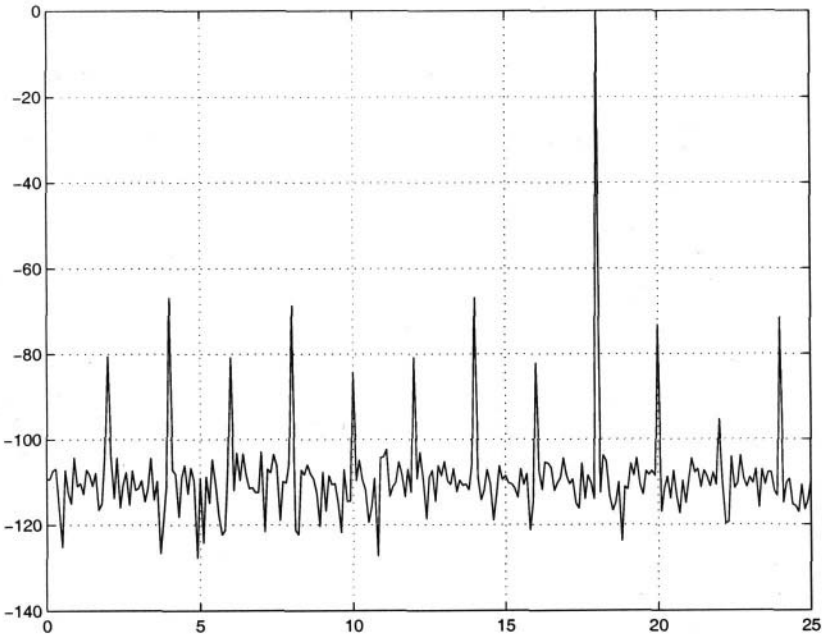


Figure 10-20 Simulated response and without phase skew. The improved sampling technique is used.

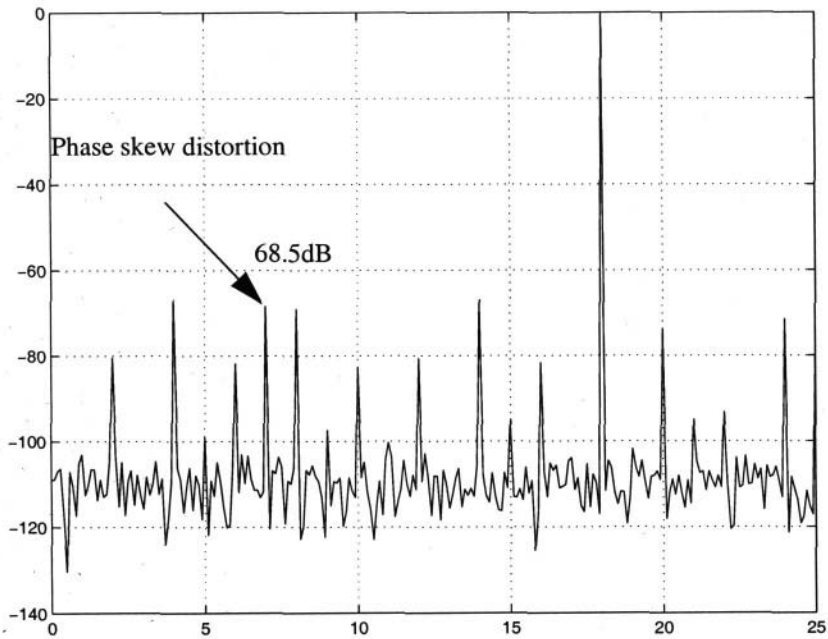


Figure 10-21 Simulated response and with 50ps phase skew. The improved sampling technique is used.

capacitor (C_{p2}). A simulation without the additional capacitor showed a harmonic distortion similar to the conventional sampling technique and a phase skew distortion of less than 80 dB. Table 10-1 summarized the results from the simulations.

Table 10-1 Phase skew distortion in simulation.

	Improved	Conventional
Phase skew = 50 ps, $C_{p2} = 50\text{fF}$	80 dB	51 dB
Phase skew = 50 ps, $C_{p2} = 150\text{fF}$	68.5 dB	51 dB

10.6 REALIZATION OF THE IMPROVED SAMPLING TECHNIQUE

In the previous section only the passive part of the S/H circuit was considered. To process the sampled analog value in the sub-ADCs, the sampled value must be held constant and for high accuracy one opamp is needed for every sub-ADC. Fig. 10-22.

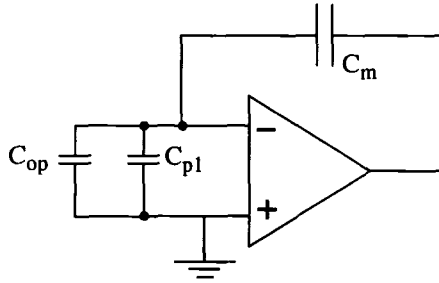


Figure 10-22 The opamp in the hold phase.

shows how the sampling capacitor is connected in the hold phase. The bottom plate parasitic capacitor C_{p1} is connected between the input of the opamp and ground as well as the input parasitic capacitance of the opamp C_{op} . Assuming a finite DC gain A of the opamp and using (10-42), the output voltage in the hold phase is given by

$$V_{out} = (V_{in}(t) \cdot (1 - \alpha) + \alpha \cdot V_{in}(t + \tau)) \cdot b \quad (10-50)$$

where

$$a = \frac{C_{p2}}{C_{p1} + C_{p2} + C_m} \text{ and } b = \frac{1}{1 - \left(1 + \frac{(C_{p1} + C_{op})}{C_m}\right) \frac{1}{A}} \quad (10-51)$$

Hence the finite gain in the opamp cause gain errors in the channels. Notice that the settling time for the operational amplifier corresponds to the slower channel sampling rate and therefore the speed requirement of the opamp is not high.

10.6.1 M-Channel S/H Circuit

Fig. 10-23. shows how the S/H circuit is implemented for M channels. The clock

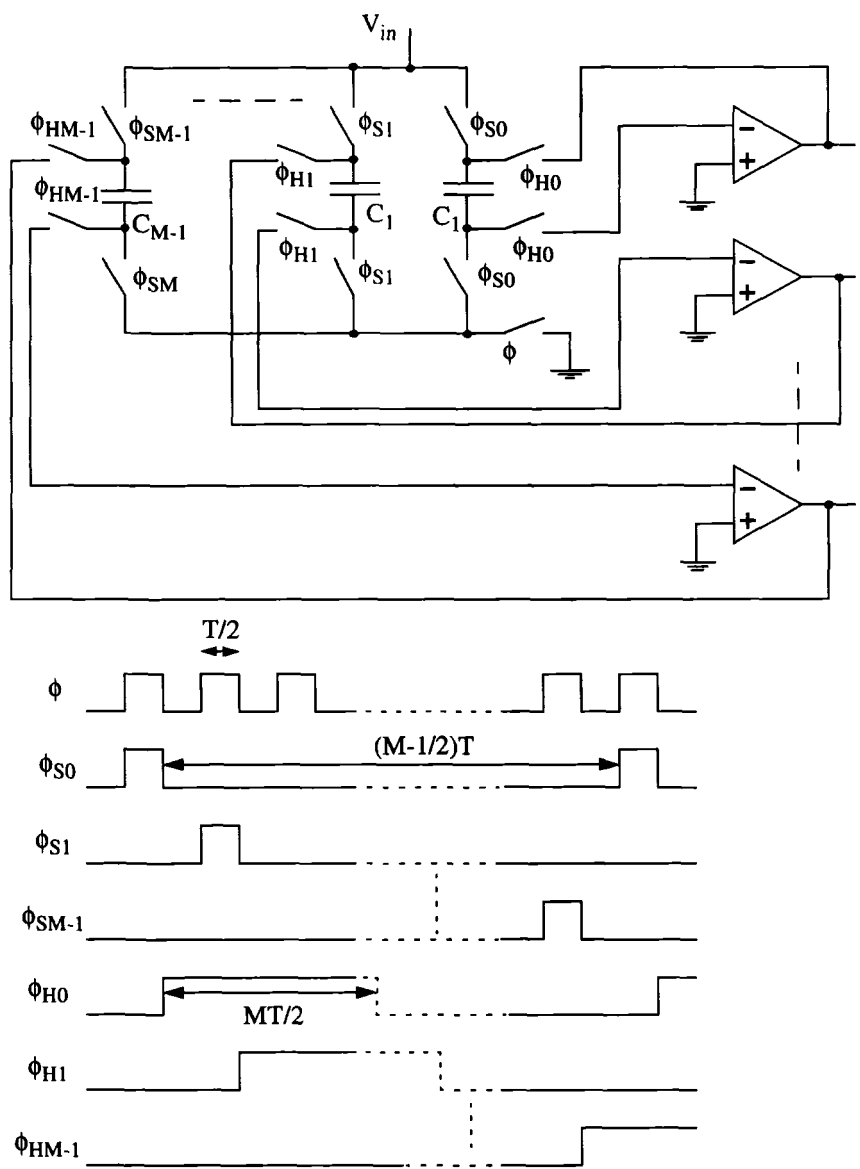


Figure 10-23 The improved sampling technique for an M -channel ADC.

phase defining the sampling instant ϕ runs at the full speed of the time-interleaved ADC, i.e. the clock period is T . The sampling phases ϕ_{Sm} , $m = 0, \dots, M - 1$ have a clock period of $T \cdot M$. The sampling phases are shown in Fig. 10-23 where the on-

time of a sampling phase is only $T/2$. The hold clock phases ϕ_{Hm} , $m = 0, \dots, M-1$ have a clock period of $M \cdot T$ but their on-time is $TM/2$. This means that the settling time for the opamps are $T \cdot M/2$. This is typically the case when the circuit is used with pipelined ADCs. If other types of converters are used it may be possible increase the hold-time to $T(M-1)$.

10.6.2 Opamp Sharing Technique

The opamps are power consuming parts of the ADC. The number of opamps can be reduced by using opamp sharing techniques. For a 2-channel ADC, one opamp can be shared by both channels [2], [20] as show in Fig. 10-24. The technique can in principle

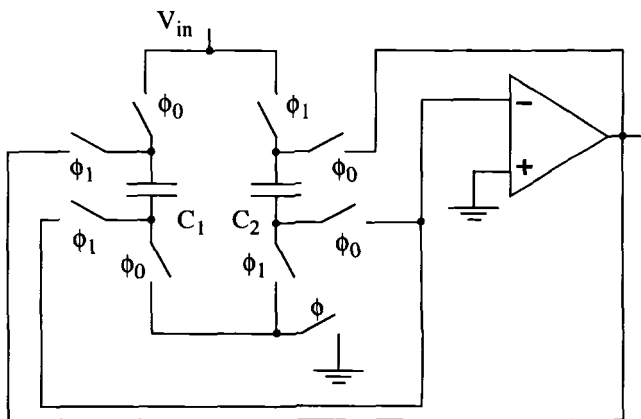


Figure 10-24 Opamp sharing for a 2-channel ADC.

be extended to a M -channel time-interleaved ADC. It can reduce the number of opamps by a factor 2.

10.7 DERIVATIONS

In this section we derive the effect of parasitic capacitors in the improved sampling technique discussed previously in this chapter.

10.7.1 Output Spectrum

According to Sec. 10.5.3 the sampled signal in channel m is

$$v_s = v(t) \cdot (1 - \alpha_m) + \alpha_m \cdot v(t + \tau_m) \quad (10-52)$$

where τ_m is the delay between the sampling switch turn-off time and the channel switch turn-off time (see Sec. 10.5.3) and

$$\alpha_m = \frac{C_{p2}}{C_{p1} + C_{p2} + C_m} \quad (10-53)$$

Due to circuit imperfections the gain factors and delays are different in the channels. The delay in channel m can be expressed as $\tau_m = t_d + r_m/T$ where t_d is the nominal delay, r_m the relative error in the delay and T the sampling period of the time-interleaved ADC. By using the results in [3], the digital spectrum for v_s can be expressed as

$$G(\omega) = G_1(\omega) - G_2(\omega) + G_3(\omega) \quad (10-54)$$

where

$$G_1(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} G_a\left(\omega - \frac{2\pi k}{T}\right), \quad (10-55)$$

$$G_2(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \left(\frac{1}{M} \sum_{m=0}^{M-1} \alpha_m e^{-jkm2\pi/M} \right) G_a\left(\omega - \frac{2\pi k}{T}\right) \quad (10-56)$$

and

$$G_3(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \left(\frac{1}{M} \sum_{m=0}^{M-1} \alpha_m \cdot e^{-j\left(\omega - \frac{2\pi k}{MT}\right)(r_m T - t_d) - jmk \frac{2\pi}{M}} \right) \cdot G_a\left(\omega - k \frac{2\pi}{M}\right) \quad (10-57)$$

$G_a(\omega)$ is the analog spectrum of the input signal $v(t)$ and $G_1(\omega)$ is the spectrum of the sampled input signal. $G_1(\omega)$ corresponds to the spectrum of a signal sampled with channel gain mismatch, i.e. $\alpha_i \cdot v(t)$. $G_3(\omega)$ is the spectrum of a signal with both gain and phase skew errors and corresponds to the last term in (10-52), $\alpha_i \cdot v(t + \tau_i)$. For a complex sinusoidal input signal $g(t) = e^{j\omega_0 t}$ the expressions can be simplified to

$$G_1(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} 2\pi \delta\left(\omega - \omega_0 - \frac{2\pi k}{T}\right) \quad (10-58)$$

$$G_2(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} A_2(k) \cdot 2\pi \delta\left(\omega - \omega_0 - \frac{2\pi k}{MT}\right) \quad (10-59)$$

$$G_3(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} A_3(k) \cdot 2\pi\delta\left(\omega - \omega_0 - \frac{2\pi k}{MT}\right) \quad (10-60)$$

where

$$A_2(k) = \frac{1}{M} \sum_{m=0}^{M-1} \alpha_m e^{-jkm2\pi/M} \quad (10-61)$$

and

$$A_3(k) = e^{j2\pi f_0 t_d} \sum_{m=0}^{M-1} \left[\frac{1}{M} \cdot \alpha_m \cdot e^{-j2\pi r_m \frac{f_0}{f_s}} \right] \cdot e^{-j2\pi \frac{km}{M}} \quad (10-62)$$

10.7.2 Phase Skew Errors

All the factors α_i are assumed to be equal, i.e. $\alpha_i = \alpha$ which gives

$$G_1(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} 2\pi\delta\left(\omega - \frac{2\pi k}{T}\right) \quad (10-63)$$

$$G_2(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \alpha \cdot 2\pi\delta\left(\omega - \frac{2\pi k}{T}\right) \quad (10-64)$$

$$G_3(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} A_3(k) \cdot 2\pi\delta\left(\omega - \frac{2\pi k}{MT}\right) \quad (10-65)$$

where

$$A_3(k) = \alpha \cdot e^{j2\pi f_0 t_d} \sum_{m=0}^{M-1} \left[\frac{1}{M} \cdot e^{-j2\pi r_m \frac{f_0}{f_s}} \right] \cdot e^{-j2\pi \frac{km}{M}} \quad (10-66)$$

The contributions from $G_1(\omega)$ and $G_2(\omega)$ to the output spectrum corresponds to a sinusoid with amplitude $1 - \alpha$. $G_3(\omega)$ is the spectrum of a non-uniformly sampled sinusoid due to the delay skews. The amplitude of the fundamental in $G_3(\omega)$ is determined by $|A_3(0)|$. Hence the total fundamental power is

$$|1 - \alpha + A_3(0)|^2 \quad (10-67)$$

The distortion power is the difference between the total power in $G_3(\omega)$ and the power of the fundamental of $G_3(\omega)$, i.e.

$$\alpha^2 - |A_3(0)|^2 \quad (10-68)$$

Hence the SNDR is limited by

$$SNDR = 10 \cdot \log \left(\frac{|1 - \alpha + A_3(0)|^2}{\alpha^2 - |A_3(0)|^2} \right) \quad (10-69)$$

The numerator can be written as

$$\begin{aligned} |1 - \alpha + A_3(0)|^2 &= (1 - \alpha + A_3(0))(1 - \alpha + A_3^*(0)) = \\ &= (1 - \alpha)^2 + (1 - \alpha)(A_3(0) + A_3^*(0)) + |A_3(0)|^2 \end{aligned} \quad (10-70)$$

If the delay skew errors are Gaussian random variables with zero mean and a standard deviation of σ_t , the following approximations can be used [3]

$$E[A_3(0)] = \alpha \cdot e^{j2\pi f_0 t_d} \cdot e^{-2\sigma_t^2 \pi^2 f_0^2} \approx \alpha \cdot e^{j2\pi f_0 t_d} \cdot (1 - 2\sigma_t^2 \pi^2 f_0^2) \quad (10-71)$$

and

$$\begin{aligned} E[|A_3(0)|^2] &= |E[A_3(0)]|^2 + \frac{1}{M} \cdot (\alpha^2 - |E[A_3(0)]|^2) \\ &\approx \alpha^2 - \alpha^2 \cdot 4\sigma_t^2 \pi^2 f_0^2 \cdot \frac{M-1}{M} \end{aligned} \quad (10-72)$$

The approximations are based on the Taylor series expansion. If we assume that $\sigma_t f_0$ is small the signal power can be approximated as

$$\begin{aligned} P_s &\approx (1 - \alpha)^2 + (1 - \alpha) \cdot a(e^{j2\pi f_0 t_d} + e^{j2\pi f_0 t_d}) + \alpha^2 \\ &= (1 - \alpha)^2 + (1 - \alpha) \cdot \alpha \cdot 2 \sin(2\pi f_0 t_d) + \alpha^2 \end{aligned} \quad (10-73)$$

The expression can be further simplified by assuming $f_0 t_d$ to be small which yields

$$P_s \approx (1 - \alpha)^2 + 2(1 - \alpha) \cdot \alpha + \alpha^2 = 1 \quad (10-74)$$

We can now approximate the SNDR as

$$\begin{aligned}
SNDR &\approx 10 \cdot \log\left(\frac{1}{\alpha^2 - |A_3(0)|^2}\right) \\
&= 10 \cdot \log\left(\frac{1}{\alpha^2 - \left(\alpha^2 - \alpha^2 \cdot 4\sigma_i^2 \pi^2 f_0^2 \cdot \frac{M-1}{M}\right)}\right) \\
&= 10 \cdot \log\left(\frac{1}{\alpha^2 \cdot 4\sigma_i^2 \pi^2 f_0^2 \cdot \frac{M-1}{M}}\right) \\
&= 20 \cdot \log\left(\frac{1}{\alpha \cdot 2\pi f_0 \sigma_i}\right) - 10 \cdot \log\left(\frac{M-1}{M}\right)
\end{aligned} \tag{10-75}$$

10.7.3 Gain Errors

We assume that there are no phase skew errors and that we have a complex sinusoidal input signal $g(t) = e^{j\omega_0 t}$. The digital spectra from section Sec. 10.7.1 can now be written as

$$G_1(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} 2\pi\delta\left(\omega - \omega_0 - \frac{2\pi k}{T}\right) \tag{10-76}$$

$$G_2(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} A_2(k) \cdot 2\pi\delta\left(\omega - \omega_0 - \frac{2\pi k}{MT}\right) \tag{10-77}$$

$$G_3(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} A_3(k) \cdot 2\pi\delta\left(\omega - \omega_0 - \frac{2\pi k}{MT}\right) \tag{10-78}$$

where

$$A_2(k) = \frac{1}{M} \sum_{m=0}^{M-1} \alpha_m e^{-jkm2\pi/M} \tag{10-79}$$

and

$$A_3(k) = \frac{e^{j2\pi f_0 t_d}}{M} \sum_{m=0}^{M-1} \alpha_m \cdot e^{-j2\pi \frac{km}{M}} \tag{10-80}$$

The output spectrum can now be written as

$$G(\omega) = G_1(\omega) + \frac{1}{T} \sum_{k=-\infty}^{\infty} A(k) \cdot 2\pi\delta\left(\omega - \omega_0 - \frac{2\pi k}{MT}\right) \quad (10-81)$$

where

$$A(k) = \frac{e^{j2\pi f_0 t_d} - 1}{M} \sum_{m=0}^{M-1} \alpha_m \cdot e^{-j2\pi \frac{km}{M}} \quad (10-82)$$

The total signal power is given by

$$\begin{aligned} P_s &= |1 + A(0)|^2 = (1 + A(0))(1 + A^*(0)) = \\ &= 1 + A(0) + A^*(0) + |A(0)|^2 \end{aligned} \quad (10-83)$$

If the gain factors are assumed to be independent Gaussian random variables with a mean of α and a standard deviation of σ_α , we have the following expectations

$$E[A(0)] = \frac{e^{j2\pi f_0 t_d} - 1}{M} \cdot E\left[\sum_{m=0}^{M-1} \alpha_m\right] = (e^{j2\pi f_0 t_d} - 1) \cdot \alpha \quad (10-84)$$

and

$$\begin{aligned} E[|A(0)|^2] &= E[A(0) \cdot A^*(0)] = \left(\frac{e^{j2\pi f_0 t_d} - 1}{M}\right)^2 \cdot \sum_{m=0}^{M-1} \sum_{n=0}^{M-1} E(\alpha_m \cdot \alpha_n) \\ &= \left(\frac{1}{M} \cdot E[\alpha_m^2] + \left(1 - \frac{1}{M}\right) \cdot E[\alpha_n] \cdot E[\alpha_m]\right) \cdot |e^{j2\pi f_0 t_d} - 1|^2 \\ &= \left(\frac{1}{M} \cdot (\sigma_\alpha^2 + \alpha^2) + \left(1 - \frac{1}{M}\right) \cdot \alpha^2\right) \cdot |e^{j2\pi f_0 t_d} - 1|^2 \\ &= \left(\frac{1}{M} \cdot \sigma_\alpha^2 + \alpha^2\right) \cdot |e^{j2\pi f_0 t_d} - 1|^2 \end{aligned} \quad (10-85)$$

where we have used that $E[\alpha_m^2] = \sigma_\alpha^2 + (E[\alpha_m])^2 = \sigma_\alpha^2 + \alpha^2$. Thus the signal power is

$$\begin{aligned} P_s &= 1 + (e^{j2\pi f_0 t_d} - 1) \cdot \alpha + (e^{-j2\pi f_0 t_d} - 1) \cdot \alpha + \left(\frac{1}{M} \cdot \sigma_\alpha^2 + \alpha^2\right) \cdot |e^{j2\pi f_0 t_d} - 1|^2 \\ &= 1 + \alpha(\sin(2\pi f_0 t_d) - 1) + \left(\frac{1}{M} \cdot \sigma_\alpha^2 + \alpha^2\right) \cdot |e^{j2\pi f_0 t_d} - 1|^2 \approx 1 \end{aligned} \quad (10-86)$$

where the last approximation is valid if $f_0 t_d$ and σ_α are small. The distortion power is given by

$$P_d = \sum_{m=0}^{M-1} |A(m)|^2 - |A(0)|^2 \quad (10-87)$$

The following expectation can be calculated (we use the fact that $A(m)$ can be interpreted as a DFT)

$$\begin{aligned} E \left(\sum_{m=0}^{M-1} |A(m)|^2 \right) &= E \left(\frac{|e^{j2\pi t_d f_o} - 1|^2}{M} \sum_{m=0}^{M-1} \alpha_m^2 \right) = \\ &= |e^{j2\pi t_d f_o} - 1|^2 \cdot (\sigma_\alpha^2 + \alpha^2) \end{aligned} \quad (10-88)$$

The SNDR can now be calculated as

$$\begin{aligned} SNDR &\approx 10 \cdot \log \left(\frac{1}{\left((\sigma_\alpha^2 + \alpha^2) - \left(\frac{1}{M} \cdot \sigma_\alpha^2 + \alpha^2 \right) \right) \cdot |e^{j2\pi t_d f_o} - 1|^2} \right) \\ &= 10 \cdot \log \left(\frac{1}{\left(\sigma_\alpha^2 - \frac{1}{M} \cdot \sigma_\alpha^2 \right) \cdot |e^{j2\pi t_d f_o} - 1|^2} \right) \\ &= 20 \cdot \log \left(\frac{1}{\sigma_\alpha} \right) - 10 \cdot \log \left(\frac{M-1}{M} \right) - 10 \cdot \log (|e^{j2\pi t_d f_o} - 1|^2) \end{aligned} \quad (10-89)$$

10.8 SUMMARY

In this chapter we have discussed the time-interleaved ADC. The performance is limited by mismatches between the channels. Gain and offset errors can be calibrated. A more severe problem is the phase skew errors which increases at high signal frequencies. The most effective way to reduce these errors is to use an input S/H circuit which usually needs an opamp that runs at the full speed of the ADC. The opamp is therefore difficult to design and power consuming. An improved global passive sampling technique that does not need an opamp was introduced. The limitations of the technique was also discussed. The phase skew distortion can be reduced by 10 to 20 dB compared to not using a global sampling technique.

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11 OVERSAMPLING A/D CONVERTERS

11.1 INTRODUCTION

Oversampling ADCs (OSADC) are commonly used in many telecommunications applications such as ADSL. Due to the use of DMT signals in ADSL applications, the requirement on the analog front end (AFE) is generally increased compared with single tone applications. It has also been shown that the linearity is more important than quantization noise for ADCs in DMT applications and OSADCs usually have higher linearity than Nyquist ADCs. An additional advantage of using OSADCs is the very low quantization noise power at low frequencies, which is beneficial for DMT applications. This is because more bits can be modulated on the lower frequency carriers where the signal is less attenuated by the telephone line. The major drawback of OSADCs is that a relatively high clock frequency must be used if the signal bandwidth is large, making the implementation difficult. Therefore techniques to reduce the oversampling ratio (OSR) of the converter are of interest. In Sec. 11.2 the basic properties of the first and second order sigma-delta converter are reviewed while sigma-delta converters for high signal bandwidths are treated in Sec. 11.3. SC implementation of sigma-delta converters are considered in Sec. 11.4 while the impact of circuit imperfections are reviewed in Sec. 11.5. Finally a design example is given in Sec. 11.6.

11.2 BASICS OF OVERSAMPLING SIGMA-DELTA CONVERTERS

The basic principle behind oversampling sigma-delta modulators is to trade signal bandwidth for resolution. The quantization noise of a low resolution ADC is high-pass filtered to yield a low quantization noise at low frequencies [1]. The noise at high frequencies is removed by a digital filter before the signal is decimated to generate the final output of the converter.

A first order sigma-delta modulator is shown in Fig. 11-1. It contains an ADC, an integrator, a DAC and a subtractor. The resolution of the ADC is low, usually only 1 bit. The digital output $y(nT)$ is fed back through the DAC and subtracted from the input signal. The output of the subtractor is accumulated in the integrator and quantized by the ADC. The integrator can be either continuous-time or discrete-time, but a discrete-time integrator is more common and will only be considered here. The performance of the modulator can be investigated by using a linear model for the modulator. We assume that the quantization error $e(nT)$ in the ADC is white noise uncorrelated to

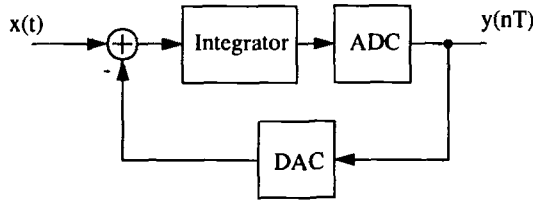


Figure 11-1 First order sigma-delta modulator.

the input signal. A linear model of the modulator is shown in Fig. 11-2. If a 1 bit quan-

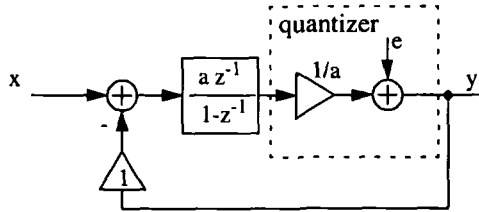


Figure 11-2 Linear model for first order sigma-delta modulator.

tizer is used, the gain in the integrator, a , will only affect the voltage swing at the output of the integrator not the output signal. Therefore this gain factor must be compensated for in the linear model of the quantizer. The z -domain output signal is given by

$$Y(z) = z^{-1} \cdot X(z) + (1 - z^{-1}) \cdot E(z) \quad (11-1)$$

Hence the input signal is only delayed while the quantization noise is high-pass filtered. If the signal bandwidth is small compared to the sampling frequency only a small portion of the quantization noise appears in the signal band and a high resolution results. The ratio between the signal bandwidth and the Nyquist frequency is called the oversampling ratio (OSR), i.e.

$$OSR = \frac{f_s/2}{BW} \quad (11-2)$$

where f_s is the sampling frequency and BW the signal bandwidth.

For the first order modulator the resolution increases by 1.5 bits for every doubling of the sampling frequency. To get a high resolution a very high OSR is needed. The required OSR can be reduced by using a second order modulator. The linear model of a second order modulator is shown in Fig. 11-3. There are now two integrators in the loop and the output signal is given by

$$Y(z) = z^{-2} \cdot X(z) + (1 - z^{-1})^2 \cdot E(z) \quad (11-3)$$

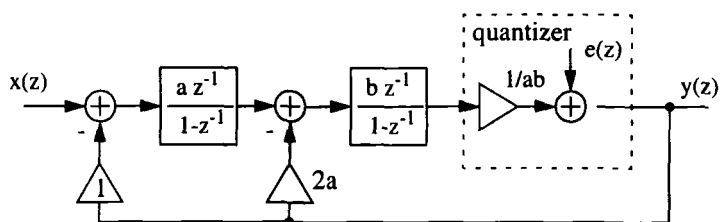


Figure 11-3 Linear model of second order sigma-delta modulator.

Hence the order of the noise shaping function is two. We now gain 2.5 bits for every doubling of the sampling frequency. An additional advantage of the second order modulator is that the assumption of white uncorrelated quantization noise is more accurate than for the first order modulator.

The order can be further increased and for an L -order modulator we gain $L + 1/2$ bits for every doubling of the sampling frequency [1]. There are however problems with increasing the order of the modulator due to stability.

11.3 OVERSAMPLED SIGMA-DELTA CONVERTERS FOR HIGH SIGNAL BANDWIDTHS

Achieving a high resolution over a large signal bandwidth is almost impossible using a second order modulator since the sampling rate would be very high. The OSR can be reduced in different ways.

One way to reduce the OSR is to increase the order of the modulator. This can be done by adding more integrators in the loop. However the modulator may be unstable unless the integrator gains and the feedback signals are scaled properly. The resulting noise shaping function is of the same order as the number of integrators but it is multiplied by a large constant [2]. This increases the noise power.

The performance can be somewhat improved by using a loop filter with feedforward and feedback of internal signals and thus introducing zeros in the transfer function. However the resulting SC implementation will have a large spread in capacitor values which makes the design difficult [2]. The resulting noise shaping function is also in this case multiplied by a large constant. Hence, we should consider other ways of decreasing the OSR.

The resolution of the modulator can also be increased by increasing the resolution of the quantizer. A problem with this solution is that nonlinearity errors in the DAC will limit the performance of the modulator unless calibration or error averaging is used [3].

Another option, avoiding the need for calibration, is to use a cascaded sigma-delta modulator [1]. This type of converter is sometimes referred to as a MASH structure. In this type of converters several low order modulators are cascaded to achieve a high

order noise shaping function. The low order quantization noise is cancelled by a digital cancellation logic. A drawback of the cascaded architecture is that mismatch between the analog and digital circuits limits the performance. The matching need not be within the full resolution of the converter and cascaded sigma-delta converters with resolutions of 15-16 bits and a signal bandwidth in the range of 1 MHz have been reported [2], [4]. Therefore this type of converters is interesting for ADSL applications. In the following we focus on architectures based on cascading low order modulators.

11.3.1 2-2 Fourth Order Cascaded Modulator

A fourth order cascaded modulator consisting of two second order modulators [5] is shown in Fig. 11-4. The analog output signal of the second integrator in the first mod-

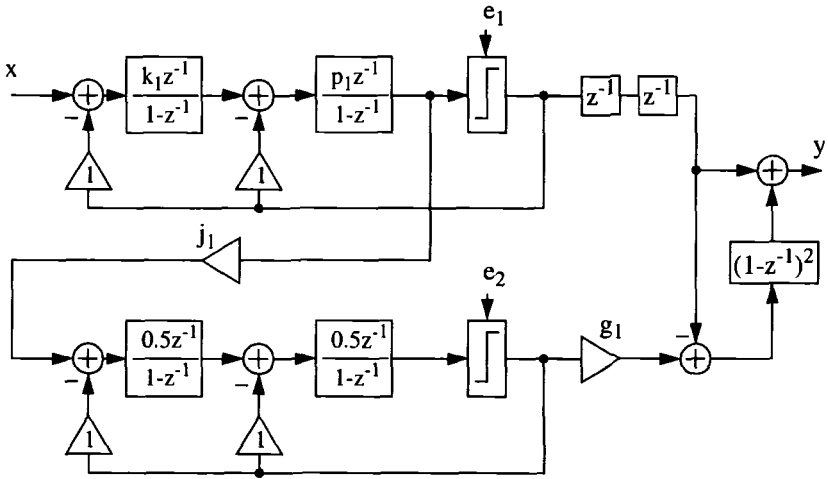


Figure 11-4 4th order cascaded sigma-delta modulator.

ulator is fed to the input of the second modulator. The digital outputs of the modulators are fed to the correction logic which generates the output y . The digital gain factor g_1 must be chosen such that the low order quantization noise is cancelled. This is achieved when the product of the gain in the first integrator ($k_1 = 0.5$), the gain in the second integrator ($p_1 = 0.5$), the gain for the signal between the two stages ($j_1 = 1$) and g_1 equals 1. Thus we have for the modulator in Fig. 11-4

$$k_1 \cdot p_1 \cdot j_1 \cdot g_1 = 1 \Rightarrow g_1 = 4 \quad (11-4)$$

The output signal is given by [5]

$$\begin{aligned} Y(z) &= z^{-4} \cdot X(z) + g_1 \cdot E_2(z) \cdot (1 - z^{-1})^4 \\ &= z^{-4} \cdot X(z) + 4 \cdot E_2(z) \cdot (1 - z^{-1})^4 \end{aligned} \quad (11-5)$$

which corresponds to a 4th-order noise shaping function. We see that g_1 should be as

small as possible to minimize the quantization noise. This is achieved by increasing the integrator gains and the gain between the two stages. However this can usually not be done for two reasons. Firstly, if the input signal to the second stage is too large the second modulator saturates and the SNR is reduced. Secondly in the actual implementation we must consider the signal swings in the circuits. The swing at the output of an opamp should not be too close to V_{dd} or V_{ss} to avoid distortion. The relation between the feedback signal swing and the swing at the output of an integrator is determined by the integrator gain. To minimize the noise it is usually desirable to have approximately the same signal swing in all the integrator outputs. The integrator gains can therefore not be chosen arbitrarily.

We must also consider the swing of the input signal. For OSADCs the maximum input signal amplitude without overloading the modulator is below the full-scale (FS) of the converter. The FS is determined by the reference levels in the DAC. For a second order modulator the maximum signal swing at the input is approximately -3 dBFS. If desirable a gain factor can be added at the input to compensate for the smaller swing of the input signal. For an SC implementation this is easily done by changing a capacitor ratio.

The signal swings in the modulator can be determined by simulating the modulator using a behavioral model. The modulator in Fig. 11-4 was simulated using MatLab. The reference levels of the DAC are +1 and -1. A sinusoidal input signal with an amplitude of FS/2 and a frequency of 118.56 kHz is used. The modulator sampling rate was 30 MS/s. A 64k-point FFT of the output signal is shown in Fig. 11-5. The

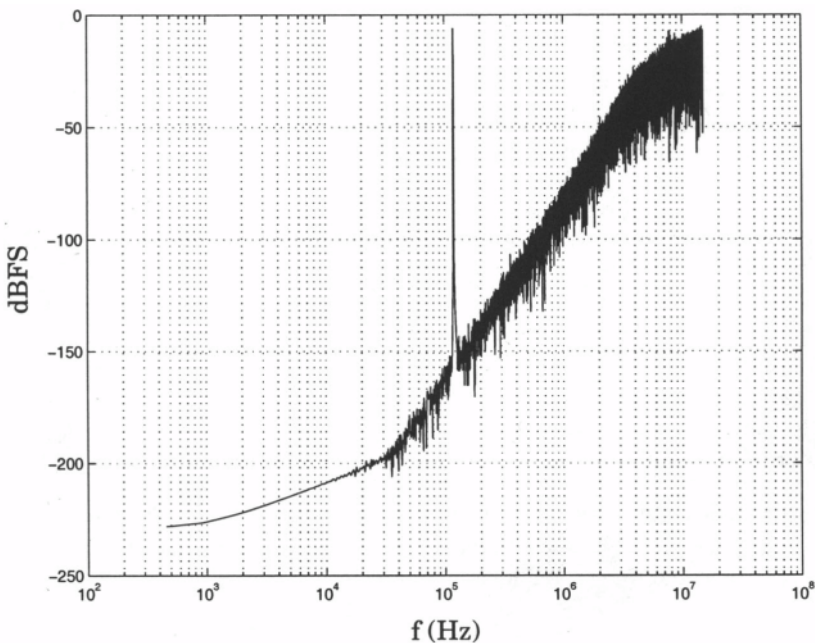


Figure 11-5 64k-point FFT of integrator output.

quantization noise increases by 80 dB/decade as expected for a 4th order noise shaping function.

The signal-to-noise ratio as a function of the input signal amplitude for different OSRs is shown in Fig. 11-6. In the figure we see that the SNR improves by slightly less than

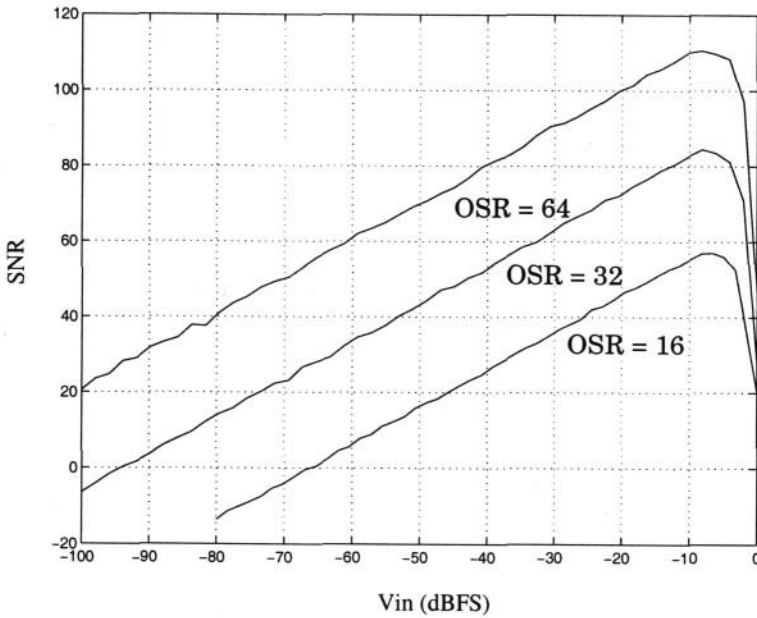


Figure 11-6 SNR vs. input amplitude.

30 dB when the OSR is doubled, which corresponds to approximately 4.5 bits. The peak SNR appears at approximately -6 dBFS which corresponds to half the full-scale amplitude.

The signal swings at the integrator outputs for the cascaded modulator in Fig. 11-4 were estimated by calculating the histogram from the MatLab simulation. The result is shown in Fig. 11-7. The swing at the output of the first integrator is approximately 1.5 times the feedback signal amplitude. The gain in the first integrator can therefore be decreased by approximately 1/3 to make the signal swing close to 1. We also see that the swing of the signal fed to the second modulator is approximately 1. This is slightly too large and the second modulator is overloaded. This explains why the peak SNR occurred at a relatively low input signal amplitude in Fig. 11-6.

From the above discussion we conclude that to improve the signal swings in the modulator it is necessary to decrease some of the gain factors. But smaller gain factors means that g_1 becomes larger, reducing the SNR.

11.3.2 Improved 2-2 Cascaded Modulator

It is possible to improve the performance of the cascaded modulator in Fig. 11-4 by modifying the architecture slightly. The architecture in Fig. 11-8 feeds only the quan-

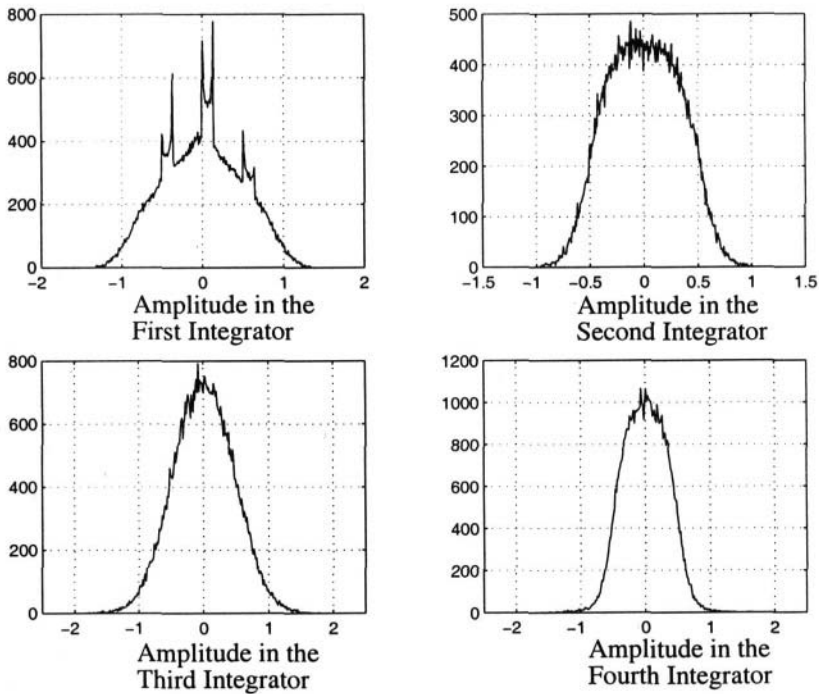


Figure 11-7 Histogram plots for integrator outputs.

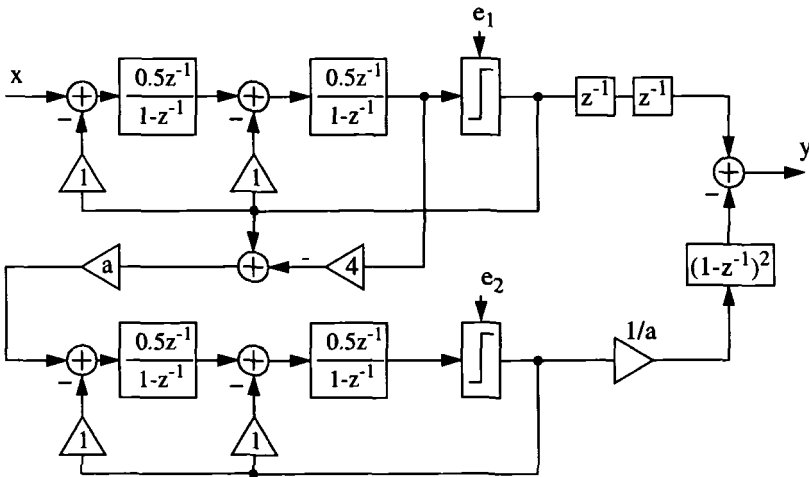


Figure 11-8 4th order cascaded sigma-delta modulator feeding only the quantization noise to the second stage.

tization noise of the first modulator to the second stage. The output signal is given by

$$Y(z) = X(z) \cdot z^{-4} + \frac{(1 - z^{-1})^4}{a} \cdot E_2(z) \tag{11-6}$$

The gain factor a should be approximately 1/4 to avoid saturation in the second stage. In this case the gain factor in the digital domain is only determined by the gain between the two stages not the integrator gains.

11.3.3 2-1-1 Cascaded Modulator

It is not necessary to use only second order modulators in a cascaded modulator. There may actually be advantages in using first order modulators instead. The modulator in the first stage is normally a second order modulator since the quantization noise is less correlated to the input signal than the noise in a first order modulator. Since the input signal to the later stages in the cascaded modulator are the quantization noise in the previous stages, the quantization error in all the stages show a small correlation to the input signal even if the later stages are of the first order. A 4th order 2-1-1 cascaded modulator [4] is shown in Fig. 11-9. The transfer function is given by [4]

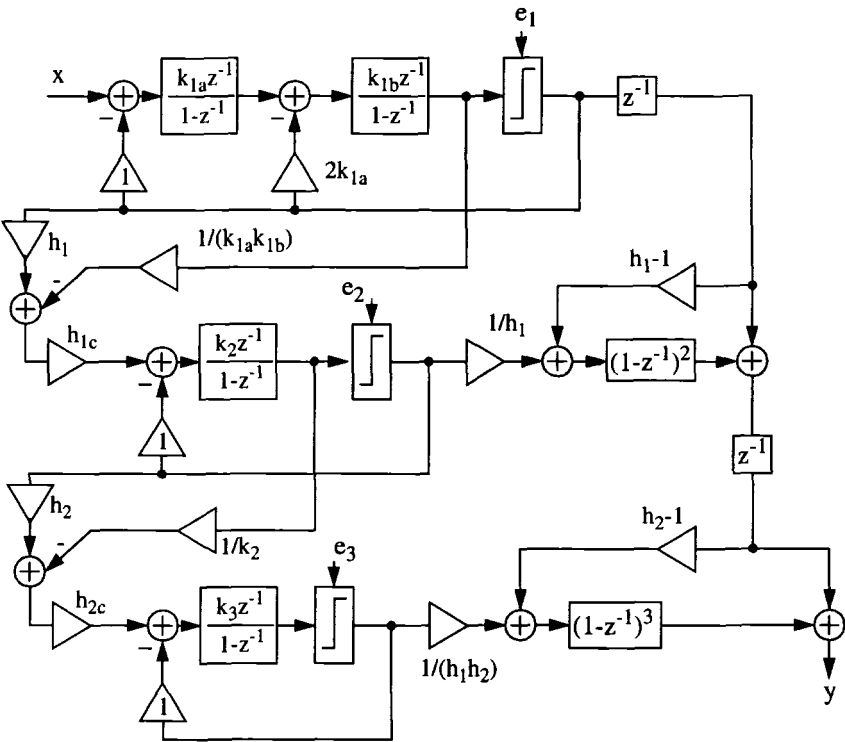


Figure 11-9 4th order 2-1-1 cascaded sigma-delta modulator.

$$Y(z) = z^{-4} \cdot X(z) + \frac{(1 - z^{-1})^4 \cdot E_3(z)}{h_{1c} \cdot h_{2c}} \quad (11-7)$$

The coefficient h_{2c} is normally chose as 1 ([4]) which means that the fourth order noise shaping function is multiplied by the gain factor at the input of stage two. This is exactly the same as for the 2-2 modulator in Sec. 11.3.2. The maximum input signal swing without causing overloading of a first order modulator is larger than for a second order modulator. The gain factor h_{1c} can therefore be larger when the second stage is a first order modulator and the quantization noise is therefore smaller. The coefficient h_{1c} is normally chosen as 3 ([4]),

11.3.4 Cascaded Modulator with Multi-Bit Quantizer

If the required dynamic range is very high and a small OSR is required due to a high signal bandwidth, a 4th order cascaded modulator may not be sufficient to meet the requirements even if the improved architectures in section Sec. 11.3.2 and Sec. 11.3.3 are used. The quantization noise can be reduced by adding more stages in cascade. The order of noise shaping function will then be increased which may be a disadvantage when considering the decimation filter that follows the modulator. If the order is high the noise increases rapidly at high frequencies making the required attenuation in this filter high which is undesirable.

Another solution is to use a multi-bit quantizer. In a single stage modulator a multi-bit quantizer is problematic since the linearity of the multi-bit DAC in the feedback path directly affects the linearity of the modulator. This is not the case in a cascaded modulator if the multi-bit quantizer is used in the final stage. The input signal to the final stage in a cascaded modulator shows a very weak correlation to the input signal which means that the errors caused by a non-ideal DAC will appear as a increased noise floor in the output signal, not harmonic distortion [1]. In addition to this the error from the DAC will be noise shaped. If the DAC error is modelled as white uncorrelated noise, e_d , the output is affected by an additional noise term $H_d(z) \cdot E_d(z)$, where $H_d(z)$ is the noise shaping function for the DAC error, A multi-bit quantizer in the second stage of a 2-2 modulator would give an output signal

$$Y(z) = z^{-4} \cdot X(z) + k_1 \cdot (1 - z^{-1})^4 \cdot E_2(z) + k_2 \cdot (1 - z^{-1})^2 \cdot E_d(z) \quad (11-8)$$

where k_1 and k_2 are constants determined by the signal scaling in the modulator. Hence the DAC errors are shaped by a second order function. The required linearity of the DAC can therefore be significantly smaller than the total resolution of the OSADC. It should also be noted the $E_2(z)$ is smaller if the resolution of the quantizer is higher.

11.4 SC IMPLEMENTATION

Since sigma-delta converters are normally used in high accuracy applications they are almost always implemented using fully differential circuits. An SC fully differential

second order modulator is shown in Fig. 11-10. There are four different clock signals

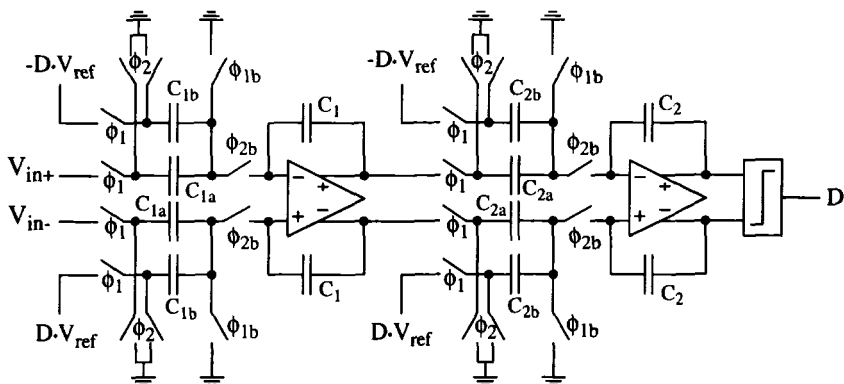


Figure 11-10 Fully differential SC implementation of second order modulator.

controlling the switches in the circuit. ϕ_1 and ϕ_2 are non-overlapping clock phases and are used to switch the capacitors to the input signals of the integrator. The clock phases ϕ_{1b} and ϕ_{2b} used for the switches at the opamp input are turned off slightly before ϕ_1 and ϕ_2 respectively to reduce the signal dependent clock feedthrough. The comparator output, D , is +1 or -1 depending on the sign of the second integrator output. It determines the sign of the reference voltage, V_{ref} , in the feedback loops. The capacitor ratios in the circuit determine the integrator gains. It may be more convenient to slightly modify the linear model of the modulator as shown in Fig. 11-11 to

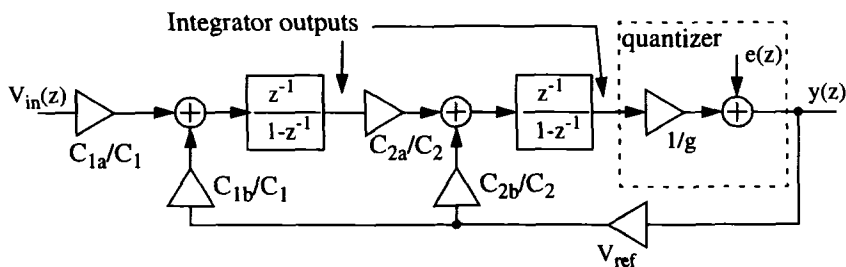


Figure 11-11 Linear model of second order Sigma-Delta Modulator.

better reflect the operation of the circuit in Fig. 11-10. The gain factor in the comparator is given by

$$g = \frac{C_{1b} \cdot C_{2a} \cdot V_{ref}}{C_1 \cdot C_2} \quad (11-9)$$

The voltage swing at the integrator outputs can now be scaled to their desired levels by changing the proper capacitance values. The swing at the first integrator can for

instance be doubled by doubling the size of C_{1a} and C_{1b} . The only restriction is that the capacitor C_{2b} must be chosen according to the relation

$$C_{2b} = 2 \cdot \frac{C_{1b} \cdot C_{2a}}{C_1} \quad (11-10)$$

The easiest way to check for proper signal swing in the circuit is to simulate the linear model in Fig. 11-10 by using MatLab.

11.4.1 Integrator Modifications

There are some modifications that can be applied to the integrator to improve the performance. If for instance the capacitors C_{1a} and C_{1b} are equal the integrator can be implemented as shown in Fig. 11-12. The same capacitor is now used to sample both

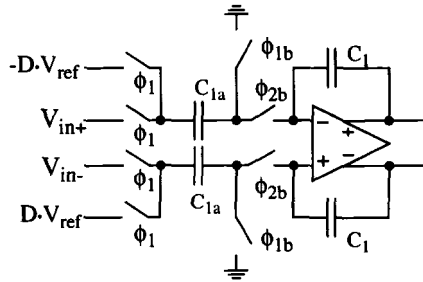


Figure 11-12 Simplified Integrator with only one.

the input signal and the reference signal [1]. In this way we can save chip area and increase the speed since the feedback factor is larger and the total load is smaller. In addition to this the noise is smaller since the noise contribution from one capacitor is removed. A potential problem with this solution is that the charge taken from the reference is signal dependent which makes the design of the reference buffers more difficult (the same problem exists for the integrator in Fig. 11-10).

There are other ways to reduce the size of the capacitors. The circuit in Fig. 11-13 uses a differencing input sampling to reduce the size of the capacitors by a factor 2 [6]. The input voltage is sampled on both clock phases but with different sign. In this way the effective voltage swing is increased by a factor of 2. Therefore it is possible to decrease the capacitor while still transferring the same amount of charge. Again both the speed and noise are improved. However, the signal dependent load on the reference buffers remains.

The signal dependent loading can be avoided by changing the switching of C_{1b} as shown in Fig. 11-14. This technique is also used in circuits with a single reference voltage [7]. The left hand side of the capacitor C_{1b} is now always switched between $+V_{ref}$ and $-V_{ref}$ making the load on the reference buffers signal independent. The sign of the reference signal is changed by connecting the right hand side of C_{1b} to the positive or negative input of the opamp. The digital signal B controlling the switches

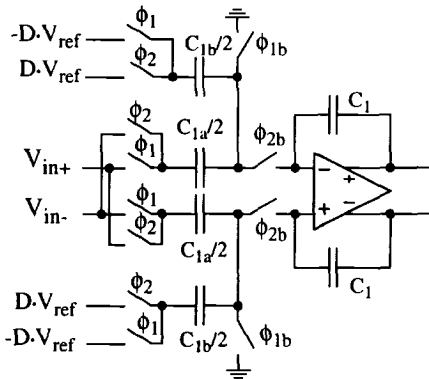


Figure 11-13 Integrator with differencing input sampling.

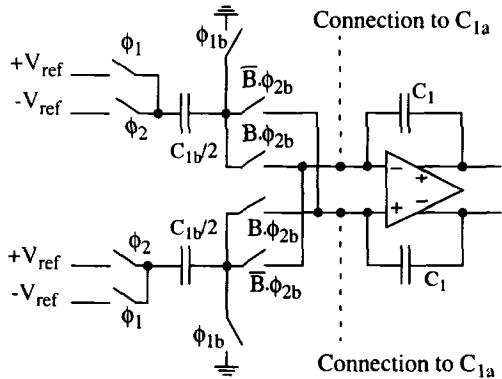


Figure 11-14 Integrator with signal independent load on reference buffers.

is

$$B = \begin{cases} 0, & D = -1 \\ 1, & D = 1 \end{cases} \tag{11-11}$$

A slightly different approach to improve the settling performance is to use double sampling at the input of the integrator [8]. This is achieved by using two sampling capacitors, one sampling the input on ϕ_1 and the other on ϕ_2 . Since it is now possible to integrate on both clock phases the whole sampling period is now available for settling, reducing the speed requirements on the opamp. A problem with this solution is that any mismatch between the double sampling capacitors cause distortion. For the input signal the distortion appears out of the signal band and is usually not a problem. However the signal coming from the quantizer must also be double sampled. Mis-

match in this signal path may cause distortion in the signal band and therefore some kind of calibration or error averaging is needed [9]. For fully differential SC circuits the problem can be avoided by using only one capacitor in the feedback path alternately connected between the two inputs of the opamp [10]. The circuit now act as a bi-linear integrator which changes the transfer function of the modulator. This change is acceptable in some applications.

11.5 NON-IDEAL EFFECTS

There are several effects that deteriorate the performance of the sigma-delta modulator, e.g. finite opamp gain, finite bandwidth, nonlinear settling and capacitor mismatch. These non-ideal effects are considered in the following.

11.5.1 Finite Opamp Gain

In chapter 6 the transfer function of an integrator with a finite DC gain and a parasitic capacitor at the opamp input was derived. The integrator is shown in Fig. 11-15. The

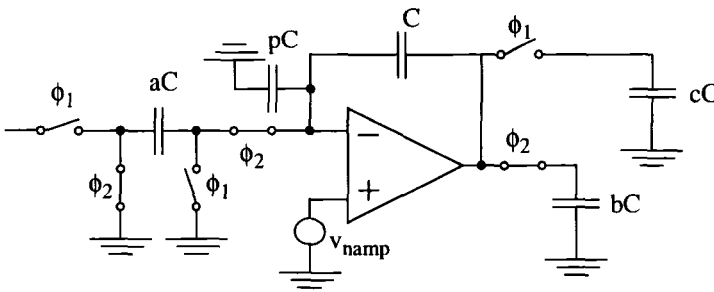


Figure 11-15 SC Integrator.

parasitic capacitor is represented by capacitor pC . According to chapter 6 the resulting transfer function is given by [4]

$$H(z) = \frac{r_2 \cdot z^{-1} \cdot a}{1 - \frac{r_2}{r_1} \cdot z^{-1}} \quad (11-12)$$

where

$$\begin{aligned}
 r_1 &= \frac{\beta_1 \cdot A_0}{1 + \beta_1 \cdot A_0} = \frac{1}{\left(\frac{p+1}{A_0} + 1\right)} \\
 r_2 &= \frac{\beta_2 \cdot A_0}{1 + \beta_2 \cdot A_0} = \frac{1}{\left(\frac{p+1+a}{A_0} + 1\right)}
 \end{aligned}
 \tag{11-13}$$

The first integrator in a sigma-delta converter is the most critical since errors in the following integrators are noise shaped. For the first integrator in the SC modulator in Fig. 11-10 the factors r_1 and r_2 are given by

$$\begin{aligned}
 r_1 &= \frac{1}{\left(\frac{C_p/C_1 + 1}{A_0} + 1\right)} \\
 r_2 &= \frac{1}{\left(\frac{C_p/C_1 + 1 + (C_{1a} + C_{1b})/C_1}{A_0} + 1\right)}
 \end{aligned}
 \tag{11-14}$$

where C_p is the parasitic capacitor at the input of the opamp. By using that $1/(1+x) \approx 1-x$ for small x and $A_0^2 \gg A_0$ we get the following approximations

$$\begin{aligned}
 r_2 &\approx 1 - \frac{C_p/C_1 + 1 + (C_{1a} + C_{1b})/C_1}{A_0} \\
 \frac{r_2}{r_1} &\approx 1 - \frac{(C_{1a} + C_{1b})/C_1}{A_0}
 \end{aligned}
 \tag{11-15}$$

Thus the relative gain error for the integrator is approximately

$$\epsilon_{gain} \approx \frac{C_p/C_1 + 1 + (C_{1a} + C_{1b})/C_1}{A_0}
 \tag{11-16}$$

while the relative error in the denominator, causing leakage in integrator, is approximated as

$$\epsilon_{leak} \approx \frac{(C_{1a} + C_{1b})/C_1}{A_0}
 \tag{11-17}$$

Usually quite large errors can be tolerated but the acceptable size of these errors depends on the architecture and the desired resolution. In for instance [4] a 4th order cascaded sigma-delta converter with $OSR = 24$ and a total resolution of 15 bits is designed. In this case a opamp gain of only 60 dB is sufficient which shows that the modulator is relatively insensitive to this kind of errors. However to provide some margin for other errors in the circuit the opamp gain should be chosen significantly larger than the lower limit (in [4] a gain of 80 dB is used). The easiest way to find the

limit on the DC gain is normally to simulate the behavioral level model of the modulator using for instance MatLab.

11.5.2 Finite Bandwidth

From chapter 6 we know that a finite opamp bandwidth introduces a gain error in the integrator. We have that

$$H(z) = \frac{(1 - \epsilon_{r,2}) \cdot a \cdot z^{-1}}{1 - z^{-1}} \quad (11-18)$$

where $\epsilon_{r,2}$ is the relative settling error in the integration phase. According to chapter 6 we have

$$\epsilon_{r,2} = e^{-\omega_{-3dB,2} \cdot t_s} = e^{-\frac{g_m}{(b \cdot (a+p+1) + (a+p)) \cdot C} \cdot t_s} \quad (11-19)$$

where t_s is the available settling time and g_m the transconductance of the input device in the opamp. For the first integrator of the modulator in Fig. 11-10 we have no load (bC) in the integration phase and hence

$$\epsilon_{r,2} = e^{-\frac{g_m}{(C_{1a} + C_{1b} + C_p)} \cdot t_s} \quad (11-20)$$

where C_p is the parasitic capacitor at the input of the opamp. It should be noted that in an actual implementation the physical bottom plate of the integrating capacitor C_1 will add a parasitic capacitor at the output of the opamp. Since C_1 may be quite large this parasitic capacitor may add a significant load to the opamp output. In a more accurate analysis this capacitor should be taken into account. In [4] it is shown that if both finite opamp gain and bandwidth are present the finite bandwidth will also affect the integrator leakage. In a simplified analysis we have [4]

$$H(z) = \frac{r_2 \cdot (1 - \epsilon_{r,2}) \cdot z^{-1}}{1 - \frac{r_2}{r_1} \cdot \left(1 - \epsilon_{r,2} \cdot \left(1 - \frac{r_1}{r_2}\right)\right) \cdot z^{-1}} \quad (11-21)$$

However the term $\epsilon_{r,2} \cdot (1 - r_1/r_2)$ corresponds to the product of the relative settling error and the error caused by finite opamp gain. Therefore it should be quite small and the transfer function with both limited gain and bandwidth can be approximated as

$$H(z) = \frac{r_2 \cdot (1 - \epsilon_{r,2}) \cdot z^{-1}}{1 - \frac{r_2}{r_1} \cdot z^{-1}} \quad (11-22)$$

As long as the settling is linear, relatively large settling errors are acceptable and the required settling accuracy is usually significantly smaller than the accuracy of the OSADC.



11.5.3 Non-linear Settling

For most practical cases the settling will not be entirely linear. For large input signals the first part of the settling is slow rate limited while the last part is linear. This means that the relative settling error is signal dependent and will therefore generate distortion. The required settling accuracy must usually be increased when the settling is non-linear. Assuming a single stage opamp the slew rate for the first integrator in Fig. 11-10 is determined by

$$SR = \frac{I_b}{C_{load}} = \frac{I_b}{\frac{(C_{1a} + C_{1b}) \cdot C_1}{C_{1a} + C_{1b} + C_1}} \quad (11-23)$$

where I_b is the bias current in the differential input stage of the opamp. For a more accurate analysis the parasitic load capacitance at the output of the opamp should be taken into account. The -3 dB bandwidth of the integrator is given by

$$\omega_{-3dB} = \frac{g_m}{C_{load}} \cdot \beta = \frac{g_m}{C_{1a} + C_{1b}} \quad (11-24)$$

where g_m is the transconductance of the input device of the opamp and β is the feedback factor. It can be shown that the output signal of the integrator at time $n \cdot T$ can be calculated as

$$V_{out}(n \cdot T) = V_{out}((n-1) \cdot T) + g(a_1 \cdot V_{in}((n-1) \cdot T) - D((n-1) \cdot T) \cdot a_2) \quad (11-25)$$

where a_1 is the gain from the input of the integrator, V_{in} , to the output, a_2 the gain of the signal from the quantizer output to the output of the integrator and D is the quantizer output (± 1). $g(x)$ is a function describing the non-linear settling of the opamp. Assuming slew rate limited single pole settling it is given by [11]

$$g(x) = \begin{cases} x(1 - e^{-\omega_{-3dB} t_s}) & |x| \leq \frac{SR}{\omega_{-3dB}} \\ x - \frac{\text{sgn}(x)SR}{\omega_{-3dB}} e^{\left(\frac{|x|}{SR} - t_s\right) \cdot \omega_{-3dB}} & \frac{SR}{\omega_{-3dB}} < |x| \leq \left(\frac{1}{\omega_{-3dB}} + t_s\right) \cdot SR \\ \text{sgn}(x) \cdot t_s \cdot SR & \left(\frac{1}{\omega_{-3dB}} + t_s\right) \cdot SR < |x| \end{cases} \quad (11-26)$$

Based on (11-25) and (11-26) a behavioral model which include the effect of non-linear settling can be simulated in for instance MatLab. In this way the required settling accuracy can be determined. This was done in [11], where the required slew rate as function of the number of settling time constants were plotted for a third order cascaded modulator. The results show that there are two regions where a high resolution can be achieved without needing a very large slew rate. This is illustrated in Fig. 11-

16. In [11] these regions are called the slow regime and the fast regime. The slow

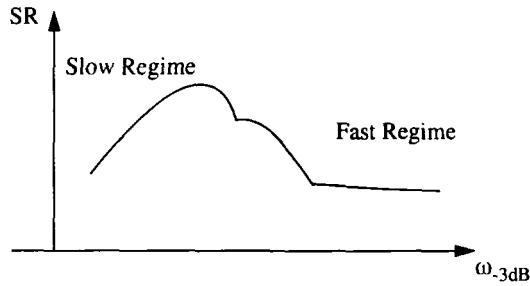


Figure 11-16 Slew rate requirement vs. bandwidth of integrator.

regime corresponds to linear settling, i.e. no slew limiting occurs. In this regime a low bandwidth is needed and the number of settling time constants need only be in the order of 2-3 to achieve more than 100dB SNDR in the converter. The number of settling time constants is defined as

$$n_{\tau} = t_s \cdot \omega_{-3dB} \quad (11-27)$$

where t_s is the available settling time and ω_{-3dB} is the -3 dB bandwidth of the integrator. It is interesting to note that if the bandwidth of the integrator is increased without increasing the slew rate the resolution of the converters is reduced. This clearly illustrates that the required settling accuracy is increased when the settling is slew rate limited. The worst case slew rate requirement may be several times larger than what is required with linear settling. If the bandwidth is increased even further the required slew rate will decrease and finally reach a value that may be as low or even lower than in the case of linear settling. This corresponds to the fast regime. Depending on the resolution of the converter the fast regime is usually entered at about 8-12 settling time constants.

It is usually not possible to directly use the result in Fig. 11-16 to design the opamp in the integrator. The reason is that the bandwidth and slew rate of the integrator can not be chosen independently of each other. They are connected through the effective gate voltage of the input devices of the opamp. Therefore it may not be possible to use the slow regime since the effective gate voltage becomes too large. If we design the integrator in the fast regime it may not be possible to use the small slew rate indicated in Fig. 11-16, since the effective gate voltage would be too small.

The above results are based on a model with only one pole in the integrator. In all practical cases there are always several poles. The parasitic poles should be at a high frequency compared to the bandwidth of the integrator to avoid that they influence the settling. Therefore the bandwidth of the integrator should not be larger than necessary. The effects of parasitic poles and switch resistance can with some effort be included in a behavioral model to get a more accurate result [12].

11.5.4 Capacitor Mismatch

The gains from the input of the modulator and from the comparator feedback signal to the output of the integrator are given by C_{1a}/C_1 and C_{1b}/C_1 respectively. Mismatch in these capacitor ratios will change the corresponding gain factors. The errors can be modelled as a gain error in the integrator and a gain error in the input signal. The gain error at the input does normally not cause any problem at all while the integrator gain error has the same effect as a finite opamp bandwidth (see previous section). A capacitor matching in the order of 0.1% or better is achievable in a modern CMOS process. This is normally sufficient to reach resolutions in the order of 15 bit in cascaded modulators ([4]).

11.5.5 Gain Errors in Cascaded Modulators

Due to capacitor mismatch the actual gains in the integrator deviate from their ideal values. This effect will degrade the performance of the cascaded modulator due to the mismatch between the analog and digital parts. For the 2-2 cascaded modulator in Fig. 11-4 the digital gain factor g_1 must meet the relation [5]

$$g_1 \cdot k_1 \cdot p_1 \cdot j_1 = 1 \quad (11-28)$$

where k_1 is the gain in the first integrator in the first modulator, p_1 the gain in the second integrator of the first modulator and j_1 the gain at the input of the second modulator. If there are gain errors in the coefficients the output signal becomes [5]

$$Y(z) = z^{-4} \cdot X(z) + g_1 \cdot (1 - z^{-1})^4 \cdot E_2(z) + \varepsilon \cdot (1 - z^{-1})^2 \cdot E_1(z) \quad (11-29)$$

where $\varepsilon = 1 - g_1 \cdot k_1 \cdot p_1 \cdot j_1$ is the gain error. With gain errors the quantization noise from the first modulator appears in the output signal. However, the noise is spectrally shaped by a second order noise-shaping function and therefore relatively large mismatch errors are acceptable unless a very high resolution is required.

11.5.6 Thermal Noise

There are two types of noise in CMOS circuits, thermal noise and 1/f noise. If the bandwidth is fairly high the thermal noise will dominate while the 1/f noise dominates at low frequencies. If the 1/f noise is large it can be reduced by circuit techniques such as correlated-double-sampling [3]. Therefore we focus on the thermal noise in this section. The input referred thermal noise of all integrators except the first in a cascaded modulator is noise shaped. The capacitor sizes in all but the first integrator are therefore usually determined by other reasons than thermal noise. From chapter 6 we have that the input referred thermal noise of an integrator can be approximated as

$$v_n^2 = \frac{4}{3} \cdot \frac{kT}{C_{load}} \cdot \beta \cdot (1 + n_t) + \frac{kT}{C_s} \quad (11-30)$$

where k is the Boltzmann's constant, T the absolute temperature, C_{load} the total

effective load capacitor at the opamp output in the integration phase, β the feedback factor in the integration phase, C_s the total sampling capacitor and n_t a noise contribution factor that depends on the opamp architecture. The first term in (11-30) correspond to the noise from the opamp while the second term corresponds to the noise from the sampling capacitor. It is worth noting that the noise only depends on the capacitor sizes in the circuit. For the integrator in Fig. 11-10 we have two input signals. It is convenient to refer all noise sources (one from opamp, one from C_{1a} and one from C_{1b}) to the input signal applied to capacitor C_{1a} . It can be shown that

$$\begin{aligned} v_{n, in}^2 &= \frac{4}{3} \cdot \frac{kT}{C_{load}} \cdot \beta \cdot (1 + n_t) \cdot \frac{(C_{1a} + C_{1b})^2}{C_{1a}^2} + \frac{kT}{C_{1a}} + \frac{kT}{C_{1b}} \cdot \frac{C_{1b}^2}{C_{1a}^2} = \\ &= \frac{kT}{C_{1a}} \cdot (1 + K_1) \end{aligned} \quad (11-31)$$

where K_1 is a factor accounting for the noise in the opamp and the sampling capacitor C_{1b} . The factor K_1 is given by

$$\begin{aligned} K_1 &= \frac{4}{3} \cdot \frac{\beta}{C_{load}} \cdot (1 + n_t) \cdot \frac{(C_{1a} + C_{1b})^2}{C_{1a}} + \frac{C_{1b}}{C_{1a}} \\ &= \frac{\frac{4}{3} \cdot (1 + n_t)}{\frac{C_{1a} C_L (C_{1a} + C_{1b} + C_1)}{(C_{1a} + C_{1b})^2 C_1} + \frac{C_{1a}}{(C_{1a} + C_{1b})}} + \frac{C_{1b}}{C_{1a}} \end{aligned} \quad (11-32)$$

where C_L is the load capacitance at the opamp output when excluding the load from the feedback network. If the improved switching in Fig. 11-12 is used C_{1b} is zero and the factor K_1 is reduced to

$$K_1 = \frac{4}{3} \cdot \frac{\beta}{C_{load}} \cdot (1 + n_t) \cdot C_{1a} = \frac{\frac{4}{3} \cdot (1 + n_t)}{\frac{C_L (C_{1a} + C_1)}{C_{1a} C_1} + 1} \quad (11-33)$$

In (11-33) we see that if the noise contribution factor n_t is small and the load capacitor, C_L is large, the factor K_1 is small and the noise is dominated by the kT/C noise from the sampling capacitor.

The noise requirement is relaxed due to the oversampling since only a small part of the noise falls in the signal band. To make the thermal noise power smaller than the quantization noise we have

$$v_{n, in}^2 < \frac{FS^2}{2^{2n_{tot}} + 2} \cdot OSR \quad (11-34)$$

where n_{tot} is the desired number of bits in the modulator and FS is the full-scale swing of the input signal.

Taking the thermal noise into account the signal-to-noise ratio of the converter can be calculated as

$$SNR = 10 \cdot \log\left(\frac{P_{signal}}{P_q + P_n}\right) \quad (11-35)$$

where P_{signal} is the signal power, P_q the quantization noise power and P_n the thermal noise power. We see that if the thermal noise power is equal to the quantization noise power the SNR will decrease by 3 dB. Therefore it is necessary to make the thermal noise significantly smaller than the quantization noise to avoid that the SNR is decreased. The thermal noise is reduced by increasing the capacitors in the circuit.

11.5.7 Comparator Errors

When using a single bit quantizer the modulator is relatively insensitive to comparator offset and hysteresis. From [4] we have that an offset in the order of 100mV and a hysteresis of 40 mV is acceptable for a total resolution of 15 bits in a 2-1-1 cascaded modulator.

11.5.8 Non-linear Effects

In addition to the gain errors discussed in the previous sections a number of non-linear phenomena may degrade the performance. As is the case for most errors sources in a sigma-delta modulator the errors in the first integrator are most critical and require the most accurate components.

In the actual implementation the capacitors are voltage dependent and this will cause distortion. This non-linearity generate a large second harmonic distortion term while higher order harmonics are smaller. For high resolutions it is therefore important to use accurately balanced differential circuits that suppress the large second harmonic.

Clock feedthrough from the switches is one more source of distortion. By using “bottom plate” sampling where the switch at the opamp input is turned off slightly before the switch at the input of the integrator a large portion of the signal dependent clock feedthrough is removed and the remaining error is mainly an offset. There is however still distortion and again it is important to use accurately balanced differential circuits to suppress even order harmonics.

A third source of distortion is the non-linear resistance of the switches. The input signal to the OSADC is normally a continuous-time signal. If the signal frequency is high and the bandwidth of the passive sampling circuit is too low the variation in bandwidth caused by signal dependent switch resistance will give large distortion. The distortion increases with signal frequency. It is therefore important to make on-resistance of the switches small enough to avoid this problem. The signal dependent on-resistance is mainly a problem for sampling circuits with continuous-time input signals. For all other switches the resistance must only be small enough to ensure that the final value

at the end of the sampling phase is within the required settling accuracy which is usually much less than the resolution of the OSADC. The resistance of the switches may also affect the settling behavior of the opamp. This must be verified by simulation.

The influence of the non-linear error sources discussed in this section can be investigated by a circuit simulators such as SPICE or SPECTRE. However, the results should not be trusted blindly since especially clock feedthrough errors are not always very accurately modelled.

11.6 DESIGN EXAMPLE

In this section we give a design example of a cascaded sigma-delta modulator. The aim is to design a 14-bit converter with a signal bandwidth of 0.5 MHz which corresponds to the requirements for G.lite. We use the 2-1-1 architecture since it gives a low quantization noise. The 2-1-1 modulator architecture is repeated in Fig. 11-17 for

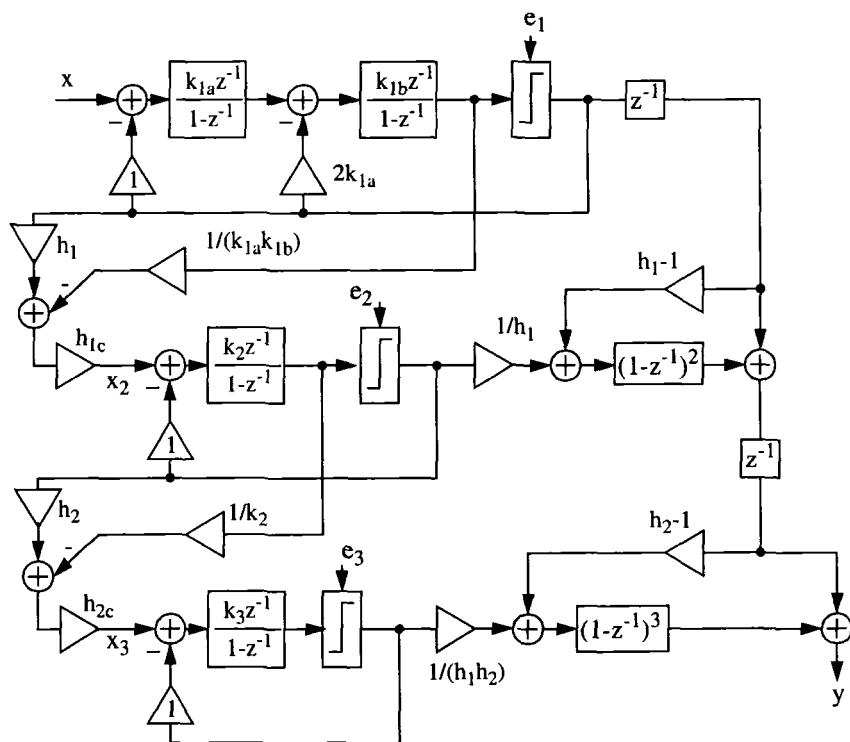


Figure 11-17 4th order 2-1-1 cascaded sigma-delta modulator.

convenience. We start by choosing the coefficients in the modulator. The transfer function of the modulator is given by [2]

$$Y(z) = z^{-4} \cdot X(z) + \frac{(1 - z^{-1})^4 \cdot E_3(z)}{h_{1c} \cdot h_{1c}} \quad (11-36)$$

where $E_3(z)$ is the quantization noise in the third stage. To make the quantization noise small, h_{1c} and h_{2c} should be as large as possible. These two coefficients also determines the gain factors in the digital cancellation logic. To simplify the implementation the factors should be powers of 2. All other gain factors in the modulator are in an SC implementation determined by capacitor ratios. The capacitor ratios should be chosen such that the circuit is easy to layout using unit-capacitors.

To make the modulator less sensitive to circuit noise the voltage swings in all the nodes of the circuit should be approximately the same. The maximum signal swing at the input of last stage can be determined by [2]

$$x_3 = h_{2c} \cdot \left(x_2 + \frac{\Delta}{2} - h_2 \frac{\Delta}{2} \right) \quad (11-37)$$

where Δ is the step size in the D/A converters and x_2 is the maximum swing at the input of the second stage. To maximize h_{2c} , h_2 should be 1. To make the swings equal we also have $h_{1c} = 1$. Coefficients h_1 and h_{1c} influence the performance of the cascaded modulator and the best choice is $h_1 = 3$ and $h_{1c} = 0.5$ ([2]). The integrator gains are chosen to give signal swings at the opamp outputs that are equal to or slightly less than the swing of the feedback signals of the DAC. Suitable values on all the coefficients are now summarized in Table 11-1 ([2]).

Table 11-1 Gain factors in the modulator.

Coeff.	Value
k_{1a}	1/4
k_{1b}	1/2
h_1	3
h_{1c}	0.5
k_2	1/4
h_2	1
h_{2c}	1
k_3	1/4

Behavioral simulations of the modulator were used to plot the SNR as function of signal amplitude for different OSR. The result is shown in Fig. 11-18. We see that an oversampling ratio of 24 is sufficient for 14 bits resolution. A closer look at the plot for **OSR = 24** reveals that the peak SNR appears for an input signal of approximately -3 dBFS. The peak SNR is approximately 86.5 dB which corresponds to a resolution of slightly more than 14 bits. Therefore we decide to use **OSR = 24**. The desired bandwidth is 0.5 MHz which makes the oversampling frequency, $f_s = 2 \cdot 0.5 \cdot 24 = 24$

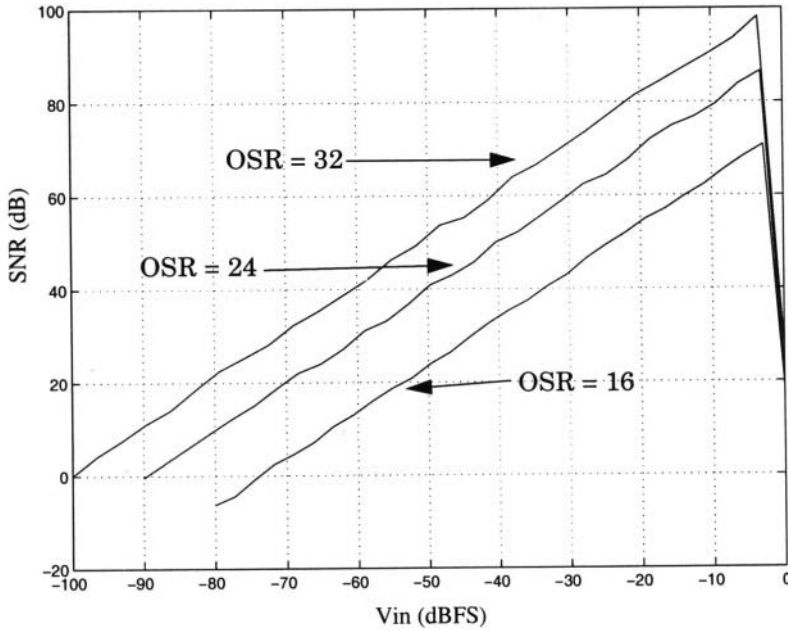


Figure 11-18 SNR vs. input amplitude.

MHz. We assume that the reference voltage levels in the DAC are $\pm 1.25\text{ V}$ and we use a fully differential SC circuits for the implementation.

We can calculate the capacitor ratios in the implementation by using the gain factors in Table 11-1. To get reasonable absolute values on the capacitors we must consider the thermal noise and capacitance matching. The thermal noise in the first integrator dominates. It can be estimated as (see Sec. 11.5.6)

$$v_{n,in}^2 = \frac{kT}{C_{1a}} \cdot (1 + K_1) \quad (11-38)$$

where K_1 is a factor accounting for the thermal noise in the opamp and the noise stored on the capacitor sampling the reference voltage. In this case it is possible to use the improved integrator in Fig. 11-12 where the same capacitor is used to sample both the input signal and the reference signal. The factor K_1 is given by

$$K_1 = \frac{4}{3} \cdot \frac{\beta}{C_{load}} \cdot (1 + n_t) \cdot C_{1a} = \frac{\frac{4}{3} \cdot (1 + n_t)}{\frac{C_L(C_{1a} + C_1)}{C_{1a}C_1} + 1} \quad (11-39)$$

The thermal noise must be smaller than the quantization noise yielding

$$\begin{aligned} v_{n, in}^2 &< \frac{FS^2}{2^{2n_{tot}+2} \cdot 3} \cdot OSR \Rightarrow C_{1b} > \frac{2^{2n_{tot}+2} \cdot 3 \cdot kT \cdot (1 + K_1)}{FS^2 \cdot OSR} = \\ &= \frac{2^2 \cdot 14 + 2 \cdot 3 \cdot 1.38 \times 10^{-23} \cdot 300 \cdot (1 + K_1)}{2.5^2 \cdot 24} \approx K_1 \cdot 90fF \end{aligned} \tag{11-40}$$

The capacitor size in (11-40) is calculated for a single ended implementation. The SNR in a fully differential implementation is at least 3dB better. To account for that the factor K_1 is larger than 1 and to give a design margin we choose a sampling capacitor in the first stage of 1 pF. For the integrators in the following stages the thermal noise is not so critical and the capacitor sizes can be scaled down to save power. Here we decrease the integration capacitors in the second integrator by a factor four, the capacitors in both the third and fourth integrators by a factor eight. The fully differential SC-implementation is shown in Fig. 11-19 ([2]). The capacitor sizes can now be determined as shown in Table 11-2. The signal swings in the integrators were simu-

Table 11-2 Capacitor values in the modulator.

Capacitor	Value (pF)
C_1	4
C_{1a}	1
C_2	1
C_{2a}	0.25
C_{2b}	0.25
C_3	0.5
C_{3a}	0.1875
C_{3b}	0.1875
C_{3c}	0.125
C_4	0.5
C_{4a}	0.25
C_{4b}	0.125
C_{4c}	0.125

lated using a behavioral model and assuming that the reference levels are ± 1.25 V. The input signal is a 0.875V sinusoid (~ -3 dBFS). The result is shown in Fig. 11-20. A closer look at this figure reveals that the peak voltage in integrator 1 is 0.95V while the peak voltage in all other integrators is approximately 0.85V. The peak SNR is 86.7 dB which corresponds to 14.1 ENOB.

The influence of capacitor mismatch was simulated. The relative capacitor errors were assumed to be random Gaussian variables. Monte Carlo analysis was used to determine the yield. The standard deviation of the errors for a 90% yield is shown in Table 11-2.

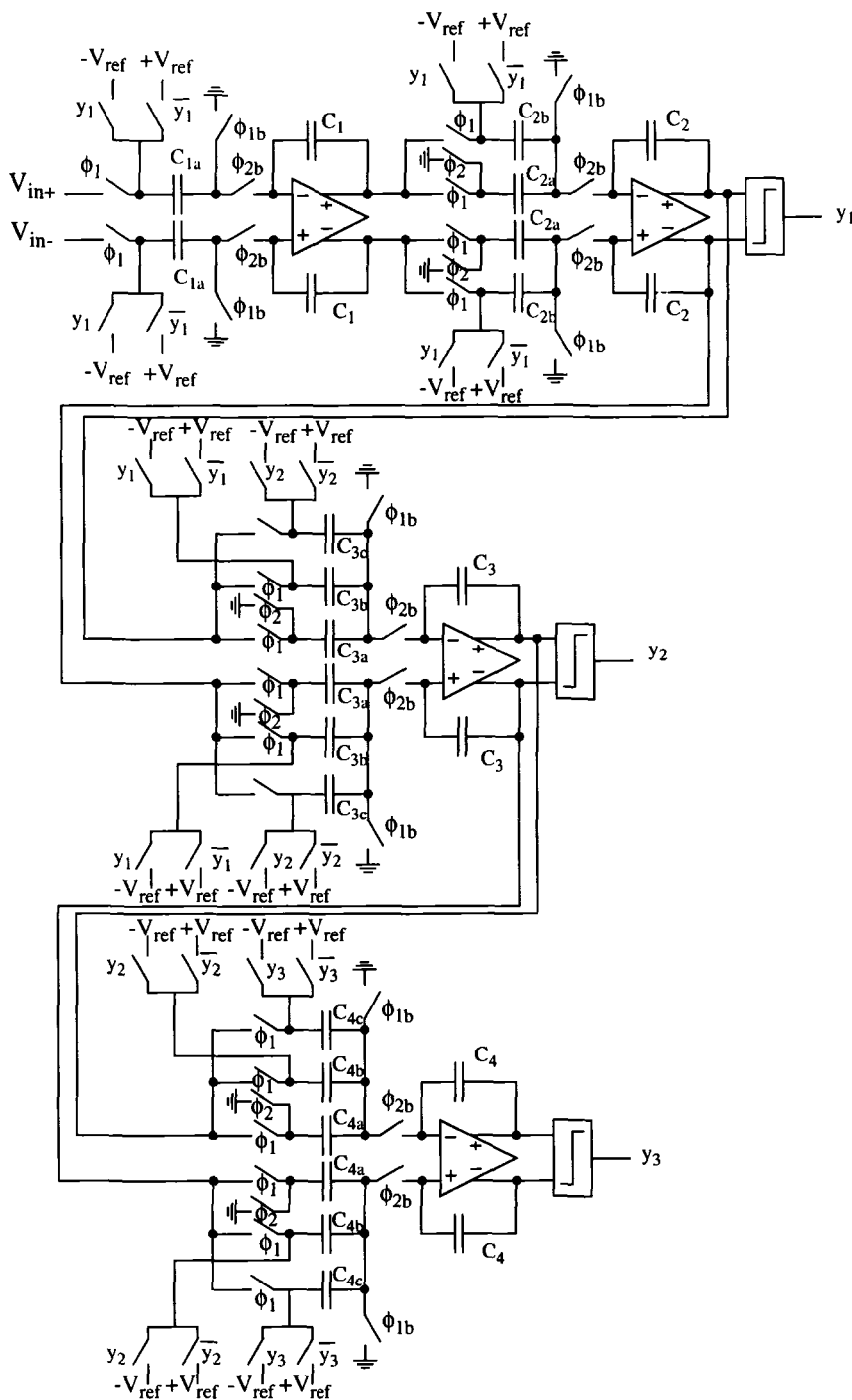


Figure 11-19 SC-implementation of 2-1-1 modulator.

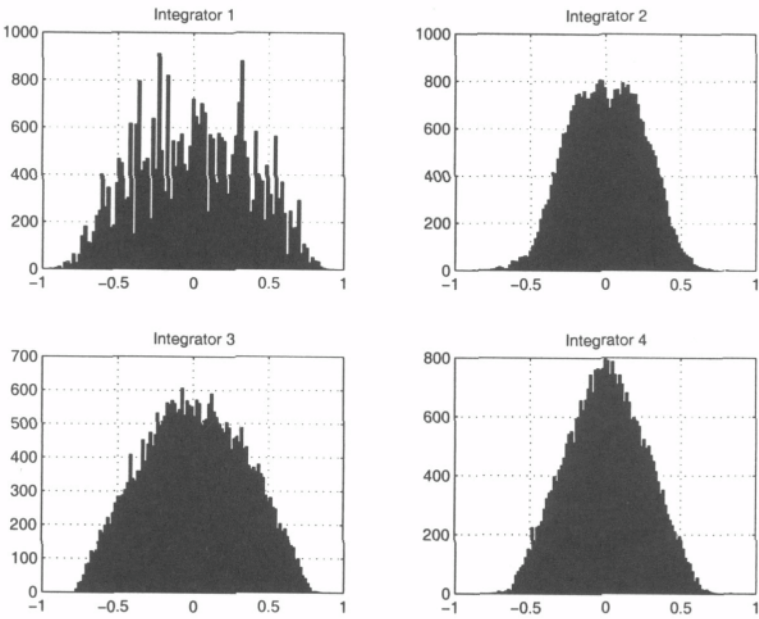


Figure 11-20 Signal swings in 2-1-1 modulator.

Table 11-3 Required relative matching error.

Integrator	Matching
1	0.25%
2	0.25%
3	0.5%
4	2%

The SNR is allowed to drop 1 dB in the yield simulation, i.e. the SNR can drop to 85.7 dB.

To determine the required opamp gain and speed of the SC circuits we must determine the capacitive loads and feedback factors of all the integrators. We must also take the parasitic capacitors into account. We assume that the parasitic capacitor at the output of a integrator is 30% of the integration capacitor The parasitic capacitor at the opamp input in the first integrator is assumed to be **0.5pF**. AH the other integrators are assumed to have a parasitic capacitor of 0.25 pF at the opamp input. The feedback factor during the integration phase of integrator *i* can be calculated as

$$\beta_i = \frac{C_i}{C_i + C_{ia} + C_{ib} + C_{ic} + C_{ip}} \tag{11-41}$$

where C_{ip} is the parasitic capacitor at the input of the opamp. The capacitive load during the integration phase can for integrator i be calculated as

$$C_{il} = \beta_i \cdot (C_{ia} + C_{ib} + C_{ic} + C_{ip}) + C_i \cdot 0.3 \quad (11-42)$$

The result of the calculation is shown in Table 11-4.

Table 11-4 Capacitors and feedback factors.

Capacitor	Value (pF)
β_1	0.72
C_{1l}	2.3 pF
C_{1p}	0.5 pF
β_2	0.57
C_{2l}	0.73 pF
C_{2p}	0.25 pF
β_3	0.4
C_{3l}	0.45 pF
C_{3p}	0.25 pF
β_4	0.4
C_{4l}	0.45 pF
C_{4p}	0.25 pF

According to the previous discussions in this chapter finite opamp gain will change the transfer function of the integrator to

$$H(z) = \frac{r_2 \cdot z^{-1} \cdot a}{1 - \frac{r_2}{r_1} \cdot z^{-1}} \quad (11-43)$$

where

$$r_1 = \frac{\beta_s \cdot A_0}{1 + \beta_s \cdot A_0} \quad r_2 = \frac{\beta_i \cdot A_0}{1 + \beta_i \cdot A_0} \quad (11-44)$$

and β_s is the feedback factor in sampling phase, β_i is the feedback factor in the integration phase and A_0 the opamp DC gain. The feedback factor for integrator i during the sampling phase can be calculated as

$$\beta_{si} = \frac{C_i}{C_i + C_{ip}} \quad (11-45)$$

The calculated values on the feedback factors are shown in Table 11-5.

Table 11-5 Feedback factors in sampling phase

Capacitor	Value (pF)
β_{s1}	0.89
β_{s2}	0.8
β_{s3}	0.67
β_{s4}	0.67

The effect of finite opamp gain was simulated using a behavioral model. The result is shown in Fig. 11-21. We see that the required opamp gain is approximately 60 dB in

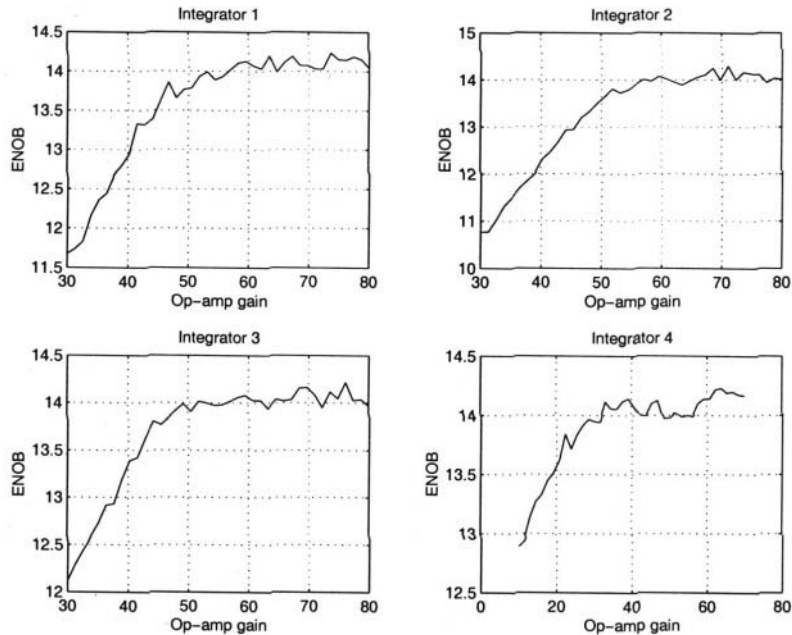


Figure 11-21 Signal swings in 2-1-1 modulator.

the two first integrators, 50 dB in the third and only 30 dB in the fourth. In the design, at least 10-20 dB should be added to the required opamp gains to provide some margin for other errors and to account for the poor modelling of the output conductance in SPICE simulations.

The finite bandwidth due to the opamp, introduces gain errors in the integrators. The equations in Sec. 11.5.2 can be used to find a lower limit on the bandwidth. However, the equations are only valid if the settling is linear. In most practical applications the settling is slew rate limited. The requirements on the bandwidth is increased when the settling is slew rate limited. Therefore the linear settling model is not so useful. It should be noted that when designing the opamp the slew rate and the bandwidth can not be chosen independently. Assuming a single pole opamp the -3 dB bandwidth is

$$\omega_{-3dB} = \frac{g_m}{C_{load}} \cdot \beta \quad (11-46)$$

where g_m is the transconductance of the input device of the opamp, C_{load} the total load at the output of the opamp and β the feedback factor. The transconductance can be calculated as

$$g_m = \frac{I_d}{V_{eff}} \quad (11-47)$$

where I_d is the bias current in the opamp and $V_{eff} = V_{gs} - V_T$ the effective gate voltage of the input device in the opamp. The slew rate is determined by

$$SR = \frac{I_d}{C_{load}} \quad (11-48)$$

Hence the ratio between the slew rate and the bandwidth is

$$\frac{SR}{\omega_{-3dB}} = \frac{V_{eff}}{\beta} \quad (11-49)$$

We see that this ratio is determined by V_{eff} . This voltage can often not change very much. If it is too small the input devices will be very large and if it is too big the voltage swing is reduced. Assuming a effective gate voltage of 0.3-0.4 V and simulating a behavioral model including slew rate (see Sec. 11.5.3) shows that the current in the first integrator must be approximately 0.3-0.4 mA. In the simulation we used typical process parameters from a 0.8u CMOS process. To provide a design margin the current must be larger than the simulated value. This current serves as a starting point for the design of the opamp. When taking all the parasitics into account the current must be increased.

11.7 SUMMARY

In this chapter we have discussed oversampling sigma-delta converters. They are suitable for high resolutions but the high oversampling ratio (OSR) limits the signal bandwidth. To increase the signal bandwidth cascaded modulators have been discussed. Gain errors in the integrators cause leakage of low order quantization noise at the output which limits the resolution of the converter. The integrator gain errors need not be as accurate as the full resolution and converters with up to 15 - 16 bit have been successfully implemented without calibration. To further reduce the quantization noise in the modulator a multi-bit quantizer can be used in the last stage of the converter. For a single stage modulator a multi-bit quantizer would require calibration since the DAC linearity directly limits linearity of the OSADC. For cascaded modulator however, the signal in the last stage has a weak correlation to the input signal and linearity errors in the multi-bit DAC will cause noise, not distortion. In addition to this the DAC linearity error is noise shaped, which significantly reduces the required linearity of the DAC.

We have also discussed SC implementations of oversampled sigma-delta converters and investigated some of the most important limitations and how they affect the performance of the modulator. Based on the discussions and design example was presented.

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12 MODELING OF NYQUIST D/A CONVERTERS

12.1 INTRODUCTION

High-speed DACs are important building blocks in communications systems. They are key components in e.g. xDSL applications. For these and similar applications, traditional static performance measures such as offset error, gain error, integral non-linearity (INL), differential non-linearity (DNL), are not sufficient. For communications applications, the dynamic performance determines the quality of DACs [1, 2]. The most important frequency-domain performance measures are spurious-free dynamic range (SFDR), inter-modulation distortion (IMD), signal-to-noise ratio (SNR), and signal-to-noise-and-distortion ratio (SNDR, or SINAD).

We present an overview of behavioral-level modeling of error sources in Nyquist-rate DACs. Prior modeling [3,4,5, 6] of DAC errors have more been focused on INL and DNL requirements, and only a few of them have discussed the impact of DAC errors on the frequency-domain measures. We show how specific circuit errors as for example matching errors, finite output impedance, and settling errors, affect the performance of the converter. The impact of DAC errors on DNL and INL is also briefly discussed in this chapter.

Since the current-steering DAC is a suitable candidate for high-speed applications, the modeling in this chapter is focused on this type of DAC. Process variations and other parasitics will influence the matching between current sources and will introduce noise and distortion. The output impedance of the current-steering DAC is dependent on the number of current sources connected to the output and is therefore signal-dependent.

The DAC and its major errors sources are outlined in Sec. 12.2. In Sec. 12.3 we show how a finite output resistance influences the performance and in Sec. 12.4 we discuss matching errors. The circuit noise, discussed in Sec. 12.5, and especially thermal noise, limits the resolution. In Sec. 12.6 we also discuss and model the influence of errors in the time-domain and dynamic errors such as nonlinear slewing, bit skew, and glitches.

In Chapter 13 we present the design of a current-steering wideband CMOS DAC for telecommunications applications where the models can be applied to the design strategy.

12.2 ERRORS IN CURRENT-STEERING DACS

The current-steering DACs are widely used. For convenience, the circuit structure of an N -bit DAC is shown in Fig. 12-1 and we will briefly recapture some of its proper-

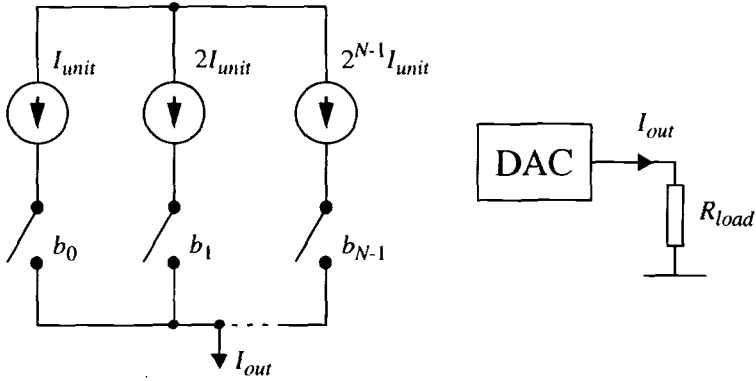


Figure 12-1 a) An N -bit binary weighted current-steering DAC. b) The output is terminated over a 50-ohm load.

ties. To improve matching, unit current sources are used to form the binary weighting. Hence for the i -th bit, we use 2^i unit current sources in parallel instead of a transistor 2^i times wider than the LSB transistor. If the matching errors are uncorrelated, this will improve the element matching and the relative error will improve by a factor of $\sqrt{2^i}$ for the i -th current source. The switches in Fig. 12-1, are controlled by the digital bits, b_i , of the code k

$$k = \sum_{l=0}^{N-1} 2^l \cdot b_l \quad (12-1)$$

where b_{N-1} is the MSB and b_0 is the LSB. The minimum value of k is 0 and the maximum value is $k = 2^N - 1$. The current associated with the i -th LSB position is given by

$$I_i = 2^i \cdot I_{unit} \quad (12-2)$$

and the total output current is given by

$$I_{out}(k(nT)) = I_{unit} \cdot k(nT) \quad (12-3)$$

where T is the sampling period and the output is held during the period T . To increase readability, we will use the notations

$$I_{out}(k) = I_{unit} \cdot k \quad (12-4)$$

12.2.1 Major Error Sources in Current-Steering DACs

Error sources that limit the current-steering DAC performance are

- *Finite output resistance.* The finite output resistance strongly effects the linearity of the converter. This is primarily due to the fact that the output resistance of the converters is signal-dependent.
- *Matching errors.* Since variations in the process cause the oxide thickness and threshold voltage, t_{ox} and V_T , to vary, the unit currents are unequal, which also affects the linearity. The matching errors are of both stochastic and deterministic nature [8].
- *Circuit noise.* A fundamental limit on the resolution is given by the signal-to-noise ratio (SNR) and we have to guarantee that the circuit noise is below the quantization noise floor.
- *Slewing and settling errors.* For DACs, the output is analog and therefore a limited settling time will introduce a settling error which may be signal dependent and hence introduce distortion. The settling errors arise due to the limited RC constant at the output of the converter.
- *Glitches.* Due to the non-ideal switches and different capacitive load on different bits, time skew between the bit switches will for a short period of time create a transition code at the output. This introduces a current or voltage step, referred to as a glitch.
- *Clockfeedthrough (CFT).* Due to capacitive coupling in the current switches between the switching signals (ϕ_i) and the current output, current spikes are added to the signal.

We will address some of these errors in this chapter. In Chapter 13, where we discuss the design of wideband current-steering DACs, we use the results. Most of the models and formulas presented can be applied and generalized to most binary weighted DACs and not only to the current-steering DAC. The models may also easily be extended to cover thermometer coded converters without any large increase of complexity.

12.3 OUTPUT RESISTANCE VARIATIONS

The output impedance and the parasitic impedance of interconnections and switches in the converter will strongly influence the performance [4]. Any non-ideal current source has a finite output resistance and can be modeled as shown in Fig. 12-2. In the figure the current source is terminated over a resistive load at the output.

When the different current sources are switched to the output, the total output resistance is changed and when only static values are considered, the AC current through the load, I_{load} , is

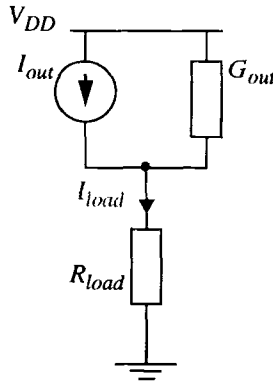


Figure 12-2 Generalized view of a current source with a non-zero output conductance.

$$I_{load} = \frac{I_{out}}{1 + R_{load} \cdot G_{out}} \quad (12-5)$$

where I_{out} is the nominal output current from the DAC given by (12-4), $G_{out} = 1/R_{out}$ is the output conductance, and R_{load} is the load resistance. From (12-5) it is seen, that if the output conductance of the DAC is constant, there is only a gain error, which does not degrade linearity. If the output conductance depends on the input, it will give rise to distortion. Using (12-5), and assuming a signal-dependent output conductance of the DAC, $G_{out}(k)$, the current delivered to the load is

$$I_{load}(k) = \frac{I_{out}(k)}{1 + G_{out}(k) \cdot R_{load}} \quad (12-6)$$

where k is the DAC's digital input given by (12-1).

The output conductance, $G_{out}(k)$, is determined by the number of parallel unit current sources that are switched to the output. The output conductance of one unit current source is assumed to be $G_{unit} = 1/R_{unit}$ and with the i -th LSB we have the corresponding conductance $G_i = 2^{i-1} \cdot G_{unit}$. The total output conductance, $G_{out}(k)$, of the converter is given by

$$G_{out}(k) = G_{unit} \cdot \sum_{l=0}^{N-1} 2^l \cdot b_l = G_{unit} \cdot k \quad (12-7)$$

We refer to the product between the unit output conductance and the load resistance as the conductance ratio, i.e.,

$$\rho = G_{unit} \cdot R_{load} \quad (12-8)$$

We also use the resistance ratio, R_{ratio} , given by

$$R_{ratio} = \frac{1}{\rho} = \frac{R_{unit}}{R_{load}} \quad (12-9)$$

Combining (12-1), (12-4), (12-6), and (12-8), we have that the load current can be written as

$$I_{load}(k) = I_{unit} \cdot \frac{k}{1 + \rho \cdot k} \quad (12-10)$$

Rewriting (12-10), we also end up at

$$I_{load}(k) = \frac{I_{unit}}{\rho} \cdot \left(1 - \frac{1}{1 + \rho \cdot k}\right) \quad (12-11)$$

The effect of the finite impedance ratio is illustrated in Fig. 12-3. The figure shows the

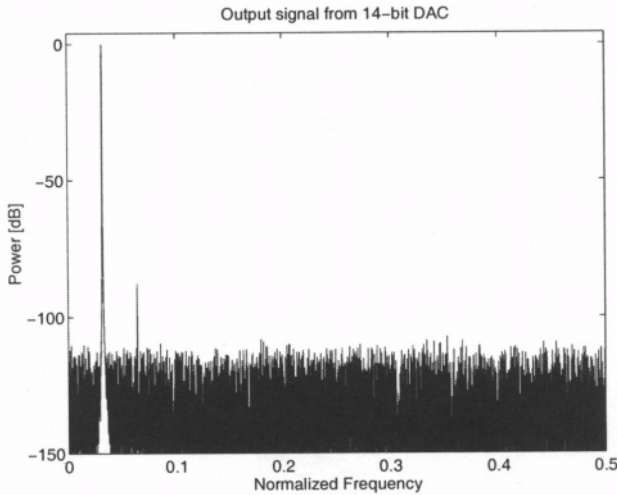


Figure 12-3 Effect of limited resistance ratio (10^8) when a full-scale sinusoid is applied to the DAC.

Matlab simulation result when applying a full-scale sinusoid to a 14-bit DAC. The maximum output impedance of one unit current source is $5 \text{ G}\Omega$ and the load resistance is $50 \text{ }\Omega$. We see in the figure that the SFDR is approximately 88 dBc.

12.3.1 DNL and INL vs. output resistance

We will now show how the nonlinearity affects the differential and integral nonlinearities. The DNL and INL definitions were given in Chapter 1 and we have

$$DNL_k = I(k) - I(k-1) - I_{unit} \quad (12-12)$$

and

$$INL_k = \sum_{i=0}^k DNL_i = I(k) - I(0) - k \cdot I_{unit} \quad (12-13)$$

k is the input code in an offset binary representation. Note that no best-fit line compensation has been done. By inserting (12-11) in (12-12) we get

$$\begin{aligned} DNL_k &= I_{unit} \cdot \left(\frac{k}{1 + \rho \cdot k} - \frac{k-1}{1 + \rho \cdot (k-1)} \right) - I_{unit} \\ &= I_{unit} \cdot \frac{k \cdot (1 + \rho \cdot (k-1)) - (k-1) \cdot (1 + \rho \cdot k)}{(1 + \rho \cdot k)(1 + \rho \cdot (k-1))} - I_{unit} \\ &= \frac{I_{unit}}{(1 + \rho \cdot k)(1 + \rho \cdot (k-1))} - I_{unit} \approx \frac{I_{unit}}{(1 + \rho \cdot k)^2} - I_{unit} \end{aligned} \quad (12-14)$$

The DNL is illustrated in Fig. 12-4a) for a 14-bit DAC with resistance ratio

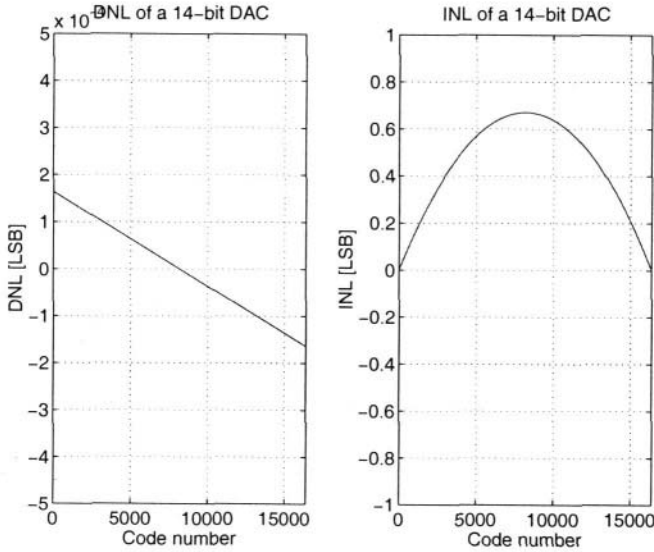


Figure 12-4 Simulated DNL and INL as a function of input code for a resistance ratio of 10^8 .

$R_{ratio} = 10^8$ and $50\text{-}\Omega$ load resistance. The DNL and INL values in the figures have been calculated with respect to a best-fit line. The best-fit offset and gradient values were approximately $0.447 \cdot I_{unit}$ and $0.9998 \cdot I_{unit}$, respectively.

We see that the DNL in (12-14) is a decreasing monotonic function, since the term

$$\frac{I_{unit}}{(1 + \rho \cdot k)^2} \quad (12-15)$$

is positive and decreasing towards zero for increasing k . This implies that DNL is always larger than -1 LSB. As long as the resistance ratio is reasonably large, monotonicity is guaranteed, and we can see in Fig. 12-4a) that the DNL is still very good, but from Fig. 12-3 we found that the $SFDR$ is low and may limit the resolution. The INL for the same DAC configuration is shown in Fig. 12-4b) where the INL is given by (12-13). Also, especially in communications, these measures cannot directly be used to analyze the true impact of the resistance error.

12.3.2 SNDR vs. Output Resistance

The error current, $\Delta I(k)$, is given by

$$\Delta I(k) = I_{load}(k) - I_{out}(k) = I_{unit} \cdot k - \frac{I_{unit}}{1 + \rho \cdot k} \cdot k = \frac{\rho \cdot k^2}{1 + \rho \cdot k} \cdot I_{unit} \quad (12-16)$$

where $I_{load}(k)$ is the actual current through the load resistance and $I_{out}(k)$ is the desired output current (the sum of the currents from the current sources). The expression in (12-16) can be compared to the ideal continuous-time current, i.e., a ramp. The error power introduced is given by the quantization noise power and the time averaged power from (12-16).

$$P_n = P_{qn} + P_\epsilon = \frac{I_{unit}^2}{12} + \overline{[\Delta I(k)]^2} \quad (12-17)$$

where

$$P_{qn} = \frac{I_{unit}^2}{12} \text{ and } P_\epsilon = \overline{[\Delta I(k)]^2} \quad (12-18)$$

Assume that the input signal is a sinusoidal signal

$$k = K_{dc} + K_{ac} \cdot \sin\theta + v \quad (12-19)$$

where K_{dc} is the DC level of the signal, K_{ac} is the amplitude of the sinusoid, θ is the normalized signal frequency times the sequence index, and v corresponds to the quantization error, which is AWGN for converters with a larger number of bits. The AC power of the sinusoid at the output is given by

$$P_s = \frac{K_{ac}^2}{2} \cdot I_{unit}^2 \quad (12-20)$$

The average error power P_ϵ is found by approximating (12-16) by using the first order Taylor expansion and the average value of $k = K_{dc}$ from (12-19), i.e.,

$$P_\epsilon \approx \{(\rho \cdot K_{dc}^2) \cdot I_{unit}\}^2 \quad (12-21)$$

Now we can find the signal-to-noise-and-distortion ratio ($SNDR$) as a function of the output impedance, hence

$$SNDR = \frac{P_s}{P_n} = \frac{P_s/P_{qn}}{1 + P_\epsilon/P_{qn}} \approx \frac{6K_{ac}^2}{1 + \{12 \cdot \rho \cdot K_{dc}^2\}^2} \quad (12-22)$$

In Fig. 12-5 we show the simulated SNDRs of 10-bit, 12-bit, and 14-bit DACs when

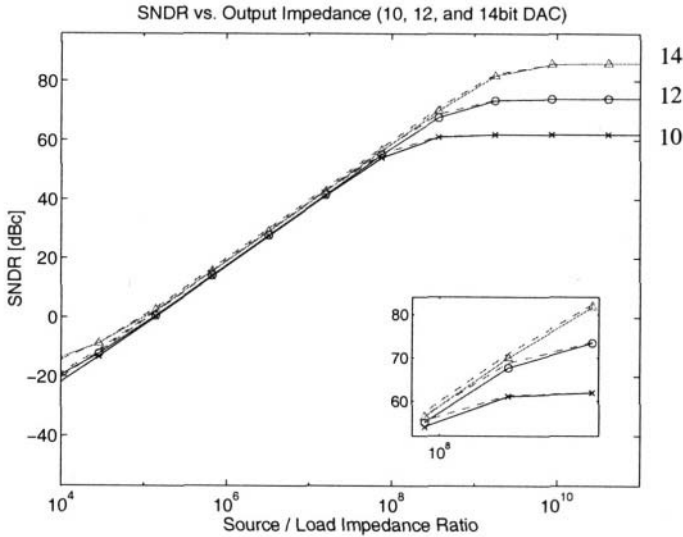


Figure 12-5 Simulated and calculated SNDR vs. resistance ratio for 10-bit, 12-bit, and 14-bit DACs. To illustrate the similarity with the calculated, a portion of the curves is magnified.

varying the resistance ratio, R_{ratio} . The input signal is a full-scale sinusoid. In the simulation we use $R_{load} = 50 \Omega$ and $I_{unit} = 1.25 \mu A$. The SNDR results is compared with the calculated results from (12-22) and the formula is verified. In the figure we see that the curves are saturated at both lower and higher resistance ratios. The lower saturation arises in the simulator (Matlab) since it becomes difficult to separate signal power from noise power at such low resistance ratios. The higher saturation arises due to the fact that the quantization noise is dominating at higher ratios and the SNDR is given by $SNDR \approx 6.02 \cdot N + 1.76 \text{ dB}$.

12.3.3 SFDR vs. Output Resistance

The size of the distortion tone in Fig. 12-3 can be found by considering the case of a single sinusoid input as given in (12-19). Then the output current as given by (12-11) becomes

$$I_{load}(k) = \frac{I_{unit}}{\rho} \cdot \left[1 - \frac{1}{1 + \rho \cdot (K_{dc} + K_{ac} \cdot \sin \theta)} \right] \quad (12-23)$$

where the noise term v has been neglected. The equation is rewritten as

$$\begin{aligned}
 I_{load}(k) &= \frac{I_{unit}}{\rho} \cdot \left(1 - \frac{1}{1 + \rho \cdot K_{dc} + \rho \cdot K_{ac} \cdot \sin\theta} \right) \\
 &= \frac{I_{unit}}{\rho} \cdot \left(1 - \frac{1}{1 + \rho \cdot K_{dc}} \cdot \frac{1}{1 + \frac{\rho \cdot K_{ac}}{1 + \rho \cdot K_{dc}} \cdot \sin\theta} \right)
 \end{aligned} \tag{12-24}$$

Examining (12-24) we find that only the second term within paranthesis contains AC frequency information, and we have

$$I_{load}(k) = \frac{I_{unit}}{\rho} \cdot \left(1 - \frac{1}{1 + \rho \cdot K_{dc}} \cdot I_{ac}(k) \right) \tag{12-25}$$

The gain factors inside and outside the parantheses can be neglected since we are considering only power *ratios* when determining the *SFDR*, and the AC signal is

$$I_{ac}(X) = \frac{1}{1 + A \cdot \sin\theta} \tag{12-26}$$

where

$$A = \frac{\rho \cdot K_{ac}}{1 + \rho \cdot K_{dc}} \tag{12-27}$$

Comparing (12-24) with (12-25) it is clear that $I_{load}(k)$ and $I_{ac}(k)$ will have the same distortion, since they only differ in offset and constant gain.

To avoid signal clipping we have that $K_{ac} < K_{dc}$ for a binary offset code, and we have

$$0 \leq A < 1 \tag{12-28}$$

and obviously that

$$|A \cdot \sin\theta| < 1 \tag{12-29}$$

This implies that we may find a converging Taylor series expansion of $I_{ac}(k)$ from (12-26)

$$\begin{aligned}
 I_{ac}(k) &= \sum_{n=0}^{\infty} (-A \cdot \sin\theta)^n \\
 &= 1 - A \sin\theta + \sum_{n=1}^{\infty} (A \sin\theta)^{2n} - \sum_{n=1}^{\infty} (A \sin\theta)^{2n+1}
 \end{aligned} \tag{12-30}$$

The DC level and the gain can once again be neglected. By using trigonometric formulas, we find that (12-30) may be written as

$$\begin{aligned}
 I_{ac}(k) = f(k) - \sin\theta \cdot A \cdot \left[1 + \sum_{n=1}^{\infty} \left(\frac{A}{2}\right)^{2n} \binom{2n+1}{n} \right] - \dots \\
 \dots - \cos 2\theta \cdot 2 \sum_{n=1}^{\infty} \left(\frac{A}{2}\right)^{2n} \binom{2n}{n-1}
 \end{aligned} \tag{12-31}$$

where $f(k)$ contains the DC component and higher order harmonics that do not influence the *SFDR*. The *SFDR* is now found in (12-31) as the power ratio between the fundamental and the 2nd harmonic as

$$SFDR = \frac{A^2}{4} \cdot \left[\frac{1 + \sum_{n=1}^{\infty} \left(\frac{A}{2}\right)^{2n} \binom{2n+1}{n}}{\sum_{n=1}^{\infty} \left(\frac{A}{2}\right)^{2n} \binom{2n}{n-1}} \right]^2 = \left(\frac{1 + \sqrt{1-A^2}}{A} \right)^2 \tag{12-32}$$

By substituting back A from (12-27) in (12-32) this becomes

$$SFDR = \left[\frac{1 + \rho \cdot K_{dc}}{\rho \cdot K_{ac}} + \sqrt{\left(\frac{1 + \rho \cdot K_{dc}}{\rho \cdot K_{ac}} \right)^2 - 1} \right]^2 \tag{12-33}$$

which may also be written as

$$SFDR = \left[\frac{R_{ratio} + K_{dc}}{K_{ac}} + \sqrt{\left(\frac{R_{ratio} + K_{dc}}{K_{ac}} \right)^2 - 1} \right]^2 \tag{12-34}$$

where $R_{ratio} = 1/\rho$. When R_{ratio} is very large, we have that (12-34) can be approximated

$$SFDR \approx \left[2 \cdot \frac{R_{ratio} + K_{dc}}{K_{ac}} \right]^2 \tag{12-35}$$

The result in (12-35) is important. If $R_{ratio} \gg K_{dc}$ we have that the *SFDR* is not strongly dependent on the DC level. On the other hand we see that the *SFDR* is strongly dependent on the AC amplitude. For the case with equal amplitude and DC level, i.e.,

$$K_{ac} = K_{dc} \tag{12-36}$$

we have that

$$SFDR = \left[1 + \frac{R_{ratio}}{K_{ac}} \cdot \left(1 + \sqrt{1 + \frac{2 \cdot K_{ac}}{R_{ratio}}} \right) \right]^2 \tag{12-37}$$

When R_{ratio} is large (12-37) becomes approximately

$$SFDR \approx 20\log R_{ratio} - 20\log \frac{K_{ac}}{2} \text{ dBc} \quad (12-38)$$

We see that by decreasing the AC amplitude, X_{ac} , of the signal we get an improvement of the linearity of the circuit. In the special case with a full-scale sinusoid, i.e.,

$$K_{ac} = K_{dc} = \frac{2^N - 1}{2} \approx 2^{N-1} \quad (12-39)$$

we get

$$SFDR = \left(1 + \frac{R_{ratio}}{2^{N-1}} \cdot \left[1 + \sqrt{1 + \frac{2^N}{R_{ratio}}} \right] \right)^2 \approx \left(2 \cdot \frac{R_{ratio} + 2^{N-1}}{2^{N-1}} \right)^2 \quad (12-40)$$

The result found in (12-40) is based on that the second harmonic is dominating the distortion [9, 10, 11]. To illustrate this, we show in Fig. 12-6 the simulated output spec-

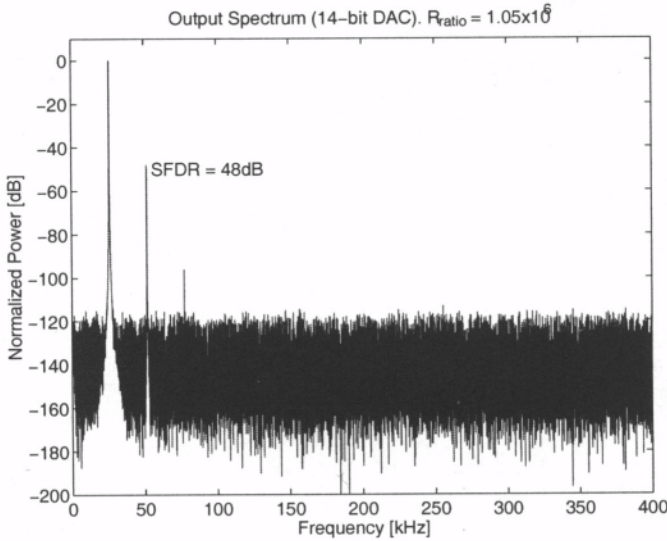


Figure 12-6 Output spectrum when applying a full-scale sinusoid on a 14-bit DAC. The resistance ratio is $R_{ratio} = 1.05 \cdot 10^6$.

trum from a 14-bit DAC when applying a full-scale sinusoid and having a resistance ratio of $R_{ratio} = 1.05 \cdot 10^6$. If $R_{ratio} \gg 2^N$, (12-40) can be approximated

$$SFDR \approx 20\log R_{ratio} - 6(N - 2) \text{ dBc} \quad (12-41)$$

From (12-41) we realize that with a doubling of the load resistance, the $SFDR$ is decreased by 6 dB. With a maintained resistance ratio, the linearity will also deteriorate with an increased nominal number of bits.

In Fig. 12-7 we show the simulated and calculated $SFDR$ vs. resistance ratio for a 10-

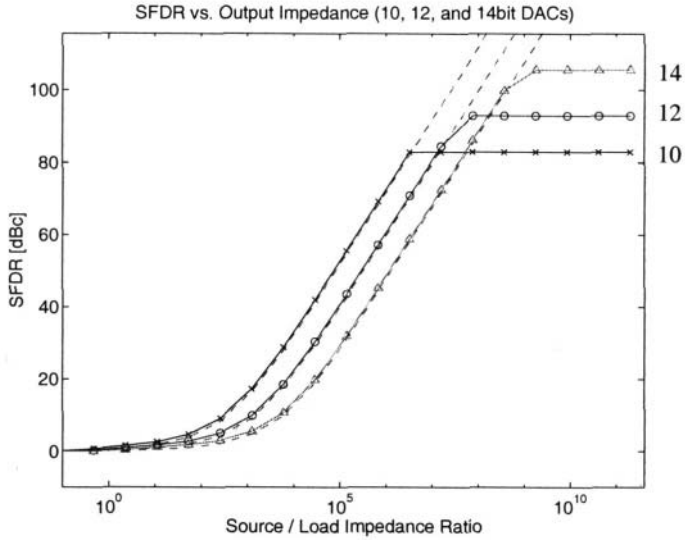


Figure 12-7 Simulated (solid) and calculated (dashed) SFDR vs. output over load resistance ratio for 10-bit (x), 12-bit (o), and 14-bit (triangle) DACs.

12-, and 14-bit DAC. At high ratios the simulated values are saturated since the spurious are hidden in the noise floor and at low ratios the distortion becomes very large (compare Fig. 12-5). It is seen that the simulated *SFDR* follows the mathematical result well.

When using differential signals, harmonics of even order will be cancelled and the 3rd harmonic dominates. It can be found [9, 10] that the distortion with respect to the 3rd harmonic is

$$SFDR = \left[1 - 2 \cdot \frac{K_{dc} + R_{ratio}}{K_{ac}} \cdot \left(\frac{K_{dc} + R_{ratio}}{K_{ac}} + \sqrt{\left(\frac{K_{dc} + R_{ratio}}{K_{ac}} \right)^2 - 1} \right) \right]^2 \quad (12-42)$$

For high ratios, this is approximately

$$SFDR \approx \left(2 \cdot \frac{K_{dc} + R_{ratio}}{K_{ac}} \right)^4 \quad (12-43)$$

For $K_{dc} = K_{ac} \approx 2^{N-1}$ and $R_{ratio} \gg K_{ac}$ we get

$$SFDR \approx 40 \log R_{ratio} - 12(N - 2) \text{ dBc} \quad (12-44)$$

From (12-44) we see that the *SFDR* with respect to the third harmonic is increasing faster with respect to increase of a nominal number of bits, N , than for the *SFDR* with respect to the second harmonic (12-41). However, it is obvious that if differential signals are used, the requirement on the output impedance can be relaxed.

12.3.4 Influence of Parasitic Resistance

The models in the previous sections only considered the load resistance. The resistance of internal wires and switches has not been considered. However, these may be incorporated as well, see the modified current source in Fig. 12-8 a). For the i -th LSB

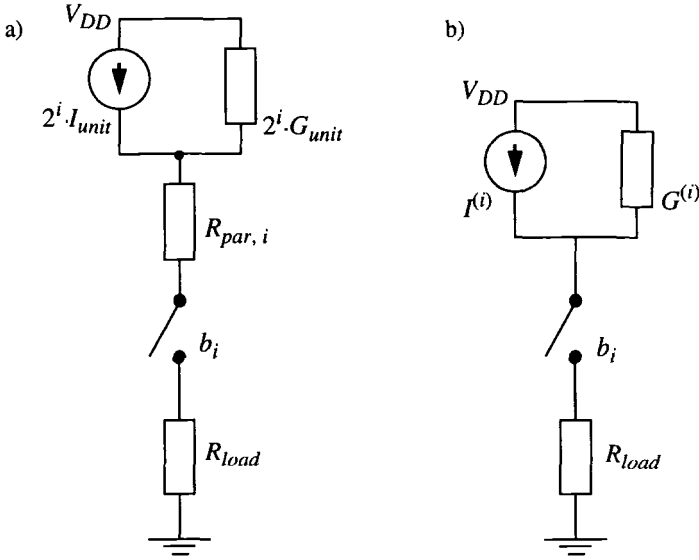


Figure 12-8 Model of the i -th LSB current source at with (a) parasitic resistance, $R_{par,i}$ from switches and internal wires and (b) modified model.

current source, we also include the parasitic resistance, $R_{par,i}$, which is associated with the interconnection wires from the current source to the output and the switch-on resistance of the switch for the bit position. We see that the switch-on resistance of the switch will introduce larger errors for the MSB current sources, since the output impedance is much smaller than for the LSB current sources.

Using Norton's theorem, we can transform the circuit to be similar to the current source as illustrated in Fig. 12-8b). Using that schematic view, we have a modified current source with the value

$$I^{(i)} = \frac{2^i \cdot I_{unit}}{1 + 2^i \cdot G_{unit} \cdot R_{par,i}} \quad (12-45)$$

and a modified output conductance

$$G^{(i)} = \frac{2^i \cdot G_{unit}}{1 + 2^i \cdot G_{unit} \cdot R_{par,i}} \quad (12-46)$$

Using superposition with the current sources of other bit positions and using the motivation leading to (12-6), we have that the total output current is

$$I_{load}(k) = \frac{\sum_i I^{(i)} \cdot b_i}{1 + R_{load} \cdot \sum_i G^{(i)} \cdot b_i} = \frac{\sum_i \frac{2^i \cdot I_{unit} \cdot b_i}{1 + 2^i \cdot G_{unit} \cdot R_{par,i}}}{1 + \sum_i \frac{2^i \cdot G_{unit} \cdot b_i \cdot R_{load}}{1 + 2^i \cdot G_{unit} \cdot R_{par,i}}} \quad (12-47)$$

Comparing (12-47) with (12-6) we see that for each bit b_i , a bit weight of

$$w_i = \frac{2^i}{1 + 2^i \cdot G_{unit} \cdot R_{par,i}} \quad (12-48)$$

is associated instead of the desired 2^i . For large i , hence for the MSBs, the weight is decreasing, hence a non-linear transfer function which introduce distortion. Assume that we are able to design the current switches to have a parasitic resistance which is exponentially lower for the MSB and higher for the LSB, as

$$R_{par,i} = \frac{R_{par,LSB}}{2^i} = \frac{R_{par}}{2^i} \quad (12-49)$$

where $R_{par} = R_{par,LSB}$ is the parasitic resistance associated with the LSB current source. The bit weight, according to (12-48), turns into

$$w_i = \frac{2^i}{1 + G_{unit} \cdot R_{par}} \quad (12-50)$$

which only gives a gain error, since using these weights, (12-47) turns into

$$\begin{aligned} I_{load}(k) &= \frac{\sum_i \frac{2^i \cdot I_{unit} \cdot b_i}{1 + G_{unit} \cdot R_{par}}}{1 + \sum_i \frac{2^i \cdot G_{unit} \cdot b_i \cdot R_{load}}{1 + G_{unit} \cdot R_{par}}} = \\ &= \frac{1}{1 + G_{unit} \cdot R_{par}} \cdot \frac{I_{unit} \cdot k}{1 + \frac{G_{unit} \cdot R_{load}}{1 + G_{unit} \cdot R_{par}} \cdot k} \end{aligned} \quad (12-51)$$

Comparing with (12-10) we see that (12-51) contains an additional gain factor and a parasitic conductance ratio as

$$\rho_p = \frac{G_{unit} \cdot R_{load}}{1 + G_{unit} \cdot R_{par}} \quad (12-52)$$

We also define the parasitic resistance ratio as

$$R_{p, ratio} = \frac{1}{\rho_p} = \frac{1 + G_{unit} \cdot R_{par}}{G_{unit} \cdot R_{load}} = \frac{R_{unit} + R_{par}}{R_{load}} \quad (12-53)$$

From (12-51) we see that for a high parasitic resistance, there is less current directed

to the output, and this gain factor itself will not influence the SFDR and SNDR, but at the same time the parasitic conductance ratio becomes smaller and the linearity is improved.

The modified ratios, (12-52) and (12-53), can be used to find the approximate SNDR and SFDR in the formulas (12-22), (12-34), and (12-44).

12.4 CURRENT SOURCE MISMATCH

A current source with a mismatch error can be modeled as an additional current source in parallel with the nominal current source, as shown in Fig. 12-9 for the current

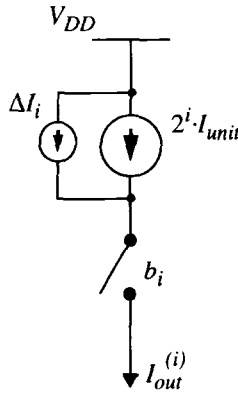


Figure 12-9 Modeling of current source with error current source, ΔI_i .

source corresponding to bit i . The actual output is then given by

$$I_{out}^{(i)} = b_i \cdot (2^i \cdot I_{unit} + \Delta I_i) \quad (12-54)$$

where an infinite output impedance is assumed. All individual error sources can be summed and modeled as one total error current source connected to the output. The output current delivered to the load, $I_{load}(k)$, can be written as the sum of the nominal output current, $I_{out}(k)$, and the error current, $\Delta I(k)$

$$I_{load}(k) = I_{out}(k) + \Delta I(k) \quad (12-55)$$

where k is the digital number given by (12-1). Considering the individual bit positions, we can rewrite (12-55) as

$$I_{load}(k) = I_{out}(k) + \sum_{i=0}^{N-1} \Delta I_i(b_i) \quad (12-56)$$

where $\Delta I_i(b_i)$ is given by

$$\Delta I_i(b_i) = 2^i \cdot b_i \cdot \delta_i(b_i) \cdot I_{unit} \quad (12-57)$$

where $\delta_i(b_i)$ is the relative error current in the i -th bit. In the static case, we assume that $\delta_i(b_i) = \delta_i$ for all different inputs, k . This gives the total output current as

$$\begin{aligned} I_{load}(k) &= I_{out}(k) + I_{unit} \cdot \sum_{i=0}^{N-1} 2^i \cdot \delta_i \cdot b_i \\ &= I_{unit} \cdot \left[k + \sum_{i=0}^{N-1} 2^i \cdot \delta_i \cdot b_i \right] = I_{unit} \cdot \sum_{i=0}^{N-1} 2^i \cdot (1 + \delta_i) \cdot b_i \end{aligned} \quad (12-58)$$

In reality, mismatch errors of transistors due to process variations include both graded linear and stochastic errors [5,8, 12]. Gradients in oxide thickness [5] and along wires or voltage drops over supply lines [4, 13] create linear matching errors which are strongly dependent on the layout of the current sources. These type of errors will not be discussed here since it can be avoided by proper layout. Correlated matching errors are briefly discussed in Sec. 12.4.3, but interested readers can find more work in [14] on linearly distributed matching errors in DACs. In this text, we focus on stochastic errors. For one specific DAC the matching error associated with one current source is fixed, but comparing a large number of chips (of same design) the matching error will be gaussian distributed and more or less uncorrelated with other the matching error in other chips.

For a CMOS current source, the β - and V_T -mismatch can be characterized by their distribution, transistor sizes, and physical mutual distance [8, 12]. This is further addressed in Chapter 13.

$E_{\delta}\{A\}$ is used to denote the expectation value of A with respect to the stochastic variable δ , and $\overline{E_{\delta}\{k\}}$ denotes the average value with respect to the input signal k , as

$$\overline{E_{\delta}\{k\}} = \lim_{P \rightarrow \infty} \frac{1}{P} \sum_{p=0}^{P-1} I(k(pT)) \quad (12-59)$$

First, we assume that the matching errors in the different unit current sources are uncorrelated. This is a very coarse assumption, and in reality, transistors close to each other will have highly correlated errors. With the unit current source $I_{unit}^{(m)}$, $m = 0, \dots, 2^N - 1$, we associate a relative error

$$\delta_{unit}^{(m)} = \delta_{unit} \quad (12-60)$$

Its mean value and standard deviation are

$$E_{\delta}\{\delta_{unit}^{(m)}\} = \mu_{unit}^{(m)} = \mu_{unit} = 0 \text{ and } E_{\delta}\{[\delta_{unit}^{(m)}]^2\} = \sigma_{unit}^{(m)} = \sigma_{unit} \quad (12-61)$$

A non-zero mean value will only give rise to a DC error in the output and can be neglected in these calculations. The expectation value, with respect to the matching error, of the output current from the i -th bit, which consists of 2^i current sources in

parallel, follows from (12-61)

$$\begin{aligned}
 E_{\delta}\{I_{out}^{(i)}\} &= E_{\delta}\left\{\sum_{m=1}^{2^i} I_{unit} \cdot (1 + \delta_{unit}^{(m)})\right\} = \dots \\
 \dots &= 2^i \cdot I_{unit} + I_{unit} \cdot \sum_{m=1}^{2^i} E_{\delta}\{\delta_{unit}^{(m)}\} \approx 2^i \cdot I_{unit}
 \end{aligned} \tag{12-62}$$

The variance of the same current is given by

$$\begin{aligned}
 E_{\delta}\{[I_{out}^{(i)} - E_{\delta}\{I_{out}^{(i)}\}]^2\} &= E_{\delta}\left\{\left[\sum_{m=1}^{2^i} I_{unit} \cdot \delta_{unit}^{(m)}\right]^2\right\} = \dots \\
 \dots &= I_{unit}^2 \cdot \sum_{m=1}^{2^i} E_{\delta}\{[\delta_{unit}^{(m)}]^2\} = I_{unit}^2 \cdot 2^i \cdot \sigma_{unit}^2
 \end{aligned} \tag{12-63}$$

From (12-62) and (12-63) we find the mean value and standard deviation for the normalized error δ_i in the current source corresponding to the i -th bit as

$$\mu_i = 0 \text{ and } \sigma_i = \frac{\sqrt{I_{unit}^2 \cdot 2^i \cdot \sigma_{unit}^2}}{I_{unit} \cdot 2^i} = \frac{\sigma_{unit}}{\sqrt{2^i}} \tag{12-64}$$

12.4.1 SNDR vs. Mismatch

Since the mismatch errors are assumed to be uncorrelated, the expectation value of the power is given by

$$\begin{aligned}
 E_{\delta}\{I_{load}^2(k)\} &= E_{\delta}\{I_{out}^2(k)\} + E_{\delta}\left\{\left(I_{unit} \cdot \sum_{i=0}^{N-1} b_i \cdot 2^i \cdot \delta_i\right)^2\right\} = \\
 &= I_{out}^2(k) + I_{unit}^2 \cdot \sum_{i=0}^{N-1} b_i^2 \cdot 2^{2i} \cdot E_{\delta}\{\delta_i^2\}
 \end{aligned} \tag{12-65}$$

For the mismatch error we have from (12-64)

$$E_{\delta}\{\delta_i^2\} = \sigma_i^2 = \frac{\sigma_{unit}^2}{2^i} \tag{12-66}$$

Since $b_i \in \{0, 1\}$, we have that $b_i = b_i^2$. Using (12-66) in (12-65) we get

$$\begin{aligned}
 E_{\delta}\{I_{load}^2(k)\} &= I_{out}^2(k) + I_{unit}^2 \cdot \sum_{i=0}^{N-1} b_i \cdot 2^i \cdot \sigma_{unit}^2 = \\
 &= I_{unit}^2 \cdot k^2 + I_{unit}^2 \cdot \sigma_{unit}^2 \cdot k
 \end{aligned} \tag{12-67}$$

which actually can be derived directly from the result in (12-63). We can now find the code-averaged power value for the expression in (12-67) (also compare with Fig. 12-5) and we have that

$$\overline{E_8\{I_{load}^2(k)\}} = I_{unit}^2 \cdot \bar{k}^2 + I_{unit}^2 \cdot \sigma_{unit}^2 \cdot K_{dc} \quad (12-68)$$

where $K_{dc} = \bar{k}$ is the average value of the input code, i.e., the DC value. We have the error power given by

$$P_\epsilon = I_{unit}^2 \cdot \sigma_{unit}^2 \cdot K_{dc} \quad (12-69)$$

and we see that by reducing the DC value of the signal, we also reduce the error power, however the DC value is mostly fixed to $K_{dc} \approx 2^{N-1}$. By using (12-18), (12-20), (12-22), and (12-69), we can calculate the SNDR as

$$SNDR = \frac{P_s}{P_{qn} + P_\epsilon} = \frac{P_s/P_{qn}}{1 + P_\epsilon/P_{qn}} = \frac{6 \cdot K_{dc}^2}{1 + 12\sigma_{unit}^2 \cdot K_{dc}} \quad (12-70)$$

With a full-scale sinusoid we have from (12-39) that the SNDR (12-70) becomes

$$SNDR = \frac{(3/2) \cdot 2^{2N}}{1 + \sigma_{unit}^2 \cdot 2^N \cdot 6} \quad (12-71)$$

In dB we have that

$$SNDR \approx 6N + 1.76 - 10\log(1 + 6 \cdot \sigma_{unit}^2 \cdot 2^N) \quad (12-72)$$

In Fig. 12-10 we show the simulated and calculated SNDR vs. mismatch for a 10-, 12-, and 14-bit DAC. The results are found by taking the average value of 1024 simulations for each mismatch value. We find that the simulated values match the calculated ones well.

12.4.2 SFDR vs. Mismatch

To find the SFDR, we have to investigate how the mismatch error power is distributed in the frequency domain. Consider the Fourier series coefficients, $C_{p,i}$, for bit b_i

$$C_{p,i} = \frac{1}{M} \sum_{m=0}^{M-1} b_i(m) \cdot e^{-2\pi j \cdot \frac{m}{M} \cdot (p+1)} \quad (12-73)$$

where m is the sequence index and M is the period in number of samples. The coefficients for the total error current, I_{Cp} , are given by using (12-58) and (12-73)

$$I_{Cp} = I_{unit} \cdot \sum_{i=0}^{N-1} 2^i \cdot \delta_i \cdot C_{p,i} \quad (12-74)$$

The power of each tone from (12-74) is given by

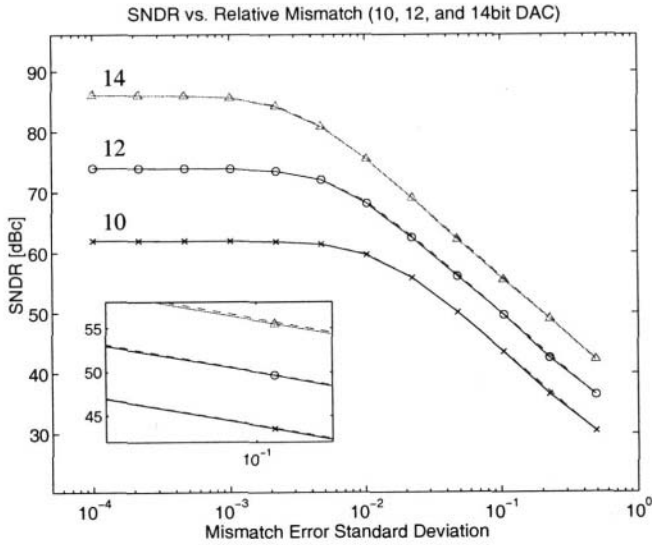


Figure 12-10 Calculated and simulated *SNDR* vs. mismatch error standard deviation for a 10-, 12-, and 14-bit DAC.

$$P_p = 2 \cdot |I_{Cp}|^2 \quad (12-75)$$

Since all mismatch errors are uncorrelated, we have that the expectation value of the power of the p -th tone is

$$E_\delta\{P_p\} = E_\delta\{2 \cdot |I_{Cp}|^2\} = 2I_{unit}^2 \cdot \sigma_{unit}^2 \cdot \sum_{i=0}^{N-1} 2^i \cdot |C_{p,i}|^2 \quad (12-76)$$

For a full-scale single-tone sinusoid the AC power is

$$P_s = \frac{1}{8} \cdot I_{unit}^2 \cdot 2^{2N} = 2^{2N-3} \cdot I_{unit}^2 \quad (12-77)$$

The harmonic distortion with respect to the p -th harmonic is approximately

$$HD_p = \frac{P_s}{P_p} = \frac{2^{2N-3} \cdot I_{unit}^2}{2I_{unit}^2 \cdot \sigma_{unit}^2 \cdot \sum_{i=0}^{N-1} 2^i \cdot |C_{p,i}|^2} \approx \frac{2^{2N}}{16\sigma_{unit}^2} \cdot \left[\sum_{i=0}^{N-1} 2^i \cdot |C_{p,i}|^2 \right]^{-1} \quad (12-78)$$

For each bit we have the error given by a pulse-shaped waveform. Now, we assume that the matching error dominates in the MSBs (compare to (12-63)). If the input signal is a full-scale sinusoid, the MSB will have a pulse width of $M/2$ and a period of M and its Fourier series coefficients are given by

$$|C_{p, N-1}|^2 = \begin{cases} 0 & p \text{ is even} \\ \frac{M^2}{\sin^2(p\pi/M)} \approx \frac{1}{(p\pi)^2} & p \text{ is odd} \end{cases} \quad (12-79)$$

where M is the period. Since in the static case M is larger than $p\pi$ the approximation holds. For the lower significant bits, the frequency spectrum is more noise-like. To find the harmonic distortion we use the approximation in (12-79) to rewrite (12-78) as

$$HD_p \approx \frac{2^{2N}}{16 \cdot \sigma_{unit}^2} \cdot \frac{(p\pi)^2}{2^N - 1} \approx \frac{2^N}{16 \cdot \sigma_{unit}^2} \cdot (p\pi)^2 \quad (12-80)$$

In dBc we find that

$$HD_p \approx 20 \log \frac{p\pi}{4} + 3N - 10 \log \sigma_{unit}^2 \quad (12-81)$$

Thereby we also find the *SFDR* for the full-scale signal to be given by the minimum value of (12-81) ($p \geq 3$), i.e.,

$$SFDR \approx 20 \log \frac{3\pi}{4} + 3N - 10 \log \sigma_{unit}^2 \quad (12-82)$$

In Fig. 12-11 we show the average simulated and calculated *SFDR* for a 10-, 12-, and

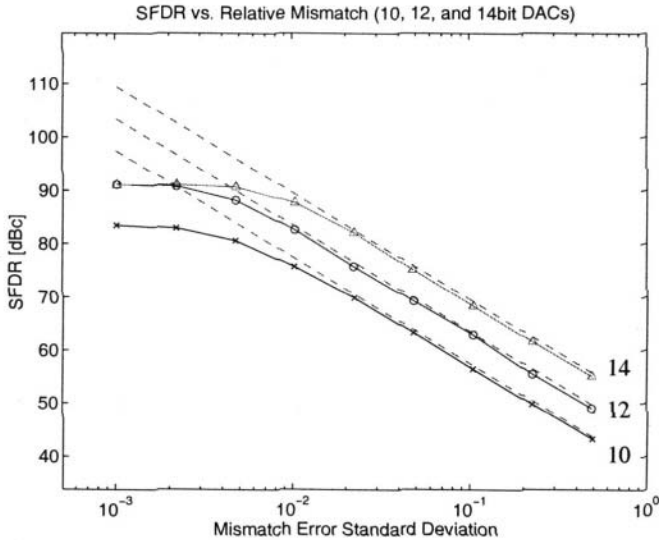


Figure 12-11 Calculated and simulated *SFDR* vs. mismatch for a 10-, 12-, and 14-bit DAC.

14-bit DAC. As we see there is a small error between the simulated and calculated values. This depends on the error in (12-80) where only the MSB has been considered. Since we are considering uncorrelated errors, the error power from all bits should be

added to the specific tone. This implies a slightly lower *SFDR*. The curves are saturated at low mismatch, since in the simulations the harmonics are hidden in the noise floor.

In Fig. 12-12 we show the typical output spectrum of a 14-bit DAC when applying

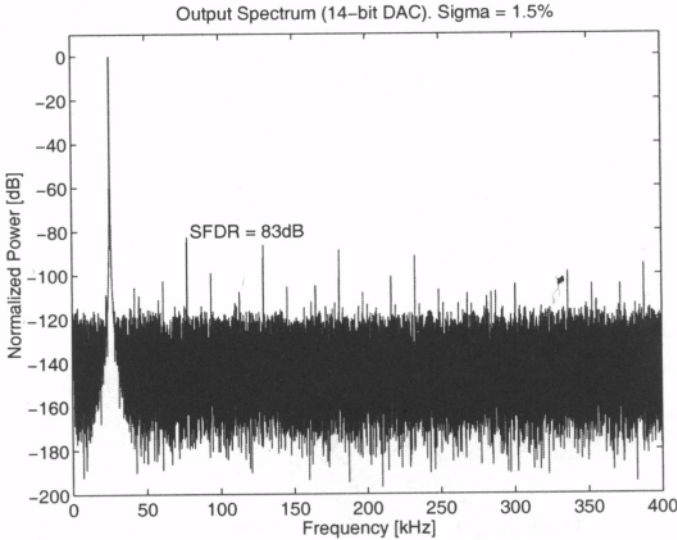


Figure 12-12 Output spectrum for a 14-bit DAC with mismatch size approximately 1.5%

matching errors with standard deviation $\sigma_{unit} \approx 1.5\%$. The *SFDR* is approximately 83 dBc for a full-scale sinusoid input.

12.4.3 Impact of Correlated and Graded Matching Errors

The matching errors are not independent Gaussian distributed variables. In reality, the matching errors in two adjacent current sources may be correlated [5,8]. We may also have a linear graded distribution of the matching errors over the array of unit current sources [5, 8, 14].

Previously, we found that if all the matching errors for the unit sources were uncorrelated, we would for the i -th bit have a relative matching error with mean and standard deviation as in (12-64), hence

$$\mu_i = 0 \text{ and } \sigma_i = \frac{\sqrt{I_{unit}^2 \cdot 2^i \cdot \sigma_{unit}^2}}{I_{unit} \cdot 2^i} = \frac{\sigma_{unit}}{\sqrt{2^i}} \quad (12-83)$$

Let the mismatch error for the m -th unit source associated with the i -th LSB be denoted $\delta_i^{(m)}$. If the errors are correlated with each other, the expectation value of the output current for bit i is still $2^i \cdot I_{unit}$. The expectation value of the squared output error current becomes

$$E_{\delta}\{\Delta I_i^2\} = E_{\delta}\left\{\left[I_{unit} \cdot \sum_{m=1}^{2^i} \delta_i^{(m)}\right]^2\right\} = I_{unit}^2 \cdot \sum_{n=1}^{2^i} \sum_{m=1}^{2^i} E_{\delta}\{\delta_i^{(m)} \cdot \delta_i^{(n)}\} \quad (12-84)$$

For the i -th bit we denote the covariance or correlation between two unit current sources as

$$r_p = E_{\delta}\{\delta_i^{(m)} \cdot \delta_i^{(m+p)}\} = E_{\delta}\{\delta_i^{(m)} \cdot \delta_i^{(m-p)}\}, \text{ and } r_{-p} = r_p \quad (12-85)$$

Using (12-85) in (12-84) we get

$$\begin{aligned} E_{\delta}\{\Delta I_i^2\} &= I_{unit}^2 \cdot \sum_{n=1}^{2^i} \sum_{m=1}^{2^i} r_{m-n} \\ &= I_{unit}^2 \cdot 2^i \cdot \sigma_{unit}^2 + 2I_{unit}^2 \cdot \sum_{m=1}^{2^i-1} (2^i - m) \cdot r_m \\ &= I_{unit}^2 \cdot 2^i \cdot \sigma_{unit}^2 \cdot \left[1 + \sum_{p=1}^{2^i-1} \frac{p}{2^i} \cdot \frac{2r_p}{\sigma_{unit}^2}\right] \end{aligned} \quad (12-86)$$

since $r_0 = \sigma_{unit}^2$. The squared value of the nominal current through the i -th binary weighted current sources is $I_i^2 = I_{unit}^2 \cdot 2^{2i}$ and we have that

$$I_i^2 + E_{\delta}\{\Delta I_i^2\} = I_{unit}^2 \cdot 2^{2i} \cdot \left[1 + \frac{\sigma_{unit}^2}{2^i} \cdot \left(1 + \sum_{p=1}^{2^i-1} \frac{p}{2^i} \cdot \frac{2r_p}{\sigma_{unit}^2}\right)\right] \quad (12-87)$$

From (12-87) we identify the standard deviation for relative matching error in the i th bit

$$\sigma_i^2 = \frac{\sigma_{unit}^2}{2^i} \cdot \left(1 + \frac{2}{2^i \cdot \sigma_{unit}^2} \cdot \sum_{p=1}^{2^i-1} p \cdot r_p\right) \quad (12-88)$$

If the correlation factors are all zero, we have the same result as in (12-64). If the correlation factors, r_m , are positive, the standard deviation σ_i is larger than the one used in the previous model. However, we need to know the correlation to be able to find an approximation for the result in (12-88). Naturally, there is a correlation of the matching errors between different bits as well and the correlation is strongly dependent on the layout style. More work on modeling and characterization of the influence of linearly graded matching errors on DAC performance is found in [5, 14, 16].

12.5 INFLUENCE OF CIRCUIT NOISE

A fundamental limit on resolution and performance is the noise found in the output signal [17]. The resolution of the converter is guaranteed as long as the circuit noise

is much less than the average quantization noise. For an ideal N -bit uniform DAC, the SNR (determined by the quantization noise) over the Nyquist band is

$$SNR \approx 6.02 \cdot N + 1.76 \text{ dB} \quad (12-89)$$

We model the noise similar to the case of a matching error as in Fig. 12-9, hence as an error current source in parallel with the unit current source. In this case, opposite to the matching error, the noise error is varying as function of time.

With the unit current source m an AWGN noise current, $i_{unit}^{(m)}$, is associated. Within a certain bandwidth, BW , its mean value and variance are given by

$$E\{i_{unit}^{(m)}\} = 0 \text{ and } E\{[i_{unit}^{(m)}]^2\} = \overline{i_{unit}^2} = \sigma_n^2 \quad (12-90)$$

The expectation of the noise power in the i -th LSB is $2^i \cdot \sigma_n^2$ if the noise in the individual noise sources is uncorrelated. The expectation of the normalized total output noise power, P_n , is given by the sum of all noise current sources

$$P_n = E\{(\sum i_{unit}^{(m)})^2\} = \sigma_n^2 \cdot \bar{k} \quad (12-91)$$

where $K_{dc} = \bar{k}$ is the mean value of the digital input, hence the number of sources that are in average connected to the output. The normalized signal power for a sinusoid is given by (12-20) and is

$$P_s \approx I_{unit}^2 \cdot \frac{K_{ac}^2}{2} \quad (12-92)$$

The SNR is found by comparing (12-91) and (12-92)

$$SNR = \frac{P_s}{P_n} = \frac{1}{2} \cdot \frac{I_{unit}^2 \cdot K_{ac}^2}{\sigma_n^2 \cdot \bar{k}} \quad (12-93)$$

Assuming that the input signal is full-scale sinusoid and that the mean value is $K_{dc} = \bar{k} = 2^{N-1}$, (12-91) turns to approximately

$$P_n = 2^{N-1} \cdot \sigma_n^2 \quad (12-94)$$

and with (12-77) we have that (12-93) becomes

$$SNR \approx \frac{1}{2} \cdot \frac{(2^{N-1})^2}{2^{N-1}} \cdot \frac{I_{unit}^2}{\sigma_n^2} = 2^{N-2} \cdot \frac{I_{unit}^2}{\sigma_n^2} \quad (12-95)$$

The result in (12-95) holds if the circuit noise is dominating. We can also add the quantization noise to the formula and we have

$$SNR \approx \frac{P_s}{P_q + P_n} = \frac{I_{unit}^2 \cdot K_{ac}^2 / 2}{\frac{I_{unit}^2}{12} + 2^{N-1} \cdot \sigma_n^2} = \frac{K_{ac}^2}{\frac{1}{6} + 2^N \cdot (\sigma_n^2 / I_{unit}^2)} \quad (12-96)$$

where we have assumed that the noise bandwidth is equal to the Nyquist range.

For a full-scale sinusoid and dominating circuit noise, the SNR in dB is

$$SNR \approx 3(N - 2) + 20\log I_{unit} - 10\log \sigma_n^2 \text{ dB} \quad (12-97)$$

If the number of bits is increased by one the signal power is increased by a factor of 6 dB, if σ_n is fixed, but at the same time, the noise is increased by a factor 3 dB. This is the essence of (12-97).

For wideband CMOS DACs it is mostly the thermal noise from the MOS transistor that dominates [17]. First we will show an example on the typical noise for a CMOS transistor and current source. The corresponding transistor thermal noise power spectral density (PSD) is

$$S_i(f) \approx \frac{8}{3}kT \cdot g_m \quad (12-98)$$

where k is the Boltzmann constant, T is the absolute temperature, and g_m is the small-signal transconductance of the current source transistor. Since the load resistance most likely is much smaller than the output impedance of the current source, (12-98) holds for both single transistor and cascoded current sources. The transconductance, g_m , is approximately [17]

$$g_m \approx \sqrt{\mu_0 C_{ox} \cdot S \cdot I_{unit}} \quad (12-99)$$

where μ_0 is the chagemobility, C_{ox} is the gate capacitance per area, and S is the transistor size aspect ratio. Using the noise and transconductance definitions in (12-98) and (12-99) and assuming that the thermal noise in the transistors dominates, the noise spectral density in the unit current source is

$$S_i(f) \approx \frac{8}{3}kT \cdot \sqrt{\mu_0 C_{ox} \cdot S \cdot I_{unit}} \quad (12-100)$$

If we assume a certain noise bandwidth of BW , the normalized total output noise power, i_{unit}^2 , from the unit current source is

$$\overline{i_{unit}^2} = S_i(f) \cdot BW = \frac{8}{3}kT \cdot BW \cdot \sqrt{\mu_0 C_{ox} \cdot S \cdot I_{unit}} \quad (12-101)$$

Substituting the values from (12-101) into (12-97) gives the approximate expression

$$SNR \approx 15\log I_{unit} + 3N - 5\log S - 10\log BW - 10\log\left(\frac{8}{3}kT \cdot \sqrt{\mu_0 C_{ox}}\right) \quad (12-102)$$

In Fig. 12-13 the simulated and calculated (from (12-97)) SNR for a 14-bit DAC vs. the current through the unit current source, I_{unit} , is shown. Data from a standard 0.6 μm CMOS process was used. It can be seen from Fig. 12-13 that for high unit currents, the thermal noise is lower than the quantization noise. If the quantization noise is added to the calculated value, as in (12-96), the curves will fit. At lower currents the

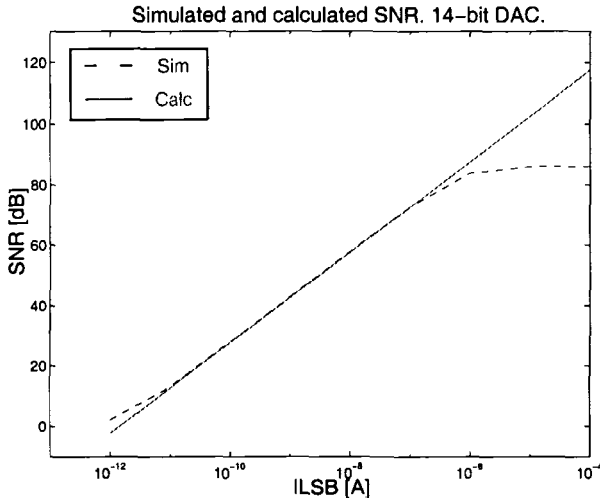


Figure 12-13 Simulated and calculated SNR vs. the I_{unit} current for a 14-bit DAC. Only thermal noise is simulated.

simulated curve saturates since the signal, due to the low SNR, cannot be separated from noise.

12.6 INFLUENCE OF DYNAMIC BEHAVIOR

In the previous sections, we have discussed how the output current is dependent on errors in static values. This implies that the derived formulas apply for low frequencies. At lower oversampling ratios – closer to the Nyquist range – the dynamic properties as for example glitching, and capacitive parts of the current sources, will influence performance. For high-speed DAC the performance is not only determined by the final values of each sample, but also determined by how each sample settles.

We will briefly discuss and model the influence of settling errors and the influence of bit skew and glitches.

12.6.1 Settling Error

Previously, we considered only the output resistance of the current source. In general, the output *impedance* of the current-steering DAC varies with the signal. Consider the modified model of the i -th LSB current source in Fig. 12-14. Parasitic capacitances have been added at the switch, to the output node of the current source, and at the load. The capacitive elements added to the current sources is further discussed in Chapter 13.

Analysing the dynamic behavior of the DAC, we can no longer only consider the settled values at the output. In this analysis we will consider the *linear* settling error. *Non-linear* settling errors, due to dominant signal-dependent capacitances, will introduce

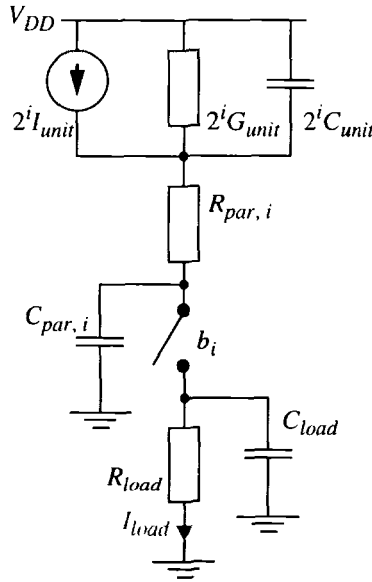


Figure 12-14 Current source in the i th LSB position with output conductance and capacitance as well as parasitics and load components.

distortion. This is not the case for linear settling errors, but this analysis gives a guidance for studies on how parasitic capacitances may influence performance.

We use a first order approximation, i.e., one dominant pole at the output. Let the start and end value errors be given by settling errors only. We can for example let the actual (or true) output current at the time instant $T_n = nT$ be denoted

$$\tilde{I}_{load}(T_n) \quad (12-103)$$

and the wanted output current through the load resistance at the same time instant is given by $I_{load}(T_n) = I_{unit} \cdot k(T_n)$. The absolute settling error found at the switching instant is given by

$$\varepsilon(T_n) = I_{load}(T_n) - \tilde{I}_{load}(T_n) \quad (12-104)$$

Let the DAC be approximated by a single-pole system with the time constant given by $\tau = RC$. At the transition from T_n to the next sampling instant, $T_n + T = T_{n+1}$, the output signal will settle towards a new end value. We have that

$$\tilde{I}_{load}(t) = \tilde{I}_{load}(T_n) + [I_{load}(T_{n+1}) - \tilde{I}_{load}(T_n)] \cdot \left(1 - e^{-\frac{t-T_n}{\tau}}\right) \quad (12-105)$$

for $T_n \leq t \leq T_{n+1}$. From the discussion above, we have that the output current at the sampling instant T_{n+1}

$$\begin{aligned}\tilde{I}_{load}(T_{n+1}) &= \tilde{I}_{load}(T_n) + [I_{load}(T_{n+1}) - \tilde{I}_{load}(T_n)] \cdot (1 - e^{-T/\tau}) \\ &= \tilde{I}_{load}(T_n) \cdot e^{-T/\tau} + I_{load}(T_{n+1}) \cdot (1 - e^{-T/\tau})\end{aligned}\quad (12-106)$$

Further using (12-106) in (12-104) we have the settling error given by

$$\begin{aligned}\epsilon(T_{n+1}) &= I_{load}(T_{n+1}) - [\tilde{I}_{load}(T_n) \cdot e^{-T/\tau} + I_{load}(T_{n+1}) \cdot (1 - e^{-T/\tau})] \\ &= -[\tilde{I}_{load}(T_n) \cdot e^{-T/\tau} + I_{load}(T_{n+1}) \cdot (-e^{-T/\tau})] \\ &= \epsilon(T_n) \cdot e^{-T/\tau} + [I_{load}(T_{n+1}) - I_{load}(T_n)] \cdot e^{-T/\tau}\end{aligned}\quad (12-107)$$

We can use the z-transform on the expression above, and we have that

$$\epsilon(z) = z^{-1} \cdot \epsilon(z) \cdot e^{-T/\tau} + [1 - z^{-1}] \cdot I_{load}(z) \cdot e^{-T/\tau}\quad (12-108)$$

This gives

$$\epsilon(z) = \frac{1 - z^{-1}}{1 - z^{-1} \cdot e^{-T/\tau}} \cdot e^{-T/\tau} \cdot I_{load}(z)\quad (12-109)$$

If we use the transfer function, $H(z)$, describing the relation between the settling error and the load current, we have

$$\epsilon(z) = H(z) \cdot I_{load}(z)\quad (12-110)$$

where

$$H(z) = \frac{1 - z^{-1}}{1 - z^{-1} \cdot e^{-T/\tau}} \cdot e^{-T/\tau}\quad (12-111)$$

If a sinusoid at the normalized frequency $\omega_0 T$ and with an amplitude $K_{ac} \cdot I_{unit}$ is fed into the DAC, the settling error signal will also be given by a sinusoid with the same frequency as

$$\epsilon(nT) = |H(e^{j\omega_0 T})| \cdot I_{unit} \cdot K_{ac} \cdot \sin(\omega_0 T \cdot n + \arg H(e^{j\omega_0 T}))\quad (12-112)$$

where n is the sequence index and $z = e^{j\omega T}$ on the unity circle. The requirement is that the maximum settling error should be less than half an LSB, hence

$$\max |\epsilon(nT)| = |H(e^{j\omega_0 T})| \cdot I_{unit} \cdot K_{ac} < \frac{1}{2} I_{unit}\quad (12-113)$$

This equation can be rewritten as

$$[|H(e^{j\omega_0 T})| \cdot K_{ac}]^2 < \frac{1}{4}\quad (12-114)$$

giving

$$|H(e^{j\omega_0 T})|^2 < \frac{1}{4 \cdot K_{ac}^2} \quad (12-115)$$

The largest settling error is found when a full-scale sinusoid is applied at the input, hence in the worst case we have

$$|H(e^{j\omega_0 T})|^2 < \frac{4}{4 \cdot (2^N - 1)^2} \approx 2^{-2N} \quad (12-116)$$

By using $z = e^{j\omega T}$ on the unit circle in (12-111) we have

$$\begin{aligned} |H(e^{j\omega T})|^2 &= \left| \frac{e^{j\omega T} - 1}{e^{j\omega T} - e^{-T/\tau}} \right|^2 \cdot e^{-\frac{2T}{\tau}} = \left| \frac{e^{\frac{j\omega T}{2}} - e^{-\frac{j\omega T}{2}}}{e^{\frac{j\omega T}{2} + \frac{T}{2\tau}} - e^{-\frac{j\omega T}{2} - \frac{T}{2\tau}}} \right|^2 \cdot e^{-T/\tau} \\ &= \frac{4 \cdot \sin^2 \frac{\omega T}{2} \cdot e^{-T/\tau}}{4 \cdot \left| \sinh \left(\frac{j\omega T}{2} + \frac{T}{2\tau} \right) \right|^2} = \frac{\sin^2 \frac{\omega T}{2} \cdot e^{-T/\tau}}{\sinh^2 \frac{T}{2\tau} \cos^2 \frac{\omega T}{2} + \cosh^2 \frac{T}{2\tau} \sin^2 \frac{\omega T}{2}} \end{aligned} \quad (12-117)$$

In the equation we can identify two important parameters, the T/τ and the $\omega_0 T$ factors. The first factor expresses the time constant related to the update time and the latter factor expresses how the signal frequency is related to the update time. Setting (12-117) in (12-116) and reordering, we have that

$$\frac{1}{|H(e^{j\omega_0 T})|^2} = \cosh^2 \frac{T}{2\tau} \cdot \left[1 + \frac{\tanh^2 \frac{T}{2\tau}}{\tan^2 \frac{\omega_0 T}{2}} \right] \cdot e^{T/\tau} > 2^{2N} \quad (12-118)$$

For a fixed value of T/τ the maximum value of the left-hand expression in the equation is found when $\omega_0 T = \pi$, hence the signal is placed at the Nyquist frequency. Therefore, we can instead investigate the worst-case as

$$\cosh^2 \frac{T}{2\tau} \cdot e^{T/\tau} = \left[\frac{e^{T/2\tau} + e^{-T/2\tau}}{2} \right]^2 \cdot e^{T/\tau} \approx \frac{e^{2T/\tau}}{4} > 2^{2N} \quad (12-119)$$

where we assume (or require) that $T/2\tau > 1$. From (12-119) we get

$$\tau < \frac{T}{(N+1) \ln 2} \approx \frac{T}{0.69 \cdot (N+1)} \quad (12-120)$$

Note that we in the frequency domain also have the sinc weighting from sample&holds at the DAC output. This will add a factor $\pi/2$ to the left-hand expression in (12-119), however, if N is large, this will not strongly influence the final result in (12-120).

Some other complexities add to these result. First, if we consider the current source in Fig. 12-14 we find that the time constant of the DAC is likely to be signal dependent,

hence $\tau = \tau(k)$, which will introduce distortion. The RC time constant of the current source itself (withouth switches and interconnection) is constant, since when connecting a number of unit current sources in parallel, the output resistance decrease with the number of sources, but the capacitance increases with the number of sources. We have that

$$\tau_{unit} = \frac{C_{unit}}{G_{unit}} = \frac{2^i \cdot C_{unit}}{2^i \cdot G_{unit}} = \tau_i \quad (12-121)$$

where τ_i is the time constant for the i -th LSB current source and τ_{unit} is the time constant associated with each unit current source.

12.6.2 Bit Skew and Glitches

As indicated in the previous section the settling time is crucial and a finite settling time introduces errors to the output signal. From (12-120) we have a limit on the possible settling time to achieve a full resolution bandwidth up to the Nyquist frequency. To meet this requirement we must also guarantee that the switching instant T_n for all bits is accurate. Errors will add voltage/current spikes to the output signal (i.e., glitches as discussed in Chapter 1) and the settling behavior will be affected in a non-linear way. For instance at the major code transition

$$011\dots 11 \rightarrow 100\dots 00$$

we may have the following transition code for a short period of time

$$111\dots 11$$

This generates a glitch of amplitude $2^{N-1} \cdot I_{unit}$. Naturally, in a good design the glitch energy – or glitch impulse – is kept as low as possible [7], by using segmentation and proper switching schemes [18]. The induction of a high current or charge also affects the settling since the start value at T_n (12-103) changes dramatically.

To model and investigate the behavior of glitches, we introduce a switching time uncertainty for the i -th LSB

$$\tau_i(k(T_n)) \quad (12-122)$$

where $T_n = nT$ and T is the sampling period. Notice that this error is dependent on both the bit position i and the input code, k . In Fig. 12-15 we show how this skew can be characterized for the i -th LSB. The ideal (dashed) voltage pulse applied to the switch is compared with a linearized actual pulse (solid) also indicated with an offset. The switching activity for the i -th LSB is denoted

$$\Delta b_i(T_n) = b_i(T_n) - b_i(T_{n-1}) \quad (12-123)$$

which expresses the number of switching bits at the time instant T_n . If the i -th LSB is switching, we will have the change in amplitude given by

$$2^i \cdot \Delta b_i(T_n) \quad (12-124)$$

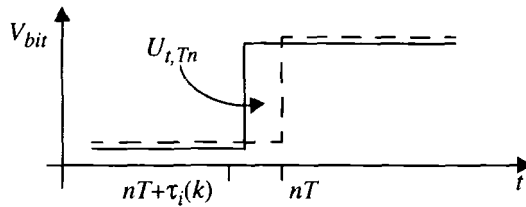


Figure 12-15 Modeling of the timing uncertainty. The ideal switching signal (---) is compared with the actual signal (—).

and for all bits, we have

$$\sum_{i=0}^{N-1} 2^i \cdot \Delta b_i(T_n) = k(T_n) - k(T_{n-1}) = \Delta k(T_n) \quad (12-125)$$

As illustrated in Fig. 12-15 we let the glitch impulse be expressed by a squared-wave pulse with the duration $\tau_i(k)$. Dependent on the sign of τ_i , the pulse is given by

$$U_{\tau_i, T_n}(t) = \begin{cases} 1 & T_n \leq t \leq T_n + \tau_i(k(T_n)) & \tau_i > 0 \\ 1 & T_n + \tau_i(k(T_n)) \leq t \leq T_n & \tau_i < 0 \\ 0 & \text{for } t \text{ outside these intervals} \end{cases} \quad (12-126)$$

An expression for the total glitch amplitude, $G(t)$, at the output is given by

$$G(t) = I_{unit} \cdot \sum_{n \geq 0} \sum_{i=0}^{N-1} U_{\tau_i, T_n}(t) \cdot 2^i \cdot \Delta b_i(T_n) \quad (12-127)$$

This is a pulse-train, where the widths and amplitudes at the time instants nT depend on the unit current I_{unit} , the switching activity, and the time uncertainty τ . An approximation of (12-127) would be to assume that the pulses U_{τ_i, T_n} are independent on the bit position. Hence, we let the glitch at time instant nT be given by the amplitude $\Delta k(T_n) \cdot I_{unit}$ and an average width $\tau(T_n)$ as

$$G(t) = I_{unit} \cdot \sum_{n \geq 0} U_{\tau, T_n}(t) \cdot \Delta k(T_n) \quad (12-128)$$

which is an rough approximation. The timing errors may also be modeled as damped sine waves instead of pulse shaped errors [6]. The timing errors are typically dependent on the voltages across the switches and mismatch of the switch sizes.

12.7 SUMMARY

We have given an overview of different models for current-steering D/A converters.

The modeling should be used as a guidance in determining circuit structure, component sizes, etc., in current steering DACs. From the models, we can also calculate the effect of matching errors.

Using the results in this chapter, we are able to simulate and predict the performance of the DAC on a high abstraction level, with for example Matlab, which saves time, over circuit-level simulations. Mostly, static requirements have been addressed but in the same results can be modified to cover AC as well.

Extending the modeling, we can also use measured result from a DAC, and extract information on the process and the parasitic resistance and capacitance in the circuit.

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13 IMPLEMENTATION OF CMOS CURRENT-STEERING D/A CONVERTERS

13.1 INTRODUCTION

One of the most suitable candidates for high-speed and high-resolution is the current-steering DAC and in this chapter an overview of different common current-steering DAC topologies is given in Sec. 13.2. We highlight the advantages and disadvantages with the different structures. For some of the topologies there are inherited good properties for high performance.

In the previous chapter we found that the DAC performance is strongly dependent on for example the finite output resistance of the current sources, R_{out} . The output resistance and circuit noise are dependent on the DC output current of the current source, I_{out} , and the DC current is dependent on the transistor size, etc. Matching is practically determined by the transistor size, aspect ratio W/L or the gate area WL . When increasing the output current with the ambition to improve the SNR, the output resistance decreases and thereby the non-linearity increase, etc. Therefore, we have a delicate relation between the three major error sources; noise, linearity and matching. The noise degrades the SNR and the non-linearity and mismatch degrade the SFDR. Usually trade-offs have to be done in order to meet the frequency-domain specifications.

Further the influence of clock feedthrough (CFT) in switches as well as their switch-on resistance are crucial for the dynamic performance. Also the influence of the interconnecting wires must not be neglected, etc. In Sec. 13.3 we discuss practical design issues for determining proper unit current sources, current switches, digital circuits, etc.

To further illustrate the design methodology, we present the design of a 1.5-V to 5-V, 10-bit to 14-bit wideband CMOS DAC chipset for xDSL and wideband radio applications in Sec. 13.4. We show design trade-offs and examples on how to implement the different circuit building blocks.

Measured and simulated results from the DAC chipset are presented and discussed in Sec. 13.5. We also compare measurement results from two similar DACs of the chipset. They have small variations in the design and layout of the current sources and interconnection wires. With a small variation in layout, an SFDR improvement of over 12 dB is achieved.

13.2 CURRENT-STEERING DAC TOPOLOGIES

The structure and operation of the binary-weighted current-steering DAC was outlined in Chapter 4 and Chapter 12 and it is recaptured in Fig. 13-1. There are N binary

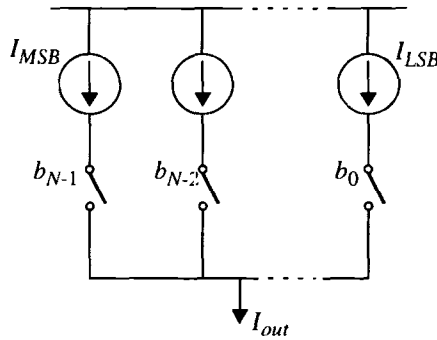


Figure 13-1 An N -bit current-steering DAC.

weighted current sources and N switches. The switches are dependent on the input code, $X_d = \{b_i\}_{i=0}^{N-1}$, and they determine which current source that should be directed to the output. The total current is terminated by an off-chip resistance or I/V converted (and filtered) using an output buffer.

As was derived in the previous chapter, the matching between current sources is crucial to the performance. Instead of using a current source with the nominal value I_{src} one should use M unit current sources in parallel, i.e., $I_{src} = M \cdot I_{unit}$. This improves the relative matching of the current sources by a factor of $\sqrt{M}$ if the matching errors are uncorrelated [1].

One should be aware of the influence of stochastic as well as linear graded matching errors (also referred to as short-distance and long-distance variations). Linear errors can often be approximately modeled by a plane [2]. Hence the size of the error is given by an expression as

$$\epsilon(x, y) = g_x \cdot x + g_y \cdot y \quad (13-1)$$

where x and y are the coordinates on the chip surface. g_x and g_y are the gradients in the respective direction and typically, they are in the same order of magnitude.

By choosing a special layout technique for the current sources, e.g., interdigitized or common-centroid, the converter can be made more or less sensitive to linear graded errors [2-6]. The stochastic error can be minimized by considering the results from Pelgrom [1] and the choice of the transistor sizes in the current sources is important.

To achieve high performance, we have to decide how to design and lay out the unit current sources and below we outline some different popular current-steering DAC structures.

13.2.1 Array Structure

The first, and perhaps simplest, approach is to follow Fig. 13-1 and use a “flat” array structure. The MSB current source is formed by placing 2^{N-1} parallel unit current sources in a row. The next MSB is formed in a similar way by placing 2^{N-2} sources in the next row, etc. Each row is interconnected and fed to a current switch as is illustrated in Fig. 13-2 for a single-ended structure.

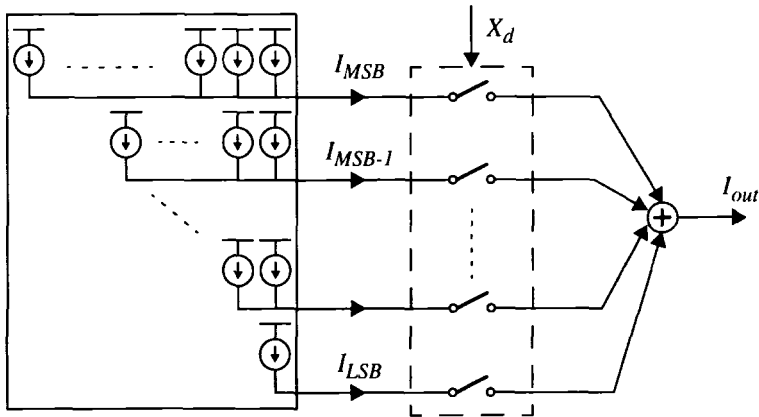


Figure 13-2 “Flat” array based layout of unit current sources.

It is obvious that the array becomes very wide for a large number of bits. The natural and obvious way to circumvent this is for the MSBs to use more than one row of current source and to use the same row for several LSBs. This also makes the converter less sensitive to linear graded mismatch errors [3].

The size of the current switch should be determined by the current flowing through it. For a large current we need a low switch-on resistance and vice versa. To make the array become more narrow, we can split the MSBs rows into several rows with less current in each row, and for each row we use one switch. Flat array structures are not suitable for resolutions over 6-8 bits, due to the large glitching and sensitivity to current source matching errors.

13.2.2 Segmented Structures

To reduce the glitching it is preferred to use a thermometer code representation. However, it is very difficult to use a full thermometer code for a high-resolution converter, since the number of switches and the size of the interconnection wires, etc., will be very large for resolutions above 7-8 bits.

The better choice is a DAC structure where the M most significant bits are thermometer coded and the $N - M$ least significant bits are binary weighted. This is referred to as a segmented structure and it is illustrated in Fig. 13-3. With the thermometer code we can use a number of equally large current sources, and hence the layout can be

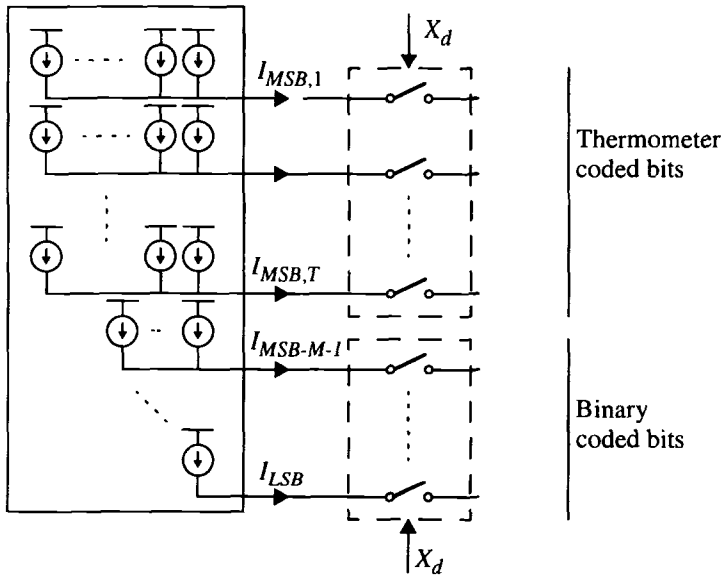


Figure 13-3 Segmented current source array. The M MSBs are thermometer coded which gives $T = 2^M - 1$.

more regular. The current switches are equally large for the thermometer coded bits and the current that flows through one switch will be smaller than for the unsegmented MSB.

The segmentation guarantees monotonicity for the MSBs and improves DNL since for an increase of the input, additional current sources are connected to the output. Matching of the thermometer coded current sources can further be improved by using dynamic randomization, averaging, or calibration techniques, as discussed in Chapter 4.

An extension to the segmented structure is to use multi-segmentation. For example, the M MSBs are thermometer coded in one cluster, the K LSBs are kept binary coded, and the $N - M - K$ intermediate bits are also thermometer coded in one separate cluster.

The key design issue in a segmented converter is to determine how many of the bits that should be thermometer coded. Roughly, the glitch energy is dependent on the number of bits that are switching in the input signal. By using a cost function, e.g., the power of the glitches or the weighted number of switching bits, we can examine how the power of the glitches vary with the number of thermometer coded bits M . Simulation results when applying a number of different signal types to a 14-bit DAC are shown in Fig. 13-4. It is found that about 4 to 6 thermometer coded bits give a large improvement in glitching performance. For more thermometer coded bits, the improvement is not that high, and more digital circuitry is also introduced, hence higher complexity, power consumption from digital circuits, and switching noise. It

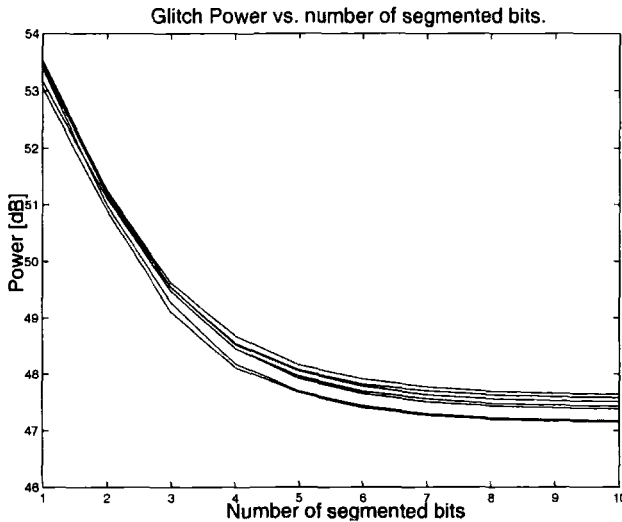


Figure 13-4 Power of glitches as a function of the number of segmented bits in a 14-bit DAC.

should be noted that in this simulation, the influence of the complexity of designing current sources with a high output resistance and switches with a low enough resistance have not been added. In the literature we find several 14-bit high-performance DACs with 6 or 7 thermometer coded bits or multi-segmented structures with similar resolutions [5, 6, 7].

13.2.3 Current Cell Matrix Structures

Another popular structure is the current cell matrix structure [8, 9, 10]. In this structure, as exemplified in Fig. 13-5, the unit current sources (or sources of a higher significance if segmentation is used) are selected from a matrix using row and column decoding logic. The select signals determine if a specific current source should be turned on or off. In for example [8] each current cell (Fig. 13-5) uses three control signals, one column and two row signals.

To reduce the effect of linear graded matching errors, such as oxide thickness variations and voltage drops over supply wires [4, 11] the cells can be laid out in such way that these errors are averaged and minimized. With modifications of the row and column decoders, also dynamic randomization techniques can be used on the entire matrix to reduce distortion (see Sec. 4.10.1).

There are some drawbacks with the current cell matrix structure. Within the unit current cell a certain amount of digital logic is needed. This requires digital supply distribution within the cell. At the same time we need the three digital control signals, analog supply, analog bias voltages, and differential output currents, i.e., two more wires. This is a total of 9-10 wires depending on the choice of current source. To decrease the amount of noise, we want to shield the analog wires from the digital as well as possible.

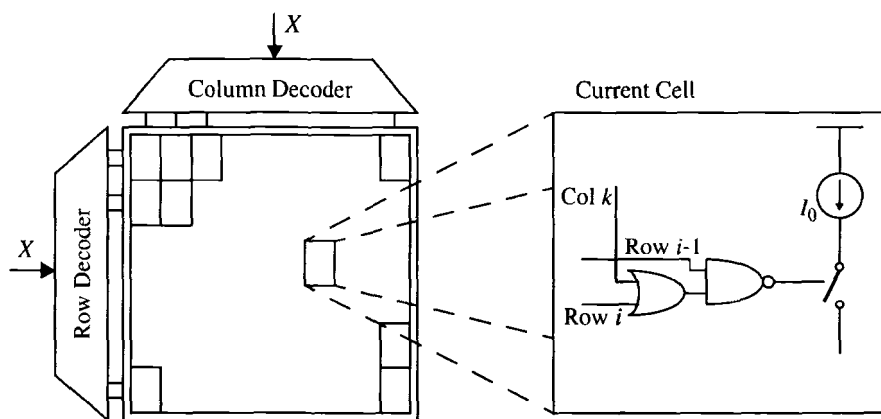


Figure 13-5 Unit current source matrix with decoding circuits.

For DAC structures with resolutions higher than 6-8 bits this technique is somewhat limited due to chip area and complexity. Hybrid DACs use this matrix approach for the MSBs and a binary weighted or thermometer coded structure for the LSBs [8, 9, 11].

13.3 PRACTICAL DESIGN CONSIDERATIONS

We present some practical design considerations concerning the design and layout of circuit elements, such as current source, switch, and digital circuits, in the current-steering DAC.

13.3.1 Implementation of Current Sources

Naturally, the unit current source to be used in the converter can be constructed in several different ways. In Fig. 13-6 we show three versions where a single PMOS transistor is used (a) together with one (b) or two cascode (c) PMOS transistors. It is the source-gate voltage applied on transistor M1 that practically sets the current through the current source. Since the ideal current source should have an infinite output impedance, the cascode transistors are used to increase the output impedance. The bulk connections of all transistors are connected to the positive supply. This decreases the gain of the cascodes, but the capacitive load at the internal nodes of the current source is reduced.

The output current, which typically is set by the DAC specification, is approximately given by the drain current of transistor M1 in the saturation region [12]

$$I_D \approx \frac{\beta}{2} \cdot (V_{SG} - V_T)^2 \cdot (1 + \lambda \cdot V_{SD}) \quad (13-2)$$

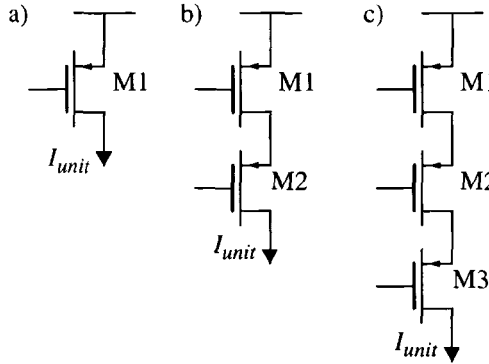


Figure 13-6 Schematic view of PMOS current sources using a) single transistor and b) single cascode and c) double cascode.

where β is the transconductance parameter, $V_{SG} = V_{DD} - V_{bias}$ is the source-gate voltage, V_T is the threshold voltage, λ is the channel length modulation factor, and V_{SD} is the source-drain voltage.

The output resistance of the current sources shown in Fig. 13-6 are given by R_a , R_b , and R_c , respectively, as

$$R_a \approx \frac{1}{\lambda_1 \cdot I_D}, R_b \approx R_a \cdot \frac{(1 + \eta_2) \sqrt{2\beta_2}}{\lambda_2 \cdot \sqrt{I_D}}, \text{ and } R_c \approx R_b \cdot \frac{(1 + \eta_3) \sqrt{2\beta_3}}{\lambda_3 \cdot \sqrt{I_D}} \quad (13-3)$$

where I_D is the drain current (the output current of the source), λ is the channel length modulation, β_i is the transconductance parameter of transistor M_i , and η_i is a parameter given by the bulk-source transconductance of the transistors. The transconductance parameter is also given by

$$\beta = \mu_0 C_{ox} \cdot \frac{W}{L} \quad (13-4)$$

where μ_0 is the charge mobility, C_{ox} is the gate capacitance per unit area, and W/L is the transistor aspect size ratio. The channel length modulation for transistor is approximately inversely proportional to the transistor channel length, hence $\lambda \sim 1/L$. This implies that, if the current is fixed by the specification, the output resistance is proportional to $\sqrt{W_2 L_2}$ and $\sqrt{W_3 L_3}$ and proportional to the length L_1 of transistor M1. Hence, with larger transistors we increase the output impedance. However, the capacitance which must be kept low, is proportional to the gate areas and the perimeters of the transistors. To illustrate this, we show in Fig. 13-7, the simulated output impedance of the different current source structures from Fig. 13-6. The supply voltage is 5 V and the DC output voltage is 0.5 V. Process parameters from a standard digital CMOS process have been used. The transistor sizes are S1 = 2u/8u, S2 = 2u/4u, S3 = 2u/1.2u. We find that the simulated output resistance (at DC) is approximately $R_a = 10 \text{ M}\Omega$, $R_b = 1 \text{ G}\Omega$, and $R_c = 50 \text{ G}\Omega$, for the respective source, with a current of 1.22 μA .

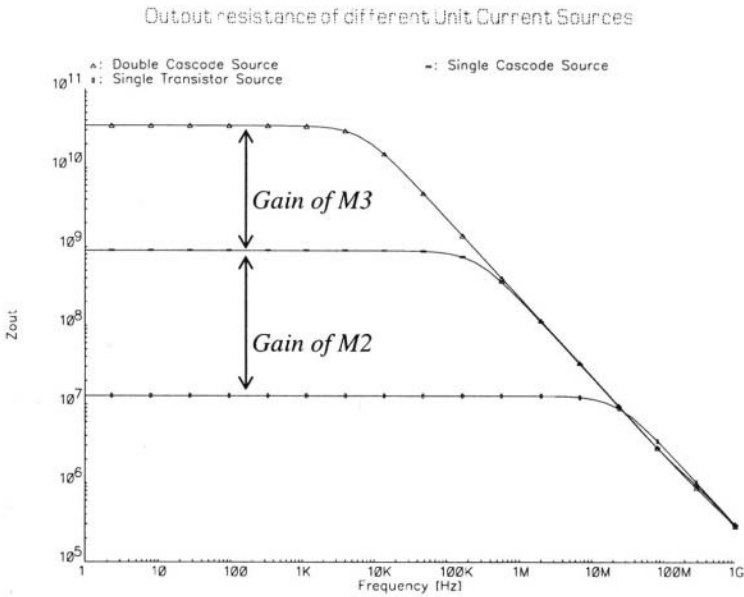


Figure 13-7 Simulated output impedance of the unit current source.

It can be seen from the figure above that the output resistance is increased with more cascodes, but the dominating pole is also lowered and at higher frequencies they basically have the same behavior. This will influence the DAC linearity at higher frequencies and is further addressed in Sec. 13.5.1. Using the results from the previous chapter, we have that for a specified SFDR the output resistance of the unit source is

$$R_{unit} \geq R_L \cdot 10^{\frac{6(N-2) + SFDR}{20}} \quad (13-5)$$

where R_L is the load resistance. With a $50\text{-}\Omega$ load, the bound on the output resistance of a unit current source for an 80-dBc SFDR is

$$R_{unit} \approx 2 \text{ G}\Omega \quad (13-6)$$

This is a lower bound, since the influence of parasitic resistance in switches and interconnection wires should be added as well. Dependent on the process parameters, we may find the proper current source structure by using the result from (13-3) and (13-6) as well as the knowledge about the output pole as illustrated in Fig. 13-7.

Note that the bandwidth of the signal is most likely set by the output load due to its dominant capacitive part. The distortion is set by the ratio between the current sources' output impedance and the load impedance.

There are several sources for transistor matching errors, i.e., size errors, threshold voltage variations, supply and bias voltages variations, oxide thickness variations, out-

put voltage variations, etc. Differentiating (13-2) with respect to four of its parameters, we have that

$$\frac{\Delta I_D}{I_D} = \frac{\Delta \beta}{\beta} + 2 \cdot \frac{\Delta V_{SG} - \Delta V_T}{V_{SG} - V_T} + \frac{\Delta \lambda \cdot V_{SD} + \lambda \cdot \Delta V_{SD}}{1 + \lambda V_{SD}} \quad (13-7)$$

where ΔP expresses the variation of the parameter P . Assuming that the deviations of the source-gate and source-drain voltages, as well as the channel length modulation, are very small, we identify the so called β - and V_T -matching errors from (13-7)

$$\frac{\Delta \beta}{\beta} \text{ and } \frac{2\Delta V_T}{V_{SG} - V_T}, \text{ respectively} \quad (13-8)$$

From studies in the literature [1, 13, 14], we know that the β - and V_T -matching can be characterized by their distribution. We have that

$$\sigma^2\left(\frac{\Delta I}{I}\right) \approx \sigma^2\left(\frac{\Delta \beta}{\beta}\right) + \frac{4 \cdot \sigma^2(\Delta V_T)}{(V_{SG} - V_T)^2} \quad (13-9)$$

where

$$\sigma^2\left(\frac{\Delta \beta}{\beta}\right) = \frac{A_\beta}{WL} + S_\beta^2 \cdot D_x \text{ and } \sigma^2(\Delta V_T) = \frac{A_{V_T}}{WL} + S_{V_T}^2 \cdot D_x \quad (13-10)$$

where A_β , A_{V_T} , S_β , and S_{V_T} are process dependent constants, WL is the gate area of the transistor, and D_x is the distance between the objects that should be matched. Hence, for good matching of two (or more) objects, we want them to be as close to each other as possible and they should be as large as possible.

From a geometrical point of view the terms in (13-10) are strongly dependent on the layout style of the transistors. For large transistors, special layout styles, e.g., common-centroid or interdigitized layout, are used to reduce the influence of these kinds of mismatch. In an array of current sources as illustrated in Fig. 13-2, the matching errors between two neighboring unit current sources may still be strongly correlated and therefore the binary weighted current sources should be constructed by using unit current sources from different positions in the array [15]. This will however increase the wire resistance and capacitance since the interconnection wires becomes longer and the routing becomes more complex.

The output resistance is dependent on the bias voltage controlling the gate voltage of the current source (13-3). These voltages may be generated using current mirrors and the reference current through the primary side of the mirror is determined by a termination resistance. But, we have to guarantee proper stable and accurate bias and supply voltages to all the unit current sources [4, 11, 16]. In Fig. 13-8 we have illustrated the effects of voltage loss over the supply distribution wire, which has been modeled as a series of resistances. This will give rise to a voltage drop along the wire, and we have $V_{DD} > V_1 > \dots > V_N$, which further influence the V_{SG} -matching of the transistors. At the nodes V_i the current sources are connected. If we assume that the supply wire

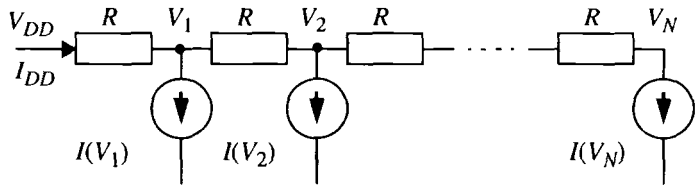


Figure 13-8 Model of the voltage supply wire connected to a number of DAC current sources.

between two adjacent current sources is equally long, the wire resistance, R , is also the same. The currents, $I(V_i)$, generated from a number of unit current sources, are determined by the square of the source-gate voltage according to (13-2), hence dependent on the squared voltages V_i . Therefore we will have currents that are deviating from their nominal values and we have to design the supply wire width and length to minimize this voltage drop.

13.3.2 Current Switches

The current switches influence the DAC performance. We have to guarantee that the switches are fast enough and that they do not introduce distortion.

Since we are switching a current source, we must ensure that the current switch at no time-instant turns the current source completely off. Otherwise, when it is turned off, the potential at the output of the current source will move towards the supply voltage. When the current source is turned on again, the potential difference between the current source output and the DAC output is large and a glitch occurs. To avoid this, we use differential switches, so that the current source always can deliver current. In Fig. 13-9a) we show a differential switch and in Fig. 13-9b) a possible MOS implementa-

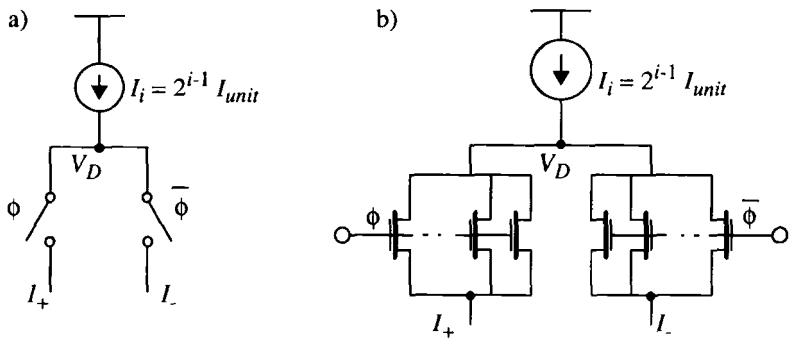


Figure 13-9 Differential current switch as a) circuit model and b) transistor implementation.

tion of the switch.

As outlined in the previous chapter, in a flat array structure (Sec. 13.2.1) the switch-on resistance is important. Especially, for larger currents through the switches, i.e., for

MSB current sources, the switch-on resistance must be low, since the output impedance of the current source is low. For an MOS implementation, this implies that the size aspect ratio of the transistors must be large. However, large switches also increase clock feedthrough (CFT) [17] due to the gate capacitances. A trade-off has to be done. The lower the switch-on resistance, the more transistors are connected in parallel, as illustrated in Fig. 13-9b).

In the linear region, the switch-on resistance of an MOS transistor is

$$R_{sw} \approx \frac{1}{\beta \cdot (V_{\phi} - V_T - V_D)} \quad (13-11)$$

where β is the transconductance parameter, V_ϕ is the gate or switch voltage, V_T is the threshold voltage, and V_D is the drain voltage. The switching voltages, $\phi, \bar{\phi}$, are usually given by the supply voltages, but for low supply voltages the switching voltage can be increased by using a charge-pump technique [18]. The charge-pump technique is also described in Sec. 8.4.

13.3.3 Digital Circuits

It is important to never turn the current source completely off and therefore the switching signals have to be properly matched to improve the glitch performance. A proper switching scheme for an NMOS implementation of the differential current switch is shown in Fig. 13-10(a) [8, 17]. The switching signals have to be slightly overlapping

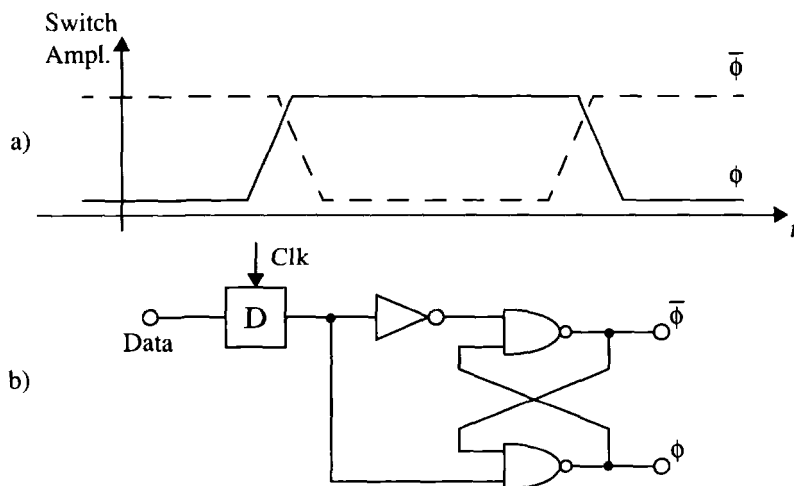


Figure 13-10 a) Wanted switch signals for the differential current switch and b) a possible circuit implementation for generation of switch signals.

and they can for example be generated with a latch as shown in Fig. 13-10b).

The input data, i.e., input bits, to the switch signal generator is aligned globally with

a latch to reduce the timing skew. Therefore, a good clock distribution is needed. In Fig. 13-11 we show two different clock distribution approaches, which are suitable for

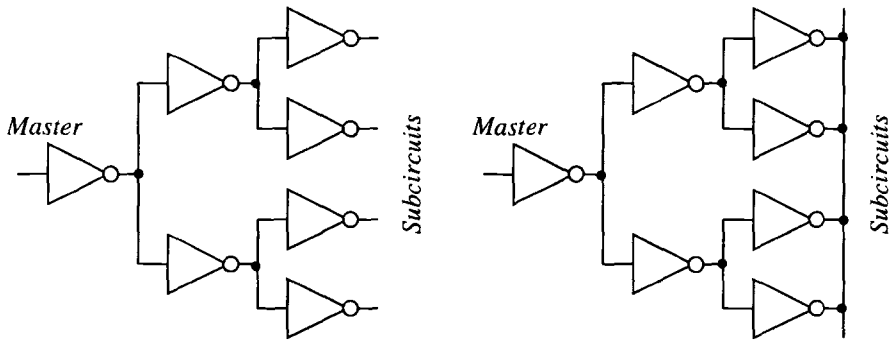


Figure 13-11 Clock tree structures to improve buffering and equal clock delay to the circuit.

high-speed and high-accuracy. Both clock trees improve the skew compared to a single-line clock distribution. The configuration in Fig. 13-11a) improves speed since it has a low capacitive load. The tree in Fig. 13-11b) has a higher capacitive load due to the additional wire capacitance, but the skew can be minimized if there are mismatch between the different inverters. The trees can be optimized for maximum speed or optimum clock edge behavior, and the tapering factor is determined by the speed requirements and process parameters.

As we have seen in the previous chapters, to improve monotonicity, to reduce the glitching, and relax the design requirements, segmented DAC structures are preferred. The most significant bits are thermometer coded, hence a binary to thermometer encoder is needed. A binary code consisting of M bits is converted into a thermometer code of $2^M - 1$ bits.

This encoding can be realized by expanding the boolean expressions and using logic. This approach becomes tedious for a higher number of input bits since the expressions and the number of outputs are increasing exponentially. Another method is similar: We use an iterative approach, since an M -to- $(2^M - 1)$ binary-to-thermometer encoder can be constructed using an $(M - 1)$ -to- $(2^{M-1} - 1)$ encoder as illustrated in Fig. 13-12. The logic depth of this encoder is equal to $M - 1$ and the propagation delay may limit the maximum achievable speed. There are 2^{M-1} AND and 2^{M-1} OR gates. With pipelining we ensure that a high sampling frequency can be reached. Due to the propagation delay through the encoder, the binary coded LSBs have to be delayed.

In mixed analog/digital designs, such as segmented DACs that have a larger digital portion, low power consumption for digital circuits and low switching noise are attractive features. Disturbances from the digital to the analog part spread along supply lines and the substrate. The substrate coupling may be strong depending on the substrate doping. It is therefore necessary to do careful designs with proper shielding, which is

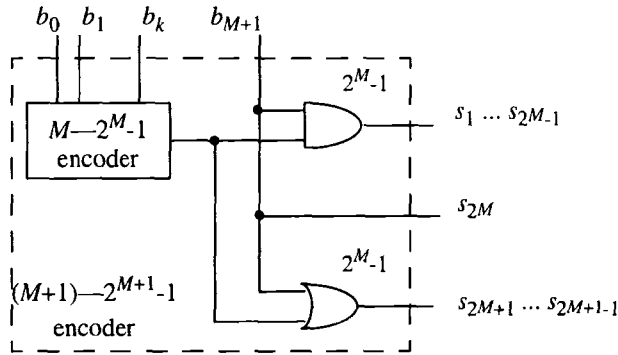


Figure 13-12 Iterative implementation of a binary-to-thermometer encoder. Note that there are $2^k - 1$ AND and OR gates.

done in several different ways [19], i.e., guard rings, grounding, etc.

It is also necessary to shield analog signals and separate analog and digital supplies [20]. The analog pins should be placed from the digital as far as possible. Grounding pins as shields should be used between analog and digital pins, etc.

13.3.4 Current Source Calibration

Some special techniques, such as calibration or trimming techniques, can be used to minimize the influence of mismatch in current-steering DACs [21, 22, 23].

Consider PMOS transistors as current sources in the DAC (Fig. 13-6). One technique is to not use the same bias voltage for all current sources. Instead we use the principle of holding the source-gate voltage, V_{SG} , during a period of time by using the charge stored on the source-gate capacitance, C_{sg} , as illustrated in Fig. 13-13. This implies that we have to charge the capacitance with a proper voltage. This is done during a calibration phase when the capacitance is charged by driving a reference current, I_{ref} , through the current source. The source-gate voltage will be set by the reference current according to the current formula (13-2) and transistor size errors of the transistor will not affect the output current, since V_{SG} is dependent on the reference current only. During the operation phase, as shown in the figure, the output current is determined by the held source-gate voltage. Due to leakage currents through the transistors and CFT, the true value cannot be guaranteed for a longer period of time and the source-gate voltage has to be refreshed on a regular basis.

This technique can be used for the thermometer coded MSBs in a segmented DAC or for the current sources in an R-2R ladder structure [21], where the current sources all are equally large. The advantage of this technique is that the influence of matching errors can be significantly reduced [17, 22]. The disadvantages are that additional switches are needed, i.e., hence a larger load and layout complexity, and for high res-

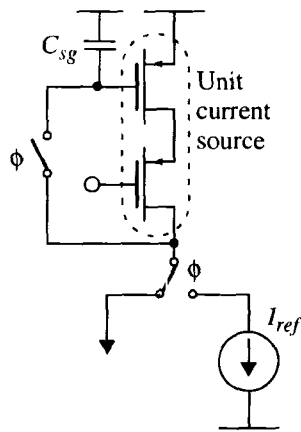


Figure 13-13 Example on circuit solution to calibrate the unit current sources during specific calibration phases.

olutions a large number of current sources has to be calibrated. This also implies that the calibration phase may need a long time, which may not be possible for certain high-speed applications. This requirement can be relaxed by using additional current sources which can be calibrated when other current sources are in operation.

13.4 A CMOS CURRENT-STEERING DAC CHIPSET

In this section we discuss the design and implementation of a CMOS DAC chipset for wideband applications. The chip's supply voltages range from 1.5 V to 5 V. The nominal resolutions are 10 bits up to 14 bits and the update frequency is specified from 10 MHz up to 100 MHz. The bandwidth should be in the range of 1 to 20 MHz. All DACs are segmented current-steering structures with the four MSBs thermometer coded. The output currents are terminated over a $50\text{-}\Omega$ load.

As a comparison, we briefly describe the implementation of three different converters in the chipset and to simplify the reference to the different converters throughout the text, we use the notations DAC A, B, and C as summarized in Table 13-1.

Measurement results and conclusions of the chip set are presented in Sec. 13.5 and we highlight the results from a closer comparison between two of the designs (A and B), where the current sources and interconnection slightly differ.

The knowledge about the fundamental limits given by the behavioral-level models in Chapter 12, was used when designing the DACs. Knowing how the output current affects transistor sizes, output impedance, and noise, lets us choose a proper value, etc. However, no process information on matching of transistors as in (13-10) was available.

The chip floor plan of the converters [24, 25, 26] is shown in Fig. 13-14 and from the floorplan we identify three different parts; the digital part consisting of clock, flip-

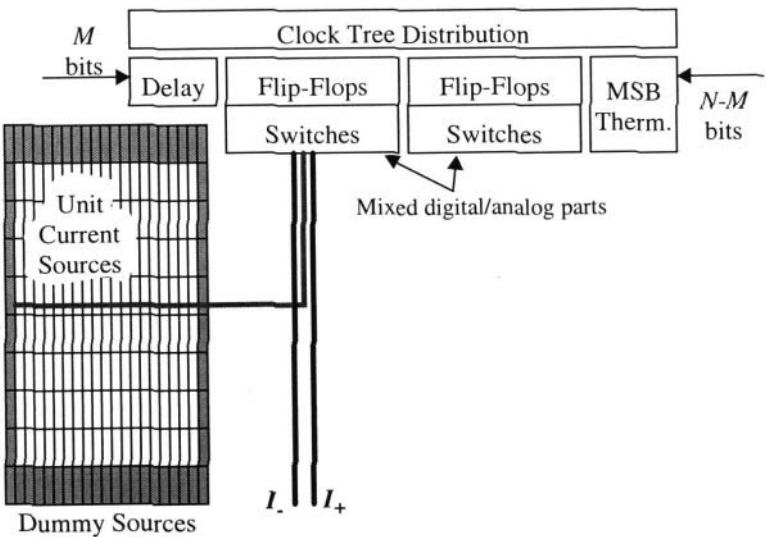


Figure 13-14 Chosen DAC segmented structure.

Table 13-1 Brief information on the DACs of the chipset.

DAC	Description
A	Segmented structure. 3.3 - 5 V supply. 14-bit resolution. 0.6um CMOS. 20 mA peak current outputs. Double and Single cascode PMOS transistors as current sources.
B	Segmented structure 3.3 - 5 V supply. 14-bit resolution. 0.6um CMOS process. 20 mA peak current outputs. Double cascode PMOS transistors as current sources.
C	Segmented structure 1.5V supply. 10-bit resolution. 0.6um CMOS process. 10 mA peak current outputs. Single cascoded PMOS transistors as current sources.

flops latching the digital input, a 4-bit binary-to-thermometer encoder for the MSBs, and a delay block for the LSBs to equalize the delay in the encoder. The analog part consists of an array of unit current sources and bias current sources, and finally the mixed analog/digital parts consisting of the current switches.

For the i -th LSB, 2^{i-1} unit current sources from the array are connected together and their output is fed to the current switch controlled by the i -th bit. The outputs are two currents and if the corresponding bit is a “1” or “0”, the current is routed to the positive, I_+ , or the negative output, I_- .

When using a thermometer code for the 4 MSB current sources in a 14-bit DAC, they are encoded into 15 equally large current sources, each one containing 1024 unit current sources. When encoding 5 MSBs we would have 31 equally large current sources each with 512 unit current sources. The segmentation introduces a delay for the MSBs

through the encoding tree and to avoid skew between all bits, a digital delay line is used for the LSBs. A clock tree as shown in Fig. 13-11a) is used to get proper clock delay and guaranteeing an alignment of the data to all flip-flops controlling the current switches.

13.4.1 Unit Current Sources and Source Array

In the chipset, both double- and single-cascode PMOS transistors were used as unit current sources. For a 5-V supply double-cascode were used and for the 1.5-V supply, single-cascode transistors were used. The layout view of a double-cascode current source is shown in Fig. 13-15. To simplify the layout and routing, all transistors have

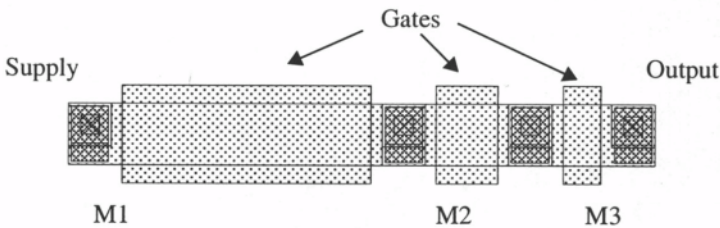


Figure 13-15 Layout view of the PMOS current source with a single cascode.

the same width. For DAC A and B the transistor sizes were given by the equal width $W = 2 \mu\text{m}$ and the lengths $L_1 = 8 \mu\text{m}$, $L_2 = 2 \mu\text{m}$, $L_3 = 1.2 \mu\text{m}$ for the transistors M1, M2, and M3, respectively. For the single-cascode source, only M1 and M2 were used. Double-cascode transistors were used to guarantee a very high output impedance needed for the 5-V 14-bit converters, and the simulated output resistance (at DC) of the double-cascode unit current source is approximately $R_{unit} \approx 30 \text{ G}\Omega$. The single-cascode unit current source has an approximate output resistance of $R_{unit} \approx 1 \text{ G}\Omega$.

The SFDR dependent on the output resistance will be determined by the odd harmonics and we can use equations (12-42) - (12-44) to find the expected SFDR performance. For a single-ended output, equations (12-38) - (12-41) apply.

In the DACs, the binary weighted current sources were formed by using unit current sources from the array by starting in one corner and use one unit current source for the LSB, the next two adjacent unit sources for the second LSB, etc. In the 14-bit DAC the array consists of 258×66 unit current sources. All edge sources are dummies and hence only used to increase the edge matching of the internal unit current sources. This layout technique was used to optimize the use of the chip area, although the matching properties are not the best, since the influence of linear graded matching errors increases.

The voltage drop across the supply line for all current sources of the array has to be avoided [16] and therefore, a power supply plane is covering the whole current source array. In all converters of the chipset, a higher-level metal layer was used. To reduce

the influence of switching noise from the digital circuits shieldings are used between analog and digital parts.

In the 14-bit DACs, the bias circuit is a cascoded current mirror formed by 1024 unit current sources. The reference current is determined by an off-chip resistor. The unit current sources used in the bias circuit are chosen from four adjacent rows of the current source array.

13.4.2 Current Switches

The layout of the current switch for a current source is shown in Fig. 13-16. The same

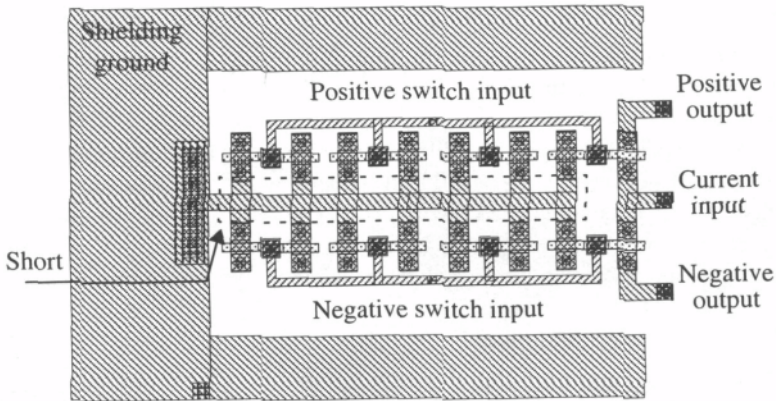


Figure 13-16 Layout view of a differential current switch.

switch is used for all weighted current sources to achieve equal capacitive load. For the LSB (as shown in the figure) only one of the NMOS transistors are used for each channel. The other transistors are shorted. For the MSBs, i.e., thermometer coded bits, all transistors are used in order to reduce the on-resistance of the switch. A shielding metal ground plane, nwell field, and substrate contacts around all switches are used to reduce noise coupling.

The simulated switch-on resistance for different switches and different supply voltages are summarized in Table 13-2 for a standard digital 0.6 μm CMOS process. In the simulations we have used the specified DC operating points such as the current flowing through the switches and the voltage outputs, which are 0.5 V for the high-voltage DACs A and B and 0.25 V for the low-voltage DAC C. The DC current through the switches are different for different switch sizes, i.e., LSB current for the smallest switch, etc. The switch on-resistance increases with a lower supply voltage since the gate-source voltage is reduced. In the 10-bit 1.5-V DAC C, the switching voltage was increased by using the charge pump circuit as discussed in Sec. 8.4.1.

13.4.3 Digital Circuits

The digital circuits are constructed in a straight forward way. The 4-to-15 binary-to-

Table 13-2 Switch-on resistance in Ω compared to the voltage supply and number of transistors.

Supply voltage [V]	Number of active NMOS transistors			
	1	2	4	8
1.5	5.25k	2.62k	1.30k	650
2.7	2.42k	1.21k	610	636
3.0	2.09k	1.05k	520	393
3.3	1.87k	930	470	314
5	1.36k	680	340	170

thermometer encoder was constructed by expanding the boolean function describing its 15 output values as functions of its inputs. The gates are realized with unlocked CMOS circuits. The total delay through the encoder is given by four gates with inverter buffers at the output. The difference in the delay is small enough to guarantee over 100 MHz operation. The LSBs are delayed with four cascaded inverters and aligned. All the data are latched before driving the bit switches. A tree-formed clock distribution (similar to the structure in Fig. 13-11a, but with a tapering factor of 3) is used to guarantee as small clock skew as possible to the current switches.

13.4.4 Chip Implementations

The chip layouts of the converters are similar as illustrated in Fig. 13-14. In Fig. 13-17 we show a die photograph of DAC A. The core size is approximately $2 \times 2 \text{ mm}^2$ and including pads, the chip area is $3.0 \times 3.4 \text{ mm}^2$. From the die photo in Fig. 13-17 we also see that by rotating the current source array 90° we can reduce the wire lengths and the core area significantly. In Table 13-3 in Sec. 13.5.1 the chip data is summarized.

13.5 CHIPSET MEASUREMENTS

There are several possible ways of measuring the DACs. However, since we are considering high-speed, high-resolution converters, the best way to measure is to use a low-jitter input data generator, a spectrum analyzer, and a high-bandwidth oscilloscope. The same principle holds for computer simulation of the converters. To perform static measurements, such as DNL or INL measurements of a 14-bit DAC with $2^{14} = 16384$ amplitude levels a lower-speed higher-resolution digitizer is used. The digital data is captured and analyzed by a computer, further reducing time and complexity [27]. This is not practical for communications and instead we apply single-tone or multi-tone inputs and examine the output spectrum. For full-scale single-tone measurements, an update frequency that is relatively prime with the signal frequency is used. Otherwise, all information about the converter's performance cannot be fully extracted. Using a prime frequency ratio also ensures that the distortion terms are not folded back onto the fundamental and interfering with other frequency components and causing inaccuracy in the measurement. For dual-tone measurements the signal

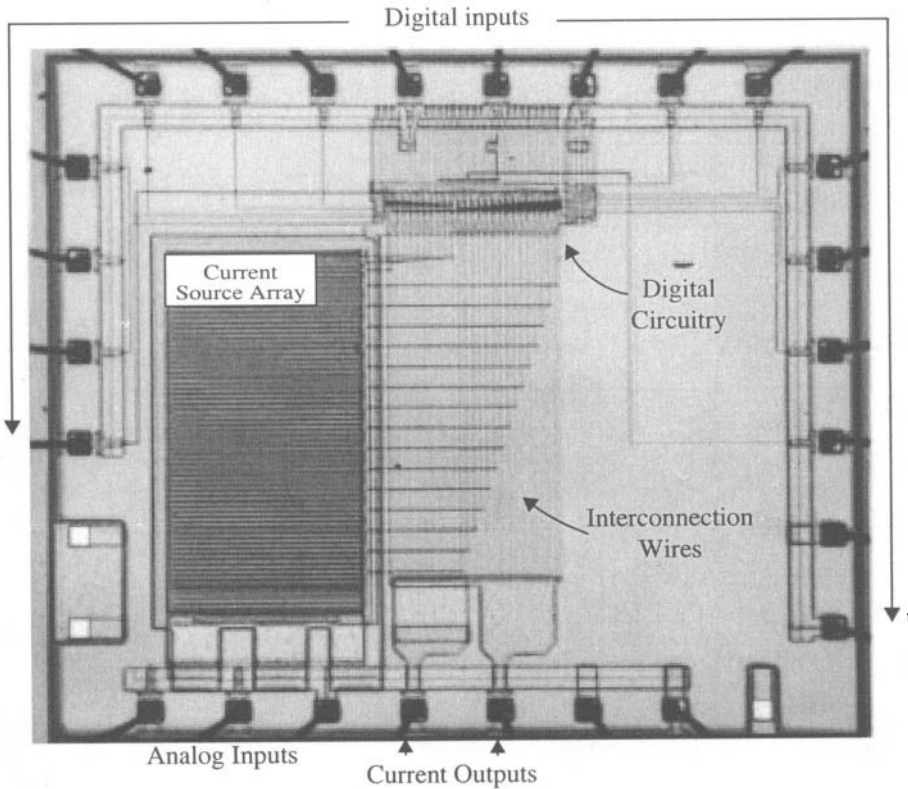


Figure 13-17 Die photograph of a 14-bit current-steering CMOS DAC (DAC B).

frequencies are also chosen to be relatively prime, and they are relatively prime with respect to the update frequency as well. For multi-tone measurements we apply a number of tones with frequencies at multiples of a fundamental frequency according to an orthogonal frequency-division multiplexing (OFDM) or discrete multi-tone (DMT) modulation.

13.5.1 Measurement Results

We present different measurement results from the converters of the chipset. No analog filtering has been applied to the output signal. The sample frequencies vary from 1 MHz up to 100 MHz.

- **Measurement of the 14-bit single- and double-cascode DAC A**

In Fig. 13-18 we show the measured output spectrum for the 14-bit single-cascode DAC A when applying a dual-tone input signal at a supply voltage of 5 V. The update frequency is 20 MHz and the signal frequencies are 3.43 MHz (f_1) and 3.51 MHz (f_2). The SFDR is 49 dBc and the IMD given by the $2f_1 - f_2$ frequency component is -54 dBc. The output is measured over a transformer, transforming the differential signal into a single-ended signal, i.e., even order harmonics are reduced and it is found

that the third harmonic limits the SFDR to 48 dBc.

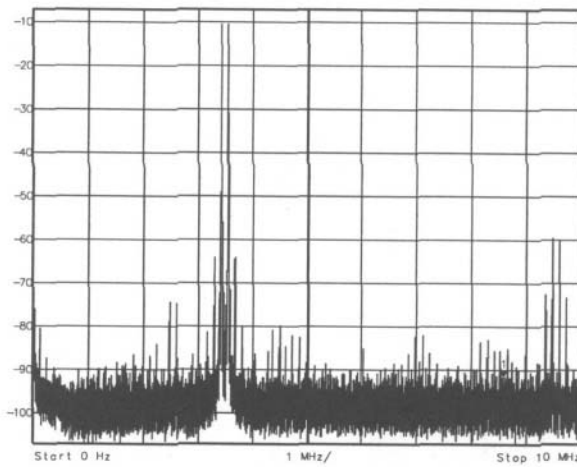


Figure 13-18 Dual tone measurement of a 14-bit double cascode DAC with a 5 V supply and an update frequency of 20 MHz.

From further measurements [31], we find that for the single-cascode 14-bit DAC A, the SFDR decreases from 49 to approximately 42 dBc when the update frequency changes from 5 to 50 MHz. The sample frequency is approximately 6 times higher than the signal frequency. The performance is poor and to find the limitations on performance we study the measurement results from DAC B and C and compare the layouts with the one of DAC A.

• *Measurement of the 14-bit double-cascode DAC B*

In Fig. 13-19 we show the summarized measurement results from DAC B. Single-tone signals at -3 dBFS are applied. The SFDR as a function of update frequency and signal frequency is shown. We find that the SFDR varies from more than 75 dBc at low frequencies down to 45 dBc for higher frequencies. In the figure we have also added measurement results from DAC A and the two lower curves (dashed-star for DAC A and solid-triangle for DAC B) are measured under the same conditions. We find that although the layouts are very similar, the improvement in SFDR is approximately 13 dB to 6 dB with increasing sample frequency and a signal frequency that is approximately 6 times lower than the sample frequency. However, in DAC B the interconnection wires have been made wider and shorter and the parasitic impedance was decreased. There were also changes to the layout of the unit current source. Internal interconnections that increase the capacitive part of the current source output impedance were removed. In Fig. 13-20 we illustrate this by displaying a partial layout view of the unit current source arrays in DAC A and B. The reduction of the internal capacitance increases the output pole frequency of the current source (compare with Fig. 13-7) and at higher frequencies we will have a higher $|Z_{out}/Z_{load}|$ ratio and therefore a higher linearity. Further, the short-distance matching is improved by removing vias and metal wires (supply connections) placed on top of the transistor gates.

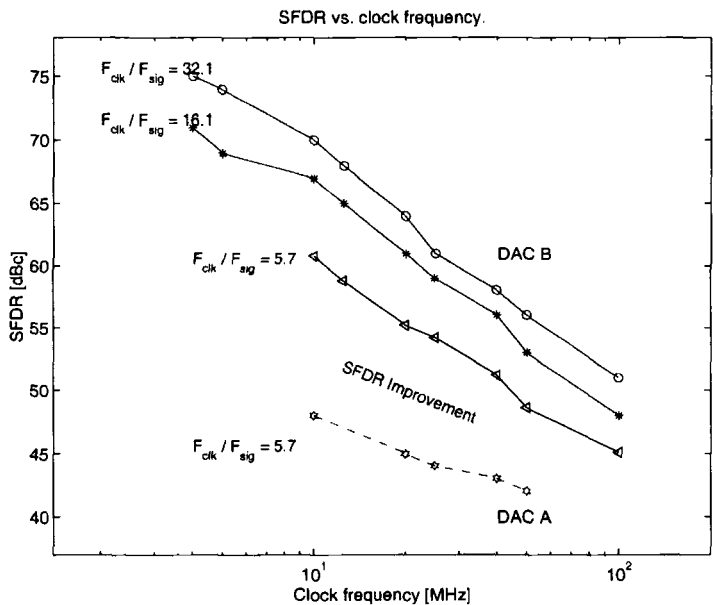


Figure 13-19 Summary of the measured result from DAC B and comparison of measured results from DAC A and B.

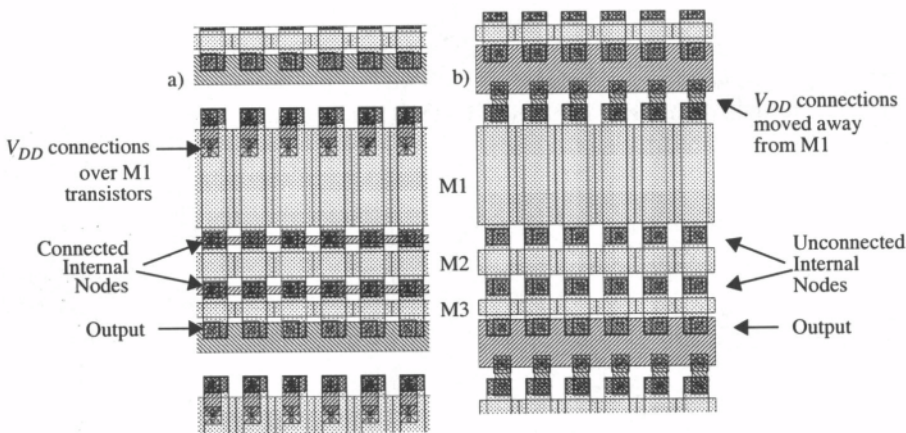


Figure 13-20 Part of current source array for DAC A (a) and DAC (b).

In Fig. 13-21 the measured output spectrum when applying an 18-tone signal to DAC B is shown. The tones have equal amplitude (-18 dBFS) and the phase of each carrier is randomized to reduce the PAR. The update frequency is 20 MHz, the supply voltage is 3.3 V, and the tone frequencies are centered around 2.2 MHz. We find the SFDR to be 57 dBc which in this case is approximately equal to the measured IMD.

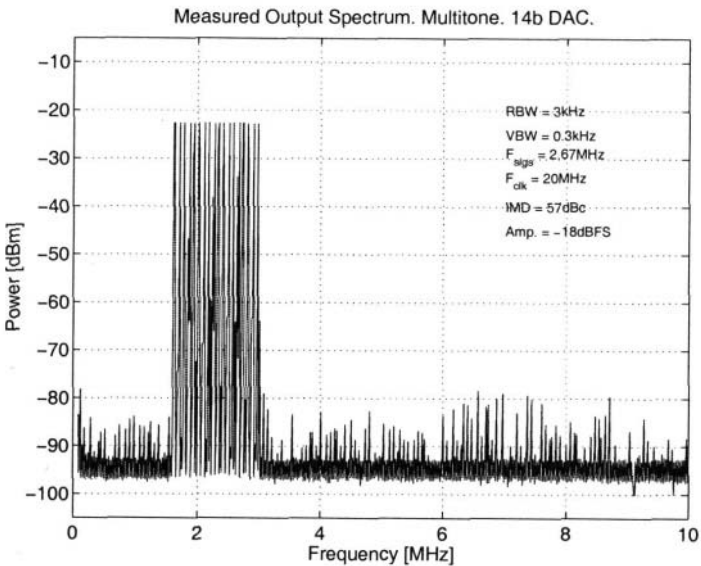


Figure 13-21 Multi-tone measurement of DAC B. 18 tones at -18dBFS are applied. The update frequency is 20 MHz.

• *Measurements of the low-voltage 10-bit single-cascode DAC C*

From the low voltage DAC C, we know that we can reach a rather high linearity. In Fig. 13-22 we show the measured single-ended output when applying a dual-tone input signal. The update frequency is 10 MHz and the signal frequencies are 1.775 MHz and 1.792 MHz. The measured IMD is approximately 52 dBc. The supply voltage is 1.5 V.

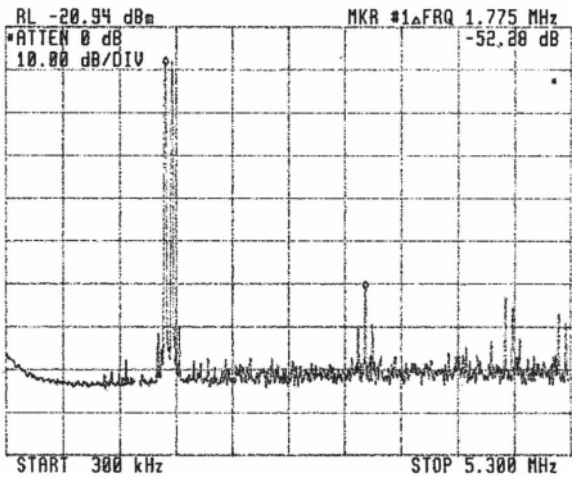


Figure 13-22 Measured dual-tone output from the 1.5 V DAC. Update frequency is 10 MHz and signal frequencies around 1.775 MHz.

In Fig. 13-23 we show the summarized single-ended measurement results from the 1.5V 10-bit DAC [18].

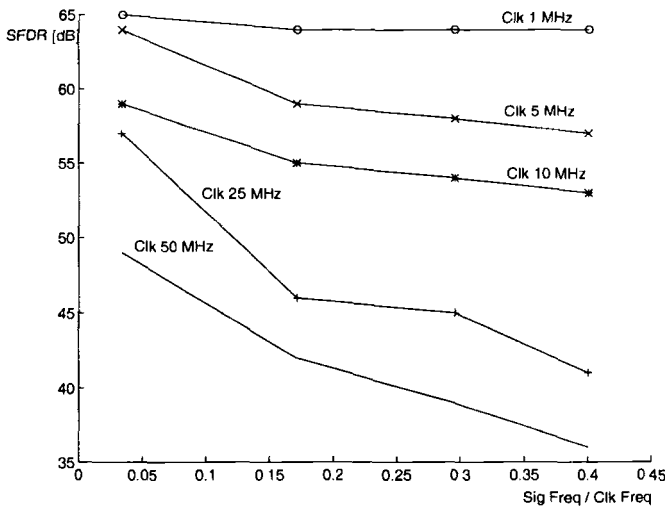


Figure 13-23 Summary of measured results from the 1.5 V DAC C.

We find that the SFDR is over 65 dBc at low frequencies and for high clock frequencies and a low clock to signal frequency ratio, the SFDR is decreasing rapidly. The improved performance over DAC A originates from the fact that the requirements on the output impedance and element matching are greatly reduced over the 14-bit versions of the converters.

13.52 Conclusions and Comparison

From the measurement results [31] we conclude:

- In DAC A and B there is no larger change in performance when changing the supply voltage from 5 V to 3.3 V.
- There are no larger variations between the single and double cascode implementations for higher frequencies. In Fig. 13-7 we see that the output impedance at higher frequencies is approximately the same for both current source versions. Hence we have to have a design for a higher pole. The bandwidth is rather low and also indicating the high capacitive part at the output of the current sources. When increasing the pole frequency we have to lower the output resistance.
- The performance can be improved with a more careful layout with respect to linear graded matching errors.
- Time-domain simulation results show low glitching due to good clock distribution and segmentation of the MSBs.

In Table 13-3 we summarize the results from the design of the chipset DACs.

Table 13-3 Chip data for the different DACs of the chipset.

DAC	Supply [V]	Nominal resolution [bit]	Chip area [mm ²]	Measured results
A	3.3 - 5	14	2x2 (core)	SFDR = 49 dBc at 10 MHz update frequency and 1.7 MHz full-scale sinusoid. 14-bit DAC.
B	3.3 - 5	14	2x2 (core)	SFDR = 75 dBc at 5 MHz update frequency and 156 kHz full-scale sinusoid. 14-bit DAC. SFDR = 61 dBc at 10 MHz update frequency and 1.7 MHz full-scale sinusoid. 14-bit DAC. IMD = 57 dBc at 20 MHz update frequency and 18-tone -18 dBFS at frequencies around 2.2 MHz.
C	1.5	10	2x2 w. pads	SFDR = 64 dBc at 1 MHz update frequency and 400 kHz full-scale sinusoid. SFDR = 57 dBc at 10 MHz update frequency and 1.7 MHz full-scale sinusoid. IMD = 52 dBc at 10 MHz update frequency and dual-tone frequencies around 1.7 MHz.

13.6 SUMMARY

In this chapter, we have discussed the design and implementation of current-steering DACs for wideband applications. Different structures have been outlined and for high-speed and high-resolution applications we have found the segmented DAC structure to be most suitable. A key design issue is to find the proper number of bits encode into a thermometer code or if multi-segmented structures are needed. We have shown how the performance of the converter depends on errors in the layout by comparing two similar DACs and highlighted the importance of high frequency poles in the output impedance of the unit current sources. We have shown that it is possible to reach an SFDR of over 75 dBc in a standard 3.3 V digital CMOS process and an SFDR of 65 dBc with a single 1.5-V supply voltage.

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